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Peter Hintz

An Introduction to Microlocal Analysis

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Peter Hintz

An Introduction to Microlocal Analysis

Springer

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To my wife and sons: Sara, Georg, and Lukas

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Preface

At this point in time, microlocal analysis is, by and large, no longer a field of research for its own sake, but rather a substantial collection of tools accompanying a powerful, versatile, and modular perspective on the analysis of linear partial differential equations (PDEs). It has found many striking applications in fields as diverse as general relativity, chaotic dynamical systems, spectral asymptotics, and inverse problems, to name just a few.

A particular focus of microlocal analysis lies on the study of the *singularities* of solutions of PDEs. Unlike local analysis, which is concerned with the regularity of distributions near points of particular interest, *micro*local analysis is concerned with regularity *in phase space*, i.e., in position *and momentum*. Phase space is also where classical Hamiltonian mechanics takes place; and every linear partial differential operator gives rise to a Hamiltonian ("energy function") on phase space. Microlocal analysis provides tools to translate information about the underlying classical mechanical system into estimates for the PDE. The foundational theorem by Hörmander on the propagation of singularities, proved in Chapter 8, is a beautiful example of this interplay: singularities of solutions of PDEs travel along trajectories of the underlying classical system. The main tool of microlocal analysis is the algebra of *pseudodifferential operators* (ps.d.o.s). Ps.d.o.s allow one to localize distributions in phase space. They also arise independently as inverses or parametrices (approximate inverses) of elliptic partial differential operators. Almost the entire first half of this book is devoted to a complete and detailed development of the theory of ps.d.o.s on Euclidean space and on manifolds.

I illustrate the utility of nearly all of the microlocal methods developed in this book in a "real-world" scenario by giving a full proof of the late-time expansion of solutions to linear wave equations on the de Sitter spacetime in Chapter 11. Many older and newer topics are beyond the scope of this book, including boundary value problems, applications to quasilinear PDEs and estimates near trapped sets (relevant especially in general relativity), microlocal analysis on noncompact and singular spaces (bounded geometry, geometric microlocal analysis, b-calculus), the propagation of singularities

and diffraction of waves on singular spaces (boundaries, corners, cones, edges), Fourier integral operators, spectral asymptotics, semiclassical defect measures, variable order spaces, analytic microlocal analysis, paradifferential techniques, etc.

The development of microlocal analysis in this book starts essentially from scratch. The only prerequisite is some basic familiarity with functional analysis. It is moreover useful to have some prior exposure to the theory of smooth manifolds, though a rapid review of the main definitions and constructions is included in Section 5.2. While at its heart microlocal analysis is a highly conceptual and modular approach to the study of PDEs, it requires some nontrivial amount of effort to develop the technical underpinnings. In this book, almost every result starting with Chapter 2 is proved with full details. I hope that this makes the treatment as accessible as possible. I have, at the same time, tried to emphasize the core ideas and main results throughout the text.

An extensive list of exercises rounds off each chapter. Some exercises ask the reader to verify certain statements made in the main text or to provide details of omitted arguments. Other exercises ask the reader to work out examples of definitions or theorems. Some exercises are more demanding and ask the reader to prove, with some support, interesting results that are frequently used in the literature but not in this book.

This book is a thoroughly revised and expanded version of a set of lecture notes that I had initially written for my graduate-level course *18.157* at MIT in the spring semester of 2019 and its repetition in 2020, and again in 2021 at ETH Zürich. A considerable part of the book can still be reasonably covered in a semester-long course for beginning graduate students, depending on what the aims and the focus of the lecturer are. For example, a natural goal would be to describe the basic theory of ps.d.o.s and wave front sets in detail and to give a proof of the propagation of singularities and sketch some applications, skipping all sections on semiclassical methods.

Later parts of the book, especially Chapters 10–11, cover more advanced topics that particularly energetic lecturers may well get to, at least in parts. I hope that in any case they will serve as a jumping board for graduate students and researchers who wish to delve deeper into the microlocal realm and also as a point of reference due to the level of detail provided here. Notes at the end of each chapter mainly provide references to original works and alternative treatments.

There exist many excellent lecture notes and textbooks on topics related to microlocal analysis, all with different foci. Among the classics, Hörmander's famous treatise [Hör03, Hör05, Hör07, Hör09] goes far beyond what is presented here, but for the most part it is certainly more of a reference work than a graduate-level textbook. The radial point estimates of Chapter 10, which have been playing an increasingly central role in scattering theory since Melrose [Mel94] and Vasy [Vas13], are, however, not covered there. Taylor's remarkable books on linear and nonlinear PDEs

[Tay81, Tay11a, Tay11b, Tay11c] cover a vast amount of material, making heavy use of pseudodifferential techniques; but microlocal analysis, in the narrower sense of analysis of singularities in phase space, plays only a minor role. Other important textbooks include Shubin [Shu87] (with a focus on spectral asymptotics), Alinhac–Gérard [AG07] (introducing ps.d.o.s on $\mathbb{R}^n$ and rapidly moving on to applications to nonlinear PDEs using fixed point theorems), and Grigis–Sjöstrand [GS94] (emphasizing Fourier integral operators and symplectic geometry).

More recent textbooks with a semiclassical microlocal flavor (involving a parameter $h \to 0$, oftentimes an inverse frequency) are those by Zworski [Zwo12] on semiclassical analysis and Dyatlov–Zworski [DZ19] on scattering theory. Unlike in the present book, where semiclassical notions are developed only insofar as they are simple variants of their classical counterparts, these two books push semiclassical methods very far for the purpose of studying spectral and scattering asymptotics at high energies. The present book contains, in particular, a detailed exposition of the material collected in [DZ19, Appendix E], and Chapter 11 is closely related to [DZ19, §5.7].

Melrose's lecture notes [Mel07] on the subject influenced my presentation, especially in the earlier parts of the book. I also drew inspiration from lecture notes by Wunsch [Wun13] and Vasy [Vas18]. The latter develop the scattering pseudodifferential calculus in great generality, with applications to scattering theory on asymptotically Euclidean spaces. In particular, propagation and radial point estimates are discussed there as well.

I had the good fortune to be exposed to microlocal analysis already during my undergraduate studies at the University of Göttingen in Prof. Dr. Ingo Witt's fantastic lecture cycle on PDEs, parts of which were strongly inspired by Hörmander's treatise. The strongest influence on me has undoubtedly been my Ph.D. advisor at Stanford University, András Vasy, whose impact on the content and mathematical style in this book will be obvious to everyone familiar with his approach to PDEs. I would like to extend my profound gratitude to him for sharing his ideas, perspectives, and insights throughout the years. I also learned a great deal from many other mentors, teachers, and colleagues, including Semyon Dyatlov, Jeffrey Galkowski, Rafe Mazzeo, Richard Melrose, Gunther Uhlmann, Jared Wunsch, and Maciej Zworski.

I am grateful to the participants of my courses and also to many colleagues for suggestions and corrections, including Yonah Borns-Weil, André Dabrowski, Jesse Gell-Redman, Vincent Hoffmann, Jure Kališnik, Cosmin Manea, Aaron Moser, Pieter-Bart Peters, Phillip Pflaum, Jossi Schütt, Franziskus Steinert, Thomas Stucker, Ethan Sussman, and Tobias Weich. I would like to extend special thanks to Jared Wunsch: firstly, he had the courage to base part of a course he taught at Northwestern University on a preliminary version of these notes; his subsequent feedback led to considerable improvements in clarity and precision. Secondly, it was he who suggested to me to turn my original lecture notes into a book.

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Declarations

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Chapter 1
Introduction



1.1 What Is Microlocal Analysis?

Microlocal analysis is concerned with the study of distributions—in particular, singularities of distributions or their absence—in *phase space*.

The main notion is that of the *wave front set* of a distribution. We illustrate this with two examples.

(1) Consider the function $u(x) = |x|^\alpha$ on $\mathbb{R}^n$ for $\alpha \in (-n, 0)$. Then $u \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ is a well-defined distribution, and u is in fact smooth at all nonzero x. However, u is not smooth at $x = 0$. Since u is rotationally symmetric, one would like to say that u should be singular at $x = 0$ *in all directions*. The δ-distribution has the same features.

(2) Let $\Omega \subset \mathbb{R}^n$ be a smoothly bounded domain, and let $v = 1_\Omega$ be the characteristic function of Ω. This function is smooth on Ω and $\mathbb{R}^n \setminus \bar{\Omega}$ and singular at $\partial\Omega$. However, at a point $x_0 \in \partial\Omega$, the function v does not behave the same way in all directions at x_0: in directions tangent to $\partial\Omega$, v is not particularly irregular, but in directions normal to $\partial\Omega$, v has a jump discontinuity.

Rather than working with "directions" (tangent vectors), one should really work with "co-directions" (cotangent vectors). Then the wave front sets of the two examples are

$$\mathrm{WF}(u) = \{0\} \times (\mathbb{R}^n \setminus \{0\}) = T^*_0\mathbb{R}^n \setminus o, \quad \mathrm{WF}(v) = N^*\partial\Omega \setminus o. \tag{1.1}$$

(Here $T^*\mathbb{R}^n = \mathbb{R}^n_x \times \mathbb{R}^n_\xi$ is the classical phase space, and $N^*\partial\Omega$ consists of all (x, ξ) with $x \in \partial\Omega$ and ξ conormal to $\partial\Omega$. Since we care about co-directions, we exclude $o = \{(x, \xi) \colon \xi = 0\}$.)

The definition of the wave front set is so elegant that we sketch it right away: a point (x_0, ξ_0) does not lie in $\mathrm{WF}(u)$—i.e., u is *microlocally smooth*

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at position x_0 and in the momentum direction determined by $\xi_0 \neq 0$—if and only if there exists a smooth compactly supported function $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(x_0) \neq 0$ such that the Fourier transform $\mathcal{F}(\phi u)$ of ϕu decays rapidly in a cone centered around ξ_0. (See Proposition 6.21.) If u is smooth near x_0, then ϕu is smooth and compactly supported, and basic Fourier analysis tells us that its Fourier transform decreases rapidly in *all* directions. Thus, $\mathrm{WF}(u)$ is a refinement of the *singular support* of u, the set of singularities of u in x-space.

The main tool for microlocal analysis is the algebra of *pseudodifferential operators* (ps.d.o.s). Much like multiplication by a function $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ allows one to localize a distribution u in configuration space (i.e., in the x-variables), a ps.d.o. allows one to *microlocalize* distributions *in phase space*: following localization in space, one localizes in momentum directions on the Fourier transform side. That is, for $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ and $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$, we map a distribution u on $\mathbb{R}^n$ to

$$v := \mathcal{F}^{-1}\Big[\psi\Big(\frac{\xi}{|\xi|}\Big)\mathcal{F}(\phi u)(\xi)\Big]. \tag{1.2}$$

Then $\mathrm{WF}(v) \subset S := \mathrm{supp}(\phi(x)\psi(\frac{\xi}{|\xi|}))$, i.e., v is *microlocalized* in S. Writing out the (inverse) Fourier transform in (1.2) expresses v as

$$\mathrm{Op}(a)u(x) = (2\pi)^{-n} \iint e^{i(x-y)\cdot\xi} a(x,y,\xi)u(y)\,\mathrm{d}y\,\mathrm{d}\xi \tag{1.3}$$

for $a(x,y,\xi) = \psi(\frac{\xi}{|\xi|})\phi(y)$. The class of operators $\mathrm{Op}(a)$ includes, besides microlocalization operators also, all differential operators (namely, when a is a polynomial in ξ). For functions a which are *symbols* (encoding certain regularity in x, y and bounds in ξ), $\mathrm{Op}(a)$ is a *pseudodifferential operator*.

While the above discussion takes place on $\mathbb{R}^n$, the notions of wave front sets and ps.d.o.s can be transferred to arbitrary smooth manifolds.

Typical theorems in microlocal analysis give information about $\mathrm{WF}(u)$ for solutions u of linear partial differential equations

$$Pu = f \tag{1.4}$$

in terms of $\mathrm{WF}(f)$ and properties of (the symbol of) P. As a first example, consider an *elliptic* operator P. (Examples include the Laplace operator on $\mathbb{R}^n$ or the Cauchy–Riemann operator on $\mathbb{C} = \mathbb{R}^2$.) Then $\mathrm{WF}(u) = \mathrm{WF}(f)$. Thus, u is microlocally smooth wherever f is. This is a sharp version of standard elliptic regularity (if f is smooth, then so is u). More generally, consider a differential operator

$$P = \sum_{|\alpha| \le m} p_\alpha(x) D^\alpha, \quad D := \frac{1}{i}\partial, \tag{1.5}$$

and define its *principal symbol* by

$$p(x,\xi) = \sigma^m(P)(x,\xi) := \sum_{|\alpha|=m} p_\alpha(x)\xi^\alpha; \tag{1.6}$$

it captures the leading order part of P. A simple non-elliptic example is the transport operator $P = D_{x_1} - D_{x_2}$, with $p(x,\xi) = \xi_1 - \xi_2$. Then the aforementioned elliptic regularity result can be generalized significantly to the *microlocal elliptic regularity* statement

$$\mathrm{WF}(f) \subset \mathrm{WF}(u) \subset \mathrm{WF}(f) \cup \mathrm{Char}(P), \quad \mathrm{Char}(P) := p^{-1}(0) \cap T^*\mathbb{R}^n \setminus o. \tag{1.7}$$

The *characteristic set* $\mathrm{Char}(P)$ encodes the locations in phase space where P is *not* elliptic and where elliptic regularity does not apply. In the case of the transport operator, $\mathrm{Char}(P)$ is a codimension 1 hypersurface, so P is in fact elliptic at most places!

A natural question is then what one can say about the properties of $\mathrm{WF}(u)$ inside of $\mathrm{Char}(P)$. For the transport operator P, for which one can produce a formula for u in terms of f by integration, one expects that singularities follow integral curves of the vector field $H_p := \partial_{x_1} - \partial_{x_2}$; and indeed $\mathrm{WF}(u) \cap \mathrm{Char}(P)$ is invariant under the flow of this vector field when f is smooth. (Singularities of f in $\mathrm{Char}(P)$ can create or destroy singularities of u.) For a more advanced example, consider the wave operator $P = -D_{x_1}^2 + \sum_{j=2}^n D_{x_j}^2$, whose principal symbol $p(x,\xi) = -\xi_1^2 + \sum_{j=2}^n \xi_j^2$ vanishes on the collection $\mathrm{Char}(P)$ of light cones in each fiber of $T^*\mathbb{R}^n \setminus o$. Explicit solution formulas suggest that singularities follow straight lines in directions $(\xi_1, \dots, \xi_n)$ with $|\xi_1| = |(\xi_2, \dots, \xi_n)|$. The precise statement is as follows: for smooth f,

$$\begin{aligned} &\mathrm{WF}(u) \cap \mathrm{Char}(P) \text{ is invariant under the flow of the vector field} \\ &\quad H_p \text{ in phase space;} \end{aligned} \tag{1.8}$$

here $H_p := -2\xi_1\partial_{x_1} + 2\sum_{j=2}^n \xi_j \partial_{x_j}$. Hörmander's remarkable theorem on the *propagation of singularities* states that this is true for general operators P with real principal symbol when H_p is defined to be the *Hamiltonian vector field* associated with p—this is our second example of a typical microlocal theorem. The remarkable connection between a partial differential operator P and the associated classical mechanical system $(T^*\mathbb{R}^n, H_p)$ is at the heart of much of the microlocal theory developed in this book.

1.2 Main Themes

We will first develop the theory of pseudodifferential operators (ps.d.o.s) before turning to microlocal analysis proper.

Theme I. Pseudodifferential Operators as Generalizations and Parametrices of Differential Operators. Even without microlocal motivation, ps.d.o.s arise naturally in the study of differential operators. As a simple example, consider the second-order differential operator

$$P = \Delta + 1, \quad \Delta = \sum_{j=1}^{n} D_{x_j}^2, \tag{1.9}$$

on $\mathbb{R}^n$. Acting on $L^2(\mathbb{R}^n)$ or the space $\mathscr{S}(\mathbb{R}^n)$ of Schwartz functions, P is invertible, and its inverse can be found using the Fourier transform: in view of $Pu = \mathcal{F}^{-1}(|\xi|^2 + 1)\mathcal{F}u$, we have

$$P^{-1}f = \mathcal{F}^{-1}(|\xi|^2 + 1)^{-1}\mathcal{F}f. \tag{1.10}$$

Writing this out explicitly shows that

$$P^{-1} = \mathrm{Op}(a), \quad a(x, y, \xi) = (|\xi|^2 + 1)^{-1}, \tag{1.11}$$

in the notation of (1.3); this is a ps.d.o. of order -2.

While in this example, P is an operator with constant coefficients, variable order operators can be handled easily: if P as in (1.5) is elliptic, i.e., its principal symbol $p(x, \xi)$ (see (1.6)) does not vanish for $\xi \neq 0$, then $Q = \mathrm{Op}(q)$, $q(x, \xi) := \chi(\xi)p(x, \xi)^{-1}$, is an approximate inverse (*parametrix*) of P where $\chi \in \mathcal{C}^\infty(\mathbb{R}^n)$ equals 0 near $\xi = 0$ and 1 for $|\xi| \geq 1$. (By parametrix, we mean here that the application of $I - QP$ and $I - PQ$ on a distribution gains regularity.) Note that $\mathrm{Op}(\chi(\xi)p(x_0, \xi)^{-1})$ is an approximate inverse of the operator $\sum_{|\alpha|\leq m} p_\alpha(x_0)D^\alpha$ arising from P by freezing its coefficients at x_0. Therefore, $\mathrm{Op}(q)$ assembles all of the "local" (approximate) inverses of the frozen coefficient models of P into a single operator.

Ps.d.o.s thus arise naturally as (approximate) inverses of certain elliptic differential operators. The utility of knowing that an operator can be (exactly or approximately) inverted by a ps.d.o. lies in the fact that ps.d.o.s define bounded maps between useful function spaces: besides spaces of smooth functions and distributions, this includes the L^2-based Sobolev spaces $H^s(\mathbb{R}^n)$ which we shall use pervasively in this book (Section 4.8).

A particularly clean setting is that of elliptic operators P on *closed* manifolds (compact, no boundary): such P are *Fredholm*, i.e., they have finite-dimensional kernel and cokernel; moreover, their generalized inverses

are ps.d.o.s (Theorem 5.52). Note that ellipticity is a condition involving only the leading order part of P. This leads us to the second theme.

Theme II. Understanding P Through Its Principal Symbol. The algebra of (pseudo)differential operators is noncommutative. On the other hand, the *principal symbol* of a product (composition) of operators is the product of principal symbols. (This underlies elliptic regularity, i.e., (1.7).) Furthermore, the principal symbol of a commutator $[P, Q]$ is $\frac{1}{i}$ times the Poisson bracket of the principal symbols of P and Q. (This underlies the propagation statement (1.8).) Many of the results proved in this book depend only on information about the principal symbol of the operator in question. The Fredholm property of elliptic operators on closed manifolds is one example in which a global property of the principal symbol determines the mapping properties of an operator, concretely the solvability and uniqueness of solutions on spaces of smooth functions or on Sobolev spaces, up to finite-dimensional obstacles (Sections 5.9 and 5.11). Another example is given by the spectral family of wave equations on de Sitter space, whose Fredholm property follows from global properties of the flow of the Hamiltonian vector field H_p (Chapter 11).

The solvability of a partial differential equation $Pu = f$ of course depends on more than merely the principal symbol of P. For example, $\Delta_g u = 1$ cannot be solved on a compact Riemannian manifold (M, g) without boundary, but $(\Delta_g + 1)u = 1$ can (Proposition 5.53 and Example 5.54). Thus, the Fredholm property of an operator is the strongest statement one can expect to deduce purely from principal symbol-based arguments (unless there are additional small or large parameters present, as in semiclassical problems).

Theme III. Modular Nature of Microlocal Results. Microlocal results are concerned with the wave front set of u in subsets S of phase space, given that $Pu = f$. For the results proved in this book, the conclusions depend only on the behavior of P near S. For example, if $S \cap \mathrm{Char}(P) = \emptyset$ (i.e., P is elliptic at S), then $\mathrm{WF}(f) \cap S = \emptyset$ implies $\mathrm{WF}(u) \cap S = \emptyset$ (see (1.7)). This highly (micro)localized character of microlocal results means that they can be used as "black box" building blocks, often by concatenating one after another, when studying the singularities of u. This also brings us to the fourth theme.

Theme IV. Tracking Singularities. To give a rough idea of how the tracking of singularities is connected to the typical goal of a PDE analyst—namely, to *solve* partial differential equations—suppose we are interested in solving the PDE $Pu = f$ on a compact manifold M, with $f \in L^2(M)$, say. Suppose that u is a distribution of a priori low regularity (say Sobolev regularity $H^{-N}(M)$). *Suppose* one can deduce from this information that in fact $u \in H^s(M)$ for some $s > -N$, and indeed that an estimate

$$\|u\|_{H^s} \leq C(\|f\|_{L^2} + \|u\|_{H^{-N}}) \tag{1.12}$$

holds. (This is a quantitative version of the statement that u has no H^s-singularities; microlocal analysts might prove this by showing that the H^s wave front set of u is empty.) Then from this estimate, one can already deduce the finite-dimensionality of the kernel of P on H^s and also the closed range property on the space of $u \in H^s$ with $Pu \in L^2$ (see Proposition 5.74). If one can prove an analogous estimate for the adjoint P^* of P, then one gets the analogous statements for P^*, and by a duality/Hahn–Banach argument, the *solvability* of $Pu = f$. In short, *knowing about the (strength of) singularities* of solutions of $Pu = f$ and $P^*u^* = f^*$ implies being able to *solve the PDE*.

Theme V. Semiclassical Problems. We will describe semiclassical analogues of most notions and results. Semiclassical operators are h-dependent *families*, $h \in (0, 1)$, of operators of the form $\sum_{|\alpha|\leq m} p_\alpha(h, x)(hD)^\alpha$ (in the case of differential operators), and one is interested in their properties when $h \to 0$, i.e., for $h < h_0$ when $h_0 > 0$ is small. The principal symbol of such operators captures more information about the family $(P_h)_{h\in(0,1)}$ than the mere collection of the principal symbols of all P_h, and its properties can be translated into rather strong statements about the underlying operators. For example, if a semiclassical operator $(P_h)_{h\in(0,1)}$ is semiclassically elliptic, then P_h is *invertible* for small h (not merely Fredholm).

Theme VI. Wave Equations. A primary purpose for microlocal analysis is the study of *non-elliptic* PDEs. Motivated by the author's own preferences, *wave equations* will provide the main source of interesting non-elliptic examples and applications of microlocal results in this book: they can be used to illustrate propagation of singularities results, and they are at the center of the grand application in Chapter 11 on asymptotics for waves on de Sitter space.

1.3 Overview

The analysis of linear PDEs is best accomplished in the setting of distribution theory. Chapter 2 provides a rapid overview of Schwartz functions, tempered distributions, the Fourier transform, and the Schwartz kernel theorem.

The remainder of the book can be roughly divided into three parts. In the *first part* (Chapters 3–5), which is fundamental for the remainder of the book, we develop the basic properties of pseudodifferential operators on $\mathbb{R}^n$ and on manifolds. Broadly speaking, this part equips the reader with technology that is particularly efficient for handling elliptic PDEs such as the Laplace equation.

- Chapter 3. We introduce *symbol classes*, which are the types of functions that we will later on be able to "quantize" using the formula (1.3).
- Chapter 4. Pseudodifferential operators on Euclidean space $\mathbb{R}^n$ are defined as "quantizations" as in (1.3). We show that ps.d.o.s can be composed and

that they act boundedly between Sobolev spaces. We define the crucial notion of the principal symbol of a ps.d.o. Furthermore, we show that the space of ps.d.o.s includes parametrices (approximate inverses) of elliptic operators and use this to prove results on the regularity of solutions to elliptic PDEs.
- Chapter 5. After showing that ps.d.o.s transform well under coordinate changes, we define ps.d.o.s on arbitrary smooth manifolds. All results on $\mathbb{R}^n$ have natural analogues on manifolds. A highlight of the theory of ps.d.o.s is the very simple proof that elliptic linear partial differential operators P on closed manifolds (compact without boundary) are Fredholm, i.e., solutions of $Pu = f$ exist, and are unique, up to finite-dimensional obstacles.

In the *second part* of the book (Chapters 6–10), we turn to microlocal analysis proper. This part provides tools that are particularly efficient for handling hyperbolic PDEs such as wave equations.

- Chapter 6. We define the central notion of the wave front set of a distribution and prove a microlocal version of elliptic regularity. As another application of wave front sets, we give rather precise criteria under which distributions can be multiplied or restricted to submanifolds.
- Chapter 7. Hyperbolic evolution equations, such as transport equations or wave equations, can be solved rather easily using the machinery of ps.d.o.s. We prove furthermore that singularities in the initial data propagate along the Hamiltonian flow of the principal symbol associated with the PDE. This is a first deep instance of the correspondence between classical mechanics and singularities of solutions to PDEs.
- Chapter 8. We prove the celebrated result on the *propagation of singularities* for solutions of real principal type equations, such as wave equations. We do this using the important method of *positive commutators*, which underlies many proofs in modern microlocal analysis.
- Chapter 9. In order to give the reader a concrete setting to hold on to, we explain how the microlocal approach can be used to solve wave-type equations on spacetimes. We develop classical energy methods (phrased in Lorentz-geometric language) as far as needed to prove a basic energy estimate. From there, microlocal analysis takes over to give a full existence, uniqueness, and regularity theory.
- Chapter 10. The propagation of singularities result proved in Chapter 8 is vacuous at *radial sets* where the Hamiltonian vector field is a multiple of the generator of fiber-dilations in phase space. Using the positive commutator method, we obtain results on the propagation of microlocal regularity near radial sets of source and sink type. Radial sets of this type appear frequently in applications in scattering theory.

Each chapter in the first two parts is primarily concerned with "standard" ps.d.o.s and microlocal analysis, where by "standard" we mean the absence

of any large (or small) parameters besides the momentum variable ξ. It is, however, not difficult to modify the definitions and proofs to incorporate a semiclassical parameter $h \to 0$. Correspondingly, each chapter, with the exception of Chapter 7, ends with a detailed description of the relevant modifications and additional proofs wherever necessary. We stress that semiclassical ps.d.o.s and microlocal analysis only play a role—but then an important one—in the final chapter of this book, where they are used to analyze scattering and prove resolvent estimates at high energies. Low-energy readers may thus safely skip all semiclassical sections.

In the (short) *third part* of the book (Chapter 11), we present an application to wave asymptotics that makes use of (almost) all the material developed thus far.

- Chapter 11. We study the long-time behavior of solutions of linear wave-type equations on *de Sitter space*, a spacetime of interest in general relativity. After Fourier transforming in time, we are led to study a family of operators, depending on a spectral parameter, on a compact manifold with boundary. We will study this operator family by combining the microlocal results of Chapters 6, 8, and 10 as well as the wave equation estimates of Chapter 9; we prove that this is a family of (non-elliptic) Fredholm operators with a meromorphic inverse. The poles of the inverse, called *resonances*, dictate the late-time behavior of solutions of the wave equation. It is chiefly for the purpose of analyzing the spectral family in the *high-energy regime* (in which the spectral parameter tends to real infinity) that we develop semiclassical methods in this book.

Chapter 2
Schwartz Functions and Tempered Distributions



The Fourier transform on $\mathbb{R}^n$ plays a central role in the development of microlocal analysis. Schwartz functions and tempered distributions are the most natural setting for it; they are introduced in Sections 2.1–2.2. The most important function spaces for us will be the L^2-based Sobolev spaces $H^s(\mathbb{R}^n)$, $s \in \mathbb{R}$; these are recalled in Section 2.3. Distributions arise in this book not only as solutions of PDEs but also as integral kernels of linear operators. The Schwartz kernel theorem characterizes bounded linear operators mapping Schwartz functions to tempered distributions as those with tempered distributional integral kernels (Section 2.4). The integral (or Schwartz) kernels of differential operators are discussed in Section 2.5. Semiclassical analogues are discussed in Section 2.6.

2.1 Basic Definitions

We write

$$\mathbb{N} = \{1, 2, 3, \ldots\}, \quad \mathbb{N}_0 = \{0\} \cup \mathbb{N}. \tag{2.1}$$

We use standard multi-index notation: for $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$, we set

$$x^\alpha := \prod_{j=1}^n x_j^{\alpha_j}, \quad \partial_x^\alpha := \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n}, \quad D_x^\alpha := D_{x_1}^{\alpha_1} \cdots D_{x_n}^{\alpha_n}, \qquad D = \frac{1}{i}\partial. \tag{2.2}$$

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When the context is clear, we shall often simply write $D^\alpha := D_x^\alpha$, and $D_j := D_{x_j}$. We also put

$$|\alpha| := \sum_{j=1}^n \alpha_j, \quad \alpha! := \prod_{j=1}^n \alpha_j!. \tag{2.3}$$

For $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$, we write $\alpha \leq \beta$ if $\alpha_j \leq \beta_j$ for all $j = 1, \dots, n$, and we write $\alpha < \beta$ if at least one of these inequalities is strict. We will moreover use the *Japanese bracket*, defined for $x \in \mathbb{R}^n$ by

$$\langle x \rangle = (1 + |x|^2)^{\frac{1}{2}}. \tag{2.4}$$

Let $k \in \mathbb{N}_0$. For an open subset $\Omega \subset \mathbb{R}^n$, we denote by $\mathcal{C}^k(\Omega)$ the space of k times continuously differentiable functions (with no growth restrictions), and $\mathcal{C}^\infty(\Omega) = \bigcap_{k\in\mathbb{N}} \mathcal{C}^k(\Omega)$. By $\mathcal{C}_b^k(\Omega) \subset \mathcal{C}^k(\Omega)$ we denote the space of all functions all of whose up to k-fold derivatives are bounded, and we let $\mathcal{C}_b^\infty(\Omega) = \bigcap_{k\in\mathbb{N}} \mathcal{C}_b^k(\Omega)$. Equipped with the norm

$$\|u\|_{\mathcal{C}_b^k(\Omega)} := \sup_{\substack{x\in\Omega \\ |\alpha|\leq k}} |\partial^\alpha u(x)|, \tag{2.5}$$

the space $\mathcal{C}_b^k(\Omega)$ is a Banach space, and the space $\mathcal{C}_b^\infty(\Omega)$ is a Fréchet space. For $k \in \mathbb{N}_0 \cup \{\infty\}$, we denote by $\mathcal{C}_c^k(\Omega)$ the subspace of compactly supported elements of $\mathcal{C}^k(\Omega)$. Unless otherwise noted, all functions will be complex-valued.

Definition 2.1 (Schwartz Space). The space $\mathscr{S}(\mathbb{R}^n)$ of *Schwartz functions* consists of all $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$ such that for all $k \in \mathbb{N}_0$,

$$\|\phi\|_k := \sup_{\substack{x\in\mathbb{R}^n \\ |\alpha|+|\beta|\leq k}} |x^\alpha D^\beta \phi(x)| < \infty. \tag{2.6}$$

Example 2.2 (Schwartz Functions). We have $\exp(-|x|^2) \in \mathscr{S}(\mathbb{R}^n)$. Moreover, we have an inclusion $\mathcal{C}_c^\infty(\mathbb{R}^n) \hookrightarrow \mathscr{S}(\mathbb{R}^n)$ with dense range. Recall that there are lots of smooth functions with compact support; indeed, when $K \subset U \subset \mathbb{R}^n$ with K compact and U open and bounded, there exists $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi \equiv 1$ on K and $\phi \equiv 0$ on $\mathbb{R}^n \setminus U$.

Equipped with the countably many seminorms $\|\cdot\|_k$, $\mathscr{S}(\mathbb{R}^n)$ is a Fréchet space. Directly from the definition, we have continuous maps

$$\begin{aligned} x_j &\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n), \qquad \phi \mapsto x_j \phi, \\ D_j &\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n), \qquad \phi \mapsto D_j \phi. \end{aligned} \tag{2.7}$$

Given $a \in \mathcal{C}_b^\infty(\mathbb{R}^n)$, pointwise multiplication by a is also continuous. Furthermore, integration is a continuous map

$$\int : \mathscr{S}(\mathbb{R}^n) \to \mathbb{C}. \tag{2.8}$$

Indeed, this follows from

$$\left| \int_{\mathbb{R}^n} \phi(x)\,\mathrm{d}x \right| = \left| \int_{\mathbb{R}^n} \langle x \rangle^{-n-1} \big(\langle x \rangle^{n+1} \phi(x) \big)\,\mathrm{d}x \right| \leq C_n \|\phi\|_{n+1}. \tag{2.9}$$

Other useful operations are the pointwise product

$$\mathscr{S}(\mathbb{R}^n) \times \mathscr{S}(\mathbb{R}^n) \ni (\phi, \psi) \mapsto \phi\psi \in \mathscr{S}(\mathbb{R}^n), \quad (\phi\psi)(x) = \phi(x)\psi(x), \tag{2.10}$$

and the exterior product

$$\mathscr{S}(\mathbb{R}^n) \times \mathscr{S}(\mathbb{R}^m) \ni (\phi, \psi) \mapsto \phi \boxtimes \psi \in \mathscr{S}(\mathbb{R}^{n+m}), \quad (\phi \boxtimes \psi)(x, y) = \phi(x)\psi(y). \tag{2.11}$$

Definition 2.3 (Tempered Distributions). The space $\mathscr{S}'(\mathbb{R}^n)$ of *tempered distributions* is the space of all continuous linear functionals $u \colon \mathscr{S}(\mathbb{R}^n) \to \mathbb{C}$, equipped with the weak topology: the seminorms are $|u|_\phi := |u(\phi)|$ for $\phi \in \mathscr{S}(\mathbb{R}^n)$. We shall usually write $\langle u, \phi \rangle := u(\phi)$.

Example 2.4 (δ-Distribution). The δ-distribution is defined by $\langle \delta, \phi \rangle := \phi(0)$. We have $\delta \in \mathscr{S}'(\mathbb{R}^n)$ since $|\langle \delta, \phi \rangle| \leq \|\phi\|_0$.

Combining (2.8) and (2.10), we can define a continuous map

$$\mathscr{S}(\mathbb{R}^n) \ni \phi \mapsto T_\phi \in \mathscr{S}'(\mathbb{R}^n), \quad \langle T_\phi, \psi \rangle = \int_{\mathbb{R}^n} \phi(x)\psi(x)\,\mathrm{d}x. \tag{2.12}$$

Proposition 2.5 (Functions as Distributions). *The map $\phi \mapsto T_\phi$ is injective and has dense range in the weak topology.*

Proof. Regarding injectivity: $\langle T_\phi, \bar{\phi} \rangle = \int_{\mathbb{R}^n} |\phi(x)|^2\,\mathrm{d}x = 0$ implies $\phi = 0$. To prove the density, it suffices to show that, given $u \in \mathscr{S}'(\mathbb{R}^n)$ and $\phi_1, \ldots, \phi_N \in \mathscr{S}(\mathbb{R}^n)$ as well as any $\epsilon > 0$, there exists $\phi \in \mathscr{S}(\mathbb{R}^n)$ such that $|\langle u, \phi_j \rangle - \langle T_\phi, \phi_j \rangle| < \epsilon$ for all $j = 1, \ldots, N$. Assuming, as one may, that the ϕ_j are orthonormal with respect to the $L^2(\mathbb{R}^n)$-inner product, this holds (with "$< \epsilon$" replaced by "$= 0$") for $\phi = \sum_{j=1}^N \langle u, \phi_j \rangle \bar{\phi}_j$. A more sensible proof, based on a mollification argument, is suggested in Exercise 2.2. □

Now on the one hand, we can extend the maps (2.7) by duality to $\mathscr{S}'(\mathbb{R}^n)$: for $u \in \mathscr{S}'(\mathbb{R}^n)$, we define $x_j u, D_j u \in \mathscr{S}'(\mathbb{R}^n)$ by

$$\langle x_j u, \phi\rangle := \langle u, x_j\phi\rangle, \quad \langle D_j u, \phi\rangle := \langle u, -D_j\phi\rangle, \qquad \phi \in \mathscr{S}(\mathbb{R}^n). \tag{2.13}$$

On the other hand, when $u \in \mathscr{S}(\mathbb{R}^n)$, then $\langle T_{x_j u}, \phi\rangle = \langle T_u, x_j\phi\rangle$ and $\langle T_{D_j u}, \phi\rangle = \langle T_u, -D_j\phi\rangle$, i.e., on the image of $\mathscr{S}(\mathbb{R}^n)$ inside of $\mathscr{S}'(\mathbb{R}^n)$, the definitions (2.13) agree with the usual definitions of multiplication and differentiation of Schwartz functions. The density statement of Proposition 2.5 then shows that (2.13) defines the *unique* continuous extensions of multiplication and differentiation from $\mathscr{S}(\mathbb{R}^n)$ to $\mathscr{S}'(\mathbb{R}^n)$.

Similarly, by duality and starting from (2.11), pointwise multiplication by a Schwartz function extends in a unique manner to a continuous map on $\mathscr{S}'(\mathbb{R}^n)$; more generally, this is true for multiplication by a function in $\mathcal{C}_b^\infty(\mathbb{R}^n)$.

Other notions, which will be significantly refined later, are:

Definition 2.6 (Support and Singular Support of Distributions). Let $u \in \mathscr{S}'(\mathbb{R}^n)$.

(1) The *support* $\operatorname{supp} u \subset \mathbb{R}^n$ of u is the complement of the set of $x \in \mathbb{R}^n$ such that there exists $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, $\chi(x) \neq 0$, such that $\chi u = 0$.
(2) The *singular support* $\operatorname{sing\,supp} u \subset \mathbb{R}^n$ of u is the complement of the set of $x \in \mathbb{R}^n$ such that there exists $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, $\chi(x) \neq 0$, such that χu is smooth, i.e., $\chi u = T_\phi$, $\phi \in \mathscr{S}(\mathbb{R}^n)$.

Example 2.7 (Support and Singular Support). We have $\operatorname{supp}\delta = \operatorname{sing\,supp}\delta = \{0\}$. For $u = \delta + e^{-|x|^2}$ and $u = |x|^\alpha \in \mathscr{S}'(\mathbb{R}^n)$ where $\alpha > -n$, $\alpha \notin \mathbb{N}_0$, we have $\operatorname{supp} u = \mathbb{R}^n$ but $\operatorname{sing\,supp} u = \{0\}$.

From now on, we shall simplify notation and denote the distribution T_ϕ by ϕ.

2.2 Fourier Transform and Its Inverse

We define the Fourier transform of $\phi \in \mathscr{S}(\mathbb{R}^n)$ by

$$(\mathcal{F}\phi)(\xi) = \hat{\phi}(\xi) := \int_{\mathbb{R}^n} e^{-ix\cdot\xi}\phi(x)\,\mathrm{d}x, \quad \xi \in \mathbb{R}^n, \tag{2.14}$$

and the inverse Fourier transform of $\psi \in \mathscr{S}(\mathbb{R}^n)$ by

$$(\mathcal{F}^{-1}\psi)(x) := (2\pi)^{-n}\int_{\mathbb{R}^n} e^{ix\cdot\xi}\psi(\xi)\,\mathrm{d}\xi, \quad x \in \mathbb{R}^n. \tag{2.15}$$

As in (2.9), one finds $\|\mathcal{F}\phi\|_0 \le C_n\|\phi\|_{n+1}$ and $\|\mathcal{F}^{-1}\phi\|_0 \le C_n\|\phi\|_{n+1}$. Moreover, we have

$$\begin{aligned} \mathcal{F}(D_{x_j}\phi) &= \xi_j\mathcal{F}\phi, & \mathcal{F}(x_j\phi) &= -D_{\xi_j}\mathcal{F}\phi, \\ \mathcal{F}^{-1}(D_{\xi_j}\phi) &= -x_j\mathcal{F}^{-1}\phi, & \mathcal{F}^{-1}(\xi_j\phi) &= D_{x_j}\mathcal{F}^{-1}\phi, \end{aligned} \tag{2.16}$$

using integration by parts for the first and third statements. Reading these from right to left shows that

$$\|\mathcal{F}\phi\|_k \le C_n\|\phi\|_{k+n+1} \quad \forall\, k \in \mathbb{N}_0, \tag{2.17}$$

similarly for $\mathcal{F}^{-1}$, and hence the (inverse) Fourier transform preserves the Schwartz space:

$$\mathcal{F},\, \mathcal{F}^{-1}\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n). \tag{2.18}$$

Note then that for $u, \psi \in \mathscr{S}(\mathbb{R}^n)$,

$$\begin{aligned} \langle \mathcal{F}u, \psi\rangle &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} e^{-ix\cdot\xi}u(x)\,\mathrm{d}x\ \psi(\xi)\,\mathrm{d}\xi = \int_{\mathbb{R}^n\times\mathbb{R}^n} e^{-ix\cdot\xi}u(x)\psi(\xi)\,\mathrm{d}x\,\mathrm{d}\xi \\ &= \langle u, \mathcal{F}\psi\rangle. \end{aligned} \tag{2.19}$$

This allows us to extend $\mathcal{F}, \mathcal{F}^{-1}$ to maps on tempered distributions,

$$\mathcal{F},\ \mathcal{F}^{-1}\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n), \tag{2.20}$$

and the formulas (2.16) remain valid for $\phi \in \mathscr{S}'(\mathbb{R}^n)$.

We recall the proof that $\mathcal{F}$ and $\mathcal{F}^{-1}$ are indeed inverses to each other:

Theorem 2.8 (Fourier Transform and Its Inverse). *We have $\mathcal{F}\circ\mathcal{F}^{-1} = \mathcal{F}^{-1}\circ\mathcal{F} = I$ on $\mathscr{S}(\mathbb{R}^n)$ and $\mathscr{S}'(\mathbb{R}^n)$.*

Proof. Let $A := \mathcal{F}^{-1}\mathcal{F}\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$. By (2.16), we have $AD_{x_j} = \mathcal{F}^{-1}\xi_j\mathcal{F} = D_{x_j}A$ and $Ax_j = \mathcal{F}^{-1}(-D_{\xi_j})\mathcal{F} = x_jA$, i.e., A commutes with differentiation in x_j and multiplication by x_j for all j. Given $\phi \in \mathscr{S}(\mathbb{R}^n)$ and $x_0 \in \mathbb{R}^n$, we can write

$$\phi(x) = \phi(x_0) + \sum_{j=1}^n \phi_j(x)(x_j - (x_0)_j), \quad \phi_j(x) = \int_0^1 (\partial_j\phi)(x_0 + t(x - x_0))\,\mathrm{d}t. \tag{2.21}$$

The fact that ϕ_j is in general not Schwartz is remedied by fixing a cutoff $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, identically 1 near x_0, and writing $\phi(x) = \chi(x)\phi(x) + (1-\chi(x))\phi(x)$, so

$$\begin{aligned}\phi(x) &= \chi(x)\phi(x_0) + \sum_{j=1}^{n} \tilde{\phi}_j(x)(x_j - (x_0)_j),\\ \tilde{\phi}_j(x) &= \chi(x)\phi_j(x) + \frac{(1-\chi(x))\phi(x)}{|x-x_0|^2}(x_j - (x_0)_j).\end{aligned} \tag{2.22}$$

Since $A(\tilde{\phi}_j(\cdot)(\cdot_j - (x_0)_j))(x_0) = ((x_j - (x_0)_j)(A\tilde{\phi}_j)(x))|_{x=x_0} = 0$, we have

$$(A\phi)(x_0) = \phi(x_0)(A\chi)(x_0). \tag{2.23}$$

Note that the constant $(A\chi)(x_0)$ here does not depend on ϕ, and not on the particular choice of χ either (since the left hand side does not involve χ at all).

The same cutoff χ can be used to evaluate $A\phi$ at points x close to x_0; but

$$D_{x_j}(A\chi)(x) = A(D_{x_j}\chi)(x) = 0 \tag{2.24}$$

for x in the interior of $\chi^{-1}(1)$, where we use (2.23) for $\phi = D_{x_j}\chi$ and with x in place of x_0. We conclude that $A = cI$ for some *constant* $c \in \mathbb{C}$. One can find c by explicitly evaluating

$$\mathcal{F}(e^{-|x|^2})(\xi) = \pi^{n/2}e^{-|\xi|^2/4}, \quad \mathcal{F}^{-1}(e^{-|\xi|^2/4})(x) = \pi^{-n/2}e^{-|x|^2}, \tag{2.25}$$

so $c = 1$ indeed. The proof that $\mathcal{F}\mathcal{F}^{-1} = I$ is completely analogous. □

Example 2.9 (Some Fourier Transforms).

(1) We have $\langle \mathcal{F}\delta, \psi\rangle = \langle \delta, \mathcal{F}\psi\rangle = \hat{\psi}(0) = \int_{\mathbb{R}^n} \psi(x)\,\mathrm{d}x$, so $\mathcal{F}\delta = 1$.
(2) One finds $\mathcal{F}1 = (2\pi)^n\delta$ from $\mathcal{F}^{-1}1 = \delta$, or directly from $\langle \mathcal{F}1, \psi\rangle = \langle 1, \hat{\psi}\rangle = \int_{\mathbb{R}^n} \hat{\psi}(\xi)\,\mathrm{d}\xi = (2\pi)^n\psi(0)$.

While the Fourier transform of a tempered distribution is, in general, again only a tempered distribution, the Fourier transform of a compactly supported distribution is more regular.

Definition 2.10 (Compactly Supported Distributions). We define

$$\mathscr{E}'(\mathbb{R}^n) := \{u \in \mathscr{S}'(\mathbb{R}^n) \colon \operatorname{supp} u \text{ is compact}\}. \tag{2.26}$$

Proposition 2.11 (Fourier Transform of Compactly Supported Distributions). *Let $u \in \mathscr{E}'(\mathbb{R}^n)$. Then $\hat{u} \in \mathscr{S}'(\mathbb{R}^n) \cap \mathcal{C}^\infty(\mathbb{R}^n)$. Moreover, there exists N such that for all $\alpha \in \mathbb{N}_0^n$ there exists a constant C_α such that*

$$|\partial_\xi^\alpha \hat{u}(\xi)| \le C_\alpha \langle \xi\rangle^N, \quad \xi \in \mathbb{R}^n. \tag{2.27}$$

Proof. Fix $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ near $\operatorname{supp} u$. Set

$$v(\xi) := \langle u, \chi e^{-i\xi\cdot}\rangle, \quad \xi \in \mathbb{R}^n. \tag{2.28}$$

The map $\mathbb{R}^n \ni \xi \mapsto \chi e^{-i\xi\cdot} \in \mathscr{S}(\mathbb{R}^n)$ is smooth; its derivative along ∂_ξ^α at $\xi \in \mathbb{R}^n$ is equal to the Schwartz function $\phi_{\xi,\alpha}(x) := (-ix)^\alpha \chi(x) e^{-i\xi x}$. Therefore, $v \in \mathcal{C}^\infty(\mathbb{R}^n)$. The continuity of u implies

$$|\partial_\xi^\alpha v(\xi)| = |\langle u, \phi_{\xi,\alpha}\rangle| \le C\|\phi_{\xi,\alpha}\|_N \le C_\alpha \langle\xi\rangle^N. \tag{2.29}$$

Since $v \in \mathcal{C}^\infty(\mathbb{R}^n)$ is thus polynomially bounded, we have $v \in \mathscr{S}'(\mathbb{R}^n)$. If $\phi \in \mathscr{S}(\mathbb{R}^n)$ denotes a test function, we can now compute

$$\langle v, \phi\rangle = \int_{\mathbb{R}^n} \langle u, \chi e^{-i\xi\cdot}\rangle \phi(\xi)\,\mathrm{d}\xi = \langle u, \chi\hat{\phi}\rangle = \langle \chi u, \hat{\phi}\rangle = \langle u, \hat{\phi}\rangle = \langle \hat{u}, \phi\rangle; \tag{2.30}$$

for the second equality, we use that $\int_{\mathbb{R}^n} e^{-i\xi\cdot x}\phi(\xi)\,\mathrm{d}\xi$ converges to $\hat{\phi}(x)$ uniformly on compact subsets of $\mathbb{R}^n_x$ with all derivatives, and thus $\chi \int_{\mathbb{R}^n} e^{-i\xi\cdot}\phi(\xi)\,\mathrm{d}\xi \to \chi\hat{\phi}$ in $\mathscr{S}(\mathbb{R}^n)$. Therefore, $\hat{u} = v$. □

A stronger result is given in Exercise 2.5.

We finally recall that $\mathcal{F}$ is an isomorphism on $L^2(\mathbb{R}^n)$; this follows from the density of $\mathscr{S}(\mathbb{R}^n)$ in $L^2(\mathbb{R}^n)$ and the following fact:

Proposition 2.12 (Plancherel's Theorem). *For $\phi \in \mathscr{S}(\mathbb{R}^n)$, we have*

$$\|\mathcal{F}\phi\|_{L^2(\mathbb{R}^n)} = (2\pi)^{n/2}\|\phi\|_{L^2(\mathbb{R}^n)}, \quad \|\phi\|_{L^2(\mathbb{R}^n)} := \left(\int_{\mathbb{R}^n} |\phi(x)|^2\,\mathrm{d}x\right)^{\frac{1}{2}}. \tag{2.31}$$

More generally, $\langle \mathcal{F}\phi, \mathcal{F}\psi\rangle_{L^2(\mathbb{R}^n)} = (2\pi)^n \langle\phi,\psi\rangle_{L^2(\mathbb{R}^n)}$ for $\phi, \psi \in \mathscr{S}(\mathbb{R}^n)$.

Proof. Analogously to (2.19), we have

$$\int_{\mathbb{R}^n} (\mathcal{F}\phi)(\xi)\bar{\psi}(\xi)\,\mathrm{d}\xi = (2\pi)^n \int_{\mathbb{R}^n} \phi(x)\overline{\mathcal{F}^{-1}\psi(x)}\,\mathrm{d}x, \quad \phi, \psi \in \mathscr{S}(\mathbb{R}^n). \tag{2.32}$$

Plugging in $\psi = \mathcal{F}\phi$ proves (2.31). □

For the purposes of this book, one can define $L^2(\mathbb{R}^n)$ to be the completion of $\mathscr{S}(\mathbb{R}^n)$ with respect to the $L^2(\mathbb{R}^n)$-norm and thereby avoid any usage of measure theoretic notions. Since $\mathscr{S}(\mathbb{R}^n) \ni \phi \mapsto T_\phi \in \mathscr{S}'(\mathbb{R}^n)$ is continuous with respect to the $L^2(\mathbb{R}^n)$-norm on $\mathscr{S}(\mathbb{R}^n)$, we can regard $L^2(\mathbb{R}^n)$ as a subspace of $\mathscr{S}'(\mathbb{R}^n)$. (Note here that if $\phi_j \in \mathscr{S}(\mathbb{R}^n)$ is a sequence with $\|\phi_j\|_{L^2(\mathbb{R}^n)} \to 0$, then for all $\phi \in \mathscr{S}(\mathbb{R}^n)$, we have $|\int_{\mathbb{R}^n} \phi_j(x)\phi(x)\,\mathrm{d}x| \le \|\phi_j\|_{L^2(\mathbb{R}^n)}\|\phi\|_{L^2(\mathbb{R}^n)} \to 0$; this shows that the map $L^2(\mathbb{R}^n) \ni \phi \mapsto T_\phi \in \mathscr{S}'(\mathbb{R}^n)$ is injective.)

2.3 Sobolev Spaces and the Schwartz Representation Theorem

Using the Fourier transform, we can define operators that differentiate a "fractional number of times":

Definition 2.13 (Fractional Derivative Operators on $\mathbb{R}^n$). For $s \in \mathbb{R}$ (or $s \in \mathbb{C}$), we define

$$\langle D\rangle^s \colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n), \quad \langle D\rangle^s := \mathcal{F}^{-1}\langle\xi\rangle^s \mathcal{F}. \tag{2.33}$$

This agrees for $s = 2k$, $k \in \mathbb{N}_0$, with the operator $(\Delta+1)^k$, and for $s = -2$ with the operator (1.10). In (2.33), we use that multiplication by $\langle\xi\rangle^s = (1+|\xi|^2)^{\frac{s}{2}}$ is continuous on $\mathscr{S}(\mathbb{R}^n)$ and thus on $\mathscr{S}'(\mathbb{R}^n)$. The definition (2.33) is a special case of $f(D) := \mathcal{F}^{-1} f(\xi)\mathcal{F} \colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ for $f \colon \mathbb{R}^n \to \mathbb{C}$ which are polynomially bounded with all derivatives; for example, for $f(\xi) = \sum_{|\alpha|\le m} f_\alpha \xi^\alpha$, $f_\alpha \in \mathbb{C}$, one has $f(D) = \sum_{|\alpha|\le m} f_\alpha D^\alpha$.

Definition 2.14 (Sobolev Spaces on $\mathbb{R}^n$). For $s \in \mathbb{R}$, the *Sobolev space of order s on $\mathbb{R}^n$* is defined by

$$H^s(\mathbb{R}^n) := \{u \in \mathscr{S}'(\mathbb{R}^n) \colon \langle D\rangle^s u \in L^2(\mathbb{R}^n)\}. \tag{2.34}$$

With the norm

$$\|u\|_{H^s(\mathbb{R}^n)} := \|\langle D\rangle^s u\|_{L^2} = (2\pi)^{-n/2}\|\langle\xi\rangle^s \mathcal{F}u\|_{L^2}, \tag{2.35}$$

it is a Hilbert space.

Example 2.15 (δ-Distribution and Sobolev Spaces). The δ-distribution at $0 \in \mathbb{R}^n$ satisfies $\delta \in H^s(\mathbb{R}^n)$ for all $s < -\frac{n}{2}$.

If $s > s'$, then $H^s(\mathbb{R}^n) \subset H^{s'}(\mathbb{R}^n)$ since $\langle\xi\rangle^{s'} = \langle\xi\rangle^s \langle\xi\rangle^{s'-s}$ and $\langle\xi\rangle^{s'-s}$ is bounded. For integer orders $s = k \in \mathbb{N}_0$, $H^k(\mathbb{R}^n)$ is equal to the space of all $u \in \mathscr{S}'(\mathbb{R}^n)$ whose distributional derivatives (often called weak derivatives in this context) up to order k are in $L^2(\mathbb{R}^n)$ (Exercise 2.7).

More useful than the H^s-inner product $\langle u, v\rangle_{H^s(\mathbb{R}^n)} = \langle\langle D\rangle^s u, \langle D\rangle^s v\rangle_{L^2(\mathbb{R}^n)}$ is often the L^2-inner product, especially due to its distinguished role in the definition of adjoints of partial differential operators (Proposition 3.3). The interaction between the $H^s(\mathbb{R}^n)$- and the L^2-inner products is as follows.

Lemma 2.16 (L^2-Dual of $H^s(\mathbb{R}^n)$). *The sesquilinear pairing*

$$\mathscr{S}(\mathbb{R}^n) \times \mathscr{S}(\mathbb{R}^n) \ni (u, v) \mapsto \langle u, v\rangle_{L^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} u(x)\overline{v(x)}\,\mathrm{d}x \in \mathbb{C} \tag{2.36}$$

extends uniquely to a continuous sesquilinear pairing

$$H^{-s}(\mathbb{R}^n) \times H^s(\mathbb{R}^n) \to \mathbb{C}. \tag{2.37}$$

The map $H^s(\mathbb{R}^n) \ni v \mapsto \langle \cdot, v\rangle_{L^2(\mathbb{R}^n)} \in (H^{-s}(\mathbb{R}^n))^*$ *is an antilinear isomorphism.*

Proof. For $u, v \in \mathscr{S}(\mathbb{R}^n)$, Plancherel's theorem implies

$$\langle u, v\rangle_{L^2} = \langle \langle D\rangle^{-s} u, \langle D\rangle^s v\rangle_{L^2}. \tag{2.38}$$

The right-hand side is well-defined for $u \in H^{-s}(\mathbb{R}^n)$ and $v \in H^s(\mathbb{R}^n)$. The claim now follows from the fact that $L^2(\mathbb{R}^n)$ is a Hilbert space. □

We recall the following important fact:

Theorem 2.17 (Rellich Compactness Theorem). *For* $s \in \mathbb{R}$ *and a compact subset* $K \subset \mathbb{R}^n$*, write*

$$\dot{H}^s(K) := \{u \in H^s(\mathbb{R}^n) \colon \operatorname{supp} u \subset K\}. \tag{2.39}$$

Let $s > s'$*. Then the inclusion map* $\dot{H}^s(K) \to \dot{H}^{s'}(K)$ *is compact.*

The standard method of proof is to wrap up a big cube containing K to a torus and then pass to Fourier series (Exercise 5.13). We instead give an entirely $\mathbb{R}^n$-based argument here.

Proof of Theorem 2.17. Let $u_j \in \dot{H}^s(K)$, $j \in \mathbb{N}$, be a bounded sequence; by rescaling, we may assume that $\|u_j\|_{H^s(\mathbb{R}^n)} \le 1$ for all j. Since $H^s(\mathbb{R}^n)$ is a separable Hilbert space, we may apply the Banach–Alaoglu theorem and replace u_j by a subsequence converging weakly to some $u \in H^s(\mathbb{R}^n)$, i.e.,

$$\langle u_j, w\rangle_{H^s(\mathbb{R}^n)} = \langle \langle D\rangle^s u_j, \langle D\rangle^s w\rangle_{L^2(\mathbb{R}^n)} \to \langle u, w\rangle_{H^s(\mathbb{R}^n)} \quad \forall\, w \in H^s(\mathbb{R}^n); \tag{2.40}$$

and $\|u\|_{H^s(\mathbb{R}^n)} \le 1$. Applying this with $w := \langle D\rangle^{-2s}\phi$ where $\phi \in \mathscr{S}(\mathbb{R}^n)$, we conclude that $u_j \to u$ in $\mathscr{S}'(\mathbb{R}^n)$. Therefore, there exist $C > 0$ and $N \in \mathbb{N}_0$ such that

$$|\langle u_j, \phi\rangle| \le C\|\phi\|_N \quad \forall\, \phi \in \mathscr{S}(\mathbb{R}^n). \tag{2.41}$$

Furthermore, for $\phi \in \mathscr{S}(\mathbb{R}^n)$ with $\operatorname{supp}\phi \cap K = \emptyset$, we have $\langle u_j, \phi\rangle = 0$ for all j and thus $\langle u, \phi\rangle = 0$. Therefore, $u \in \dot{H}^s(K)$.

We claim that $u_j \to u$ in $H^{s'}(\mathbb{R}^n)$. Let $\epsilon > 0$. Then for all $j \in \mathbb{N}$ we have

$$\int_{|\xi|\ge R} \langle\xi\rangle^{2s'} |\widehat{u_j}(\xi) - \hat{u}(\xi)|^2 \,\mathrm{d}\xi \le 2\langle R\rangle^{-2(s-s')} \int_{|\xi|\ge R} \langle\xi\rangle^{2s} \big(|\widehat{u_j}(\xi)|^2 + |\hat{u}(\xi)|^2\big) \,\mathrm{d}\xi$$

$$\leq 4\langle R\rangle^{-2(s-s')} < \frac{\epsilon}{2} \tag{2.42}$$

for sufficiently large $R > 0$.

In order to bound $|\widehat{u_j}(\xi) - \hat{u}(\xi)|$ for low frequencies $|\xi| \leq R$, we must use the compact support property of the u_j. Fix $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ near K. Then the proof of Proposition 2.11 shows that $\widehat{u_j} \in \mathscr{S}'(\mathbb{R}^n) \cap \mathcal{C}^\infty(\mathbb{R}^n)$ and

$$\widehat{u_j}(\xi) = \langle u_j, \chi e^{-i\xi\cdot}\rangle, \quad \xi \in \mathbb{R}^n. \tag{2.43}$$

Let $B := \{\xi \in \mathbb{R}^n \colon |\xi| \leq R\}$. The smoothness of $\xi \mapsto \chi e^{-i\xi\cdot} \in \mathscr{S}(\mathbb{R}^n)$ and the uniform estimate (2.41) imply

$$\sup_{j\in\mathbb{N}} \|\widehat{u_j}\|_{\mathcal{C}^k(B)} < \infty \tag{2.44}$$

for all k. We only use this for $k = 1$. The Arzelà–Ascoli theorem then implies that $\widehat{u_j}|_B$ has a subsequence converging in $\mathcal{C}^0(B)$. But for each $\xi \in B$, the limit must equal $\lim_{j\to\infty}\langle u_j, \chi e^{-i\xi\cdot}\rangle = \langle u, \chi e^{-i\xi\cdot}\rangle = \hat{u}(\xi)$. Therefore, $\widehat{u_j}|_B \to \hat{u}|_B$ in $\mathcal{C}^0(B)$. Thus, there exists $j_0 \in \mathbb{N}$ such that for all $j \geq j_0$,

$$\int_{|\xi|\leq R} \langle\xi\rangle^{2s'} |\widehat{u_j}(\xi) - \hat{u}(\xi)|^2 \,\mathrm{d}\xi \leq C_R \int_B |\widehat{u_j}(\xi) - \hat{u}(\xi)|^2 \,\mathrm{d}\xi < \frac{\epsilon}{2}. \tag{2.45}$$

Combining this with (2.42), we conclude that $\|u_j - u\|^2_{H^{s'}(\mathbb{R}^n)} < \epsilon$ for $j \geq j_0$. □

Since multiplication by $\langle x\rangle^r$ is continuous on $\mathscr{S}'(\mathbb{R}^n)$ for every $r \in \mathbb{R}$, we can define *weighted Sobolev spaces* by

$$\langle x\rangle^r H^s(\mathbb{R}^n) := \{u \in \mathscr{S}'(\mathbb{R}^n) \colon \langle x\rangle^{-r} u \in H^s(\mathbb{R}^n)\}. \tag{2.46}$$

These are Sobolev spaces with squared norms

$$\|u\|^2_{\langle x\rangle^r H^s(\mathbb{R}^n)} := \|\langle x\rangle^{-r} u\|^2_{H^s(\mathbb{R}^n)}. \tag{2.47}$$

The L^2-pairing (2.36) induces an identification $\langle x\rangle^r H^s(\mathbb{R}^n) \cong (\langle x\rangle^{-r} H^{-s}(\mathbb{R}^n))^*$.

The second part of the following is one version of the *Schwartz representation theorem*:

Theorem 2.18 (Schwartz Representation Theorem). *We have*

$$\mathscr{S}(\mathbb{R}^n) = \bigcap_{s,r\in\mathbb{R}} \langle x\rangle^r H^s(\mathbb{R}^n), \quad \mathscr{S}'(\mathbb{R}^n) = \bigcup_{s,r\in\mathbb{R}} \langle x\rangle^r H^s(\mathbb{R}^n). \tag{2.48}$$

Proof. See Exercises 2.8 and 2.16. □

This easily implies (using Sobolev embedding, Exercise 2.8) that every tempered distribution is a finite sum of distributions of the form $x^\alpha D^\beta a$, $a \in \mathcal{C}_b^0(\mathbb{R}^n)$.

Useful intermediate spaces between $\mathscr{S}(\mathbb{R}^n)$ and $\mathscr{S}'(\mathbb{R}^n)$ are $H^\infty(\mathbb{R}^n) := \bigcap_{s\in\mathbb{R}} H^s(\mathbb{R}^n)$ and $H^{-\infty}(\mathbb{R}^n) := \bigcup_{s\in\mathbb{R}} H^s(\mathbb{R}^n)$.

2.4 The Schwartz Kernel Theorem

The Schwartz kernel theorem is a philosophically useful fact establishing a 1–1 correspondence between the "most general" operators K in the present context—mapping Schwartz functions to tempered distributions—and distributional integral kernels O_K, also called *Schwartz kernels*. To state this, we note that any distribution $K \in \mathscr{S}'(\mathbb{R}^{n+m})$ induces a bounded linear operator $O_K \colon \mathscr{S}(\mathbb{R}^m) \to \mathscr{S}'(\mathbb{R}^n)$ by integration along the $\mathbb{R}^m$ factor, to wit,

$$(O_K\phi)(\psi) := \langle K, \psi \boxtimes \phi\rangle, \quad \phi \in \mathscr{S}(\mathbb{R}^m),\ \psi \in \mathscr{S}(\mathbb{R}^n). \tag{2.49}$$

Formally, one writes

$$(O_K\phi)(x) = \int_{\mathbb{R}^m} K(x,y)\phi(y)\,\mathrm{d}y, \tag{2.50}$$

and then (2.49) formally reads $(O_K\phi)(\psi) = \int_{\mathbb{R}^n}(\int_{\mathbb{R}^m} K(x,y)\phi(y)\,\mathrm{d}y)\psi(x)\,\mathrm{d}x$.

Theorem 2.19 (Schwartz Kernel Theorem: Euclidean Case). *The map $K \mapsto O_K$ is a bijection between $\mathscr{S}'(\mathbb{R}^{n+m})$ and the space of continuous linear operators $\mathscr{S}(\mathbb{R}^m) \to \mathscr{S}'(\mathbb{R}^n)$.*

Proof. See Exercises 2.17 and 2.18. □

Example 2.20 (Schwartz Kernel of I). The Schwartz kernel of the identity operator $I \colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ followed by the inclusion $\mathscr{S}(\mathbb{R}^n) \hookrightarrow \mathscr{S}'(\mathbb{R}^n)$ is given by

$$K(x,y) = \delta(x-y), \quad x, y \in \mathbb{R}^n. \tag{2.51}$$

2.5 Differential Operators

Given $a_\alpha \in \mathcal{C}_b^\infty(\mathbb{R}^n)$ for $\alpha \in \mathbb{N}_0^n$, $|\alpha| \le m$, we can define the m-th order differential operator

$$A = \sum_{|\alpha| \le m} a_\alpha(x) D^\alpha. \tag{2.52}$$

Since multiplication by a_α is continuous on $\mathscr{S}(\mathbb{R}^n)$, A defines a continuous linear operator on $\mathscr{S}(\mathbb{R}^n)$. Moreover, A extends (uniquely) to a continuous linear operator on $\mathscr{S}'(\mathbb{R}^n)$.

Definition 2.21 (Differential Operators). We denote by $\mathrm{Diff}^m(\mathbb{R}^n)$ the space of all operators $A = \sum_{|\alpha|\le m} a_\alpha(x) D^\alpha$, $a_\alpha \in \mathcal{C}_b^\infty(\mathbb{R}^n)$.

Multiplication by an element of $\mathcal{C}_b^\infty(\mathbb{R}^n)$ defines a bounded linear map on $H^s(\mathbb{R}^n)$ for all $s \in \mathbb{R}$. This can be proved for $s \in \mathbb{N}_0$ by direct differentiation. For general $s \in \mathbb{R}$, this is a special case of the mapping properties of ps.d.o.s (Corollary 4.38). For an elementary argument, see Exercise 2.11. (One can alternatively prove this using duality and complex interpolation arguments, as done in Taylor [Tay11a, Chapter 4.2].) Therefore, every $A \in \mathrm{Diff}^m(\mathbb{R}^n)$ defines a continuous linear map $A\colon H^s(\mathbb{R}^n) \to H^{s-m}(\mathbb{R}^n)$ for all $s \in \mathbb{R}$.

Given A as in (2.52), let us define the *full symbol* of A by

$$\sigma(A)(x,\xi) := a(x,\xi) := \sum_{|\alpha|\le m} a_\alpha(x)\xi^\alpha. \tag{2.53}$$

Then, in view of (2.16), we can write

$$\begin{aligned}(Au)(x) &= (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix\cdot\xi} a(x,\xi)\hat{u}(\xi)\,\mathrm{d}\xi \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(x,\xi) u(y)\,\mathrm{d}y \right) \mathrm{d}\xi,\end{aligned} \tag{2.54}$$

which we read as an iterated integral. On the other hand, the Schwartz kernel K of A can be obtained from that of the identity operator (see Example 2.20) by applying $\sum_{|\alpha|\le m} a_\alpha(x) D_x^\alpha$, so

$$K(x,y) = \sum_{|\alpha|\le m} a_\alpha(x)(D^\alpha\delta)(x-y). \tag{2.55}$$

Comparing this with (2.54), we therefore (formally) have

$$K(x,y) = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(x,\xi)\,\mathrm{d}\xi. \tag{2.56}$$

This is indeed (rigorously) correct if one reads this as the Fourier transform of a in ξ.

The formula (2.54) will be the starting point for the development of pseudodifferential operators on $\mathbb{R}^n$ in Chapter 4.

Proposition 2.22 (Locality and Pseudolocality of Differential Operators). *Let $A \in \mathrm{Diff}^m(\mathbb{R}^n)$. Then A is* local*, that is,*

$$\operatorname{supp} Au \subset \operatorname{supp} u, \quad u \in \mathscr{S}'(\mathbb{R}^n), \tag{2.57}$$

and A is pseudolocal*, that is,*

$$\operatorname{sing\,supp} Au \subset \operatorname{sing\,supp} u, \quad u \in \mathscr{S}'(\mathbb{R}^n). \tag{2.58}$$

The proof is straightforward. From the perspective of the Schwartz kernel K of A, (2.57) is due to the fact that $K(x,y)$ is supported on the *diagonal* $x = y$, while (2.58) is related to the fact that $K(x,y)$ is smooth away from $x = y$; see Exercise 2.21. That is, adding to K an element of $\mathscr{S}(\mathbb{R}^{2n})$ preserves (2.58) but destroys (2.57). Since in this book we are mainly interested in *singularities*, it is the property (2.58) that we care about most; and this will persist when A is a pseudodifferential operator. On the other hand, the *only* continuous linear operators $A\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ satisfying the property (2.57) are differential operators (Exercise 2.22).

2.6 Semiclassical Setting

We now consider h-dependent operators, $h \in (0,1)$, for which the basic derivative is hD instead of D. The coefficients may have further h-dependence; we only require uniform bounds in $\mathcal{C}^\infty_b$, but no regularity in h.

Definition 2.23 (Semiclassical Differential Operators). Let $k \in \mathbb{R}$ and $m \in \mathbb{N}_0$. By $h^{-k}\mathrm{Diff}^m_h(\mathbb{R}^n)$, we denote the space of all families of operators $A = (A_h)_{h\in(0,h_0)}$, where $h_0 \in (0,1]$, such that

$$A_h = \sum_{|\alpha|\le m} a_\alpha(h,x)(hD)^\alpha, \qquad a_\alpha(h,\cdot) \in \mathcal{C}^\infty_b(\mathbb{R}^n), \tag{2.59}$$

and where $\{h^k a_\alpha(h,\cdot)\colon h \in (0,h_0)\}$ is a bounded subset of $\mathcal{C}^\infty_b(\mathbb{R}^n)$. We call m the *differential order* and k the *semiclassical order*. For $k = 0$, we write $\mathrm{Diff}^m_h(\mathbb{R}^n)$ simply. We denote by $h^{-k}\mathrm{Diff}^m_{h,\mathrm{cl}}(\mathbb{R}^n) \subset h^{-k}\mathrm{Diff}^m_h(\mathbb{R}^n)$ the subspace of *classical* semiclassical operators, consisting of all A of this form for which, moreover, $h^k a_\alpha \in \mathcal{C}^\infty([0,h_0);\mathcal{C}^\infty_b(\mathbb{R}^n))$. Finally, we write $\mathcal{V}_h(\mathbb{R}^n)$ for the space of (real or complex) *semiclassical vector fields*, consisting of all operators of the form $\sum_{|\alpha|=1} a_\alpha(h,x)(h\partial)^\alpha$ with $\{a_\alpha(h,\cdot)\colon h \in (0,h_0)\} \subset \mathcal{C}^\infty_b(\mathbb{R}^n)$ bounded (and real- or complex-valued).

We use here the notational convention that spaces containing semiclassical objects have a subscript "$\hbar$," while a particular member of a semiclassical

family has a subscript "h" (which is sometimes suppressed). One often abuses notation by writing $A = \sum_{|\alpha|\leq m} a_\alpha(h,x)(hD)^\alpha$, thus not making the h-dependence of the operator explicit. Furthermore, one can consider families $A = (A_h)_{h\in I}$ for arbitrary subsets $I \subset (0,1)$; a typical scenario is $I = \{h_0, h_1, \ldots\}$ where $1 \geq h_0 > h_1 > \cdots \searrow 0$; we leave the required notational modifications to the reader.

The semiclassical character of A only arises in the limit $h \searrow 0$. While every family $(A_h)_{h\in(0,h_0)} \in \mathrm{Diff}_h^m(\mathbb{R}^n)$ is a uniformly bounded family of elements of $\mathrm{Diff}^m(\mathbb{R}^n)$ (in the sense that the coefficients $h^{|\alpha|}a_\alpha(h,\cdot)$ of D^α are uniformly bounded in $\mathcal{C}_b^\infty(\mathbb{R}^n)$), the converse is of course not true: for example, $A_h = D_{x_1} = h^{-1}hD_{x_1}$ is uniformly bounded in $\mathrm{Diff}^1(\mathbb{R}^n)$ but defines an element of $h^{-k}\mathrm{Diff}_h^1(\mathbb{R}^n)$ if and only if $k \geq 1$.

Example 2.24 (Spectral Family at High Energy). Fix $z \in \mathbb{C}$. Then

$$A_h = h^2\Delta - z = \sum_{j=1}^n (hD_j)^2 - z \tag{2.60}$$

defines an element of $\mathrm{Diff}_h^2(\mathbb{R}^n)$. (More precisely, we have $(A_h)_{h\in(0,1)} \in \mathrm{Diff}_h^2(\mathbb{R}^n)$.) This operator arises from the spectral family $\Delta - \lambda$ of the Laplacian on $\mathbb{R}^n$ in the high energy limit $\lambda = h^{-2}z \to \infty$. That is, fixing $z \in \mathbb{C}$, $|z| = 1$, we have $|\lambda| \to \infty$ when $h \searrow 0$ and $h^2(\Delta - h^{-2}z) = h^2\Delta - z \in \mathrm{Diff}_h^2(\mathbb{R}^n)$ (and thus $\Delta - h^{-2}z \in h^{-2}\mathrm{Diff}_h^2(\mathbb{R}^n)$). These operators are in fact *classical* semiclassical differential operators.

The full (semiclassical) symbol of a semiclassical operator is obtained by formally replacing hD by ξ. Thus, the symbol of $A = (A_h)$ as in (2.59) is the h-dependent function

$$a(h,x,\xi) := \sum_{|\alpha|\leq m} a_\alpha(h,x)\xi^\alpha. \tag{2.61}$$

(For fixed $h > 0$, this can alternatively be obtained from the full symbol of A_h by replacing ξ with ξ/h.)

Example 2.25 (Full Symbols). The full semiclassical symbol of the operator (2.60) is given by $(h,x,\xi) \mapsto |\xi|^2 - z = \sum_{j=1}^n \xi_j^2 - z$. A slightly more interesting example is given by the semiclassical rescaling $A = h^2(\Delta + V - h^{-2}z) = h^2\Delta + h^2V - z$ of the spectral family of $\Delta + V$, $V \in \mathcal{C}_b^\infty(\mathbb{R}^n)$, at spectral parameter $h^{-2}z$: the full symbol of A is $(h,x,\xi) \mapsto |\xi|^2 + h^2V(x) - z$.

Sobolev spaces adapted to the semiclassical rescaling of derivatives are defined as follows.

Definition 2.26 (Semiclassical Sobolev Spaces on $\mathbb{R}^n$). For $s,k \in \mathbb{R}$, we define $h^k H_h^s(\mathbb{R}^n) := H^s(\mathbb{R}^n)$ as a vector space and equip it with the norm

$$\|u\|_{h^k H^s_h(\mathbb{R}^n)} := h^{-k}\|\langle hD\rangle^s u\|_{L^2(\mathbb{R}^n)}, \qquad \langle hD\rangle^s := \mathcal{F}^{-1}\langle h\xi\rangle^s \mathcal{F}. \tag{2.62}$$

For $k = 0$, we denote the space and its norm by $H^s_h(\mathbb{R}^n)$ and $\|\cdot\|_{H^s_h(\mathbb{R}^n)}$, respectively.

Every $A = (A_h)_{h\in(0,h_0)} \in \mathrm{Diff}^m_h(\mathbb{R}^n)$ is bounded between the appropriate semiclassical Sobolev spaces, by which we mean that there exists a constant C such that

$$\|A_h u\|_{H^{s-m}_h(\mathbb{R}^n)} \le C\|u\|_{H^s_h(\mathbb{R}^n)}, \qquad h \in (0, h_0). \tag{2.63}$$

This is thus a *uniform* estimate for all $h \in (0, h_0)$. (An analogous uniform estimate holds for weighted operators on weighted spaces.) This is a special case of the much more general Theorem 4.51. (It can also be proved using duality and complex interpolation.)

An interesting relationship between standard and semiclassical Sobolev spaces arises from a partial Fourier transform:

Proposition 2.27 (Sobolev Spaces and Fourier Transform). *Write $x = (x_1, x') \in \mathbb{R}\times\mathbb{R}^{n-1}$. Let $s \in \mathbb{R}$. Then the Fourier transform in x_1,*

$$(\mathcal{F}_1 u)(\xi_1, x') := \int_{\mathbb{R}} e^{-ix_1\xi_1} u(x_1, x')\,\mathrm{d}x_1, \tag{2.64}$$

extends from $\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ to an isomorphism

$$H^s(\mathbb{R}\times\mathbb{R}^{n-1}) \to L^2\big(\mathbb{R}_{\xi_1}; \langle\xi_1\rangle^{-s} H^s_{\langle\xi_1\rangle^{-1}}(\mathbb{R}^{n-1})\big), \tag{2.65}$$

where the space on the right is defined as the completion of $\mathscr{S}(\mathbb{R}\times\mathbb{R}^{n-1})$ with respect to the norm

$$\|v\|^2_{L^2\left(\mathbb{R}_{\xi_1};\langle\xi_1\rangle^{-s}H^s_{\langle\xi_1\rangle^{-1}}(\mathbb{R}^{n-1})\right)} := \int_{\mathbb{R}} \|v(\xi_1,\cdot)\|^2_{\langle\xi_1\rangle^{-s}H^s_{\langle\xi_1\rangle^{-1}}(\mathbb{R}^{n-1})}\,\mathrm{d}\xi_1. \tag{2.66}$$

This is useful when solving PDEs whose coefficients are independent of x_1 via the Fourier transform in x_1 and plays an important role in Chapter 11.

Proof of Proposition 2.27. For $v = \mathcal{F}_1 u$, the right-hand side of (2.66) is equal to

$$\int_{\mathbb{R}} \langle\xi_1\rangle^{2s} \int_{\mathbb{R}^{n-1}} \big\langle\langle\xi_1\rangle^{-1}\xi'\big\rangle^{2s} |\hat{u}(\xi_1,\xi')|^2\,\mathrm{d}\xi'\,\mathrm{d}\xi_1. \tag{2.67}$$

Since $\langle\xi_1\rangle\langle\langle\xi_1\rangle^{-1}\xi'\rangle = \langle\xi\rangle$ for $\xi = (\xi_1, \xi')$, the desired conclusion follows from the definition of $H^s(\mathbb{R}^n)$ and the density of $\mathscr{S}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n)$. □

Notes

The theory of distributions was pioneered by Schwartz [Sch52]. A self-contained presentation of the main notions and results is given in Hörmander [Hör03] and Friedlander–Joshi [Fri98]. Trèves [Trè67] contains a detailed account of topological aspects of spaces of test functions and distributions $(\mathcal{C}_c^\infty(\mathbb{R}^n),\ \mathscr{D}'(\mathbb{R}^n))$. (We will use $\mathcal{C}_c^\infty$ and $\mathscr{D}'$ when working on general manifolds in Chapter 5.)

Exercises

Exercise 2.1 (Shifted Laplacian). Recall $\Delta = \sum_{j=1}^n D_{x_j}^2$.

(1) Show that $\Delta + 1 \colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is an isomorphism.
(2) Find a nontrivial solution $u \in \mathcal{C}^\infty(\mathbb{R}^n)$ of $(\Delta + 1)u = 0$. Why does this not contradict the first part?

Exercise 2.2 (Density of $\mathcal{C}_c^\infty$ in Tempered Distributions). We will prove in a constructive manner that $\mathcal{C}_c^\infty(\mathbb{R}^n) \subset \mathscr{S}'(\mathbb{R}^n)$ (or more precisely the image of $\mathcal{C}_c^\infty(\mathbb{R}^n)$ under the map (2.12)) is dense.

(1) Let $\chi \in \mathscr{S}(\mathbb{R}^n)$, $\chi(0) = 1$. Let $\phi \in \mathscr{S}(\mathbb{R}^n)$. Put $\phi_\epsilon(x) = \chi(\epsilon x)\phi(x)$. Show that $\phi_\epsilon \to \phi$ in $\mathscr{S}(\mathbb{R}^n)$ as $\epsilon \searrow 0$. Show that the space $\mathscr{E}'(\mathbb{R}^n)$ of compactly supported distributions is dense in $\mathscr{S}'(\mathbb{R}^n)$.
(2) Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ and put $\tilde{\psi}(x) = \psi(-x)$. For $\phi \in \mathscr{S}(\mathbb{R}^n)$, define the *convolution* of ϕ with $\tilde{\psi}$ by

$$(\phi * \tilde{\psi})(x) := \int_{\mathbb{R}^n} \phi(x-y)\tilde{\psi}(y)\,\mathrm{d}y. \tag{2.68}$$

Show that $\phi * \tilde{\psi} \in \mathscr{S}(\mathbb{R}^n)$. Define the convolution of $u \in \mathscr{S}'(\mathbb{R}^n)$ with ψ by $\langle u * \psi, \phi\rangle := \langle u, \phi * \tilde{\psi}\rangle$. Check that this is the correct definition when $u \in \mathscr{S}(\mathbb{R}^n)$.
(3) Let now $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, $\int_{\mathbb{R}^n} \chi(x)\,\mathrm{d}x = 1$, and set $\chi_\epsilon(x) := \epsilon^{-n}\chi(\epsilon^{-1}x)$. Show that $\chi_\epsilon * \phi \to \phi$ in $\mathscr{S}(\mathbb{R}^n)$ as $\epsilon \searrow 0$.
(4) When $u \in \mathscr{E}'(\mathbb{R}^n)$ and $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, show that $u * \psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, that is, there exists $v \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ such that $u * \psi = T_v$.
(5) Combine the previous parts to conclude that $\mathcal{C}_c^\infty(\mathbb{R}^n) \subset \mathscr{S}'(\mathbb{R}^n)$ is dense.

Exercise 2.3 (Dual of the Space of Tempered Distributions). Suppose $F \colon \mathscr{S}'(\mathbb{R}^n) \to \mathbb{C}$ is a continuous linear functional *on the space of tempered distributions*. Show that there exists a (unique) $\phi \in \mathscr{S}(\mathbb{R}^n)$ such that $F(u) = \langle u, \phi\rangle$ for all $u \in \mathscr{S}'(\mathbb{R}^n)$.

Exercise 2.4 (Compactly Supported Distributions). Show that $\mathscr{E}'(\mathbb{R}^n)$ is the space of all continuous linear functionals $\mathcal{C}^\infty(\mathbb{R}^n) \to \mathbb{C}$. More precisely, show that $u \in \mathscr{S}'(\mathbb{R}^n)$ lies in $\mathscr{E}'(\mathbb{R}^n)$ if and only if $u\colon \mathscr{S}(\mathbb{R}^n) \to \mathbb{C}$ is the restriction to $\mathscr{S}(\mathbb{R}^n)$ of a continuous linear map $\mathcal{C}^\infty(\mathbb{R}^n) \to \mathbb{C}$.

Exercise 2.5 (Fourier Transform of Compactly Supported Distributions). Let $u \in \mathscr{E}'(\mathbb{R}^n)$. Sharpen Proposition 2.11 by showing that $\mathcal{F}u$ extends to an entire function on $\mathbb{C}^n$ for which there exist $C, N \in \mathbb{R}$ such that $|(\mathcal{F}u)(\xi)| \leq C\langle\xi\rangle^N e^{C|\operatorname{Im}\xi|}$ for all $\xi \in \mathbb{C}^n$.

Exercise 2.6 (Some Homogeneous Distributions). For $s \in \mathbb{C}$ with $\operatorname{Re} s > -1$, we define the distributions $(x \pm i0)^s$, $x_\pm^s \in \mathscr{S}'(\mathbb{R})$ by

$$\langle (x \pm i0)^s, \phi\rangle := \int_{-\infty}^{\infty} \lim_{\epsilon \searrow 0} (x \pm i\epsilon)^s \phi(x)\,\mathrm{d}x, \quad \langle x_\pm^s, \phi\rangle := \int_0^\infty x^s \phi(\pm x)\,\mathrm{d}x \tag{2.69}$$

for $\phi \in \mathscr{S}(\mathbb{R})$. Here $z^s := \exp(s \log z)$ for $z \in \mathbb{C} \setminus (-\infty, 0]$, with log being the principal branch (i.e., $\log z \in \mathbb{R}$ for $z > 0$, and the branch cut is along $(-\infty, 0]$).

(1) Show that $\frac{\mathrm{d}}{\mathrm{d}x} x_+^s = s x_+^{s-1}$ for $\operatorname{Re} s > 0$. Use this to define x_+^s for $s \in \mathbb{C} \setminus (-\mathbb{N})$ as a holomorphic family of distributions (i.e., $\mathbb{C} \setminus (-\mathbb{N}) \ni s \mapsto \langle x_+^s, \phi\rangle \in \mathbb{C}$ is holomorphic for all $\phi \in \mathscr{S}(\mathbb{R})$).
(2) For $\operatorname{Re} s \in (-1, 0)$, prove the identity

$$\mathcal{F}(x_\pm^s) = \Gamma(s+1) e^{\mp \frac{i\pi}{2}(s+1)} (\xi \mp i0)^{-s-1} \tag{2.70}$$

where $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1}\,\mathrm{d}t$, $\operatorname{Re} z > 0$, is the Γ-function.
(3) Prove $\Gamma(z+1) = z\Gamma(z)$. Deduce that $\frac{1}{\Gamma(z)}$ can be continued to an entire function with simple zeros at $z \in -\mathbb{N}_0$. Show that

$$\chi_\pm^s := \frac{1}{\Gamma(s+1)} x_\pm^s \tag{2.71}$$

defines an entire family of tempered distributions. Compute $\chi_\pm^{-k}$ for $k \in \mathbb{N}$.
(4) Show that $(x \pm i0)^s$ can be continued to an entire family of tempered distributions with

$$\mathcal{F}((x \pm i0)^s) = 2\pi e^{\pm \frac{i\pi}{2} s} \chi_\pm^{-s-1}. \tag{2.72}$$

(5) Show that $(x \pm i\epsilon)^s \in \mathscr{S}'(\mathbb{R}^n)$ converges to $(x \pm i0)^s$ in $\mathscr{S}'(\mathbb{R}^n)$ as $\epsilon \searrow 0$.
(6) Prove the identity $(x - i0)^{-1} - (x + i0)^{-1} = 2\pi i \delta(x)$.

Exercise 2.7 (Sobolev Spaces and Weak Derivatives). Let $u \in \mathscr{S}'(\mathbb{R}^n)$, $k \in \mathbb{N}_0$. Show that the following are equivalent.

(1) For all $\alpha \in \mathbb{N}_0^n$, $|\alpha| \le k$, we have $\partial^\alpha u \in L^2(\mathbb{R}^n)$.
(2) $u \in H^k(\mathbb{R}^n)$.

Exercise 2.8 (Sobolev Embedding). Let $s > \frac{n}{2}$.

(1) Prove that there exists a constant $C_s < \infty$ such that for $\phi \in \mathscr{S}(\mathbb{R}^n)$, the estimate

$$\|\phi\|_{L^\infty(\mathbb{R}^n)} := \sup_{x\in\mathbb{R}^n} |\phi(x)| \le C_s \|\phi\|_{H^s(\mathbb{R}^n)}. \tag{2.73}$$

holds. Deduce that $H^s(\mathbb{R}^n) \subset \mathcal{C}_b^0(\mathbb{R}^n)$.
(2) Show more generally that $H^s(\mathbb{R}^n) \subset \mathcal{C}_b^k(\mathbb{R}^n)$ for $s > \frac{n}{2} + k$.
(3) Prove the first equality in Theorem 2.18.

Exercise 2.9 (Algebra Properties of Sobolev Spaces). Let $n \in \mathbb{N}$.

(1) Let $u, v \in \mathscr{S}(\mathbb{R}^n)$ and recall that their convolution is defined by $(u * v)(x) = \int_{\mathbb{R}^n} u(y)v(x-y)\,\mathrm{d}y$. Show that $\mathcal{F}(u * v) = \mathcal{F}(u)\mathcal{F}(v)$. Use this to find a formula for $\mathcal{F}(uv)$.
(2) Let $s > \frac{n}{2}$. Define the function

$$a(\xi,\eta) := \frac{\langle\xi\rangle^{2s}}{\langle\eta\rangle^{2s}\langle\xi-\eta\rangle^{2s}}, \qquad \xi,\eta \in \mathbb{R}^n. \tag{2.74}$$

Show that $\sup_{\xi\in\mathbb{R}^n} \int_{\mathbb{R}^n} a(\xi,\eta)\,\mathrm{d}\eta < \infty$.
(3) Let $s > \frac{n}{2}$ and $u, v \in H^s(\mathbb{R}^n)$. Show that $uv \in H^s(\mathbb{R}^n)$, and prove the estimate $\|uv\|_{H^s(\mathbb{R}^n)} \le C\|u\|_{H^s(\mathbb{R}^n)}\|v\|_{H^s(\mathbb{R}^n)}$ for some constant C which is independent of u, v.

Exercise 2.10 (Equivalent Characterization of Sobolev Spaces). Let $s \in (0,1)$. Write $e_j = (0,\dots,1,\dots,0) \in \mathbb{R}^n$ for the j-th standard basis vector. Show that

$$\begin{aligned} &|||u|||^2_{H^s(\mathbb{R}^n)} \\ &\quad := \|u\|^2_{L^2(\mathbb{R}^n)} + \sum_{j=1}^n \int_0^\infty \Big\| \frac{\tau_h^j u - u}{h^s} \Big\|^2_{L^2(\mathbb{R}^n)} \frac{\mathrm{d}h}{h}, \quad (\tau_h^j u)(x) := u(x + he_j), \end{aligned} \tag{2.75}$$

defines an equivalent norm on $H^s(\mathbb{R}^n)$.

Exercise 2.11 (Multiplication on Sobolev Spaces). For $\alpha \in (0,1]$, write $\mathcal{C}_b^{0,\alpha}(\mathbb{R}^n)$ for the Hölder space consisting of all $u = u(x)$ with

$$\|u\|_{\mathcal{C}_b^{0,\alpha}(\mathbb{R}^n)} := \sup_{x\in\mathbb{R}^n} |u(x)| + \sup_{x\neq y} \frac{|u(x)-u(y)|}{|x-y|^\alpha} < \infty. \tag{2.76}$$

For $k \in \mathbb{N}_0$, set $\mathcal{C}_b^{k,\alpha}(\mathbb{R}^n) = \{u \in \mathcal{C}_b^k(\mathbb{R}^n) \colon \partial^\beta u \in \mathcal{C}_b^{0,\alpha}(\mathbb{R}^n)\ \forall\, \beta \in \mathbb{N}_0^n,\ |\beta| \le k\}$. Use Exercise 2.10 to show that

$$k + \alpha > |s|,\ a \in \mathcal{C}_b^{k,\alpha}(\mathbb{R}^n),\ u \in H^s(\mathbb{R}^n) \implies au \in H^s(\mathbb{R}^n). \tag{2.77}$$

(In particular, for all $a \in \mathcal{C}_b^\infty(\mathbb{R}^n)$ and $u \in H^s(\mathbb{R}^n)$, one has $au \in H^s(\mathbb{R}^n)$.)

Exercise 2.12 (Rellich Compactness for Weighted Sobolev Spaces). Let $s > s'$ and $r < r'$. Show that the inclusion map $\langle x\rangle^r H^s(\mathbb{R}^n) \to \langle x\rangle^{r'} H^{s'}(\mathbb{R}^n)$ is compact.

Exercise 2.13 (An Interpolation Inequality). Let $s_- < s < s_+$ and $\epsilon > 0$. Show that there exists a constant C_ϵ such that for all $u \in H^{s_+}(\mathbb{R}^n)$,

$$\|u\|_{H^s(\mathbb{R}^n)} \le \epsilon \|u\|_{H^{s_+}(\mathbb{R}^n)} + C_\epsilon \|u\|_{H^{s_-}(\mathbb{R}^n)}. \tag{2.78}$$

Exercise 2.14 (A Paley–Wiener Theorem). Define the spaces

$$\begin{aligned} \dot{\mathscr{S}}([0,\infty)) &:= \{u \in \mathscr{S}(\mathbb{R}) \colon \operatorname{supp} u \subset [0,\infty)\}, \\ \dot{\mathscr{S}}'([0,\infty)) &:= \{u \in \mathscr{S}'(\mathbb{R}) \colon \operatorname{supp} u \subset [0,\infty)\}. \end{aligned} \tag{2.79}$$

(1) Show that for $u \in \mathscr{S}(\mathbb{R})$, we have $u \in \dot{\mathscr{S}}([0,\infty))$ if and only if $\hat{u} \in \mathscr{S}(\mathbb{R})$ is the boundary value $\hat{u}|_{\mathbb{R}}$ of a continuous function $\hat{u} \colon \{\zeta \in \mathbb{C} \colon \operatorname{Im}\zeta \le 0\} \to \mathbb{C}$ with the following properties: $\hat{u}$ is holomorphic in $\operatorname{Im}\zeta < 0$, and for all $N \in \mathbb{R}$, there exists a constant C_N such that

$$|\hat{u}(\zeta)| \le C_N \langle \operatorname{Re}\zeta\rangle^{-N}, \quad \operatorname{Im}\zeta \le 0. \tag{2.80}$$

(2) Show that for $u \in \mathscr{S}'(\mathbb{R})$, we have $u \in \dot{\mathscr{S}}'([0,\infty))$ if and only if $\hat{u}$ is the boundary value $\hat{u} = v(0)$ of a continuous function $v \colon (-\infty, 0]_\eta \to \mathscr{S}'(\mathbb{R}_\xi)$ with the following properties: the map $(-\infty,0) \times \mathbb{R} \ni (\eta,\xi) \mapsto v(\eta)(\xi)$ is smooth, and indeed $\zeta = \xi - i\eta \mapsto v(\zeta) := v(\eta)(\xi)$ is holomorphic; and there exists $N \in \mathbb{R}$ such that for all $\epsilon > 0$, there exists a constant C_ϵ with

$$|v(\zeta)| \le C_\epsilon e^{\epsilon|\operatorname{Im}\zeta|} \langle \operatorname{Re}\zeta\rangle^N, \quad \operatorname{Im}\zeta < -\epsilon. \tag{2.81}$$

Exercise 2.15 (Trace Theorem). Let $s > \frac{1}{2}$. Show that the restriction map $\mathscr{S}(\mathbb{R}_{x_1} \times \mathbb{R}^{n-1}) \ni u \mapsto u|_{x_1=0} \in \mathscr{S}(\mathbb{R}^{n-1})$ extends by continuity and density to a bounded surjective map

$$H^s(\mathbb{R} \times \mathbb{R}^{n-1}) \to H^{s-\frac{1}{2}}(\mathbb{R}^{n-1}). \tag{2.82}$$

Remark. Iterating this yields an alternative proof of the Sobolev embedding $H^s(\mathbb{R}^n) \subset \mathcal{C}^0_b(\mathbb{R}^n)$, $s > \frac{n}{2}$, from Exercise 2.8.

Exercise 2.16 (Schwartz Representation Theorem). Prove the second equality in Theorem 2.18. You may proceed as follows. Let $u \in \mathscr{S}'(\mathbb{R}^n)$.

(1) Show that there exist C, k such that $|\langle u, \phi\rangle| \le C\|\phi\|_k$.
(2) Let $R_q = \langle x\rangle^{-q}\langle D\rangle^{-q}$. Show that R_q is an isomorphism on $\mathscr{S}(\mathbb{R}^n)$ and $\mathscr{S}'(\mathbb{R}^n)$. Show moreover that there exists q such that $\|R_q\phi\|_k \le C'\|\phi\|_{L^2(\mathbb{R}^n)}$. (*Hint.* Use Exercise 2.8. It may be convenient to take s there and q here to be even integers.)
(3) Denoting $R_q^\dagger = \langle D\rangle^{-q}\langle x\rangle^{-q}$, deduce that $R_q^\dagger u \in L^2(\mathbb{R}^n)$, and conclude that $u \in \langle x\rangle^q H^{-q}(\mathbb{R}^n)$.

Exercise 2.17 (Schwartz Kernel Theorem, I). Prove the injectivity claim of Theorem 2.19. (*Hint.* Let $K \in \mathscr{S}'(\mathbb{R}^{n+m})$ be given with $O_K = 0$. Given $\phi \in \mathscr{S}(\mathbb{R}^{n+m})$, you need to show that $\langle K, \phi\rangle = 0$. You know that this is true when ϕ is a finite linear combination of exterior products $\psi_1 \boxtimes \psi_2$, $\psi_1 \in \mathscr{S}(\mathbb{R}^n)$, $\psi_2 \in \mathscr{S}(\mathbb{R}^m)$. Use Fourier series to approximate ϕ by such linear combinations. It helps to first reduce to the case that $\operatorname{supp} K$ is compact.)

Exercise 2.18 (Schwartz Kernel Theorem, II). Let $A\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^m)$ be continuous. Prove the surjectivity claim of Theorem 2.19. You may proceed as follows.

(1) Show that for all $\psi \in \mathscr{S}(\mathbb{R}^n)$ there exists $N \in \mathbb{N}$ such that $|\langle A\phi, \psi\rangle| \le N\|\phi\|_N$ for all $\phi \in \mathscr{S}(\mathbb{R}^m)$.
(2) Show that there exist $N, M \in \mathbb{R}$ such that A extends by continuity to a bounded operator

$$A\colon \langle x\rangle^{-M} H^M(\mathbb{R}^m) \to \langle x\rangle^N H^{-N}(\mathbb{R}^n). \tag{2.83}$$

(*Hint.* An estimate from Exercise 2.16 will come in handy, in the form $\|\psi\|_k \le C_k\|\psi\|_{\langle x\rangle^{-M}H^M(\mathbb{R}^n)}$ for given k and sufficiently large M.)
(3) Show that the operator

$$A' := \langle D\rangle^{-N-\frac{n}{2}-1}\langle x\rangle^{-N} A \langle D\rangle^{-M-\frac{m}{2}-1}\langle x\rangle^{-M} \tag{2.84}$$

is bounded from $H^{-\frac{m}{2}-1}(\mathbb{R}^m)$ to $\mathcal{C}^0_b(\mathbb{R}^n)$.
(4) Evaluate $A'\delta_y$ for $y \in \mathbb{R}^m$ and deduce that the Schwartz kernel K' of A' satisfies $K' \in \mathcal{C}^0_b(\mathbb{R}^{n+m})$.
(5) By relating the Schwartz kernels of A' and A, prove that $A = O_K$ for some $K \in \mathscr{S}'(\mathbb{R}^{n+m})$.

Exercise 2.19 (Schwartz Kernel of Differential Operators). Check that (2.55) indeed equals the Schwartz kernel of the operator A in (2.52)

by verifying that (2.49) defines an operator O_K with $(O_K\phi)(\psi) = \int_{\mathbb{R}^n}(A\phi)(x)\psi(x)\,\mathrm{d}x$.

Exercise 2.20 (Operators with Schwartz Schwartz Kernels). Let $A\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ be continuous, and denote by $K \in \mathscr{S}'(\mathbb{R}^{2n})$ its Schwartz kernel. Show that $K \in \mathscr{S}(\mathbb{R}^{2n})$ if and only if A maps $\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ and extends by continuity to a bounded map $\mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$.

Exercise 2.21 (Singular Support). Suppose $A\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is a continuous linear map, and denote by $K \in \mathscr{S}'(\mathbb{R}^n \times \mathbb{R}^n)$ its Schwartz kernel.

(1) Suppose that $\operatorname{sing\,supp} K \subset \operatorname{diag}_{\mathbb{R}^n} := \{(x,y) \in \mathbb{R}^n \times \mathbb{R}^n \colon x = y\}$. Show that $\operatorname{sing\,supp} Au \subseteq \operatorname{supp} u$ for all $u \in \mathscr{S}'(\mathbb{R}^n)$.
(2) Give an example of such an operator A with $\operatorname{sing\,supp} K \subset \operatorname{diag}_{\mathbb{R}^n}$ which is *not* pseudolocal, i.e., for some $u \in \mathscr{S}'(\mathbb{R}^n)$, $\operatorname{sing\,supp} Au \not\subset \operatorname{sing\,supp} u$.
(3) Suppose that A is pseudolocal. Show that $\operatorname{sing\,supp} K \subset \operatorname{diag}_{\mathbb{R}^n}$.

Exercise 2.22 (Peetre's Theorem). Let $A\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ be a continuous linear operator, and suppose for all $u \in \mathscr{S}(\mathbb{R}^n)$, we have $\operatorname{supp} Au \subset \operatorname{supp} u$. Prove that A is a differential operator. (*Hint.* Show that the Schwartz kernel K of A has support in the diagonal $\{x = y\}$. Then show that it must be a locally finite linear combination of (differentiated) δ-distributions with smooth coefficients. To prove that A is a differential operator of *finite* order, exploit that K is a *tempered* distribution.)

Exercise 2.23 (Semiclassical and Non-semiclassical Sobolev Spaces). Let $s \in \mathbb{R}$ and write $s_\pm := \max(\pm s, 0)$. Show that

$$h^{s_-}\|u\|_{H^s_h(\mathbb{R}^n)} \le \|u\|_{H^s(\mathbb{R}^n)} \le h^{-s_+}\|u\|_{H^s_h(\mathbb{R}^n)}, \quad h \in (0,1). \tag{2.85}$$

Chapter 3
Symbols



To any function

$$a(x,\xi) = \sum_{|\alpha|\le m} a_\alpha(x)\xi^\alpha, \quad a_\alpha \in \mathcal{C}_b^\infty(\mathbb{R}^n), \tag{3.1}$$

we can associate a differential operator by setting

$$(\mathrm{Op}(a)u)(x) := \sum_{|\alpha|\le m} a_\alpha(x)D^\alpha u(x). \tag{3.2}$$

The map Op thus defined is called a *quantization map*. We have already seen in Definition 2.13 that it is useful to be able to quantize more general functions $a(x,\xi)$ such as $a(x,\xi) = \langle\xi\rangle^s$ (with $\mathrm{Op}(a) = \langle D\rangle^s$) or, say,

$$a(x,\xi) = a_s(x)\langle\xi\rangle^s, \quad a_s \in \mathcal{C}_b^\infty(\mathbb{R}^n). \tag{3.3}$$

While no longer a polynomial in ξ, this function shares many important properties with polynomials: it gains a power of decay as $|\xi| \to \infty$ upon each differentiation in ξ, and its bounds as $|\xi| \to \infty$ do not get worse when differentiating in x. General classes of *symbols* with such properties are introduced and analyzed in Section 3.1.

While knowledge of the full symbol $a(x,\xi)$ is needed to determine a corresponding operator $\mathrm{Op}(a)$, it is unnecessary for many purposes to keep track of the full symbol. The *principal symbol*, which in the case of differential operators $A = \mathrm{Op}(a)$ as in (3.2) only keeps the terms in (3.1) with $|\alpha| = m$, controls key properties of A such as its ellipticity.

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Definition 3.1 (Principal Symbol). Given a differential operator $A = \sum_{|\alpha|\le m} a_\alpha(x)D^\alpha$, $m \in \mathbb{N}_0$, its *principal symbol* is defined as

$$\sigma^m(A)(x,\xi) := \sum_{|\alpha|=m} a_\alpha(x)\xi^\alpha. \tag{3.4}$$

One says that A is *elliptic* if $\sigma^m(A)(x,\xi) \neq 0$ for all $x \in \mathbb{R}^n$ and $\xi \neq 0$. Examples are the Cauchy–Riemann operator $A = \frac{1}{2}(\partial_{x_1} + i\partial_{x_2}) \in \mathrm{Diff}^1(\mathbb{R}^2)$ and the operator $A = \Delta - z \in \mathrm{Diff}^2(\mathbb{R}^n)$ for any $z \in \mathbb{C}$. The notion of ellipticity for general symbols is discussed in Section 3.2.

Remark 3.2 (Order). The principal symbol depends on a choice of m. For example, one may regard an operator $A \in \mathrm{Diff}^{m-1}(\mathbb{R}^n)$ as an operator of order m, and as such its principal symbol $\sigma^m(A)$ vanishes. Put differently, for $A \in \mathrm{Diff}^m(\mathbb{R}^n)$, we have $\sigma^m(A) = 0$ if and only if $A \in \mathrm{Diff}^{m-1}(\mathbb{R}^n)$.

For further motivation for the principal symbol, we note that it behaves in a very simple manner under standard operations on differential operators such as taking adjoints, compositions, and changing coordinates:

Proposition 3.3 (Behavior of the Principal Symbol). *Let* $A \in \mathrm{Diff}^m(\mathbb{R}^n)$.

(1) *Define the adjoint* A^* *of* A *by* $\int_{\mathbb{R}^n}(A^*u)(x)\overline{v(x)}\,\mathrm{d}x = \int_{\mathbb{R}^n} u(x)\overline{(Av)(x)}\,\mathrm{d}x$, $u, v \in \mathcal{C}_c^\infty(\mathbb{R}^n)$. *Then* $A^* \in \mathrm{Diff}^m(\mathbb{R}^n)$ *and*

$$\sigma^m(A^*)(x,\xi) = \overline{\sigma^m(A)(x,\xi)}. \tag{3.5}$$

(2) *Let* $B \in \mathrm{Diff}^{m'}(\mathbb{R}^n)$. *Then* $A \circ B \in \mathrm{Diff}^{m+m'}(\mathbb{R}^n)$ *and*

$$\sigma^{m+m'}(A \circ B)(x,\xi) = \sigma^m(A)(x,\xi)\sigma^{m'}(B)(x,\xi). \tag{3.6}$$

(3) *Let* $\kappa\colon \mathbb{R}^n \to \mathbb{R}^n$ *be a diffeomorphism which is the identity outside of a compact set. Define* $A_\kappa\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ *by* $(A_\kappa u)(y) := (A(u \circ \kappa^{-1}))(\kappa(y))$. *Then* $A_\kappa \in \mathrm{Diff}^m(\mathbb{R}^n)$ *and*

$$\sigma^m(A_\kappa)(y,\eta) = \sigma^m(A)\big(\kappa(y), (\kappa'(y)^T)^{-1}\eta\big). \tag{3.7}$$

Thus, loosely speaking, the principal symbol maps map the non-commutative algebra $\mathrm{Diff}(\mathbb{R}^n) = \bigcup_{m\in\mathbb{N}_0}\mathrm{Diff}^m(\mathbb{R}^n)$ of operators into a *commutative* algebra of functions. Property (3) shows that the principal symbol is well-defined as a function on the cotangent bundle $T^*\mathbb{R}^n$; this invariant perspective is discussed at length in Chapter 5.

Proof of Proposition 3.3. It suffices to consider $A = a(x)D^\alpha$, $|\alpha| = m$, and $B = b(x)D^\beta$, $|\beta| = m'$. Then $A^* = D^\alpha\overline{a} = \overline{a}D^\alpha + [D^\alpha, \overline{a}]$. The first part thus follows from $[D^\alpha, \overline{a}] \in \mathrm{Diff}^{m-1}(\mathbb{R}^n)$. For the second part, we note that

$$A \circ B = abD^{\alpha+\beta} + a[D^\alpha, b]D^\beta, \tag{3.8}$$

with the second summand lying in $\mathrm{Diff}^{m+m'-1}(\mathbb{R}^n)$. For the final part, write $y = \kappa^{-1}(x)$. The chain rule gives $\partial_x = ((\kappa^{-1})'|_x)^T \partial_y = (\kappa'(y)^T)^{-1}\partial_y$ and thus

$$(A_\kappa u)(y) = a(\kappa(y))\big((\kappa'(y)^T)^{-1}D_y\big)^\alpha u. \tag{3.9}$$

Expanding the derivatives, all terms where at least one y-derivative falls on $(\kappa'(y)^T)^{-1}$ contribute differential operators of order $\leq m-1$ acting on u. Thus, A_κ has principal symbol $a(\kappa(y))((\kappa'(y)^T)^{-1}\eta)^\alpha$, as claimed. □

Many symbols arising in applications share the following similarity with (full) symbols of differential operators: they are, for $|\xi| > 1$, sums of terms which are homogeneous with respect to dilations $\xi \mapsto \lambda\xi$, $\lambda \geq 1$. (The only homogeneous functions that are smooth at $\xi = 0$ are polynomials; see Exercise 3.5.) For example, the symbol $\langle\xi\rangle^s$, for $s \notin 2\mathbb{N}_0$, can be written as a sum $\sum_{j\geq 0} c_j|\xi|^{s-j}$ for suitable $c_j \in \mathbb{R}$; this is an example of a *classical symbol*; see Section 3.3 and Exercise 3.3. The convergence of this sum is actually irrelevant for all microlocal purposes: symbols of order $-\infty$ (i.e., with $|\xi|^{-N}$ decay as $|\xi| \to \infty$, for all N) will correspond, via quantization, to *smoothing* operators (see Lemma 4.17). We thus discuss the important notion of *asymptotic summation* in Section 3.4: this allows one to sum infinitely many symbols of increasingly negative order (i.e., more and more decay as $|\xi| \to \infty$), modulo symbols of order $-\infty$ (so-called *residual symbols*), without any convergence requirements. Asymptotic summation arises frequently in the construction of full symbols of auxiliary operators, such as approximate inverses of elliptic operators (Section 4.7), in which one constructs symbols by iteratively adding correction terms of increasingly lower order.

Symbols depending on an additional (semiclassical) parameter $h \in (0, 1)$ are discussed in Section 3.5.

3.1 Basic Definitions

The main definition of this chapter reads as follows.

Definition 3.4 (Symbols). Let $m \in \mathbb{R}$, $n, N \in \mathbb{N}_0$. Then the space of *(uniform) symbols of order* m

$$S^m(\mathbb{R}^n; \mathbb{R}^N) \subset \mathcal{C}^\infty(\mathbb{R}^n \times \mathbb{R}^N) \tag{3.10}$$

consists of all functions $a = a(x, \xi)$ which for all $\alpha \in \mathbb{N}_0^n$, $\beta \in \mathbb{N}_0^N$ satisfy the estimate

$$|\partial_x^\alpha \partial_\xi^\beta a(x,\xi)| \le C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|}. \tag{3.11}$$

for some constants $C_{\alpha\beta}$. We also write

$$S^m(\mathbb{R}^N) := S^m(\mathbb{R}^0;\mathbb{R}^N) \tag{3.12}$$

for symbols depending only on ξ.

The gain of decay upon differentiation in the ξ-variables is often called *symbolic behavior* (in ξ), and ξ is the *symbolic variable*.

Remark 3.5 (Alternative Notation). Sometimes these symbol classes are denoted $S^m_\infty(\mathbb{R}^n;\mathbb{R}^N)$, the subscript "$\infty$" indicating the uniform boundedness of the "coefficients" in the x-variables.

Equipped with the norms given by the best constants in (3.11), or more concisely

$$\|a\|_{m,k} := \sup_{(x,\xi)\in\mathbb{R}^n\times\mathbb{R}^N} \max_{|\alpha|+|\beta|\le k} \langle\xi\rangle^{-m+|\beta|}|\partial_x^\alpha\partial_\xi^\beta a(x,\xi)|, \quad k\in\mathbb{N}_0, \tag{3.13}$$

the space $S^m(\mathbb{R}^n;\mathbb{R}^N)$ is a Fréchet space. Directly from the definition, we have continuous linear maps

$$\begin{aligned} D_x^\alpha &: \ S^m(\mathbb{R}^n;\mathbb{R}^N) \to S^m(\mathbb{R}^n;\mathbb{R}^N), \\ D_\xi^\beta &: \ S^m(\mathbb{R}^n;\mathbb{R}^N) \to S^{m-|\beta|}(\mathbb{R}^n;\mathbb{R}^N). \end{aligned} \tag{3.14}$$

Example 3.6 (Symbols, I). Full symbols of differential operators of order $m\in\mathbb{N}_0$ on $\mathbb{R}^n$, see (2.53), lie in $S^m(\mathbb{R}^n;\mathbb{R}^n)$. A special case of this is: given $a\in\mathcal{C}^\infty_b(\mathbb{R}^n)$, the function $(x,\xi)\mapsto a(x)$ lies in $S^0(\mathbb{R}^n;\mathbb{R}^N)$ (for any N).

Example 3.7 (Symbols, II). Let $m\in\mathbb{R}$. Then $\langle\xi\rangle^m \in S^m(\mathbb{R}^n)$ (Exercise 3.3).

Proposition 3.8 (Multiplication of Symbols). *Pointwise multiplication of symbols is a continuous bilinear map*

$$S^m(\mathbb{R}^n;\mathbb{R}^N)\times S^{m'}(\mathbb{R}^n;\mathbb{R}^N) \to S^{m+m'}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.15}$$

Proof. This follows from the Leibniz rule: for $a\in S^m(\mathbb{R}^n;\mathbb{R}^N)$, $b\in S^{m'}(\mathbb{R}^n;\mathbb{R}^N)$, and $\alpha\in\mathbb{N}_0^n$, $\beta\in\mathbb{N}_0^N$, we have

$$
\begin{aligned}
|\partial_x^\alpha \partial_\xi^\beta (a \cdot b)| &= \Big| \sum_{\substack{\alpha'+\alpha''=\alpha \\ \beta'+\beta''=\beta}} \binom{\alpha}{\alpha'} \binom{\beta}{\beta'} (\partial_x^{\alpha'} \partial_\xi^{\beta'} a)(\partial_x^{\alpha''} \partial_\xi^{\beta''} b) \Big| \\
&\leq \sum_{\substack{\alpha'+\alpha''=\alpha \\ \beta'+\beta''=\beta}} C'_{\alpha'\beta'} C''_{\alpha''\beta''} \langle \xi \rangle^{m+m'-|\beta'|-|\beta''|} \\
&\leq C_{\alpha\beta} \langle \xi \rangle^{m+m'-|\beta|},
\end{aligned}
$$

as required. □

We note the trivial continuous inclusion

$$
m \leq m' \quad \Longrightarrow \quad S^m(\mathbb{R}^n;\mathbb{R}^N) \subseteq S^{m'}(\mathbb{R}^n;\mathbb{R}^N), \tag{3.16}
$$

and hence the spaces $S^m(\mathbb{R}^n;\mathbb{R}^N)$ give a filtration of the space $\bigcup_{m\in\mathbb{R}} S^m(\mathbb{R}^n;\mathbb{R}^N)$ of all symbols. In the other direction, we define the space of *residual symbols* by

$$
S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N) := \bigcap_{m\in\mathbb{R}} S^m(\mathbb{R}^n;\mathbb{R}^N). \tag{3.17}
$$

Equipped with the norms $\|\cdot\|_{m,k}$, $m \in \mathbb{Z}$, $k \in \mathbb{N}_0$, this is again a Fréchet space.

Example 3.9 (Symbols, III). We have $\mathscr{S}(\mathbb{R}^n \times \mathbb{R}^N) \subset S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$. Moreover, $\mathscr{S}(\mathbb{R}^N) = S^{-\infty}(\mathbb{R}^N)$, and more generally $\mathcal{C}_b^\infty(\mathbb{R}^n, \mathscr{S}(\mathbb{R}^N)) = S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$. Finally, given a cutoff $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^N)$, its pullback along $\mathbb{R}^n \times \mathbb{R}^N \ni (x,\xi) \mapsto \xi$, i.e., $(x,\xi) \mapsto \chi(\xi)$, is a residual symbol.

While the inclusion (3.16) never has dense range for $m < m'$ (cf. Exercise 3.7), there is a satisfactory replacement:

Proposition 3.10 (Density Properties of Symbol Spaces). *Let $m < m'$. Then $S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$ is a dense subspace of $S^m(\mathbb{R}^n;\mathbb{R}^N)$ in the topology of $S^{m'}(\mathbb{R}^n;\mathbb{R}^N)$. In fact, a stronger statement is true: for any $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$, there exists a sequence $a_j \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$ which is uniformly bounded in $S^m(\mathbb{R}^n;\mathbb{R}^N)$ and converges to a in the topology of $S^{m'}(\mathbb{R}^n;\mathbb{R}^N)$.*

Proof. Fix a cutoff function $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^N) \subset S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$ (see Example 3.9) which is identically 1 for $|\xi| \leq 1$ and 0 for $|\xi| \geq 2$ and satisfies $\chi(\xi) \in [0,1]$ for all ξ. By Proposition 3.8, we have

$$
a_j(x,\xi) := a(x,\xi)\chi(\xi/j) \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.18}
$$

In view of Proposition 3.8, it suffices to show that

$$
\chi_j(\xi) := \chi(\xi/j) \tag{3.19}
$$

is bounded in $S^0(\mathbb{R}^N)$ and converges to 1 in the topology of $S^\epsilon(\mathbb{R}^N)$ for all $\epsilon > 0$. Regarding the former, we have $|\chi_j(\xi)| \le \|\chi\|_{0,0}$ for all j, while for $|\beta| \ge 1$, we have $\partial_\xi^\beta \chi_j(\xi) = 0$ for $|\xi| \le j$ and

$$|\xi|^{|\beta|}\partial_\xi^\beta \chi_j(\xi) = \chi_\beta(\xi/j), \quad \chi_\beta(\xi) := |\xi|^{|\beta|}(\partial_\xi^\beta \chi)(\xi) \in \mathcal{C}_c^\infty(\mathbb{R}^N). \tag{3.20}$$

Regarding the latter, we note that $\operatorname{supp}(\chi_j - 1) \subset \{|\xi| \ge j\}$. Therefore,

$$|\chi(\xi/j) - 1| \le j^{-\epsilon}|\xi|^\epsilon \le j^{-\epsilon}\langle\xi\rangle^\epsilon. \tag{3.21}$$

For derivatives, we note that the support observation and (3.20) give

$$|\xi|^{|\beta|-\epsilon}|\partial_\xi^\beta(\chi_j(\xi) - 1)| = |\xi|^{|\beta|-\epsilon}|\partial_\xi^\beta \chi_j(\xi)| \le j^{-\epsilon}|\chi_\beta(\xi/j)|. \tag{3.22}$$

Thus, $\|\chi_j - 1\|_{\epsilon,k} \le C_{\epsilon,k} j^{-\epsilon} \to 0$ as $j \to \infty$. □

3.2 Ellipticity

We now generalize a key property of the symbol of the operator $\Delta+1$ in (1.9).

Definition 3.11 (Elliptic Symbols). Let $m \in \mathbb{R}$. A symbol $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$ is *(uniformly) elliptic* if there exists a symbol $b \in S^{-m}(\mathbb{R}^n;\mathbb{R}^N)$ such that $ab - 1 \in S^{-1}(\mathbb{R}^n;\mathbb{R}^N)$.

A simple example with $m = 2$ is $a(x,\xi) = |\xi|^2 - z$ where $z \in \mathbb{C}$ is arbitrary (e.g., $z = -1$).

Proposition 3.12 (Equivalent Formulations of Ellipticity). *Let $m \in \mathbb{R}$, $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$. Then the following are equivalent:*

(1) *a is elliptic.*
(2) *There exist constants $C, c > 0$ such that*

$$|\xi| \ge C \quad \Longrightarrow \quad |a(x,\xi)| \ge c|\xi|^m. \tag{3.23}$$

(3) *There exist constants $C', c' > 0$ such that*

$$|a(x,\xi)| \ge c'|\xi|^m - C'|\xi|^{m-1}, \quad |\xi| \ge 1. \tag{3.24}$$

Proof. If a is elliptic, then in the notation of Definition 3.11, we have

$$1 - C\langle\xi\rangle^{-1} \le |a(x,\xi)||b(x,\xi)| \le C|a(x,\xi)|\langle\xi\rangle^{-m} \tag{3.25}$$

for some constant $C > 0$, that is,

$$|a(x,\xi)| \geq \frac{1}{C}\langle\xi\rangle^m - \langle\xi\rangle^{m-1}. \tag{3.26}$$

This implies (3.24) since $\frac{\langle\xi\rangle}{|\xi|} \in (1, \sqrt{2}]$ for $|\xi| \geq 1$. This in turn implies (3.23) with $c = \frac{c'}{2}$ and $C = \frac{2C'}{c'}$ since then $|\xi| \geq C$ implies $C'|\xi|^{m-1} \leq \frac{c'}{2}|\xi|^m$.

Finally, if (3.23) holds, choose a cutoff $\chi \in \mathcal{C}^\infty(\mathbb{R}^n)$, $\chi(\xi) = 0$ for $|\xi| \leq C$, $\chi(\xi) = 1$ for $|\xi| \geq 2C$. Then (Exercise 3.4)

$$b(x,\xi) := \chi(\xi)/a(x,\xi) \in S^{-m}(\mathbb{R}^n;\mathbb{R}^N), \tag{3.27}$$

and $a(x,\xi)b(x,\xi) - 1 = \chi(\xi) - 1 \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$. □

Note that if $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$ is elliptic, then so is $a + a'$ for every $a' \in S^{m-1}(\mathbb{R}^n;\mathbb{R}^N)$. Thus, ellipticity is only a condition on the equivalence class

$$[a] \in S^m(\mathbb{R}^n;\mathbb{R}^N)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.28}$$

For full symbols of differential operators, we can identify $[a]$ with the leading order, homogeneous of degree m, part of a. This holds more generally for *classical symbols*, which we introduce next.

3.3 Classical Symbols

An important subclass of symbols mimics those of differential operators (which are sums of functions that are homogeneous in ξ). More precisely, we call a function $a(x,\xi)$, defined for $\xi \neq 0$, (positively) homogeneous of degree $m \in \mathbb{R}$ if and only if

$$a(x,\lambda\xi) = \lambda^m a(x,\xi), \quad \lambda > 0. \tag{3.29}$$

Definition 3.13 (Homogeneous Symbols). Let $m \in \mathbb{R}$. Then $S^m_{\mathrm{hom}}(\mathbb{R}^n;\mathbb{R}^N\setminus\{0\})$ is the space of all functions $a(x,\xi) \in \mathcal{C}^\infty(\mathbb{R}^n \times (\mathbb{R}^N\setminus\{0\}))$, positively homogeneous of degree m in ξ, such that for all $\alpha \in \mathbb{N}_0^n$, $\beta \in \mathbb{N}_0^N$, there exists a constant $C_{\alpha\beta}$ with

$$|\partial_x^\alpha \partial_\xi^\beta a(x,\xi)| \leq C_{\alpha\beta}|\xi|^{m-|\beta|}, \quad \xi \neq 0. \tag{3.30}$$

Examples of homogeneous symbols are the homogeneous components of the full symbol $a(x,\xi) = \sum_{|\alpha|\leq m} a_\alpha(x)\xi^\alpha$ of a differential operator, to wit,

$$\sum_{|\alpha|=j} a_\alpha(x)\xi^\alpha \in S^j_{\mathrm{hom}}(\mathbb{R}^n;\mathbb{R}^n \setminus \{0\}). \tag{3.31}$$

These homogeneous symbols of course extend smoothly to $\xi = 0$. Homogeneous polynomials are the *only* homogeneous symbols with this property (Exercise 3.5). If $a \in S^m_{\hom}(\mathbb{R}^n;\mathbb{R}^N \setminus \{0\})$, we can thus typically only fashion an element of $S^m(\mathbb{R}^n;\mathbb{R}^N)$ out of a by multiplying a with $1-\chi$ where $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^N)$ is equal to 1 near 0. Taking sums of such symbols leads to the following definition (which foreshadows the notion of *asymptotic summation*, discussed in the next section):

Definition 3.14 (Classical Symbols). Let $m \in \mathbb{R}$, and fix a cutoff $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^N)$ which is identically 1 near 0. A symbol $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$ is called a *classical symbol of order* m if there exist functions $a_{m-j} \in S^{m-j}_{\hom}(\mathbb{R}^n;\mathbb{R}^N \setminus \{0\})$, $j \in \mathbb{N}_0$, such that for all $J \in \mathbb{N}$, we have

$$a - \sum_{j=0}^{J-1} (1-\chi) a_{m-j} \in S^{m-J}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.32}$$

The space of classical symbols of order m is denoted $S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$. Finally, we put

$$S^{-\infty}_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N) := S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.33}$$

Instead of the symbol class membership (3.32), one can show that it is sufficient to merely require pointwise bounds (Exercise 3.9).

Lemma 3.15 (Homogeneous Components of Classical Symbols). *The homogeneous terms a_{m-j} in (3.32) are uniquely determined by a.*

The proof is left to the reader (Exercise 3.11).

Equipped with the seminorms of a_{m-j} and the remainders $a - \sum_{j=0}^{J-1}(1-\chi)a_{m-j}$ in the respective spaces, $S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$ is a Fréchet space. Proposition 3.10 fails dramatically for classical symbols; indeed, one readily sees that

$$S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N) \subset S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N) \quad \text{is closed for all } m \in \mathbb{R}. \tag{3.34}$$

For $a \in S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$ as in Definition 3.14, we can identify the equivalence class $[a] \in S^m(\mathbb{R}^n;\mathbb{R}^N)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^N)$ with the leading order homogeneous part a_m (cf. (3.4)).

3.4 Asymptotic Summation

There is a (general) "converse" to (3.32) that is very useful when performing iterative constructions of symbols (classical or not) involving increasingly lower-order corrections:

Proposition 3.16 (Existence and Uniqueness of Asymptotic Sums). *Let $a_j \in S^{m_j}(\mathbb{R}^n;\mathbb{R}^N)$, $j \geq 0$, and suppose $\limsup_{j\to\infty} m_j = -\infty$. Let $\bar{m}_j := \sup_{j'\geq j} m_{j'}$ and $m = \bar{m}_0$. Then there exists a symbol $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$ such that for all $J \in \mathbb{N}$,*

$$a - \sum_{j=0}^{J-1} a_j \in S^{\bar{m}_J}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.35}$$

Moreover, a is unique modulo $S^{-\infty}(\mathbb{R}^n;\mathbb{R}^N)$.

We call a "the" asymptotic sum of the a_j and write

$$a \sim \sum_{j=0}^{\infty} a_j. \tag{3.36}$$

Proof of Proposition 3.16. This is similar to Borel's lemma concerning the existence of a smooth function with prescribed Taylor series at 0. Uniqueness is clear, since any two asymptotic sums a, a' satisfy $a - a' \in S^{\bar{m}_J}(\mathbb{R}^n;\mathbb{R}^N)$ for all J. Since $\bar{m}_J \to -\infty$, the difference $a - a'$ is residual indeed.

For existence, we may partially sum finitely many of the a_j and thereby reduce to the case that $a_j \in S^{m-j}(\mathbb{R}^n;\mathbb{R}^N)$, $j \geq 0$, and $\bar{m}_j = m - j$. Fix a cutoff $\chi \in \mathcal{C}^\infty(\mathbb{R}^N)$, equal to 0 for $|\xi| \leq 1$ and equal to 1 for $|\xi| \geq 2$. With $\epsilon_j > 0$, $\epsilon_j \to 0$, to be determined, we set

$$a(x,\xi) := \sum_{j=0}^{\infty} \chi(\epsilon_j \xi) a_j(x,\xi). \tag{3.37}$$

This sum is locally finite, so $a \in \mathcal{C}^\infty(\mathbb{R}^n \times \mathbb{R}^N)$. For a more careful choice of ϵ_j, we can arrange that

$$\|\chi(\epsilon_j \cdot) a_j\|_{m-j',j'} \leq 2^{-j}, \quad j > j' \geq 0. \tag{3.38}$$

Indeed, for fixed j, j', we can choose $\epsilon_j > 0$ such that this holds since $\chi(\epsilon_j \cdot) a_j \to 0$ in $S^{m-j'}(\mathbb{R}^n;\mathbb{R}^N)$ as $\epsilon_j \to 0$ (see the proof of Proposition 3.10); but for any fixed j, (3.38) gives a *finite* number of conditions on ϵ_j, one for each $0 \leq j' < j$.

But then $\chi(\epsilon_J \xi) a_J(x,\xi) + \sum_{j=J+1}^{\infty} \chi(\epsilon_j \xi) a_j(x,\xi)$ converges in $S^{m-J}(\mathbb{R}^n;\mathbb{R}^N)$. Thus, the series (3.37) converges in $S^m(\mathbb{R}^n;\mathbb{R}^N)$, and we have

$$a(x,\xi) - \sum_{j=0}^{J-1} a_j(x,\xi) = \sum_{j=0}^{J-1} (\chi(\epsilon_j\xi) - 1) a_j(x,\xi) + \sum_{j=J}^{\infty} \chi(\epsilon_j\xi) a_j(x,\xi) \in S^{m-J}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.39}$$

The proof is complete. □

The space $S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$ can be characterized as the space of symbols in $S^m(\mathbb{R}^n;\mathbb{R}^N)$ which are asymptotic sums of symbols which in $|\xi| \geq 1$ are positively homogeneous of degree $m-j$, $j \in \mathbb{N}_0$.

For completeness and later use, we refine the previous result to ensure the continuous dependence of a on the sequence (a_j).

Proposition 3.17 (Continuous Asymptotic Summation). *Denote by*

$$\ell S^m(\mathbb{R}^n;\mathbb{R}^N) := \prod_{j=0}^{\infty} S^{m-j}(\mathbb{R}^n;\mathbb{R}^N) \tag{3.40}$$

the space of all sequences $(a_0, a_1, \ldots)$ of symbols $a_j \in S^{m-j}(\mathbb{R}^n;\mathbb{R}^N)$. Equip $\ell S^m(\mathbb{R}^n;\mathbb{R}^N)$ with the topology generated by the seminorms $\|(a_j)\|_J := \max_{0\leq k\leq J} \|a_k\|_{m-k,J}$. Then there exists a continuous (nonlinear) map

$$\sum\nolimits_{\mathcal{A}} \colon \ell S^m(\mathbb{R}^n;\mathbb{R}^N) \to S^m(\mathbb{R}^n;\mathbb{R}^N) \tag{3.41}$$

with the property that $\sum_{\mathcal{A}}((a_j)_{j\in\mathbb{N}_0}) \sim \sum_{j=0}^{\infty} a_j$.

Remark 3.18 (Comparison with $\mathcal{C}^\infty(\mathbb{R}^n)$). The topology on $\ell S^m(\mathbb{R}^n;\mathbb{R}^N)$ is akin to the standard topology on $\mathcal{C}^\infty(\mathbb{R}^n)$ which is induced by the seminorms $\|\cdot\|_{\mathcal{C}^k(B(0,k))}$. The convergence of a sequence of elements of $\ell S^m(\mathbb{R}^n;\mathbb{R}^N)$ is equivalent to the convergence of the first J terms of each element in the respective symbol space, for every $J \in \mathbb{N}$.

Proof of Proposition 3.17. Fix $\chi \in \mathcal{C}^\infty(\mathbb{R}^N)$ with $\chi(\xi) = 0$ for $|\xi| \leq 1$ and $\chi(\xi) = 1$ for $|\xi| \geq 2$. As in the previous proof, we shall set, for $a = (a_j)_{j\in\mathbb{N}_0} \in \ell S^m(\mathbb{R}^n;\mathbb{R}^N)$,

$$\Big(\sum\nolimits_{\mathcal{A}} a\Big)(x,\xi) := \sum_{j=0}^{\infty} \chi(\epsilon_j(a)\xi) a_j(x,\xi), \tag{3.42}$$

where $\epsilon_j(a)$, as in (3.38), is chosen such that for all $j \in \mathbb{N}$

$$\max_{0\leq j'\leq j-1} \|\chi(\epsilon_j(a)\xi) a_j(x,\xi)\|_{m-j',j'} \leq 2^{-j}, \tag{3.43}$$

and we set $\epsilon_0(a) = 1$. We now need to make a concrete choice of $\epsilon_j(a)$. To this end, we note that for $|\alpha| + |\beta| \le j' \le j - 1$,

$$\begin{aligned}
&\max_{0\le j'\le j-1} \langle\xi\rangle^{-m+j'+|\beta|} \left|\partial_x^\alpha \partial_\xi^\beta \big(\chi(\epsilon_j(a)\xi) a_j(x,\xi)\big)\right| \\
&\quad \le \max_{0\le j'\le j-1} C_j \langle\xi\rangle^{-m+j'} \langle\xi\rangle^{m-j} \|a_j\|_{m-j,j'} 1_{|\xi|\ge \epsilon_j(a)^{-1}} \\
&\quad \le C_j \epsilon_j(a) \|a_j\|_{m-j,j},
\end{aligned} \tag{3.44}$$

where C_j only depends on χ (and j of course). Therefore, (3.43) holds, provided we take

$$\epsilon_j(a) := 2^{-j}\big(1 + C_j \|a_j\|_{m-j,j}\big)^{-1}. \tag{3.45}$$

With this choice, $\sum_{\mathcal{A}} a$ is well-defined, and $\sum_{\mathcal{A}} a \sim \sum_{j=0}^\infty a_j$.

We now check continuity. Define

$$\chi_j(q,\xi) := \chi(2^{-j}(1 + C_j q)^{-1}\xi), \tag{3.46}$$

so $(\sum_{\mathcal{A}} a)(x,\xi) = \sum_{j=0}^\infty \chi_j(\|a_j\|_{m-j,j},\xi) a_j(x,\xi)$. Fix $a = (a_j)_{j\in\mathbb{N}_0} \in \ell S^m(\mathbb{R}^n;\mathbb{R}^N)$, and fix $k \in \mathbb{N}_0$, $\epsilon > 0$. We need to show that there exist $\delta > 0$ and $J \in \mathbb{N}$ such that

$$a' \in \ell S^m(\mathbb{R}^n;\mathbb{R}^N),\ \|a - a'\|_J \le \delta \implies \Big\|\sum\nolimits_{\mathcal{A}} a - \sum\nolimits_{\mathcal{A}} a'\Big\|_{m,k} < \epsilon, \tag{3.47}$$

which holds provided

$$\sum_{j=0}^\infty \big\|\chi_j(\|a_j\|_{m-j,j},\xi) a_j - \chi_j(\|a_j'\|_{m-j,j},\xi) a_j'\big\|_{m,k} < \epsilon. \tag{3.48}$$

We first record for $q, q' \ge 0$ the estimate

$$\|\chi_j(q,\xi) - \chi_j(q',\xi)\|_{0,k} \le C_j' |q - q'|. \tag{3.49}$$

We only prove this for $k = 0$. Without loss, we may assume $q \le q'$. Then $\chi_j(q,\xi) - \chi_j(q',\xi) = 0$ unless $2^j(1 + C_j q) \le |\xi| \le 2^{j+1}(1 + C_j q')$. When $|\xi|$ lies in this range, we can estimate

$$
\begin{aligned}
|\chi_j(q,\xi)-\chi_j(q',\xi)| &= \Big|\chi\Big(\frac{\xi}{2^j(1+C_jq)}\Big)-\chi\Big(\frac{\xi}{2^j(1+C_jq')}\Big)\Big| \\
&\le \|\nabla\chi\|_{L^\infty}\frac{|\xi|}{2^j}\Big(\frac{1}{1+C_jq}-\frac{1}{1+C_jq'}\Big) \\
&\le 2\|\nabla\chi\|_{L^\infty}\Big(\frac{1+C_jq'}{1+C_jq}-1\Big) \\
&\le 2C_j\|\nabla\chi\|_{L^\infty}(q'-q).
\end{aligned}
\tag{3.50}
$$

The j-th summand in (3.48) can now be estimated by

$$
\begin{aligned}
&\|\chi_j(\|a_j\|_{m-j,j},\xi)(a_j-a_j')\|_{m,k} \\
&\quad+\big\|\big(\chi_j(\|a_j\|_{m-j,j},\xi)-\chi_j(\|a_j'\|_{m-j,j},\xi)\big)a_j'\big\|_{m,k} \\
&\qquad\le \tilde{C}_j\big(\|a_j-a_j'\|_{m,k}+\big|\|a_j\|_{m-j,j}-\|a_j'\|_{m-j,j}\big|\|a_j'\|_{m,k}\big) \\
&\qquad\le \tilde{C}_j(1+\|a_j'\|_{m,k})\|a-a'\|_{\max(j,k)}.
\end{aligned}
\tag{3.51}
$$

This tends to zero as $a'\to a$ in $\ell S^m(\mathbb{R}^n;\mathbb{R}^N)$.

We estimate the tail of the sum (3.48) using (3.43): this gives the upper bound

$$
\|\chi_j(\|a_j\|_{m-j,j},\xi)a_j\|_{m,k}+\|\chi_j(\|a_j'\|_{m-j,j},\xi)a_j'\|_{m,k}\le 2^{-j}+2^{-j}=2^{-j+1}
\tag{3.52}
$$

provided $j>k$. Thus, we first choose $J_0\in\mathbb{N}$, $J_0>k$, such that $\sum_{j=J_0}^\infty 2^{-j+1}<\epsilon/2$, and then $\delta>0$, $J\in\mathbb{N}$ such that $a'\in\ell S^m(\mathbb{R}^n;\mathbb{R}^N)$, $\|a-a'\|_J<\delta$ implies that the j-th summand in (3.48) is bounded by $\epsilon/(2J_0)$ for $j=0,\dots,J_0-1$. This achieves (3.47). □

3.5 Semiclassical Symbols

The following definition generalizes Definition 3.4 in the same way that Definition 2.23 generalizes Definition 2.21.

Definition 3.19 (Semiclassical Symbols). Let $k,m\in\mathbb{R}$ and $n,N\in\mathbb{N}_0$. We then let

$$
\begin{aligned}
h^{-k}S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N) := \big\{&a=(a_h)_{h\in(0,h_0)}\colon h_0\in(0,1], \\
&\{h^ka_h\}\text{ is a bounded subset of }S^m(\mathbb{R}^n;\mathbb{R}^N)\big\}.
\end{aligned}
\tag{3.53}
$$

For $k=0$, we denote this space by $S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N)$ simply. The subspace $h^{-k}S^m_{\hbar,\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$ (for $k=0$ denoted $S^m_{\hbar,\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$) of *classical* semiclassical

symbols is defined by the requirement that $h^k a_h \in \mathcal{C}^\infty([0,h_0); S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N))$ (i.e., there exists $b \in \mathcal{C}^\infty([0,h_0); S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N))$ such that $h^k a_h(x,\xi) = b(h,x,\xi)$ for all $h \in (0,h_0)$, $x \in \mathbb{R}^n$, $\xi \in \mathbb{R}^N$). Finally, we write

$$h^\infty S^{-\infty}_\hbar(\mathbb{R}^n;\mathbb{R}^N) := \bigcap_{k,m\in\mathbb{R}} h^{-k} S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N). \tag{3.54}$$

We put a topology on the space of semiclassical symbols defined for $h < h_0$ for fixed $h_0 \in (0,1]$ using the family of seminorms $\sup_{h\in(0,h_0)} \|h^k a_h\|_{m,j}$, $j \in \mathbb{N}_0$, where the S^m-seminorms $\|\cdot\|_{m,j}$ were defined in (3.13).

The full symbols in Example 2.25 are classical semiclassical symbols: they lie in $S^2_{\hbar,\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^n)$. Classicality of a semiclassical symbol means not only that each a_h is a classical symbol but also that the a_h are smooth down to $h = 0$. To relate this to Definition 3.14, note that this smoothness in particular implies the existence of $b_j \in S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$, $j \in \mathbb{N}_0$, with the property $a_h(x,\xi) - \sum_{j=0}^{J-1} h^j b_j(x,\xi) \in h^J S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N)$ for all $J \in \mathbb{N}$.

In order to identify the correct notions of ellipticity and principal symbols, consider the composition of $A = a(h,x)(hD)^\alpha$ and $B = b(h,x)(hD)^\beta$ where $a(h,\cdot), b(h,\cdot)$ are uniformly bounded in $\mathcal{C}^\infty_b(\mathbb{R}^n)$: it is given by

$$A \circ B = ab(hD)^{\alpha+\beta} + a[(hD)^\alpha, b](hD)^\beta. \tag{3.55}$$

Expanding the commutator, one finds that $[(hD)^\alpha, b] = h\sum_{\gamma<\alpha} b_\gamma (hD)^\gamma$, with $b_\gamma(h,\cdot)$ uniformly bounded in $\mathcal{C}^\infty_b(\mathbb{R}^n)$. Therefore, the second term in (3.55) is not only of lower order in the sense of semiclassical differential order (it involves at most $|\alpha| + |\beta| - 1$ many semiclassical derivatives) but also in the sense of decay as $h \searrow 0$ (due to the prefactor h). This suggests that the principal symbol of a semiclassical operator should be an element of the quotient space

$$h^{-k} S^m_\hbar(\mathbb{R}^n;\mathbb{R}^n)/h^{-k+1} S^{m-1}_\hbar(\mathbb{R}^n;\mathbb{R}^n). \tag{3.56}$$

We confirm that this is the appropriate space when we discuss semiclassical pseudodifferential operators in Section 4.9. For special classes of semiclassical operators, one can avoid working with the quotient space (3.56), as the following example shows.

Example 3.20 (Semiclassical Principal Symbol of Certain Differential Operators). Suppose $A_h = \sum_{|\alpha|\le m} a_\alpha(h,x)(hD)^\alpha \in \mathrm{Diff}^m_{\hbar,\mathrm{cl}}(\mathbb{R}^n)$ has the property that $\sum_{|\alpha|=m} a_\alpha(h,x)\xi^\alpha$ is independent of h. Then we can identify its semiclassical principal symbol, i.e., the equivalence class of its full semiclassical symbol $a(h,x,\xi) = \sum_{|\alpha|\le m} a_\alpha(h,x)\xi^\alpha$ in $S^m_\hbar(\mathbb{R}^n;\mathbb{R}^n)/hS^{m-1}_\hbar(\mathbb{R}^n;\mathbb{R}^n)$, with the function

$$\sum_{|\alpha|\le m} a_\alpha(0,x)\xi^\alpha \in S^m(\mathbb{R}^n;\mathbb{R}^n). \tag{3.57}$$

(The point is that $a(h,x,\xi)-\sum_{|\alpha|\leq m} a_\alpha(0,x)\xi^\alpha \in hS^{m-1}_\hbar$.) For example, the semiclassical principal symbol of $h^2(\Delta+V)-z$ in this sense is $|\xi|^2-z$.

Definition 3.21 (Semiclassical Ellipticity). Let $k,m\in\mathbb{R}$. A semiclassical symbol $a\in h^{-k}S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N)$ is *(uniformly) semiclassically elliptic* if there exist a symbol $b\in h^kS^{-m}_\hbar(\mathbb{R}^n;\mathbb{R}^N)$ and a number $h_0>0$ such that $ab-1\in hS^{-1}_\hbar(\mathbb{R}^n;\mathbb{R}^N)$ in $h<h_0$.

The inclusion of a (possibly small) $h_0>0$ is made to account for the fact that one is typically only interested in semiclassical symbols (and operators) in the limit $h\searrow 0$. Thus, one would not care about the failure of ellipticity of $a(h,\cdot)$ for $h=\frac{1}{2}$, say.

Example 3.22 (Ellipticity Versus Semiclassical Ellipticity). The semiclassical symbol $|\xi|^2-z$ from Example 3.20 is elliptic if and only if $z\notin[0,\infty)$. Carefully note that the term z, despite being of lower order in the sense of growth as $|\xi|\to\infty$, does matter here since it is not of lower order (compared to $|\xi|^2$) in the sense of decay as $h\searrow 0$.

The condition in Definition 3.21 is equivalent to a seemingly stronger condition:

Proposition 3.23 (Equivalent Characterization of Semiclassical Ellipticity). *Let $a\in h^{-k}S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N)$. Then the following are equivalent:*

(1) *a is semiclassically elliptic.*
(2) *There exists $h_0>0$ such that a does not vanish for $h<h_0$, and $\frac{1}{a}\in h^kS^{-m}_\hbar(\mathbb{R}^n;\mathbb{R}^N)$.*

Proof. If a is semiclassically elliptic, there exist $b\in h^kS^{-m}_\hbar$ and $r\in hS^{-1}_\hbar$ such that $ab=1-r$ for $h<h_0$. For sufficiently small $h_0>0$, we have $|1-r|\geq\frac{1}{2}$ for $h<h_0$. Therefore, the function $\frac{1}{1-r}$ is well-defined, and indeed $\frac{1}{1-r}=1+r'$ with $r'=\frac{r}{1-r}\in hS^{-1}_\hbar$. Thus, $ab(1+r')=1$ for $h<h_0$; that is, $\frac{1}{a}=b(1+r')\in h^kS^{-m}_\hbar$. The converse is obvious. □

For classical symbols, one can give a more concrete characterization of ellipticity; we only discuss this for semiclassical orders $k=0$ for brevity.

Proposition 3.24 (Semiclassical Ellipticity: Classical Symbols). *Let $m\in\mathbb{R}$ and $a\in S^m_{\hbar,\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$. Then a is semiclassically elliptic if and only if $a|_{h=0}\in S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$ is elliptic and nowhere vanishing.*

Proof. If a is semiclassically elliptic, there exist $b\in S^{-m}_\hbar(\mathbb{R}^n;\mathbb{R}^N)$ and $h_0>0$ with $ab=1$ for $h<h_0$. This implies that $a\neq 0$ for $h<h_0$, and

$$\left|\frac{1}{a}\right|=|b|\leq C\langle\xi\rangle^{-m},\quad h<h_0. \tag{3.58}$$

Therefore, $a|_{h=0} \in S^m_{\mathrm{cl}}$ is elliptic. Conversely, if $a|_{h=0} \in S^m_{\mathrm{cl}}$ is elliptic and nowhere vanishing, set $b := (a|_{h=0})^{-1} \in S^{-m}_{\mathrm{cl}}$. Then

$$a(h,x,\xi)b(x,\xi) = 1 - hr(h,x,\xi), \quad r \in S^0_{\hbar,\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.59}$$

Therefore, there exists $h_0 > 0$ such that $|hr(h,\cdot)| < \frac{1}{2}$ for $h < h_0$. By direct differentiation, one then finds that $\frac{1}{1-hr} = 1 + hr'$ where $r' = \frac{r}{1-hr} \in S^0_{\hbar,\mathrm{cl}}$, and therefore $ab(1+hr') = 1$ for $h < h_0$. □

Finally, we discuss the asymptotic summation of semiclassical symbols.

Proposition 3.25 (Existence and Uniqueness of Asymptotic Sums). *Let $a_j \in h^{-k_j} S^{m_j}_{\hbar}(\mathbb{R}^n;\mathbb{R}^N)$, defined for $h < h_0$, and suppose that $\limsup_{j\to\infty} k_j = -\infty$ and $\limsup_{j\to\infty} m_j = -\infty$. Let $\bar{k}_j := \sup_{j'\geq j} k_{j'}$, $\bar{m}_j := \sup_{j'\geq j} m_{j'}$, and $k = \bar{k}_0$, $m = \bar{m}_0$. Then there exists a symbol $a \in h^{-k} S^m_{\hbar}(\mathbb{R}^n;\mathbb{R}^N)$ such that for all $J \in \mathbb{N}$,*

$$a - \sum_{j=0}^{J-1} a_j \in h^{-\bar{k}_J} S^{\bar{m}_J}_{\hbar}(\mathbb{R}^n;\mathbb{R}^N), \quad h < h_0. \tag{3.60}$$

Moreover, a is unique modulo $h^\infty S^{-\infty}_{\hbar}(\mathbb{R}^n;\mathbb{R}^N)$. We write $a \sim \sum_{j=0}^\infty a_j$.

Proof. The proof is very similar to that of Proposition 3.16. We only prove the existence of an asymptotic sum. Without loss of generality, we may assume that $k_j = k - j$ and $m_j = m - j$. For $\chi \in \mathcal{C}^\infty([0,\infty))$, equal to 1 on $[0,1]$ and equal to 0 on $[2,\infty)$, and a suitable sequence $\epsilon_j > 0$, $\epsilon_j \to 0$, we set

$$a(h,x,\xi) := \sum_{j=0}^\infty \chi\Big(\frac{h\langle\xi\rangle^{-1}}{\epsilon_j}\Big) a_j(h,x,\xi). \tag{3.61}$$

For each $h \in (0,h_0)$, the sum is locally finite on $\mathbb{R}^n_x \times \mathbb{R}^N_\xi$, and thus $a(h,\cdot) \in \mathcal{C}^\infty(\mathbb{R}^n_x \times \mathbb{R}^N_\xi)$. We choose ϵ_j such that

$$\sup_{h\in(0,h_0)} h^{k-j'} \Big\| \chi\Big(\frac{h\langle\xi\rangle^{-1}}{\epsilon_j}\Big) a_j(h,x,\xi) \Big\|_{m-j',j'} \leq 2^{-j}, \quad j > j' \geq 0. \tag{3.62}$$

Then for each $h \in (0,h_0)$, the sum in (3.61) converges in $S^m(\mathbb{R}^n;\mathbb{R}^N)$, and every seminorm of the limit is bounded by an h-independent constant times h^{-k}. □

We leave it to the reader to establish the analogue of Proposition 3.17; see Exercise 3.15.

Notes

Classical symbols were introduced by Kohn–Nirenberg [KN65] in the same paper that introduced pseudo-differential operators in their modern form. (Classical symbols are also sometimes called "one-step polyhomogeneous," but this is not the author's preferred terminology.) Besides $S^m(\mathbb{R}^n;\mathbb{R}^N)$ (or more generally $S^m(U;\mathbb{R}^N)$ where the constants $C_{\alpha\beta}$ in (3.11) are allowed to depend on the compact subset $K \subset U$ in which x is allowed to vary), there exist many other symbol classes. We only mention two examples here. Symbols of type (ρ,δ) , where $m-|\beta|$ in (3.11) is replaced by $m-\rho|\beta|+\delta|\alpha|$, are discussed in Hörmander [Hör71a, Definition 1.1.1]. (Among other uses, they are important for the study of variable order operators and Sobolev spaces; see Vasy [Vas18] for applications in scattering theory.) Symbols that have symbolic behavior also in x arise in the study of problems on asymptotically Euclidean or conic spaces in the context of *scattering* pseudodifferential operators; they were introduced by Parenti [Par72], Shubin [Shu71], and Melrose [Mel94] (see also Exercises 4.14–4.22). Upon quantization—as discussed for the symbols of Definition 3.4 in the next chapter—they give rise to algebras of pseudodifferential operators, each with different properties and applications. More general symbol classes are discussed in Zworski [Zwo12] and Lerner [Ler10].

Exercises

Exercise 3.1 (Principal Symbol via Oscillatory Testing). Show that the principal symbol $\sigma^m(A)$ of $A \in \mathrm{Diff}^m(\mathbb{R}^n)$ captures the "high-frequency behavior" of A in the following sense: for $x_0, \xi_0 \in \mathbb{R}^n$, we have

$$\sigma^m(A)(x_0,\xi_0) = \lim_{\lambda\to\infty} \lambda^{-m}(e^{-i\lambda\xi_0\cdot}Ae^{i\lambda\xi_0\cdot})(x_0), \tag{3.63}$$

where $e^{i\lambda\xi_0\cdot}$ is the function $x \mapsto e^{i\lambda\xi_0\cdot x}$.

Exercise 3.2 (Characterization of Symbol Classes). Let $\mathcal{A} := \{\xi \in \mathbb{R}^N : 1 \le |\xi| \le 4\}$. Let $a \in \mathcal{C}^\infty(\mathbb{R}^N)$ and set

$$a_j \in \mathcal{C}^\infty(\mathcal{A}),\ a_j(\xi) := 2^{-jm}a(2^j\xi), \tag{3.64}$$

for $j \in \mathbb{N}$. Show that $a \in S^m(\mathbb{R}^N)$ if and only if $\{a_j : j \in \mathbb{N}\} \subset \mathcal{C}^\infty(\mathcal{A})$ is bounded. (Here we use the supremum norm of a function and its derivatives as seminorms on $\mathcal{C}^\infty(\mathcal{A})$.)

Exercise 3.3 (Symbols and Classical Symbols).

(1) Let $m \in \mathbb{R}$. Show that $\langle \xi \rangle^m \in S^m_{\mathrm{cl}}(\mathbb{R}^N)$.
(2) Show that $\mathrm{Re}\langle \xi \rangle^i \in S^0(\mathbb{R}^N)$ is *not* a classical symbol. (Here $i = \sqrt{-1}$.)

Exercise 3.4 (Inverses of Elliptic Symbols).

(1) Show that if $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$ satisfies (3.23), and $\chi \in S^0(\mathbb{R}^N)$ vanishes for $|\xi| \leq 2C$, then $\chi/a \in S^{-m}(\mathbb{R}^n; \mathbb{R}^N)$.
(2) If in addition a and χ are classical symbols, show that χ/a is classical as well.

Exercise 3.5 (Smooth Homogeneous Symbols).

(1) Let $\beta \in \mathbb{N}_0^N$. Show that

$$\partial_\xi^\beta \colon S^m_{\mathrm{hom}}(\mathbb{R}^n; \mathbb{R}^N \setminus \{0\}) \to S^{m-|\beta|}_{\mathrm{hom}}(\mathbb{R}^n; \mathbb{R}^N \setminus \{0\}). \tag{3.65}$$

(2) Suppose $a \in S^m_{\mathrm{hom}}(\mathbb{R}^n; \mathbb{R}^N \setminus \{0\})$, $m < 0$. If a extends smoothly to $\mathbb{R}^n \times \mathbb{R}^N$, show that a vanishes identically.
(3) Let $m \in \mathbb{R}$, and suppose that $a \in S^m_{\mathrm{hom}}(\mathbb{R}^n; \mathbb{R}^N \setminus \{0\})$ is not identically 0 and extends smoothly to $\mathbb{R}^n \times \mathbb{R}^N$. Show that $m \in \mathbb{N}_0$, and $a(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x)\xi^\alpha$ for some $a_\alpha \in \mathcal{C}^\infty_b(\mathbb{R}^n)$.

Exercise 3.6 (Classical Symbol Spaces). Let $m, m' \in \mathbb{R}$ with $m - m' \notin \mathbb{Z}$. Show that $S^m_{\mathrm{cl}}(\mathbb{R}^n; \mathbb{R}^N) \cap S^{m'}_{\mathrm{cl}}(\mathbb{R}^n; \mathbb{R}^N) = S^{-\infty}(\mathbb{R}^n; \mathbb{R}^N)$.

Exercise 3.7 (Closure of Symbol Spaces).

(1) Let $a \in S^0(\mathbb{R}^N)$ and suppose that $\lim_{|\xi|\to\infty} |a(\xi)| = 0$. Show that $\lim_{|\xi|\to\infty} \langle \xi \rangle^{|\beta|} |\partial^\beta a(\xi)| = 0$ for all $\beta \in \mathbb{N}_0^N$.
(2) Show that the closure of $S^{-\infty}(\mathbb{R}^N)$ in $S^0(\mathbb{R}^N)$ is equal to the space

$$\Big\{ a \in S^0(\mathbb{R}^N) \colon \lim_{|\xi|\to\infty} |a(\xi)| = 0 \Big\}. \tag{3.66}$$

Exercise 3.8 (Residual Symbols via Pointwise Bounds). Let $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$. Suppose that $|a(x, \xi)| \leq C_N \langle \xi \rangle^{-N}$ for all N. Show that $a \in S^{-\infty}(\mathbb{R}^n; \mathbb{R}^N)$.

Exercise 3.9 (Classical Symbols via Pointwise Remainder Bounds). Let $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$. Show that $a \in S^m_{\mathrm{cl}}(\mathbb{R}^n; \mathbb{R}^N)$ if and only if there exist $a_{m-j} \in S^{m-j}_{\mathrm{hom}}(\mathbb{R}^n; \mathbb{R}^N \setminus \{0\})$, $j \in \mathbb{N}_0$, such that for all $J \in \mathbb{N}$ there exists a constant C_J such that

$$\Big| a(x, \xi) - \sum_{j=0}^{J-1} a_{m-j}(x, \xi) \Big| \leq C_J |\xi|^{m-J} \quad \forall\, x \in \mathbb{R}^n,\ \xi \in \mathbb{R}^N,\ |\xi| \geq 1. \tag{3.67}$$

Exercise 3.10 (Compositions of Functions with Symbols).

(1) Let $a \in S^0(\mathbb{R}^n;\mathbb{R}^N)$, and let $I = [\inf a, \sup a]$. Let $f \in \mathcal{C}^\infty(I)$. Then also $f \circ a \in S^0(\mathbb{R}^n;\mathbb{R}^N)$.
(2) Show that if $a \in S^0(\mathbb{R}^n;\mathbb{R}^N)$ is elliptic and positive for large $\xi \in \mathbb{R}^N$, then there exists $b \in S^0(\mathbb{R}^n;\mathbb{R}^N)$ such that $a - b^2 \in S^{-1}(\mathbb{R}^n;\mathbb{R}^N)$.

Exercise 3.11 (Homogeneous Components of Classical Symbols). Prove Lemma 3.15.

Exercise 3.12 (Classical Symbols via Testing). Let $m \in \mathbb{R}$ and $a \in S^m(\mathbb{R}^n_x;\mathbb{R}^N_\xi)$. Define the vector field $V := \xi \cdot \partial_\xi = \sum_{i=1}^n \xi_i \partial_{\xi_i}$. Show that $a \in S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^N)$ if and only, for all $J \in \mathbb{N}$,

$$(V - (m - J + 1)) \cdots (V - (m-1))(V - m)a \in S^{m-J}(\mathbb{R}^n;\mathbb{R}^N). \tag{3.68}$$

Exercise 3.13 (Nonlinear Character of Asymptotic Summation). Show that there does not exist a map $\sum_{\mathcal{A}} \colon \ell S^m(\mathbb{R}^n;\mathbb{R}^N) \to S^m(\mathbb{R}^n;\mathbb{R}^N)$ with $\sum_{\mathcal{A}}((a_j)_{j\in\mathbb{N}_0}) \sim \sum_{j=0}^\infty a_j$ that is both continuous and linear.

Exercise 3.14 (Semiclassical Symbols). Construct a semiclassical symbol $a \in S^0_\hbar(\mathbb{R}^N)$ such that $\lim_{h \searrow 0} a(h,\xi) = 1$ uniformly on every compact subset of $\mathbb{R}^N_\xi$, but a is not semiclassically elliptic.

Exercise 3.15 (Continuous Asymptotic Summation of Semiclassical Symbols). In the following, all semiclassical symbols are required to be defined for $h \in (0, h_0)$. Equip the space

$$\ell S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N) := \prod_{j=0}^\infty h^j S^{m-j}_\hbar(\mathbb{R}^n;\mathbb{R}^N) \tag{3.69}$$

with the topology generated by the seminorms

$$\|(a_j)\|_J := \max_{0\le k\le J} \sup_{h\in(0,h_0)} h^{-k} \|a_k(h,\cdot)\|_{m-k,J}. \tag{3.70}$$

Prove that there exists a continuous map

$$\sum\nolimits_{\mathcal{A}} \colon \ell S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N) \to S^m_\hbar(\mathbb{R}^n;\mathbb{R}^N) \tag{3.71}$$

with the property that $\sum_{\mathcal{A}}((a_j)_{j\in\mathbb{N}_0}) \sim \sum_{j=0}^\infty a_j$.

Chapter 4
Pseudodifferential Operators

Recall the formula (2.54) for the quantization of a symbol $a = a(x, \xi)$ which is a polynomial in ξ: it defines the differential operator $a(x, D)$. When $a \in S^m(\mathbb{R}^n; \mathbb{R}^n)$ is a general symbol, it defines a *pseudodifferential operator*. It is important for the development of the theory of ps.d.o.s to allow for symbols $a = a(x, y, \xi) \in S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, as these naturally arise when studying adjoints and compositions. It is in fact useful to consider slightly more general symbols still, namely, those in the class

$$\langle x - y \rangle^w S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi) = \{\langle x - y \rangle^w \tilde{a} \colon \tilde{a} \in S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)\}, \tag{4.1}$$

where $w \in \mathbb{R}$. Our goal in Section 4.1 will be to make sense of the following definition.

Definition 4.1 (Quantization). Let $m, w \in \mathbb{R}$ and $a \in \langle x - y \rangle^w S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$. Then we define its *quantization* $\operatorname{Op}(a)$ by

$$(\operatorname{Op}(a)u)(x) := (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(x, y, \xi) u(y) \, \mathrm{d}y \, \mathrm{d}\xi, \quad u \in \mathscr{S}(\mathbb{R}^n). \tag{4.2}$$

Previously, see (2.54), we only considered the special case of the *left quantization* of a *left symbol* $a \in S^m(\mathbb{R}^n_x; \mathbb{R}^n_\xi)$, i.e., a symbol independent of y:

$$(\operatorname{Op}_L(a)u)(x) := (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(x, \xi) u(y) \, \mathrm{d}y \, \mathrm{d}\xi; \tag{4.3}$$

this immediately makes sense as an iterated integral for $u \in \mathscr{S}(\mathbb{R}^n)$ and should be thought of as "differentiate first, then multiply by coefficients." Dually, we can consider the *right quantization* of a *right symbol* $a \in S^m(\mathbb{R}^n_y; \mathbb{R}^n_\xi)$,

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$$(\mathrm{Op}_R(a)u)(x) := (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(y,\xi) u(y)\, \mathrm{d}y\, \mathrm{d}\xi, \tag{4.4}$$

which requires a moment's reflection to make sense of for $u \in \mathscr{S}(\mathbb{R}^n)$; this should be thought of as "multiply by coefficients, then differentiate." Indeed, for $a(z,\xi) = \xi^\alpha a_\alpha(z)$ with $a_\alpha \in \mathcal{C}_b^\infty(\mathbb{R}^n)$, we have

$$(\mathrm{Op}_L(a)u)(x) = a_\alpha(x) D_x^\alpha u(x), \qquad (\mathrm{Op}_R(a)u)(x) = D_x^\alpha (a_\alpha(x) u(x)). \tag{4.5}$$

The quantization map (4.2) should be read as "multiply (y), then differentiate (ξ), then multiply (x)." (Try this with $a(x,y,\xi) = a_1(x)\xi^\alpha a_2(y)$.) We shall see in Section 4.2 that every operator $\mathrm{Op}(a)$ can be written as $\mathrm{Op}(a) = \mathrm{Op}_L(a_L) = \mathrm{Op}_R(a_R)$ for suitable left and right symbols a_L and a_R of the same order as a; this argument utilizes the flexibility afforded by the weight w in Definition 4.1 (see (4.59)).

We will show that the space of m-th order ps.d.o.s is equal to $\Psi^m(\mathbb{R}^n) = \mathrm{Op}_L(S^m(\mathbb{R}^n;\mathbb{R}^n))$. The space of all ps.d.o.s forms a *-algebra under adjoints (Corollary 4.18) and compositions (Section 4.4). As first applications of ps.d.o.s and the associated principal symbol calculus (Section 4.5), we prove elliptic regularity results on spaces of distributions (Section 4.7) and also on L^2-based Sobolev spaces—after proving the boundedness of ps.d.o.s on these (Section 4.8). (The boundedness of ps.d.o.s on weighted Sobolev spaces is the second place where the weight w is important.) Finally, we discuss the analogues of these results in the semiclassical setting (Section 4.9).

4.1 Quantization

An immediate issue with Definition 4.1 is that the integral does not converge absolutely for general w, m. The strategy to make sense of Definition 4.1 is as follows:

(1) We show that absolute convergence holds when m is sufficiently negative (Lemma 4.2).
(2) We demonstrate that $\mathrm{Op}(a)u$ is Schwartz when u is, and that moreover the Schwartz seminorms of $\mathrm{Op}(a)u$, for residual a, can be controlled by those of u and the *S^m-seminorms of a for any fixed $m \in \mathbb{R}$* (Proposition 4.4).
(3) We use a density argument based on Proposition 3.10 to extend Definition 4.1 to general orders $m \in \mathbb{R}$ (see (4.24)).

Lemma 4.2 (Quantization of Symbols of Very Negative Order). *Let $w \in \mathbb{R}$, $m < -n$, and let $a = \langle x-y\rangle^w \tilde{a}$, $\tilde{a} \in S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$. Then the integral (4.2) is absolutely convergent and defines a continuous operator*

$$\mathrm{Op}(a)\colon \mathscr{S}(\mathbb{R}^n) \to \langle x\rangle^w \mathcal{C}_b^0(\mathbb{R}^n) := \{\langle x\rangle^w f\colon f \in \mathcal{C}_b^0(\mathbb{R}^n)\}. \tag{4.6}$$

More precisely, for $M > n + |w|$, there exists a constant $C < \infty$ such that

$$\|\operatorname{Op}(a)u\|_{\langle x\rangle^w \mathcal{C}^0_b(\mathbb{R}^n)} := \|\langle\cdot\rangle^{-w} \operatorname{Op}(a)u\|_{\mathcal{C}^0_b(\mathbb{R}^n)} \le C\|\tilde{a}\|_{m,0}\|u\|_M, \quad u \in \mathscr{S}(\mathbb{R}^n). \tag{4.7}$$

(We use the notation (2.6).)

For the proof, we need a simple lemma:

Lemma 4.3 (Peetre's Inequality). *Let $w \in \mathbb{R}$. Then $\langle x + y\rangle^w \le 2^{|w|/2}\langle x\rangle^w\langle y\rangle^{|w|}$.*

Proof. By the triangle and Cauchy–Schwarz inequalities, we have

$$1 + |x + y|^2 \le 1 + 2|x|^2 + 2|y|^2 \le 2(1 + |x|^2)(1 + |y|^2). \tag{4.8}$$

If $w > 0$, then taking this to the power $w/2$ proves the lemma. For $w = 0$, the lemma is the inequality $1 \le 1$. For $w < 0$, so $-w > 0$, we obtain, analogously to (4.8),

$$\langle x\rangle^{-w} \le 2^{-w/2}\langle x + y\rangle^{-w}\langle y\rangle^{-w}, \tag{4.9}$$

which upon multiplication by $\langle x\rangle^w\langle x + y\rangle^w$ gives the desired result. □

Proof of Lemma 4.2. Since u is Schwartz, we have $|u(y)| \le C\|u\|_M\langle y\rangle^{-M}$. Therefore, the integrand in (4.2) satisfies

$$\begin{aligned} |e^{i(x-y)\cdot\xi}a(x, y, \xi)u(y)| &\le C\langle x - y\rangle^w\|\tilde{a}\|_{m,0}\langle\xi\rangle^m \cdot \|u\|_M\langle y\rangle^{-M} \\ &\le C'\langle x\rangle^w \cdot \langle\xi\rangle^m\langle y\rangle^{|w|-M} \cdot \|\tilde{a}\|_{m,0}\|u\|_M. \end{aligned} \tag{4.10}$$

This is integrable in (y, ξ) provided $m < -n$ and $|w| - M < -n$. □

Proposition 4.4 (Bounds on Quantizations of Residual Symbols). *Let $w \in \mathbb{R}$ and $a = \langle x - y\rangle^w\tilde{a}$, $\tilde{a} \in S^{-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$. Then $\operatorname{Op}(a)$ maps $\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ continuously. In fact, for all $k \in \mathbb{N}_0$, $m \in \mathbb{R}$, there exist $N \in \mathbb{N}$ and a constant C such that*

$$\|\operatorname{Op}(a)u\|_k \le C\|\tilde{a}\|_{m,N}\|u\|_N. \tag{4.11}$$

Lemma 4.5 (Differentiation of Weighted Symbols). *D^α_x and D^α_y are continuous maps $\langle x - y\rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n) \to \langle x - y\rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$. More precisely,*

$$\|\langle x - y\rangle^{-w} D^\alpha_x a\|_{m,k} \le C\|\langle x - y\rangle^{-w} a\|_{m,k+|\alpha|}, \tag{4.12}$$

likewise for $D^\alpha_y a$.

Proof. It suffices to prove the claim for D_{x_1}. For $a(x,y,\xi) = \langle x-y\rangle^w \tilde{a}(x,y,\xi)$, $\tilde{a} \in S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, we have

$$\partial_{x_1} a = \langle x-y\rangle^w (\partial_{x_1}\tilde{a}) + w\langle x-y\rangle^{w-2}(x_1-y_1)\tilde{a}. \tag{4.13}$$

The first summand lies in $\langle x-y\rangle^w S^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$, and the second summand even lies in the smaller space $\langle x-y\rangle^{w-1}S^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$. □

Proof of Proposition 4.4. The key is that for $\xi \neq 0$, the phase $(x-y)\cdot\xi$ has no critical points in y. We exploit this by writing

$$(1-\xi\cdot D_y)e^{i(x-y)\cdot\xi} = \langle\xi\rangle^2 e^{i(x-y)\cdot\xi}. \tag{4.14}$$

For $N \in \mathbb{N}$, we thus have

$$\begin{aligned}
&\operatorname{Op}(a)u(x)\\
&\quad = (2\pi)^{-n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}((1-\xi\cdot D_y)^N e^{i(x-y)\cdot\xi})\langle\xi\rangle^{-2N}a(x,y,\xi)u(y)\,\mathrm{d}y\,\mathrm{d}\xi\\
&\quad = (2\pi)^{-n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi}(1+\xi\cdot D_y)^N\big(\langle\xi\rangle^{-2N}a(x,y,\xi)u(y)\big)\,\mathrm{d}y\,\mathrm{d}\xi.
\end{aligned} \tag{4.15}$$

By the Leibniz rule, we have

$$(1+\xi\cdot D_y)^N\big(\langle\xi\rangle^{-2N}a(x,y,\xi)u(y)\big) = \sum_{|\gamma|\le N} a_\gamma(x,y,\xi)\cdot D_y^\gamma u, \tag{4.16}$$

where

$$a_\gamma(x,y,\xi) = \sum_{|\delta|,|\epsilon|\le N} c_{\gamma\delta\epsilon}\langle\xi\rangle^{-2N}\xi^\delta D_y^\epsilon a(x,y,\xi) \tag{4.17}$$

for some combinatorial constants $c_{\gamma\delta\epsilon}$. By Lemma 4.5, we have $\tilde{a}_\gamma := \langle x-y\rangle^{-w}a_\gamma \in S^{-\infty}(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$, and setting $\tilde{a} := \langle x-y\rangle^{-w}a$, we have, for any $m\in\mathbb{R}$,

$$\|\tilde{a}_\gamma\|_{m-N,0} \le C\|\tilde{a}\|_{m,N}. \tag{4.18}$$

Thus, if $N > m+n$, Lemma 4.2 applies, giving

$$\|\operatorname{Op}(a_\gamma)D^\gamma u\|_{\langle x\rangle^w\mathcal{C}^0(\mathbb{R}^n)} \le C\|\tilde{a}_\gamma\|_{m-N,0}\|D^\gamma u\|_M, \quad M > n+|w|, \tag{4.19}$$

and therefore

$$\|\operatorname{Op}(a)u\|_{\langle x\rangle^w\mathcal{C}^0(\mathbb{R}^n)} \le C\|\tilde{a}\|_{m,N}\|u\|_M, \quad M > n+N+|w|. \tag{4.20}$$

We next prove higher regularity and decay. Let $\alpha, \beta \in \mathbb{N}_0^n$. Then

$$\begin{aligned} &x^\alpha D_x^\beta \operatorname{Op}(a)u(x) \\ &\quad = (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \big((D_\xi + y)^\alpha e^{i(x-y)\cdot\xi}\big)(\xi + D_x)^\beta a(x,y,\xi)u(y)\,\mathrm{d}y\,\mathrm{d}\xi \\ &\quad = (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi}(y - D_\xi)^\alpha (\xi + D_x)^\beta \big(a(x,y,\xi)u(y)\big)\,\mathrm{d}y\,\mathrm{d}\xi. \end{aligned} \tag{4.21}$$

This can be expanded using the Leibniz rule; note that powers of y are acceptable since u is Schwartz. We thus obtain

$$\|x^\alpha D_x^\beta \operatorname{Op}(a)u\|_{\langle x\rangle^w \mathcal{C}^0(\mathbb{R}^n)} \le C\|\tilde{a}\|_{m,N}\|u\|_N \tag{4.22}$$

for N sufficiently large (depending on m, n, α, β). Thus, $\operatorname{Op}(a)u \in \mathscr{S}(\mathbb{R}^n)$. □

This shows that the map

$$\langle x-y\rangle^w S^{-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n) \times \mathscr{S}(\mathbb{R}^n) \ni (a,u) \mapsto \operatorname{Op}(a)u \in \mathscr{S}(\mathbb{R}^n) \tag{4.23}$$

is a continuous bilinear map when putting the topology of $\langle x-y\rangle^w S^{m'}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$ on the first factor (for any $m' \in \mathbb{R}$). By Proposition 3.10, it thus extends uniquely to a continuous bilinear map

$$\langle x-y\rangle^w S^{m}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n) \times \mathscr{S}(\mathbb{R}^n) \ni (a,u) \mapsto \operatorname{Op}(a)u \in \mathscr{S}(\mathbb{R}^n). \tag{4.24}$$

Identifying $\operatorname{Op}(a)$ with its Schwartz kernel, we thus get a continuous map

$$\operatorname{Op}\colon \langle x-y\rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n \times \mathbb{R}^n), \tag{4.25}$$

which is given by

$$\operatorname{Op}(a)(x,y) = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(x,y,\xi)\,\mathrm{d}\xi, \tag{4.26}$$

interpreted as the limit in $\mathscr{S}'(\mathbb{R}^n \times \mathbb{R}^n)$ along a sequence of residual symbols. (This is of course much weaker than (4.24).)

Remark 4.6 (Quantization via an Explicit Limit). Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ be identically 1 near 0. Given $a \in \langle x-y\rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, (the proof of) Proposition 3.10 implies that

$$\operatorname{Op}(a)u(x) = \lim_{j\to\infty} (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} \chi(\xi/j) a(x,y,\xi)u(y)\,\mathrm{d}y\,\mathrm{d}\xi \tag{4.27}$$

for every $u \in \mathscr{S}(\mathbb{R}^n)$, with convergence in $\mathscr{S}(\mathbb{R}^n)$.

Definition 4.7 (Pseudodifferential Operators). Let $m \in \mathbb{R}$. The space of *(uniform) pseudodifferential operators of order* m,

$$\Psi^m(\mathbb{R}^n), \tag{4.28}$$

is the space of all operators of the form $\mathrm{Op}(a)\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ where $a \in \langle x-y\rangle^w S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$ and $w \in \mathbb{R}$. (As we show in the next section, one can take $w = 0$. See Exercise 4.1 for the case of differential operators.) The space of *residual ps.d.o.s* is

$$\Psi^{-\infty}(\mathbb{R}^n) := \bigcap_{m\in\mathbb{R}} \Psi^m(\mathbb{R}^n). \tag{4.29}$$

Note that a priori it is not clear that $\Psi^{-\infty}(\mathbb{R}^n)$ is equal to the space of quantizations of residual symbols (it certainly contains the latter); we show this in Proposition 4.13.

By duality, we can define the action of $A = \mathrm{Op}(a) \in \Psi^m(\mathbb{R}^n)$ on tempered distributions: for $u, v \in \mathscr{S}(\mathbb{R}^n)$ and $a \in \langle x-y\rangle^w S^{-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, we have

$$\begin{aligned}
\langle \mathrm{Op}(a)u, v\rangle &= (2\pi)^{-n} \iiint_{\mathbb{R}^{3n}} e^{i(x-y)\cdot\xi} a(x,y,\xi)u(y)v(x)\,\mathrm{d}y\,\mathrm{d}\xi\,\mathrm{d}x \\
&= (2\pi)^{-n} \iiint_{\mathbb{R}^{3n}} e^{i(x-y)\cdot\xi} a(y,x,-\xi)v(y)u(x)\,\mathrm{d}y\,\mathrm{d}\xi\,\mathrm{d}x \\
&= \langle u, \mathrm{Op}(a^\dagger)v\rangle,
\end{aligned} \tag{4.30}$$

where we put

$$a^\dagger(x,y,\xi) := a(y,x,-\xi). \tag{4.31}$$

Since $a \mapsto a^\dagger$ is an isomorphism on $\langle x-y\rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, the equality

$$\mathrm{Op}(a)^\dagger = \mathrm{Op}(a^\dagger), \quad \text{that is,} \quad \langle \mathrm{Op}(a)u, v\rangle = \langle u, \mathrm{Op}(a^\dagger)v\rangle, \tag{4.32}$$

continues to hold for $a \in \langle x-y\rangle^w S^m(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$. By the density $\mathscr{S}(\mathbb{R}^n) \subset \mathscr{S}'(\mathbb{R}^n)$, we can thus uniquely extend $\mathrm{Op}(a)$ to an operator on $\mathscr{S}'(\mathbb{R}^n)$ via (4.32).

While not needed for our further development of pseudodifferential operators, we close this section by giving the interested reader a feel for Schwartz kernels of ps.d.o.s. We first prove:

Proposition 4.8 (Symbols and Fourier Transforms).

(1) *Let $a \in S^m(\mathbb{R}^n_\xi)$ and set $b := \mathcal{F}^{-1}a \in \mathscr{S}'(\mathbb{R}^n_x)$. Then* $\operatorname{sing\,supp} b \subset \{0\}$. *Moreover, for all $\alpha, \beta \in \mathbb{N}_0^n$, $|\beta| \le |\alpha|$, and $x \neq 0$, we have*

$$|x^\alpha \partial^\beta b(x)| \le C_{\alpha\beta} \cdot \begin{cases} |x|^{-m-n}, & m > -n, \\ 1 + |\log|x||, & m = -n, \\ 1, & m < -n. \end{cases} \tag{4.33}$$

In particular, b decays rapidly with all derivatives as $|x| \to \infty$.

(2) *Let $m < 0$. Suppose that $b \in L^1_{\mathrm{loc}}(\mathbb{R}^n_x)$ is smooth for $x \neq 0$ and satisfies*

$$|x^\alpha \partial^\beta b(x)| \le C_{\alpha\beta} |x|^{-m-n} \quad \forall\, \alpha, \beta \in \mathbb{N}_0^n,\ |\beta| \le |\alpha|, \quad x \neq 0. \tag{4.34}$$

Then $a := \mathcal{F}b \in S^m(\mathbb{R}^n_\xi)$.

For $m \in (-n, 0)$, this fully characterizes symbols in terms of pointwise bounds on their inverse Fourier transforms. For $m \notin (-n, 0)$, such straightforward characterizations are no longer available. For $m \ge 0$, the pointwise bounds (4.33) do not suffice to capture the singular behavior of b at $x = 0$; for example, $a = 1 \in S^0(\mathbb{R}^n)$ has inverse Fourier transform $b = \delta$. For $m < -n$ (resp. $m = -n$) on the other hand, the bounds (4.34) force b to vanish (resp. be bounded) at $x = 0$, which does not hold for inverse Fourier transforms of all $a \in S^m(\mathbb{R}^n)$; a counterexample for $m \le -n$ is provided by $a(\xi) = \langle \xi \rangle^m$.

Proof of Proposition 4.8. For part (1), suppose first that $m < -n$. Using $(\frac{x}{|x|^2} \cdot D_\xi) e^{ix\cdot\xi} = e^{ix\cdot\xi}$ and repeated integrations by parts, we can write

$$b(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix\cdot\xi} \Big(-\frac{x}{|x|^2} \cdot D_\xi \Big)^N a(\xi)\, \mathrm{d}\xi. \tag{4.35}$$

For general m, the integral converges provided that $m - N < -n$, and it defines a continuous function of $x \in \mathbb{R}^n \setminus \{0\}$. Writing $a \in S^m(\mathbb{R}^N)$ as a limit of symbols of order $< -n$ in the topology of $S^{m'}$, with $m' > m$ and $m' - N < -n$, shows that $\mathcal{F}^{-1}a$ is continuous on $\mathbb{R}^n \setminus \{0\}$ and given by the formula (4.35) for $x \neq 0$. Since N can be taken arbitrarily large, we moreover conclude from (4.35) that $|b(x)| \le C_N |x|^{-N}$ for all N and $|x| \ge 1$.

To bound b for $|x| \le 1$, we split the integral into two pieces depending on the relative sizes of $|\xi|$ and $|x|^{-1}$. We first estimate

$$\begin{aligned} \Big| \int_{|\xi| \ge |x|^{-1}} e^{ix\cdot\xi} \Big(-\frac{x}{|x|^2} \cdot D_\xi \Big)^N a(\xi)\, \mathrm{d}\xi \Big| &\le C|x|^{-N} \int_{|\xi| \ge |x|^{-1}} \|a\|_{m,N} \langle \xi \rangle^{m-N}\, \mathrm{d}\xi \\ &\le C'|x|^{-N} \int_{|x|^{-1}}^\infty r^{m-N} r^{n-1}\, \mathrm{d}r \\ &\le C''|x|^{-n-m}, \end{aligned} \tag{4.36}$$

again choosing N such that $m - N < -n$. For the integral over $|\xi| \le |x|^{-1}$, we undo the integration by parts to obtain

$$\begin{aligned}
&\Big|\int_{|\xi|\le|x|^{-1}} e^{ix\cdot\xi}\Big(-\frac{x}{|x|^2}\cdot D_\xi\Big)^N a(\xi)\,\mathrm{d}\xi\Big| \\
&\quad\le \int_{|\xi|\le|x|^{-1}} |a(\xi)|\,\mathrm{d}\xi \\
&\qquad + \sum_{j=0}^{N-1}\Big|\int_{|\xi|=|x|^{-1}} e^{ix\cdot\xi}\Big(-\frac{x}{|x|^2}\cdot D_\xi\Big)^j a(\xi)\Big(\frac{x}{|x|^2}\cdot\frac{\xi}{|\xi|}\Big)\,\mathrm{d}\xi\Big| \\
&\quad\le C\int_0^{|x|^{-1}} \langle r\rangle^m r^{n-1}\,\mathrm{d}r + C\sum_{j=0}^{N-1}|x|^{-j}(|x|^{-1})^{m-j}|x|^{-1}(|x|^{-1})^{n-1}.
\end{aligned} \tag{4.37}$$

The sum evaluates to $CN|x|^{-m-n}$, whereas the integral is bounded by a constant times $|x|^{-m-n}$ when $m+n>0$, a constant times $1+\log|x|^{-1}$ when $m+n=0$, and a constant when $m+n<0$. This proves (4.33) for $\alpha=\beta=0$. The case of general α,β now follows from $x^\alpha D_x^\beta \mathcal{F}^{-1}a = \mathcal{F}^{-1}((-D_\xi)^\alpha \xi^\beta a)$ since $(-D_\xi)^\alpha\xi^\beta a\in S^m(\mathbb{R}^n)$ when $|\beta|\le|\alpha|$.

For part (2), note first that b is integrable, and hence a is bounded. Consider thus $\xi\in\mathbb{R}^n$ with $|\xi|\ge 1$. We use $(-\frac{\xi}{|\xi|^2}\cdot D_x)e^{-ix\cdot\xi} = e^{-ix\cdot\xi}$ to write

$$a(\xi) = \int_{|x|\le|\xi|^{-1}} e^{-ix\cdot\xi}b(x)\,\mathrm{d}x + \int_{|x|\ge|\xi|^{-1}}\Big[\Big(-\frac{\xi}{|\xi|^2}\cdot D_x\Big)^N e^{-ix\cdot\xi}\Big]b(x)\,\mathrm{d}x. \tag{4.38}$$

The first integral is bounded in absolute value by $C\int_0^{|\xi|^{-1}} r^{-m-n}r^{n-1}\,\mathrm{d}r = C'|\xi|^m$. In the second integral, we integrate by parts N times; the boundary terms are bounded in absolute value by constants times

$$|\xi|^{-j}\Bigg(\sup_{\substack{|\gamma|=j\\ |x|=|\xi|^{-1}}}|\partial^\gamma b(x)|\Bigg)|\xi|^{-1}(|\xi|^{-1})^{n-1}\le C|\xi|^m,\quad j=0,\dots,N-1. \tag{4.39}$$

The remaining bulk term can be estimated by

$$\begin{aligned}
&\Big|\int_{|x|\ge|\xi|^{-1}} e^{-ix\cdot\xi}\Big(\frac{\xi}{|\xi|^2}\cdot D_x\Big)^N b(x)\,\mathrm{d}x\Big| \\
&\qquad\le C\int_{|\xi|^{-1}}^\infty |\xi|^{-N}r^{-m-n-N}r^{n-1}\,\mathrm{d}r = C'|\xi|^m,
\end{aligned} \tag{4.40}$$

provided we choose N such that $-m-N<0$. We have now shown that $|a(\xi)|\le C\langle\xi\rangle^m$. Symbol bounds for derivatives of a are a consequence of (2.16). □

Corollary 4.9 (Pointwise Bounds for Schwartz Kernels of Ps.d.o.s).

(1) *Let $A \in \Psi^m(\mathbb{R}^n)$, and denote by $K \in \mathscr{S}'(\mathbb{R}^n \times \mathbb{R}^n)$ its Schwartz kernel. Then* $\operatorname{sing\,supp} K \subset \operatorname{diag}_{\mathbb{R}^n}$. *Moreover, for all $\alpha, \beta, \gamma \in \mathbb{N}_0^n$, $N \in \mathbb{R}$, and $x \neq y$, we have*

$$|(\partial_x + \partial_y)^\gamma \partial_x^\alpha \partial_y^\beta K(x,y)| \leq C_{\gamma\alpha\beta N} \langle x-y \rangle^{-N} \cdot \begin{cases} |x-y|^{-m-|\alpha|-|\beta|-n}, & m+|\alpha|+|\beta| > -n, \\ 1 + |\log|x-y||, & m+|\alpha|+|\beta| = -n, \\ 1, & m+|\alpha|+|\beta| < -n. \end{cases} \tag{4.41}$$

(2) *Let $m < 0$. If $K \in L^1_{\mathrm{loc}}(\mathbb{R}^n_x \times \mathbb{R}^n_y)$ is smooth for $x \neq y$ and satisfies*

$$|(\partial_x+\partial_y)^\gamma \partial_x^\alpha \partial_y^\beta K(x,y)| \leq C_{\gamma\alpha\beta N} \langle x-y \rangle^{-N} |x-y|^{-m-|\alpha|-|\beta|-n} \tag{4.42}$$

for all $\alpha, \beta, \gamma \in \mathbb{N}_0^n$, $N \in \mathbb{R}$, and $x \neq y$, then K is the Schwartz kernel of a ps.d.o. $A \in \Psi^m(\mathbb{R}^n)$.

Proof. For part (1), write $A = \operatorname{Op}(a)$, $a \in \langle x-y \rangle^w S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$. For $x \neq y$ and for all $N \in \mathbb{N}_0$ with $m + |\alpha| + |\beta| - N < -n$, we have

$$\begin{aligned} &(D_x + D_y)^\gamma D_x^\alpha D_y^\beta K(x,y) \\ &\quad = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} \Big(-\frac{x-y}{|x-y|^2} \cdot D_\xi \Big)^N a_{\gamma\alpha\beta}(x,y,\xi)\, \mathrm{d}\xi, \\ &a_{\gamma\alpha\beta}(x,y,\xi) := (D_x + D_y)^\gamma (D_x + \xi)^\alpha (D_y - \xi)^\beta a(x,y,\xi) \\ &\quad \in \langle x-y \rangle^w S^{m+|\alpha|+|\beta|}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n). \end{aligned} \tag{4.43}$$

Repeating the arguments in the proof of Proposition 4.8(1), we obtain (4.41).

For part (2), we define

$$a(x,\xi) := \int_{\mathbb{R}^n} e^{-iz\cdot\xi} K(x, x-z)\, \mathrm{d}z. \tag{4.44}$$

Repeating the arguments in the proof of Proposition 4.8(2), we obtain $a \in S^m(\mathbb{R}^n_x; \mathbb{R}^n_\xi)$. The Schwartz kernel of $A = \operatorname{Op}(a)$ is equal to K by the Fourier inversion formula. □

Example 4.10 (Schwartz Kernels of Ps.d.o.s). We identify operators with their Schwartz kernels as in (4.26).

(1) An illustration of Corollary 4.9 for $n = 3$ and $m = -2$ is given by

$$\mathrm{Op}((|\xi|^2+\mu^2)^{-1})(x,y)=\frac{e^{-\mu|x-y|}}{4\pi|x-y|},\quad \mu>0. \tag{4.45}$$

This is the inverse of $\Delta+\mu^2$ on $\mathscr{S}(\mathbb{R}^3)$.

(2) An example in the case $n=1$ and $m=-1$ is

$$\mathrm{Op}((i\xi+1)^{-1})(x,y)=e^{-(x-y)}H(x-y) \tag{4.46}$$

where H is the Heaviside function. This is the inverse of ∂_x+1 on $\mathscr{S}(\mathbb{R})$.

(3) Toward a concrete example with $m=-n$, $n=2$, with a logarithmic singularity at $x=y$, recall that $u(x):=-\frac{1}{2\pi}\log|x|$, $x\in\mathbb{R}^2$, is a fundamental solution of Δ in $\mathbb{R}^2$, i.e., $\Delta u=\delta$, and therefore $|\xi|^2\hat{u}(\xi)=1$ in $\mathscr{S}'(\mathbb{R}^2)$. If $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^2)$ equals 1 near 0, one can then check that $a:=\mathcal{F}(\chi u)=(2\pi)^2\hat{\chi}*\hat{u}\in S^{-2}(\mathbb{R}^2)$ (and this indeed equals $|\xi|^{-2}+r(\xi)$ for $|\xi|\geq 1$ where $r\in S^{-\infty}(\mathbb{R}^2)$). Therefore, $\mathrm{Op}(a)\in\Psi^{-2}(\mathbb{R}^2)$ has Schwartz kernel

$$-\frac{1}{2\pi}\chi(x-y)\log|x-y|. \tag{4.47}$$

See Exercise 4.6 for a generalization of this construction.

(4) For $n\in\mathbb{N}$, the operator $A_t:=e^{-t\Delta}:=\mathrm{Op}(e^{-t|\xi|^2})$, $t>0$, arising in the solution of the heat equation defines a residual ps.d.o., i.e., an element of $\Psi^{-\infty}(\mathbb{R}^n)$. Its Schwartz kernel is $(4\pi t)^{-\frac{n}{2}}\exp(-\frac{|x-y|^2}{4t})$.

4.2 Left/Right Reduction, Adjoints

In this section, we shall prove:

Theorem 4.11 (Left/Right Reduction). *Let $a\in\langle x-y\rangle^w S^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$. Then there exists a unique left symbol $a_L\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ such that*

$$\mathrm{Op}(a)=\mathrm{Op}_L(a_L), \tag{4.48}$$

and a unique right symbol $a_R\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ such that

$$\mathrm{Op}(a)=\mathrm{Op}_R(a_R). \tag{4.49}$$

The symbols a_L,a_R depend continuously on a. Modulo residual symbols, they are given by asymptotic sums

$$a_L(x,\xi)\sim\sum_{\alpha\in\mathbb{N}_0^n}\frac{1}{\alpha!}\big(\partial_\xi^\alpha D_y^\alpha a(x,y,\xi)\big)\big|_{y=x}, \tag{4.50}$$

$$a_R(y,\xi) \sim \sum_{\alpha\in\mathbb{N}_0^n} \frac{(-1)^{|\alpha|}}{\alpha!}\big(\partial_\xi^\alpha D_x^\alpha a(x,y,\xi)\big)\big|_{x=y}. \tag{4.51}$$

(The summands are ordered by increasing $|\alpha|$.)

Definition 4.12 (Left/Right Reduction). In the notation of Theorem 4.11, we call a_L, resp. a_R the *left*, resp. *right reduction* of the full symbol a. Writing $A = \mathrm{Op}(a)$, we write

$$\sigma_L(A) := a_L, \quad \sigma_R(A) := a_R. \tag{4.52}$$

We begin with a description of residual operators, i.e., elements of $\Psi^{-\infty}(\mathbb{R}^n)$:

Proposition 4.13 (Characterization of Residual Operators). *An operator $A\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n)$ is a residual operator if and only if its Schwartz kernel $K(x,y)$ is smooth and satisfies*

$$|\partial_x^\alpha \partial_y^\beta K(x,y)| \le C_{\alpha\beta N}\langle x-y\rangle^{-N} \quad \forall\, \alpha,\beta,N. \tag{4.53}$$

Moreover, any such A can be written as $A = \mathrm{Op}_L(a_L) = \mathrm{Op}_R(a_R)$ for unique symbols $a_L, a_R \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$.

Proof. If $A \in \Psi^{-N}(\mathbb{R}^n)$ for all $N \in \mathbb{R}$, we can write $A = \mathrm{Op}(a_N)$ with $a_N = \langle x-y\rangle^{w_N}\tilde{a}_N$, $\tilde{a}_N \in S^{-N}(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$, for some $w_N \in \mathbb{R}$. Taking $N > n$, the Schwartz kernel K of A is then given by the absolutely convergent integral

$$K(x,y) = (2\pi)^{-n}\int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a_N(x,y,\xi)\,\mathrm{d}\xi. \tag{4.54}$$

Let $M \in \mathbb{N}_0$. For $|x-y| < 1$ and $\alpha,\beta \in \mathbb{N}_0^n$ with $|\alpha|+|\beta| \le M$, and taking $N > n+M$, we can thus bound

$$|\partial_x^\alpha\partial_y^\beta K(x,y)| \le C_{\alpha\beta}\|\tilde{a}_N\|_{-N,M} \tag{4.55}$$

using the triangle inequality. This gives (4.53) in this region.

To obtain the bounds (4.53) in the region $|x-y| > 1$, we use a simple version of the arguments used in the proof of Proposition 4.8. Fix $N = n+1$. We use $(\frac{x-y}{|x-y|^2}\cdot D_\xi)e^{i(x-y)\cdot\xi} = e^{i(x-y)\cdot\xi}$ and repeated integrations by parts to deduce that

$$\begin{aligned}
|K(x,y)| &= (2\pi)^{-n}\left|\int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi}\Big(-\frac{x-y}{|x-y|^2}\cdot D_\xi\Big)^M a_{n+1}(x,y,\xi)\,\mathrm{d}\xi\right| \\
&\le C_M|x-y|^{-M}\langle x-y\rangle^{w_{n+1}}\int_{\mathbb{R}^n}\langle\xi\rangle^{-n-1-M}\,\mathrm{d}\xi \\
&\le C'_M\langle x-y\rangle^{-M+w_{n+1}}.
\end{aligned} \tag{4.56}$$

Since M is arbitrary, this proves (4.53) for $\alpha = \beta = 0$. Up to k-fold derivatives in x, y are estimated in the same way, except one now works with a_{n+1+k} instead of a_{n+1}.

For the converse, note that if K satisfies (4.53), we can define

$$a_L(x,\xi) = \int_{\mathbb{R}^n} e^{-iz\cdot\xi} K(x, x-z)\,\mathrm{d}z. \tag{4.57}$$

Then $\mathrm{Op}_L(a_L)$ has Schwartz kernel K by the Fourier inversion formula, and the estimates (4.53) imply $a_L \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$. Similarly, the operator $\mathrm{Op}_R(a_R)$ has Schwartz kernel K for

$$a_R(y,\xi) = \int_{\mathbb{R}^n} e^{-iz\cdot\xi} K(y+z, y)\,\mathrm{d}z. \tag{4.58}$$

The proof is complete. □

Remark 4.14 (Continuity of Left/Right Reduction for Residual Operators). Define seminorms on the space of all $K \in \mathcal{C}^\infty(\mathbb{R}^n_x \times \mathbb{R}^n_y)$ satisfying the estimates (4.53) to be the optimal constants: $\|K\|_{\alpha\beta N} := \sup_{x,y\in\mathbb{R}^n} \langle x-y\rangle^N |\partial_x^\alpha \partial_y^\beta K(x,y)|$. Then the proof of Proposition 4.13 shows that the maps $K \mapsto a_{L/R} \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$ and $S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n) \ni a \mapsto K = \mathrm{Op}_{L/R}(a)$ are continuous.

To handle the case of general orders $m \in \mathbb{R}$, we first note that integration by parts in ξ implies the equality of Schwartz kernels

$$\begin{aligned}
\mathrm{Op}((y-x)^\alpha a)(x,y) &= (2\pi)^{-n} \int ((-D_\xi)^\alpha e^{i(x-y)\cdot\xi}) a(x,y,\xi)\,\mathrm{d}\xi \\
&= (2\pi)^{-n} \int e^{i(x-y)\cdot\xi} D_\xi^\alpha a(x,y,\xi)\,\mathrm{d}\xi \\
&= \mathrm{Op}(D_\xi^\alpha a)(x,y),
\end{aligned} \tag{4.59}$$

first for $a \in \langle x-y\rangle^w S^{-\infty}(\mathbb{R}^n \times \mathbb{R}^n;\mathbb{R}^n)$ and then for symbols of order m by density and continuity. The additional off-diagonal growth of $(y-x)^\alpha a$ compared to that of a is an important reason for working with the symbol class (4.1).

Proof of Theorem 4.11. Let $N \in \mathbb{N}$. We expand $a(x,y,\xi)$ in Taylor series around $y = x$:

$$\begin{aligned}
a(x,y,\xi) &= \sum_{|\alpha|<N} \frac{1}{\alpha!}(y-x)^\alpha \big(\partial_y^\alpha a(x,y,\xi)\big)|_{y=x} + r_N(x,y,\xi), \\
r_N(x,y,\xi) &= \sum_{|\alpha|=N} \frac{N}{\alpha!}(y-x)^\alpha \int_0^1 (1-t)^{N-1} (\partial_y^\alpha a)(x, x+t(y-x),\xi)\,\mathrm{d}t.
\end{aligned} \tag{4.60}$$

Using the identity (4.59), we have

$$\mathrm{Op}\left(a - \sum_{|\alpha|<N} \frac{1}{\alpha!}(D_\xi^\alpha \partial_y^\alpha a)|_{y=x}\right) = \mathrm{Op}(\tilde{r}_N) \in \Psi^{m-N}(\mathbb{R}^n), \tag{4.61}$$

where

$$\tilde{r}_N(x,y,\xi) = \sum_{|\alpha|=N} \frac{N}{\alpha!} \int_0^1 (1-t)^{N-1} (D_\xi^\alpha \partial_y^\alpha a)(x, x+t(y-x), \xi)\, \mathrm{d}t. \tag{4.62}$$

We check that $\tilde{r}_N$ is a symbol: in view of the symbolic estimates for a, the function $\tilde{r}_N$ satisfies the estimate

$$|\partial_x^\beta \partial_y^\gamma \partial_\xi^\delta \tilde{r}_N(x,y,\xi)| \leq C_{\beta\gamma\delta N} \langle x-y\rangle^{w_+} \langle \xi\rangle^{m-N-|\delta|} \tag{4.63}$$

where we set $w_+ = \max(w, 0)$. Hence,

$$\tilde{r}_N \in \langle x-y\rangle^{w_+} S^{m-N}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n). \tag{4.64}$$

Note that for $|\alpha| = k$, we have $D_\xi^\alpha \partial_y^\alpha a|_{y=x} \in S^{m-k}(\mathbb{R}^n_x; \mathbb{R}^n_\xi)$. Thus, we can let $b \in S^m(\mathbb{R}^n; \mathbb{R}^n)$ be an asymptotic sum

$$b \sim \sum_\alpha \frac{1}{\alpha!}(D_\xi^\alpha \partial_y^\alpha a)|_{y=x} \tag{4.65}$$

(or equivalently $b \sim \sum_\alpha \frac{1}{\alpha!}(\partial_\xi^\alpha D_y^\alpha a)|_{y=x}$), and then

$$\begin{aligned} R := \mathrm{Op}(a-b) &= \mathrm{Op}(\tilde{r}_N) - \mathrm{Op}\left(b - \sum_{|\alpha|<N} \frac{1}{\alpha!}(D_\xi^\alpha \partial_y^\alpha a)|_{y=x}\right) \\ &\in \bigcap_{N\in\mathbb{N}} \Psi^{m-N}(\mathbb{R}^n) = \Psi^{-\infty}(\mathbb{R}^n). \end{aligned} \tag{4.66}$$

By Proposition 4.13, we then have $R = \mathrm{Op}_L(r)$ for some $r \in S^{-\infty}(\mathbb{R}^n; \mathbb{R}^n)$. Therefore,

$$A = \mathrm{Op}_L(a_L), \quad a_L := b + r. \tag{4.67}$$

The continuous dependence of a_L on a follows by using the explicit asymptotic summation procedure of Proposition 3.17 to define b (which thus depends continuously on a) and then noting that the optimal constants for the Schwartz kernel K of R in (4.53), and thus the $S^{-\infty}(\mathbb{R}^n; \mathbb{R}^N)$-seminorms of r (see Remark 4.14), depend continuously on a, b.

Reduction to a right symbol is proved analogously. Instead of going through the argument, one can instead use duality as in (4.30), the idea being that the real adjoint of a left quantization is a right quantization (and vice versa). Namely, using (4.31), we write the real adjoint of $\mathrm{Op}(a)$ as $\mathrm{Op}(a)^\dagger = \mathrm{Op}(a^\dagger) = \mathrm{Op}_L(a'_L)$ for $a'_L \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, and then

$$\mathrm{Op}(a) = \mathrm{Op}(a^\dagger)^\dagger = (\mathrm{Op}_L(a'_L))^\dagger = \mathrm{Op}_R((a'_L)^\dagger) = \mathrm{Op}_R(a_R), \tag{4.68}$$

where $a_R(y,\xi) = a'_L(y,-\xi)$. The left reduction formula applied to $a^\dagger$ gives

$$a'_L(x,\xi) \sim \sum_\alpha \frac{1}{\alpha!}((-\partial_\xi)^\alpha D_x^\alpha a)(y,x,-\xi)|_{y=x}, \tag{4.69}$$

yielding the asymptotic description (4.51) of a_R.

It remains to prove the uniqueness of a_L, a_R. Suppose that $\mathrm{Op}_L(a_L) = 0$. Then for all $\chi \in \mathscr{S}(\mathbb{R}^n)$, $\epsilon > 0$, and $\xi_0 \in \mathbb{R}^n$, we have

$$0 = \mathrm{Op}_L(a_L)(\chi(\epsilon\cdot)e^{i\xi_0\cdot}). \tag{4.70}$$

Since the Fourier transform of $x \mapsto \chi(\epsilon x)e^{i\xi_0\cdot x}$ is given by $\xi \mapsto \epsilon^{-n}\hat\chi(\epsilon^{-1}(\xi - \xi_0))$, this means

$$(2\pi)^{-n}\int_{\mathbb{R}^n} e^{ix\cdot\xi}a_L(x,\xi)\epsilon^{-n}\hat\chi\Big(\frac{\xi-\xi_0}{\epsilon}\Big)\,\mathrm{d}\xi = 0 \tag{4.71}$$

for all $x \in \mathbb{R}^n$. If we require $\chi(0) = 1$ and thus $\int_{\mathbb{R}^n}\hat\chi(\xi)\,\mathrm{d}\xi = (2\pi)^n$, then upon taking $\epsilon \searrow 0$ we obtain $a_L(x,\xi_0) = 0$. Since $x, \xi_0 \in \mathbb{R}^n$ are arbitrary, this proves $a_L = 0$. One may alternatively argue as follows: a left symbol a_L can be viewed as an element $a_L \in \mathcal{C}^\infty(\mathbb{R}^n_x;\mathscr{S}'(\mathbb{R}^n_\xi))$, and the Schwartz kernel of $\mathrm{Op}(a_L)$ is

$$\mathrm{Op}(a_L)(x,x-z) = (\mathcal{F}_2^{-1}a_L)(x,z), \tag{4.72}$$

with $\mathcal{F}_2^{-1}$ denoting the inverse Fourier transform in the second argument. Since $\mathcal{F}_2^{-1}$ is an isomorphism of $\mathcal{C}^\infty(\mathbb{R}^n;\mathscr{S}'(\mathbb{R}^n))$, $\mathrm{Op}(a_L) = 0$ implies $a_L = 0$. The proof for a_R is similar. □

Corollary 4.15 (Ps.d.o.s as Left/Right Quantizations). *Let $m \in \mathbb{R}$ or $m = -\infty$. Then $\Psi^m(\mathbb{R}^n) = \mathrm{Op}_{L/R}(S^m(\mathbb{R}^n;\mathbb{R}^n))$.*

Another application of the left reduction formula is the following:

Proposition 4.16 (Pseudolocality of Ps.d.o.s). *Let $A \in \Psi^m(\mathbb{R}^n)$. Then*

$$\operatorname{sing\,supp} Au \subset \operatorname{sing\,supp} u, \quad u \in \mathscr{S}'(\mathbb{R}^n). \tag{4.73}$$

We first prove a quantitative version of the statement that residual operators A are smoothing, i.e., $\operatorname{sing\,supp} Au = \emptyset$.

Lemma 4.17 (Residual Operators Acting on Tempered Distributions). *A residual operator $A \in \Psi^{-\infty}(\mathbb{R}^n)$ is continuous as a map*

$$A\colon \mathscr{S}'(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n) \cap \mathcal{C}^\infty(\mathbb{R}^n). \tag{4.74}$$

More precisely, for any $u \in \mathscr{S}'(\mathbb{R}^n)$ we have $Au \in \langle x\rangle^N \mathcal{C}_b^\infty(\mathbb{R}^n)$ for some N (depending on u).

Proof. Let K denote the Schwartz kernel of A; recall that it satisfies the estimates (4.53). In particular, $K(x,\cdot) \in \mathscr{S}(\mathbb{R}^n)$ for all $x \in \mathbb{R}^n$. For $u \in \mathscr{S}'(\mathbb{R}^n)$, we then have, for some $N \in \mathbb{N}$ and $C > 0$,

$$\begin{aligned} |(Au)(x)| = |\langle u, K(x,\cdot)\rangle| &\le C\|K(x,\cdot)\|_N = C \sup_{\substack{y\in\mathbb{R}^n \\ |\alpha|+|\beta|\le N}} |y^\alpha D_y^\beta K(x,y)| \\ &\le C \sup_{\substack{y\in\mathbb{R}^n \\ |\beta|\le N}} |\langle y\rangle^N D_y^\beta K(x,y)| \\ &= C \sup_{\substack{y\in\mathbb{R}^n \\ |\beta|\le N}} \Big(\langle y\rangle^N \langle x-y\rangle^{-N} |\langle x-y\rangle^N D_y^\beta K(x,y)|\Big). \end{aligned} \tag{4.75}$$

Lemma 4.3 gives $\langle y\rangle^N \langle x-y\rangle^{-N} \le C_N \langle x\rangle^N$, and hence

$$|(Au)(x)| \le C\langle x\rangle^N. \tag{4.76}$$

Derivatives in x are estimated analogously, so $Au \in \mathcal{C}^\infty(\mathbb{R}^n)$ and

$$|\partial_x^\alpha (Au)(x)| \le C_\alpha \langle x\rangle^N. \tag{4.77}$$

This completes the proof. □

Proof of Proposition 4.16. Suppose $x \notin \operatorname{sing\,supp} u$. There exist cutoffs $\chi, \widetilde{\chi} \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ such that

$$\chi(x) \neq 0, \quad \widetilde{\chi} \equiv 1 \text{ on } \operatorname{supp}\chi, \quad \widetilde{\chi} u \in \mathcal{C}_c^\infty(\mathbb{R}^n). \tag{4.78}$$

Then

$$\chi A u = \chi A(\widetilde{\chi} u) + \chi A(1-\widetilde{\chi})u. \tag{4.79}$$

Since A acts on $\mathscr{S}(\mathbb{R}^n)$, we have $\chi A(\widetilde{\chi}u) \in \mathscr{S}(\mathbb{R}^n)$. For the second term, note that $\chi A(1-\widetilde{\chi}) = \operatorname{Op}(\chi(x)\sigma_L(A)(x,\xi)(1-\widetilde{\chi}(y)))$. Since χ and $1-\widetilde{\chi}$ have disjoint supports, we have

$$\sigma_L(\chi A(1-\widetilde{\chi}))(x,\xi) \sim \sum_\alpha \frac{1}{\alpha!} \chi(x) \partial_\xi^\alpha \sigma_L(A)(x,\xi) D_y^\alpha (1-\widetilde{\chi}(y))|_{y=x} = 0, \tag{4.80}$$

which implies

$$\chi A(1-\widetilde{\chi}) \in \Psi^{-\infty}(\mathbb{R}^n). \tag{4.81}$$

By Lemma 4.17, we conclude that $\chi A(1-\widetilde{\chi})u \in \mathcal{C}^\infty(\mathbb{R}^n)$, finishing the proof. □

Returning to the observation (4.81), note that if $A = \mathrm{Op}(a)$ has Schwartz kernel $K \in \mathscr{S}'(\mathbb{R}^n \times \mathbb{R}^n)$, then the Schwartz kernel of $\chi A(1-\widetilde{\chi})$ is $\chi(x)(1-\widetilde{\chi}(y))K(x,y)$. Thus, (4.81) constitutes another proof of the first statement in Corollary 4.9(1).

We record two other technical consequences of Corollary 4.15.

Corollary 4.18 (Adjoints of Ps.d.o.s). *Let $A \in \Psi^m(\mathbb{R}^n)$, then*

$$\int_{\mathbb{R}^n} (A^*u)(x)\overline{v(x)}\,\mathrm{d}x = \int_{\mathbb{R}^n} u(x)\overline{(Av)(x)}\,\mathrm{d}x, \quad u, v \in \mathscr{S}(\mathbb{R}^n), \tag{4.82}$$

defines an operator $A^* \in \Psi^m(\mathbb{R}^n)$. If $A = \mathrm{Op}(a)$, then $A^* = \mathrm{Op}(a^*)$, $a^*(x,y,\xi) = \bar{a}(y,x,\xi)$. If $A = \mathrm{Op}_L(a_L)$, then $A^* = \mathrm{Op}_L(a_L^*)$ with

$$a_L^*(x,\xi) \sim \sum_{\alpha \in \mathbb{N}_0^n} \frac{1}{\alpha!} \partial_\xi^\alpha D_x^\alpha \overline{a_L}(x,\xi). \tag{4.83}$$

Proof. A variant of (4.30) gives the first part; the second part is an immediate application of Theorem 4.11. □

Corollary 4.19 (Asymptotic Summation of Ps.d.o.s). *Let $A_j \in \Psi^{m_j}(\mathbb{R}^n)$, $j \geq 0$, and suppose $\limsup_{j\to\infty} m_j = -\infty$. Let $\bar{m}_j := \sup_{j' \geq j} m_{j'}$ and $m = \bar{m}_0$. Then there exists an operator $A \in \Psi^m(\mathbb{R}^n)$ such that for all $J \in \mathbb{N}$,*

$$A - \sum_{j=0}^{J-1} A_j \in \Psi^{\bar{m}_j}(\mathbb{R}^n). \tag{4.84}$$

Moreover, A is unique modulo $\Psi^{-\infty}(\mathbb{R}^n)$. One writes $A \sim \sum_{j=0}^\infty A_j$.

Proof. This follows from Corollary 4.15 and Proposition 3.16. □

4.3 Topology on Spaces of Pseudodifferential Operators

Let $m \in \mathbb{R}$ or $m = -\infty$. Since $\mathrm{Op}_L \colon S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ is an isomorphism of vector spaces, we simply transport the Fréchet space structure of $S^m(\mathbb{R}^n;\mathbb{R}^n)$ to $\Psi^m(\mathbb{R}^n)$ via Op_L. For instance:

Lemma 4.20 (Mollifiers). *Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n_\xi)$ be identically 1 near 0, and put $J_\epsilon = \mathrm{Op}(\chi(\epsilon\cdot))$, $\epsilon > 0$. Then $J_\epsilon \in \Psi^{-\infty}(\mathbb{R}^n)$ is uniformly bounded in $\Psi^0(\mathbb{R}^n)$ and converges to the identity operator $I = \mathrm{Op}(1)$ in the topology of $\Psi^\eta(\mathbb{R}^n)$ for any $\eta > 0$.*

Proof. This is equivalent to the main part of (the proof of) Proposition 3.10. □

It is reassuring to note that one can equally well define the topology on $\Psi^m(\mathbb{R}^n)$ using the right quantization:

Proposition 4.21 (Topology on $\Psi^m(\mathbb{R}^n)$). *Let $m \in \mathbb{R}$ or $m = -\infty$. Then the isomorphism of vector spaces $\mathrm{Op}_R \colon S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ is an isomorphism of Fréchet spaces.*

Proof. Taking the right symbol σ_R is the inverse of Op_R. By definition of the Fréchet space structure of $\Psi^m(\mathbb{R}^n)$, the claim is thus equivalent to the continuity of $\sigma_R \circ \mathrm{Op}_L$—which is part of Theorem 4.11. □

4.4 Composition

Proving that composition of ps.d.o.s produces another ps.d.o. is now straightforward:

Theorem 4.22 (Composition of Ps.d.o.s). *Let $A \in \Psi^m(\mathbb{R}^n)$, $B \in \Psi^{m'}(\mathbb{R}^n)$. Then $A \circ B \colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$ is a pseudodifferential operator,*

$$A \circ B \in \Psi^{m+m'}(\mathbb{R}^n), \tag{4.85}$$

and its left symbol is given as an asymptotic sum

$$\sigma_L(A \circ B) \sim \sum_{\alpha \in \mathbb{N}_0^n} \frac{1}{\alpha!} \partial_\xi^\alpha \sigma_L(A) \cdot D_x^\alpha \sigma_L(B). \tag{4.86}$$

The bilinear map $(A, B) \mapsto A \circ B$ is continuous.

Proof. Write $A = \mathrm{Op}_L(a)$ and $B = \mathrm{Op}_R(b_R)$. Assume first that $A, B \in \Psi^{-\infty}(\mathbb{R}^n)$, then for $u, v \in \mathscr{S}(\mathbb{R}^n)$, we have

$$\begin{aligned} Av(x) &= (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix\cdot\xi} a(x,\xi) \hat{v}(\xi)\, \mathrm{d}\xi, \\ \widehat{Bu}(\xi) &= \int_{\mathbb{R}^n} e^{-iy\cdot\xi} b_R(y,\xi) u(y)\, \mathrm{d}y. \end{aligned} \tag{4.87}$$

Thus,

$$ABu(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(x,\xi) b_R(y,\xi) u(y)\, \mathrm{d}y\, \mathrm{d}\xi, \tag{4.88}$$

giving $A \circ B = \mathrm{Op}(c)$, $c(x,y,\xi) = a(x,\xi)b_R(y,\xi)$. (This is another reason for considering such general symbols.) By density and continuity, this continues to hold for A, B as in the statement of the theorem.

To get the asymptotic expansion (4.86), let us write $a = \sigma_L(A)$, $b = \sigma_L(B)$. Since for asymptotic sums it suffices to consider only those terms that have symbolic order bigger than some fixed but arbitrary number, there are no convergence or rearrangement issues. We thus have

$$\begin{aligned}
\sigma_L(A \circ B)(x,\xi) &\sim \sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha \big(a(x,\xi) D_y^\alpha b_R(y,\xi)|_{y=x} \big) \\
&\sim \sum_{\beta,\gamma} \frac{1}{\beta!\gamma!} \partial_\xi^\beta a(x,\xi) \cdot \partial_\xi^\gamma D_x^{\beta+\gamma} \left(\sum_\delta \frac{(-1)^{|\delta|}}{\delta!} \partial_\xi^\delta D_x^\delta b(x,\xi) \right) \\
&\sim \sum_\beta \frac{1}{\beta!} \partial_\xi^\beta a(x,\xi) \cdot D_x^\beta \left(\sum_\epsilon \frac{1}{\epsilon!} \partial_\xi^\epsilon D_x^\epsilon b(x,\xi) \sum_{\gamma+\delta=\epsilon} \frac{\epsilon!}{\gamma!\delta!} (-1)^{|\delta|} \right).
\end{aligned} \tag{4.89}$$

For $\epsilon = 0$, the final sum evaluates to 1, while for $\mathbb{N}_0^n \ni \epsilon \neq 0$,

$$\sum_{\gamma+\delta=\epsilon} \frac{\epsilon!}{\gamma!\delta!} (-1)^{|\delta|} = \prod_{\epsilon_j \neq 0} (1-1)^{\epsilon_j} = 0. \tag{4.90}$$

The proof is complete. □

4.5 Principal Symbols

As in the case of differential operators (Proposition 3.3), the "leading order part" of the left or right symbol of a pseudodifferential operator $A \in \Psi^m(\mathbb{R}^n)$ has particularly simple properties.

Definition 4.23 (Principal Symbol of Ps.d.o.s). Let $m \in \mathbb{R}$. The *principal symbol* $\sigma^m(A)$ of a ps.d.o. $A \in \Psi^m(\mathbb{R}^n)$ is the equivalence class

$$\sigma^m(A) := [\sigma_L(A)] \in S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n). \tag{4.91}$$

We shall often omit passage to the equivalence class from the notation.

Directly from the definition, this gives a short exact sequence for every $m \in \mathbb{R}$:

$$0 \to \Psi^{m-1}(\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n) \xrightarrow{\sigma^m} S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n) \to 0. \tag{4.92}$$

The surjectivity of σ^m is clear: given a representative $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ of an equivalence class of symbols, we have $\sigma^m(\mathrm{Op}_L(a)) = [a]$. The following result is an immediate consequence of Theorem 4.11, Corollary 4.18, and Theorem 4.22:

Proposition 4.24 (Properties of the Principal Symbol).

(1) *For $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, we have $\sigma^m(\mathrm{Op}_R(a)) = [a]$, i.e., using the right symbol in (4.91) gives the same principal symbol map.*
(2) *For $A \in \Psi^m(\mathbb{R}^n)$, we have $\sigma^m(A^*) = \overline{\sigma^m(A)}$.*
(3) *For $A \in \Psi^m(\mathbb{R}^n)$, $B \in \Psi^{m'}(\mathbb{R}^n)$, we have $\sigma^{m+m'}(A \circ B) = \sigma^m(A)\sigma^{m'}(B)$.*

(The behavior under changes of variables will be discussed in Chapter 5, Section 5.1.) Notice that the principal symbol map translates operator composition (a non-commutative operation) to the multiplication of (equivalence classes of) functions (a commutative operation), though of course at what seems to be an enormous loss of information compared to the full expansion (4.86) (which itself gives up all information on the "residual part" of $A \circ B$). However, in most situations, the principal symbol and sometimes a "subprincipal" part of the full symbol dominate the behavior of the operator, while lower-order parts are irrelevant. A first example (elliptic operators) is discussed in Section 4.7.

One crucial calculation is the following: for $A \in \Psi^m(\mathbb{R}^n)$, $B \in \Psi^{m'}(\mathbb{R}^n)$, note that $\sigma^{m+m'}(A \circ B) = \sigma^m(A)\sigma^{m'}(B) = \sigma^{m+m'}(B \circ A)$, so

$$\sigma^{m+m'}([A,B]) = 0, \quad [A,B] = A \circ B - B \circ A. \tag{4.93}$$

In view of (4.92), we thus have $[A,B] \in \Psi^{m+m'-1}(\mathbb{R}^n)$, and it is natural to inquire about its principal symbol as an operator of order $m + m' - 1$. It turns out that it can be computed solely in terms of the principal symbols of A and B:

Proposition 4.25 (Principal Symbols of Commutators). *For $A \in \Psi^m(\mathbb{R}^n)$, $B \in \Psi^{m'}(\mathbb{R}^n)$, we have*

$$\sigma^{m+m'-1}(i[A,B]) = \{\sigma^m(A), \sigma^{m'}(B)\}, \tag{4.94}$$

where the Poisson bracket *of $a, b \in \mathcal{C}^\infty(\mathbb{R}^n_x \times \mathbb{R}^n_\xi)$ is defined as*

$$\{a,b\} := \sum_{j=1}^{n} (\partial_{\xi_j} a)(\partial_{x_j} b) - (\partial_{x_j} a)(\partial_{\xi_j} b). \tag{4.95}$$

This will be the key connection between "quantum mechanics" (quantizations of symbols, noncommutative algebra of operators, commutators) and "classical mechanics" (symbols themselves, commutative algebra of functions, Poisson bracket), which will play a central role from Chapter 8 onward.

Proof of Proposition 4.25. We leave it to the reader to verify that (4.94) is well-defined, i.e., that the image of the right-hand side in the quotient space $S^{m+m'-1}/S^{m+m'-2}$ does not depend on the choice of representatives of the principal symbols of A and B.

The proof is an immediate application of (4.86). Let $a = \sigma_L(A)$, $b = \sigma_L(B)$. Working modulo $S^{m+m'-2}(\mathbb{R}^n;\mathbb{R}^n)$, we have

$$\sigma_L(A\circ B)\equiv ab+\frac{1}{i}\sum_{j=1}^{n}(\partial_{\xi_j}a)(\partial_{x_j}b),\quad \sigma_L(B\circ A)\equiv ab+\frac{1}{i}\sum_{j=1}^{n}(\partial_{\xi_j}b)(\partial_{x_j}a), \tag{4.96}$$

and (4.94) follows. □

4.6 Classical Operators

To classical symbols (Definition 3.14) corresponds the subclass of classical operators:

Definition 4.26 (Classical Ps.d.o.s). For $m\in\mathbb{R}$, we define the space of *classical pseudodifferential operators of order m* by

$$\Psi^m_{\rm cl}(\mathbb{R}^n):=\mathrm{Op}_L(S^m_{\rm cl}(\mathbb{R}^n;\mathbb{R}^n))\subset\Psi^m(\mathbb{R}^n), \tag{4.97}$$

equipped with the structure of a Fréchet space which makes Op_L into an isomorphism. We put $\Psi^{-\infty}_{\rm cl}(\mathbb{R}^n):=\Psi^{-\infty}(\mathbb{R}^n)$.

The symbol expansions in Theorem 4.22 and Corollary 4.18 imply that compositions and adjoints of classical operators are still classical:

Proposition 4.27 (Compositions and Adjoints of Classical Ps.d.o.s). *Composition of ps.d.o.s restricts to a continuous bilinear map*

$$\Psi^m_{\rm cl}(\mathbb{R}^n)\times\Psi^{m'}_{\rm cl}(\mathbb{R}^n)\ni(A,B)\mapsto A\circ B\in\Psi^{m+m'}_{\rm cl}(\mathbb{R}^n). \tag{4.98}$$

Moreover, the map

$$\Psi^m_{\rm cl}(\mathbb{R}^n)\ni A\mapsto A^*\in\Psi^m_{\rm cl}(\mathbb{R}^n) \tag{4.99}$$

is a continuous antilinear map.

For a classical operator $A = \mathrm{Op}_L(a)$, $a \in S^m_{\mathrm{cl}}(\mathbb{R}^n;\mathbb{R}^n)$, we can identify the principal symbol $\sigma^m(A)$ with the homogeneous leading order part of a, as discussed after Lemma 3.15. The corresponding short exact sequence is

$$0 \to \Psi^{m-1}_{\mathrm{cl}}(\mathbb{R}^n) \to \Psi^m_{\mathrm{cl}}(\mathbb{R}^n) \to S^m_{\mathrm{hom}}(\mathbb{R}^n;\mathbb{R}^n \setminus \{0\}) \to 0. \tag{4.100}$$

Virtually all pseudodifferential operators arising in practice (including those given as examples in this book) are classical. For most practical purposes, it is however not necessary to keep track of classicality; the most precise information we need on any ps.d.o. in this book, specifically in Chapters 8 and 10, is that its *principal symbol* is homogeneous, i.e., admits a representative that is homogeneous for $|\xi| \geq 1$.

4.7 Elliptic Parametrix

Recall Definition 3.11 and the discussion around (3.28). Then:

Definition 4.28 (Elliptic Ps.d.o.s). We call an operator $A \in \Psi^m(\mathbb{R}^n)$ *(uniformly) elliptic* if its principal symbol $\sigma^m(A)$ is elliptic.

As a first application of the symbol calculus we have developed above, we construct *parametrices* (approximate inverses) of uniformly elliptic operators.

Theorem 4.29 (Elliptic Parametrix). *Let $A \in \Psi^m(\mathbb{R}^n)$ be uniformly elliptic. Then there exists an operator $B \in \Psi^{-m}(\mathbb{R}^n)$ that is unique modulo $\Psi^{-\infty}(\mathbb{R}^n)$ such that*

$$AB - I,\ BA - I \in \Psi^{-\infty}(\mathbb{R}^n). \tag{4.101}$$

We call an operator B satisfying (4.101) a *parametrix of* A.

Lemma 4.30 (Asymptotic Neumann Series). *Let $R \in \Psi^{-\delta}(\mathbb{R}^n)$, $\delta > 0$. Let $R' \in \Psi^{-\delta}(\mathbb{R}^n)$ with $R' \sim \sum_{j=1}^{\infty} R^j$. Then $(I-R)(I+R') = I + E$ where $E \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. For any $N \in \mathbb{N}$, we have

$$\begin{aligned}(I-R)(I+R') - I &= (I-R)\Big(I + \sum_{j=1}^{N} R^j\Big) - I + (I-R)\Big(R' - \sum_{j=1}^{N} R^j\Big) \\ &= -R^{N+1} + (I-R)\Big(R' - \sum_{j=1}^{N} R^j\Big).\end{aligned} \tag{4.102}$$

The first term on the right lies in $\Psi^{-(N+1)\delta}(\mathbb{R}^n)$, and so does the second term since $R' - \sum_{j=1}^{N} R^j \in \Psi^{-(N+1)\delta}(\mathbb{R}^n)$. □

Proof of Theorem 4.29. Let $b \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$ be such that $\sigma^m(A)b - 1 = 0 \in S^0/S^{-1}$. Put $B_0 = \mathrm{Op}(b) \in \Psi^{-m}(\mathbb{R}^n)$, then

$$A \circ B_0 = I - R, \quad R \in \Psi^{-1}(\mathbb{R}^n). \tag{4.103}$$

Indeed, this follows from $\sigma^0(AB_0 - I) = 0$. Choosing $R' \sim \sum_{j=1}^{\infty} R^j \in \Psi^{-1}(\mathbb{R}^n)$ and setting

$$B := B_0(I + R') \in \Psi^{-m}(\mathbb{R}^n), \tag{4.104}$$

we conclude using Lemma 4.30 that $AB = I + E$, as desired.

An analogous argument produces $B' \in \Psi^{-m}(\mathbb{R}^n)$ with $B'A = I + E'$, $E' \in \Psi^{-\infty}(\mathbb{R}^n)$. But then abstract "group theory" gives

$$B = IB = (B'A - E')B = B'AB - E'B = B'(I+E) - E'B = B' + (B'E - E'B). \tag{4.105}$$

Therefore $B - B' \in \Psi^{-\infty}(\mathbb{R}^n)$. In particular, any two parametrices differ by an element of $\Psi^{-\infty}(\mathbb{R}^n)$. □

Example 4.31 (Laplacian). An inverse of the Laplacian $\Delta = \sum_{j=1}^{n} D_{x_j}^2 \in \mathrm{Diff}^2(\mathbb{R}^n)$ is given by convolution with

$$E_n(x) := c_n |x|^{-n+2}, \ n \neq 2, \quad E_2(x) := c_2 \log |x|, \tag{4.106}$$

where c_n is a dimension-dependent constant. (Concretely, $c_2^{-1} = -2\pi$ and $c_n^{-1} = n(n-2)\mathrm{vol}(B(0,1))$, $n \geq 3$. Here we use the notation $B(x_0, r)$ for the r-ball around x_0.) The Schwartz kernel of this convolution operator is $K(x,y) = E_n(x-y)$. This does *not* lie in $\Psi^{-2}(\mathbb{R}^n)$ because it does not decay rapidly as $|x-y| \to \infty$. If however $\chi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$ equals 1 near 0, then $K(x,y)\chi(x-y)$ *is* (the Schwartz kernel of) an element $B \in \Psi^{-2}(\mathbb{R}^n)$; and $I - \Delta B, I - B\Delta \in \Psi^{-\infty}(\mathbb{R}^n)$. (These statements follow from Corollary 4.9(2).)

See Exercise 4.6 for further examples.

Elliptic regularity is a simple consequence of the existence of a parametrix:

Proposition 4.32 (Elliptic Regularity: Smooth Case). *Let $A \in \Psi^m(\mathbb{R}^n)$ be uniformly elliptic, and suppose that*

$$u \in \mathscr{S}'(\mathbb{R}^n), \quad Au = f \in \mathcal{C}^{\infty}(\mathbb{R}^n). \tag{4.107}$$

Then $u \in \mathcal{C}^{\infty}(\mathbb{R}^n)$. More generally, we have

$$\operatorname{sing\,supp} u = \operatorname{sing\,supp} Au, \quad u \in \mathscr{S}'(\mathbb{R}^n). \tag{4.108}$$

Proof. We prove (4.108). Let $B \in \Psi^{-m}(\mathbb{R}^n)$ be a parametrix of A, so $BA = I - R$, $R \in \Psi^{-\infty}(\mathbb{R}^n)$. Then Proposition 4.16 and Lemma 4.17 give

$$\begin{aligned} \operatorname{sing\,supp} u &= \operatorname{sing\,supp}(BAu + Ru) \\ &= \operatorname{sing\,supp} BAu \subseteq \operatorname{sing\,supp} Au \subseteq \operatorname{sing\,supp} u. \end{aligned} \tag{4.109}$$

Therefore, equality must hold at each step. □

Example 4.33 (Elliptic Operators). Examples to which Proposition 4.32 applies are the Laplacian $\Delta \in \Psi^2(\mathbb{R}^n)$ and the Cauchy–Riemann operator $\bar\partial = \frac{1}{2}(\partial_{x_1} + i\partial_{x_2}) \in \Psi^1(\mathbb{R}^2)$ where we identify $\mathbb{R}^2 \cong \mathbb{C}$ via $(x_1, x_2) \mapsto x_1 + ix_2$. For the latter operator, we deduce that if $\bar\partial u = 0$ for $u \in \mathscr{S}'(\mathbb{R}^2)$, then $u \in \mathcal{C}^\infty(\mathbb{R}^2)$. (In complex analysis one learns that u is, in fact, *analytic*; here we are only developing microlocal analysis in the *smooth* category however and hence do not recover this stronger conclusion.)

4.8 Boundedness on Sobolev Spaces

In practice, one typically uses function spaces other than $\mathscr{S}(\mathbb{R}^n)$ and $\mathscr{S}'(\mathbb{R}^n)$, such as Hölder or L^p spaces. Here, we focus on function spaces related to L^2, in parts because they are the most natural for the study of non-elliptic operators in Chapter 8.

As usual, we first consider residual operators:

Proposition 4.34 (L^2-Boundedness of Residual Operators). *Let $A \in \Psi^{-\infty}(\mathbb{R}^n)$. Then A extends by continuity and density from $\mathscr{S}(\mathbb{R}^n)$ to a bounded linear operator $A\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$.*

This will follow from the Schwartz kernel estimates (4.53) and *Schur's lemma*:

Lemma 4.35 (Schur's Lemma). *Let (X, μ) and (Y, ν) be measure spaces. Suppose $K(x, y)$ is measurable on $X \times Y$ and*

$$\int_X |K(x,y)|\,\mathrm{d}\mu(x) \le C_1, \quad \int_Y |K(x,y)|\,\mathrm{d}\nu(y) \le C_2 \tag{4.110}$$

for almost all $y \in Y$ and $x \in X$, respectively. Let

$$Tu(x) = \int_Y K(x,y)u(y)\,\mathrm{d}\nu(y). \tag{4.111}$$

Then $T\colon L^2(Y) \to L^2(X)$ is bounded. Quantitatively,

$$\|Tu\|_{L^2(X)} \le (C_1 C_2)^{\frac{1}{2}} \|u\|_{L^2(Y)}. \tag{4.112}$$

Proof. Let $u \in L^2(Y)$ and $v \in L^2(X)$, then by Cauchy–Schwarz

$$\begin{aligned}
&\left| \int_X \int_Y K(x,y) u(y) \overline{v(x)} \, \mathrm{d}\nu(y) \, \mathrm{d}\mu(x) \right| \\
&\quad \le \left(\int_{X\times Y} |K(x,y)||u(y)|^2 \, \mathrm{d}\mu(x) \, \mathrm{d}\nu(y) \right)^{\frac{1}{2}} \\
&\qquad \times \left(\int_{X\times Y} |K(x,y)||v(x)|^2 \, \mathrm{d}\mu(x) \, \mathrm{d}\nu(y) \right)^{\frac{1}{2}} \\
&\quad \le C_1^{\frac{1}{2}} \|u\|_{L^2(Y)} \cdot C_2^{\frac{1}{2}} \|v\|_{L^2(X)}.
\end{aligned}$$

This completes the proof. □

Proof of Proposition 4.34. The Schwartz kernel K of A satisfies $|K(x,y)| \le C\langle x-y\rangle^{-n-1}$, and hence

$$\int_{\mathbb{R}^n} |K(x,y)| \, \mathrm{d}x \le C \int_{\mathbb{R}^n} \langle z\rangle^{-n-1} \, \mathrm{d}z =: C_1 < \infty, \quad \int_{\mathbb{R}^n} |K(x,y)| \, \mathrm{d}y \le C_2 < \infty. \tag{4.113}$$

The claim then follows from Lemma 4.35. □

Using "Hörmander's square root trick," we can now prove:

Theorem 4.36 (L^2-Boundedness of Zeroth Order Ps.d.o.s). *Let $A \in \Psi^0(\mathbb{R}^n)$. Then*

$$A \colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n) \tag{4.114}$$

is bounded. The linear map $\Psi^0(\mathbb{R}^n) \to \mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n))$ thus defined is continuous; here we write $\mathcal{L}(X,Y)$ for the space of bounded linear operators between normed spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$.

Proof. By Corollary 4.18 and Theorem 4.22, we have $A^*A \in \Psi^0(\mathbb{R}^n)$. With $a = \sigma^0(A) \in S^0(\mathbb{R}^n;\mathbb{R}^n)$ (i.e., a is any representative of $\sigma^0(A)$), we have $\sigma^0(A^*A) = |a|^2$, which is real, non-negative, and bounded. Thus, for $C > \sup_{x,\xi\in\mathbb{R}^n} |a|^2$, the symbol $C - |a|^2 \in S^0(\mathbb{R}^n;\mathbb{R}^n)$ is elliptic and positive. By Exercise 3.10, it has an approximate square root $0 < b_0 \in S^0(\mathbb{R}^n;\mathbb{R}^n)$, i.e., $C - |a|^2 - b_0^2 \in S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$. For later use, we note that b_0 is elliptic. Let $B_0 = \mathrm{Op}(b_0)$. Then $\sigma^0(C - A^*A - B_0^*B_0) = 0$; thus

$$C - A^*A = B_0^*B_0 + R_1, \quad R_1 \in \Psi^{-1}(\mathbb{R}^n). \tag{4.115}$$

Assume inductively that we have found $B_j \in \Psi^{-j}(\mathbb{R}^n)$, $j = 0, \dots, k-1$, such that

$$R_k := C - A^*A - (B_0 + \cdots + B_{k-1})^*(B_0 + \cdots + B_{k-1}) \in \Psi^{-k}(\mathbb{R}^n), \tag{4.116}$$

with (4.115) being the case $k = 1$. In order to find the next correction $B_k = \mathrm{Op}(b_k) \in \Psi^{-k}(\mathbb{R}^n)$, we compute

$$\begin{aligned} R_{k+1} &= C - A^*A - (B_0 + \cdots + B_k)^*(B_0 + \cdots + B_k) \\ &= R_k - (B_k^*(B_0 + \cdots + B_{k-1}) \\ &\qquad + (B_0 + \cdots + B_{k-1})^* B_k + B_k^* B_k) \in \Psi^{-k}(\mathbb{R}^n). \end{aligned} \tag{4.117}$$

Thus, the requirement $R_{k+1} \in \Psi^{-k-1}(\mathbb{R}^n)$ is equivalent to a principal symbol condition,

$$\overline{b_k} b_0 + b_0 b_k = \sigma^{-k}(R_k) \quad (\text{in } S^{-k}(\mathbb{R}^n;\mathbb{R}^n)/S^{-k-1}(\mathbb{R}^n;\mathbb{R}^n)). \tag{4.118}$$

Since $R_k = R_k^*$, the principal symbol $\sigma^{-k}(R_k)$ is real, and hence we can take $b_k = \frac{1}{2}\sigma^{-k}(R_k)/b_0$. We have $b_k \in S^{-k}(\mathbb{R}^n;\mathbb{R}^n)$ since b_0 is elliptic.

Finally, we let $B \in \Psi^0(\mathbb{R}^n)$ be the asymptotic sum

$$B \sim \sum_{k=0}^{\infty} B_k. \tag{4.119}$$

We have then arranged

$$R := C - A^*A - B^*B \in \Psi^{-\infty}(\mathbb{R}^n). \tag{4.120}$$

(Thus, we have constructed a square root, modulo residual operators, of $C - A^*A$.)

Given $u \in \mathscr{S}(\mathbb{R}^n)$, we then have

$$\begin{aligned} \|Au\|^2_{L^2(\mathbb{R}^n)} &= \langle A^*Au, u\rangle_{L^2(\mathbb{R}^n)} \\ &= C\|u\|^2_{L^2(\mathbb{R}^n)} - \|Bu\|^2_{L^2(\mathbb{R}^n)} - \langle Ru, u\rangle \\ &\le C\|u\|^2_{L^2(\mathbb{R}^n)} + \|Ru\|_{L^2(\mathbb{R}^n)}\|u\|_{L^2(\mathbb{R}^n)} \\ &\le C'\|u\|^2_{L^2(\mathbb{R}^n)}, \end{aligned} \tag{4.121}$$

using Proposition 4.34 in the last step. Thus, A extends by continuity to a bounded operator on $L^2(\mathbb{R}^n)$.

The second part of the theorem can be proved constructively as follows. For A in a neighborhood $\mathcal{U}$ of $0 \in \Psi^0(\mathbb{R}^n)$, we can take the constant C above to be equal to 1. We can choose the operators B_k, $k \in \mathbb{N}_0$, to depend continuously (albeit nonlinearly) on A, and then also B in (4.119) can be chosen to depend continuously on A in view of Proposition 3.17. Therefore also $R \in \Psi^{-\infty}(\mathbb{R}^n)$ depends continuously on A. The estimate (4.121) reads $\|A\|^2_{\mathcal{L}(L^2(\mathbb{R}^n),L^2(\mathbb{R}^n))} \le 1 + \|R\|_{\mathcal{L}(L^2(\mathbb{R}^n),L^2(\mathbb{R}^n))}$ for $A \in \mathcal{U}$. Now, $\|R\|_{\mathcal{L}(L^2(\mathbb{R}^n),L^2(\mathbb{R}^n))}$ is bounded by some fixed continuous seminorm of R; indeed, writing $R = \mathrm{Op}_L(r)$, one

can take this seminorm to be $C'\|r\|_{-n-1,n+1}$ for some universal constant C' (Exercise 4.2). By the continuity of $\mathcal{U} \ni A \mapsto R \in \Psi^{-\infty}(\mathbb{R}^n)$, there exists a neighborhood $\mathcal{V} \subset \mathcal{U}$ of $0 \in \Psi^0(\mathbb{R}^n)$ such that this seminorm remains uniformly bounded for $A \in \mathcal{V}$. Therefore, $\|A\|_{\mathcal{L}(L^2(\mathbb{R}^n),L^2(\mathbb{R}^n))}$ remains uniformly bounded for $A \in \mathcal{V}$. (See Exercise 4.3 for an alternative argument using the Closed Graph theorem.) □

Proposition 4.37 (Compactness Property of Certain Negative Order Ps.d.o.s). *Let $m < 0$, and let $A \in \Psi^m(\mathbb{R}^n)$. Suppose that the Schwartz kernel of A has compact support in $\mathbb{R}^n \times \mathbb{R}^n$. Then $A\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is compact.*

Proof. Write $A = \mathrm{Op}_L(a)$ where $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, and let $B \subset \mathbb{R}^n$ be a bounded open ball such that the support of the Schwartz kernel of A is contained in $B \times B$. Since for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\operatorname{supp}\phi \cap B = \emptyset$ we have $0 = \phi A = \mathrm{Op}_L(\phi(x)a(x,\xi))$, the uniqueness part of Theorem 4.11 implies that $\phi(x)a(x,\xi) = 0$; that is, $a(x,\xi) = 0$ for all $x \notin B$. The proof of Proposition 3.10 produces a sequence $a_j \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$, $j \in \mathbb{N}_0$, of symbols that are uniformly bounded in $S^m(\mathbb{R}^n;\mathbb{R}^n)$, converge to a in $S^0(\mathbb{R}^n;\mathbb{R}^n)$ as $j \to \infty$, and satisfy $a_j(x,\xi) = 0$ for all $x \notin B$. Now, $\mathrm{Op}(a_j)$ is compact on $L^2(\mathbb{R}^n)$, as it can be factored as

$$L^2(\mathbb{R}^n) \xrightarrow{\mathrm{Op}(a_j)} \mathcal{C}^\infty_{\mathrm{c}}(B) \subset \mathcal{C}^1(\bar{B}) \hookrightarrow \mathcal{C}^0(\bar{B}) \hookrightarrow L^2(\mathbb{R}^n), \tag{4.122}$$

where the penultimate inclusion map is compact by the Arzelà–Ascoli theorem, and the final map is extension by 0. By Theorem 4.36, $\mathrm{Op}(a)$ is the limit of $\mathrm{Op}(a_j)$ in $\mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n))$ and therefore compact as well. □

The boundedness of ps.d.o.s on Sobolev spaces is a straightforward consequence of Theorem 4.36 and the composition properties of ps.d.o.s.

Corollary 4.38 (Boundedness on Sobolev Spaces). *Let $s, m \in \mathbb{R}$, $A \in \Psi^m(\mathbb{R}^n)$. Then $A\colon H^s(\mathbb{R}^n) \to H^{s-m}(\mathbb{R}^n)$ is bounded. The linear map $\Psi^m(\mathbb{R}^n) \to \mathcal{L}(H^s(\mathbb{R}^n), H^{s-m}(\mathbb{R}^n))$ thus defined is continuous.*

Proof. Recall the operators $\langle D\rangle^\sigma = \mathcal{F}^{-1}\langle\xi\rangle^\sigma\mathcal{F}$ for $\sigma \in \mathbb{R}$ from Definition 2.13; note that $\langle D\rangle^\sigma \in \Psi^\sigma(\mathbb{R}^n)$. Moreover, $\langle D\rangle^{-s}\colon L^2(\mathbb{R}^n) \to H^s(\mathbb{R}^n)$ and $\langle D\rangle^{s-m}\colon H^{s-m}(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ are isometric isomorphisms. Now

$$\langle D\rangle^{s-m} A \langle D\rangle^{-s} \in \Psi^0(\mathbb{R}^n) \tag{4.123}$$

is bounded on $L^2(\mathbb{R}^n)$ by Theorem 4.36, which is equivalent to the statement of the corollary. The second part of the corollary follows from the continuity of compositions of ps.d.o.s proved in Theorem 4.22. □

One can now sharpen Proposition 4.32 to an elliptic regularity statement on the scale of Sobolev spaces. We omit the statement of such a result and

instead postpone a further generalization to weighted Sobolev spaces until Corollary 4.41.

The following technical result concerning the relationship between the approximation of symbols in Proposition 3.10 and the mapping properties of the associated ps.d.o.s is frequently useful.

Lemma 4.39 (Strong Convergence of Ps.d.o.s). *Let $m \in \mathbb{R}$. Let $A_\epsilon \in \Psi^m(\mathbb{R}^n)$, $\epsilon \in (0,1]$, be a uniformly bounded family of ps.d.o.s, and let $A \in \Psi^m(\mathbb{R}^n)$. Suppose that for some $m' \geq m$, the convergence $A_\epsilon \to A$, $\epsilon \to 0$, holds in the topology of $\Psi^{m'}(\mathbb{R}^n)$. Then A_ϵ converges strongly to A in $\mathcal{L}(H^s(\mathbb{R}^n), H^{s-m}(\mathbb{R}^n))$: for all $u \in H^s(\mathbb{R}^n)$, we have $A_\epsilon u \to Au$ in $H^{s-m}(\mathbb{R}^n)$. (When $m' = m$, the convergence $A_\epsilon \to A$ holds in operator norm.)*

Proof. We only consider the case $m' > m$. Let $s' := s + (m' - m)$. If $u \in H^{s'}$, then the convergence $A_\epsilon u \to Au$ in H^{s-m} follows from Corollary 4.38. Given $u \in H^s$ and $\delta > 0$, choose $u' \in H^{s'}$ with $\|u - u'\|_{H^s} < \delta$. Let then $\epsilon_0 > 0$ such that for $\epsilon \in (0, \epsilon_0)$, we have $\|A_\epsilon u' - Au'\|_{H^{s-m}} < \delta$. Let $C := \sup_{\epsilon \in (0,1]} \|A_\epsilon\|_{\mathcal{L}(H^s, H^{s-m})}$, $C' := \|A\|_{\mathcal{L}(H^s, H^{s-m})}$. Then for $\epsilon < \epsilon_0$, we have

$$\begin{aligned}\|A_\epsilon u - Au\|_{H^{s-m}} &\leq \|A_\epsilon(u - u')\|_{H^{s-m}} \\ &\quad + \|A_\epsilon u' - Au'\|_{H^{s-m}} + \|A(u - u')\|_{H^{s-m}} \\ &\leq C\delta + \delta + C'\delta,\end{aligned}$$

as desired. □

As a simple example, the family of *Friedrichs mollifiers*

$$J_\epsilon = \phi(\epsilon D), \quad \phi \in \mathscr{S}(\mathbb{R}^n),\ \phi(0) = 1, \tag{4.124}$$

is uniformly bounded in $\Psi^0(\mathbb{R}^n)$ and converges to I in $\Psi^{m'}(\mathbb{R}^n)$ for all $m' > 0$. A concrete example is when $\phi(\xi) = e^{-|\xi|^2}$, so $J_\epsilon = e^{-\epsilon^2 |D|^2}$. (The strong convergence $J_\epsilon \to I$ in $\mathcal{L}(H^s(\mathbb{R}^n))$ can of course easily be checked by hand in this case.)

We finally generalize some of the above results to *weighted* Sobolev spaces.

Theorem 4.40 (Boundedness Between Weighted Sobolev Spaces). *Let $s, m, r \in \mathbb{R}$, $A \in \Psi^m(\mathbb{R}^n)$. Then $A\colon \langle x\rangle^r H^s(\mathbb{R}^n) \to \langle x\rangle^r H^{s-m}(\mathbb{R}^n)$ is bounded. The linear map $\Psi^m(\mathbb{R}^n) \to \mathcal{L}(\langle x\rangle^r H^s(\mathbb{R}^n), \langle x\rangle^r H^{s-m}(\mathbb{R}^n))$ thus defined is continuous.*

Proof. Since the operators $\langle x\rangle^r \langle D\rangle^{-s}\colon L^2(\mathbb{R}^n) \to \langle x\rangle^r H^s(\mathbb{R}^n)$ and $\langle D\rangle^{s-m} \langle x\rangle^{-r}\colon \langle x\rangle^r H^{s-m}(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ are isomorphisms, it suffices to show that

$$A' := \langle D\rangle^{s-m} \langle x\rangle^{-r} \circ A \circ \langle x\rangle^r \langle D\rangle^{-s} \in \Psi^0(\mathbb{R}^n). \tag{4.125}$$

If $a = \sigma_L(A)$, then the full symbol $a^\sharp(x,y,\xi)$ of $A^\sharp := \langle x\rangle^{-r} \circ A \circ \langle x\rangle^r$ is given by $a^\sharp(x,y,\xi) = \langle x\rangle^{-r}\langle y\rangle^r a(x,\xi)$. By Lemma 4.3, we have

$$|a^\sharp(x,y,\xi)| \le 2^{|r|/2}\langle x-y\rangle^{|r|}|a(x,\xi)| \le C\langle x-y\rangle^{|r|}\langle\xi\rangle^m, \tag{4.126}$$

which is the first step toward showing that $a^\sharp \in \langle x-y\rangle^{|r|} S^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$; it remains to consider derivatives. The essence of this is contained in

$$\begin{aligned} |\partial_{y_j} a^\sharp(x,y,\xi)| &= |r\langle x\rangle^{-r}\langle y\rangle^{r-2} y_j a(x,\xi)| \\ &\le C\langle x-y\rangle^{|r|}\frac{y_j}{\langle y\rangle^2}\langle\xi\rangle^m \\ &\le C\langle x-y\rangle^{|r|}\langle\xi\rangle^m. \end{aligned} \tag{4.127}$$

We conclude that $A^\sharp \in \Psi^m(\mathbb{R}^n)$, and hence $A' \in \Psi^0(\mathbb{R}^n)$. □

In view of the Schwartz representation theorem (Theorem 2.18), we thus obtain another proof of Lemma 4.17. Indeed, a residual operator maps $\mathscr{S}'(\mathbb{R}^n) = \bigcup_{r,s}\langle x\rangle^r H^s(\mathbb{R}^n)$ into $\bigcup_r \langle x\rangle^r H^\infty(\mathbb{R}^n) = \bigcup_r \langle x\rangle^r \mathcal{C}_b^\infty(\mathbb{R}^n)$; this uses Sobolev embedding (Exercise 2.8).

We finally sharpen the elliptic regularity result, Proposition 4.32:

Corollary 4.41 (Elliptic Regularity: Weighted Sobolev Case). *Let $A \in \Psi^m(\mathbb{R}^n)$ be uniformly elliptic, and suppose $u \in \langle x\rangle^r H^{-N}(\mathbb{R}^n)$ for some $r, N \in \mathbb{R}$. If $Au = f \in \langle x\rangle^r H^{s-m}(\mathbb{R}^n)$, then $u \in \langle x\rangle^r H^s(\mathbb{R}^n)$. In particular, if $u \in H^{-N}(\mathbb{R}^n)$ and $f \in H^{s-m}(\mathbb{R}^n)$, then $u \in H^s(\mathbb{R}^n)$.*

Proof. With $B \in \Psi^{-m}(\mathbb{R}^n)$ denoting a parametrix of A, so $I = BA + R$, $R \in \Psi^{-\infty}(\mathbb{R}^n)$, we have

$$u = BAu + Ru = Bf + Ru, \tag{4.128}$$

with $Bf \in \langle x\rangle^r H^s(\mathbb{R}^n)$ and $Ru \in \langle x\rangle^r \bigcup_{\sigma\in\mathbb{R}} H^\sigma(\mathbb{R}^n)$. □

Remark 4.42 (A Priori Membership in a Weighted Space). It is important that the assumption on u already has the weight factor $\langle x\rangle^r$. Indeed, the conclusion would be false in general if we merely assumed $u \in \langle x\rangle^{r'} H^{-N}(\mathbb{R}^n)$ for some $r' < r$. (Convince yourself of this. For example, take $A = \Delta$, the Laplacian on $\mathbb{R}^n$, $u = 1$, and suitable r, r'.)

4.9 Semiclassical Pseudodifferential Operators

In the semiclassical setting, we wish to quantize a symbol $a = a(h,x,\xi)$ as $a(h,x,hD) =: \mathrm{Op}_h(a)$. This is the same as the standard quantization $\mathrm{Op}(\tilde{a}_h)$ of the rescaled symbol $\tilde{a}_h\colon (x,\xi) \mapsto a(h,x,h\xi)$, so

$$(\mathrm{Op}_h(a)u)(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi} a(h, x, h\xi) u(y)\, \mathrm{d}y\, \mathrm{d}\xi. \tag{4.129}$$

This is more commonly rewritten by replacing ξ with ξ/h. The analogue of Definition 4.1, for symbols depending on two sets of base variables, is then the following.

Definition 4.43 (Semiclassical Quantization). Let $k, m, w \in \mathbb{R}$, and let $a \in h^{-k}\langle \frac{x-y}{h}\rangle^w S^m_\hbar(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$ be defined for $h < h_0$. (The notation means that $a(h, x, y, \xi) = h^{-k}\langle \frac{x-y}{h}\rangle^w \tilde{a}(h, x, y, \xi)$ where $\tilde{a} \in S^m_\hbar(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$.) Then we define its *semiclassical quantization* $\mathrm{Op}_\hbar(a) = (\mathrm{Op}_h(a))_{h\in(0,h_0)}$ by

$$(\mathrm{Op}_h(a)u)(x) := (2\pi h)^{-n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi/h} a(h, x, y, \xi) u(y)\, \mathrm{d}y\, \mathrm{d}\xi. \tag{4.130}$$

For $a \in h^{-k} S^m_\hbar(\mathbb{R}^n; \mathbb{R}^n_\xi)$, we define the *semiclassical left/right quantization* by $\mathrm{Op}_{h,L/R}(a) = \mathrm{Op}_h(a_{L/R})$ where $a_L \colon (h, x, y, \xi) \mapsto a(h, x, \xi)$ and $a_R \colon (h, x, y, \xi) \mapsto a(h, y, \xi)$. Finally, we write $\mathrm{Op}_\hbar(a) := (\mathrm{Op}_h(a))_{h\in(0,h_0)}$ and $\mathrm{Op}_{\hbar,L/R}(a) := (\mathrm{Op}_{h,L/R}(a))_{h\in(0,h_0)}$.

For clarity, we sometimes write $\mathrm{Op}_h(a(h,\cdot))$ for $\mathrm{Op}_h(a)$. Since $\mathrm{Op}_h(a) = \mathrm{Op}(\tilde{a}_h)$ for $\tilde{a}_h(x, y, \xi) := a(h, x, y, h\xi)$, we have $\mathrm{Op}_h(a) \in \Psi^m(\mathbb{R}^n)$ for all $h \in (0, h_0)$. The form $\langle \frac{x-y}{h}\rangle^w$ of the off-diagonal weight is chosen here for a largely aesthetic reason: Schwartz kernels of semiclassical ps.d.o.s decay rapidly as $|\frac{x-y}{h}| \to \infty$ (Exercise 4.9), which is a stronger statement than rapid decay as $|x - y| \to \infty$.

Definition 4.44 (Semiclassical Pseudodifferential Operators). Let $k, m \in \mathbb{R}$. We then define the space of *semiclassical pseudodifferential operators* (of differential order m and semiclassical order k) by

$$h^{-k}\Psi^m_\hbar(\mathbb{R}^n) := \Big\{ (\mathrm{Op}_h(a))_{h\in(0,h_0)} : a \in h^{-k}\Big\langle \frac{x-y}{h}\Big\rangle^w S^m_\hbar(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n),\ w \in \mathbb{R} \Big\}. \tag{4.131}$$

Equivalently (see below),

$$h^{-k}\Psi^m_\hbar(\mathbb{R}^n) := \big\{ (\mathrm{Op}_{h,L/R}(a))_{h\in(0,h_0)} : a \in h^{-k} S^m_\hbar(\mathbb{R}^n; \mathbb{R}^n) \big\}. \tag{4.132}$$

For $k = 0$, we denote this space by $\Psi^m_\hbar(\mathbb{R}^n)$ simply. We moreover define by $h^\infty \Psi^{-\infty}_\hbar(\mathbb{R}^n) := \bigcap_{m,k\in\mathbb{R}} h^{-k}\Psi^m_\hbar(\mathbb{R}^n)$ the space of *residual semiclassical ps.d.o.s.*

Example 4.45 (Order Shifting Operators). Let $a(h, x, y, \xi) = h^{-k}\langle\xi\rangle^m$, then $\mathrm{Op}_h(a) = h^{-k}\langle hD\rangle^m \in h^{-k}\Psi^m_\hbar(\mathbb{R}^n)$. This operator is an (isometric) isomorphism $h^l H^s_h(\mathbb{R}^n) \to h^{l-k} H^{s-m}_h(\mathbb{R}^n)$ for all $l, s \in \mathbb{R}$.

Example 4.46 (Microlocalizers). Let $a(h,x,y,\xi) = \chi(x)\psi(\xi)$ where $\chi,\psi \in \mathcal{C}^\infty_c(\mathbb{R}^n)$. Then $A := \mathrm{Op}_h(a) \in \Psi^{-\infty}_\hbar(\mathbb{R}^n)$, which acts via $u \mapsto \chi\mathcal{F}^{-1}(\psi(h\xi)\hat{u}(\xi))$. Note that A is *not* residual unless $\chi = 0$ or $\psi = 0$.

We proceed to prove the equivalence of (4.131) and (4.132) via a left/right reduction argument. For brevity, we only consider the case $k = 0$. Let thus

$$a \in \left\langle \frac{x-y}{h} \right\rangle^w S^m_\hbar(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi), \quad a = a(h,x,y,\xi),\ h < h_0. \tag{4.133}$$

While Theorem 4.11 implies that there exists a unique $a_{h,L} \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ such that $\mathrm{Op}(\tilde{a}_h) = \mathrm{Op}_L(\tilde{a}_{h,L})$ where $\tilde{a}_h(x,y,\xi) = a(h,x,y,h\xi)$ and $\tilde{a}_{h,L}(x,\xi) = a_{h,L}(x,h\xi)$ (and therefore $\mathrm{Op}_h(a) = \mathrm{Op}_{h,L}(a_{h,L})$), we cannot immediately read off the desired conclusion $(a_{h,L})_{h\in(0,h_0)} \in S^m_\hbar(\mathbb{R}^n_x;\mathbb{R}^n_\xi)$: the behavior as $h \to 0$ of the remainder terms implicit in the asymptotic sum formulas (4.50)–(4.51) is at issue. We shall follow the proof of Theorem 4.11 and point out the modifications caused by the presence of the semiclassical parameter h. We begin with the analogue of Proposition 4.13.

Lemma 4.47 (Residual Semiclassical Operators). *A family $A = (A_h)_{h\in(0,h_0)}$, $A_h\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n)$, is a residual semiclassical operator if and only if the Schwartz kernels $K_h(x,y)$ of A_h are smooth and satisfy*

$$|(h\partial_x)^\alpha (h\partial_y)^\beta K_h(x,y)| \le C_{\alpha\beta MN} h^N \left\langle \frac{x-y}{h} \right\rangle^{-M} \qquad \forall\, \alpha,\beta,M,N. \tag{4.134}$$

Moreover, any such A can be written as $A_h = \mathrm{Op}_{h,L}(a_L) = \mathrm{Op}_{h,R}(a_R)$ for unique $a_L, a_R \in h^\infty S^{-\infty}_\hbar(\mathbb{R}^n;\mathbb{R}^n)$. In particular,

$$h^\infty \Psi^{-\infty}_\hbar(\mathbb{R}^n) = \{\mathrm{Op}_{h,L}(a)\colon a \in h^\infty S^{-\infty}_\hbar(\mathbb{R}^n;\mathbb{R}^n)\}. \tag{4.135}$$

Proof. Since $K_h = \mathrm{Op}_h(a_N)$ for $a_N = h^N \langle \frac{x-y}{h}\rangle^{w_N} \tilde{a}_N$, $\tilde{a}_N \in S^{-N}_\hbar(\mathbb{R}^n \times \mathbb{R}^n;\mathbb{R}^n)$, we can estimate

$$K_h(x,y) = (2\pi h)^{-n} \int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi/h} a_N(h,x,y,\xi)\,\mathrm{d}\xi \tag{4.136}$$

in absolute value by $|K_h(x,y)| \le C_N h^{-n} h^N \|\tilde{a}_N(h,\cdot)\|_{-n-1,0}$ for $|x-y| \le h$. More generally, $|(h\partial_x)^\alpha(h\partial_y)^\beta K_h(x,y)| \le C_{N\alpha\beta} h^{-n} h^N \|\tilde{a}_N(h,\cdot)\|_{-N,M}$ when $M \ge |\alpha| + |\beta|$ and $N > n + M$. For $|x-y| \ge h$, we integrate by parts as in (4.56) and estimate

$$\begin{aligned} |K_h(x,y)| &= (2\pi h)^{-n} \left| \int_{\mathbb{R}^n} \left[\left(\frac{x-y}{|x-y|^2} \cdot h D_\xi \right)^M e^{i(x-y)\cdot\xi/h} \right] a_N(h,x,y,\xi)\,\mathrm{d}\xi \right| \\ &\le C_N \left| \frac{x-y}{h} \right|^{-M} h^{-n+N} \left\langle \frac{x-y}{h} \right\rangle^{w_N} \|\tilde{a}_N\|_{-n-1,M}. \end{aligned} \tag{4.137}$$

Derivatives of K_h can be estimated in the same fashion. This proves (4.134).

For the converse, we only construct a_L. Rescaling ξ in (4.57) by h, we set

$$a_L(h,x,\xi) = \int_{\mathbb{R}^n} e^{-iz\cdot\xi/h} K_h(x, x-z)\,\mathrm{d}z; \tag{4.138}$$

this satisfies $\mathrm{Op}_{h,L}(a_L) = A_h$ indeed, and the membership $a_L \in h^\infty S^{-\infty}_\hbar(\mathbb{R}^n;\mathbb{R}^n)$ is an easy consequence of the estimates (4.134). □

Theorem 4.48 (Semiclassical Left/Right Reduction). *Let $a \in \langle\frac{x-y}{h}\rangle^w S^m_\hbar(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$, defined for $h<h_0$. Then there exists a unique left, resp. right symbol $a_L, a_R \in S^m_\hbar(\mathbb{R}^n;\mathbb{R}^n)$ such that $\mathrm{Op}_h(a) = \mathrm{Op}_{h,L}(a_L) = \mathrm{Op}_{h,R}(a_R)$ for $h<h_0$. The symbols a_L, a_R depend continuously on a. Modulo residual symbols (i.e., symbols of class $h^\infty S^{-\infty}_\hbar(\mathbb{R}^n;\mathbb{R}^n)$), they are given by asymptotic sums*

$$a_L(h,x,\xi) \sim \sum_{\alpha\in\mathbb{N}_0^n} \frac{h^{|\alpha|}}{\alpha!}\big(\partial_\xi^\alpha D_y^\alpha a(h,x,y,\xi)\big)\big|_{y=x}, \tag{4.139}$$

$$a_R(h,y,\xi) \sim \sum_{\alpha\in\mathbb{N}_0^n} \frac{(-h)^{|\alpha|}}{\alpha!}\big(\partial_\xi^\alpha D_x^\alpha a(h,x,y,\xi)\big)\big|_{x=y}. \tag{4.140}$$

Proof. We only consider the left reduction. The semiclassical analogue of (4.59) reads

$$\begin{aligned}\mathrm{Op}_h((y-x)^\alpha a)(x,y) &= (2\pi h)^{-n}\int_{\mathbb{R}^n}((-hD_\xi)^\alpha e^{i(x-y)\cdot\xi/h})a(h,x,y,\xi)\,\mathrm{d}\xi\\ &= \mathrm{Op}_h((hD_\xi)^\alpha a).\end{aligned} \tag{4.141}$$

Analogously to (4.60)–(4.64), we thus have

$$\mathrm{Op}_h\left(a - \sum_{|\alpha|<N}\frac{h^{|\alpha|}}{\alpha!}(D_\xi^\alpha\partial_y^\alpha a)|_{y=x}\right) = \mathrm{Op}_h(\tilde r_N), \tag{4.142}$$

where

$$\begin{aligned}\tilde r_N(h,x,y,\xi) &= \sum_{|\alpha|=N}\frac{Nh^{|\alpha|}}{\alpha!}\int_0^1 (1-t)^{N-1}(D_\xi^\alpha\partial_y^\alpha a)(h,x,x+t(y-x),\xi)\,\mathrm{d}t\\ &\in h^N\left\langle\frac{x-y}{h}\right\rangle^{w_+} S^{m-N}_\hbar(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n).\end{aligned} \tag{4.143}$$

Letting $b\in S^m_\hbar(\mathbb{R}^n;\mathbb{R}^n)$ be an asymptotic sum $b\sim\sum_{\alpha\in\mathbb{N}_0^n}\frac{h^{|\alpha|}}{\alpha!}(D_\xi^\alpha\partial_y^\alpha a)|_{y=x}$, we thus have

$$\mathrm{Op}_h(a-b) =: R_h, \quad (R_h)_{h\in(0,h_0)} \in \bigcap_{N\in\mathbb{N}} h^N \Psi_\hbar^{m-N}(\mathbb{R}^n) = h^\infty \Psi_\hbar^{-\infty}(\mathbb{R}^n). \tag{4.144}$$

By Lemma 4.47, the residual operator R can be written as $R = \mathrm{Op}_{h,L}(r)$ for a unique $r \in h^\infty S_\hbar^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$. □

Note that the $|\alpha| = 0$ terms of (4.139)–(4.140) are equal to $a(h,x,x,\xi)$ and $a(h,y,y,\xi)$, respectively; all remaining terms on the other hand are of class $hS_\hbar^{m-1}$. Matching the discussion leading up to (3.56), we thus set:

Definition 4.49 (Semiclassical Principal Symbol). Let $m,k \in \mathbb{R}$. The *semiclassical principal symbol* $\sigma_\hbar^{m,k}(A)$ of a semiclassical ps.d.o. $A = (\mathrm{Op}_h(a))_{h\in(0,h_0)} \in h^{-k}\Psi_\hbar^m(\mathbb{R}^n)$ is the equivalence class

$$\sigma_\hbar^{m,k}(A) := [\sigma_{\hbar,L}(A)] \in h^{-k}S_\hbar^m(\mathbb{R}^n;\mathbb{R}^n)/h^{-k+1}S_\hbar^{m-1}(\mathbb{R}^n;\mathbb{R}^n), \tag{4.145}$$

where we write $\sigma_{\hbar,L}(A)$ for the left-reduced symbol of A. For $k = 0$, we denote this by $\sigma_\hbar^m(A) \in S_\hbar^m/hS_\hbar^{m-1}$. If $\sigma_\hbar^m(A)$ admits an h-independent representative $a_0 \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ (which is then unique), we shall also refer to a_0 as the semiclassical principal symbol of A.

Theorem 4.48 shows that $\sigma_\hbar^m(A) = [a(h,x,x,\xi)]$ when $A = (\mathrm{Op}_h(a))_{h\in(0,h_0)}$ for $a = a(h,x,y,\xi) \in S_\hbar^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$ or more generally $a \in \langle\frac{x-y}{h}\rangle^w S_\hbar^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$. The analogue of the short exact sequence (4.92) is

$$0 \to h\Psi_\hbar^{m-1}(\mathbb{R}^n) \to \Psi_\hbar^m(\mathbb{R}^n) \xrightarrow{\sigma_\hbar^m} S_\hbar^m(\mathbb{R}^n;\mathbb{R}^n)/hS_\hbar^{m-1}(\mathbb{R}^n;\mathbb{R}^n) \to 0. \tag{4.146}$$

Thus, the semiclassical principal symbol not only captures a semiclassical ps.d.o. to leading order in the differential order sense but also in the semiclassical order sense, i.e., modulo operators whose "coefficients" have an extra order h of vanishing as $h \searrow 0$.

From here on, it is a simple matter to prove the basic properties of semiclassical ps.d.o.s. We only state the results and leave the proofs to the reader. For notational simplicity, we restrict ourselves to semiclassical orders $k = 0$.

Theorem 4.50 (Adjoints, Compositions, and Commutators of Semiclassical Ps.d.o.s). *Let $A \in \Psi_\hbar^m(\mathbb{R}^n)$, $B \in \Psi_\hbar^{m'}(\mathbb{R}^n)$.*

(1) *(Adjoints.) The adjoint A^* of A, defined (for each h) by the formula (4.82), defines an element $A^* \in \Psi_\hbar^m(\mathbb{R}^n)$, and*

$$\sigma_{\hbar,L}(A^*)(h,x,\xi) \sim \sum_{\alpha\in\mathbb{N}_0} \frac{h^{|\alpha|}}{\alpha!}\partial_\xi^\alpha D_x^\alpha \overline{\sigma_{\hbar,L}(A)}(h,x,\xi). \tag{4.147}$$

In particular, $\sigma_\hbar^m(A^*) = \overline{\sigma_\hbar^m(A)}$.

(2) *(Compositions.) We have* $A \circ B \in \Psi_\hbar^{m+m'}(\mathbb{R}^n)$ *and*

$$\sigma_{\hbar,L}(A \circ B) \sim \sum_{\alpha \in \mathbb{N}_0} \frac{h^{|\alpha|}}{\alpha!} \partial_\xi^\alpha \sigma_{\hbar,L}(A) \cdot D_x^\alpha \sigma_{\hbar,L}(B). \tag{4.148}$$

In particular, $\sigma_\hbar^{m+m'}(A \circ B) = \sigma_\hbar^m(A)\sigma_\hbar^{m'}(B)$.

(3) *(Commutators.) We have* $[A, B] \in h\Psi_\hbar^{m+m'-1}(\mathbb{R}^n)$ *and (in the notation (4.95))*

$$\sigma_\hbar^{m+m'-1}\Big(\frac{i}{h}[A, B]\Big) = \{\sigma_\hbar^m(A), \sigma_\hbar^{m'}(B)\}. \tag{4.149}$$

Theorem 4.51 (Boundedness on Semiclassical Sobolev Spaces). *Let* $s, m \in \mathbb{R}$ *and* $A = (A_h)_{h \in (0,h_0)} \in \Psi_\hbar^m(\mathbb{R}^n)$. *Then there exists a constant* C *such that*

$$\|A_h u\|_{H_h^{s-m}(\mathbb{R}^n)} \leq C \|u\|_{H_h^s(\mathbb{R}^n)}, \quad h \in (0, h_0). \tag{4.150}$$

Proof. Exercise 4.10. □

The semiclassical principal symbol controls a semiclassical ps.d.o. A modulo error terms with an extra power of h, and these error terms have a vanishing operator norm in the limit $h \to 0$. For example, if $A \in \Psi_\hbar^0(\mathbb{R}^n)$ has $\sigma_\hbar^0(A) = 1$, then A is h-close to the identity operator *as an element of* $\mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n))$ and thus *invertible* for small h. Thus, appropriate properties of the principal symbol of a semiclassical ps.d.o. imply its invertibility for small h. The following result is the prototypical example.

Theorem 4.52 (Invertibility of Elliptic Semiclassical Ps.d.o.s). *Let* $A \in \Psi_\hbar^m(\mathbb{R}^n)$ *be semiclassically elliptic, i.e., its principal symbol is semiclassically elliptic in the sense of Definition 3.21. Then there exists* $h_0 > 0$ *such that for all* $h \in (0, h_0)$ *and* $s \in \mathbb{R}$, *the operator* $A_h \colon H_h^s(\mathbb{R}^n) \to H_h^{s-m}(\mathbb{R}^n)$ *(or equivalently* $H^s(\mathbb{R}^n) \to H^{s-m}(\mathbb{R}^n)$*) is invertible. Moreover, the inverse can be written as*

$$A_h^{-1} = B_h + R_h, \tag{4.151}$$

where $B = (B_h)_{h \in (0,h_0)} \in \Psi_\hbar^{-m}(\mathbb{R}^n)$, *while the remainder term* R_h *satisfies* $R_h \colon H_h^{-N}(\mathbb{R}^n) \to H_h^N(\mathbb{R}^n)$ *and has operator norm bounded by* $C_N h^N$, *for all* N.

The main ingredient is an elliptic parametrix construction in the semiclassical algebra.

Theorem 4.53 (Semiclassical Elliptic Parametrix). *Let* $A = (A_h)_{h\in(0,h_0)} \in \Psi_\hbar^m(\mathbb{R}^n)$ *be semiclassically elliptic. Then there exist* $h_1 \in (0, h_0]$ *and an operator* $B = (B_h)_{h\in(0,h_1)} \in \Psi_\hbar^{-m}(\mathbb{R}^n)$*, unique modulo* $h^\infty\Psi_\hbar^{-\infty}(\mathbb{R}^n)$*, such that*

$$AB - I,\ BA - I \in h^\infty\Psi_\hbar^{-\infty}(\mathbb{R}^n). \tag{4.152}$$

The proof is completely analogous to that of Theorem 4.29 and left to the reader.

Proof of Theorem 4.52. With $h_1 > 0$, $B \in \Psi_\hbar^{-m}(\mathbb{R}^n)$ as in Theorem 4.53, set

$$R = (R_h)_{h\in(0,h_1)} := I - BA \in h^\infty\Psi_\hbar^{-\infty}(\mathbb{R}^n). \tag{4.153}$$

Since $\|R_h\|_{\mathcal{L}(H_h^m(\mathbb{R}^n),H_h^m(\mathbb{R}^n))} \le Ch$ for some $C < \infty$, we can fix $h_0 \in (0, h_1]$ such that $\|R_h\|_{\mathcal{L}(H_h^m(\mathbb{R}^n),H_h^m(\mathbb{R}^n))} \le \frac{1}{2}$ for all $h \in (0, h_0)$. Therefore, $I - R_h$ is invertible on $H_h^m(\mathbb{R}^n)$ using a Neumann series. This implies that $(I - R)^{-1}BA = I$ on $H_h^m(\mathbb{R}^n)$ for $h < h_0$, and therefore $A\colon H_h^m(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is left invertible. Similarly, from

$$R' := I - AB \in h^\infty\Psi_\hbar^{-\infty}(\mathbb{R}^n), \tag{4.154}$$

we deduce that $AB(I - R')^{-1} = I$ on $L^2(\mathbb{R}^n)$, so $A\colon H_h^m(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is right invertible; therefore, it is invertible. For clarity, we write its inverse *acting between these spaces* as

$$A_{h,L^2}^{-1}\colon L^2(\mathbb{R}^n) \to H_h^m(\mathbb{R}^n), \quad h \in (0, h_0), \tag{4.155}$$

so $A_h A_{h,L^2}^{-1} = I$ on L^2 and $A_{h,L^2}^{-1}A_h = I$ on H_h^m.

It remains to show that A is invertible, for the same range of h, acting between any other compatible pair of Sobolev spaces. We first write

$$A_{h,L^2}^{-1} = A_{h,L^2}^{-1}(AB + R') = B + (BA + R)A_{h,L^2}^{-1}R' = B + BR' + RA_{h,L^2}^{-1}R'. \tag{4.156}$$

Since $B \in \Psi_\hbar^{-m}$ and $BR' \in h^\infty\Psi_\hbar^{-\infty}$, the first two terms on the right lie in $\Psi_\hbar^{-m}$. By Theorem 4.51, the last term is bounded in operator norm as a map $H_h^{-N}(\mathbb{R}^n) \to H_h^N(\mathbb{R}^n)$ by $C_N h^N$ for all N. The right-hand side of (4.156) defines a bounded operator $Q\colon H_h^{s-m}(\mathbb{R}^n) \to H_h^s(\mathbb{R}^n)$ for all $s \in \mathbb{R}$. From the identities

$$AQ = I \text{ on } L^2(\mathbb{R}^n), \quad QA = I \text{ on } H_h^m(\mathbb{R}^n) \tag{4.157}$$

for $h \in (0, h_0)$ we deduce that $AQ = I$ and $QA = I$ on $\mathscr{S}(\mathbb{R}^n)$ and hence by density also on $H_h^{s-m}(\mathbb{R}^n)$ and $H_h^s(\mathbb{R}^n)$, respectively, for all $s \in \mathbb{R}$. The decomposition (4.151) thus arises from (4.156) by using $B_h + B_h R_h'$ and $R_h A_{h,L^2}^{-1} R_h'$ for B_h and R_h, respectively. □

For an application of Theorem 4.53, see Exercise 4.11.

Notes

Pseudodifferential operators were introduced in Kohn–Nirenberg [KN65]. Our development in the early parts of this chapter follows Melrose [Mel07]. For a detailed account of the semiclassical case, see, for example, Zworski [Zwo12], Dimassi–Sjöstrand [DS99] (with a focus on spectral theory at high energies), and Martinez [Mar01] (who systematically uses FBI transform techniques). The "square root trick" in the proof of L^2-boundedness (Theorem 4.36) is taken from [Hör71a, Theorem 2.2.1].

Exercises

Exercise 4.1 (Quantization of Polynomials in ξ). Let $m \in \mathbb{N}_0$, and let $a \in S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi)$ be a polynomial in the symbolic variable ξ.

(1) Show, starting from the definition as a limit of quantizations of residual symbols, that $\mathrm{Op}(a) \in \mathrm{Diff}^m(\mathbb{R}^n)$.
(2) Let $w \in \mathbb{R}$. Prove that $\mathrm{Op}(\langle x - y\rangle^w a) \in \mathrm{Diff}^m(\mathbb{R}^n)$.

Exercise 4.2 (Bounds on Very Negative Order Ps.d.o.s). Let $n \in \mathbb{N}$ and fix $m < -n$.

(1) Show that the Schwartz kernel $K = K(x, y)$ of $A = \mathrm{Op}_L(a) \in \Psi^m(\mathbb{R}^n)$ is a continuous function of $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$. Show moreover that for all $N \in \mathbb{N}$ there exists a constant $C \in \mathbb{R}$ such that

$$|K(x, y)| \leq C \|a\|_{m,N} \langle x - y\rangle^{-N}. \tag{4.158}$$

(2) Deduce that there exists $C \in \mathbb{R}$ such that $\|A\|_{\mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n))} \leq C\|a\|_{m,n+1}$.

Exercise 4.3 (Zeroth Order Ps.d.o.s as Linear Maps on L^2). Denote by

$$\Phi \colon \Psi^0(\mathbb{R}^n) \to \mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n)) \tag{4.159}$$

the linear map assigning to $A \in \Psi^0(\mathbb{R}^n)$ the bounded linear operator $\Phi(A)\colon u \mapsto Au$, $u \in L^2(\mathbb{R}^n)$. Use the Closed Graph theorem to show that Ψ is continuous.

Exercise 4.4 (A Classical Ps.d.o.). Let $k \in \mathcal{C}_b^\infty(\mathbb{R})$. Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ be identically 1 near 0. Show that the operator

$$Au(x) = \lim_{\epsilon \to 0} \int_{|x-y|\geq\epsilon} \chi(x-y)k(x)(x-y)^{-1}u(y)\,\mathrm{d}y, \quad u \in \mathcal{C}_c^\infty(\mathbb{R}^n), \tag{4.160}$$

is well-defined and defines an element $A \in \Psi^0_{\mathrm{cl}}(\mathbb{R})$. Compute its principal symbol.

Exercise 4.5 (Gårding's Inequality). Let $A \in \Psi^{2m}(\mathbb{R}^n)$, $a = \sigma^{2m}(A) \in S^{2m}(\mathbb{R}^n;\mathbb{R}^n)$. Let $c \in \mathbb{R}$, and suppose that $\operatorname{Re}\sigma^{2m}(A) \geq c\langle\xi\rangle^{2m}$ for all $x, \xi \in \mathbb{R}^n$ with $|\xi| \geq C_0$ for some $C_0 > 0$. Show that for all $\epsilon > 0$ and $N \in \mathbb{R}$, there exists a constant C such that

$$\operatorname{Re}\langle Au, u\rangle_{L^2(\mathbb{R}^n)} \geq (c-\epsilon)\|u\|^2_{H^m(\mathbb{R}^n)} - C\|u\|^2_{H^{-N}(\mathbb{R}^n)}, \quad u \in \mathscr{S}(\mathbb{R}^n). \tag{4.161}$$

Remark. The *sharp Gårding inequality* states that (4.161) holds for $\epsilon = 0$, but then with $-N = m - \frac{1}{2}$; see Hörmander [Hör03, Theorem 18.1.14] for a textbook account. This can be further refined to the *Fefferman–Phong* inequality [FP78] which allows $-N = m - 1$.

Exercise 4.6 (Inverses and Parametrices).

(1) Suppose $A \in \mathrm{Diff}^m(\mathbb{R}^n)$ has constant coefficients, so $A = \sum_{|\alpha|\leq m} a_\alpha D^\alpha$, $a_\alpha \in \mathbb{C}$. Suppose $u \in \mathscr{S}'(\mathbb{R}^n)$ is a fundamental solution, i.e., $Au = \delta$. Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ be equal to 1 near 0. Show that the operator B with Schwartz kernel

$$K_B(x,y) = u(x-y)\chi(x-y) \tag{4.162}$$

is a (classical) parametrix of A, i.e., $B \in \Psi^{-m}_{\mathrm{cl}}(\mathbb{R}^n)$ and $AB - I, BA - I \in \Psi^{-\infty}(\mathbb{R}^n)$, if and only if A is elliptic. *Remark.* The Malgrange–Ehrenpreis theorem [Mal56, Ehr54] states that every constant coefficient operator $A \in \mathrm{Diff}^m(M)$ has a fundamental solution u.

(2) Find an explicit parametrix $B \in \Psi^{-1}(\mathbb{R}^2)$ of the Cauchy–Riemann operator $\bar\partial = \frac{1}{2}(\partial_{x_1} + i\partial_{x_2})$.

In the next exercise, we discuss a symmetric alternative to left/right quantization.

Exercise 4.7 (Weyl Quantization). Define the *Weyl quantization* of a symbol $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ by

$$\mathrm{Op}^{\mathrm{w}}(a) := \mathrm{Op}\Big(a\Big(\frac{x+y}{2}, \xi\Big)\Big) \in \Psi^m(\mathbb{R}^n). \tag{4.163}$$

(1) Show that for all $A \in \Psi^m(\mathbb{R}^n)$ there exists a unique $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ such that $A = \mathrm{Op}^{\mathrm{w}}(a)$. We call a the *(full) Weyl symbol* of A and write $a = \sigma^{\mathrm{w}}(A)$. Show moreover for $a \in S^m(\mathbb{R}^n_x \times \mathbb{R}^n_y;\mathbb{R}^n_\xi)$ that

$$\sigma^{\mathrm{w}}(\mathrm{Op}(a))(x,\xi) \sim \sum_{\alpha\in\mathbb{N}_0^n} \frac{(-1)^{|\alpha|}}{\alpha!2^{|\alpha|}} \partial_\xi^\alpha \big((D_x - D_y)^\alpha a(x,y,\xi)\big)\big|_{y=x}\,, \tag{4.164}$$

as well as

$$\sigma_L(\mathrm{Op}^{\mathrm{w}}(a)) \sim \sum_\alpha \frac{1}{\alpha!2^{|\alpha|}} \partial_\xi^\alpha D_x^\alpha a, \quad \sigma^{\mathrm{w}}(\mathrm{Op}_L(a)) \sim \sum_\alpha \frac{(-1)^{|\alpha|}}{\alpha!2^{|\alpha|}} \partial_\xi^\alpha D_x^\alpha a. \tag{4.165}$$

(2) Let $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$. Show that $\mathrm{Op}^{\mathrm{w}}(a)^* = \mathrm{Op}^{\mathrm{w}}(\bar{a})$.
(3) Let $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, $b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$. Show that

$$\sigma^{\mathrm{w}}\big(\mathrm{Op}^{\mathrm{w}}(a)\circ\mathrm{Op}^{\mathrm{w}}(b)\big) \equiv ab + \frac{1}{2i}\{a,b\} \bmod S^{m+m'-2}(\mathbb{R}^n;\mathbb{R}^n). \tag{4.166}$$

Exercise 4.8 (Strong Convergence in Families). Let $k \in \mathbb{N}_0 \cup \{\infty\}$, $s, m \in \mathbb{R}$.

(1) Let H be a Hilbert space, and let $A_j \in \mathcal{L}(H)$, $j \in \mathbb{N}$, be a uniformly bounded family converging strongly to $A \in \mathcal{L}(H)$ as $j \to \infty$. Let $\phi \in \mathcal{C}^k([0,1];H)$. Show that $A_j\phi\colon t \mapsto A_j(\phi(t))$ converges to $A\phi$ in $\mathcal{C}^k([0,1];H)$ as $j \to \infty$.
(2) Suppose $A_\epsilon \in \Psi^m(\mathbb{R}^n)$ is uniformly bounded for $\epsilon \in (0,1]$ and converges to $A \in \Psi^m(\mathbb{R}^n)$ in the topology of $\Psi^{m'}(\mathbb{R}^n)$ for some $m' \geq m$ as $\epsilon \to 0$. Let $\phi \in \mathcal{C}^k([0,1];H^s(\mathbb{R}^n))$. Show that $A_\epsilon\phi\colon t \mapsto A_\epsilon(\phi(t))$ converges to $A\phi$ in $\mathcal{C}^k([0,1];H^{s-m}(\mathbb{R}^n))$ as $\epsilon \to 0$.

Exercise 4.9 (Decay of Schwartz Kernels of Semiclassical Ps.d.o.s). Let $A = (A_h)_{h\in(0,h_0)} \in \Psi_\hbar^m(\mathbb{R}^n)$, and denote by K_h the Schwartz kernel of A_h. Show that for all $\epsilon > 0$, $N \in \mathbb{R}$, $\alpha, \beta \in \mathbb{N}_0^n$ there exists a constant C such that

$$|(h\partial_x)^\alpha (h\partial_y)^\beta K_h(x,y)| \leq C\Big|\frac{x-y}{h}\Big|^{-N}, \quad \Big|\frac{x-y}{h}\Big| > \epsilon. \tag{4.167}$$

As a stronger result, establish an analogue of Corollary 4.9.

Exercise 4.10 (Boundedness of Ps.d.o.s on Semiclassical Sobolev Spaces).

(1) Let $A = (A_h)_{h\in(0,h_0)} \in \Psi_\hbar^{-\infty}(\mathbb{R}^n)$. Show that there exists a constant $C < \infty$ such that $\|A_h u\|_{L^2(\mathbb{R}^n)} \leq C\|u\|_{L^2(\mathbb{R}^n)}$ for all $h \in (0,h_0)$ and $u \in \mathscr{S}(\mathbb{R}^n)$.

(2) Using the "square root trick" as in the proof of Theorem 4.36, show that every $A = (A_h)_{h\in(0,h_0)} \in \Psi^0_\hbar(\mathbb{R}^n)$ defines a uniformly bounded family of operators $A_h \colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$.
(3) Prove Theorem 4.51. State and prove a generalization to operators and spaces with general semiclassical orders.
(4) Let $A = (A_h)_{h\in(0,h_0)} \in \Psi^0_\hbar(\mathbb{R}^n)$, and let $C_A := \limsup_{h\to 0} \sup_{x,\xi\in\mathbb{R}^n} |\sigma^0_\hbar(A)(x,\xi)|$. Show that for all $\epsilon > 0$ there exists a constant $C_\epsilon < \infty$ such that

$$\|A\|_{\mathcal{L}(L^2(\mathbb{R}^n),L^2(\mathbb{R}^n))} \leq C_A + \epsilon + C_\epsilon h, \quad h \in (0,h_0). \tag{4.168}$$

Thus, in the limit $h \searrow 0$, the operator norm of a semiclassical ps.d.o. is controlled by the sup norm of its principal symbol. *Remark.* The conclusion is valid also for $\epsilon = 0$; see Zworski [Zwo12, Theorem 13.13].

Exercise 4.11 (Inversion of a Spectral Family at High Frequencies, I). Let $Q \in \mathrm{Diff}^1(\mathbb{R}^n)$. Let moreover $(g^{ij})_{i,j=1,\dots,n}$ be a matrix of smooth functions $g^{ij} \in \mathcal{C}^\infty_b(\mathbb{R}^n)$ for which there exists a constant $c > 0$ such that $\sum_{i,j=1}^n g^{ij} v_i v_j \geq c|v|^2$ for all $v \in \mathbb{R}^n$. Define the operator

$$P(\sigma) := \sum_{i,j=1}^n g^{ij} D_i D_j + Q - \sigma^2, \quad \sigma \in \mathbb{C}. \tag{4.169}$$

Show that for all $c > 0$ there exists $C > 0$ such that for all $\sigma \in \mathbb{C}$ with $|\mathrm{Im}\,\sigma| > c|\mathrm{Re}\,\sigma|$ and $|\mathrm{Re}\,\sigma| > C$, the operator $P(\sigma) \colon H^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is invertible.

Exercise 4.12 (Inversion of a Spectral Family at High Frequencies, II). Let $A \in \mathrm{Diff}^m(\mathbb{R}^n)$ be an elliptic differential operator, and denote by $a(x,\xi) = \sigma^m(A)(x,\xi)$ its principal symbol. Suppose $z_0 \in \mathbb{C}$, $|z_0| = 1$, is such that $\inf_{x,\xi} |a(x,\xi) - z_0| > 0$. Show that there exist $\lambda_0, \epsilon > 0$ such that the operator $A - \lambda \colon H^m(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is invertible for all $\lambda \in \mathbb{C}$ with $|\lambda| > \lambda_0$ and $|\frac{\lambda}{|\lambda|} - z_0| < \epsilon$.

Exercise 4.13 (Conjugation of Semiclassical Operators). Let $a \in S^m_\hbar(\mathbb{R}^n;\mathbb{R}^n)$ and $A = (A_h)_{h\in(0,h_0)}$, $A_h = \mathrm{Op}_h(a)$.

(1) Let $\xi_0 \in \mathbb{R}^n$ and set $B_h := e^{i\xi_0\cdot/h} A_h e^{-i\xi_0\cdot/h}$. Show that $\sigma_{\hbar,L}(B)(x,\xi) = \sigma_{\hbar,L}(A)(x,\xi-\xi_0)$.
(2) Let $\phi \in \mathcal{C}^\infty_b(\mathbb{R}^n)$. Show that $\sigma^m_\hbar(e^{i\phi/h} A e^{-i\phi/h})(x,\xi) = \sigma^m_\hbar(A)(x,\xi - \nabla\phi(x))$.

The following series of exercises introduces the basic properties of *scattering pseudodifferential operators* on $\mathbb{R}^n$.

Exercise 4.14 (Scattering Symbols). For $m, r_1, r_2 \in \mathbb{R}$, define the space of symbols

$$S^{m,r_1,r_2}(\mathbb{R}^n_x \times \mathbb{R}^n_y; \mathbb{R}^n_\xi) \tag{4.170}$$

to consist of all $a \in \mathcal{C}^\infty(\mathbb{R}^{3n})$ such that the seminorms

$$\begin{aligned}\|a\|_{m,r_1,r_2,k} := \sup_{\substack{x,y,\xi\in\mathbb{R}^n \\ |\alpha_1|+|\alpha_2|+|\beta|\le k}} &\langle x\rangle^{-r_1+|\alpha_1|}\langle y\rangle^{-r_2+|\alpha_2|}\langle\xi\rangle^{-m+|\beta|} \\ &\times |\partial_x^{\alpha_1}\partial_y^{\alpha_2}\partial_\xi^\beta a(x,y,\xi)|\end{aligned} \tag{4.171}$$

are finite for all $k \in \mathbb{N}_0$. (Such a are symbolic not only in ξ but also in x and y.) Let

$$S^{-\infty,-\infty,-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n) := \bigcap_{m,r_1,r_2\in\mathbb{R}} S^{m,r_1,r_2}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n). \tag{4.172}$$

(1) Prove that $S^{-\infty,-\infty,-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n) \subset S^{m,r_1,r_2}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$ is dense in the topology of $S^{m',r_1',r_2'}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$ whenever $m < m'$, $r_1 < r_1'$, $r_2 < r_2'$.

(2) Prove the following variant of Proposition 3.16: given $a_j \in S^{m-j,r_1-j,r_2-j}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, there exists $a \in S^{m,r_1,r_2}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, unique modulo $S^{-\infty,-\infty,-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, such that $a - \sum_{j=0}^{J-1} a_j \in S^{m-J,r_1-J,r_2-J}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$ for all $J \in \mathbb{N}$.

Exercise 4.15 (Scattering Ps.d.o.s, I: Boundedness). Let $m, r_1, r_2 \in \mathbb{R}$. Prove that

$$\mathrm{Op}(a)\colon \mathscr{S}(\mathbb{R}^n) \to \mathscr{S}(\mathbb{R}^n), \quad a \in S^{m,r_1,r_2}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n). \tag{4.173}$$

Prove this more generally for $a \in \langle x-y\rangle^w S^{m,r_1,r_2}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, $w \in \mathbb{R}$.

Exercise 4.16 (Scattering Ps.d.o.s, II: Residual Operators). Let $r_1 \in \mathbb{R}$. Show that an operator A can be written as $A = \mathrm{Op}(a_N)$, $a_N \in S^{-N,r_1,-N}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$, for all N if and only if its Schwartz kernel $K = K(x,y)$ satisfies $K \in \mathscr{S}(\mathbb{R}^n_x \times \mathbb{R}^n_y)$. Show that in this case there exist unique $a_L, a_R \in S^{-\infty,-\infty}(\mathbb{R}^n; \mathbb{R}^n)$ such that $A = \mathrm{Op}_L(a_L) = \mathrm{Op}_R(a_R)$.

Exercise 4.17 (Scattering Ps.d.o.s, III: Reduction). We write

$$S^{m,r}(\mathbb{R}^n; \mathbb{R}^n) \tag{4.174}$$

for the space of all $a = a(x,\xi) \in \mathcal{C}^\infty(\mathbb{R}^{2n})$ which satisfy $|\partial_x^\alpha \partial_\xi^\beta a(x,\xi)| \le C_{\alpha\beta}\langle x\rangle^{r-|\alpha|}\langle\xi\rangle^{m-|\beta|}$ for all $\alpha, \beta \in \mathbb{N}_0^n$. Let $A = \mathrm{Op}(a)$, $a \in S^{m,r_1,r_2}(\mathbb{R}^n \times$

$\mathbb{R}^n;\mathbb{R}^n)$. Prove that there exists a unique left symbol $a_L \in S^{m,r_1+r_2}(\mathbb{R}^n;\mathbb{R}^n)$ such that $A = \mathrm{Op}_L(a_L)$. (*Attention:* the control of the remainder term of the Taylor expansion requires some care.)

Exercise 4.18 (Scattering Ps.d.o.s, IV: Algebra). Define the space of scattering ps.d.o.s by

$$\Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n) := \mathrm{Op}(S^{m,r}(\mathbb{R}^n;\mathbb{R}^n)). \tag{4.175}$$

(1) Prove that $A \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$ implies $A^* \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$.
(2) Suppose $A \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$, $B \in \Psi_{\mathrm{sc}}^{m',r'}(\mathbb{R}^n)$. Prove that

$$A \circ B \in \Psi_{\mathrm{sc}}^{m+m',r+r'}(\mathbb{R}^n). \tag{4.176}$$

Exercise 4.19 (Scattering Ps.d.o.s, V: Principal Symbol). Define the principal symbol of $A = \mathrm{Op}_L(a_L) \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$ by

$${}^{\mathrm{sc}}\sigma^{m,r}(A) := [a_L] \in S^{m,r}(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1,r-1}(\mathbb{R}^n;\mathbb{R}^n). \tag{4.177}$$

State and prove the analogue of Proposition 4.24 for scattering ps.d.o.s. (Thus, the principal symbol is more powerful in the scattering world: it not only captures the high frequency, i.e., large ξ, behavior of an operator, but also the large x behavior.)

Exercise 4.20 (Scattering Ps.d.o.s, VI: Ellipticity). Suppose $A = \mathrm{Op}_L(a_L) \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$ is elliptic, that is, there exists $b \in S^{-m,-r}(\mathbb{R}^n;\mathbb{R}^n)$ such that $a_L b - 1 \in S^{-1,-1}(\mathbb{R}^n;\mathbb{R}^n)$.

(1) Prove that there exists $B \in \Psi_{\mathrm{sc}}^{-m,-r}(\mathbb{R}^n)$ such that $BA - I \in \Psi_{\mathrm{sc}}^{-\infty,-\infty}(\mathbb{R}^n;\mathbb{R}^n) = \bigcap_{m,r\in\mathbb{R}} \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$.
(2) Suppose $u \in \mathscr{S}'(\mathbb{R}^n)$, and $Au = f \in \mathscr{S}(\mathbb{R}^n)$. Prove that $u \in \mathscr{S}(\mathbb{R}^n)$. *Remark.* Notice the difference to the statements of Proposition 4.32 or Corollary 4.41. For example, the Laplacian $\Delta \in \Psi^2(\mathbb{R}^n)$ is uniformly elliptic, but $\Delta u = 0$ for $u = 1$, $u = x_1 x_2$, etc. However, Δ is *not* elliptic as an element of $\Psi_{\mathrm{sc}}^{2,0}(\mathbb{R}^n)$. (Check this.) What about $\Delta + 1$?

Exercise 4.21 (Scattering Ps.d.o.s, VII: Boundedness on Sobolev Spaces).

(1) Prove that elements of $\Psi_{\mathrm{sc}}^{0,0}(\mathbb{R}^n)$ are bounded as maps on $L^2(\mathbb{R}^n)$.
(2) Show that $\Lambda_{m,r} := \langle x\rangle^r \langle D\rangle^m \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$ and $\Lambda'_{m,r} := \langle D\rangle^m \langle x\rangle^r \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$.
(3) Let $A \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$. Show that for all $\rho, \sigma \in \mathbb{R}$, A defines a bounded operator

$$A\colon \langle x\rangle^\rho H^\sigma(\mathbb{R}^n) \to \langle x\rangle^{\rho+r} H^{\sigma-m}(\mathbb{R}^n). \tag{4.178}$$

Exercise 4.22 (Scattering Ps.d.o.s, VIII: Elliptic Scattering Ps.d.o.s Are Fredholm). This exercise builds on Exercise 2.12.

(1) Let $A \in \Psi_{\mathrm{sc}}^{m,r}(\mathbb{R}^n)$ be elliptic (Exercise 4.20). Show that for all $\rho, \sigma \in \mathbb{R}$, the operator

$$A \colon \langle x \rangle^{\rho} H^{\sigma}(\mathbb{R}^n) \to \langle x \rangle^{\rho+r} H^{\sigma-m}(\mathbb{R}^n) \tag{4.179}$$

is a Fredholm operator.

(2) Show that the Fredholm index of A in (4.179), defined by $\operatorname{ind} A := \dim \ker A - \dim \operatorname{coker} A$, is independent of ρ, σ.

Chapter 5
Pseudodifferential Operators on Manifolds



We now show how the ps.d.o. algebra on $\mathbb{R}^n$ can be transferred to smooth manifolds by using local coordinate charts. The key input is the invariance of the class of m-th order ps.d.o.s under changes of coordinates on $\mathbb{R}^n$ (Section 5.1). We recall manifolds and differential operators on them in Sections 5.2–5.3. We define ps.d.o.s on manifolds and discuss their algebra and symbolic properties in Sections 5.4–5.8; we explain this passage to manifolds in considerable detail.

In the definition of ps.d.o.s on manifolds, one has two options. The first option is to require all ps.d.o.s to be properly supported (Definition 5.30), in which case any two ps.d.o.s can be composed; the drawback is that simple ps.d.o.s even on $\mathbb{R}^n$, such as $\langle D\rangle^s$ for $s \notin 2\mathbb{N}_0$, would not qualify as ps.d.o.s. The second option is to not impose any restrictions on the supports of Schwartz kernels (Definition 5.21); the drawback is that the composition of two such ps.d.o.s is, in general, not defined (see however Theorem 5.32). Unlike Euclidean space $\mathbb{R}^n$, general manifolds M do not come naturally equipped with a distance function that one could use to measure the off-diagonal decay of Schwartz kernels (cf. Corollary 4.9), and thus there is no natural third option. To maintain a maximum level of generality, we go with the second option here. (On compact manifolds, the two options are of course one and the same.)

As a first application of the ps.d.o. calculus, we can give an elegant treatment of elliptic operators on compact manifolds and their Fredholm properties, first in the smooth setting (Section 5.9) and then, following the introduction of Sobolev spaces on manifolds (Section 5.10), in the Sobolev space setting (Sections 5.11–5.12). As a simple nonlinear application, we discuss a local version of the uniformization theorem (Section 5.13). As a small preparation for the analysis of non-elliptic operators, we revisit the principal symbol of commutators and the relationship with the symplectic

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structure of T^*M (Section 5.14). Finally, we describe how to adapt the theory developed up to this point to the semiclassical setting (Section 5.15).

5.1 Local Coordinate Invariance

We prove an analogue of the final part of Proposition 3.3 for ps.d.o.s.

Definition 5.1 (Ps.d.o.s with Restricted Schwartz Kernels). Let $\Omega \subset \mathbb{R}^n$ be an open set. Then

$$\Psi_{\mathrm{c}}^m(\Omega) := \{A \in \Psi^m(\mathbb{R}^n) \colon \operatorname{supp} K_A \text{ is a compact subset of } \Omega \times \Omega\}, \tag{5.1}$$

where $K_A \in \mathscr{S}'(\mathbb{R}^{2n})$ denotes the Schwartz kernel of A.

Theorem 5.2 (Ps.d.o.s Under Changes of Variables). *Suppose $\Omega, \Omega' \subset \mathbb{R}^n$ are open and $\kappa \colon \Omega \to \Omega'$ is a diffeomorphism. Given $A \in \Psi_{\mathrm{c}}^m(\Omega')$, define $A_\kappa u := \kappa^* A((\kappa^{-1})^*(u|_\Omega))$ for $u \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$. Then $A_\kappa \in \Psi_{\mathrm{c}}^m(\Omega)$, and the map $\Psi_{\mathrm{c}}^m(\Omega') \ni A \mapsto A_\kappa \in \Psi_{\mathrm{c}}^m(\Omega)$ is bijective. Moreover,*

$$\sigma^m(A_\kappa)(x,\xi) = \sigma^m(A)(\kappa(x), (\kappa'(x)^T)^{-1}\xi). \tag{5.2}$$

See Figure 5.1.

Proof of Theorem 5.2. Write $A = \operatorname{Op}_L(a)$, $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$. Choose $\psi \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega')$ such that $\psi(x)\psi(y) = 1$ on $\operatorname{supp} K_A$; thus $K_A(x,y) = \psi(x)K_A(x,y)\psi(y)$, and therefore

$$A = \operatorname{Op}\big(\psi(x)a(x,\xi)\psi(y)\big). \tag{5.3}$$

We localize near the diagonal: for $\epsilon > 0$ (to be determined), let $\chi_\epsilon(x,y) \in \mathcal{C}^\infty(\mathbb{R}^{2n})$ be such that $\chi_\epsilon(x,y) = 1$ for $|x-y| < \epsilon$ and $\chi_\epsilon(x,y) = 0$ for $|x-y| > 2\epsilon$. Then

$$K_{A_\epsilon} := \operatorname{Op}(a_\epsilon), \quad a_\epsilon(x,y,\xi) := \chi_\epsilon(x,y)\psi(x)\psi(y)a(x,\xi), \tag{5.4}$$

is the Schwartz kernel of an operator $A_\epsilon \in \Psi_{\mathrm{c}}^m(\Omega')$. The difference

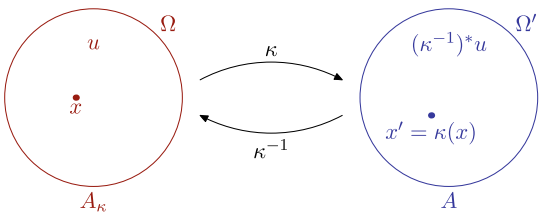


Figure 5.1 Coordinate change for ps.d.o.s.

$$R_\epsilon := A - A_\epsilon \tag{5.5}$$

is a ps.d.o. with Schwartz kernel supported away from the diagonal, thus $R_\epsilon \in \Psi^{-\infty}(\mathbb{R}^n)$, and its Schwartz kernel satisfies $K_{R_\epsilon} \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega' \times \Omega')$. We then have

$$\begin{aligned}(R_\epsilon)_\kappa u(x) &= \int_{\Omega'} K_{R_\epsilon}(\kappa(x), y') u(\kappa^{-1}(y'))\,\mathrm{d}y' \\ &= \int_{\Omega} K_{R_\epsilon}(\kappa(x), \kappa(y))|\det \kappa'(y)|u(y)\,\mathrm{d}y.\end{aligned} \tag{5.6}$$

Therefore, the Schwartz kernel of $(R_\epsilon)_\kappa$ is $K_{(R_\epsilon)_\kappa}(x,y) = K_{R_\epsilon}(\kappa(x), \kappa(y))|\det \kappa'(y)|$ for $x, y \in \Omega$, and 0 otherwise. Thus, $(R_\epsilon)_\kappa \in \Psi_{\mathrm{c}}^{-\infty}(\Omega)$.

It remains to show that $(A_\epsilon)_\kappa \in \Psi_{\mathrm{c}}^m(\Omega)$. Suppose first that $a \in S^{-\infty}(\mathbb{R}^n; \mathbb{R}^n)$. Then

$$\begin{aligned}(A_\epsilon)_\kappa u(x) &= (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\Omega'} e^{i(\kappa(x)-y')\cdot\xi'} a_\epsilon(\kappa(x), y', \xi') u(\kappa^{-1}(y'))\,\mathrm{d}y'\,\mathrm{d}\xi' \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} \int_{\Omega} e^{i(\kappa(x)-\kappa(y))\cdot\xi'} a_\epsilon(\kappa(x), \kappa(y), \xi') \\ &\qquad \times |\det \kappa'(y)| u(y)\,\mathrm{d}y\,\mathrm{d}\xi',\end{aligned} \tag{5.7}$$

and thus the Schwartz kernel of $(A_\epsilon)_\kappa$ is

$$\begin{aligned}K_{(A_\epsilon)_\kappa}(x,y) &= (2\pi)^{-n} \int e^{i(\kappa(x)-\kappa(y))\cdot\xi'} b_\epsilon(x,y,\xi')\,\mathrm{d}\xi', \\ b_\epsilon(x,y,\xi') &:= a_\epsilon(\kappa(x), \kappa(y), \xi')|\det \kappa'(y)| \\ &= \chi_\epsilon(\kappa(x), \kappa(y))\psi(\kappa(x))\psi(\kappa(y))a(\kappa(x), \xi)|\det \kappa'(y)|.\end{aligned} \tag{5.8}$$

To recognize $K_{(A_\epsilon)_\kappa}$ as the Schwartz kernel of a ps.d.o., we use the "Kuranishi trick." We Taylor expand the exponent: denoting by κ_j the j-th component of κ, we have

$$\kappa_j(x) - \kappa_j(y) = \sum_{k=1}^n \Phi_{jk}(x,y)(x_k - y_k), \quad \Phi_{jk}(x,y) = \int_0^1 \partial_{x_k}\kappa_j(y + t(x-y))\,\mathrm{d}t, \tag{5.9}$$

and therefore

$$(\kappa(x) - \kappa(y)) \cdot \xi' = \langle \Phi(x,y)(x-y), \xi' \rangle = \langle x - y, \Phi(x,y)^T \xi' \rangle, \tag{5.10}$$

where $\Phi(x,y) = (\Phi_{jk}(x,y))_{j,k=1,\ldots,n}$, and $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{R}^n$. Note now that

$$\Phi(x,x) = \kappa'(x) \tag{5.11}$$

is invertible for $x \in \Omega$ since κ is a diffeomorphism. For

$$(x,y) \in \operatorname{supp} \chi_\epsilon(\kappa(x),\kappa(y))\psi(\kappa(x))\psi(\kappa(y)), \tag{5.12}$$

we have $(x,y) \in \kappa^{-1}(\operatorname{supp}\psi) \times \kappa^{-1}(\operatorname{supp}\psi)$ (which is a compact subset of $\Omega \times \Omega$) and $|\kappa(x) - \kappa(y)| \le 2\epsilon$. Therefore, we can choose $\epsilon > 0$ such that $\Phi(x,y)$ is invertible for these (x,y). In (5.8), we can then make the change of variables $\xi' = (\Phi(x,y)^T)^{-1}\xi$, so

$$\begin{aligned}(A_\epsilon)_\kappa = \operatorname{Op}(c_\epsilon), \quad c_\epsilon(x,y,\xi) &= b_\epsilon(x,y,(\Phi(x,y)^T)^{-1}\xi)|\det \Phi(x,y)|^{-1} \\ &= a_\epsilon(\kappa(x),\kappa(y),(\Phi(x,y)^T)^{-1}\xi) \\ &\quad \times |\det \Phi(x,y)|^{-1}|\det \kappa'(y)|. \end{aligned} \tag{5.13}$$

We claim that $c_\epsilon \in S^{-\infty}(\mathbb{R}^n \times \mathbb{R}^n; \mathbb{R}^n)$; more generally, if $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, we claim that c_ϵ defined by (5.13) satisfies $c_\epsilon \in S^m(\mathbb{R}^n \times \mathbb{R}^n;\mathbb{R}^n)$ and depends continuously on a. We can drop the (smooth) Jacobian factors. We then compute

$$\begin{aligned}&\partial_x^\alpha \partial_y^\beta \partial_\xi^\gamma \big(a_\epsilon(\kappa(x),\kappa(y),(\Phi(x,y)^T)^{-1}\xi)\big) \\ &\quad = \sum_{\substack{|\alpha'|+|\alpha''|\le|\alpha| \\ |\beta'|+|\beta''|\le|\beta|}} F^{\beta\beta'\beta''}_{\alpha\alpha'\alpha''\gamma}(x,y)\xi^{\alpha''+\beta''} \\ &\qquad\qquad \times (\partial_x^{\alpha'}\partial_y^{\beta'}\partial_\xi^{\gamma+\alpha''+\beta''}a_\epsilon)(\kappa(x),\kappa(y),(\Phi(x,y)^T)^{-1}\xi)\end{aligned} \tag{5.14}$$

for some $F^{\beta\beta'\beta''}_{\alpha\alpha'\alpha''\gamma} \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega\times\Omega)$ depending only on κ. Since $\Phi(x,y)^T$ and its inverse are uniformly bounded on $\operatorname{supp} a_\epsilon$, there exist $c, C > 0$ such that $c|\xi| \le |(\Phi(x,y)^T)^{-1}\xi| \le C|\xi|$ on $\operatorname{supp} a_\epsilon$. Therefore, (5.14) is bounded by $\langle\xi\rangle^{m-|\gamma|}$ times a continuous seminorm of a, as desired. Finally, given $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, we select $a^{(j)} \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$, $j \in \mathbb{N}$, which are uniformly bounded in $S^m(\mathbb{R}^n;\mathbb{R}^n)$ and converge to a in $S^{m+1}(\mathbb{R}^n;\mathbb{R}^n)$ as $j \to \infty$. Then the corresponding symbols $c_\epsilon^{(j)}$ are uniformly bounded in $S^m(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$ and converge to c_ϵ in $S^{m+1}(\mathbb{R}^n\times\mathbb{R}^n;\mathbb{R}^n)$. Setting $A_\epsilon^{(j)} = \operatorname{Op}(a_\epsilon^{(j)})$ with $a_\epsilon^{(j)}$ defined in terms of $a^{(j)}$ as in (5.4), we have $(A_\epsilon)_\kappa u = \lim_{j\to\infty}(A_\epsilon^{(j)})_\kappa u$ for all $u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, and since $(A_\epsilon^{(j)})_\kappa u = \operatorname{Op}(c_\epsilon^{(j)})u \to \operatorname{Op}(c_\epsilon)u$ as $j \to \infty$ by (4.24), we have established (5.13) in general.

As for the principal symbol, we have $\sigma^m(A_\kappa) = \sigma^m((A_\epsilon)_\kappa) = \sigma^m(\operatorname{Op}(c_\epsilon))$, which can be read off from the (first term of the) reduction formula (4.50): using (5.11), it is given by the equivalence class in $S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n)$ of

$$\begin{aligned} c_\epsilon(x,x,\xi) &= b_\epsilon(x,x,(\kappa'(x)^T)^{-1}\xi)|\det\kappa'(x)|^{-1} \\ &= a_\epsilon(\kappa(x),\kappa(x),(\kappa'(x)^T)^{-1}\xi) \\ &= a(\kappa(x),(\kappa'(x)^T)^{-1}\xi). \end{aligned}$$

This finishes the proof. □

5.2 Manifolds, Smooth Functions, Bundles, Densities

We recall:

Definition 5.3 (Smooth Manifolds). Let $n \in \mathbb{N}$. A *smooth manifold of dimension* n is a second countable Hausdorff space M such that:

(1) For each point $p \in M$, there exist an open neighborhood $U_p \ni p$ and a homeomorphism $F_p \colon U_p \to B(0,1) = \{x \in \mathbb{R}^n \colon |x| < 1\} \subset \mathbb{R}^n$.
(2) For all $p, q \in M$ such that $U_p \cap U_q \neq \emptyset$, the transition map

$$F_p \circ F_q^{-1}|_{U_p \cap U_q} \colon F_q(U_p \cap U_q) \to F_p(U_p \cap U_q) \tag{5.15}$$

is smooth (as a map between open subsets of $\mathbb{R}^n$).

Definition 5.4 (Smooth Structure). Let M be a smooth manifold. An *atlas* on M is a collection $\{(U_\alpha, F_\alpha)\}$ of pairs (U_α, F_α), with $U_\alpha \neq \emptyset$ and $M = \bigcup_\alpha U_\alpha$, such that $F_\alpha \colon U_\alpha \to \mathbb{R}^n$ is a diffeomorphism onto an open subset $F_\alpha(U_\alpha)$ of $\mathbb{R}^n$. A *maximal atlas*, or *smooth structure*, is an atlas with the property that any other atlas is contained in it. An element of the maximal atlas is called a *(local coordinate) chart*.

On any given smooth manifold, there always exists a unique maximal atlas.

We shall also encounter *manifolds with boundary* and, more generally, *manifolds with corners*: these are defined in the same fashion, except that coordinate charts map into $\mathbb{R}^n$ or $[0,\infty) \times \mathbb{R}^{n-1}$ for manifolds with boundary, and $[0,\infty)^k \times \mathbb{R}^{n-k}$ (for $k \in \{0,\dots,n\}$ depending on the point $p \in M$) for manifolds with corners. Given a smooth manifold M, a *smooth submanifold with corners* $\Omega \subset M$ is then a subset such that for all $p \in \Omega$ there exists a smooth coordinate chart $F \colon U \to F(U) \subset \mathbb{R}^n$ on M such that

$$\Omega \cap U = F^{-1}([0,\infty)^k \times \mathbb{R}^{n-k}) \tag{5.16}$$

for some k (depending on p); a smooth submanifold with boundary is the special case when only $k = 0, 1$ are allowed.

For $k \in \mathbb{N}_0 \cup \{\infty\}$, we write $\mathcal{C}^k(M)$ for the space of all continuous functions $f \colon M \to \mathbb{C}$ for which $f \circ F_p^{-1} \colon B(0,1) \to \mathbb{C}$ defines an element of $\mathcal{C}^k(B(0,1))$ for all p. We endow $\mathcal{C}^k(M)$ with its standard locally convex topology as follows. Recall first that for open subsets $\Omega \subset \mathbb{R}^n$ such as $\Omega = B(0,1)$, the

space $\mathcal{C}^k(\Omega)$ can be given the structure of a Fréchet space by picking any exhaustion $\Omega = \bigcup_{i=1}^{\infty} K_i$ by compact sets $K_1 \subset K_2 \subset \cdots$, $K_i \subset K_{i+1}^\circ$, and using the seminorms

$$|u|_{i,k} := \max_{|\alpha| \le k} \max_{x \in K_i} |\partial^\alpha u(x)|. \tag{5.17}$$

Also $\mathcal{C}^\infty(\Omega)$ becomes a Fréchet space if we use the seminorms $|\cdot|_{i,k}$, $i \in \mathbb{N}$, $k \in \mathbb{N}_0$. Fixing a countable collection of points $\{p_j\}$ such that $M = \bigcup_j U_{p_j}$, we then use the $\mathcal{C}^k(B(0,1))$-seminorms of the functions $f \circ F_{p_j}^{-1} \in \mathcal{C}^k(B(0,1))$ to define seminorms on $\mathcal{C}^k(M)$.

For $k \in \mathbb{N}_0 \cup \{\infty\}$, we write $\mathcal{C}_{\mathrm{c}}^k(M) \subset \mathcal{C}^k(M)$ for the subspace of compactly supported functions. The locally convex topology on this space is defined as follows. First, for compact $K \subset M$ we define the Fréchet space

$$\dot{\mathcal{C}}^k(K) = \{u \in \mathcal{C}^k(M) \colon \operatorname{supp} u \subset K\}. \tag{5.18}$$

Pick now any exhaustion $M = \bigcup_{i=1}^{\infty} K_i$ of M by compact sets. Then $\mathcal{C}_{\mathrm{c}}^k(M) = \bigcup_i \dot{\mathcal{C}}^k(K_i)$ carries the inductive limit topology; see [RS72, Chapter V.4]. This can be defined in concrete terms as follows. Pick $\phi_i \in \mathcal{C}_{\mathrm{c}}^\infty(K_{i+1}^\circ)$ which equals 1 near K_i; set $\phi_0 := 0$. Then $\psi_i := \phi_i - \phi_{i-1} \in \mathcal{C}_{\mathrm{c}}^\infty(K_{i+1}^\circ \setminus K_{i-1})$, and $\sum_i \psi_i = 1$ on M. The topology on $\mathcal{C}_{\mathrm{c}}^k(M)$ is then induced by the seminorms

$$|u|_{(w_i),(p_i)} := \sup_i w_i p_i(\psi_i u) \tag{5.19}$$

where $(w_i)_{i \in \mathbb{N}}$ is a sequence of positive real numbers, and each p_i is a continuous seminorm on $\dot{\mathcal{C}}^k(K_{i+1} \setminus K_{i-1}^\circ)$. See Exercises 5.1–5.3 for more on the topology of these spaces.

Given an index set I and a cover $M = \bigcup_{i \in I} U_i$ of a smooth manifold M (possibly with boundary or corners) by open sets U_i, we recall that there exists a *partition of unity* $\{\chi_i \colon i \in I\}$ *subordinate to the cover* $\{U_i\}$. This means that $\chi_i \in \mathcal{C}^\infty(M)$ and $\operatorname{supp} \chi_i \subset U_i$ for all $i \in I$, the collection $\{\operatorname{supp} \chi_i\}$ is locally finite (i.e., for every compact subset $K \subset M$, one has $\operatorname{supp} \chi_i \cap K \neq \emptyset$ only for finitely many $i \in I$), and $\sum_{i \in I} \chi_i = 1$ on M. One can arrange for each χ_i to take values in the interval $[0,1]$, and we shall always assume that this is the case when using partitions of unity in the sequel. We also note that one can pick a refinement $\{V_j \colon j \in J\}$ of the cover U_i, i.e., J is another index set and for each $j \in J$ there exists $i \in I$ such that $V_j \subset U_i$, such that each V_j is precompact; a partition of unity $\{\phi_j \colon j \in J\}$ subordinate to $\{V_j\}$ then has the additional property that each $\phi_j \in \mathcal{C}_{\mathrm{c}}^\infty(M)$ has compact support.

The "hands-on" definition of *vector bundles* is the following.

Definition 5.5 (Vector Bundles). Let M be a smooth n-dimensional manifold. A *real vector bundle of rank k over M* is a triple (π, E, M)

(sometimes written $\pi\colon E \to M$, $E \xrightarrow{\pi} M$, or simply $E \to M$) with the following properties:

(1) E is a smooth $(n+k)$-dimensional manifold, and $\pi\colon E \to M$ is smooth.
(2) For all $p \in M$, there exist an open neighborhood $U_p \subset M$, $p \in U_p$, and a diffeomorphism $\tau_p\colon \pi^{-1}(U_p) \to U_p \times \mathbb{R}^k$ such that $\pi(\tau_p^{-1}(q,v)) = q$ is the projection onto the first factor.
(3) For all $p, q \in M$ such that $U_p \cap U_q \neq \emptyset$, the transition map

$$\tau_{pq} := \tau_p \circ \tau_q^{-1}|_{(U_p \cap U_q)\times \mathbb{R}^k}\colon (U_p \cap U_q) \times \mathbb{R}^k \to (U_p \cap U_q) \times \mathbb{R}^k \tag{5.20}$$

takes the form $\tau_{pq}(r,v) = (r, \Phi_{pq}(r)v)$, where $\Phi_{pq}\colon U_p \cap U_q \to GL(k)$ is smooth.

The *fiber of E over $p \in M$* is denoted $E_p := \pi^{-1}(p)$; it carries the structure of a k-dimensional real vector space. The *zero section* o of E is the submanifold of E given locally in $\pi^{-1}(U_p)$ by $\tau_p^{-1}(U_p \times \{0\})$. A *smooth section of E* is a smooth map $s\colon M \to E$ such that $\pi \circ s = \mathrm{Id}_M$. The space of smooth sections is denoted $\mathcal{C}^\infty(M; E)$. *Complex vector bundles* are defined analogously, with $\mathbb{C}^k$ replacing $\mathbb{R}^k$.

We call the maps τ_p *local trivializations*. If U_p is the domain of a coordinate chart $F\colon U_p \to F(U_p) \subset \mathbb{R}^n$, we also call $(F \times \mathrm{Id}) \circ \tau_p\colon \pi^{-1}(U_p) \to F(U_p) \times \mathbb{R}^k$ a local trivialization.

Replacing $\mathbb{R}^k$ with its linear structure by a smooth manifold, one obtains the following notion.

Definition 5.6 (Fiber Bundles). Let M be a smooth n-dimensional manifold; let F be a smooth k-dimensional manifold. A *fiber bundle over M with typical fiber F* is a triple (π, E, M) with the following properties:

(1) E is a smooth $(n+k)$-dimensional manifold, and $\pi\colon E \to M$ is smooth.
(2) For all $p \in M$, there exist an open neighborhood $U_p \subset M$, $p \in U_p$, and a diffeomorphism $\tau_p\colon \pi^{-1}(U_p) \to U_p \times F$ such that $\pi(\tau_p^{-1}(q,f)) = q$.
(3) for all $p, q \in M$ such that $U_p \cap U_q \neq \emptyset$, the transition map

$$\tau_{pq} := \tau_p \circ \tau_q^{-1}|_{(U_p \cap U_q)\times F}\colon (U_p \cap U_q) \times F \to (U_p \cap U_q) \times F \tag{5.21}$$

takes the form $\tau_{pq}(r,f) = (r, \Phi_{pq}(r,f))$ where $\Phi_{pq}(r,\cdot)\colon F \to F$ is a diffeomorphism for all $r \in U_p \cap U_q$.

Returning to vector bundles, we note that in order to specify a real rank k vector bundle uniquely (up to vector bundle isomorphisms), it suffices to have the following data and conditions:

(1) A cover $\{U_\alpha\}$ of M by open non-empty subsets.
(2) For all α, β with $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, a map $\tau_{\alpha\beta}\colon U_{\alpha\beta} \times \mathbb{R}^k \to U_{\alpha\beta} \times \mathbb{R}^k$ of the form $\tau_{\alpha\beta}(p,v) = (p, \Phi_{\alpha\beta}(p)v)$ with $\Phi_{\alpha\beta}\colon U_{\alpha\beta} \to GL(k)$ smooth.
(3) $\tau_{\alpha\alpha}(p,v) = (p,v)$ for all $p \in U_\alpha$, $v \in \mathbb{R}^k$.

(4) The *cocycle condition* holds: for α, β, γ with $U_{\alpha\beta\gamma} := U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$, we have $\tau_{\gamma\beta} \circ \tau_{\beta\alpha} = \tau_{\gamma\alpha}$ on $U_{\alpha\beta\gamma} \times \mathbb{R}^k$.

Indeed, one can then set

$$E := \Big(\bigsqcup_\alpha U_\alpha \times \mathbb{R}^k \Big) / \sim, \tag{5.22}$$

where we define the equivalence relation $\sim$ by

$$U_\alpha \times \mathbb{R}^k \ni (p, v) \sim (q, w) \in U_\beta \times \mathbb{R}^k \iff p = q,\ \tau_{\alpha\beta}(q, w) = (p, v). \tag{5.23}$$

(The cocycle condition guarantees that this is transitive, while reflexivity follows from the cocycle condition together with $\tau_{\alpha\alpha} = \mathrm{Id}$.) We denote the equivalence classes by $[U_\alpha, (p, v)] \in E$. Note that the same point p can lie in two (or more) open sets U_α and U_β, but typically $[U_\alpha, (p, v)] \neq [U_\beta, (p, v)]$ for $v \in \mathbb{R}^k$. The projection map $\pi \colon E \to M$ is simply given by $\pi([U_\alpha, (p, v)]) = p$. As local trivializations, we can take

$$\tau_\alpha \colon \{[U_\alpha, (p, v)] \colon p \in U_\alpha\} \mapsto (p, v) \in U_\alpha \times \mathbb{R}^k. \tag{5.24}$$

Example 5.7 (Tangent Bundle). Taking a cover of an n-dimensional manifold M by coordinate charts $F_i \colon U_i \to \mathbb{R}^n$, consider $\tau_{ij}(p, v) = (p, (F_i \circ F_j^{-1})'|_{F_j(p)} v)$. The resulting vector bundle is the *tangent bundle of* M, denoted TM. Note that a chart $F \colon U \to \mathbb{R}^n$ induces a trivialization of $T_U M := \pi^{-1}(U)$ via $U \times \mathbb{R}^n \ni (p, v) \mapsto [U, (p, v)] \in T_U M$. (For maximum precision, one should record not only U but also F, i.e., write $[U, F, (p, v)] \in T_U M$.) A tangent vector $V = [U, (p, v)] \in T_p M$ has several interpretations.

(1) V is a directional derivative on M at p. That is, it gives a map

$$\mathcal{C}^\infty(M) \ni f \mapsto V(f) := \frac{\mathrm{d}}{\mathrm{d}s} \Big(\big((F^{-1})^* f|_U \big)(F(p) + sv) \Big) \Big|_{s=0}. \tag{5.25}$$

(See also Exercise 5.4.)

(2) V is the tangent vector of a smooth curve on M. This comes from the following construction: consider the set of all smooth curves $\gamma \colon I \to M$ with $\gamma(0) = p$, where $I \subset \mathbb{R}$ is an open interval containing 0. An equivalence relation on this set is defined as follows: $\gamma_1 \sim \gamma_2$ if and only if, in any local coordinate system $F \colon U \to \mathbb{R}^n$ with $p \in U$, we have $\frac{\mathrm{d}}{\mathrm{d}s} F(\gamma_1(s))|_{s=0} = \frac{\mathrm{d}}{\mathrm{d}s} F(\gamma_2(s))|_{s=0}$. (Check that this condition is independent of the choice of F.) Then $T_p M$ is the set of equivalence classes of curves. The tangent vector V is identified with the equivalence class of the curve $s \mapsto F^{-1}(F(p) + sv)$.

Functorial operations on vector spaces give corresponding operations on vector bundles. For instance, given a linear map $A \colon V \to W$ between two

vector spaces, the adjoint is $A^T\colon W^* \to V^*$; if A is invertible, then this gives a map $(A^T)^{-1}\colon V^* \to W^*$. In the notation of Example 5.7, we may thus take

$$\tau_{ij}(p,v) = \big(p, ((\kappa'_{ij}|_{F_j(p)})^T)^{-1}v\big), \quad \kappa_{ij} := F_i \circ F_j^{-1}. \tag{5.26}$$

The resulting vector bundle is the *cotangent bundle*, denoted $\pi\colon T^*M \to M$. Note that formula (5.26) appears in (5.2) (except in the latter we also use/change local coordinates on the base M via κ_{ij}). We recall then that given a smooth function $f \in \mathcal{C}^\infty(M)$, we can define its *exterior derivative* $\mathrm{d}f \in \mathcal{C}^\infty(M; T^*M)$ as follows: if $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ is a coordinate chart, we define

$$\tau_i(\mathrm{d}f(p)) := \big(p, \big(\partial_{x_j}((F_i^{-1})^* f)\big)_{j=1,\dots,n}\big) \in U_i \times \mathbb{R}^n, \quad p \in U_i. \tag{5.27}$$

That this indeed gives a well-defined section of T^*M follows from the change of variables formula.

As in Example 5.7, a local coordinate system $F\colon U \to \mathbb{R}^n$ induces a trivialization $T^*U = \pi^{-1}(U) \cong U \times \mathbb{R}^n$. The standard coordinates on the $\mathbb{R}^n$ factor are called *canonical momentum variables*. The relationship between T^*M and TM is as follows: for any $p \in M$, we have an isomorphism of vector spaces

$$T_p^*M \cong \mathrm{Hom}(T_pM, \mathbb{R}) \tag{5.28}$$

which is defined independently of any choices. This isomorphism can be described in local trivializations of TM and T^*M induced by the same local coordinate system F: if $\omega = [U,(p,\xi)] \in T_p^*M$ and $V = [U,(p,v)] \in T_pM$ where $\xi, v \in \mathbb{R}^n$, we set $\omega(V) = \langle \xi, v\rangle$ (Euclidean inner product). See Exercise 5.5.

A natural choice for f in a chart F_i near p is $f = F_i^* x_k$ (or $f = x_k$ when the chart is clear from the context), in which case (5.27) defines the differential $\mathrm{d}x_k$ with $\tau_i(\mathrm{d}x_k) = (p, (0,\dots,1,\dots,0))$ (with the 1 in the k-th slot). The exterior derivative $\mathrm{d}f$ is then usually written as $\mathrm{d}f = \sum_{j=1}^n (\partial_{x_j} f)\,\mathrm{d}x^j$ (i.e., one drops the coordinate and trivialization maps from the notation).

Example 5.8 (Operations on Vector Bundles). Let $E \to M$ and $F \to M$ denote two vector bundles.

(1) The fiberwise direct sum of vector spaces produces the vector bundle $E \oplus F \to M$, with fibers $(E \oplus F)_p = E_p \oplus F_p$.
(2) Likewise, taking the fiberwise tensor product gives $E \otimes F \to M$, with fibers $(E \otimes F)_p = E_p \otimes F_p$.
(3) The vector bundle $\mathrm{Hom}(E,F) \to M$ has fibers $\mathrm{Hom}(E_p, F_p)$. We have $\mathrm{Hom}(E,F) = E \otimes F^*$.
(4) Let $q \in \mathbb{N}$. The fiberwise q-th exterior power of E produces the vector bundle $\Lambda^q E \to M$. In the special case $E = T^*M$, one often writes $\Lambda^q M := \Lambda^q T^*M$.

Another useful operation on vector bundles is the following.

Definition 5.9 (Pullback of Vector Bundles). Let M, N be smooth manifolds (not necessarily of the same dimension), and let $f\colon M \to N$ be smooth. If $\pi\colon E \to N$ is a vector bundle, then the *pullback of E by f* is the vector bundle

$$\tilde{\pi}\colon f^*E \to M \tag{5.29}$$

given by $f^*E := \{(p, e) \in M \times E\colon f(p) = \pi(e)\}$, with projection map $\tilde{\pi}(p, e) = p$, and with linear structure on $(f^*E)_p = E_{f(p)}$ equal to that on $E_{f(p)}$.

We next introduce an important vector bundle that is closely related to the top exterior power $\Lambda^n T^*M$ of the cotangent bundle of an n-dimensional manifold M:

Definition 5.10 (Density Bundles). Let $\alpha \in \mathbb{R}$. In the notation of Example 5.7, the *α-density bundle on M* is the rank 1 vector bundle

$$\Omega^\alpha M \to M \tag{5.30}$$

with transition functions $\tau_{ij}(p, v) = (p, |\det \kappa'_{ij}|_{F_j(p)}|^{-\alpha} v)$, $\kappa_{ij} = F_i \circ F_j^{-1}$. We also write

$$\Omega M := \Omega^1 M. \tag{5.31}$$

Since the transition functions τ_{ij} act in the second argument (v) via multiplication by a positive function, there is a well-defined notion of *positive α-densities*: these are densities that in the trivialization of $\Omega^\alpha M$ over a coordinate chart on M are given by positive smooth functions. A *volume density* is a smooth positive 1-density.

Remark 5.11 (Functorial Perspective on Density Bundles). $\Omega^\alpha M \to M$ arises functorially from the following operation on vector spaces, applied to TM: given a real n-dimensional vector space V, we define

$$\Omega^\alpha V := \{\omega\colon \Lambda^n V \to \mathbb{R}\colon \omega(\mu v) = |\mu|^\alpha \omega(v),\ v \in \Lambda^n V,\ \mu \in \mathbb{R}\}. \tag{5.32}$$

To see the relationship, note first that $\Lambda^n V$ is one-dimensional. Then, given another n-dimensional vector space W and a linear map $\kappa\colon V \to W$, let us fix bases $e_1, \ldots, e_n$ of V and $f_1, \ldots, f_n$ of W. Consider, as a warm-up, the top exterior powers: $e_1 \wedge \cdots \wedge e_n$ and $f_1 \wedge \cdots \wedge f_n$ are bases of $\Lambda^n V$ and $\Lambda^n W$, and the map $\Lambda^n \kappa\colon \Lambda^n V \to \Lambda^n W$ is given by

$$\Lambda^n \kappa\colon e_1 \wedge \cdots \wedge e_n \mapsto \kappa(e_1) \wedge \cdots \wedge \kappa(e_n) = (\det \kappa) f_1 \wedge \cdots \wedge f_n, \tag{5.33}$$

where $\det \kappa$ is the determinant of the matrix of κ in these bases. Similarly, $\Omega^\alpha V$ and $\Omega^\alpha W$ are one-dimensional, with basis elements $\omega_V : \mu e_1 \wedge \cdots \wedge e_n \mapsto |\mu|^\alpha$ and $\omega_W : \mu f_1 \wedge \cdots \wedge f_n \mapsto |\mu|^\alpha$. For invertible κ, the map

$$\Omega^\alpha \kappa \colon \Omega^\alpha V \to \Omega^\alpha W \tag{5.34}$$

is given by

$$\Omega^\alpha \kappa(\omega)(f_1 \wedge \cdots \wedge f_n) = \omega(\kappa^{-1}(f_1) \wedge \cdots \wedge \kappa^{-1}(f_n)), \tag{5.35}$$

and hence $\Omega^\alpha \kappa(\omega_V) = |\det \kappa|^{-\alpha} \omega_W$.

The proof of the following simple lemma is left to the reader (Exercise 5.9).

Lemma 5.12 (Properties of Density Bundles). *Let $\alpha, \beta \in \mathbb{R}$. Then*

(1) $(\Omega^\alpha M)^* = \Omega^{-\alpha} M$.
(2) $\Omega^\alpha M \otimes \Omega^\beta M = \Omega^{\alpha+\beta} M$.
(3) $\Omega^0 M = M \times \mathbb{R}$.

If $x \in \mathbb{R}^n$ denotes local coordinates on a manifold M, then a typical α-density is

$$|\mathrm{d}x|^\alpha \colon \mu \partial_{x_1} \wedge \cdots \wedge \partial_{x_n} \mapsto |\mu|^\alpha. \tag{5.36}$$

Similarly to differential forms, α-densities can be pulled back by smooth maps (Exercise 5.13):

Lemma 5.13 (Pullback of Densities). *Let $f \colon M \to N$ be a smooth map between smooth manifolds of the same dimension; assume that f has no critical points. In local coordinates x, y on M, N and for an α-density $u(y) = u_0(y)|\mathrm{d}y|^\alpha$, define $(f^* u)(x) = u_0(f(x))|\det f'(x)|^\alpha |\mathrm{d}x|^\alpha$. Then f^* is a well-defined map*

$$f^* \colon \mathcal{C}^\infty(N; \Omega^\alpha N) \to \mathcal{C}^\infty(M; \Omega^\alpha M). \tag{5.37}$$

Remark 5.14 (Invariant Formulation). If $V \in T_p M$, define its pushforward along a smooth map $f \colon M \to N$ to be the tangent vector $f_* V \in T_{f(p)} N$ which, as a directional derivative, satisfies $(f_* V)(g) = V(f^* g)$ for all $g \in \mathcal{C}^\infty(N)$. Then in the notation of Lemma 5.13, the pullback of an α-density u is given by $(f^* u)|_p (V_1 \wedge \cdots \wedge V_n) = u|_{f(p)}(f_* V_1 \wedge \cdots \wedge f_* V_n)$, where $n = \dim M$.

For us, 1-densities are the most useful: sections of ΩM can be invariantly integrated. On $\mathbb{R}^n$, we write for $u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n; \Omega \mathbb{R}^n)$, $u = u_0(x)|\mathrm{d}x|$:

$$\int_{\mathbb{R}^n} u := \int_{\mathbb{R}^n} u_0(x)\, \mathrm{d}x. \tag{5.38}$$

Let $\{\phi_i\}$ be a partition of unity on M subordinate to a cover by coordinate systems $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ with $\overline{U_i}$ compact. Define the map

$$\int_M : \mathcal{C}_c^\infty(M;\Omega M) \to \mathbb{R}, \quad u \mapsto \sum_i \int_{\mathbb{R}^n} (F_i^{-1})^*(\phi_i u). \tag{5.39}$$

(Here, $(F_i^{-1})^*(\phi_i u) \in \mathcal{C}_c^\infty(\mathbb{R}^n;\Omega\mathbb{R}^n)$ is defined by Lemma 5.13.)

Proposition 5.15 (Integration of 1-Densities). *The map (5.39) is independent of the choice of local coordinates and the partition of unity.*

Proof. First, suppose $u \in \mathcal{C}_c^\infty(M;\Omega M)$ is supported in the intersection of two coordinate charts, with local coordinates $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^n$ and transition function $x = \kappa(y)$, then

$$u(x) = u_0(x)|\mathrm{d}x| = u_1(y)|\mathrm{d}y|. \tag{5.40}$$

At $x = \kappa(y)$, we have $(\Omega^1\kappa)|\mathrm{d}y| = |\det \kappa'(y)|^{-1}|\mathrm{d}x|$, so $|\mathrm{d}x| = |\det \kappa'(y)||\mathrm{d}y|$. Therefore, $u_1(y) = u_0(\kappa(y))|\det \kappa'(y)|$, and thus

$$\int_{\mathbb{R}^n} u_1(y)\,\mathrm{d}y = \int_{\mathbb{R}^n} u_0(\kappa(y))|\det \kappa'(y)|\,\mathrm{d}y = \int_{\mathbb{R}^n} u_0(x)\,\mathrm{d}x. \tag{5.41}$$

The proposition follows easily from this: if $\{\psi_j\}$ is another partition of unity subordinate to a cover by coordinate systems $G_j\colon V_j \to G_j(U_j) \subset \mathbb{R}^n$, then $\int_M u = \sum_j \int_M \psi_j u$, and thus

$$\begin{aligned}
\int_M u &= \sum_j \int_M \psi_j u \\
&= \sum_{i,j} \int_{\mathbb{R}^n} (F_i^{-1})^*(\phi_i\psi_j u) \\
&= \sum_{i,j} \int_{\mathbb{R}^n} (G_j \circ F_i^{-1})^*\big((G_j^{-1})^*(\psi_j\phi_i u)\big) \\
&= \sum_{i,j} \int_{\mathbb{R}^n} (G_j^{-1})^*(\psi_j\phi_i u) \\
&= \sum_j \int_{\mathbb{R}^n} (G_j^{-1})^*\psi_j u.
\end{aligned}$$

This completes the proof. □

We can now define:

Definition 5.16 (Distributions on Manifolds). The space $\mathscr{D}'(M)$ of (complex-valued) *distributions on* M consists of all continuous linear maps $\mathcal{C}^\infty_{\mathrm{c}}(M;\Omega M) \to \mathbb{C}$. More generally, if $E \to M$ is a complex (or real) vector bundle, then the space of *E-valued distributions* $\mathscr{D}'(M;E)$ consists of all continuous linear functionals $\mathcal{C}^\infty_{\mathrm{c}}(M;E^* \otimes \Omega M) \to \mathbb{C}$ (or $\mathbb{R}$). The space $\mathscr{E}'(M;E)$ consists of all continuous linear functionals $\mathcal{C}^\infty(M;E^* \otimes \Omega M) \to \mathbb{C}$ (or $\mathbb{R}$).

Thus, $\mathcal{C}^\infty(M;E) \hookrightarrow \mathscr{D}'(M;E)$ via

$$\mathcal{C}^\infty(M;E) \times \mathcal{C}^\infty_{\mathrm{c}}(M;E^* \otimes \Omega M) \ni (u,\phi) \mapsto \int_M \langle u(p), \phi(p)\rangle, \tag{5.42}$$

where $\langle \cdot,\cdot\rangle \colon E^* \times E \to \mathbb{R}$ is the dual pairing; note that $\langle u,\phi\rangle \in \mathcal{C}^\infty(M;\Omega M)$ can indeed be invariantly integrated by Proposition 5.15.

The continuity of a linear map $\mathcal{C}^\infty_{\mathrm{c}}(M;\Omega M) \to \mathbb{C}$ is equivalent to the sequential continuity of all of its restrictions $\dot{\mathcal{C}}^\infty(K;\Omega M) \to \mathbb{C}$ where $K \subset M$ is compact (Exercise 5.2). The support and singular support of a distribution are defined analogously to the local ($\mathbb{R}^n$) case (Definition 2.6). The space $\mathscr{E}'(M;E) \subset \mathscr{D}'(M;E)$ is the space of distributions with compact support. (Without further structure, there is no natural analogue of the space of Schwartz functions or tempered distributions on a general smooth manifold.)

Example 5.17 (δ-Distribution on a Manifold). Let $p \in M$, then $\delta_p \in \mathscr{E}'(M;\Omega M)$ is the distribution defined by mapping $\phi \in \mathcal{C}^\infty(M)$ to $\phi(p)$. Note here that the space of test functions for an element of $\mathscr{E}'(M;\Omega M)$ is equal to the space of smooth sections of the bundle $(\Omega M)^* \otimes \Omega M = M \times \mathbb{R}$ by Lemma 5.12, i.e., the space of smooth functions.

Distributions are characterized by their local behavior: a linear map $u \colon \mathcal{C}^\infty_{\mathrm{c}}(M;\Omega M) \to \mathbb{C}$ is a distribution if and only if for all charts $F \colon U \to F(U) \subset \mathbb{R}^n$ and smooth functions $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(F(U))$, the map

$$\mathscr{S}(\mathbb{R}^n) \ni \phi \mapsto \langle u, F^*(\chi\phi\,|\mathrm{d}x|)\rangle \in \mathbb{C} \tag{5.43}$$

defines a tempered distribution (with compact support) on $\mathbb{R}^n$.

In order to state the Schwartz kernel theorem in this general context, we define the projections

$$\begin{aligned} \pi_L &\colon M^2 \to M, \quad (p,q) \mapsto p, \\ \pi_R &\colon M^2 \to M, \quad (p,q) \mapsto q. \end{aligned} \tag{5.44}$$

Theorem 5.18 (Schwartz Kernel Theorem: Manifold Case). *Let M be a smooth n-dimensional manifold, and let $E, F \to M$ be two vector bundles. Then there is a one-to-one correspondence between continuous linear operators $A \colon \mathcal{C}^\infty_{\mathrm{c}}(M;E) \to \mathscr{D}'(M;F)$ and distributional Schwartz kernels*

$K \in \mathscr{D}'(M \times M; \pi_L^* F \otimes \pi_R^*(E^* \otimes \Omega M))$. This correspondence is given by assigning to K the operator $O_K \colon \mathcal{C}_c^\infty(M;E) \to \mathscr{D}'(M;F)$, defined as

$$(O_K\phi)(\psi) = \langle K, \pi_L^*\psi \otimes \pi_R^*\phi\rangle, \quad \phi \in \mathcal{C}_c^\infty(M;E),\ \psi \in \mathcal{C}_c^\infty(M;F^* \otimes \Omega M). \tag{5.45}$$

The reader is asked to supply a proof of a generalization of Theorem 5.18 in Exercise 5.15.

5.3 Differential Operators on Manifolds

Let M be a smooth n-dimensional manifold. Before we talk about ps.d.o.s on M, let us think about *differential* operators.

Definition 5.19 (Vector Fields on Manifolds). The space of smooth (real) vector fields on M is $\mathcal{V}(M) := \mathcal{C}^\infty(M;TM)$.

In local coordinates $x_1, \dots, x_n$ on M, we can express every $V \in \mathcal{V}(M)$ in the form $V = \sum_{j=1}^n V_j(x)\partial_{x_j}$ for smooth real-valued V_j, $j = 1, \dots, n$. We will encounter also *complex vector fields* that are of this form in local coordinates for complex-valued V_j. (Equivalently, a complex vector field is a section of the complexified tangent bundle $\mathbb{C} \otimes TM$.)

An element $V \in \mathcal{V}(M)$ can be regarded as a differential operator by assigning

$$\mathcal{C}^\infty(M) \ni f \mapsto Vf \in \mathcal{C}^\infty(M), \quad (Vf)(p) = \mathrm{d}f(p)(V(p)). \tag{5.46}$$

Definition 5.20 (Differential Operators on Manifolds).

(1) We define $\mathrm{Diff}^0(M) = \mathcal{C}^\infty(M)$, with $A \in \mathrm{Diff}^0(M) = \mathcal{C}^\infty(M)$ acting by multiplication.
(2) We define $\mathrm{Diff}^1(M)$ as the space of all operators $A \colon \mathcal{C}_c^\infty(M) \to \mathcal{C}_c^\infty(M)$ of the form $Au = Vu + fu$ with $V \in \mathcal{V}(M)$, $f \in \mathcal{C}^\infty(M)$.
(3) Let $m \in \mathbb{N}_0$. Then $\mathrm{Diff}^m(M)$ is the space of all operators $A \colon \mathcal{C}_c^\infty(M) \to \mathcal{C}_c^\infty(M)$ such that for all compact subsets $K \subset M$, there exist $N \in \mathbb{N}$ and $A_{kj} \in \mathrm{Diff}^1(M)$, $k = 1, \dots, N$, $j = 1, \dots, N_k \le m$, with

$$\begin{gathered} Au = \sum_{k=1}^N A_{k1} \cdots A_{kN_k} u \quad \forall\, u \in \dot{\mathcal{C}}^\infty(K), \\ \dot{\mathcal{C}}^\infty(K) := \{u \in \mathcal{C}^\infty(M) \colon \operatorname{supp} u \subset K\}. \end{gathered} \tag{5.47}$$

(Check that this agrees with the standard local coordinate definition.) Of course, differential operators also map $\mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$, $\mathscr{D}'(M) \to \mathscr{D}'(M)$,

$\mathscr{E}'(M) \to \mathscr{E}'(M)$. What are the Schwartz kernels of differential operators? The Schwartz kernel K_I of the identity operator $I \in \mathrm{Diff}^0(M)$ should be

$$K_I(x,y) = \delta(x-y). \tag{5.48}$$

We aim to make sense of this. Using the right projection π_R from (5.44), we define the *right density bundle* by

$$\Omega_R := \pi_R^*(\Omega M). \tag{5.49}$$

Thus, integration in the second variable is a well-defined map $\mathcal{C}_c^\infty(M^2;\Omega_R) \to \mathcal{C}_c^\infty(M)$. More generally, the following map is well-defined:

$$\mathscr{D}'(M^2;\Omega_R) \times \mathcal{C}_c^\infty(M) \ni (K,u) \mapsto \int_M K(\cdot,y)u(y) \in \mathscr{D}'(M). \tag{5.50}$$

By the Schwartz kernel theorem, every continuous map $\mathcal{C}_c^\infty(M) \to \mathscr{D}'(M)$ is represented by some $K \in \mathscr{D}'(M^2;\Omega_R)$. Thus, (5.48) must be regarded as an element

$$K_I \in \mathscr{D}'(M^2;\Omega_R). \tag{5.51}$$

As such, it acts on test functions of class

$$\mathcal{C}_c^\infty(M^2;\Omega(M^2)\otimes(\Omega_R)^*) = \mathcal{C}_c^\infty(M^2;\Omega_L), \quad \Omega_L := \pi_L^*(\Omega M), \tag{5.52}$$

and indeed maps $u \in \mathcal{C}_c^\infty(M^2;\Omega_L)$ to $\int_M u(x,x)$, defined by Proposition 5.15. We use here that $\Omega(M^2) = \Omega_L \otimes \Omega_R$; moreover, we use that $(\Omega_L)_{(x,y)} = \Omega_x M$, and thus restriction to the diagonal defines a map $\mathcal{C}_c^\infty(M^2;\Omega_L) \to \mathcal{C}_c^\infty(M;\Omega M)$.

Given $A \in \mathrm{Diff}^m(M)$, its Schwartz kernel $K_A \in \mathscr{D}'(M^2;\Omega_R)$ is then given by

$$K_A = (\pi_L^* A)K_I, \tag{5.53}$$

where $\pi_L^* A$ denotes the lift of A to the first factor. (The reader should check that if $\pi\colon E \to M$ is a vector bundle and $K \in \mathscr{D}'(M^2;\pi_R^*E)$, then

$$(\pi_L^*A)K \in \mathscr{D}'(M^2;\pi_R^*E), \quad ((\pi_L^*A)K)(\cdot,y) := A(K(\cdot,y)),\ y \in M, \tag{5.54}$$

is well-defined; here $K(\cdot,y)$ takes values in E_y and A acts component-wise upon passing to any x-independent basis of E_y.)

5.4 Definition of $\Psi^m(M)$

We continue to denote by M a smooth n-dimensional manifold and use the notation π_L, π_R from (5.44) and Ω_R from (5.49). The following definition captures what we would like pseudodifferential operators on a manifold (not necessarily compact) to be: their Schwartz kernels should, near the diagonal, be Schwartz kernels of ps.d.o.s on $\mathbb{R}^n$ in local coordinates, while away from the diagonal, they should be smooth (cf. Corollary 4.9).

Definition 5.21 (Pseudodifferential Operators on Manifolds). Let M be a smooth n-dimensional manifold. Let $m \in \mathbb{R} \cup \{-\infty\}$. Then $\Psi^m(M)$ is the space of linear operators

$$A \colon \mathcal{C}_{\mathrm{c}}^\infty(M) \to \mathcal{C}^\infty(M) \tag{5.55}$$

with the following properties:

(1) If $F \colon U \subset M \to F(U) \subset \mathbb{R}^n$ is a coordinate chart and $\psi \in \mathcal{C}_{\mathrm{c}}^\infty(U)$, then there exists $B \in \Psi_{\mathrm{c}}^m(F(U)) \subset \Psi^m(\mathbb{R}^n)$ (see Definition 5.1) such that on U, we have

$$\psi A(\psi u) = F^*\Big(B\big((F^{-1})^* u\big)\Big), \quad u \in \mathcal{C}_{\mathrm{c}}^\infty(M). \tag{5.56a}$$

(2) If $\phi, \psi \in \mathcal{C}^\infty(M)$, $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$, then there exists $K \in \mathcal{C}^\infty(M^2; \Omega_R)$ such that

$$\phi A(\psi u) = \int_M K(\cdot, y) u(y), \quad u \in \mathcal{C}_{\mathrm{c}}^\infty(M). \tag{5.56b}$$

See Figure 5.2.

Remark 5.22 (Ps.d.o.s on the Manifold $\mathbb{R}^n$). Taking as the smooth manifold $M = \mathbb{R}^n$, the space $\Psi^m(M)$ defined here is much larger than the space $\Psi^m(\mathbb{R}^n)$ of uniform pseudodifferential operators defined in Chapter 4. (One reason is that we do not constrain the size of Schwartz kernels away

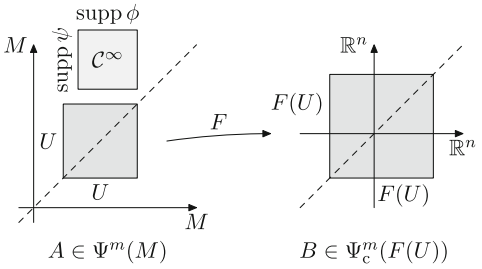


Figure 5.2 Schwartz kernels of ps.d.o.s on manifolds.

from the diagonal $\operatorname{diag}_M = \{(p,p)\colon p \in M\}$.) To avoid confusion, one should denote the latter space by $\Psi^m_\infty(\mathbb{R}^n)$. When working on $\mathbb{R}^n$, we shall, in this book, only ever employ operators in the uniform algebra; therefore, we shall omit the subscript "∞."

Remark 5.23 (Residual Operators). Directly from the definition, the space $\Psi^{-\infty}(M)$ consists of all operators that have a Schwartz kernel in $\mathcal{C}^\infty(M^2;\Omega_R)$. Equivalently, $\Psi^{-\infty}(M)$ is the space of all bounded linear operators $\mathscr{E}'(M) \to \mathcal{C}^\infty(M)$.

Remark 5.24 (Topology). Already at this point we can put a locally convex topology on the space $\Psi^m(M)$ by taking the seminorms of all B and K arising in (5.56a) and (5.56b). The space $\Psi^m(M)$ is then complete (Exercise 5.17). Theorem 5.29 implies that it suffices to take only the seminorms of operators B in (5.56a) arising from a locally finite open cover (which is thus in particular countable) and appropriate cutoffs ψ; thus $\Psi^m(M)$ is in fact a Fréchet space.

Example 5.25 (Inverse of the Laplacian). Let $M = \mathbb{R}^n$, $n \neq 2$. Then the operator A with Schwartz kernel $K(x,y) = |x-y|^{-n+2}\,|\mathrm{d}y|$ defines an element $A \in \Psi^{-2}(M)$. (This is not a uniform ps.d.o. on $\mathbb{R}^n$.) We have $\Delta \circ A = c_n^{-1} I$ in the notation of Example 4.31. Related examples are discussed after (5.127) and in Proposition 5.53.

Example 5.26 (An Operator on the Circle). Consider the one-dimensional manifold $\mathbb{S}^1 = \mathbb{R}_\theta/2\pi\mathbb{Z}$. Define $f\colon \mathbb{S}^1 \to \mathbb{R}$ as the 2π-periodic continuation of

$$(-\pi,\pi] \ni \theta \mapsto -\frac{1}{2}|\sinh\theta| + \frac{\frac{1}{2}\sinh(2\pi)}{\cosh(2\pi)-1}\cosh\theta. \tag{5.57}$$

Then the operator A with Schwartz kernel $K(\theta,\theta') = f(\theta-\theta')\,|\mathrm{d}\theta'|$ defines an element $A \in \Psi^{-2}(\mathbb{S}^1)$. We have $(D_\theta^2+1)\circ A = I$.

Further examples of ps.d.o.s on manifolds that are not *differential* operators are discussed in Exercises 5.19, 5.22, 5.23, 5.24, and 5.25 (which significantly generalizes Example 5.26).

Ps.d.o.s act on distributions with compact support. We give a direct proof here and defer a "better" proof based on duality (in the spirit of (4.32)) to Corollary 5.45.

Proposition 5.27 (Boundedness on Distributions). *Let $A \in \Psi^m(M)$. Then A extends by continuity from $\mathcal{C}^\infty_{\mathrm{c}}(M)$ to a bounded linear operator*

$$A\colon \mathscr{E}'(M) \to \mathscr{D}'(M). \tag{5.58}$$

Proof. Fix a cover of M by coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ with $\overline{U_i}$ compact, and let $\{\phi_i\}$, $\phi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$, be a subordinate partition of unity.

Fix $\tilde{\phi}_i \in \mathcal{C}_c^\infty(U_i)$ with $\tilde{\phi}_i = 1$ near $\operatorname{supp}\phi_i$. By (5.56a), we can write

$$\tilde{\phi}_i A \phi_i = \tilde{\phi}_i A \tilde{\phi}_i \phi_i = F_i^* B_i (F_i^{-1})^* \phi_i, \quad B_i \in \Psi_c^m(F_i(U_i)). \tag{5.59}$$

Let now $u \in \mathscr{E}'(M)$, then $\phi_i u \neq 0$ only for finitely many i. We then set

$$\tilde{A}u := \sum_i F_i^* B_i (F_i^{-1})^* (\phi_i u) + \sum_i (1 - \tilde{\phi}_i) A \phi_i u. \tag{5.60}$$

Each one of the finitely many nonzero summands in the first sum is a pullback from $\mathbb{R}^n$ of a tempered distribution with compact support and hence lies in $\mathscr{E}'(M)$. The second (also finite) sum lies in $\mathcal{C}^\infty(M)$ by (5.56b).

For $u \in \mathcal{C}_c^\infty(M)$, we clearly have $\tilde{A}u = Au$. Since $\mathcal{C}_c^\infty(M) \subset \mathscr{E}'(M)$ is dense, (5.60) defines the unique continuous extension of A to $\mathscr{E}'(M)$. □

To get a more manageable characterization of $\Psi^m(M)$, we first prove:

Lemma 5.28 (Ps.d.o.s Defined in a Chart). *Let M be an n-dimensional manifold, and let $F\colon U \to F(U) \subset \mathbb{R}^n$ be a coordinate chart. If $B \in \Psi_c^m(F(U)) \subset \Psi^m(\mathbb{R}^n)$, then the operator $A\colon \mathcal{C}_c^\infty(M) \to \mathcal{C}^\infty(M)$ defined by*

$$Au := F^* B (F^{-1})^* (u|_U), \quad u \in \mathcal{C}_c^\infty(M), \tag{5.61}$$

on U, and $Au := 0$ in $M \setminus U$, defines an element $A \in \Psi^m(M)$.

Proof. We first check (5.56a). Suppose $G\colon V \to G(V) \subset \mathbb{R}^n$ is another coordinate patch, and let $\chi \in \mathcal{C}_c^\infty(M)$, $\operatorname{supp}\chi \subset V$. Then

$$B_1 := ((F^{-1})^* \chi) B ((F^{-1})^* \chi) \in \Psi_c^m(F(U \cap V)). \tag{5.62}$$

Denote the change of coordinates by $\kappa = F \circ G^{-1}\colon G(U \cap V) \to F(U \cap V)$; then

$$B_2 := (B_1)_\kappa = \kappa^* B_1 (\kappa^{-1})^* \in \Psi_c^m(G(U \cap V)) \tag{5.63}$$

by Theorem 5.2. Therefore,

$$\begin{aligned} \chi A(\chi u) &= F^* B_1 (F^{-1})^* u|_{U\cap V} = G^* \kappa^* B_1 (\kappa^{-1})^* (G^{-1})^* u|_{U\cap V} \\ &= G^* B_2 (G^{-1})^* u|_{U\cap V}, \end{aligned} \tag{5.64}$$

as desired.

As for (5.56b): given $\phi, \psi \in \mathcal{C}^\infty(M)$ with $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$, we have for $u \in \mathcal{C}_c^\infty(M)$

$$\begin{aligned} \phi A(\psi u) &= F^* B' (F^{-1})^* (u|_U), \\ B' &:= ((F^{-1})^* \phi) B ((F^{-1})^* \psi) \in \Psi_c^{-\infty}(F(U)) \subset \Psi^{-\infty}(\mathbb{R}^n), \end{aligned} \tag{5.65}$$

where we used that $\operatorname{supp}((F^{-1})^*\phi)\cap\operatorname{supp}((F^{-1})^*\psi)=\emptyset$. Since the Schwartz kernel of B' is smooth, we obtain (5.56b). □

This already implies that there are lots of pseudodifferential operators on M, given by locally finite sums of operators of the form (5.61). This gives almost (namely, up to operators with smooth integral kernels) all of $\Psi^m(M)$:

Theorem 5.29 (Characterization of Ps.d.o.s via Charts). *Let M be an n-dimensional manifold, and let $M=\bigcup_i U_i$ be a locally finite open cover by coordinate charts $F_i\colon U_i\to F_i(U_i)\subset\mathbb{R}^n$ with $\overline{U_i}$ compact. Let $A\colon \mathcal{C}_c^\infty(M)\to\mathscr{D}'(M)$ be a linear operator. Then $A\in\Psi^m(M)$ if and only if there exist operators $B_i\in\Psi_c^m(F_i(U_i))$ and a section $K\in\mathcal{C}^\infty(M^2;\Omega_R)$ such that*

$$A=K+\sum_i F_i^* B_i (F_i^{-1})^*. \tag{5.66}$$

(We identify operators with their Schwartz kernels here.)

Proof. If A is of the form (5.66), then $A\in\Psi^m(M)$ by the previous lemma. Conversely, suppose $A\in\Psi^m(M)$. Let $\{\phi_i\}$, $\phi_i\in\mathcal{C}_c^\infty(U_i)$, be a partition of unity subordinate to $\{U_i\}$, i.e., $\operatorname{supp}\phi_i\subset U_i$, and $\sum_i\phi_i(x)=1$ for all $x\in M$. Choose $\tilde\phi_i\in\mathcal{C}_c^\infty(U_i)$ with $\operatorname{supp}\tilde\phi_i\subset U_i$ and $\tilde\phi_i\equiv 1$ near $\operatorname{supp}\phi_i$. For $u\in\mathcal{C}_c^\infty(M)$, we then have

$$Au=\sum_i\tilde\phi_i A\phi_i u+\sum_i(1-\tilde\phi_i)A\phi_i u. \tag{5.67}$$

By definition, $(1-\tilde\phi_i)A\phi_i$ has a smooth Schwartz kernel $K_i\in\mathcal{C}^\infty(M^2;\Omega_R)$; since $\operatorname{supp}K_i$ is locally finite, we can define

$$K:=\sum_i K_i\in\mathcal{C}^\infty(M^2;\Omega_R). \tag{5.68}$$

Considering a term $\tilde\phi_i A\phi_i$ in the first sum in (5.67), note that

$$\tilde\phi_i A(\phi_i u)=\tilde\phi_i A\tilde\phi_i(\phi_i u). \tag{5.69}$$

But $\tilde\phi_i A\tilde\phi_i=F_i^*B_i'(F_i^{-1})^*$ for some $B_i'\in\Psi_c^m(F_i(U_i))$, and therefore

$$\tilde\phi_i A\phi_i=F_i^*B_i(F_i^{-1})^*,\quad B_i v:=B_i'\big(((F_i^{-1})^*\phi_i)v\big),\ v\in\mathcal{C}_c^\infty(F_i(U_i)), \tag{5.70}$$

with $B_i\in\Psi_c^m(F_i(U_i))$, as desired. □

When M is not compact, one can in general not compose two ps.d.o.s, even when both are of order $-\infty$, since they only act on $\mathcal{C}_c^\infty(M)$, but not on $\mathcal{C}^\infty(M)$ in general, the problem being that the Schwartz kernel may grow away from the diagonal. The simplest cure is to place an additional assumption on the Schwartz kernels:

Definition 5.30 (Properly Supported Ps.d.o.s). We say that $A \in \Psi^m(M)$, with Schwartz kernel $K \in \mathscr{D}'(M^2; \Omega_R)$, is *properly supported* if the projection maps $\pi_L \colon \operatorname{supp} K \to M$ and $\pi_R \colon \operatorname{supp} K \to M$ are *proper*, i.e., preimages of compact sets are compact.

Every ps.d.o. on a *compact* manifold is properly supported. On a general manifold, every ps.d.o. is the sum of a properly supported operator and a residual operator (Exercise 5.16). In situations where one does not care about order $-\infty$ errors, one can thus work entirely with properly supported operators.

Thus, properly supported operators are bounded on $\mathcal{C}^\infty_{\mathrm{c}}(M)$ and $\mathscr{E}'(M)$. Using partition of unity arguments, one can show that they are also bounded on $\mathcal{C}^\infty(M)$, $\mathscr{D}'(M)$. For a proof using a duality argument, see Corollary 5.45. Properly supported residual operators map $\mathscr{D}'(M)$ to $\mathcal{C}^\infty(M)$ and $\mathscr{E}'(M)$ to $\mathcal{C}^\infty_{\mathrm{c}}(M)$.

Remark 5.31 (Properly Supported and Uniform Ps.d.o.s on $\mathbb{R}^n$). Complementing Remark 5.22, the subspace of $\Psi^m(M)$, $M = \mathbb{R}^n$, consisting of *properly supported* operators does not have a simple relationship with $\Psi^m_\infty(\mathbb{R}^n)$: on the one hand, Schwartz kernels of elements of $\Psi^m_\infty(\mathbb{R}^n)$ may have full support in $\mathbb{R}^n \times \mathbb{R}^n$, and hence are not properly supported; on the other hand, properly supported elements of $\Psi^m(M)$ may not be elements of $\Psi^m_\infty(\mathbb{R}^n)$ since membership in the latter space requires *uniform* bounds off the diagonal (cf. Corollary 4.9).

Theorem 5.32 (Composition of Ps.d.o.s on Manifolds). *Let $A \in \Psi^m(M)$ and $B \in \Psi^{m'}(M)$, and assume that at least one of the two operators A, B is properly supported. Then $A \circ B \colon \mathcal{C}^\infty_{\mathrm{c}}(M) \to \mathcal{C}^\infty(M)$ is a ps.d.o.: $A \circ B \in \Psi^{m+m'}(M)$. If both A and B are properly supported, then so is $A \circ B$.*

For the proof, we will use the description of Theorem 5.29 for a particular kind of open cover:

Lemma 5.33 (Special Covers). *Let M be a smooth manifold. There exists a locally finite open cover $\{U_i\}$ of M such that whenever $U_i \cap U_j \neq \emptyset$, then there exists a local coordinate chart $F \colon U \to F(U) \subset \mathbb{R}^n$ with $U \supset U_i \cup U_j$.*

Proof. M is metrizable; this follows either by Urysohn's metrization theorem or from basic Riemannian geometry. Denote a fixed metric on M by d, and denote metric balls by $B(p, r) = \{q \in M \colon d(p, q) < r\}$. For each $p \in M$, let

$$r_0(p) := \sup\{r \in [0, 1] \colon B(p, r) \text{ is contained in a coordinate chart}\}. \tag{5.71}$$

Since M is a manifold, we have $r_0(p) > 0$ for all $p \in M$. For $p \in M$, define the open set

$$V_p := B\Big(p, \frac{r_0(p)}{10}\Big). \tag{5.72}$$

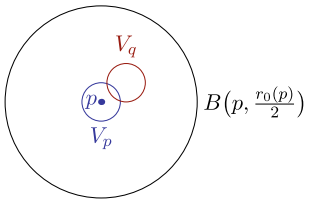


Figure 5.3 Charts $B(p, \frac{r_0(p)}{2})$ and the smaller open sets V_p, V_q with $V_p \cap V_q \neq \emptyset$.

Suppose $V_p \cap V_q \neq 0$; then $d(p,q) \leq \frac{1}{10}(r_0(p) + r_0(q)) \leq \frac{1}{5}\max(r_0(p), r_0(q))$. By symmetry, we may assume $r_0(p) \geq r_0(q)$. If $z \in V_p \cup V_q$, then

$$d(p,z) < \max\Big(\frac{r_0(p)}{10}, d(p,q) + \frac{r_0(q)}{10}\Big) \leq \max\Big(\frac{r_0(p)}{10}, \frac{r_0(p)}{5} + \frac{r_0(p)}{10}\Big) < \frac{r_0(p)}{2}. \tag{5.73}$$

Therefore, $V_p \cup V_q \subset B(p, \frac{r_0(p)}{2})$ is contained in a coordinate chart. Any locally finite refinement $\{U_i\}$ of the cover $\{V_p \colon p \in M\}$ of M satisfies the desired properties. See Figure 5.3. □

***Proof of Theorem* 5.32**. By the previous lemma, we can fix an open cover $M = \bigcup_i U_i$ of M by coordinate charts $F_i \colon U_i \to F_i(U_i) \subset \mathbb{R}^n$, with $\overline{U_i}$ compact, and such that for any i, j with $U_i \cap U_j \neq \emptyset$, the union $U_i \cup U_j$ is contained in a coordinate chart $F_{ij} \colon U_{ij} \to F_{ij}(U_{ij}) \subset \mathbb{R}^n$.

Let us assume that A is properly supported. (The case that B is properly supported is handled similarly.) Write

$$\begin{aligned} A &= K + \sum_i F_i^* A_i (F_i^{-1})^*, \quad A_i \in \Psi_{\mathrm{c}}^m(F_i(U_i)),\ K \in \mathcal{C}^\infty(M^2; \Omega_R), \\ B &= K' + \sum_i F_i^* B_i (F_i^{-1})^*, \quad B_i \in \Psi_{\mathrm{c}}^m(F_i(U_i)),\ K' \in \mathcal{C}^\infty(M^2; \Omega_R). \end{aligned} \tag{5.74}$$

Since A and all the $F_i^* A_i (F_i^{-1})^*$ are properly supported, so is K.

We consider the composition $A \circ B$ term by term and keep track of the supports of the Schwartz kernels of the various pieces.

We first prove that $K \circ K' \in \Psi^{-\infty}(M)$. We have $(K \circ K')(x,y) = \int K(x,z)K'(z,y)$. (More precisely, denote by $\pi_F, \pi_S \colon M^3 \to M^2$ the projections $\pi_F(x,y,z) = (x,z)$, $\pi_S(x,y,z) = (z,y)$, and consider $(\pi_F^* K) \otimes (\pi_S^* K')$: this is a section of $\pi_2^*(\Omega M) \otimes \pi_3^*(\Omega M) \to M^3$ where $\pi_2, \pi_3 \colon M^3 \to M$ are the projections $\pi_2(x,y,z) = y$, $\pi_3(x,y,z) = z$. The expression $\int K(x,z)K'(z,y)$ is then defined as the integration of $(\pi_F^* K) \otimes (\pi_S^* K')$ along the middle factor of $M \times M \times M$; it is thus a section of $\pi_R^*(\Omega M) = \Omega_R \to M \times M$.) For every compact set $K_1 \subset M$, there exists $K_2 \subset M$ such that in fact

$$(K \circ K')(x,y) = \int_{K_2} K(x,z)K'(z,y), \quad x \in K_1. \tag{5.75}$$

(This holds for $K_2 = \pi_R(\operatorname{supp} K \cap \pi_L^{-1}(K_1))$.) Thus, the Schwartz kernel of $K \circ K'$ lies in $\mathcal{C}^\infty(M^2;\Omega_R)$.

Consider next the composition $K_i := K \circ F_i^* B_i (F_i^{-1})^*$. This maps $u \in \mathscr{E}'(M)$ into $\mathcal{C}^\infty(M)$ (in fact, into $\mathcal{C}_c^\infty(M)$); and if $\operatorname{supp} u \cap U_i = \emptyset$, then $K_i u = 0$. Thus, by Remark 5.23,

$$K_i \in \mathcal{C}^\infty(M^2;\Omega_R), \quad \operatorname{supp} K_i \subset M \times U_i. \tag{5.76}$$

(In fact, $\operatorname{supp} K_i \subset \pi_L(\operatorname{supp} K \cap \pi_R^{-1}(U_i)) \times U_i$ is compact, but we do not need this information.) Similarly, one shows that

$$K_i' := F_i^* A_i (F_i^{-1})^* \circ K' \in \mathcal{C}^\infty(M^2;\Omega_R), \quad \operatorname{supp} K_i' \subset U_i \times M. \tag{5.77}$$

(Note that its Schwartz kernel is not necessarily compactly supported since K' is not properly supported.) Finally, we need to consider the composition

$$C_{ij} := F_i^* A_i (F_i^{-1})^* \circ F_j^* B_j (F_j^{-1})^* \colon \mathcal{C}_c^\infty(M) \to \mathcal{C}^\infty(M). \tag{5.78}$$

When $U_i \cap U_j = \emptyset$, this composition is 0. When $U_i \cap U_j \neq \emptyset$, we can use Lemma 5.28 and write (5.78) equivalently as

$$F_{ij}^* A_{ij} (F_{ij}^{-1})^* \circ F_{ij}^* B_{ij} (F_{ij}^{-1})^* = F_{ij}^* (A_{ij} \circ B_{ij}) (F_{ij}^{-1})^*, \tag{5.79}$$

for appropriate $A_{ij} \in \Psi_c^m(F_{ij}(U_i))$, $B_{ij} \in \Psi_c^{m'}(F_{ij}(U_j))$. But then $A_{ij} \circ B_{ij} \in \Psi_c^{m+m'}(F_{ij}(U_i \cup U_j))$, and hence (5.79) lies in $\Psi^{m+m'}(M)$, with Schwartz kernel supported in $(U_i \cup U_j) \times (U_i \cup U_j)$.

The proof is complete once we show that the supports of the Schwartz kernels of K_i, K_i', and C_{ij} are locally finite. Take a point $(p,q) \in M^2$, and choose i_0, j_0 such that $p \in U_{i_0}$ and $q \in U_{j_0}$. Then $U := U_{i_0} \times U_{j_0}$ has nonempty intersection with only finitely many of these supports, as follows immediately from the local finiteness of $\{U_i\}$. □

Corollary 5.34 (Composition on Compact Manifolds). *If M is a compact manifold, then $\Psi^m(M) \circ \Psi^{m'}(M) \subset \Psi^{m+m'}(M)$.*

5.5 Principal Symbol

Motivated by Theorem 5.2, in particular formula (5.2), we want to define the principal symbol of $A \in \Psi^m(M)$ as an equivalence class of symbols on T^*M.

Definition 5.35 (Symbol Spaces on Vector Bundles). Let M be a manifold and $\pi\colon E \to M$ a rank k vector bundle. For $m \in \mathbb{R}$, we define $S^m(E) \subset \mathcal{C}^\infty(E)$ as the subspace of all $a \in \mathcal{C}^\infty(E)$ having the following property: for each coordinate chart $F\colon U \to F(U) \subset \mathbb{R}^n$ on M on which E is trivialized by $\tau\colon \pi^{-1}(U) \to U \times \mathbb{R}^k \xrightarrow{F\times \mathrm{Id}} F(U) \times \mathbb{R}^k$, set

$$b(x,v) := (\tau^{-1})^*(a|_{\pi^{-1}(U)})(x,v) = a(\tau^{-1}(x,v)) \in \mathcal{C}^\infty(F(U)\times\mathbb{R}^k). \tag{5.80}$$

Then for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(F(U)) \subset \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, we have $\phi(x)b(x,v) \in S^m(\mathbb{R}^n;\mathbb{R}^k)$.

We give $S^m(E)$ the structure of a Fréchet space by using the symbol seminorms of the symbols $\phi(x)b(x,v)$ where ϕ ranges over a partition of unity subordinate to a countable cover of M by coordinate charts on which E is trivialized.

The key to making this a useful definition is the following analogue of Lemma 5.28.

Lemma 5.36 (Symbols Defined in a Chart). *In the notation of Definition 5.35, let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(F(U))$, $b \in S^m(\mathbb{R}^n;\mathbb{R}^k)$. Then $a := \tau^*(\phi b) \in S^m(E)$.*

Proof. The expression for a in another coordinate system $F'\colon U' \to F'(U') \subset \mathbb{R}^n$ on M and a trivialization of E on U' is

$$b'(x,v) = \phi(\kappa(x))b(\kappa(x),\Phi(x)v), \quad x \in F'(U\cap U'), \tag{5.81}$$

for $\kappa := F\circ(F')^{-1}\colon F'(U\cap U') \to F(U\cap U')$ and a smooth map $\Phi\colon F'(U\cap U') \to GL(k,\mathbb{R})$. Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(F'(U'))$. Then $\chi(x) := \psi(x)\phi(\kappa(x)) \in \mathcal{C}^\infty_{\mathrm{c}}(F'(U\cap U'))$, and we need to check that

$$\chi(x)b(\kappa(x),\Phi(x)v) \in S^m(\mathbb{R}^n;\mathbb{R}^k). \tag{5.82}$$

This follows from the same type of calculation as (5.14). □

In analogy with Theorem 5.29, we have:

Corollary 5.37 (Characterization of Symbols via Charts). *Let $M = \bigcup_i U_i$ be a locally finite open cover by coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$, with $\overline{U_i}$ compact, over each of which E has a trivialization $\tau_i\colon \pi^{-1}(U_i) \to F_i(U_i)\times\mathbb{R}^k$. Let $a \in \mathcal{C}^\infty(E)$. Then $a \in S^m(E)$ if and only if there exist symbols $b_i \in S^m(\mathbb{R}^n;\mathbb{R}^k)$ and $\chi_i \in \mathcal{C}^\infty_{\mathrm{c}}(F_i(U_i))$ such that*

$$a = \sum_i \tau_i^*(\chi_i b_i). \tag{5.83}$$

Now, given an operator $A \in \Psi^m(M)$, we expect its principal symbol to be an element of the quotient space $S^m(T^*M)/S^{m-1}(T^*M)$. We first define it locally. Let $F\colon U \to F(U) \subset \mathbb{R}^n$ be a coordinate chart with $\bar{U}$ compact,

and let $V \subset U$ be open with $\bar{V} \subset U$. Denote by $\tau\colon T^*_U M \to F(U) \times \mathbb{R}^n$ the trivialization induced by F. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ be such that $\chi = 1$ on $\bar{V}$. Then we put

$$a_V := \tau^*\big(\sigma_L\big((F^{-1})^*\chi A \chi F^*\big)\big)|_{T^*_V M} \in S^m(T^*_V M). \tag{5.84}$$

By the local ($\mathbb{R}^n$) theory, and in particular by Theorem 5.2, the equivalence class

$$[a_V] \in S^m(T^*_V M)/S^{m-1}(T^*_V M) \tag{5.85}$$

is independent of the choice of χ and of the coordinate system F covering a neighborhood of $\bar{V}$. Moreover, if $V' \subset V$, then restriction to V' gives $[a_V]|_{V'} = [a_{V'}]$.

Definition 5.38 (Principal Symbol of Ps.d.o.s on Manifolds). The *principal symbol* of $A \in \Psi^m(M)$ is the unique element

$$\sigma^m(A) \in S^m(T^*M)/S^{m-1}(T^*M) \tag{5.86}$$

with the following property: if $a \in S^m(T^*M)$ is any representative of $\sigma^m(A)$ and V is as above, then $[a|_{T^*_V M}] = [a_V] \in S^m(T^*_V M)/S^{m-1}(T^*_V M)$.

This definition entails the claim that such an element $\sigma^m(A)$ exists and is unique. We start by proving existence. This follows easily from the properties of the $[a_V]$. Indeed, taking a locally finite subcover $\{V_i\}$ of the cover of M by all sets V as above, and a subordinate partition of unity $\{\phi_i\}$, we have $a = \sum_i \phi_i a_{V_i} \in S^m(T^*M)$ by Corollary 5.37; we then put

$$\sigma^m(A) := [a]. \tag{5.87}$$

We check that this satisfies the property required in Definition 5.38. Given V open as above, it suffices to show that for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(V)$, we have $[\phi a|_V] = [\phi a_V]$. Now $\phi\phi_i a_{V_i} = \phi\phi_i a_V + e_i$ for some $e_i \in S^{m-1}(T^*M)$ with support in $T^*_{V_i \cap V} M$. Let $\tilde{\phi}_i \in \mathcal{C}^\infty_{\mathrm{c}}(V_i)$ be equal to 1 on $\operatorname{supp}\phi_i$, and with $\operatorname{supp}\tilde{\phi}_i$ locally finite; then on V we have

$$\phi a = \sum_i \tilde{\phi}_i(\phi_i \phi a_{V_i}) = \sum_i \tilde{\phi}_i(\phi_i \phi a_V + e_i) = \phi a_V + \sum_i \tilde{\phi}_i e_i \tag{5.88}$$

as desired (since $\sum_i \tilde{\phi}_i e_i \in S^{m-1}(T^*M)$).

We now turn to uniqueness; it suffices to show that if $a \in S^m(T^*M)$ is such that $a|_{T^*_V M} \in S^{m-1}(T^*_V M)$ for open sets $V \subset M$ as above, then $a \in S^{m-1}(T^*M)$. But this follows by writing $a = \sum_i \phi_i a|_{T^*_{V_i} M}$, where ϕ_i, V_i are as above.

Proposition 5.39 (Principal Symbol Short Exact Sequence). *The principal symbol map gives a short exact sequence*

$$0 \to \Psi^{m-1}(M) \to \Psi^m(M) \xrightarrow{\sigma^m} S^m(T^*M)/S^{m-1}(T^*M) \to 0. \tag{5.89}$$

Proof. We only prove the surjectivity of σ^m. Take a partition of unity $\{\phi_i\}$ subordinate to a locally finite cover of M by coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ with $\overline{U_i}$ compact. Writing any $a \in S^m(T^*M)$ as $a = \sum_i \phi_i a$, it suffices to show that there exists an operator $A_i \in \Psi^m(M)$ with Schwartz kernel supported in $U_i \times U_i$ such that $\sigma^m(A_i) = [\phi_i a]$, as we can then take $A = \sum_i A_i$ (which is a locally finite sum). This is easy: if $\tilde{\phi}_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$, $\tilde{\phi}_i = 1$ on $\operatorname{supp}\phi_i$, then the operator

$$A_i = F_i^* \operatorname{Op}\Big(\big((F_i^{-1})^*\phi_i\big)(x)a(x,\xi)\big((F_i^{-1})^*\tilde{\phi}_i\big)(y)\Big)(F_i^{-1})^* \tag{5.90}$$

does the job. □

An immediate consequence of the $\mathbb{R}^n$ result, Proposition 4.24, is:

Proposition 5.40 (Multiplicativity of the Principal Symbol). *If $A \in \Psi^m(M)$, $B \in \Psi^{m'}(M)$, with at least one of them properly supported, then*

$$\sigma^{m+m'}(A \circ B) = \sigma^m(A)\sigma^{m'}(B). \tag{5.91}$$

The analogue of Proposition 4.25 concerning the principal symbol of commutators will be discussed in Section 5.14.

5.6 Quantization

There is no completely natural way, in general, to quantize symbols on T^*M to pseudodifferential operators on M. However, we do have the following useful construction: fix a locally finite open cover $M = \bigcup_i U_i$ by coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ with $\overline{U_i}$ compact, and let $\{\phi_i\}$, $\phi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$, be a partition of unity subordinate to $\{U_i\}$. Fix $\tilde{\phi}_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$ with $\tilde{\phi}_i = 1$ near $\operatorname{supp}\phi_i$. Given $a \in S^m(T^*M)$, define then

$$\operatorname{Op}(a) := \sum_i F_i^* \operatorname{Op}\Big(\big((F_i^{-1})^*\phi_i\big)(x)\big((\tau_i^{-1})^*a\big)(x,\xi)\big((F_i^{-1})^*\tilde{\phi}_i\big)(y)\Big)(F_i^{-1})^*, \tag{5.92}$$

where $\tau_i\colon T^*_{U_i}M \to F_i(U_i) \times \mathbb{R}^n$ is the trivialization induced by F_i. By Theorem 5.29, the formula (5.92) defines a map

$$\operatorname{Op}\colon S^m(T^*M) \to \Psi^m(M). \tag{5.93}$$

Proposition 5.41 (Properties of the Quantization Map). *The map* Op *in* (5.93) *is continuous and linear and takes values in the subspace of properly supported operators. We have* $\mathrm{Op}(1) = I$ *(the identity operator). Moreover,* $\mathrm{Op}\colon S^m(T^*M) \to \Psi^m(M)$ *is surjective modulo* $\Psi^{-\infty}(M)$*; that is,* $\Psi^m(M) = \mathrm{Op}(S^m(T^*M)) + \Psi^{-\infty}(M)$.

Proof. We only sketch a proof of the final claim. In the notation of (5.66), we can write any given $A \in \Psi^m(M)$ in the form $A := A_m + K$ where $A_m = \sum_i F_i^* B_i (F_i^{-1})^*$ and $K \in \Psi^{-\infty}(M)$. We add the pullbacks of the symbols of the operators $B_i \in \Psi_{\mathrm{c}}^m(F_i(U_i))$ along the trivializations τ_i to obtain a symbol $a_m \in S^m(T^*M)$; by the coordinate invariance of the principal symbol, we then have

$$A_{m-1} := A_m - \mathrm{Op}(a_m) \in \Psi^{m-1}(M). \tag{5.94}$$

We may then apply Theorem 5.29 to A_{m-1}. An inductive argument thus produces $a_{m-j} \in S^{m-j}(T^*M)$, $j \in \mathbb{N}$, such that $A_{m-j-1} := A_{m-j} - \mathrm{Op}(a_{m-j}) \in \Psi^{m-j-1}(M)$. Letting $a \in S^m(T^*M)$ be an asymptotic sum of the a_{m-j}, $j \in \mathbb{N}_0$, we therefore have

$$A = A_m + K = \mathrm{Op}(a) + K + R, \quad R \in \Psi^{-\infty}(M). \tag{5.95}$$

The sum $K + R$ is a residual operator. □

5.7 Operators Acting on Sections of Vector Bundles

The reader might ask why we have not discussed adjoints of $A \in \Psi^m(M)$ (or even $A \in \mathrm{Diff}^m(M)$) yet. Since we do not have an invariant way of integrating *functions* on M, the only sensible way to define A^* is by

$$\int_M (Au)(x)\overline{v(x)} = \int_M u(x)\overline{A^* v(x)}, \quad u \in \mathcal{C}_{\mathrm{c}}^\infty(M),\ v \in \mathcal{C}_{\mathrm{c}}^\infty(M; \Omega M), \tag{5.96}$$

that is, A^* should be an operator acting on *1-densities*. We leave it to the reader to define the space of m-th order differential operators $\mathrm{Diff}^m(M; E, F)$ mapping sections of E to sections of F and go straight for the pseudodifferential version.

Definition 5.42 (Ps.d.o.s Acting on Sections of Vector Bundles). Let M be a smooth manifold, and let $\pi_E\colon E \to M$, $\pi_F\colon F \to M$ denote two real vector bundles of ranks k_E, k_F. (The complex case is completely analogous.) Then $\Psi^m(M; E, F)$ is the space of linear operators

$$A\colon \mathcal{C}_{\mathrm{c}}^\infty(M; E) \to \mathcal{C}^\infty(M; F) \tag{5.97}$$

with the following properties:

(1) Let $U \subset M$ be any open set, $G\colon U \to G(U) \subset \mathbb{R}^n$ a diffeomorphism, and let $\tau_E\colon \pi_E^{-1}(U) \to G(U) \times \mathbb{R}^{k_E}$, $\tau_F\colon \pi_F^{-1}(U) \to G(U) \times \mathbb{R}^{k_F}$ be local trivializations of E, F. Using τ_E, identify smooth sections of E over U with k_E-tuples of smooth functions on U, likewise for sections of F. If $\psi \in \mathcal{C}_c^\infty(U)$, then there exists a $k_F \times k_E$ matrix $B = (B_{ij})$ of ps.d.o.s $B_{ij} \in \Psi_c^m(G(U))$ such that on U

$$(\psi A(\psi u))_i = \sum_{j=1}^{k_E} G^*\Big(B_{ij}\big((G^{-1})^* u\big)_j\Big), \quad u \in \mathcal{C}_c^\infty(M; E),\ i = 1, \dots, k_F. \tag{5.98}$$

(2) If $\phi, \psi \in \mathcal{C}^\infty(M)$, $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$, then there exists $K \in \mathcal{C}^\infty(M^2; \pi_L^* F \otimes \pi_R^*(E^* \otimes \Omega M))$ such that $\phi A \psi = K$.

In the special case $F = E$, we write $\Psi^m(M; E) = \Psi^m(M; E, E)$.

In local coordinates and trivializations, the symbol of $A \in \Psi^m(M; E, F)$ is a symbol with values in linear maps from $\mathbb{R}^{k_E}$ to $\mathbb{R}^{k_F}$. The invariant definition is as follows. Denote by $\pi\colon T^*M \to M$ the projection. Given a vector bundle $G \to M$, we can consider its pullback $\pi^*G \to T^*M$ (Definition 5.9); a trivialization of G over an open set $U \subset M$, so $G|_U \cong U \times \mathbb{R}^{k_G}$, then induces a trivialization of π^*G over T_U^*M which is "constant in the fibers of T^*M," namely,

$$(\pi^*G)|_{T_U^*M} \cong T^*U \times \mathbb{R}^{k_G}, \tag{5.99}$$

by identifying $(\pi^*G)_{(x,\xi)} = G_x \cong \mathbb{R}^{k_G}$ using the local trivialization. We then denote by

$$S^m(T^*M; \pi^*G) \subset \mathcal{C}^\infty(T^*M; \pi^*G) \tag{5.100}$$

the space of all smooth functions that in local coordinates and in a trivialization of G (which induces a trivialization of π^*G as in (5.99)) are k_G-vectors of symbols on $\mathbb{R}^n$ of order m. Invariantly then,

$$\sigma^m(A) \in (S^m/S^{m-1})(T^*M; \pi^* \operatorname{Hom}(E, F)). \tag{5.101}$$

This means that a representative of $\sigma^m(A)$ is a map assigning to $(x, \xi) \in T^*M$ an element of $\operatorname{Hom}(E_x, F_x)$. We have a short exact sequence

$$\begin{aligned} 0 \to \Psi^{m-1}(M; E, F) &\to \Psi^m(M; E, F) \\ &\xrightarrow{\sigma^m} (S^m/S^{m-1})(T^*M; \pi^* \operatorname{Hom}(E, F)) \to 0. \end{aligned} \tag{5.102}$$

The results from Sections 5.4–5.5 carry over; moreover, one can define adjoints:

Proposition 5.43 (Compositions and Adjoints). *Let M be a smooth manifold, and let $E, F, G \to M$ denote three vector bundles.*

(1) Let $A \in \Psi^m(M; F, G)$ and $B \in \Psi^{m'}(M; E, F)$, with at least one of them properly supported. Then $A \circ B \in \Psi^{m+m'}(M; E, G)$ and $\sigma^{m+m'}(A \circ B) = \sigma^m(A) \circ \sigma^{m'}(B)$.

(2) Let $A \in \Psi^m(M; E, F)$. Then the (real) adjoint A^T, defined by

$$\int_M (Au)v = \int_M u(A^T v), \quad u \in \mathcal{C}_c^\infty(M; E),\ v \in \mathcal{C}_c^\infty(M; F^* \otimes \Omega M), \tag{5.103}$$

is a pseudodifferential operator

$$A^T \in \Psi^m(M; F^* \otimes \Omega M, E^* \otimes \Omega M) \tag{5.104}$$

with $\sigma^m(A^T) = \sigma^m(A)^T$. It is properly supported if A is.

The proof is left to the reader.

Remark 5.44 (Bundles with Extra Structure).

(1) If E, F are complex vector bundles with an antilinear involution ("complex conjugation"), then one can define the adjoint A^* similarly to (5.103), but with complex conjugation of the second factor; one then has

$$A^* \in \Psi^m(M; F^* \otimes \Omega M, E^* \otimes \Omega M), \quad \sigma^m(A^*) = \sigma^m(A)^*. \tag{5.105}$$

This in particular applies to the case that $E = F = M \times \mathbb{C}$, so sections of E, F are simply complex-valued functions on M.

(2) An inner product on E induces an isomorphism $E^* \cong E$ (antilinear when the inner product is sesquilinear). If one moreover chooses a trivialization of ΩM, then $A^T \in \Psi^m(M; F, E)$ (and $A^* \in \Psi^m(M; E, F)$ in the complex case).

A consequence of (5.104) is the following extension of Proposition 5.27:

Corollary 5.45 (Action of Ps.d.o.s on Distributions and Smooth Functions). *Let $A \in \Psi^m(M; E, F)$. Then A extends to a bounded linear operator $A\colon \mathscr{E}'(M; E) \to \mathscr{D}'(M; F)$. If A is properly supported, then A also maps $\mathscr{D}'(M; E) \to \mathscr{D}'(M; F)$, and by restriction $\mathcal{C}^\infty(M; E) \to \mathcal{C}^\infty(M; F)$.*

Proof. A^T is a bounded map $\mathcal{C}_c^\infty(M; F^* \otimes \Omega M) \to \mathcal{C}^\infty(M; E^* \otimes \Omega M)$. Therefore we can define $A\colon \mathscr{E}'(M; E) \to \mathscr{D}'(M; F)$ by duality using (5.104); this agrees with the original operator A when restricted to $\mathcal{C}_c^\infty(M; E)$.

If A is properly supported, then A^T maps $\mathcal{C}_c^\infty(M; F^* \otimes \Omega M) \to \mathcal{C}_c^\infty(M; E^* \otimes \Omega M)$, and hence we can now define $A\colon \mathscr{D}'(M; E) \to \mathscr{D}'(M; F)$

by duality. Finally, note that $\mathcal{C}^\infty(M;E) \subset \mathscr{D}'(M;E)$; it thus remains to check that $Au \in \mathcal{C}^\infty(M;F)$ when $u \in \mathcal{C}^\infty(M;E)$. But using a partition of unity $1 = \sum_i \phi_i$, with $\operatorname{supp}\phi_i$ compact, we have

$$Au = A\left(\sum_i \phi_i u\right) = \sum_i A(\phi_i u), \tag{5.106}$$

with convergence in $\mathscr{D}'(M;F)$. Since A is properly supported, the final sum is a locally finite sum of smooth terms and hence smooth. □

5.8 Special Classes of Operators

Let M be a manifold, and let $E, F \to M$ denote two vector bundles of rank k_E, k_F.

Definition 5.46 (Classical Ps.d.o.s on Manifolds). Let $m \in \mathbb{R}$. The subspace $\Psi^m_{\mathrm{cl}}(M;E,F) \subset \Psi^m(M;E,F)$ of *classical pseudodifferential operators* consists of those operators whose full symbol in a local coordinate chart and in local trivializations of E, F is a $k_F \times k_E$ matrix of classical symbols of order m. The principal symbol map on $\Psi^m_{\mathrm{cl}}(M;E,F)$ records the leading order homogeneous part,

$$\sigma^m \colon \Psi^m_{\mathrm{cl}}(M;E,F) \to S^m_{\mathrm{hom}}(T^*M \setminus o; \pi^* \operatorname{Hom}(E,F)). \tag{5.107}$$

This is a sensible definition since classicality is preserved under local coordinate transformations; this follows from the proof of Theorem 5.2, in particular equation (5.13). One easily checks that the composition of two classical ps.d.o.s (at least one of which is properly supported) is again a classical ps.d.o. and that taking adjoints preserves classicality as well. We have

$$\operatorname{Diff}^m(M;E,F) \subset \Psi^m_{\mathrm{cl}}(M;E,F). \tag{5.108}$$

An important special class of operators acting on sections of vector bundles is the following (see, e.g., Example 5.55):

Definition 5.47 (Principally Scalar Operators). Let $m \in \mathbb{R}$ and $A \in \Psi^m(M;E)$. Then A is *principally scalar* if its principal symbol is multiplication by scalars on the fibers of E, that is, if there exists a symbol $a \in S^m(T^*M)$ such that $\sigma^m(A)(x,\xi) = a(x,\xi)\operatorname{Id}_{E_x}$.

5.9 Elliptic Operators on Compact Manifolds, Fredholm Theory

Let M be an n-dimensional manifold without boundary, and let $E, F \to M$ denote two vector bundles.

Definition 5.48 (Ellipticity). We say that $A \in \Psi^m(M; E, F)$, with principal symbol $a = \sigma^m(A)$, is *elliptic* if there exists a symbol $b \in S^{-m}(T^*M; \pi^* \operatorname{Hom}(F, E))$ such that $ab - 1 \in S^{-1}(T^*M; \pi^* \operatorname{End}(F))$, $ba - 1 \in S^{-1}(T^*M; \pi^* \operatorname{End}(E))$.

Remark 5.49 (Equivalent Definition of Ellipticity). By "abstract group theory" as in (4.105), the following definition is equivalent to the ellipticity of A: there exist symbols $b, b' \in S^{-m}(T^*M; \pi^* \operatorname{Hom}(F, E))$ with the property that $ab - 1 \in S^{-1}(T^*M; \pi^* \operatorname{End}(F))$ and $b'a - 1 \in S^{-1}(T^*M; \pi^* \operatorname{End}(E))$.

The elliptic parametrix construction, Theorem 4.29, works in this setting as well; the proof is left to the reader.

Theorem 5.50 (Elliptic Parametrix). *Let $A \in \Psi^m(M; E, F)$ be elliptic. Then there exists a properly supported operator $B \in \Psi^{-m}(M; F, E)$, unique modulo $\Psi^{-\infty}(M; F, E)$, such that $A \circ B - I \in \Psi^{-\infty}(M; F)$, $B \circ A - I \in \Psi^{-\infty}(M; E)$. If A is classical, then so is B.*

Example 5.26 provides an example of the *inverse* of an elliptic differential operator on a compact manifold. An example of a parametrix of an operator that is not invertible is the following.

Example 5.51 (Derivative on the Circle). Let $M = \mathbb{S}^1 := \mathbb{R}_\theta / 2\pi\mathbb{Z}$ and $A = D_\theta \in \Psi^1(M)$. This is not invertible on any function space containing nonzero constant functions. A fundamental solution of D_θ on $\mathbb{R}_\theta$ is given by $iH(\theta)$. A parametrix B for A on $\mathbb{S}^1$ can be constructed from this using a cutoff away from $\theta = 0$. Concretely, if $\chi \in \mathcal{C}_{\mathrm{c}}^\infty((-\pi, \pi))$ equals 1 near 0, then we can take B to have Schwartz kernel

$$K_B(\theta, \theta') := -\frac{1}{2}|\theta - \theta'|\chi(\theta - \theta')|\mathrm{d}\theta'|, \quad \theta, \theta' \in (-\pi, \pi), \tag{5.109}$$

which is an element of $\mathscr{D}'(M^2; \Omega_R)$. The reader should check that this defines $B \in \Psi_{\mathrm{cl}}^{-1}(M)$.

Theorem 5.52 (Fredholm Properties and Generalized Inverses of Elliptic Ps.d.o.s). *Let M be a compact manifold. Let $A \in \Psi^m(M; E, F)$ be an elliptic operator.*

(1) *Then $A \colon \mathcal{C}^\infty(M; E) \to \mathcal{C}^\infty(M; F)$ is Fredholm, that is,* $\ker A$ *is finite-dimensional, and* $\operatorname{ran} A$ *is closed and has finite codimension. Furthermore,*

$$\begin{aligned}\operatorname{ran} A = \{f \in \mathcal{I}(M;F)\colon\ &\langle f^*, f\rangle = 0\\ &\forall\, f^* \in \mathcal{I}(M;F^*\otimes \Omega M),\ A^T f^* = 0\},\end{aligned} \tag{5.110}$$

where $A^T \in \Psi^m(M;F^*\otimes\Omega M, E^*\otimes\Omega M)$.

(2) *Let* $0 < \nu \in \mathcal{C}^\infty(M;\Omega M)$ *be a volume density on* M*, and fix positive definite fiber inner products on* E, F*. A linear operator* $G\colon \mathcal{C}^\infty(M;F) \to \mathcal{C}^\infty(M;E)$ *is uniquely determined by*

$$\begin{aligned}&Gf = u \quad \text{if } f \in \operatorname{ran} A,\ Au = f,\ u \perp \ker A \text{ in } L^2(M;E;\nu),\\ &Gf = 0 \quad \text{if } f \perp \operatorname{ran} A \text{ in } L^2(M;F;\nu).\end{aligned} \tag{5.111}$$

It is called the generalized inverse *of* A*. It satisfies* $G \in \Psi^{-m}(M;F,E)$ *and*

$$GA = I - \pi_N, \quad AG = I - \pi_R, \tag{5.112}$$

where $\pi_N\colon L^2(M;E;\nu) \to L^2(M;E;\nu)$ *is the orthogonal projection onto* $\ker A$ *and* $\pi_R\colon L^2(M;F;\nu) \to L^2(M;F;\nu)$ *is the orthogonal projection onto* $\ker A^* = (\operatorname{ran} A)^\perp$*. We have* $\pi_N \in \Psi^{-\infty}(M;E)$ *and* $\pi_R \in \Psi^{-\infty}(M;F)$*. In particular, if* A *is invertible, then* $G = A^{-1} \in \Psi^{-m}(M;F,E)$.

Proof. We fix a volume density on M and positive definite fiber inner products on E, F. Let $B \in \Psi^{-m}(M;F,E)$ be an elliptic parametrix of A, so

$$R_1 := AB - I \in \Psi^{-\infty}(M;F), \quad R_2 := BA - I \in \Psi^{-\infty}(M;E). \tag{5.113}$$

We show that $\dim\ker A < \infty$. First, note that

$$u \in \mathscr{D}'(M;E),\ Au = 0 \implies u = (BA - R_2)u = -R_2 u \in \mathcal{C}^\infty(M;E). \tag{5.114}$$

We look at this from the point of view that the identity map on $\ker A \subset L^2(M;E;\nu)$ can be written as $I = BA - R_2 = -R_2$. Now, R_2 maps $L^2(M;E;\nu) \to \mathcal{C}^\infty(M;E)$ and hence is compact as a map $L^2(M;E;\nu) \to L^2(M;E;\nu)$ by the Arzelà–Ascoli theorem. Therefore, the unit ball in the closed subspace $\ker A \subset L^2(M;E;\nu)$ is compact; thus, $\ker A \subset L^2(M;E;\nu)$ is finite-dimensional.

Next, we show that $\operatorname{ran} A$ is closed. Suppose $f_j = Au_j \to f \in \mathcal{C}^\infty(M;F)$, $u_j \in \mathcal{C}^\infty(M;E)$. We may change u_j by an element of $\ker A$ to ensure that $u_j \perp \ker A$. We have

$$u_j = BAu_j - R_2 u_j = Bf_j - R_2 u_j. \tag{5.115}$$

Suppose that, along some subsequence, $\|u_j\|_{L^2} \to \infty$. Then

$$\frac{u_j}{\|u_j\|} = B\Big(\frac{f_j}{\|u_j\|}\Big) - R_2\Big(\frac{u_j}{\|u_j\|}\Big). \tag{5.116}$$

This is bounded in $\mathcal{C}^\infty(M;E)$, and hence we can pass to a subsequence which converges in $L^2(M;E;\nu)$, say $u_j/\|u_j\| \to u \in L^2(M;E;\nu)$. Then $Au = \lim_{j\to\infty} f_j/\|u_j\| = 0$, so $u \in \ker A$, but also $u \perp \ker A$ by construction. Since $\|u\|_{L^2} = 1$, this is a contradiction.

Therefore, $\|u_j\|_{L^2}$ is bounded. Equation (5.115) then shows that u_j is bounded in $\mathcal{C}^\infty(M;E)$ and hence has a subsequence converging to $u \in \mathcal{C}^\infty(M;E)$. Replacing u_j by this subsequence, we obtain

$$Au = \lim_{j\to\infty} Au_j = \lim_{j\to\infty} f_j = f. \tag{5.117}$$

Since A^* is elliptic, the finite-codimensionality of $\operatorname{ran} A$ is a consequence of (5.110). Now, (5.110) is a consequence of two facts.

(1) Since $R = \operatorname{ran} A \subset \mathcal{C}^\infty(M;F)$ is a closed subspace of a Fréchet space, we have $R = {}^\perp(R^\perp) := \{f \in \mathcal{C}^\infty(M;F)\colon \langle f^*, f\rangle = 0\ \forall\, f^* \in R^\perp\}$ where

$$R^\perp = \{f^* \in \mathscr{D}'(M;F^*\otimes \Omega M)\colon \langle f^*, f\rangle = 0\ \forall\, f \in R\} \tag{5.118}$$

is the annihilator of R. Indeed, the inclusion $R \subset {}^\perp(R^\perp)$ is clear. For the converse inclusion, suppose that $f_0 \in \mathcal{C}^\infty(M;F) \setminus R$. The Hahn–Banach theorem, using the closedness of R, produces a continuous linear functional $f^*\colon \mathcal{C}^\infty(M;F) \to \mathbb{C}$, i.e., $f^* \in \mathscr{D}'(M;F^*\otimes\Omega M)$, which vanishes on R (thus $f^* \in R^\perp$) but $\langle f^*, f_0\rangle = 1$, so $f_0 \notin {}^\perp(R^\perp)$.

(2) We have $f^* \in R^\perp$ if and only if for all $u \in \mathcal{C}^\infty(M;E)$ we have $0 = \langle f^*, Au\rangle = \langle A^*f^*, u\rangle$, i.e., if and only if $f^* \in \ker A^*$.

Since A^* is elliptic, we have $R^\perp \subset \mathcal{C}^\infty(M;F^*\otimes\Omega M)$, and (5.110) follows.

Fixing an orthonormal basis $\{u_1,\dots,u_J\} \subset \mathcal{C}^\infty(M;E)$ of $\ker A$, the orthogonal projection π_N onto $\ker A$ is given by

$$\pi_N = \sum_{j=1}^{J} u_j \langle \cdot, u_j\rangle_{L^2(M;E;\nu)}. \tag{5.119}$$

Therefore, it has a smooth Schwartz kernel. An analogous argument shows that the orthogonal projection $\pi_R\colon L^2(M;F;\nu) \to L^2(M;F;\nu)$ onto $(\operatorname{ran} A)^\perp = \ker A^* \subset \mathcal{C}^\infty(M;F)$ has a smooth Schwartz kernel. Therefore, $\pi_N, \pi_R \in \Psi^{-\infty}$.

The statement $G \in \Psi^{-m}(M; F, E)$ for the generalized inverse (5.111) now follows by writing

$$\begin{aligned} G &= G(AB - R_1) \\ &= (I - \pi_N)B - GR_1 \\ &= (I - \pi_N)B - (BA - R_2)GR_1 \\ &= (I - \pi_N)B - B(I - \pi_R)R_1 + R_2GR_1. \end{aligned} \tag{5.120}$$

Indeed, the first summand lies in $\Psi^m(M; F, E)$ and the second in $\Psi^{-\infty}(M; F, E)$, and the last one is a smoothing operator and hence lies in $\Psi^{-\infty}(M; F, E)$ as well. □

We revisit the proof of the existence of the generalized inverse using L^2-techniques in Section 5.11.

As a typical example, we discuss the *Laplace operator* on a compact n-dimensional manifold M, which we assume to be connected for convenience. Denote by S^2T^*M the second symmetric tensor power of T^*M. Let $g \in \mathcal{C}^\infty(M; S^2T^*M)$ be a Riemannian metric, so in local coordinates

$$g = \sum_{i,j=1}^{n} g_{ij}(x) \cdot \frac{1}{2}(\mathrm{d}x^i \otimes \mathrm{d}x^j + \mathrm{d}x^j \otimes \mathrm{d}x^i), \quad g_{ij}(x) = g_{ji}(x), \tag{5.121}$$

with $(g_{ij}(x))_{i,j=1,\dots,n}$ positive definite. Write $g^{ij}(x) = g^{-1}(x)_{ij}$ and $|g| = |\det(g_{ij})|$. Then the (scalar) Laplace operator is

$$\Delta_g u = \sum_{i,j=1}^{n} |g|^{-\frac{1}{2}} D_{x_i}\big(|g|^{\frac{1}{2}} g^{ij} D_{x_j} u\big). \tag{5.122}$$

It satisfies

$$\Delta_g \in \mathrm{Diff}^2(M), \quad \sigma^2(\Delta_g)(x, \xi) = \sum_{i,j=1}^{n} g^{ij}(x)\xi_i\xi_j =: |\xi|^2_{g_x^{-1}}. \tag{5.123}$$

Thus, Δ_g is elliptic. By Theorem 5.52, the kernel and cokernel of Δ_g are finite-dimensional.

Moreover, Δ_g is a symmetric operator with respect to the inner product on $L^2(M; |\mathrm{d}g|)$, where $|\mathrm{d}g| \in \mathcal{C}^\infty(M; \Omega M)$ is the metric volume density, defined in local coordinates by

$$|\mathrm{d}g| = |g(x)|^{\frac{1}{2}} \mathrm{d}x. \tag{5.124}$$

(This is well-defined independently of the choice of local coordinates.) Thus, $\ker \Delta_g = (\operatorname{ran} \Delta_g)^\perp$; and if $u \in \ker \Delta_g$, then

$$0 = \int_M (\Delta_g u)\bar{u} \, |\mathrm{d}g| = \int_M |\nabla u|_g^2 \, |\mathrm{d}g| \tag{5.125}$$

where $|\nabla u|_g^2 := \sum_{i,j=1}^n g^{ij}(\partial_{x_i} u)(\partial_{x_j} \bar{u})$, so u is constant. In the notation of Theorem 5.52, the projection operators

$$\pi_N = \frac{1}{\operatorname{vol}(M)} \langle \cdot, 1 \rangle 1 = \pi_R \tag{5.126}$$

are thus both equal to the orthogonal projection onto the space of constant functions.

Let us study

$$\Delta_g u = f, \quad f \in \mathscr{D}'(M). \tag{5.127}$$

Let $G \in \Psi^{-2}(M)$ denote the generalized inverse of Δ_g. In the notation of Theorem 5.52, we then have

$$u = (G\Delta_g + \pi_N)u = Gf + \pi_N u. \tag{5.128}$$

This solves (5.127) if and only if $f = \Delta_g u = \Delta_g G f + \Delta_g \pi_N u = (I - \pi_R) f$. This shows:

Proposition 5.53 (Laplace Equation on Compact Manifolds). *The equation (5.127) has a solution $u \in \mathscr{D}'(M)$ if and only if $\langle f, 1 \rangle = 0$, and in this case u is unique up to additive constants. If $f \in \mathcal{C}^\infty(M)$, then $u \in \mathcal{C}^\infty(M)$.*

Example 5.54 (Shifted Laplacian). For the operator $A = \Delta_g + 1 \in \Psi^2(M)$ on a compact Riemannian manifold, one has $\ker A = \{0\} = (\operatorname{ran} A)^\perp$. Therefore, one can always solve $Au = f$ given $f \in \mathcal{C}^\infty(M)$, resp. $f \in \mathscr{D}'(M)$, with the unique solution satisfying $u \in \mathcal{C}^\infty(M)$, resp. $u \in \mathscr{D}'(M)$.

Example 5.55 (Hodge Laplacian). One can define natural generalizations of Δ_g which act on vector bundles rather than functions. Let $\mathrm{d}_k \in \operatorname{Diff}^1(M; \Lambda^k T^*M; \Lambda^{k+1} T^*M)$ denote the exterior derivative, and denote by $\delta_k \in \operatorname{Diff}^1(M; \Lambda^k T^*M; \Lambda^{k-1} T^*M)$ the adjoint of d_{k-1}. Let $\mathrm{d}_n = 0$ and $\delta_0 = 0$. Then the *Hodge Laplacian* acting on k-forms is

$$\Delta_k := \delta_{k+1}\mathrm{d}_k + \mathrm{d}_{k-1}\delta_k \in \operatorname{Diff}^2(M; \Lambda^k T^*M). \tag{5.129}$$

Its principal symbol is *scalar*, i.e., at each $(x, \xi) \in T^*M$ a multiple of the identity operator on $(\pi^*\Lambda^k T^*M)_{(x,\xi)}$; in fact $\sigma^2(\Delta_k)(x, \xi) = |\xi|^2_{g_x^{-1}} \operatorname{Id}$. (The expression (5.122) is the special case $k = 0$.) Again Δ_k is symmetric with

respect to the fiber inner product and volume density induced by g. Its kernel and orthocomplement of the range are finite-dimensional and can be identified with the de Rham cohomology group $H^k(M;\mathbb{C}) := \ker d_k / \operatorname{ran} d_{k-1}$ by Hodge theory. For a general version of Hodge theory, see Exercise 5.39.

5.10 Sobolev Spaces on Manifolds

We need two key facts about the Sobolev spaces $H^s(\mathbb{R}^n)$ for the generalization of Sobolev spaces to manifolds. For an open set $\Omega \subset \mathbb{R}^n$, we define

$$H^s_{\mathrm{c}}(\Omega) := \{u \in H^s(\mathbb{R}^n) \colon \operatorname{supp} u \subset \Omega\}. \tag{5.130}$$

Lemma 5.56 (Sobolev Spaces Under Localizations and Coordinate Changes). *Sobolev spaces on $\mathbb{R}^n$ have the following properties:*

(1) *Let $a \in \mathcal{C}^\infty_b(\mathbb{R}^n)$. Then multiplication by a is a bounded linear map $H^s(\mathbb{R}^n) \to H^s(\mathbb{R}^n)$ for all $s \in \mathbb{R}$.*
(2) *Suppose $\kappa \colon \Omega \to \Omega'$ is a diffeomorphism of the open subsets $\Omega, \Omega' \subset \mathbb{R}^n$. Then $\kappa^* \colon H^s_{\mathrm{c}}(\Omega') \to H^s_{\mathrm{c}}(\Omega)$. Here, the pullback of a distribution $u \in \mathscr{D}'(\mathbb{R}^n)$ with support in Ω' is defined via duality using the formula*

$$\langle \kappa^* u, \phi \rangle = \langle u, |\det(\kappa^{-1})'| (\kappa^{-1})^* \phi \rangle, \quad \phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega). \tag{5.131}$$

Proof. The first claim is proved in Exercise 2.11. With the machinery of Chapter 4 at hand, we can instead just observe that $a \in \Psi^0(\mathbb{R}^n)$ and appeal to Corollary 4.38.

The second claim is clear for $s = 0$. We shall prove it for general $s \in \mathbb{R}$ using our ps.d.o. machinery. Given $u \in H^s_{\mathrm{c}}(\Omega') \subset \mathscr{E}'(\Omega')$, we certainly have $\kappa^* u \in \mathscr{E}'(\Omega)$. Let $A \in \Psi^s(\mathbb{R}^n)$ be elliptic, and let $\phi, \tilde{\phi} \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ be such that $\phi = \tilde{\phi} = 1$ near $\operatorname{supp}(\kappa^* u)$ and $\tilde{\phi} = 1$ in a neighborhood of $\operatorname{supp} \phi$. By localizing the Schwartz kernel of A near the diagonal (which does not affect its ellipticity property), we may arrange that

$$A(\kappa^* u) = \tilde{\phi} A \phi (\kappa^* u). \tag{5.132}$$

Note that $\tilde{\phi} A \phi \in \Psi^s_{\mathrm{c}}(\Omega)$. Therefore, by Theorem 5.2,

$$A(\kappa^* u) = \kappa^*(A' u), \quad A' = (\kappa^{-1})^* \tilde{\phi} A \phi \kappa^* \in \Psi^s_{\mathrm{c}}(\Omega'). \tag{5.133}$$

Therefore $A'u \in L^2(\Omega')$, and hence $\kappa^*(A'u) \in L^2(\Omega)$, so $A(\kappa^* u) \in L^2(\mathbb{R}^n)$. Since A is elliptic, Corollary 4.41 implies that $\kappa^* u \in H^s(\mathbb{R}^n)$, as desired. □

The "local coordinate" definition of Sobolev spaces on manifolds is then:

Definition 5.57 (Sobolev Spaces on Manifolds). Let M be an n-dimensional manifold, $s \in \mathbb{R}$.

(1) We define $H^s_{\mathrm{loc}}(M)$ as the space of all $u \in \mathscr{D}'(M)$ such that for all coordinate charts $F\colon U \to F(U) \subset \mathbb{R}^n$ and all $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$, the distribution $(F^{-1})^*(\chi u)\colon \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n) \ni \phi \mapsto \langle u, \chi F^*(\phi |\mathrm{d}x|)\rangle$ is an element of $H^s(\mathbb{R}^n)$.
(2) We define $H^s_{\mathrm{c}}(M) = \{u \in H^s_{\mathrm{loc}}(M)\colon \operatorname{supp} u \subset M \text{ is compact}\}$.

If M is compact, we write

$$H^s(M) = H^s_{\mathrm{loc}}(M) = H^s_{\mathrm{c}}(M). \tag{5.134}$$

Lemma 5.56 shows that if $u \in H^s_{\mathrm{c}}(F(U))$ for some coordinate chart $F\colon U \to F(U) \subset \mathbb{R}^n$ on M, then $F^*u \in H^s_{\mathrm{c}}(M)$.

We equip $H^s_{\mathrm{loc}}(M)$ with the structure of a Fréchet space by using the seminorms

$$\|(F_i^{-1})^*\phi_i u\|_{H^s(\mathbb{R}^n)} \tag{5.135}$$

for any fixed cover of M by countably many coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$, $\overline{U_i}$ compact, and a subordinate partition of unity $\{\phi_i\}$, $\phi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$. The resulting topology is independent of the cover and the partition of unity. The topology on $H^s_{\mathrm{c}}(M)$ is the inductive limit topology arising from $H^s_{\mathrm{c}}(M) = \bigcup_{K_i} \dot{H}^s(K_i)$ where $K_1 \subset K_2 \subset \cdots$ is an exhaustion of M by compact sets and

$$\dot{H}^s(K_i) = \{u \in H^s_{\mathrm{loc}}(M)\colon \operatorname{supp} u \subset K_i\}. \tag{5.136}$$

Note that $\dot{H}^s(K_i) \subset H^s_{\mathrm{loc}}(M)$ is itself a Fréchet space. We can equivalently define the locally convex topology on $H^s_{\mathrm{c}}(M)$ using explicit seminorms analogously to (5.19).

The space $\dot{H}^s(K)$ for compact $K \subset M$ can can be given the structure of a Hilbert space:

Proposition 5.58 (Hilbert Space Structure). *Let $K \subset M$ be a compact subset. Then $\dot{H}^s(K)$ is Hilbertizable, i.e., there exists an inner product $\langle \cdot, \cdot \rangle_{\dot{H}^s(K)}$ such that the topology on $\dot{H}^s(K)$ induced by the associated norm is equal to the subspace topology of $\dot{H}^s(K) \subset H^s_{\mathrm{loc}}(M)$.*

Proof. Cover K by a finite number of coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ of M, and let $\{\phi_i\}$, $\phi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$, $i = 1, \ldots, N$, denote a subordinate partition of unity (i.e., we require $\sum_{i=1}^N \phi_i = 1$ on K only). We then define

$$\|u\|^2_{\dot{H}^s(K)} := \sum_{i=1}^N \|(F_i^{-1})^* \phi_i u\|^2_{H^s(\mathbb{R}^n)}, \tag{5.137}$$

and $\langle \cdot, \cdot \rangle_{\dot{H}^s(K)}$ by polarization.

To show that this induces the subspace topology of $\dot{H}^s(K) \subset H^s_{\mathrm{loc}}(M)$, consider a coordinate chart $F\colon U \to F(U) \subset \mathbb{R}^n$ and let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$; then

$$(F^{-1})^* \phi u = \sum_{i=1}^N (F^{-1})^* \phi F_i^* (F_i^{-1})^* \phi_i u = \sum_{i=1}^N (F_i F^{-1})^* (F_i^{-1})^* (\phi \phi_i u). \tag{5.138}$$

If we fix $\tilde{\phi}_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$ to be equal to 1 on $\operatorname{supp} \phi_i$, we can thus estimate

$$\|(F^{-1})^* \phi u\|_{H^s(\mathbb{R}^n)} \leq C \sum_{i=1}^N \|(F_i^{-1})^* (\phi \tilde{\phi}_i) \cdot (F_i^{-1})^* (\phi_i u)\|_{\dot{H}^s(\mathbb{R}^n)} \leq C' \|u\|_{\dot{H}^s(K)}, \tag{5.139}$$

where we used the two parts of Lemma 5.56 for the two inequalities. □

The proof of Lemma 5.56 suggests a more intrinsic definition of Sobolev spaces on M. Note first that the spaces $L^2_{\mathrm{loc}}(M)$ and $L^2_{\mathrm{c}}(M)$ are well-defined, independently of a choice of integration measure on M. (On the other hand, the space $L^2(M)$, even as a set, is *not* well-defined when M is non-compact unless a measure is specified on M.)

Proposition 5.59 (Boundedness of Ps.d.o.s on Sobolev Spaces: I). *Let $u \in \mathscr{D}'(M)$.*

(1) *Suppose that $u \in H^s_{\mathrm{c}}(M)$. Then $Au \in L^2_{\mathrm{loc}}(M)$ for all $A \in \Psi^s(M)$. If A is properly supported, then $A\colon H^s_{\mathrm{c}}(M) \to L^2_{\mathrm{c}}(M)$, $H^s_{\mathrm{loc}}(M) \to L^2_{\mathrm{loc}}(M)$.*
(2) *If $Au \in L^2_{\mathrm{loc}}(M)$ for some properly supported elliptic operator $A \in \Psi^s(M)$, then $u \in H^s_{\mathrm{loc}}(M)$.*

Proof. Let $F\colon U \to F(U) \subset \mathbb{R}^n$ be a coordinate chart on M, and let $\phi, \tilde{\phi} \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ with $\tilde{\phi} = 1$ in a neighborhood of $\operatorname{supp} \phi$.

The first claim follows by writing

$$\phi A u = \phi A \tilde{\phi} u + \phi A (1 - \tilde{\phi}) u. \tag{5.140}$$

The second summand lies in $\mathcal{C}^\infty(M) \subset L^2_{\mathrm{loc}}(M)$. The first summand can be evaluated in local coordinates and lies in $L^2_{\mathrm{c}}(M)$ by Theorem 4.36.

Turning to the second claim, we need to show that $(F^{-1})^*(\phi u) \in \mathscr{E}'(F(U))$ lies in $H^s(\mathbb{R}^n)$. Let $B \in \Psi^s(\mathbb{R}^n)$ be elliptic; we claim that $B(F^{-1})^*(\phi u) \in L^2(\mathbb{R}^n)$. We can arrange for the Schwartz kernel of B to be supported so close to the diagonal that

$$(1 - (F^{-1})^*\tilde{\phi})B((F^{-1})^*\phi) = 0. \tag{5.141}$$

By elliptic regularity, and noting that $(F^{-1})^*(\phi u) \in H^{-N}(\mathbb{R}^n)$ for some N, it suffices to show that

$$B'u \in L^2_{\mathrm{c}}(M), \quad B' := F^*((F^{-1})^*\tilde{\phi})B((F^{-1})^*\phi)(F^{-1})^* \in \Psi^s(M). \tag{5.142}$$

Since A is elliptic, there exists a properly supported parametrix $Q \in \Psi^{-s}(M)$ with $I = QA + R$, where $R \in \Psi^{-\infty}(M)$ is then also properly supported. Therefore,

$$B'u = B'(QA + R)u = (B'Q)(Au) + B'Ru. \tag{5.143}$$

Now $B'Q \in \Psi^0(M)$ is bounded on $L^2_{\mathrm{loc}}(M)$, so $(B'Q)(Au) \in L^2_{\mathrm{loc}}(M)$, while $B'R \in \Psi^{-\infty}(M)$, so $B'Ru \in \mathcal{C}^\infty(M)$. Therefore, $B'u \in L^2_{\mathrm{loc}}(M)$. □

Corollary 5.60 (Boundedness of Ps.d.o.s on Sobolev Spaces: II). *Let $A \in \Psi^m(M)$. Then A is a bounded linear operator*

$$A\colon H^s_{\mathrm{c}}(M) \to H^{s-m}_{\mathrm{loc}}(M). \tag{5.144}$$

If A is properly supported, then $A\colon H^s_{\mathrm{c}}(M) \to H^{s-m}_{\mathrm{c}}(M)$, $H^s_{\mathrm{loc}}(M) \to H^{s-m}_{\mathrm{loc}}(M)$.

Proof. We only prove (5.144). Let $\Lambda \in \Psi^{s-m}(M)$ be properly supported and elliptic. By the second part of Proposition 5.59, it suffices to show that $\Lambda \circ A\colon H^s_{\mathrm{c}}(M) \to L^2_{\mathrm{loc}}(M)$; but this follows from $\Lambda \circ A \in \Psi^s(M)$ and the first part of Proposition 5.59. □

Proposition 5.61 (Compactness of Negative Order Ps.d.o.s). *Let M be a compact n-dimensional manifold. Let $A \in \Psi^m(M)$, $m < 0$. Then $A\colon L^2(M) \to L^2(M)$ is compact.*

Proof. Decomposing A into finitely many terms as in (5.66), this follows from the compactness on $L^2(\mathbb{R}^n)$ of negative order ps.d.o.s on $\mathbb{R}^n$ with compactly supported Schwartz kernels (Proposition 4.37) and the compactness of residual operators with Schwartz kernels $K \in \mathcal{C}^\infty(M^2; \Omega_R)$. (The latter follows from the fact that such operators are continuous $L^2(M) \to \mathcal{C}^\infty(M)$, while $\mathcal{C}^\infty(M) \subset \mathcal{C}^1(M) \to \mathcal{C}^0(M) \hookrightarrow L^2(M)$ is compact by the Arzelà–Ascoli theorem.) □

Adding vector bundles to this discussion requires only notational changes. Namely, if $E \to M$ is a real/complex rank k vector bundle, we say that $u \in \mathscr{D}'(M; E)$ lies in $H^s_{\mathrm{loc}}(M; E)$ if and only if in local trivializations of E over coordinate charts on M, ϕu is a k-vector of real-valued/complex-valued elements of $H^s(\mathbb{R}^n)$, where ϕ is smooth and compactly supported in the chart. We let $H^s_{\mathrm{c}}(M; E) = H^s_{\mathrm{loc}}(M; E) \cap \mathscr{E}'(M; E)$ as usual. We leave the statements

and proofs of the generalizations of Proposition 5.59, Corollary 5.60, and also of Theorem 5.63 and Proposition 5.64 to the reader.

Example 5.62 (δ-Distribution). If M is n-dimensional and $p \in M$, then δ_p (see Example (5.17)) satisfies $\delta_p \in H^s_{\mathrm{c}}(M;\Omega M)$ for all $s < -\frac{n}{2}$.

We have the following important duality result:

Theorem 5.63 (Duality). *Let $s \in \mathbb{R}$. Then the sesquilinear pairing*

$$\mathcal{C}^\infty(M) \times \mathcal{C}^\infty_{\mathrm{c}}(M;\Omega M) \ni (u,v) \mapsto \langle u,v\rangle_{L^2} := \int_M u(x)\overline{v(x)} \in \mathbb{C} \tag{5.145}$$

extends by continuity and density to a sesquilinear pairing

$$H^s_{\mathrm{loc}}(M) \times H^{-s}_{\mathrm{c}}(M;\Omega M) \to \mathbb{C} \tag{5.146}$$

which identifies $H^s_{\mathrm{loc}}(M) \cong H^{-s}_{\mathrm{c}}(M;\Omega M)^*$ and $H^{-s}_{\mathrm{c}}(M;\Omega M) \cong H^s_{\mathrm{loc}}(M)^*$ via $u \mapsto \langle u,\cdot\rangle_{L^2}$ and $v \mapsto \langle \cdot,\bar{v}\rangle_{L^2}$, respectively.

A choice of volume density on M trivializes ΩM and thus identifies $H^{-s}_{\mathrm{c}}(M)^* = H^s_{\mathrm{loc}}(M)$ and $H^s_{\mathrm{loc}}(M)^* = H^{-s}_{\mathrm{c}}(M)$.

Proof of Theorem 5.63. For notational simplicity, we trivialize ΩM using a volume density. The subspaces $\mathcal{C}^\infty(M) \subset H^s_{\mathrm{loc}}(M)$ and $\mathcal{C}^\infty_{\mathrm{c}}(M) \subset H^s_{\mathrm{c}}(M)$ are dense (Exercise 5.28). The well-definedness of (5.146) for $s = 0$ follows from the Cauchy–Schwarz inequality. Let now $s \in \mathbb{R}$ be arbitrary. Let $A \in \Psi^s(M)$ be elliptic and properly supported; let $B \in \Psi^{-s}(M)$ be a properly supported parametrix with

$$I = BA + R, \quad I = AB + R', \quad R, R' \in \Psi^{-\infty}(M). \tag{5.147}$$

For $u \in \mathcal{C}^\infty(M)$ and $v \in \mathcal{C}^\infty_{\mathrm{c}}(M)$, we then have

$$\begin{aligned} \langle u,v\rangle_{L^2} &= \langle Au, B^*v\rangle_{L^2} + \langle Ru, v\rangle_{L^2} \\ &= \langle Au, B^*v\rangle_{L^2} + \langle A^*Ru, Bv\rangle_{L^2} + \langle Ru, R'v\rangle_{L^2}. \end{aligned} \tag{5.148}$$

The right-hand side is well-defined for $u \in H^s_{\mathrm{loc}}(M)$ and $v \in H^{-s}_{\mathrm{c}}(M)$; this gives (5.146).

We first identify the dual spaces in the case $s = 0$. Given $\lambda \in L^2_{\mathrm{loc}}(M)^*$, there exists a compact subset $K \subset M$ such that $|\lambda(u)| \le C\|u\|_{L^2(K)}$ for all $u \in L^2_{\mathrm{loc}}(M)$. Therefore, λ induces a continuous linear functional on the Hilbert space $L^2(K)$, and therefore there exists $v \in L^2(K)$ such that $\lambda(u) = \langle u,v\rangle_{L^2(K)}$. Extending v by 0 to $M \setminus K$ gives an element of $L^2_{\mathrm{c}}(M)$ which we still denote by v; then $\lambda(u) = \langle u,v\rangle_{L^2}$.

Similarly, if $\lambda \in L^2_{\mathrm{c}}(M)^*$, then λ defines a continuous functional on $L^2(K)$ for all compact $K \subset M$ and thus determines functions $u_K \in L^2(K)$ for all

K, with $u_K = u_{K'}$ on the overlap $K \cap K'$ of any two compact $K, K' \subset M$. Therefore, we can define $u \in L^2_{\mathrm{loc}}(M)$ to be equal to u_K on K, and then $\lambda(v) = \overline{\langle u, v\rangle_{L^2}}$. Thus $L^2_{\mathrm{c}}(M)^* = L^2_{\mathrm{loc}}(M)$.

For general $s \in \mathbb{R}$, let $\lambda \in H^s_{\mathrm{loc}}(M)^*$. Using (5.147), we write $\lambda(u) = \lambda(BAu) + \lambda(Ru)$. Since $\lambda \circ B \in L^2_{\mathrm{loc}}(M)^*$, there exists $v_0 \in L^2_{\mathrm{c}}(M)$ such that

$$\lambda(BAu) = \langle Au, v_0\rangle_{L^2} = \langle u, A^* v_0\rangle_{L^2}. \tag{5.149}$$

Similarly, $\lambda \circ R \in L^2_{\mathrm{loc}}(M)^*$ is given by $\lambda(Ru) = \langle u, v_1\rangle_{L^2}$ for some $v_1 \in L^2_{\mathrm{c}}(M)$. We can improve this latter statement by writing

$$\begin{aligned} \lambda(Ru) &= \lambda(BA(Ru)) + \lambda(R(Ru)) \\ &= \langle Ru, A^* v_0\rangle_{L^2} + \langle Ru, v_1\rangle_{L^2} = \langle u, v_2\rangle_{L^2} \end{aligned} \tag{5.150}$$

where now $v_2 := R^* A^* v_0 + R^* v_1 \in \mathcal{C}^\infty_{\mathrm{c}}(M)$. Altogether, we have now shown that $\lambda(u) = \langle u, v\rangle_{L^2}$ where $v := A^* v_0 + v_2 \in H^{-s}_{\mathrm{c}}(M)$; this gives $H^s_{\mathrm{loc}}(M)^* = H^{-s}_{\mathrm{c}}(M)$. We leave the proof of $H^{-s}_{\mathrm{c}}(M)^* = H^s_{\mathrm{loc}}(M)$ to the reader. □

On a compact manifold M, the space $H^s(M)$ can be given the structure of a Hilbert space:

Proposition 5.64 ($H^s(M)$ as a Hilbert Space). *Let M be compact, and let $s \in \mathbb{R}$. Fix a volume density on M. Then there exists $A \in \Psi^s(M)$ such that*

$$\langle u, v\rangle_{H^s(M)} := \langle Au, Av\rangle_{L^2(M)}, \quad \|u\|^2_{H^s(M)} := \langle u, u\rangle_{H^s(M)}, \tag{5.151}$$

gives $H^s(M)$ the structure of a Hilbert space. That is, the topology on $H^s(M)$ is equal to the norm topology of $(H^s(M), \|\cdot\|_{H^s(M)})$.

Proof. Consider first the case $s \geq 0$. Fix a smooth fiber metric $|\cdot|$ on T^*M (e.g., induced by a Riemannian metric on M), and let $\Lambda' \in \Psi^{\frac{s}{2}}_{\mathrm{cl}}(M)$ be an operator with $\sigma^{\frac{s}{2}}(\Lambda')(x,\xi) = |\xi|^{\frac{s}{2}}$. Then Λ' is elliptic, and so is

$$\Lambda_s := I + (\Lambda')^* \Lambda' \in \Psi^s(M). \tag{5.152}$$

By Theorem 5.52, $\Lambda_s \colon \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$ is Fredholm. We claim that Λ_s is invertible on $\mathcal{C}^\infty(M)$. Indeed, $\Lambda_s u = 0$ implies $\|u\|^2_{L^2(M)} + \|\Lambda' u\|^2_{L^2(M)} = 0$, and hence $u = 0$. Since Λ_s is symmetric (i.e., $\langle \Lambda_s u, f\rangle_{L^2(M)} = \langle u, \Lambda_s f\rangle_{L^2(M)}$ for $u, f \in \mathcal{C}^\infty(M)$), this also implies that Λ_s is surjective. The second part of Theorem 5.52 then implies that

$$\Lambda_{-s} := \Lambda_s^{-1} \in \Psi^{-s}(M). \tag{5.153}$$

Using Proposition 5.59, we conclude that $\Lambda_s\colon H^s(M) \to L^2(M)$ and $\Lambda_{-s}\colon H^{-s}(M) \to L^2(M)$ are isomorphisms. For $s \in \mathbb{R}$, we can thus take $A = \Lambda_s$. □

In particular, Theorem 5.63 identifies the $L^2(M)$-dual of $H^s(M)$ as

$$H^s(M)^* \cong H^{-s}(M). \tag{5.154}$$

Remark 5.65 (The Case $H^{2k}(M)$, $k \in \mathbb{N}_0$). For $s = 2k$, $k \in \mathbb{N}$, one can take $\Lambda_{2k} = (\Delta_g + 1)^k$ for any Riemannian metric g on M. (In fact, this is true for any $k \in \mathbb{R}$ by a theorem of Seeley [See67] which states, as a special case, that $(\Delta_g + 1)^s \in \Psi^{2s}(M)$ for any $s \in \mathbb{R}$. This operator is defined using the functional calculus for self-adjoint operators; cf. Theorem 5.71.)

5.11 Elliptic Operators on Compact Manifolds, Revisited

Throughout this section, we denote by M a *compact* manifold.

Proposition 5.66 (Rellich Compactness Theorem). *Let $s' < s$. Then the inclusion*

$$H^s(M) \hookrightarrow H^{s'}(M) \tag{5.155}$$

is compact.

Proof. One can prove this by localizing in coordinate charts and using Theorem 2.17. In the spirit of using ps.d.o. techniques to establish properties of Sobolev spaces, one can alternatively argue as follows. Fixing invertible ps.d.o.s $\Lambda_\sigma \in \Psi^\sigma(M)$ for $\sigma = s, s'$ as in Proposition 5.64, we can factor the inclusion (5.155) as

$$H^s(M) \xrightarrow{\Lambda_s} L^2(M) \xrightarrow{\Lambda_{s'} \circ \Lambda_s^{-1}} L^2(M) \xrightarrow{\Lambda_{s'}^{-1}} H^{s'}(M), \tag{5.156}$$

where the first and last arrows are isomorphisms. The middle arrow is $\Lambda_{s'} \circ \Lambda_s^{-1} \in \Psi^{s'-s}(M)$, i.e., a ps.d.o. of negative order. The result then follows from Proposition 5.61. □

We can now refine Theorem 5.52:

Theorem 5.67 (Fredholm Properties of Elliptic Ps.d.o.s on Sobolev Spaces). *Let $A \in \Psi^m(M; E, F)$ be an elliptic operator. Then for all $s \in \mathbb{R}$,*

$$A\colon H^s(M; E) \to H^{s-m}(M; F) \tag{5.157}$$

is Fredholm. Its kernel $\ker A$ *is independent of* s *and satisfies* $\ker A \subset \mathcal{C}^\infty(M;E)$. *Moreover, if we fix a volume density on* M *and positive definite fiber inner products on* E, F, *the cokernel* $\operatorname{coker} A$ *can be identified with the subset* $\ker A^* \subset \mathcal{C}^\infty(M;F)$ *which is independent of* s; *that is,* $f \in H^{s-m}(M;F)$ *lies in* $\operatorname{ran} A$ *if and only if* $\langle f, g\rangle_{L^2(M;F)} = 0$ *for all* $g \in \ker A^*$.

Proof. If $B \in \Psi^m(M;F,E)$ denotes an elliptic parametrix, then $AB = I + R_1$ and $BA = I + R_2$ with $R_1, R_2 \in \Psi^{-\infty}$ as in (5.113). By Proposition 5.66, the errors $R_1 \colon H^s(M;F) \to \mathcal{C}^\infty(M;F) \hookrightarrow H^s(M;F)$ and $R_2 \colon H^{s-m}(M;E) \to \mathcal{C}^\infty(M;E) \hookrightarrow H^{s-m}(M;E)$ are compact operators. Therefore, A is Fredholm. The regularity statement $\ker A \subset \mathcal{C}^\infty(M;E)$ is a special case of (5.114). The solvability claim follows from Theorem 5.52 and the identity $u = BAu - R_2 u$ which gives $u \in H^s(M;E)$ when $f \in H^{s-m}(M;F)$. □

In fact, this theorem has a converse: if $A \in \Psi^m(M;E,F)$ is such that (5.157) is Fredholm for some $s \in \mathbb{R}$, then A is elliptic. A special case of this result is the subject of Exercise 5.33.

Remark 5.68 (Fredholm Index). Theorem 5.67 also shows that the index $\operatorname{ind} A = \dim \ker A - \dim \operatorname{coker} A$ is independent of s. Simple functional analytic arguments show that $\operatorname{ind} A = \operatorname{ind}(A + B)$ for any $B \in \Psi^{m-1}(M;E,F)$; thus, $\operatorname{ind} A$ only depends on the principal symbol $\sigma^m(A)$. The Atiyah–Singer index theorem [AS63] gives a formula to compute $\operatorname{ind} A$ in terms of $\sigma^m(A)$.

We can now give a more transparent perspective on the generalized inverse of Theorem 5.52(2). Indeed, by Theorem 5.67, $\operatorname{ran}_{H^m(M;E)} A := A(H^m(M;E)) \subset L^2(M;F)$ is a closed subspace, and therefore $L^2(M;F) = \operatorname{ran}_{H^m(M;E)} A \oplus (\operatorname{ran}_{H^m(M;E)} A)^\perp$. Thus, (5.111) for $f \in L^2(M;F)$ defines a linear operator $G \colon L^2(M;F) \to H^m(M;E)$. The membership $G \in \Psi^m(M;F,E)$ now again follows from (5.120).

Example 5.69 (Spectral Family of the Laplace–Beltrami Operator). Expanding upon Example 5.54, the operator $A = \Delta_g + \lambda \in \Psi^2(M)$ on a compact Riemannian manifold (M,g) is invertible as a map $H^s(M) \to H^{s-2}(M)$, $s \in \mathbb{R}$, whenever $\lambda \notin (-\infty, 0]$. Indeed, for $u \in \ker A$ one has $u \in \mathcal{C}^\infty(M)$ by elliptic regularity, and therefore $0 = \langle Au, u\rangle_{L^2(M;|\mathrm{d}g|)} = \int_M |\nabla u|_g^2 \, |\mathrm{d}g| + \lambda \|u\|^2_{L^2(M;|\mathrm{d}g|)} = 0$ which implies $u = 0$; one similarly shows that $\ker A^* = \ker(\Delta_g + \bar\lambda)$ is trivial.

The final observation in the proof of Theorem 5.67 is so important that we record it again in a slightly generalized form:

Theorem 5.70 (Elliptic Regularity). *Let* M *be a smooth manifold (not necessarily compact). Let* $A \in \Psi^m(M;E,F)$ *be elliptic. Let* $u \in \mathscr{E}'(M;E)$ *(or* $u \in \mathscr{D}'(M;E)$ *in case* A *is properly supported), and suppose that* $Au \in H^{s-m}_{\mathrm{loc}}(M;F)$, $s \in \mathbb{R}$. *Then* $u \in H^s_{\mathrm{loc}}(M;E)$.

Proof. Let $B \in \Psi^{-m}(M; F, E)$ be a properly supported parametrix of A, so $BA = I - R$, $R \in \Psi^{-\infty}(M; E)$. Then

$$u = BAu + Ru. \tag{5.158}$$

The first term lies in $H^s_{\mathrm{loc}}(M; E)$ and the second in $\mathcal{C}^\infty(M; E)$. □

An interesting application of elliptic theory on compact manifolds concerns the spectral theory of symmetric ps.d.o.s.

Theorem 5.71 (Self-adjointness of Elliptic Symmetric Ps.d.o.s). *Fix a volume density on M and a positive definite fiber inner product on $E \to M$. Let $m > 0$, and let $A \in \Psi^m(M; E)$ be elliptic and symmetric, that is, $\langle Au, v\rangle = \langle u, Av\rangle$ for $u, v \in \mathcal{C}^\infty(M; E)$, where $\langle \cdot, \cdot \rangle$ is the inner product on $L^2(M; E)$. Then A is an unbounded self-adjoint operator on $L^2(M; E)$ with domain $H^m(M; E)$. Its spectrum* $\operatorname{spec} A \subset \mathbb{R}$ *is discrete and accumulates only at ∞. There exists an orthonormal basis of $L^2(M; E)$ consisting of eigenfunctions of A, all of which are smooth.*

Proof. We first show that the operators $A \pm i\colon H^m(M; E) \to L^2(M; E)$ are surjective. By Theorem 5.67, their ranges are closed. Furthermore, any element $u \in (\operatorname{ran}(A \pm i))^\perp \subset L^2(M; E)$ lies in $\ker(A \pm i)^* = \ker(A \mp i) \subset \mathcal{C}^\infty(M; E)$, so

$$0 = \operatorname{Im}\langle (A \mp i)u, u\rangle = \mp \|u\|^2_{L^2(M;E)} \quad \Longrightarrow \quad u = 0. \tag{5.159}$$

By [RS72, Theorem VIII.3], this implies that A is self-adjoint with domain $\mathcal{D}(A) = H^m(M; E)$.

One can also argue directly: $v \in L^2(M; E)$ lies in $\mathcal{D}(A^*)$ if and only if $\mathcal{D}(A) \ni u \mapsto \langle Au, v\rangle$ satisfies a bound $|\langle Au, v\rangle| \le C\|u\|_{L^2}$ for some C. But $\langle Au, v\rangle = \langle u, A^*v\rangle$, and hence we conclude that $A^*v \in L^2(M; E)$. Elliptic regularity gives $v \in H^m(M; E)$; thus $\mathcal{D}(A^*) \subset \mathcal{D}(A)$. The converse is clear since A is symmetric.

To prove the discreteness of the spectrum, note first that $(A + i)^{-1}\colon L^2(M; E) \to H^m(M; E) \hookrightarrow L^2(M; E)$ is a compact operator, and hence its spectrum is discrete and can only accumulate at 0. Therefore, there exists a complex number $\lambda \in \mathbb{C}$ such that $i - \lambda^{-1} \in \mathbb{R}$ and $(A + i)^{-1} - \lambda$ is invertible on $L^2(M; E)$. But since

$$(A + i)^{-1} - \lambda = -\lambda(A + i)^{-1}(A + i - \lambda^{-1}), \tag{5.160}$$

this implies that the spectral family $A - \mu\colon H^m(M; E) \to L^2(M; E)$ is invertible for the spectral parameter $\mu := \lambda^{-1} - i \in \mathbb{R}$. Therefore, $(A - \mu)^{-1}\colon L^2(M; E) \to L^2(M; E)$ is compact *and self-adjoint*, and the spectral theorem produces an orthonormal basis of $L^2(M; E)$ consisting of eigenfunctions of $(A-\mu)^{-1}$ corresponding to a sequence of eigenvalues tending

to 0. But $(A-\mu)^{-1}\phi = \lambda\phi$ implies

$$A\phi = (\mu + \lambda^{-1})\phi, \tag{5.161}$$

and hence $\operatorname{spec} A$ accumulates only at ∞. Finally, if $\phi \in L^2(M;E)$ solves this eigenvalue equation, then since the right-hand side of (5.161) lies in $L^2(M;E)$, Theorem 5.70 implies $u \in H^m(M;E)$. This in turn implies that the right-hand side of (5.161) lies in $H^m(M;E)$, and Theorem 5.70 gives $u \in H^{2m}(M;E)$. Continuing in this fashion implies $u \in \bigcap_{k\in\mathbb{N}_0} H^{km}(M;E) = \mathcal{C}^\infty(M;E)$ by Sobolev embedding. □

Example 5.72 (Laplacian). Theorem 5.71 applies to the Laplacian Δ_g on any compact Riemannian manifold (M,g), acting on functions or differential forms.

Example 5.73 (Non-self-adjoint Case). There exist elliptic non-self-adjoint operators whose spectrum is the entire complex plane. In fact, there exists an elliptic operator $A \in \Psi^1(\mathbb{S}^1)$ with index 1 (or any other integer) (Exercise 5.24). For such A and by Remark 5.68, $A-\lambda$ is not invertible for *any* $\lambda \in \mathbb{C}$.

5.12 Elliptic Theory via Fredholm Estimates

We describe a slight variant of the proof of the Fredholm property of elliptic ps.d.o.s on closed manifolds which starts from *estimates* rather than parametrices. The key functional analytic result is the following.

Proposition 5.74 (Fredholm Estimates). *Let $(X, \|\cdot\|_X), (Y, \|\cdot\|_Y)$ be two Banach spaces, and suppose $A\colon X \to Y$ is a bounded linear map.*

(1) *Let $(Z, \|\cdot\|_Z)$ be a Banach space, and suppose $\iota\colon X \to Z$ is a compact inclusion (continuous and injective). Suppose there exists $C > 0$ such that*

$$\|u\|_X \le C(\|Au\|_Y + \|\iota u\|_Z), \quad u \in X. \tag{5.162}$$

Then $\ker A \subset X$ is finite-dimensional and $\operatorname{ran} A \subset Y$ is closed.

(2) *Let $(\tilde{Z}, \|\cdot\|_{\tilde{Z}})$ be a Banach space, and suppose $\tilde{\iota}\colon Y^* \hookrightarrow \tilde{Z}$ is a compact inclusion. Suppose there exists $C > 0$ such that*

$$\|v\|_{Y^*} \le C(\|A^*v\|_{X^*} + \|\tilde{\iota}v\|_{\tilde{Z}}), \quad v \in Y^*. \tag{5.163}$$

Assuming also the validity of (5.162), $A\colon X \to Y$ is then a Fredholm operator, and $\operatorname{ran} A = {}^\perp(\ker A^*) := \{f \in Y\colon \langle v, f\rangle = 0 \ \forall\, v \in \ker A^*\}$.

Proof. Due to its importance in Chapter 11, we present the full proof of this standard result. We drop the inclusions $\iota, \tilde{\iota}$ from the notation unless necessary for clarity.

For $u \in V := \ker A$, we have $\|u\|_X \le C\|u\|_Z$. Thus, $(V, \|\cdot\|_Z)$ is complete. The identity map $(V, \|\cdot\|_X) \to (V, \|\cdot\|_Z)$ is compact, while $(V, \|\cdot\|_Z) \to (V, \|\cdot\|_X)$ is continuous. By composing these two maps, we conclude that the identity map on $(V, \|\cdot\|_X)$ is compact. This implies $\dim V < \infty$.

Let now $W \subset X$ be a complementary subspace to V. We claim that there exists $C' < \infty$ such that

$$\|u\|_X \le C'\|Au\|_Y, \quad u \in W. \tag{5.164}$$

Assuming the contrary, there exists a sequence $u_k \in W$, $\|u_k\|_X = 1$, with $\|Au_k\|_Y \to 0$. Since $\{\iota u_k\} \subset Z$ is precompact, we may assume without loss of generality that ιu_k is a Cauchy sequence in Z. By (5.162), we then have

$$\|u_j - u_k\|_X \le C(\|Au_k - Au_j\|_Y + \|u_k - u_j\|_Z) \xrightarrow{j,k\to\infty} 0, \tag{5.165}$$

so u_k is a Cauchy sequence in X. Denote by $u \in X$ its limit. Since W is closed, $u \in W$. On the other hand, by continuity of A, we have $Au = \lim_{k\to\infty} Au_k = 0$, so $u \in V$, and therefore $u \in V \cap W = \{0\}$. But

$$\|u\|_Z = \lim_{k\to\infty} \|u_k\|_Z \ge \limsup_{k\to\infty} C^{-1}(\|u_k\|_X - \|Au_k\|) = C^{-1} > 0, \tag{5.166}$$

which is a contradiction.

It is now straightforward to conclude that $\operatorname{ran} A$ is closed: if $Au_k \to f$ in Y, then we may add to u_k suitable elements of V so as to arrange that $u_k \in W$. By (5.164), u_k is a Cauchy sequence in X, and its limit u satisfies $Au = f$. This proves part (1).

For part (2), we now know that $\operatorname{ran} A = \overline{\operatorname{ran} A}$ is closed and $\ker A^*$ is finite-dimensional. But for any bounded linear operator, we have $\overline{\operatorname{ran} A} = \{f \in Y : \langle v, f\rangle = 0 \ \forall\, v \in \ker A^*\}$. □

Another Proof of Theorem 5.67. We drop the bundles E, F from the notation. Since $A \in \Psi^m(M)$ is elliptic, we have *elliptic estimates*

$$\begin{aligned} \|u\|_{H^s} &\le C(\|Au\|_{H^{s-m}} + \|u\|_{H^{-N}}), \\ \|v\|_{H^{-s+m}} &\le C(\|A^*v\|_{H^{-s}} + \|v\|_{H^{-N}}). \end{aligned} \tag{5.167}$$

Here N is arbitrary but fixed; we choose it to satisfy $-N < s$ and $-N < -s + m$. The first estimate follows from applying $I = BA + R$, where $B \in \Psi^{-m}(M)$ is a parametrix and $R \in \Psi^{-\infty}(M)$, to u and taking H^s-norms, similarly for the second estimate. When $s = \frac{m}{2}$ (so $(H^{s-m})^* = H^s$), we are directly in the setting of Proposition 5.74. For general s, one can repeat the

arguments in the proof of Proposition 5.74 to conclude the Fredholm property of $A\colon H^s(M) \to H^{s-m}(M)$. □

Remark 5.75 (Estimates Versus Parametrices). Since the proof of the crucial estimates (5.167) uses the existence of a (pseudodifferential) parametrix, one may wonder what has been gained from this slight change in perspective. The point is that in *non-elliptic* problems, it is often much easier to prove estimates than to construct parametrices: all a priori estimates for non-elliptic operators that we will discuss in this book (Chapters 8 and 10) are proved using the algebra of ps.d.o.s as a tool, whereas parametrices would be more complicated objects. (The impatient reader may also take a peek at Exercises 8.7, 9.7, and 9.8.)

We proceed to use the language of Fredholm operators to study a variant of Theorem 5.71 that does not require self-adjointness. (Spectral theory outside of the confines of self-adjointness arises in many applications, such as in Chapter 11. The following discussion may be considered a warm-up.) An important tool for finding eigenvalues in non-self-adjoint settings is the *analytic Fredholm theorem*. To state it, we need the following notion.

Definition 5.76 (Finite-Meromorphic Functions). Let X, Y denote two Banach spaces. Let $\Omega \subset \mathbb{C}$ be an open set. Then we say that $B(\sigma)\colon X \to Y$, $\sigma \in \Omega$, is *finite-meromorphic* if there exists a closed discrete subset $D = \{\sigma_1, \sigma_2, \ldots\} \subset \Omega$ such that:

(1) $B(\sigma)$ is holomorphic on $\Omega \setminus D$.
(2) For all j, there exists $k_j \in \mathbb{N}$ such that

$$B(\sigma) = \sum_{k=1}^{k_j} (\sigma - \sigma_j)^{-k} B_{jk} + \tilde{B}_j(\sigma), \tag{5.168}$$

where $\tilde{B}_j(\sigma)\colon X \to Y$ is holomorphic near $\sigma = \sigma_j$, and $B_{jk}\colon X \to Y$, $1 \le k \le k_j$, is a finite rank operator.

Theorem 5.77 (Analytic Fredholm Theorem). *Let X, Y be Banach spaces, let $\Omega \subset \mathbb{C}$ be open and connected, and suppose $A(\sigma)\colon X \to Y$, $\sigma \in \Omega$, is an analytic family of Fredholm operators. Then either $A(\sigma)$ is not invertible for any $\sigma \in \Omega$ or $A(\sigma)^{-1}$ is finite-meromorphic.*

Proof. Suppose the (open) set $\Omega' \subset \Omega$ of σ for which $A(\sigma)$ is invertible is non-empty; then $A(\sigma)$ has index 0 for $\sigma \in \Omega'$, and hence for all $\sigma \in \Omega$.

If $\Omega' \neq \Omega$, let $\sigma_0 \in \Omega \cap \partial\Omega'$. Consider $A(\sigma_0)\colon X \to Y$. Let $X_2 = \ker A(\sigma_0)$ and $R_1 = \operatorname{ran} A(\sigma_0)$. Pick closed subspaces $X_1 \subset X$ and $Y_2 \subset Y$ with

$$X = X_1 \oplus X_2, \quad Y = Y_1 \oplus Y_2. \tag{5.169}$$

Since $\operatorname{ind} A(\sigma_0) = 0$, $\dim X_2 = \dim Y_2 =: N < \infty$. We write $A(\sigma)$ as a block matrix in the decomposition (5.169),

$$A(\sigma) = \begin{pmatrix} P(\sigma) & Q(\sigma) \\ S(\sigma) & T(\sigma) \end{pmatrix}, \tag{5.170}$$

where $P(\sigma_0)\colon X_1 \to Y_1$ is invertible and $Q(\sigma_0)$, $S(\sigma_0)$, $T(\sigma_0) = 0$. Thus, $P(\sigma)\colon X_1 \to Y_1$ is invertible for $|\sigma - \sigma_0| < \epsilon$ for some $\epsilon > 0$. By the Schur complement formula (i.e., block-wise inversion of $A(\sigma)$), $A(\sigma)$ is invertible for $|\sigma - \sigma_0| < \epsilon$ if and only if

$$Z(\sigma) := T(\sigma) - S(\sigma)P(\sigma)^{-1}Q(\sigma)\colon X_2 \to Y_2 \tag{5.171}$$

is invertible; in this case, we have

$$A(\sigma)^{-1} = \begin{pmatrix} P^{-1} + P^{-1}QZ^{-1}SP^{-1} & -P^{-1}QZ^{-1} \\ -Z^{-1}SP^{-1} & Z^{-1} \end{pmatrix}. \tag{5.172}$$

But $Z(\sigma)$ is a holomorphic $N \times N$ *matrix* near σ_0, and it is invertible for some σ arbitrarily close to σ_0 due to our choice of σ_0. Hence, fixing a basis of X_2 and Y_2, its determinant $\det Z(\sigma)$ is a nonzero holomorphic function which vanishes at $\sigma = \sigma_0$; therefore, $\det Z(\sigma)^{-1}$ is meromorphic, and thus so is $Z(\sigma)^{-1}$. Therefore, $A(\sigma)$ is invertible in a punctured neighborhood of σ_0. The conclusion is now immediate from (5.172). □

We give a simple application that is as much as possible in the spirit of Theorem 5.71:

Theorem 5.78 (Finite-meromorphy of Resolvent Families). *Let $m > 0$. Suppose $A \in \Psi^m(M; E)$ is elliptic, and assume that there exists $\lambda_0 \in \mathbb{C}$ such that $A - \lambda_0\colon \mathcal{C}^\infty(M; E) \to \mathcal{C}^\infty(M; E)$ is invertible. Then the resolvent*

$$R(\lambda) := (A - \lambda)^{-1} \in \mathcal{L}(L^2(M; E), H^m(M; E)) \tag{5.173}$$

is finite-meromorphic. A number $\lambda_0 \in \mathbb{C}$ is a pole of $R(\lambda)$ if and only if there exists $0 \neq u \in \mathcal{C}^\infty(M; E)$ with $(A - \lambda)u = 0$.

Proof. Due to the ellipticity of A, the operator $A - \lambda\colon H^m(M; E) \to L^2(M; E)$ is Fredholm for all λ, and it depends holomorphically on λ. The meromorphicity of $R(\lambda)$ is now immediate from Theorem 5.77.

Since $A - \lambda_0$ has index 0, so does $A - \lambda$ for all $\lambda \in \mathbb{C}$, and hence the invertibility of $A - \lambda$ is equivalent to its injectivity. □

Example 5.73 shows that the existence of λ_0 is a nontrivial assumption. An analogue of Exercise 4.12 on compact manifolds gives a class of examples

$A \in \mathrm{Diff}^m(M;E)$ for which the existence of λ_0 follows provided $\sigma^m(A)$ avoids a sector in the complex plane; this is left to the reader.

5.13 A Nonlinear Example

As a simple and weak but instructive nonlinear application of the elliptic theory developed thus far, we shall solve a nonlinear elliptic equation on a compact two-dimensional manifold M without boundary. If g is a smooth Riemannian metric on M, we denote the Gauss curvature of M by $K_g \in \mathcal{C}^\infty(M)$. If $\phi \in \mathcal{C}^\infty(M)$, then the metric $g'(x) = e^{2\phi(x)}g(x)$ is said to be *conformal* to g. The Gauss curvature of g' is given by

$$K_{g'} = e^{-2\phi}(K_g + \Delta_g \phi). \tag{5.174}$$

We recall the expression for K_g in local coordinates $(x_1, x_2) \in \mathbb{R}^2$. Write $\partial_i := \partial_{x_i}$ and $g_{ij} = g(\partial_i, \partial_j)$; write g^{ij} for the components of the inverse matrix $(g_{ij})^{-1}$, and $\Gamma^k_{ij} = \frac{1}{2}g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$ for the Christoffel symbols of g. Then

$$K_g := \frac{1}{2}\sum_{i,j,k} g^{jk} R^i{}_{jik}, \quad R^i{}_{jik} := \partial_i \Gamma^i_{jk} - \partial_k \Gamma^i_{ij} + \Gamma^i_{il}\Gamma^l_{jk} - \Gamma^i_{kl}\Gamma^l_{ij}. \tag{5.175}$$

Theorem 5.79 (Local Version of the Uniformization Theorem in Negative Curvature). *Suppose (M, g) has constant Gauss curvature $K_g \equiv -1$. Let $\tilde{g} \in \mathcal{C}^\infty(M; S^2T^*M)$ be a Riemannian metric with $\|g - \tilde{g}\|_{H^4(M;S^2T^*M)} < \epsilon$, $\epsilon > 0$ small. (Here we use g to define the fiber inner product on S^2T^*M.) Then there exists $\phi \in \mathcal{C}^\infty(M)$ such that $e^{2\phi}\tilde{g}$ has constant Gauss curvature -1.*

This is a local version of the uniformization theorem in negative curvature. The assumptions require that M is a surface of genus ≥ 2; the conclusion then in fact holds for *all* $\tilde{g}$. A full proof is given in Taylor [Tay11c, Chapter 14.2]. (For $M \cong \mathbb{S}^2$, one can always find a conformal multiple $e^{2\phi}\tilde{g}$ of $\tilde{g}$ with constant curvature $+1$; for $M \cong \mathbb{T}^2$, one can always find one with constant curvature 0.)

In our proof of Theorem 5.79, we will use the algebra properties of Sobolev spaces of sufficiently high orders on manifolds (Exercise 5.29): since M is two-dimensional, $H^s(M)$ is an algebra under pointwise multiplication for $s > 1$, with

$$\|uv\|_{H^s(M)} \leq C_s \|u\|_{H^s(M)} \|v\|_{H^s(M)} \tag{5.176}$$

for some constant C_s. Furthermore, we recall that $H^s(M) \hookrightarrow \mathcal{C}^k(M)$ when $s > 1 + k$.

Proof of Theorem 5.79. Metrics $g \in H^4(M; S^2T^*M)$ are twice continuously differentiable, and therefore their Gauss curvature is a continuous function on M. More precisely, in smooth local coordinates $(x_1, x_2) \in U$ on M, we have $g_{ij} \in H^4_{\mathrm{loc}}(U)$, therefore $\det(g_{ij}) \in H^4_{\mathrm{loc}}(U)$ as well and thus also $g^{ij} \in H^4_{\mathrm{loc}}(U)$. This implies $\Gamma^k_{ij} \in H^3_{\mathrm{loc}}(U)$ and $R^i{}_{jik} \in H^2_{\mathrm{loc}}(U)$. By (5.175), K_g lies in $H^2_{\mathrm{loc}}(U)$ in local coordinates. Globally on M, we have thus shown that $K_g \in H^2(M)$; and the map

$$H^4(M; S^2T^*M) \ni g \mapsto K_g \in H^2(M; S^2T^*M) \tag{5.177}$$

is continuous, and in fact Lipschitz.

We want to solve the equation

$$-1 = K_{e^{2\phi}\tilde{g}} = e^{-2\phi}(K_{\tilde{g}} + \Delta_{\tilde{g}}\phi), \tag{5.178}$$

or equivalently

$$\Delta_{\tilde{g}}\phi + e^{2\phi} - 1 = -(K_{\tilde{g}} + 1). \tag{5.179}$$

Since $\|K_{\tilde{g}} + 1\|_{H^2(M)} = \|K_{\tilde{g}} - K_g\|_{H^2(M)}$ is small (namely, smaller than a constant times ϵ), we expect $\phi \in H^4(M)$ to be small. (We remark that for $\phi \in H^4(M)$, the series $\sum_{j=0}^{\infty} \frac{1}{j!}(2\phi)^j$ converges in $H^4(M)$ in view of (5.176), so $e^{2\phi} \in H^4(M)$ is well-defined.) This suggests Taylor expanding around $\phi = 0$, which turns (5.179) into

$$\begin{aligned} &A\phi = E(\phi) - N(\phi), \\ &A = \Delta_g + 2, \quad E(\phi) = -(K_{\tilde{g}} + 1) - (\Delta_{\tilde{g}} - \Delta_g)\phi, \quad N(\phi) = e^{2\phi} - 1 - 2\phi. \end{aligned} \tag{5.180}$$

We solve this using the contraction mapping principle, i.e., by iterating the map

$$T\colon H^4(M) \ni \phi \mapsto A^{-1}(E(\phi) - N(\phi)) \in H^4(M). \tag{5.181}$$

Recall from Example 5.69 that A is invertible as a map $H^s(M) \to H^{s-2}(M)$ for all $s \in \mathbb{R}$.

Now, the local coordinate expression (5.122) for the Laplace operator, which can be rewritten as $\Delta_g u = -\sum_{i,j}(g^{ij}\partial_i\partial_j - \sum_k \Gamma^k_{ij}\partial_k)u$, shows that $\|\Delta_{\tilde{g}} - \Delta_g\|_{\mathcal{L}(H^4(M), H^2(M))} \leq C\epsilon$ for some constant C, thus

$$\|E(\phi)\|_{H^2} \leq C\epsilon(1 + \|\phi\|_{H^4(M)}). \tag{5.182}$$

Moreover,

$$\|N(\phi)\|_{H^2(M)} \le \|N(\phi)\|_{H^4(M)} \le \sum_{j=2}^{\infty} \frac{1}{j!} \|(2\phi)^j\|_{H^4(M)} \le C\|\phi\|^2_{H^4(M)} \tag{5.183}$$

for $\|\phi\|_{H^4(M)} \le 1$. Therefore, if $\|\phi\|_{H^4(M)} \le \delta$ where $\delta \in (0,1]$, then

$$\|T\phi\|_{H^4(M)} \le C'(C\epsilon(1+\delta)+C\delta^2), \quad C' = \|A^{-1}\|_{\mathcal{L}(H^2(M),H^4(M))}. \tag{5.184}$$

Fix $\delta_0 > 0$ such that $C'C\delta_0 \le \frac{1}{2}$; then for $0 < \delta \le \min(\delta_0, 1)$, we obtain $\|T\phi\|_{H^4(M)} \le \delta$ provided $\epsilon > 0$ is sufficiently small. Thus, T maps the δ-ball in $H^4(M)$ into itself.

Shrinking δ further (and then also ϵ), the map T is moreover a contraction on the δ-ball in $H^4(M)$ since we can then arrange for

$$\begin{aligned} \|T\phi - T\psi\|_{H^4(M)} &\le C'\big(C\epsilon\|\phi-\psi\|_{H^4(M)} \\ &\qquad + C\|\phi-\psi\|_{H^4(M)}(\|\phi\|_{H^4(M)} + \|\psi\|_{H^4(M)})\big) \\ &\le C'(C\epsilon + 2C\delta)\|\phi-\psi\|_{H^4(M)} \\ &\le \frac{1}{2}\|\phi-\psi\|_{H^4(M)}. \end{aligned} \tag{5.185}$$

Here we use $N(\phi) - N(\psi) = \sum_{j=2}^{\infty} \frac{1}{j!} 2^j (\phi - \psi) \sum_{k=0}^{j-1} \phi^k \psi^{j-1-k}$ and the triangle inequality.

Let now $\phi \in H^4(M)$, $\|\phi\|_{H^4(M)} \le \delta$, denote the unique fixed point of T; then ϕ solves (5.179). We rewrite this one last time as

$$\Delta_{\tilde{g}}\phi = -K_{\tilde{g}} - e^{2\phi}. \tag{5.186}$$

Suppose we already know $\phi \in H^k(M)$, $k \ge 4$. Then the right-hand side of this equation lies in $H^k(M)$, so by elliptic regularity (Theorem 5.70), we conclude that $\phi \in H^{k+2}(M)$. Therefore, $\phi \in \bigcap_k H^k(M) = \mathcal{C}^\infty(M)$. □

5.14 Commutators and Symplectic Geometry

We tie up a loose end and describe, invariantly, the principal symbol of the commutator of two ps.d.o.s. Key is the *symplectic structure* of the cotangent bundle T^*M.

Definition 5.80 (Canonical 1-Form and Symplectic Form on T^*M). Let M be an n-dimensional manifold. The *canonical 1-form* on T^*M is the section $\alpha \in \mathcal{C}^\infty(T^*M; T^*(T^*M))$ defined by

$$\alpha_{(x,\xi)}(v) := \xi(\pi_* v), \quad x \in M,\ \xi \in T_x^* M,\ v \in T_{(x,\xi)}(T^* M), \tag{5.187}$$

where $\pi\colon T^*M \to M$ is the projection. The *canonical symplectic form* on T^*M is

$$\omega := -\mathrm{d}\alpha \in \mathcal{C}^\infty(T^*M; \Lambda^2(T^*M)). \tag{5.188}$$

In local coordinates $x \in \mathbb{R}^n$ and corresponding canonical momentum coordinates $\xi \in \mathbb{R}^n$ on the fibers of T^*M, we have $\pi_*(\sum_k a_k \partial_{x_k} + b_k \partial_{\xi_k}) = \sum_k a_k \partial_{x_k}$ and therefore

$$\alpha = \sum_{k=1}^n \xi_k \,\mathrm{d}x_k, \quad \omega = \sum_{k=1}^n \mathrm{d}x_k \wedge \mathrm{d}\xi_k. \tag{5.189}$$

This is a non-degenerate 2-form: contraction $T(T^*M) \ni v \mapsto v \lrcorner \omega = \omega(v, -) \in T^*(T^*M)$ is an isomorphism and identifies vector fields and 1-forms on T^*M:

$$\sum_{k=1}^n a_k \partial_{x_k} + b_k \partial_{\xi_k} \quad \overset{\lrcorner\omega}{\longmapsto} \quad \sum_{k=1}^n -b_k \,\mathrm{d}x_k + a_k \,\mathrm{d}\xi_k. \tag{5.190}$$

Definition 5.81 (Hamiltonian Vector Field). Let $p \in \mathcal{C}^\infty(T^*M)$. Then the *Hamiltonian vector field* of p is the unique $H_p \in \mathcal{V}(T^*M)$ such that

$$H_p \lrcorner \omega = \mathrm{d}p. \tag{5.191}$$

The *Poisson bracket* of $p, q \in \mathcal{C}^\infty(T^*M)$ is defined as

$$\{p, q\} := H_p q = -H_q p. \tag{5.192}$$

In local coordinates, we deduce from (5.190) that

$$H_p = \sum_{k=1}^n (\partial_{\xi_k} p)\partial_{x_k} - (\partial_{x_k} p)\partial_{\xi_k}. \tag{5.193}$$

Thus, the ad hoc definition (4.95) in fact makes invariant sense on T^*M. As a consequence of the local $\mathbb{R}^n$ result (Proposition 4.25), we thus deduce:

Corollary 5.82 (Principal Symbols of Commutators). *Let $A \in \Psi^m(M)$, $B \in \Psi^{m'}(M)$, at least one of which is properly supported. Then $[A, B] \in \Psi^{m+m'-1}(M)$ and*

$$\sigma^{m+m'-1}(i[A, B]) = \{\sigma^m(A), \sigma^{m'}(B)\}. \tag{5.194}$$

5.15 Semiclassical Ps.d.o.s

Adapting the definitions introduced thus far to the semiclassical setting is straightforward; we only state the definitions and results and leave the proofs to the reader. The starting point is the following analogue of Theorem 5.2.

Theorem 5.83 (Semiclassical Ps.d.o.s Under Changes of Variables). *Suppose $\Omega, \Omega' \subset \mathbb{R}^n$ are open and $\kappa\colon \Omega \to \Omega'$ is a diffeomorphism. Write $\Psi^m_{\hbar,\mathrm{c}}(\Omega)$ for the space of $B = (B_h)_{h\in(0,h_0)} \in \Psi^m_\hbar(\mathbb{R}^n)$ for which there exists a compact subset $K \subset \Omega$ with the property that $\operatorname{supp} K_{B_h} \subset K \times K$ for all $h \in (0, h_0)$, where $K_{B_h} \in \mathscr{S}'(\mathbb{R}^{2n})$ denotes the Schwartz kernel of B_h. Then the map*

$$\Psi^m_{\hbar,\mathrm{c}}(\Omega') \ni A \mapsto A_\kappa := \kappa^* A(\kappa^{-1})^* \in \Psi^m_{\hbar,\mathrm{c}}(\Omega) \tag{5.195}$$

is a bijection. Moreover, $\sigma^m_\hbar(A_\kappa)(x,\xi) = \sigma^m_\hbar(A)(\kappa(x), (\kappa'(x)^T)^{-1}\xi)$.

Definition 5.84 (Semiclassical Ps.d.o.s on Manifolds). Let M be a smooth n-dimensional manifold.

(1) Let $m \in \mathbb{R} \cup \{-\infty\}$. Then $\Psi^m_\hbar(M)$ is the space of families $A = (A_h)_{h\in(0,h_0)}$ (for some $h_0 > 0$) of linear operators $A_h\colon \mathcal{C}^\infty_\mathrm{c}(M) \to \mathcal{C}^\infty(M)$ with the following properties:

  (i) If $F\colon U \to F(U) \subset \mathbb{R}^n$ is a coordinate chart and $\psi \in \mathcal{C}^\infty_\mathrm{c}(U)$, then there exists $B = (B_h)_{h\in(0,h_0)} \in \Psi^m_{\hbar,\mathrm{c}}(F(U)) \subset \Psi^m_\hbar(\mathbb{R}^n)$ such that on U, we have $\psi A_h(\psi u) = F^*(B_h((F^{-1})^*(\psi u)))$ for all $u \in \mathcal{C}^\infty_\mathrm{c}(M)$.

  (ii) If $\phi, \psi \in \mathcal{C}^\infty(M)$, $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$, then there exists $K = (K_h)_{h\in(0,h_0)}$ with $K_h \in \mathcal{C}^\infty(M^2; \Omega_R)$ (see (5.49)) such that $\phi A_h(\psi u) = \int_M K_h(\cdot, y)u(y)$ for all $u \in \mathcal{C}^\infty_\mathrm{c}(M)$, and $\{h^{-N}K_h\colon h \in (0,h_0)\}$ is a bounded subset of $\mathcal{C}^\infty(M^2;\Omega_R)$ for all $N \in \mathbb{R}$.

(2) For $k \in \mathbb{R}$, we set $h^{-k}\Psi^m_\hbar(M) = \{(h^{-k}A_h)_{h\in(0,h_0)}\colon (A_h)_{h\in(0,h_0)} \in \Psi^m_\hbar(M)\}$. The space of *residual semiclassical ps.d.o.s* is

$$h^\infty \Psi^{-\infty}_\hbar(M) = \bigcap_{k\in\mathbb{R}} h^{-k}\Psi^{-\infty}_\hbar(M). \tag{5.196}$$

(3) We call an element $A = (A_h)_{h\in(0,h_0)} \in \Psi^m_\hbar(M)$ *properly supported* if for every compact $K \subset M$ there exists a compact set $K' \subset M$ such that $\pi_L^{-1}(K) \cap \operatorname{supp} K_{A_h} \subset K \times K'$ and $\pi_R^{-1}(K) \cap \operatorname{supp} K_{A_h} \subset K' \times K$ for all $h \in (0,h_0)$; here $\pi_{L/R}\colon M \times M \to M$ is the left/right projection.

Analogously to Remark 5.22, Definition 5.84 produces a larger class of semiclassical ps.d.o.s in the case $M = \mathbb{R}^n$ than previously defined in Definition 4.44. The notational clash will not cause any confusion in this book,

and hence we shall not adorn the space of Definition 4.44 with a subscript "∞" (for "uniform" semiclassical ps.d.o.s). Elements $R = (R_h)_{h\in(0,h_0)} \in h^\infty \Psi_\hbar^{-\infty}(M)$ are precisely those families of operators for which $h^{-N} K_{R_h}$ is bounded in $\mathcal{C}^\infty(M^2;\Omega_R)$ for all N; here K_{R_h} denotes the Schwartz kernel of R_h. Operators acting on sections of vector bundles can be defined in an analogous fashion as in Definition 5.42.

A simple class of examples of semiclassical ps.d.o.s is given by semiclassical *differential* operators, i.e., $\mathrm{Diff}_\hbar^m(M) \subset \Psi_\hbar^m(M)$ for $m \in \mathbb{N}_0$; here $\mathrm{Diff}_\hbar^m(M)$ consists of all those families of differential operators on M that, when cut off to a coordinate chart, are given by an element of $\mathrm{Diff}_\hbar^m(\mathbb{R}^n)$. Inside of $\mathrm{Diff}_\hbar^1(M)$ is the space $\mathcal{V}_\hbar(M)$ of *semiclassical vector fields* (given by elements of $\mathcal{V}_\hbar(\mathbb{R}^n)$ in each chart).

The proof of Theorem 5.32 carries over to show:

Theorem 5.85 (Compositions of Semiclassical Ps.d.o.s). *Let $A \in \Psi_\hbar^m(M)$, $B \in \Psi_\hbar^{m'}(M)$, with at least one of them properly supported. Then $A \circ B \in \Psi_\hbar^{m+m'}(M)$.*

Turning to symbols, consider first a general vector bundle $E \to M$; we then set

$$S_\hbar^m(E) := \{\text{bounded families } (a_h)_{h\in(0,h_0)} \text{ of elements } a_h \in S^m(E)\}, \tag{5.197}$$

and more generally $h^{-k} S_\hbar^m(E) = \{(h^{-k} a_h)_{h\in(0,h_0)} \colon (a_h) \in S_\hbar^m(E)\}$. A construction completely analogous to the one leading up to Definition 5.38, now relying on Theorem 5.83, allows one to define the principal symbol of semiclassical ps.d.o.s as an element

$$\sigma_\hbar^m(A) \in S_\hbar^m(T^*M)/hS_\hbar^{m-1}(T^*M), \quad A \in \Psi_\hbar^m(M), \tag{5.198}$$

fitting into the short exact sequence

$$0 \to h\Psi_\hbar^{m-1}(M) \hookrightarrow \Psi_\hbar^m(M) \xrightarrow{\sigma_\hbar^m} S_\hbar^m(T^*M)/hS_\hbar^{m-1}(T^*M) \to 0. \tag{5.199}$$

A right inverse of the third arrow can be constructed using a quantization map, which can be defined via (5.92), except now using $\mathrm{Op}_{\hbar,L}$ instead of Op_L. This map has the following properties:

$$\begin{aligned} \mathrm{Op}_\hbar \colon S_\hbar^m(T^*M) \to \Psi_\hbar^m(M) &\text{ is continuous for all } m \in \mathbb{R}, \\ &\text{surjective modulo } h^\infty \Psi^{-\infty}(M), \\ &\mathrm{Op}_\hbar(1) = I. \end{aligned} \tag{5.200}$$

Example 5.86 (Semiclassical Pseudodifferential Operators). If $F \colon U \subset M \to F(U) \subset \mathbb{R}^n$ is a coordinate chart on the smooth manifold M, and if $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(U)$ and $\tilde{\chi} \in \mathcal{C}_{\mathrm{c}}^\infty(F(U))$, then the operator

$$A_h \colon u \mapsto F^*(\tilde{\chi}\langle hD\rangle^s (F^{-1})^*(\chi u)) \tag{5.201}$$

defines an element $A = (A_h)_{h\in(0,1)} \in \Psi_h^s(M)$. The sum of such operators when F, U ranges over a cover of M by coordinate charts, χ over a subordinate partition of unity, and $\tilde{\chi} \in \mathcal{C}_c^\infty(F(U))$ is chosen to be equal to 1 on $F(\operatorname{supp}\chi)$ for all such F, U, χ, defines an elliptic element $\Lambda = (\Lambda_h)_{h\in(0,1)} \in \Psi_h^s(M)$ whose semiclassical principal symbol is semiclassically elliptic. In the special case that $M = \mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$ is the n-torus, the Fourier multiplier $e^{ik\cdot x} \mapsto \langle hk\rangle^s e^{ik\cdot x}$, $k \in \mathbb{Z}^n$, defines an element of $\Psi_h^s(M)$ (Exercise 5.25).

Next, the properties of the semiclassical principal symbol in Theorem 4.50 carry over to the manifold setting analogously to Proposition 5.40 and Corollary 5.82 (and Proposition 5.43, though we do not state the bundle-valued results here for brevity):

Proposition 5.87 (Semiclassical Principal Symbol). *Let $A \in \Psi_h^m(M)$, $B \in \Psi_h^{m'}(M)$. If at least one of the operators A, B is properly supported, then*

$$\sigma_h^{m+m'}(A\circ B) = \sigma_h^m(A)\sigma_h^{m'}(B), \quad \sigma_h^{m+m'-1}\Big(\frac{i}{h}[A,B]\Big) = \{\sigma_h^m(A), \sigma_h^{m'}(B)\}. \tag{5.202}$$

Furthermore, the (formal) adjoint $A^* \in \Psi_h^m(M;\Omega M)$ has principal symbol $\sigma_h^m(A^*) = \overline{\sigma_h^m(A)}$.

A standard application of the semiclassical calculus is the following result whose proof is left to the reader.

Theorem 5.88 (Semiclassical Elliptic Parametrix). *Let $A \in \Psi_h^m(M)$ be semiclassically elliptic. Then there exists a properly supported operator $B \in \Psi_h^{-m}(M)$ such that $A\circ B - I$, $B\circ A - I \in h^\infty \Psi_h^{-\infty}(M)$.*

To state estimates for semiclassical ps.d.o.s, we next introduce semiclassical Sobolev spaces. The key point, as already in Definition 2.26, is to capture uniform bounds as $h \searrow 0$; this means that we need to introduce seminorms depending in a controlled fashion on the parameter $h > 0$.

Definition 5.89 (Semiclassical Local Sobolev Spaces). Let M be an n-dimensional manifold, $s \in \mathbb{R}$.

(1) For a coordinate chart $F\colon U \to F(U) \subset \mathbb{R}^n$ on M and a cutoff function $\chi \in \mathcal{C}_c^\infty(U)$, we define for $u \in H^s_{\mathrm{loc}}(M)$ the h-dependent seminorm

$$\begin{aligned} |u|_{F,\chi,h} &:= \|u_{F,\chi}\|_{H_h^s(\mathbb{R}^n)}, \\ u_{F,\chi} &:= (F^{-1})^*(\chi u)\colon \mathcal{C}_c^\infty(\mathbb{R}^n) \ni \phi \mapsto \langle u, \chi F^*(\phi\,|\mathrm{d}x|)\rangle. \end{aligned} \tag{5.203}$$

We write $H^s_{h,\mathrm{loc}}(M)$ for the space $H^s_{\mathrm{loc}}(M)$ equipped with these seminorms.

(2) A family $u = (u_h)_{h\in(0,h_0)}$ is *uniformly bounded* in $H^s_{h,\mathrm{loc}}(M)$ if $\sup_{h\in(0,h_0)} |u_h|_{F,\chi,h} < \infty$ for all charts $F\colon U \to F(U) \subset \mathbb{R}^n$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(F(U))$. More generally, we say that u is uniformly bounded in $h^k H^s_{h,\mathrm{loc}}(M)$ if $(h^{-k}u_h)_{h\in(0,h_0)}$ is uniformly bounded in $H^s_{h,\mathrm{loc}}(M)$.

We leave it to the interested reader to define, building on (5.19), the appropriate h-dependent seminorms on $H^s_{\mathrm{c}}(M)$ which give rise to a space $H^s_{h,\mathrm{c}}(M)$. When M is compact, we can give $H^s_h(M) := H^s_{h,\mathrm{loc}}(M)$ the structure of a Hilbert space: the squared norm of an element $u \in H^s_h(M)$ is the sum of $\|u_{F,\chi}\|^2_{H^s_h(\mathbb{R}^n)}$ where F runs over a finite cover of M by coordinate charts $F\colon U \to F(U) \subset \mathbb{R}^n$ and χ runs over a subordinate partition of unity. We denote this family of Hilbert spaces by

$$(H^s_h(M), \|\cdot\|_{H^s_h(M)}), \quad h \in (0,1). \tag{5.204}$$

More generally, if K is a compact subset of the (possibly non-compact) manifold M, then we can define the family of Hilbert spaces

$$(\dot H^s_h(K), \|\cdot\|_{\dot H^s_h(K)}), \quad h \in (0,1), \tag{5.205}$$

in an analogous fashion; here $\dot H^s_h(K) = \dot H^s(K)$ as a vector space, but we equip this space with an h-dependent norm.

Example 5.90 (δ-Distribution). Consider the h-independent distribution $u = \delta \in \mathscr{D}'(\mathbb{R}^n)$. Then $\hat u(\xi) = 1$ and thus for $s < -\frac{n}{2}$

$$\|u\|^2_{H^s_h(\mathbb{R}^n)} = \int_{\mathbb{R}^n} \langle h\xi\rangle^{2s}\,\mathrm{d}\xi = Ch^{-n}, \quad C = \int_{\mathbb{R}^n} \langle\zeta\rangle^{2s}\,\mathrm{d}\zeta < \infty. \tag{5.206}$$

Therefore, u is uniformly bounded in $h^{-\frac{n}{2}} H^s_h(\mathbb{R}^n)$.

Proposition 5.91 (Boundedness of Semiclassical Ps.d.o.s). *Let $A = (A_h)_{h\in(0,h_0)} \in \Psi^m_h(M)$. Then A_h is uniformly bounded as a map $H^s_{h,\mathrm{c}}(M) \to H^{s-m}_{h,\mathrm{loc}}(M)$, i.e., for every compact $K \subset M$, the restriction*

$$A_h\colon \dot H^s_h(K) \to H^{s-m}_{h,\mathrm{loc}}(M) \tag{5.207}$$

is uniformly bounded. If A is properly supported, then also $A_h\colon H^s_{h,\mathrm{loc}}(M) \to H^{s-m}_{h,\mathrm{loc}}(M)$ is uniformly bounded. When M is compact, then $A_h\colon H^s_h(M) \to H^{s-m}_h(M)$ is uniformly bounded.

Proof. This follows from a simple modification of the proofs of Proposition 5.59 and Corollary 5.60, which we leave to the reader. □

As an application, we can prove the invertibility of semiclassically elliptic ps.d.o.s on compact manifolds; this is an amalgamation of Theorems 4.52 and 5.67:

Theorem 5.92 (Invertibility of Elliptic Semiclassical Ps.d.o.s). *Let M be a compact manifold, and let $A \in \Psi_\hbar^m(M)$ be semiclassically elliptic. Then there exists $h_0 > 0$ such that for all $h \in (0, h_0]$ and $s \in \mathbb{R}$, the operator $A_h \colon H_h^s(M) \to H_h^{s-m}(M)$ (or equivalently $H^s(M) \to H^{s-m}(M)$) is invertible. Moreover, the inverse is itself a semiclassical pseudodifferential operator,*

$$(A_h^{-1})_{h\in(0,h_0)} \in \Psi_\hbar^{-m}(M). \tag{5.208}$$

Proof. Let $B \in \Psi_\hbar^{-m}(M)$ be a semiclassical elliptic parametrix, so $R := I - BA$, $R' := I - AB \in h^\infty \Psi_\hbar^{-\infty}(M)$. Then the operator norms of R_h, R_h' on $L^2(M)$ are less than $\frac{1}{2}$ for all $h \leq h_0$ when h_0 is sufficiently small. Therefore, $(I-R)^{-1}BA = I$ on $H_h^m(M)$ and $AB(I-R')^{-1} = I$ on $L^2(M)$, giving the existence of an inverse $A_{h,L^2}^{-1} \colon L^2(M) \to H_h^m(M)$ of A. We can express A_{h,L^2}^{-1} using the formula (4.156) as the sum of an element in $\Psi_\hbar^{-m}(M)$ and an operator $R'' = (R_h'')_{h\in(0,h_0)}$, where $R_h'' \colon H_h^{-N}(M) \to H_h^N(M)$ is bounded in operator norm by $C_N h^N$ for all N. This implies the invertibility of $A_h \colon H_h^s(M) \to H_h^{s-m}(M)$ for all $s \in \mathbb{R}$ by a continuity and density argument. Lemma 5.93 implies that $R'' \in h^\infty \Psi_\hbar^{-\infty}(M)$, which finishes the proof also of (5.208). □

To complete the argument, we need the following result.

Lemma 5.93 (Residual Operators via Mapping Properties). *Let M be compact, and let $R = (R_h)_{h\in(0,h_0)}$ be a family of operators with the property that for all $N \in \mathbb{R}$, there exists a constant C_N such that the operator norm of $R_h \colon H_h^{-N}(M) \to H_h^N(M)$ is bounded by $C_N h^N$ for all $h \in (0, h_0)$. Then $R \in h^\infty \Psi_\hbar^{-\infty}(M)$.*

Proof. We trivialize the density bundle ΩM. For $y \in M$, consider the δ-distribution at y; this satisfies $\|\delta_y\|_{H_h^s(M)} \leq C h^{-\frac{n}{2}}$ for any fixed $s < -\frac{n}{2}$ (Example 5.90). Therefore, $K_h(\cdot, y) := R_h \delta_y$ is uniformly bounded in $H_h^N(M)$ by $C_N h^N$ for all N. This implies analogous uniform bounds also in $H^N(M)$ for all N (Exercise 2.23) and thus by Sobolev embedding uniform $\mathcal{C}^\infty$-bounds by $C_N h^N$. Furthermore, δ_y is continuous in y, so we conclude that $K_h \in \mathcal{C}^\infty(M; \mathcal{C}^0(M))$, with all seminorms bounded by $C_N h^N$. Applying the same arguments to derivatives of δ_y shows that all $\mathcal{C}^\infty(M^2)$-seminorms of K_h are bounded by $C_N h^N$. Since K_h is the Schwartz kernel of R_h we conclude that $R \in h^\infty \Psi_\hbar^{-\infty}(M)$ indeed. □

Example 5.94 (Order Shifting Operators). Theorem 5.92 applies to $\Lambda \in \Psi_\hbar^s(M)$ from Example 5.86; here $s \in \mathbb{R}$, and M is compact. Fixing

any volume density on M to define $L^2(M)$, we can thus define the norm on $H_h^s(M)$ for all sufficiently small $h > 0$ by

$$\|u\|_{H_h^s(M)} := \|\Lambda_h u\|_{L^2(M)}. \tag{5.209}$$

Example 5.95 (Spectral Family). Let g be a Riemannian metric on the compact manifold M, and let $V \in \mathcal{C}^\infty(M)$ and $z \in \mathbb{C}$. Then $A = h^2(\Delta_g + V) - z \in \mathrm{Diff}_h^2(M)$ has semiclassical principal symbol $|\xi|_{g^{-1}}^2 - z$. Thus, A is semiclassically elliptic if and only if $z \notin [0, \infty)$. For fixed $z \notin [0, \infty)$, A_h is thus invertible when h is sufficiently small (depending on z). On the other hand, on $M = (\mathbb{R}/2\pi\mathbb{Z})^n$ with the Euclidean metric $g = \sum_{j=1}^n (\mathrm{d}x^j)^2$, and for $V = 0$, the eigenvalues of Δ_g are given by $|k|^2$ where $k \in \mathbb{Z}^n$; thus $h^2\Delta_g - 1$ (which is *not* semiclassically elliptic) is not invertible for the sequence $h = |k|^{-1}$, $k \in \mathbb{Z}^n \setminus \{0\}$, of values of h tending to 0.

Notes

The "Kuranishi trick" used to prove the local coordinate invariance of ps.d.o.s is taken from Hörmander [Hör71a]. A general reference for ps.d.o.s on $\mathbb{R}^n$ and manifolds is Hörmander [Hör07, Chapter XVIII]. In particular, a concise formula for the full symbol of A_κ in Theorem 5.2, at the level of asymptotic sums, is given in [Hör07, Theorem 18.1.17]. Our subsequent development of the ps.d.o. calculus on manifolds fills in many details that are typically omitted.

There exist other, more global, approaches to the definition of ps.d.o.s on manifolds. A particularly geometric approach was pioneered by Melrose who emphasizes the nature of Schwartz kernels of ps.d.o.s as conormal distributions on $M \times M$ at the diagonal. Compositions are then analyzed using results on the pullback and push-forward of conormal distributions along projection and inclusion maps. General results can be found in Melrose [Mel92, Mel96] (see also Grieser [Gri01]). For early concrete instances of the power and versatility of this approach, which is now called *geometric microlocal analysis*, see, for example, [MM87, Maz91].

On noncompact manifolds M, Definition 5.21 is often too imprecise for useful applications to PDEs. The situation can be improved when additional structures are present on M, as in the case $M = \mathbb{R}^n$ in Remark 5.22. An important example is when M is a *manifold with bounded geometry*; a good overview is given in Shubin [Shu92, Appendix 1].

A beautiful application of ps.d.o. techniques which is beyond the scope of this book concerns boundary value problems for elliptic PDEs; see Hörmander [Hör07, Chapter 20.1].

Exercises

Exercise 5.1 (Convergence in $\mathcal{C}^k_c(M)$). Let $(u_i)_{i\in\mathbb{N}}$ be a sequence in $\mathcal{C}^k_c(M)$. Show that $(u_i)_{i\in\mathbb{N}}$ is a Cauchy sequence (i.e., $|u_j - u_k|_{(w_i),(p_i)} \to 0$ as $j,k \to \infty$ for all seminorms (5.19)) if and only if there exists a compact set $K \subset M$ such that $\operatorname{supp} u_i \subset K$ and u_i is a Cauchy sequence in $\dot{\mathcal{C}}^k(K)$ (and thus converges to a limit in $\dot{\mathcal{C}}^k(K) \subset \mathcal{C}^k_c(M)$).

Exercise 5.2 (Continuity of Maps on $\mathcal{C}^k_c(M)$). Let X be a topological space.

(1) Show that a map $\Phi\colon \mathcal{C}^k_c(M) \to X$ is continuous if and only if the restriction $\Phi|_{\dot{\mathcal{C}}^k(K)}\colon \dot{\mathcal{C}}^k(K) \to X$ is continuous for all compact $K \subset M$.
(2) Show that a map $\Phi\colon \mathcal{C}^k_c(M) \to X$ is continuous if and only if it is sequentially continuous.

Exercise 5.3 (Inductive Limit Topology).

(1) Show that the topology induced by the seminorms (5.19) is independent of the choice of exhaustion and of the functions ϕ_i used to define the functions ψ_i.
(2) Suppose $U \subset \mathcal{C}^k_c(M)$ is open. Show that $U \cap \dot{\mathcal{C}}^k(K) \subset \dot{\mathcal{C}}^k(K)$ is open for all compact $K \subset M$.
(3) Suppose $U \subset \mathcal{C}^k_c(M)$ is a convex subset with the property that $U \cap \dot{\mathcal{C}}^k(K) \subset \dot{\mathcal{C}}^k(K)$ is open for all compact $K \subset M$. Show that U is open.

Remark. Parts (2)–(3) show that the locally convex topology on $\mathcal{C}^k_c(M)$ defined by the seminorms (5.19) is equal to the inductive limit topology; see [RS72, Theorem V.15].

Exercise 5.4 (Tangent Vectors as Directional Derivatives). Prove that the map (5.25) is well-defined, i.e., does not depend on the choice of coordinate system.

Exercise 5.5 (Cotangent Bundle as the Dual of the Tangent Bundle). Prove that the definition of the isomorphism (5.28) given in the subsequent paragraph is independent of the choice of the local coordinate chart.

Exercise 5.6 (Derivatives and Differentials). Let $V \in \mathcal{C}^\infty(M;TM)$ denote a vector field.

(1) For a smooth function $f \in \mathcal{C}^\infty(M)$, define $(Vf)(p) := V(p)f$ for $p \in M$ as the directional derivative of f along $V(p)$ (see Example 5.7). Show that $Vf \in \mathcal{C}^\infty(M)$. Show moreover that the map $f \mapsto Vf$ is a derivation, i.e., it satisfies the Leibniz rule

$$V(fg) = fV(g) + gV(f). \tag{5.210}$$

(2) Given $f \in \mathcal{C}^\infty(M)$, note that $\mathrm{d}f \in \mathcal{C}^\infty(M, T^*M)$. Show that $\mathrm{d}f(V(p)) = V(p)f$, where the left-hand side is the dual pairing between T_p^*M and T_pM (see (5.28)).

Exercise 5.7 (Quotient Vector Bundle). Let $E \to M$ be a vector bundle. Let $F \to M$ be a subbundle of E, i.e., a vector bundle over M with the property that $F_x \subset E_x$ for all $x \in M$. Give a construction of the quotient vector bundle $E/F \to M$ whose fibers are the quotient vector spaces E_x/F_x.

Exercise 5.8 (Normal and Conormal Bundles). Given a smooth submanifold $S \subset M$ of a manifold M, define its normal bundle as the quotient bundle $NS = T_SM/TS$. Show that the duality between TM and T^*M induces a duality between NS and the conormal bundle $N^*S \subset T_S^*M$ consisting of all covectors that annihilate TS.

Exercise 5.9 (Properties of Density Bundles). Prove Lemma 5.12.

Exercise 5.10 (Pullback of Densities). Prove Lemma 5.13 and the statement in Remark 5.14.

The next three exercises concern distributions on the torus: these can be studied using Fourier series.

Exercise 5.11 (Distributions on the Torus). We work on $\mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$ and use the Fourier transform

$$(\mathcal{F}u)(k) := (2\pi)^{-n} \int_{\mathbb{T}^n} e^{-i\theta\cdot k} u(\theta)\, \mathrm{d}\theta, \quad k \in \mathbb{Z}^n, \tag{5.211}$$

and the inverse Fourier transform

$$(\mathcal{F}^{-1}v)(\theta) := \sum_{k\in\mathbb{Z}^n} e^{i\theta\cdot k} v(k), \quad \theta \in \mathbb{T}^n. \tag{5.212}$$

(1) Define the space

$$s(\mathbb{Z}^n) := \{a = (a_k)_{k\in\mathbb{Z}^n} : \|a\|_m < \infty \,\forall\, m \in \mathbb{N}_0\}, \quad \|a\|_m := \sup_{k\in\mathbb{Z}^n} \langle k\rangle^m |a_k|. \tag{5.213}$$

Show that $\mathcal{F}\colon \mathcal{C}^\infty(\mathbb{T}^n) \to s(\mathbb{Z}^n)$ is an isomorphism with inverse $\mathcal{F}^{-1}\colon s(\mathbb{Z}^n) \to \mathcal{C}^\infty(\mathbb{T}^n)$.

(2) Show that the dual space of $s(\mathbb{Z}^n)$ is equal to

$$s'(\mathbb{Z}^n) := \{(b_k)_{k\in\mathbb{Z}^n} : \exists\, m \in \mathbb{N}_0,\ C < \infty \text{ s.t. } |b_k| \le C\langle k\rangle^m \ \forall\, k \in \mathbb{Z}^n\}, \tag{5.214}$$

in the sense that every continuous linear functional $\ell\colon s(\mathbb{Z}^n) \to \mathbb{C}$ is of the form $\ell((a_k)_{k\in\mathbb{Z}^n}) = \sum_{k\in\mathbb{Z}^n} a_k b_k$ for some $(b_k)_{k\in\mathbb{Z}^n} \in s'(\mathbb{Z}^n)$.

(3) Equip $s'(\mathbb{Z}^n)$ with the weak topology using the seminorms $|a|_b := |\sum_{k\in\mathbb{Z}^n} a_k b_k|$ where $b \in s(\mathbb{Z}^n)$. Show that $\mathcal{F}$ extends uniquely to an isomorphism

$$\mathcal{F}\colon \mathscr{D}'(\mathbb{T}^n) \to s'(\mathbb{Z}^n) \tag{5.215}$$

with inverse $\mathcal{F}^{-1}\colon s'(\mathbb{Z}^n) \to \mathscr{D}'(\mathbb{T}^n)$.

Exercise 5.12 (Poisson Summation Formula). We work on $\mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$.

(1) Show that the map $\mathcal{C}^\infty(\mathbb{T}^n) \ni u \mapsto U \in \mathcal{C}^\infty(\mathbb{R}^n)$ given by 2π-periodic extension in each coordinate, i.e., $U(x) := u(x + (2\pi\mathbb{Z})^n)$, extends to a continuous map $\mathscr{D}'(\mathbb{T}^n) \to \mathscr{S}'(\mathbb{R}^n)$.

(2) Show that $\hat{U}(\xi) = (2\pi)^n \sum_{k\in\mathbb{Z}^n} (\mathcal{F}u)(k)\delta(\xi - k)$.

(3) Deduce from the special case $u = \delta_{0+(2\pi\mathbb{Z})^n}$ the *Poisson summation formula*

$$(2\pi)^{-n} \sum_{k\in\mathbb{Z}^n} \hat{\phi}(k) = \sum_{m\in\mathbb{Z}^n} \phi(2\pi m), \quad \phi \in \mathscr{S}(\mathbb{R}^n). \tag{5.216}$$

Exercise 5.13 (Sobolev Spaces on the Torus). For $s \in \mathbb{R}$, define

$$h^s(\mathbb{Z}^n) := \left\{ a = (a_k)_{k\in\mathbb{Z}^n}\colon \|a\|^2_{h^s(\mathbb{Z}^n)} := \sum_{k\in\mathbb{Z}^n} \langle k\rangle^{2s} |a_k|^2 < \infty \right\}. \tag{5.217}$$

(1) Show that the Fourier transform (5.211) defines an isomorphism $H^s(\mathbb{T}^n) \to h^s(\mathbb{Z}^n)$.

(2) Let $s > s'$. Give a direct proof of the following statement: the inclusion map $h^s(\mathbb{Z}^n) \to h^{s'}(\mathbb{Z}^n)$ is compact. Deduce that the inclusion $H^s(\mathbb{T}^n) \to H^{s'}(\mathbb{T}^n)$ is compact.

(3) Let $K \subset (0, 2\pi)^n$ be compact. Write $\dot{\mathcal{C}}^\infty(K) := \{u \in \mathcal{C}^\infty(\mathbb{R}^n)\colon \operatorname{supp} u \subset K\}$. Show that the map $\dot{\mathcal{C}}^\infty(K) \ni u \mapsto U \in \mathcal{C}^\infty(\mathbb{T}^n)$, $U(x + (2\pi\mathbb{Z})^n) := u(x)$ where $x \in [0, 2\pi)^n$, extends to a continuous map $\dot{H}^s(K) \to H^s(\mathbb{T}^n)$. Show conversely that if $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, then the map $\mathcal{C}^\infty(\mathbb{T}^n) \ni U \mapsto u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $u(x) := \chi(x)U(x + (2\pi\mathbb{Z})^n)$, extends to a continuous map $H^s(\mathbb{T}^n) \to H^s(\mathbb{R}^n)$.

(4) Combine the first three parts to give another proof of Theorem 2.17.

Exercise 5.14 (Distributions and Sobolev Spaces on Manifolds).

(1) Let M be a manifold (compact or non-compact), and let $u \in \mathscr{E}'(M)$. Show that there exists $s \in \mathbb{R}$ such that $u \in H^s_{\mathrm{c}}(M)$. (Thus, $\mathscr{E}'(M) = \bigcup H^s_{\mathrm{c}}(M)$.)
(2) Suppose M is non-compact. Show that there exists $u \in \mathscr{D}'(M)$ such that $u \notin H^s_{\mathrm{loc}}(M)$ for any $s \in \mathbb{R}$. (Thus, $\mathscr{D}'(M) \supsetneq \bigcup H^s_{\mathrm{loc}}(M)$.)

Exercise 5.15 (Schwartz Kernel Theorem: Manifold Case). Prove the following generalization of Theorem 5.18: if $E \to M$ and $F \to N$ are vector bundles over the smooth manifolds M and N, then there is a one-to-one correspondence between continuous linear operators $A\colon \mathcal{C}^\infty_{\mathrm{c}}(M;E) \to \mathscr{D}'(N;F)$ and distributional Schwartz kernels $K \in \mathscr{D}'(N\times M; \pi_L^* F \otimes \pi_R^*(E^* \otimes \Omega M))$, where $\pi_L\colon N \times M \to N$ and $\pi_R\colon N \times M \to M$ are the projection maps.

Exercise 5.16 (Ps.d.o.s Are Properly Supported Modulo Residual Operators). Let $A \in \Psi^m(M)$. Show that there exists a properly supported operator $A_0 \in \Psi^m(M)$ with $A - A_0 \in \Psi^{-\infty}(M)$.

Exercise 5.17 (Locally Convex Topology on $\Psi^m(M)$). Prove that $\Psi^m(M)$, with the seminorms as in Remark 5.24, is complete. Prove the same for the subspace of $\Psi^m(M)$ consisting of all properly supported operators—which seminorms should one use here?

Exercise 5.18 (Asymptotic Summation on Manifolds). Let M be a manifold, and let $m \in \mathbb{R}$.

(1) Given a sequence of symbols $a_j \in S^{m-j}(T^*M)$, $j \in \mathbb{N}_0$, show that there exists a symbol $a \in S^m(T^*M)$ such that for all $N \in \mathbb{N}_0$, we have $a - \sum_{j=0}^{N-1} a_j \in S^{m-N}(T^*M)$.
(2) Given a sequence of operators $A_j \in \Psi^{m-j}(M)$, $j \in \mathbb{N}_0$, show that there exists a properly supported operator $A \in \Psi^m(M)$ such that for all $N \in \mathbb{N}_0$, we have $A - \sum_{j=0}^{N-1} A_j \in \Psi^{m-N}(M)$.

Exercise 5.19 (Fractional Laplacians). Let $n \in \mathbb{N}$, and let $\alpha \in \mathbb{R}$, $\alpha > -\frac{n}{2}$. Set $\Delta^\alpha := \mathcal{F}^{-1}|\xi|^{2\alpha}\mathcal{F}$.

(1) Show that Δ^α is a well-defined operator $\mathscr{S}(\mathbb{R}^n) \to \mathscr{S}'(\mathbb{R}^n) \cap \mathcal{C}^\infty(\mathbb{R}^n)$.
(2) Show that Δ^α is a classical pseudodifferential operator of order 2α on $M = \mathbb{R}^n$ in the sense of Definitions 5.21 and 5.46. Compute its principal symbol and show that Δ^α is elliptic.
(3) Show that Δ^α is *not* a uniform ps.d.o. on $\mathbb{R}^n$ in the sense of Definition 4.7 unless $\alpha \in \mathbb{N}_0$.

Exercise 5.20 (Exterior Derivative on k-Forms). Let M denote a smooth manifold, and denote by

$$\mathrm{d}\colon \mathcal{C}^\infty(M;\Lambda^k T^*M) \to \mathcal{C}^\infty(M;\Lambda^{k+1}T^*M) \tag{5.218}$$

the exterior derivative. Show that $\mathrm{d} \in \mathrm{Diff}^1(M; \Lambda^k T^*M, \Lambda^{k+1} T^*M)$ and compute its principal symbol.

Exercise 5.21 (Covariant Derivative). Let (M, g) denote a smooth Riemannian manifold, and denote by

$$\nabla \colon \mathcal{I}(M; TM) \to \mathcal{I}(M; T^*M \otimes TM), \quad V \mapsto (\nabla V \colon X \mapsto \nabla_X V), \tag{5.219}$$

the covariant derivative on vector fields. Show that ∇ is a first-order differential operator. Compute its principal symbol.

Exercise 5.22 (A Classical ps.d.o.). Let $\Gamma \subset \mathbb{C}$ be a smooth, simple, closed curve. Let $K \in \mathcal{C}^\infty(\Gamma \times \Gamma)$. Prove that

$$Au(t) := \lim_{\epsilon \to 0} \int_{|t-s| \geq \epsilon} \frac{K(t,s)}{t-s} u(s)\, \mathrm{d}s, \quad u \in \mathcal{C}^\infty(\Gamma), \tag{5.220}$$

is well-defined and defines an element $A \in \Psi^0_{\mathrm{cl}}(\Gamma)$. Here, $t, s \in \Gamma \subset \mathbb{C}$ are complex numbers, and the division here is *division by a complex number*. Compute the principal symbol of A.

In the next two exercises, we identify $\mathbb{S}^1 = \{z \in \mathbb{C} \colon |z| = 1\}$.

Exercise 5.23 (A Projection Operator). Define a continuous linear operator $A \colon L^2(\mathbb{S}^1) \to L^2(\mathbb{S}^1)$ by

$$A(e^{inx}) = \begin{cases} 0, & n \leq -1, \\ e^{inx}, & n \geq 0. \end{cases} \tag{5.221}$$

Show that $A \in \Psi^0_{\mathrm{cl}}(\mathbb{S}^1)$ and compute its principal symbol.

Exercise 5.24 (An Elliptic Operator with Nonzero Index). Define a continuous linear operator $A \colon L^2(\mathbb{S}^1) \to L^2(\mathbb{S}^1)$ by

$$A(e^{inx}) = \begin{cases} e^{inx}, & n \leq -1, \\ e^{i(n+1)x}, & n \geq 0. \end{cases} \tag{5.222}$$

Show that $A \in \Psi^0_{\mathrm{cl}}(\mathbb{S}^1)$ is elliptic and has Fredholm index -1. Compute its principal symbol.

Exercise 5.25 (Fourier Multipliers on the Torus). *This exercise builds on Exercises 5.11 and 5.12.* Let $a \in S^m(\mathbb{R}^n)$.

(1) Define $A := \mathrm{Op}_{\mathbb{T}^n}(a)$ via

$$\mathcal{F}(\mathrm{Op}_{\mathbb{T}^n}(a)u)(k) := a(k)(\mathcal{F}u)(k), \quad u \in \mathcal{C}^\infty(\mathbb{T}^n),\ k \in \mathbb{Z}^n. \tag{5.223}$$

Show that $A \in \Psi^m(\mathbb{T}^n)$ and $\sigma^m(A)(\theta, \xi) = a(\xi)$.

(2) For $h \in (0,1)$, define $A_h := \mathrm{Op}_{h,\mathbb{T}^n}(a)$ via

$$\mathcal{F}(\mathrm{Op}_{h,\mathbb{T}^n}(a)u)(k) := a(hk)(\mathcal{F}u)(k), \quad u \in \mathcal{C}^\infty(\mathbb{T}^n),\ k \in \mathbb{Z}^n. \tag{5.224}$$

Show that $A := (A_h)_{h\in(0,1)} \in \Psi_\hbar^m(\mathbb{T}^n)$ and $\sigma_\hbar^m(A)(\theta,\xi) = a(\xi)$.

Exercise 5.26 (Operators on Half-densities and the Subprincipal Symbol). *This exercise builds on Exercise 4.7.* Let M be a smooth manifold.

(1) Suppose $\Omega, \Omega' \subset \mathbb{R}^n$ are open and $\kappa\colon \Omega \to \Omega'$ is a diffeomorphism. Denote points in these sets by $x \in \Omega$, $x' = \kappa(x) \in \Omega'$. Let $A \in \Psi_{\mathrm{c}}^m(\Omega')$. We define an operator B acting on $\frac{1}{2}$-densities on $\mathbb{R}^n$ by

$$B\big(u'\,|\mathrm{d}x'|^{\frac{1}{2}}\big) := (Au')\,|\mathrm{d}x'|^{\frac{1}{2}}. \tag{5.225}$$

Define $B_\kappa u := \kappa^* B((\kappa^{-1})^* u|_\Omega)$ for $u \in \mathcal{C}^\infty(\mathbb{R}^n; \Omega^{\frac{1}{2}}\mathbb{R}^n)$, and define A_κ via

$$B_\kappa\big(u\,|\mathrm{d}x|^{\frac{1}{2}}\big) =: (A_\kappa u)\,|\mathrm{d}x|^{\frac{1}{2}}, \quad u \in \mathcal{C}^\infty(\mathbb{R}^n). \tag{5.226}$$

Show that $A_\kappa \in \Psi_{\mathrm{c}}^m(\Omega)$ and

$$\sigma^{\mathrm{w}}(A_\kappa)(x,\xi) \equiv \sigma^{\mathrm{w}}(A)(\kappa(x), (\kappa'(x)^T)^{-1}\xi) \bmod S^{m-2}(\mathbb{R}^n;\mathbb{R}^n). \tag{5.227}$$

(2) Let $A \in \Psi^m(M; \Omega^{\frac{1}{2}}M)$. Define the *Weyl principal symbol* of A as an element

$$\sigma^{\mathrm{w,m}}(A) \in S^m(T^*M)/S^{m-2}(T^*M). \tag{5.228}$$

That is, *the Weyl symbol of an operator acting on half-densities is invariantly defined not merely modulo S^{m-1}, but modulo S^{m-2}.*

(3) Let $A \in \Psi^m(M; \Omega^{\frac{1}{2}}M)$. Denote by $a = a(x,\xi)$ the full *left symbol* of A in local coordinates x on M. Show that

$$a - \frac{1}{2i}\sum_{j=1}^n \partial_{x_j}\partial_{\xi_j} a \tag{5.229}$$

is well-defined (i.e., independent of the choice of coordinates) as an element of $S^m(T^*M)/S^{m-2}(T^*M)$. *Remark.* For *classical* $A \in \Psi_{\mathrm{cl}}^m(M; \Omega^{\frac{1}{2}}M)$, with a an asymptotic sum of cut-off homogeneous symbols $a_{m-j} \in S_{\mathrm{hom}}^{m-j}(\mathbb{R}^n; \mathbb{R}^n \setminus \{0\})$, this shows that

$$a_{m-1} - \frac{1}{2i}\sum_{j=1}^n \partial_{x_j}\partial_{\xi_j} a_m \in S_{\mathrm{hom}}^{m-1}(\mathbb{R}^n; \mathbb{R}^n \setminus \{0\}) \tag{5.230}$$

is well-defined independently of coordinates; it is called the subprincipal symbol. Thus, every $A \in \Psi^m_{\mathrm{cl}}(M;\Omega^{\frac{1}{2}}M)$ has an invariantly defined principal symbol in $S^m_{\hom}(T^*M \setminus o)$ as well as a *subprincipal symbol* in $S^{m-1}_{\hom}(T^*M \setminus o)$.

Exercise 5.27 (Regularity of the Green's Function). Let M be a smooth compact n-dimensional manifold, and fix a volume density on M. Let $A \in \Psi^m(M)$ be an elliptic ps.d.o., and assume that $A\colon \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$ is invertible. For $p \in M$, put $G(p,\cdot) := A^{-1}(\delta_p)$. Determine the set of all $s \in \mathbb{R}$ such that $G(p,\cdot) \in H^s(M)$.

Exercise 5.28 (Density). Let $s \in \mathbb{R}$. Show that $\mathcal{C}^\infty(M) \subset H^s_{\mathrm{loc}}(M)$ and $\mathcal{C}^\infty_{\mathrm{c}}(M) \subset H^s_{\mathrm{c}}(M)$ are dense.

Exercise 5.29 (Algebra Properties of Sobolev Spaces). Let M be a smooth n-dimensional manifold, and let $s > \frac{n}{2}$. *This exercise builds on Exercises 2.8 and 2.9.*

(1) Let $k \in \mathbb{N}_0$ be such that $s - \frac{n}{2} > k$. Show that $H^s_{\mathrm{loc}}(M) \hookrightarrow \mathcal{C}^k(M)$.
(2) If $u, v \in H^s_{\mathrm{loc}}(M)$, show that $uv \in H^s_{\mathrm{loc}}(M)$.
(3) When M is compact, show that for any fixed choice of norm on $H^s(M)$ there exists a constant C_s such that $\|uv\|_{H^s(M)} \le C_s \|u\|_{H^s(M)} \|v\|_{H^s(M)}$ for all $u, v \in H^s(M)$.

Exercise 5.30 (Very Rough Estimates for Eigenfunctions). Let (M,g) be a compact Riemannian manifold, $n = \dim M$. Show that for all $\epsilon > 0$ there exists a constant C_ϵ such that for all eigenfunctions $\phi \in \mathcal{C}^\infty(M)$ of Δ_g, with $\Delta_g \phi = \lambda\phi$ and $\|\phi\|_{L^2} = 1$, we have

$$\|\phi\|_{L^\infty} \le C_\epsilon (1+\lambda)^{\frac{n}{4}+\epsilon}. \tag{5.231}$$

Remark. Exercise 6.26 proves this estimate for $\epsilon = 0$.

Exercise 5.31 (Compact Operators). Let M be a closed manifold. Let $A \in \Psi^0(M)$, $a = \sigma^0(A)$. Assume that $\lim_{|\xi|\to\infty} a(x,\xi) = 0$ for all $x \in M$, where $|\cdot|$ is a norm on the fibers of T^*M. Show that $A\colon L^2(M) \to L^2(M)$ is compact. (*Hint.* Exercise 3.7.) *Remark.* The converse is true as well (Exercise 6.7).

Exercise 5.32 (Principal Symbol via Oscillatory Testing). Let M be a smooth manifold, and let $A \in \mathrm{Diff}^m(M)$. Let $x_0 \in M$ and $0 \neq \xi_0 \in T^*_{x_0}M$.

(1) Show that there exists a smooth function $u \in \mathcal{C}^\infty(M)$ with $(\mathrm{d}u)(x_0) = \xi_0$.
(2) Let $\phi \in \mathcal{C}^\infty(M)$ be equal to 1 at x_0. Prove that

$$\sigma^m(A)(x_0,\xi_0) = \lim_{\lambda\to\infty} \lambda^{-m} e^{-i\lambda u(x_0)} A(e^{i\lambda u}\phi)(x_0). \tag{5.232}$$

(3) State and prove analogous results for $A \in \mathrm{Diff}^m(M; E, F)$ where $E, F \to M$ are two vector bundles.
(4*) Prove analogous results for $A \in \Psi^m_{\mathrm{cl}}(M; E, F)$. (This requires the use of the *stationary phase lemma*, which we do not discuss in this book.)

Exercise 5.33 (Ellipticity and the Fredholm Property). Let M be a closed manifold and let $A \in \mathrm{Diff}^m(M)$, $m \geq 1$. Show that A is elliptic if and only if $A\colon H^m(M) \to H^0(M)$ is Fredholm. (*Hint.* If A is Fredholm, prove the validity of an estimate $\|u\|_{H^m} \leq C(\|Au\|_{H^0(M)} + \|u\|_{H^0})$. Plug in highly oscillatory functions, as in the previous exercise, multiplied with cutoff functions to neighborhoods of points in M, into this estimate to obtain lower bounds on $|\sigma^m(A)|$.)

Exercise 5.34 (Symbol and Spectral Bounds). Let M be a closed manifold; fix a volume density on M. Let $m > 0$, and let $A \in \Psi^m(M)$ be elliptic and symmetric. Assume moreover that there exists $c > 0$ such that $\sigma^m(A)(x, \xi) \geq c|\xi|^m$ for all $x \in M$, $\xi \in T^*_x M$ where $|\cdot|$ denotes a norm on the fibers of T^*M. Show that A has only finitely many eigenvalues in $(-\infty, 0]$.

The next four exercises concern underdetermined-elliptic and overdetermined-elliptic operators.

Exercise 5.35 (Over- and Underdetermined-Elliptic Operators: I). Let M be a compact manifold, let $E, F \to M$ denote two vector bundles, and let $A \in \Psi^m(M; E, F)$.

(1) Suppose there exists $b \in S^{-m}(T^*M; \mathrm{Hom}(F, E))$ such that $b\sigma^m(A) - 1 \in S^{-1}(T^*M; \mathrm{End}(E))$. (If A is not elliptic, one says in this case that A is *overdetermined-elliptic*.) Show that $A\colon H^s(M; E) \to H^{s-m}(M; F)$ has finite-dimensional kernel and closed range.
(2) Suppose there exists $b \in S^{-m}(T^*M; \mathrm{Hom}(F, E))$ such that $\sigma^m(A)b - 1 \in S^{-1}(T^*M; \mathrm{End}(F))$. (If A is not elliptic, one says in this case that A is *underdetermined-elliptic*.) Show that $A\colon H^s(M; E) \to H^{s-m}(M; F)$ has closed range and finite-dimensional cokernel.
(3) Show that if A has a homogeneous principal symbol $\sigma^m(A)$ (so in particular when A is a classical operator), the assumption in part (1) is equivalent to the injectivity of $\sigma^m(A)$ on $T^*M \setminus o$ and the assumption in part (2) to the surjectivity of $\sigma^m(A)$.

Exercise 5.36 (Over- and Underdetermined-Elliptic Operators: II). Let M be a compact manifold without boundary, let $E, F \to M$ denote two vector bundles, and let $A \in \Psi^m(M; E, F)$ be elliptic, or over- or underdetermined-elliptic. Fix a positive smooth density on M and fiber inner products on E, F.

(1) Establish the $L^2(M; E)$-orthogonal splitting

$$L^2(M; E) = \ker_{L^2(M;E)} A \oplus A^*(H^m(M; F)). \tag{5.233}$$

(2) For $s \in \mathbb{R}$, show that $H^s(M;E) = \ker_{H^s(M;E)} A \oplus A^*(H^{s+m}(M;F))$.
(3) Show that $\mathcal{C}^\infty(M;E) = \ker_{\mathcal{C}^\infty(M;E)} A \oplus A^*(\mathcal{C}^\infty(M;F))$. That is, show that every $u \in \mathcal{C}^\infty(M;E)$ can be written as $u = u_0 + A^*u_1$ for smooth u_0, u_1, with u_0 and A^*u_1 unique, and prove that $A^*(\mathcal{C}^\infty(M;F)) \subset \mathcal{C}^\infty(M;E)$ is closed.

Exercise 5.37 (Underdetermined-Elliptic Operators). Let M be a compact manifold, let $E, F \to M$ denote two vector bundles, and let $A \in \mathrm{Diff}^m(M;E,F)$ be underdetermined-elliptic, i.e., $\sigma^m(A)$ is surjective but not injective. (An example is $A = \mathrm{div}_g$, the divergence operator on vector fields on a Riemannian manifold (M,g) of dimension ≥ 2.)

(1) Show that there exists a constant $C \in \mathbb{R}$ such that for all $u \in A^*(H^{s+m}(M;F))$ one has $\|u\|_{H^s(M;E)} \leq C\|Au\|_{H^{s-m}(M;F)}$.
(2) Show that $\ker_{H^s(M;E)} A$ is infinite-dimensional. (*Hint.* Use ideas from Exercise 5.32.)
(3) Show that $\ker_{\mathcal{C}^\infty(M;E)} A$ is infinite-dimensional.

Exercise 5.38 (Helmholtz Decomposition). Let (M,g) be a compact connected Riemannian manifold without boundary, denote by $\mathrm{d}\colon \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M;T^*M)$ the exterior derivative acting on functions, and denote by $\delta_g = \mathrm{d}^*$ its adjoint. Let $\omega \in H^s(M;T^*M)$ be a 1-form. Prove that there exist $u \in H^{s+1}(M)$ and $\eta \in H^s(M;T^*M)$ such that

$$\omega = \mathrm{d}u + \eta, \quad \delta_g \eta = 0. \tag{5.234}$$

Show moreover that η is unique and that u is unique up to additive constants.

Exercise 5.39 (Elliptic Complexes). Let M be a compact manifold without boundary, let $E_i \to M$, $i = 0, \dots, N$, be complex vector bundles, and let $\mathrm{d}_i \in \mathrm{Diff}^1(M;E_i,E_{i+1})$, $i = 0, \dots, N-1$. Suppose they form a *complex of differential operators*

$$\mathcal{C}^\infty(M;E_0) \xrightarrow{\mathrm{d}_0} \mathcal{C}^\infty(M;E_1) \xrightarrow{\mathrm{d}_1} \cdots \xrightarrow{\mathrm{d}_{N-1}} \mathcal{C}^\infty(M;E_N); \tag{5.235}$$

this means that for each $i = 0, \dots, N-2$,

$$\mathrm{d}_{i+1} \circ \mathrm{d}_i = 0 \in \mathrm{Diff}^2(M;E_i,E_{i+2}). \tag{5.236}$$

Assume moreover that the complex (5.235) is *elliptic*; this means that the symbol complex

$$\mathcal{C}^\infty(T^*M \setminus o; \pi^*E_0) \xrightarrow{\sigma^1(\mathrm{d}_0)} \mathcal{C}^\infty(T^*M \setminus o; \pi^*E_1) \xrightarrow{\sigma^1(\mathrm{d}_1)} \cdots \xrightarrow{\sigma^1(\mathrm{d}_{N-1})} \mathcal{C}^\infty(T^*M \setminus o; \pi^*E_N) \tag{5.237}$$

is *exact*, that is, $\operatorname{ran}\sigma^1(\mathrm{d}_{i-1})(x,\xi) = \ker\sigma^1(\mathrm{d}_i)(x,\xi) \subset (E_i)_x$ for all $x \in M$, $\xi \in T^*_x M$, and $i = 1, \dots, N-1$. (Here $\pi\colon T^*M \to M$ is the projection.) The goal of this exercise is to study the cohomology groups

$$H^i(E_\bullet) := \ker \mathrm{d}_i / \operatorname{ran} \mathrm{d}_{i-1}, \quad i = 1, \dots, N-1, \tag{5.238}$$

using PDE theory. Equip M with a volume density and the E_i with Hermitian fiber inner products; define $\delta_i \in \operatorname{Diff}^1(M; E_i, E_{i-1})$ to be the adjoint of d_{i-1} for $i = 1, \dots, N$.

(1) Show that the "Laplacian"

$$\Delta_i := \mathrm{d}_{i-1} \circ \delta_i + \delta_{i+1} \circ \mathrm{d}_i \in \operatorname{Diff}^2(M; E_i), \quad 1 \le i \le N-1, \tag{5.239}$$

is elliptic and symmetric.

(2) Show that

$$\ker \Delta_i = \{u \in \mathcal{C}^\infty(M; E_i)\colon \mathrm{d}_i u = 0,\ \delta_i u = 0\}. \tag{5.240}$$

(3) Show that the inclusion $\ker \Delta_i \hookrightarrow \ker \mathrm{d}_i$ induces an isomorphism of vector spaces

$$\ker \Delta_i \cong H^i(E_\bullet). \tag{5.241}$$

(4) Prove the *Hodge theorem*: if (M, g) is a compact Riemannian manifold without boundary and $\Delta_i \in \operatorname{Diff}^2(M; \Lambda^i T^*M)$ is the Hodge Laplacian on i-forms (Example 5.55), then $\ker \Delta_i \cong H^i(M)$. Here $H^i(M) := \ker \mathrm{d}_i / \operatorname{ran} \mathrm{d}_{i-1}$ denotes the i-th de Rham cohomology group of M (with complex coefficients), with d_i being the exterior derivative on i-forms.

Chapter 6
Microlocalization

We now turn to the study of finer—*microlocal*—properties of distributions and, closely related, to non-elliptic phenomena. We develop the notion of *wave front set* from the following observation about the local nature of full symbolic expansions. Recall the formula (4.86) for the full symbols, modulo $S^{-\infty}$, of compositions of ps.d.o.s: if $A = \mathrm{Op}_L(a)$, $B = \mathrm{Op}_L(b)$ are ps.d.o.s on $\mathbb{R}^n$, then

$$\sigma_L(A \circ B)(x,\xi) \sim \sum_{\alpha \in \mathbb{N}_0^n} \frac{1}{\alpha!} \partial_\xi^\alpha a(x,\xi) \cdot D_x^\alpha b(x,\xi). \tag{6.1}$$

A key feature, which so far we have only exploited at the principal symbol level, is that this formula is *local* in (x,ξ). We would like to say that if A is "trivial" at or near (x,ξ), then $A \circ B$ (for any B) is trivial there as well. Since the expression (6.1) is an *asymptotic* sum and thus has no content at any fixed point (x,ξ), the correct notion of "triviality" must only depend on the behavior of symbols as $|\xi| \to \infty$ in conic sets, and it must be insensitive to modifications by symbols of order $-\infty$. Thus, "triviality" refers to *rapid decay* as $|\xi| \to \infty$ in conic subsets of $\mathbb{R}^n_\xi \setminus \{0\}$ (and near points $x \in \mathbb{R}^n$). This leads to the notions of *essential supports* of symbols (Definition 6.1) and *operator wave front sets* of ps.d.o.s (Definition 6.3): the essential support of a symbol a consists of all $(x,\xi) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$ such that a is *not* trivial near x and in directions close to ξ; the operator wave front set of a ps.d.o. is the essential support of its left symbol.

Instead of asking for the "triviality" of a symbol $a \in S^m(\mathbb{R}^n_x; \mathbb{R}^n_\xi)$ near $x \in \mathbb{R}^n$ and in a cone centered around $\xi \in \mathbb{R}^n \setminus \{0\}$, one can also ask for the opposite, namely for lower bounds $|a(x,\xi) \geq c|\xi|^m$, $c > 0$, as $|\xi| \to \infty$. This gives rise to the notion of *elliptic sets* of symbols and ps.d.o.s (Section 6.2). The *wave front set* of a distribution u (introduced in Section 6.3) then consists of all points x and directions ξ such that u is *not* "microlocally smooth" near

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(x,ξ), in the sense that it does not get mapped to a smooth function by any ps.d.o. that is elliptic at (x,ξ). The idea is that elliptic operators preserve *local* singularities (i.e., singularities near points $x \in \mathbb{R}^n$) (Proposition 4.32), and this definition of wave front sets leads to the same preservation of *microlocal* singularities (i.e., singularities near points x *in frequency directions* $\xi \in \mathbb{R}^n \setminus \{0\}$).

Connecting this to the operator wave front set, it is then a simple consequence of (6.1) that Au is microlocally smooth (i.e., has no wave front set) near all (x,ξ) where A is trivial (i.e., has no operator wave front set). One should thus think of A as a smoothing operator microlocally near all (x,ξ) which do *not* lie in the operator wave front set of A.

The wave front set can be used to give rather precise conditions under which distributions can be multiplied, paired (in an L^2-sense), or restricted to submanifolds (Section 6.4).

Up to this point, all microlocal notions lead to conic subsets of $\mathbb{R}^n_x \times (\mathbb{R}^n_\xi \setminus \{0\})$, or on manifolds to subsets of $T^*M \setminus o$ which are conic in the fibers. An efficient way to encode such subsets is via the cosphere bundle, which arises in a particularly useful fashion as the boundary at infinity of the radial compactification $\overline{T^*}M$ of phase space (Section 6.5).

We end with a discussion of microlocal notions in the semiclassical context (Section 6.6).

From now on, all ps.d.o.s shall either be properly supported or elements of the uniform ps.d.o. algebra on $\mathbb{R}^n$ (or, in Section 6.6, its semiclassical version), unless stated otherwise.

6.1 Operator Wave Front Set

In light of the discussion of (6.1), we introduce:

Definition 6.1 (Essential Support). Let $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$. Then a point $(x_0,\xi_0) \in \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$ does *not* lie in the *essential support*

$$\operatorname{ess\,supp} a \subset \mathbb{R}^n_x \times (\mathbb{R}^N_\xi \setminus \{0\}) \tag{6.2}$$

if and only if a is a symbol of order $-\infty$ near x_0 and in a conic neighborhood of ξ_0 (see Figure 6.1); that is, there exists $\epsilon > 0$ such that for all $\alpha \in \mathbb{N}_0^n$, $\beta \in \mathbb{N}_0^N$, $k \in \mathbb{R}$, we have

$$\begin{aligned} |\partial_x^\alpha \partial_\xi^\beta a(x,\xi)| \le C_{\alpha\beta k}\langle\xi\rangle^{-k} \quad & \forall\, (x,\xi) \in \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\}), \\ & |x - x_0| + \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \epsilon. \end{aligned} \tag{6.3}$$

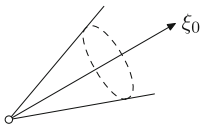


Figure 6.1 A conic neighborhood of $\xi_0 \in \mathbb{R}^n \setminus \{0\}$.

For a symbol $a \in S^m(E)$ on a vector bundle $E \to M$ over a smooth manifold M, we define the conic subset $\operatorname{ess\,supp} a \subset E \setminus o$ (with $o \subset E$ denoting the zero section) in the same fashion in local trivializations of E.

It suffices, in fact, to assume (6.3) only for $\alpha = \beta = 0$; the estimates for general α, β are then automatic (Exercise 6.1).

By definition, $\operatorname{ess\,supp} a$ is a *closed* subset of $\mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$. Moreover, $\operatorname{ess\,supp} a$ is *conic* in ξ, that is, $(x, \xi) \in \operatorname{ess\,supp} a$ implies $(x, \lambda\xi) \in \operatorname{ess\,supp} a$ for all $\lambda > 0$.

Remark 6.2 (Terminology). We caution the reader not to conflate ess supp with the measure theoretic notion. Some authors avoid this clash by calling the set introduced in Definition 6.1 the "conic support" of a symbol.

Definition 6.3 (Operator Wave Front Set: $\mathbb{R}^n$ Case). Let $A = \operatorname{Op}_L(a) \in \Psi^m(\mathbb{R}^n)$, $m \in \mathbb{R}$. Then we define the *operator wave front set of* A as the closed and conic (in ξ) set

$$\operatorname{WF}'(A) := \operatorname{ess\,supp}(a) \subset \mathbb{R}^n_x \times (\mathbb{R}^n_\xi \setminus \{0\}). \tag{6.4}$$

The following follows immediately from (4.83), (4.86), the formulas (4.50)–(4.51) for left and right reductions, and the formula (5.13) in the proof of the local coordinate invariance of ps.d.o.s:

Proposition 6.4 (Properties of the Operator Wave Front Set). *Let A, B be ps.d.o.s on $\mathbb{R}^n$.*

(1) *Suppose the Schwartz kernel of A has compact support. Then $\operatorname{WF}'(A) = \emptyset$ if and only if $A \in \Psi^{-\infty}(\mathbb{R}^n)$.*
(2) *Let $A = \operatorname{Op}_R(a')$. Then $\operatorname{WF}'(A) = \operatorname{ess\,supp} a'$.*
(3) $\operatorname{WF}'(A + B) \subset \operatorname{WF}'(A) \cup \operatorname{WF}'(B)$.
(4) $\operatorname{WF}'(A \circ B) \subset \operatorname{WF}'(A) \cap \operatorname{WF}'(B)$.
(5) $\operatorname{WF}'(A^*) = \operatorname{WF}'(A)$.
(6) *If $\Omega, \Omega' \subset \mathbb{R}^n$ are open, $\kappa \colon \Omega \to \Omega'$ is a diffeomorphism, $A \in \Psi^m_{\mathrm{c}}(\Omega')$ (recall Definition 5.1), and $A_\kappa := \kappa^* A (\kappa^{-1})^* \in \Psi^m_{\mathrm{c}}(\Omega)$, then*

$$\kappa_*\big(\operatorname{WF}'(A_\kappa)\big) = \operatorname{WF}'(A), \tag{6.5}$$

where we define $\kappa_*(x, \xi) = (\kappa(x), (\kappa'(x)^T)^{-1}\xi)$.

Property (4) was our motivation for the introduction of ess supp and WF'.

Remark 6.5 (Empty Operator Wave Front Set). The assumption on the Schwartz kernel in part (1) is made so as to avoid scenarios where $A = \mathrm{Op}(a)$ has empty operator wave front set, but the constants in the estimate (6.3) do not remain bounded when $|x_0|$ gets large. An example is given by $a(x,\xi) = \chi(\langle\xi\rangle\langle x\rangle^{-1})$ where $0 \neq \chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$: this is a uniform symbol of order 0 on $\mathbb{R}^n$ but not of any negative order, even though $\mathrm{WF}'(\mathrm{Op}(a)) = \emptyset$ since $\psi a \in S^{-\infty}$ for all $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$.

The formula (6.5) implies that the operator wave front set can be defined invariantly for operators on manifolds:

Definition 6.6 (Operator Wave Front Set: Manifold Case). Let M be a manifold and $A \in \Psi^m(M)$ (not necessarily properly supported). Then $\mathrm{WF}'(A) \subset T^*M \setminus o$ is the closed conic subset (i.e., invariant under dilations in the fibers of T^*M) given near T^*_pM, $p \in M$, by $\mathrm{WF}'(A_0)$ where $A_0 := (F^{-1})^*\psi A \psi F^*$, with $F\colon U \to F(U) \subset \mathbb{R}^n$ a chart with $p \in U$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ equal to 1 near p, and in the local trivialization of T^*M induced by F.

One thinks of $\mathrm{WF}'(A) \subset T^*M \setminus o$ as the set in phase space where A is *microlocally nontrivial*. Properties (1) and (3)–(5) in Proposition 6.4 hold for ps.d.o.s on manifolds as well. On manifolds, property (1) in fact holds without any assumptions on the Schwartz kernel of A.

Example 6.7 (Elliptic Ps.d.o.s). If $A \in \Psi^m(M)$ is elliptic, then $\mathrm{WF}'(A) = T^*M \setminus o$.

Example 6.8 (Differential Operators). Let $A = \sum_{|\alpha|\leq m} a_\alpha(x) D_x^\alpha \in \mathrm{Diff}^m(\mathbb{R}^n)$. Then

$$\mathrm{WF}'(A) = \left(\bigcup_{|\alpha|\leq m} \operatorname{supp} a_\alpha \right) \times (\mathbb{R}^n \setminus \{0\}). \tag{6.6}$$

Thus, *differential* operators never have "interesting" operator wave front sets in that for every $x_0 \in \mathbb{R}^n$, the intersection $\mathrm{WF}'(A) \cap (T^*_{x_0}\mathbb{R}^n \setminus \{0\})$ equals either $T^*_{x_0}\mathbb{R}^n \setminus \{0\}$ or $\emptyset$.

Example 6.9 (Fourier Multipliers). Let $\chi \in S^m(\mathbb{R}^n_\xi)$ and consider the Fourier multiplier $A = \chi(D) := \mathrm{Op}(\chi)$. Then $\mathrm{WF}'(A) = \mathbb{R}^n_x \times \operatorname{ess\,supp} \chi$. Here we regard $\operatorname{ess\,supp} \chi \subset \mathbb{R}^n_\xi \setminus \{0\}$.

Example 6.10 (Microlocalizers). We combine Examples 6.8 (for $m = 0$) and 6.9. Let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and $\chi \in S^0(\mathbb{R}^n_\xi)$. Then

$$A := \chi(D) \circ \phi(x) = \mathrm{Op}_R(\chi(\xi)\phi(y)) \in \Psi^0(\mathbb{R}^n) \tag{6.7}$$

has $\mathrm{WF}'(A) = \operatorname{supp} \phi \times \operatorname{ess\,supp} \chi$.

Example 6.11 (Quantization). The quantization map (5.92)–(5.93) which is the subject of Proposition 5.41 has the additional property that $\mathrm{WF}'(\mathrm{Op}(a)) = \operatorname{ess\,supp} a$ for all $a \in S^m(T^*M)$.

Working with conic sets is sometimes tedious (and wasteful, as a whole ray can be identified with any one of its points). In most circumstances, one can simplify notation by working on the *cosphere bundle*

$$S^*M := (T^*M \setminus o)/\mathbb{R}_+, \tag{6.8}$$

with fibers given by $S^*_pM = (T^*_pM \setminus o)/\mathbb{R}_+$ where $\mathbb{R}_+$ acts by dilations in the fibers. Thus, S^*M is a fiber bundle with typical fiber $\mathbb{S}^{n-1}$. We can identify conic subsets of $T^*M \setminus o$ with their images in S^*M. For instance, if $A \in \Psi^m(M)$, then for $\alpha \in S^*_pM$, the condition $\alpha \in \mathrm{WF}'(A)$ means that $(p,\xi) \in \mathrm{WF}'(A)$ where $\alpha = [\xi]$ (i.e., $\alpha = \mathbb{R}_+\xi$). Note that a compact subset of S^*M is identified with a conic subset of $T^*M \setminus o$ whose cross section (i.e., intersection with $|\xi| = 1$ for some choice of fiber metric on T^*M) is compact. The projection map is denoted

$$\pi\colon S^*M \to M. \tag{6.9}$$

The following technical result states that one can construct partitions of unity *microlocally*:

Lemma 6.12 (Microlocal Partitions of Unity). *Suppose $S^*M = \bigcup_i U_i$ is an open cover. Then there exist operators $A_i \in \Psi^0(M)$ such that:*

(1) *The supports of the Schwartz kernels of A_i are locally finite.*
(2) $\mathrm{WF}'(A_i) \subset U_i$.
(3) $\sum_i A_i = I$.

Proof. It suffices to prove this for a locally finite refinement of the cover, which we shall denote by $\{U_i\}$ still, for which moreover each U_i lies over a coordinate chart. That is, $U_i \subset S^*_{V_i}M = \pi^{-1}(V_i)$ where $F_i\colon V_i \to F_i(V_i) \subset \mathbb{R}^n$ is a coordinate chart and $\{V_i\}$ is locally finite.

Pick a partition of unity $\{\chi_i\}$ subordinate to the cover $\{U_i\}$; fix $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}(F_i(V_i))$ such that $\psi_i \equiv 1$ near $F_i(\pi(\operatorname{supp}\chi_i))$. Put

$$A_i' := F_i^*\,\psi_i\,\mathrm{Op}(\chi_i)\psi_i\,(F_i^{-1})^*. \tag{6.10}$$

Then (1) and (2) are satisfied for the A_i'. Since $\sum_i \sigma^0(A_i') = 1$, we have $\sum_i A_i' = I - R'$, $R' \in \Psi^{-1}(M)$. Fix a properly supported operator $B \in \Psi^0(M)$ with $B \sim \sum_{j=0}^\infty (R')^j$ and put $A_i := A_i'B$. This still satisfies (1) (since B is properly supported) and (2) by Proposition 6.4(4); and $\sum_i A_i = I - R$, $R \in \Psi^{-\infty}(M)$. Replacing any single one of the A_i by $A_i + R$, we are done. □

Corollary 6.13 (Microlocalizers). *Let $K \subset U \subset S^*M$, with K closed and U open. Then there exists $A \in \Psi^0(M)$ such that* $\mathrm{WF}'(A) \subset U$ *and* $\mathrm{WF}'(I-A) \cap K = \emptyset$.

We say that A is *microlocally equal to I on K*.

Proof of Corollary 6.13. $S^*M = U \cup (S^*M \setminus K)$ is an open cover of S^*M. Lemma 6.12 produces a partition of unity $I = A + B$ with $\mathrm{WF}'(A) \subset U$ and $\mathrm{WF}'(I-A) \cap K = \mathrm{WF}'(B) \cap K = \emptyset$. □

6.2 Elliptic Set, Characteristic Set

We next refine the notion of ellipticity of symbols and operators in a microlocal manner.

Definition 6.14 (Elliptic Set). Let $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$. Then the *elliptic set* of a,

$$\mathrm{Ell}(a) \subset \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\}), \tag{6.11}$$

consists of all $(x_0, \xi_0) \in \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$ in a conic neighborhood of which a is elliptic; that is, there exist $c, C > 0$ and $\epsilon > 0$ such that

$$|a(x,\xi)| \geq c|\xi|^m, \quad |\xi| \geq C, \ |x - x_0| + \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \epsilon. \tag{6.12}$$

The complement of $\mathrm{Ell}(a)$ is the *characteristic set*

$$\mathrm{Char}(a) := (\mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})) \setminus \mathrm{Ell}(a). \tag{6.13}$$

For a symbol $a \in S^m(T^*M)$ on a smooth manifold M, we define $\mathrm{Ell}(a) \subset T^*M \setminus o$ in the same manner in local charts. For $A \in \Psi^m(\mathbb{R}^n)$ or $A \in \Psi^m(M)$, we define $\mathrm{Ell}(A) := \mathrm{Ell}(\sigma^m(A))$ and $\mathrm{Char}(A) := \mathrm{Char}(\sigma^m(A))$.

The sets $\mathrm{Ell}(A)$ and $\mathrm{Char}(A)$ depend on the order m, so for maximum clarity, one can write $\mathrm{Ell}^m(A)$ and $\mathrm{Char}^m(A)$. An equivalent definition of $\mathrm{Ell}(a)$ for $a \in S^m(\mathbb{R}^n; \mathbb{R}^N)$, closer to Definition 3.11, is that there exists $b \in S^{-m}(\mathbb{R}^n; \mathbb{R}^N)$ such that $ab-1$ is a symbol of order -1 in a conic neighborhood of (x_0, ξ_0). Note that $\mathrm{Ell}(A)$ is *open* and *conic*. Moreover,

$$\mathrm{Ell}^{m+m'}(A \circ B) = \mathrm{Ell}^m(A) \cap \mathrm{Ell}^{m'}(B), \quad A \in \Psi^m(M), \ B \in \Psi^{m'}(M). \tag{6.14}$$

Example 6.15 (Elliptic Operators). Elliptic operators on M have elliptic sets equal to $T^*M \setminus o$. Their characteristic sets are empty.

Example 6.16 (Wave Operator). On $\mathbb{R}^n = \mathbb{R}_t \times \mathbb{R}^{n-1}_x$, with canonical momentum variables denoted $\sigma \in \mathbb{R}$, $\xi \in \mathbb{R}^{n-1}$, the wave operator $\Box = -D_t^2 + \sum_{j=1}^{n-1} D_{x_j}^2$ has principal symbol $\sigma^2(\Box)(t,x,\sigma,\xi) = -\sigma^2 + |\xi|^2$. Hence, its characteristic set is a double cone (without the vertex) in each cotangent fiber:

$$\operatorname{Char}(\Box) = \{(t,x,\sigma,\xi)\colon\ -\sigma^2 + |\xi|^2 = 0,\ (\sigma,\xi) \neq (0,0)\}. \tag{6.15}$$

This is a conic 1-codimensional submanifold of $T^*\mathbb{R}^n \setminus o$.

The elliptic parametrix construction (Theorem 4.29) can be microlocalized:

Proposition 6.17 (Microlocal Elliptic Parametrix). *Let $A \in \Psi^m(M)$. Let $K \subset \operatorname{Ell}(A) \subset S^*M$ be a closed subset. Then there exists a* microlocal parametrix *for A on K, that is, an operator $B \in \Psi^{-m}(M)$ such that*

$$K \cap \operatorname{WF}'(AB - I) = \emptyset, \quad K \cap \operatorname{WF}'(BA - I) = \emptyset. \tag{6.16}$$

Proof. By Corollary 6.13, we can pick $Q, \tilde{Q} \in \Psi^0(M)$ with

$$\operatorname{WF}'(I-Q)\cap K = \emptyset,\ \operatorname{WF}'(I-\tilde{Q})\cap \operatorname{WF}'(Q) = \emptyset,\ \operatorname{WF}'(Q), \operatorname{WF}'(\tilde{Q}) \subset \operatorname{Ell}(A). \tag{6.17}$$

Pick then $B_0 \in \Psi^{-m}(M)$ with $\sigma^{-m}(B_0) = \sigma^0(\tilde{Q})/\sigma^m(A)$ and write

$$AB_0 = I - R = I - QR - (I-Q)R, \quad R \in \Psi^0(M). \tag{6.18}$$

We have $\operatorname{WF}'((I-Q)R) \cap K = \emptyset$, while $\sigma^0(QR) = \sigma^0(Q)(1 - \sigma^0(\tilde{Q})) = 0$, so $QR \in \Psi^{-1}(M)$. We then improve the situation near K using a Neumann series argument (cf. Lemma 4.30). That is, let

$$\Psi^0(M) \ni B' \sim \sum_{j=0}^{\infty} (QR)^j \tag{6.19}$$

and put $B := B_0 B' \in \Psi^{-m}(M)$. Then

$$AB = I - R' - (I-Q)RB', \quad R' \in \Psi^{-\infty}(M),\ \operatorname{WF}'((I-Q)RB') \cap K = \emptyset, \tag{6.20}$$

so $\operatorname{WF}'(AB - I) \cap K = \emptyset$.

A microlocal left parametrix, i.e., $\tilde{B} \in \Psi^0(M)$ with $\operatorname{WF}'(\tilde{B}A - I) \cap K = \emptyset$, can be constructed in a similar fashion. Then, modulo operators with WF' disjoint from K, we have

$$B \equiv (\tilde{B}A)B = \tilde{B}(AB) \equiv \tilde{B}. \tag{6.21}$$

The validity of (6.16) is unchanged if one adds to B an operator whose operator wave front set is disjoint from K. Therefore, every microlocal left parametrix is also a right parametrix and vice versa. □

The notation used in (6.21) has a natural generalization: one says that $A \equiv B$ *microlocally on* $K \subset S^*M$ if $\mathrm{WF}'(A-B) \cap K = \emptyset$.

One would like to use Proposition 6.17 to sharpen elliptic regularity theory (Proposition 4.32) to the following type of statement: if $Au = f$, then on $\mathrm{Ell}(A)$, u is "smooth" wherever f is. This leads to the notion of *wave front set*, which we discuss next.

6.3 Wave Front Set of Distributions: Microlocal Elliptic Regularity

Let $u \in \mathscr{S}'(\mathbb{R}^n)$ and $A \in \Psi^m(\mathbb{R}^n)$. Then $Au \in \mathscr{S}'(\mathbb{R}^n)$ is "trivial" outside of $\mathrm{WF}'(A)$: all information about singularities of u is lost in $(T^*\mathbb{R}^n \setminus o) \setminus \mathrm{WF}'(A)$. Indeed, if $B \in \Psi^0(\mathbb{R}^n)$ is such that $\mathrm{WF}'(B) \cap \mathrm{WF}'(A) = \emptyset$, we have $B(Au) \in \mathcal{C}^\infty(\mathbb{R}^n)$ by Proposition 6.4(1). The precise notion of "triviality" is the absence of wave front set. We define this directly on manifolds:

Definition 6.18 (Wave Front Set). Let $u \in \mathscr{D}'(M)$. Then $\alpha \in S^*M$ does *not* lie in the *wave front set* of u,

$$\alpha \notin \mathrm{WF}(u) \subset S^*M, \tag{6.22}$$

if and only if there exists a neighborhood $U \subset S^*M$ of α such that

$$A \in \Psi^0(M),\ \mathrm{WF}'(A) \subset U \implies Au \in \mathcal{C}^\infty(M). \tag{6.23}$$

One can equivalently regard $\mathrm{WF}(u)$ as a conic subset of $T^*M \setminus o$.

If $\alpha \notin \mathrm{WF}(u)$, one says that u is *microlocally smooth near* α. By definition, $\mathrm{WF}(u) \subset S^*M$ is closed (equivalently, $\mathrm{WF}(u) \subset T^*M \setminus o$ is conic and closed). We leave to the reader the simple verification that for $M = \mathbb{R}^n$, $\mathrm{WF}(u)$ for $u \in \mathscr{S}'(\mathbb{R}^n)$ can be defined equivalently by testing with uniform ps.d.o.s $A \in \Psi^0(\mathbb{R}^n)$.

We proceed to give several equivalent definitions of $\mathrm{WF}(u)$. We begin by reducing the number of operators for which one needs to check (6.23) to *one*.

Lemma 6.19 (Wave Front Set: Equivalent Definition #1). *Let $u \in \mathscr{D}'(M)$. Then $\alpha \notin \mathrm{WF}(u)$ if and only if there exists $A \in \Psi^0(M)$, elliptic at α, such that $Au \in \mathcal{C}^\infty(M)$.*

Proof. The direction '$\implies$' is obvious. To prove '$\impliedby$,' we take $U := \mathrm{Ell}(A)$ which is thus a neighborhood of α. Let $B \in \Psi^0(M)$, $\mathrm{WF}'(B) =: K \subset U$; we

claim that $Bu \in \mathcal{C}^\infty(M)$. Proposition 6.17 produces a microlocal parametrix $Q \in \Psi^0(M)$ of A on K, so $QA = I - R$, $R \in \Psi^0(M)$, $\mathrm{WF}'(R) \cap K = \emptyset$. Therefore,

$$Bu = B(QA + R)u = (BQ)(Au) + BRu \in \mathcal{C}^\infty(M). \tag{6.24}$$

Indeed, $Au \in \mathcal{C}^\infty(M)$, and hence, the first summand is smooth; and $\mathrm{WF}'(BR) \subset \mathrm{WF}'(B) \cap \mathrm{WF}'(R) \subset K \cap \mathrm{WF}'(R) = \emptyset$, and hence $BR \in \Psi^{-\infty}(M)$, so $BRu \in \mathcal{C}^\infty(M)$ as well. □

Corollary 6.20 (Wave Front Set: Equivalent Definition #2). *Let $u \in \mathscr{D}'(M)$. Then*

$$\mathrm{WF}(u) = \bigcap_{\substack{A \in \Psi^0(M) \\ Au \in \mathcal{C}^\infty(M)}} \mathrm{Char}(A). \tag{6.25}$$

This leads to the following concrete description of wave front sets.

Proposition 6.21 (Wave Front Set: Equivalent Definition #3). *Let $u \in \mathscr{D}'(\mathbb{R}^n)$, $(x_0, \xi_0) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$.*

(1) *If there exist $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $\phi(x_0) \neq 0$, and $\epsilon > 0$ such that for all $N \in \mathbb{R}$ we have*

$$|\widehat{\phi u}(\xi)| \leq C_N |\xi|^{-N}, \quad \xi \in \mathbb{R}^n,\ |\xi| \geq 1,\ \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \epsilon, \tag{6.26}$$

then $(x_0, \xi_0) \notin \mathrm{WF}(u)$.

(2) *If for all $\epsilon > 0$ there exist $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(B(x_0, \epsilon))$ and $N \in \mathbb{R}$ such that (6.26) does* not *hold for any constant C_N, then $(x_0, \xi_0) \in \mathrm{WF}(u)$.*

In (6.26), we use that the Fourier transform of $\phi u \in \mathscr{E}'(\mathbb{R}^n)$ is a continuous function (Proposition 2.11). We spell out the contrapositive of Proposition 6.21(2):

(2') *Suppose that $(x_0, \xi_0) \notin \mathrm{WF}(u)$. Then there exists $\epsilon > 0$ such that for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(B(x_0, \epsilon))$ and $N \in \mathbb{R}$, there exists a constant C_N such that (6.26) holds.*

For distributions on manifolds, the same result applies in any local coordinate chart. An advantage of our invariant approach to the wave front set in Definition 6.18 is that it implies for free that the wave front set is a subset of $T^*M \setminus o$. By contrast, this is not immediately apparent from the characterization in Proposition 6.21. (We encourage the reader to try and give a direct proof of the coordinate invariance of the conditions in Proposition 6.21.)

Proof of Proposition 6.21. Suppose that (6.26) holds. Let $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ have support in an ϵ-ball around $\frac{\xi_0}{|\xi_0|}$, with $\psi(\frac{\xi_0}{|\xi_0|}) \neq 0$. Fix moreover $\chi \in \mathcal{C}^\infty(\mathbb{R}^n_\xi)$ with $\chi(\xi) = 0$ for $|\xi| \leq 1$ and $\chi(\xi) = 1$ for $|\xi| \geq 2$. Set

$$a(y,\xi) := \chi(\xi)\psi\Big(\frac{\xi}{|\xi|}\Big)\phi(y) \in S^0(\mathbb{R}^n;\mathbb{R}^n). \tag{6.27}$$

Then $|\mathcal{F}(\mathrm{Op}_R(a)u)(\xi)| \leq C_N\langle\xi\rangle^{-N}$, and therefore $\mathrm{Op}_R(a)u \in \mathcal{C}^\infty(\mathbb{R}^n)$. Since a is elliptic at (x_0,ξ_0), this implies $(x_0,\xi_0) \notin \mathrm{WF}(u)$.

For part (2), suppose that $(x_0,\xi_0) \notin \mathrm{WF}(u)$. Pick $B \in \Psi^0(\mathbb{R}^n)$, elliptic at (x_0,ξ_0), such that $Bu \in \mathcal{C}^\infty(\mathbb{R}^n)$. Let $\epsilon > 0$ be such that $|x - x_0| < 2\epsilon$ and $|\frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|}| < 2\epsilon$ imply $(x,\xi) \in \mathrm{Ell}(B)$. Suppose $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(B(x_0,\epsilon))$ and $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ are given, with $\psi(\omega) = 1$ for all $\omega \in \mathbb{S}^{n-1}$ with $|\omega - \frac{\xi_0}{|\xi_0|}| < \epsilon$, and $|\omega - \frac{\xi_0}{|\xi_0|}| < 2\epsilon$ for all $\omega \in \operatorname{supp}\psi$. Define $a(y,\xi)$ by (6.27) and set $A := \mathrm{Op}_R(a)$. Then

$$\mathrm{WF}'(A) \subset \mathrm{Ell}(B). \tag{6.28}$$

(See Example 6.10.) Therefore, $Au \in \mathcal{C}^\infty(\mathbb{R}^n)$. We claim that in fact

$$Au \in \mathscr{S}(\mathbb{R}^n), \tag{6.29}$$

which proves the validity of (6.26) for all N upon taking the inverse Fourier transform of Au. To prove (6.29), let $\tilde\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ be identically 1 near $\operatorname{supp}\phi$. Then $\tilde\phi(Au) \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, while $(1-\tilde\phi)Au = \mathrm{Op}(a')u$ where

$$a'(x,y,\xi) := (1-\tilde\phi(x))a(y,\xi) \in S^0(\mathbb{R}^n \times \mathbb{R}^n;\mathbb{R}^n). \tag{6.30}$$

Note then that $a'(x,y,\xi) = 0$ near $x = y$. Therefore, $\mathrm{Op}(a') \in \Psi^{-\infty}(\mathbb{R}^n)$. (This follows from Corollary 4.9 and Proposition 4.13.) Since the Schwartz kernel of $\mathrm{Op}(a')$ has support in $\mathbb{R}^n \times B(x_0,\epsilon)$ and decays rapidly as $|x-y| \to \infty$, it lies in $\mathscr{S}(\mathbb{R}^n \times \mathbb{R}^n)$, and hence, $\mathrm{Op}(a')u = \mathrm{Op}(a')(\tilde\phi u) \in \mathscr{S}(\mathbb{R}^n)$ (Exercise 2.20). □

Remark 6.22 (Interpretation). In the notation of (6.26), consider the formula for the inverse Fourier transform of $\widehat{\phi u}$,

$$(\phi u)(x) = (2\pi)^{-n}\int_{\mathbb{R}^n} e^{ix\cdot\xi}\widehat{\phi u}(\xi)\,\mathrm{d}\xi. \tag{6.31}$$

This reconstructs ϕu as a superposition of plane waves $e^{ix\cdot\xi}$ with amplitudes $\widehat{\phi u}(\xi)$. If $\widehat{\phi u}(\xi)$ does *not* decay rapidly for $\frac{\xi}{|\xi|}$ near $\frac{\xi_0}{|\xi_0|}$, the integral over $\xi \in \mathbb{R}^n$ with $|\xi| \geq 1$ and $|\frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|}| < \epsilon$ thus superimposes plane waves with rapid oscillations and "nontrivial" amplitudes in directions $\approx \frac{\xi_0}{|\xi_0|}$. Note that

$\frac{\xi_0}{|\xi_0|}$ is conormal to the level sets of the plane wave $x \mapsto e^{ix\cdot\xi_0}$. This motivates the terminology "wave front set."

We next show that WF is a significant refinement of sing supp:

Theorem 6.23 (Wave Front Set and Singular Support). *Denote by* $\pi\colon T^*M \to M$ *the projection. Let* $u \in \mathscr{D}'(M)$. *Then*

$$\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp} u. \tag{6.32}$$

Proof. If $x_0 \notin \operatorname{sing\,supp} u$, then there exists $\chi \in \mathcal{C}_c^\infty(M)$ with $\chi(x_0) \neq 0$ such that $\chi u \in \mathcal{C}_c^\infty(M)$. But $A_\alpha \in \Psi^0_{\mathrm{cl}}(M)$ is elliptic at (x_0, ξ_0) for all $0 \neq \xi_0 \in T^*_{x_0}M$; hence $(T^*_{x_0}M \setminus o) \cap \mathrm{WF}(u) = \emptyset$.

To prove the converse, suppose that $x_0 \notin \pi(\mathrm{WF}(u))$. Then for each $\alpha \in S^*_{x_0}M$ there exists $A_\alpha \in \Psi^0_{\mathrm{cl}}(M)$, elliptic at α, such that $A_\alpha u \in \mathcal{C}^\infty(M)$. Set $U_\alpha := \mathrm{Ell}(A_\alpha) \cap S^*_{x_0}M$. Then U_α is an open cover of the *compact* set $S^*_{x_0}M$; thus we can pick a finite subcover,

$$S^*_{x_0}M = \bigcup_{i=1}^N U_{\alpha_i}, \quad \alpha_i \in S^*_{x_0}M,\ i = 1, \dots, N. \tag{6.33}$$

Fix a volume density on M and consider the operator

$$A := \sum_{i=1}^N A^*_{\alpha_i} A_{\alpha_i} \in \Psi^0(M). \tag{6.34}$$

It is elliptic on $S^*_{x_0}M$ since $\sigma^0(A) = \sum_{i=1}^N |\sigma^0(A_{\alpha_i})|^2$, and for every $\alpha \in S^*_{x_0}M$ there exists an index i such that $\sigma^0(A_{\alpha_i}) \neq 0$ at α. Furthermore, $Au \in \mathcal{C}^\infty(M)$. For all $\chi \in \mathcal{C}_c^\infty(M)$ with $S^*_{\operatorname{supp}\chi}M \subset \mathrm{Ell}(A)$, we then have $\chi u \in \mathcal{C}^\infty(M)$ by the proof of Lemma 6.19. Taking such a χ with $\chi(x_0) \neq 0$ shows that $x_0 \notin \operatorname{sing\,supp} u$. □

Corollary 6.24 (Wave Front Set and Smoothness). *Let* $u \in \mathscr{D}'(M)$. *Then* $\mathrm{WF}(u) = \emptyset$ *if and only if* $u \in \mathcal{C}^\infty(M)$.

It is now time to give some examples.

Example 6.25 (δ-Distributions).

(1) Consider $\delta \in \mathscr{D}'(\mathbb{R}^n)$. We claim that

$$\mathrm{WF}(\delta) = \{(0, \xi) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})\colon \xi \neq 0\}. \tag{6.35}$$

Since $\operatorname{sing\,supp}\delta = \{0\}$, Theorem 6.23 implies that $\mathrm{WF}(\delta) \subseteq \{(0,\xi)\colon \xi \neq 0\}$. Given $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(0) \neq 0$, we have $\widehat{\phi\delta}(\xi) = \phi(0)$, which is not rapidly decaying in the conic neighborhood of any $\xi_0 \neq 0$. The claim now

follows from Proposition 6.21. (One can also give an ad hoc argument based on Corollary 6.24: since $\delta \notin \mathcal{C}^\infty(\mathbb{R}^n)$, we must have $\mathrm{WF}(\delta) \neq \emptyset$. But δ is rotationally symmetric, so $(0,\xi) \in \mathrm{WF}(\delta)$ implies $(0, R\xi) \in \mathrm{WF}(\delta)$ for all $R \in SO(n)$. This implies (6.35).)

(2) More generally, if we split $\mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^{n-k}_y$, $k \leq n-1$, then for $u(x,y) = \delta(y)$ we have

$$\mathrm{WF}(u) = \{(x,y,\xi,\eta)\colon y = 0,\ \xi = 0,\ \eta \neq 0\} = N^*\{y=0\} \setminus o. \tag{6.36}$$

Example 6.26 (Characteristic Function of the Half Space). Write points in $\mathbb{R}^n = \mathbb{R} \times \mathbb{R}^{n-1}$ as $x = (x_1, x')$. Let $u(x_1,x') = H(x_1)$ where H is the Heaviside function. Then

$$\mathrm{WF}(u) = \{(x_1,x',\xi_1,\xi')\colon x_1 = 0,\ \xi' = 0,\ \xi_1 \neq 0\} = N^*\{x_1=0\} \setminus o. \tag{6.37}$$

We verify this using Proposition 6.21. First, note that $\operatorname{sing\,supp} u = \{x_1 = 0\}$. Let now $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ be equal to 1 near 0, and let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{n-1})$. Then

$$\mathcal{F}\big(\phi(x_1)\chi(x')u(x_1,x')\big)(\xi_1,\xi') = \hat\chi(\xi') \int_0^\infty e^{-ix_1\xi_1}\phi(x_1)\,\mathrm{d}x_1. \tag{6.38}$$

The integral equals $\mathcal{F}(H\phi) = (2\pi)^{-1}\hat H * \hat\phi$. From $\hat\phi \in \mathscr{S}(\mathbb{R})$, $\int_{\mathbb{R}} \hat\phi(\xi_1)\,\mathrm{d}\xi_1 = 2\pi$, and $\hat H(\xi_1) = -i(\xi_1 - i0)^{-1}$ (recalling Exercise 2.6) it follows that $\mathcal{F}(H\phi)(\xi_1) = -i\xi_1^{-1} + r(\xi_1)$ where $|r(\xi_1)| \leq C_N|\xi_1|^{-N}$ for all N and $|\xi_1| \geq 1$. The first conclusion is the upper bound

$$|\mathcal{F}(\phi\chi u)(\xi_1,\xi')| \leq C_N\langle\xi_1\rangle^{-1}\langle\xi'\rangle^{-N} \tag{6.39}$$

for all N. This decays rapidly in sufficiently small conic neighborhoods of all points $(\xi_1,\xi') \neq (\pm 1, 0)$. The second conclusion is

$$\mathcal{F}(\phi\chi u)(\xi_1,0) = \frac{1}{i\xi_1}\hat\chi(0) + r(\xi_1), \quad |\xi_1| \geq 1, \tag{6.40}$$

where r decays rapidly as $|\xi_1| \to \infty$. This does not decay rapidly as $|\xi_1| \to \infty$, and hence, $(0,x',\xi_1,0) \in \mathrm{WF}(u)$.

Example 6.27 (Characteristic Functions of Smooth Domains). A curved analogue of Example 6.26 is the following. Let $\Omega \subset \mathbb{R}^n$ be a smoothly bounded domain. Then $\mathrm{WF}(1_\Omega) = N^*\partial\Omega \setminus o$ (Exercise 6.3).

Example 6.28 (Characteristic Function of a Domain with Corners). Write points in $\mathbb{R}^2$ as $x = (x_1,x_2)$. Let $u(x_1,x_2) = H(x_1)H(x_2)$. Then

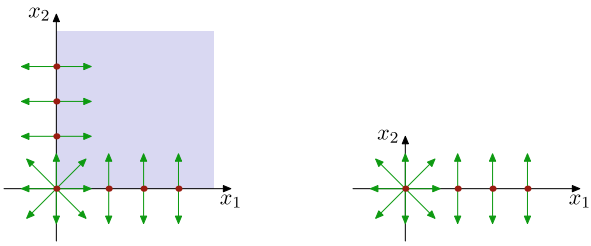


Figure 6.2 *On the left:* illustration of the wave front set of $H(x_1)H(x_2)$ (Example 6.28). *On the right:* illustration of the wave front set of $H(x_1)\delta(x_2)$ (Example 6.29).

$$\begin{aligned}\mathrm{WF}(u) = (N^*\{x_1 = 0,\ x_2 > 0\} \setminus o) \cup (N^*\{x_2 = 0,\ x_1 > 0\} \setminus o) \\ \cup\, (N^*\{x_1 = x_2 = 0\} \setminus o).\end{aligned} \tag{6.41}$$

See Figure 6.2 (left panel). The reader can verify this near $x_1 = x_2 = 0$ using a variant of the computation (6.38).

Example 6.29 (Cutoff δ-Distribution). In the notation of the previous example, consider $u(x_1, x_2) = H(x_1)\delta(x_2)$; by this we mean the distribution $\mathscr{S}(\mathbb{R}^2) \ni \phi \mapsto \int_0^\infty \phi(x_1, 0)\, \mathrm{d}x_1$. Then

$$\mathrm{WF}(u) = (N^*\{x_1 > 0,\ x_2 = 0\} \setminus o) \cup (N^*\{x_1 = x_2 = 0\} \setminus o). \tag{6.42}$$

See Figure 6.2 (right panel). We leave the detailed verification to the reader.

Example 6.30 ($+i0$ Distributions). Recall the distribution $(x+i0)^{-1} = \lim_{\epsilon\to 0}(x + i\epsilon)^{-1} \in \mathscr{S}'(\mathbb{R})$ from Exercise 2.6. Then $\mathrm{WF}((x + i0)^{-1}) = \{(0, \xi)\colon \xi > 0\}$. This can be deduced from $\mathcal{F}((x + i0)^{-1}) = -2\pi i H$. See Exercises 6.4 and 6.10 for generalizations.

Example 6.31 (Schwartz Kernel of Ps.d.o.s). If $A \in \Psi^m(\mathbb{R}^n)$ is a ps.d.o. with Schwartz kernel K, then (Exercise 6.9)

$$\mathrm{WF}(K) = \{(x, x, \xi, -\xi)\colon (x, \xi) \in \mathrm{WF}'(A)\}. \tag{6.43}$$

For general $K \in \mathscr{S}'(\mathbb{R}^n \times \mathbb{R}^m)$, one writes $\mathrm{WF}'(K) := \{(x, y, \xi, -\eta)\colon (x, y, \xi, \eta) \in \mathrm{WF}(K)\}$, the prime indicating the reflection in the second set of momentum variables (η); see, e.g., Hörmander [Hör71a, Theorem 2.5.14]. This is the origin of the notation $\mathrm{WF}'(A)$ for the operator wave front set.

We next study the relationship of wave front sets and PDEs. In order to analyze wave front sets of solutions of PDEs, it is typically neither necessary nor useful to work with hands-on characterizations such as Proposition 6.21. (This is analogous to how one can work with ps.d.o.s without knowing much about their Schwartz kernels.) We first prove:

Proposition 6.32 (Microlocality of Pseudodifferential Operators). *Let $A \in \Psi^m(M)$, $u \in \mathscr{D}'(M)$. Then*

$$\mathrm{WF}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u). \tag{6.44}$$

In view of Theorem 6.23, this is a significant strengthening of the pseudolocality property of ps.d.o.s (Proposition 4.16 and its generalization to manifolds).

***Proof of Proposition* 6.32**. If $\alpha \notin \mathrm{WF}'(A)$, then there exists $B \in \Psi^0(M)$, elliptic at α, with $\mathrm{WF}'(B) \cap \mathrm{WF}'(A) = \emptyset$. Thus $B(Au) = (BA)u \in \mathcal{C}^\infty(M)$ since $BA \in \Psi^{-\infty}(M)$, and hence, $\alpha \notin \mathrm{WF}(Au)$.

If $\alpha \notin \mathrm{WF}(u)$, then there exists $B \in \Psi^0(M)$, elliptic at α, such that $Bu \in \mathcal{C}^\infty(M)$. Let $\tilde{B} \in \Psi^0(M)$ be elliptic at α and such that $\mathrm{WF}'(\tilde{B}) \subset \mathrm{Ell}(B)$. Let $Q \in \Psi^0(M)$ be a microlocal parametrix of B on $\mathrm{WF}'(\tilde{B})$, that is, $QB = I - R$, $R \in \Psi^0(M)$, $\mathrm{WF}'(\tilde{B}) \cap \mathrm{WF}'(R) = \emptyset$. Then

$$\tilde{B}(Au) = \tilde{B}A(QB + R)u = \tilde{B}AQ(Bu) + (\tilde{B}AR)u. \tag{6.45}$$

The first summand is smooth since Bu is; the second summand is smooth since $\tilde{B}AR \in \Psi^{-\infty}(M)$. □

Moreover, we have the following regularity result, which substantially sharpens Proposition 4.32:

Proposition 6.33 (Microlocal Elliptic Regularity). *Let $A \in \Psi^m(M)$, $u \in \mathscr{D}'(M)$. Then*

$$\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{Char}(A). \tag{6.46}$$

In particular, if A is elliptic, then $\mathrm{WF}(u) = \mathrm{WF}(Au)$.

Proof. Suppose $\alpha \notin \mathrm{WF}(Au)$ and $\alpha \in \mathrm{Ell}(A)$. Then there exists $B \in \Psi^0(M)$, elliptic at α, such that $B(Au) \in \mathcal{C}^\infty(M)$. Let $\Lambda \in \Psi^{-m}(M)$ be elliptic. Then $\alpha \in \mathrm{Ell}(\Lambda BA)$ by (6.14), and hence $\Lambda BAu \in \mathcal{C}^\infty(M)$ implies $\alpha \notin \mathrm{WF}(u)$ by Corollary 6.20. (We introduce the operator Λ here to shift orders: $\Lambda BA \in \Psi^0(M)$.)

The claim about elliptic A follows from (6.44) and (6.46) since $\mathrm{Char}(A) = \emptyset$. □

Example 6.34 (Wave Front Set Bounds from PDEs).

(1) Consider $\delta \in \mathscr{D}'(\mathbb{R}^n)$. Then $x_j\delta = 0$, $j = 1, \dots, n$. Proposition 6.33 with $A = x_j$, $u = \delta$ gives

$$\mathrm{WF}(\delta) \subset \bigcap_{j=1}^{n} \mathrm{Char}(x_j) = \{(0, \xi) \colon \xi \neq 0\}. \tag{6.47}$$

Equality holds by Example 6.25.

(2) Consider $u(x_1, x_2) = H(x_1)\delta(x_2) \in \mathscr{D}'(\mathbb{R}^2)$. Then $\partial_{x_1} u = \delta(x_1, x_2)$ and $x_2 u = 0$. In view of $\operatorname{supp} u \subset \{x_1 \geq 0\}$, we thus obtain

$$\begin{aligned} \mathrm{WF}(u) &\subset \big(\mathrm{WF}(\delta) \cup \mathrm{Char}(\partial_{x_1})\big) \cap \mathrm{Char}(x_2) \cap \{x_1 \geq 0\} \\ &= \{(x_1, 0, \xi_1, \xi_2) \colon x_1 = 0 \text{ or } (x_1 > 0 \text{ and } \xi_1 = 0), \\ &\qquad\qquad (\xi_1, \xi_2) \neq (0, 0)\}. \end{aligned} \tag{6.48}$$

Equality holds by Example 6.29.

(3) Consider $u(t, x) = \frac{1}{2} H(t-x)H(t+x) = \frac{1}{2} H(t) 1_{|x| \leq t} \in \mathscr{D}'(\mathbb{R}^2_{t,x})$. Let $A = -D_t^2 + D_x^2$. Then $Au = \delta(t, x)$. Writing σ, ξ for the canonical momentum variables, we therefore have

$$\begin{aligned} \mathrm{WF}(u) \subset \{(t, x, \sigma, \xi) \colon &-\sigma^2 + \xi^2 = 0,\ (\sigma, \xi) \neq (0, 0)\} \\ &\cup \{(0, 0, \sigma, \xi) \colon (\sigma, \xi) \neq (0, 0)\}. \end{aligned} \tag{6.49}$$

One can compute $\mathrm{WF}(u)$ using Example 6.28 via a change of coordinates. From the non-elliptic equation $Au = \delta$, one can in fact extract much more information with stronger tools (Example 8.6 below).

The wave front set studied earlier is more specifically the *smooth wave front set* or $\mathcal{C}^\infty$ *wave front set*: it measures the lack of smoothness of a distribution. In applications, the following more refined notion is more useful:

Definition 6.35 (H^s Wave Front Set). Let $s \in \mathbb{R}$, $u \in \mathscr{D}'(M)$. Then the H^s *wave front set* of u is

$$\mathrm{WF}^s(u) := \bigcap_{\substack{A \in \Psi^0(M) \\ Au \in H^s_{\mathrm{loc}}(M)}} \mathrm{Char}(A). \tag{6.50}$$

That is, its complement is the set of all $\alpha \in S^*M$ for which there exists $A \in \Psi^0(M)$, elliptic at α, such that $Au \in H^s_{\mathrm{loc}}(M)$.

This is equivalent to an alternative definition paralleling Definition 6.18. If $\alpha \notin \mathrm{WF}^s(u)$, one says that u is *microlocally in* H^s *near* α. We collect results analogous to those for the $\mathcal{C}^\infty$ wave front set; the proofs are left to the reader. We have

$$\mathrm{WF}^s(u) = \emptyset \iff u \in H^s_{\mathrm{loc}}(M). \tag{6.51}$$

The analogue of Proposition 6.21 is the following:

Proposition 6.36 (H^s Wave Front Set: Equivalent Definition). *Let* $u \in \mathscr{D}'(\mathbb{R}^n)$, $(x_0, \xi_0) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$.

(1) *If there exist* $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, $\phi(x_0) \neq 0$, *and* $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$, $\psi(\frac{\xi_0}{|\xi_0|}) \neq 0$, *such that*

$$\langle \xi \rangle^s \psi\Big(\frac{\xi}{|\xi|}\Big)\widehat{\phi u}(\xi) \in L^2(\mathbb{R}^n_\xi), \tag{6.52}$$

then $(x_0, \xi_0) \notin \mathrm{WF}^s(u)$.

(2) *If for all* $\epsilon > 0$ *there exist* $\phi \in \mathcal{C}_c^\infty(B(x_0, \epsilon))$, $\phi(x_0) \neq 0$, *and* $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ *with* $\psi(\frac{\xi_0}{|\xi_0|}) \neq 0$ *and* $|\omega - \frac{\xi_0}{|\xi_0|}| < \epsilon$ *for* $\omega \in \operatorname{supp}\psi$ *such that* (6.52) *does* not *hold, then* $(x_0, \xi_0) \in \mathrm{WF}^s(u)$.

Example 6.37 (Sobolev Wave Front Sets).

(1) Consider $\delta \in \mathscr{D}'(\mathbb{R}^n)$. Then $\mathrm{WF}^s(\delta) = \emptyset$ for $s < -\frac{n}{2}$, and $\mathrm{WF}^s(\delta) = T_0^*\mathbb{R}^n \setminus o$ for $s \geq -\frac{n}{2}$.
(2) Let $\Omega \subset \mathbb{R}^n$ be a smoothly bounded domain. Then $\mathrm{WF}^s(1_\Omega) = \emptyset$ for $s < \frac{1}{2}$, and $\mathrm{WF}^s(1_\Omega) = N^*\partial\Omega \setminus o$ for $s \geq \frac{1}{2}$.
(3) $\mathrm{WF}^s((x+i0)^{-1}) = \emptyset$ for $s < -\frac{1}{2}$, and $\mathrm{WF}^s((x+i0)^{-1}) = \{(0,\xi)\colon \xi > 0\}$ for $s \geq -\frac{1}{2}$.

The sharpening of Propositions 6.32 and 6.33 is:

Proposition 6.38 (Microlocality and Microlocal Elliptic Regularity: H^s Version). *Let* $A \in \Psi^m(M)$, $u \in \mathscr{D}'(M)$. *Then*

$$\mathrm{WF}^{s-m}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}^s(u), \tag{6.53}$$

$$\mathrm{WF}^s(u) \subset \mathrm{WF}^{s-m}(Au) \cup \mathrm{Char}^m(A). \tag{6.54}$$

In particular, if A *is elliptic, then* $\mathrm{WF}^s(u) = \mathrm{WF}^{s-m}(Au)$.

A *quantitative* version of the *qualitative* statement (6.54) on microlocal elliptic regularity reads as follows.

Proposition 6.39 (Microlocal Elliptic Regularity Estimate). *Let* $A \in \Psi^m(M)$. *Suppose* $B, G \in \Psi^0(M)$ *are such that*

$$\mathrm{WF}'(B) \subset \mathrm{Ell}(G), \quad \mathrm{WF}'(B) \subset \mathrm{Ell}(A). \tag{6.55}$$

Assume that the Schwartz kernels B *and* G *are supported in* $K \times K$ *where* $K \subset M$ *is compact, and let* $\tilde{K} \subset M$ *be a compact set containing* K *such that* $\operatorname{supp} K_A \cap (K \times M) \subset K \times \tilde{K}$ *where* K_A *denotes the Schwartz kernel of* A. *(See Figure 6.3.) Let* $\chi \in \mathcal{C}_c^\infty(M)$ *be identically* 1 *near* $\tilde{K}$. *Then for all* $s, N \in \mathbb{R}$, *there exists* $C > 0$ *such that*

$$\|Bu\|_{H^s} \leq C\big(\|GAu\|_{H^{s-m}} + \|\chi u\|_{H^{-N}}\big). \tag{6.56}$$

Here $\|\cdot\|_{H^s}$ *denotes a norm on* $\dot{H}^s(K')$ *where* K' *is the compact set* $K' := \operatorname{supp}\chi$ *(see Proposition 5.58). The estimate* (6.56) *holds in the strong sense*

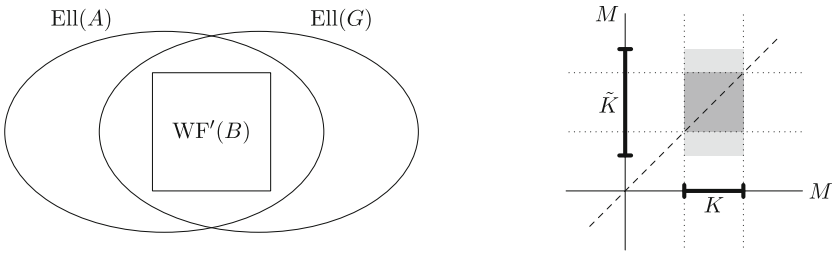


Figure 6.3 *On the left:* illustration of the microlocal elliptic regularity estimate (6.56). *On the right:* the Schwartz kernels of B, G are supported in the dark square, and the support of the Schwartz kernel of A intersects the strip over K inside of $K \times \tilde{K}$. The dashed line is the diagonal in $M \times M$.

that if $u \in \mathscr{D}'(M)$ is such that the right-hand side is finite, then so is the left-hand side, and the estimate holds.

For *differential* operators A, we can of course take $\tilde{K} = K$. We stress that Proposition 6.39 is *better* than an a priori estimate, as microlocal H^s-membership of u is *concluded* (with estimates) rather than merely *assumed* (and estimated).

Proof of Proposition 6.39. Since $\mathrm{WF}'(B) \subset \mathrm{Ell}(GA)$, Proposition 6.17 produces $Q \in \Psi^{-m}(M)$ with $QGA = I - R$, $R \in \Psi^0(M)$, $\mathrm{WF}'(B) \cap \mathrm{WF}'(R) = \emptyset$. Therefore,

$$Bu = B\chi u = BQGA\chi u + BR\chi u = BQ(GAu) + BR(\chi u). \tag{6.57}$$

If $\chi u \in H^{-N}$ and $GAu \in H^{s-m}$, this expresses Bu as a sum of an element of H^s and an element of $\mathcal{C}^\infty$; therefore, $Bu \in H^s$. Taking H^s-norms of (6.57) and using Corollary 5.60 gives the estimate (6.56). □

The dependence of the constant C in (6.56) on A, B, G, s, m, N, χ can in principle be tracked. A simple but often sufficient observation is that for fixed s, m, N, χ, the constant C can be taken to be a function of fixed seminorms of A, B, G.

One can also regard the quantitative Proposition 6.39 as a consequence of the qualitative Proposition 6.38 and the Closed Graph theorem (Exercise 6.13). While conceptually rather attractive, this abstract approach has two main disadvantages. First, the constant C in (6.56) cannot be tracked. Second, the most natural proof of the qualitative statement (6.54) proceeds via the identity (6.57), which immediately implies the quantitative statement (6.56); that is, the application of the Closed Graph theorem merely recovers quantitative bounds which one has forgotten to write down when noting (6.54).

We end this section by recording the relationship between $\mathcal{C}^\infty$ and H^s wave front sets:

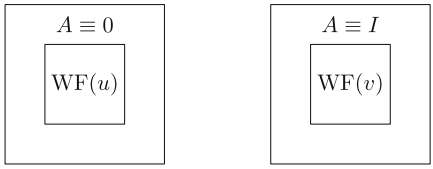


Figure 6.4 Wave front sets of u, v and microlocal properties of A in Proposition 6.41.

Proposition 6.40 ($\mathcal{C}^\infty$ and H^s Wave Front Sets). *Let $u \in \mathscr{D}'(M)$. Then*

$$\mathrm{WF}(u) = \overline{\bigcup_{s\in\mathbb{R}} \mathrm{WF}^s(u)}. \tag{6.58}$$

It is easy to see that $\bigcup_{s\in\mathbb{R}} \mathrm{WF}^s(u)$ is, in general, a proper subset of $\mathrm{WF}(u)$.

Proof of Proposition 6.40. Clearly $\mathrm{WF}^s(u) \subset \mathrm{WF}(u)$; this shows "$\supseteq$" in (6.58). To prove "$\subseteq$," suppose $\alpha \in S^*M$ has an open neighborhood $U \subset S^*M$ such that $U \cap \mathrm{WF}^s(u) = \emptyset$ for all $s \in \mathbb{R}$. Then if $A \in \Psi^0(M)$, $\mathrm{WF}'(A) \subset U$, is elliptic at α, then $Au \in \bigcap_{s\in\mathbb{R}} H^s_{\mathrm{loc}}(M) = \mathcal{C}^\infty(M)$, and hence $\alpha \notin \mathrm{WF}(u)$. □

6.4 Pairings, Products, Restrictions

We now use the wave front set to give fairly precise answers to questions such as: when is the product of two distributions well-defined? When can distributions be restricted to submanifolds? *For notational simplicity, we work on* $\mathbb{R}^n$ for the most part, but all results have analogues on manifolds.

We first consider generalizations of the $L^2(\mathbb{R}^n)$-inner product

$$\langle u, v\rangle_{L^2} = \int_{\mathbb{R}^n} u(x)\overline{v(x)}\,\mathrm{d}x. \tag{6.59}$$

Proposition 6.41 (L^2-Pairings with Wave Front Set Conditions). *Suppose $u, v \in \mathscr{E}'(\mathbb{R}^n)$ satisfy $\mathrm{WF}(u) \cap \mathrm{WF}(v) = \emptyset$. If $A \in \Psi^0(\mathbb{R}^n)$ is such that*

$$\mathrm{WF}(u) \cap \mathrm{WF}'(A) = \emptyset, \quad \mathrm{WF}(v) \cap \mathrm{WF}'(I - A) = \emptyset \tag{6.60}$$

(see Figure 6.4), then

$$\langle u, v\rangle_{L^2} := \langle Au, v\rangle_{L^2} + \langle u, (I - A^*)v\rangle_{L^2} \tag{6.61}$$

is well-defined; and it is independent of the choice of A.

Proof. Proposition 6.32 gives $Au, (I - A^*)v \in \mathcal{C}^\infty(\mathbb{R}^n)$, so (6.61) is well-defined.

Suppose $B \in \Psi^0(\mathbb{R}^n)$ satisfies the conditions on A in (6.60). Then

$$\langle u, v\rangle'_{L^2} := \langle Bu, v\rangle_{L^2} + \langle u, (I - B^*)v\rangle_{L^2} \tag{6.62}$$

is well-defined, too, and we want to show that the difference

$$\langle u, v\rangle'_{L^2} - \langle u, v\rangle_{L^2} = \langle (A - B)u, v\rangle_{L^2} - \langle u, (A^* - B^*)v\rangle_{L^2} \tag{6.63}$$

vanishes. If $u, v \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, this follows from integration by parts. When u, v are merely distributions, we need to use an approximation argument. Choose $v_j \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ such that $v_j \to v$ in $\mathscr{E}'(\mathbb{R}^n)$; then

$$\langle (A-B)u, v\rangle_{L^2} = \lim_{j\to\infty} \langle (A-B)u, v_j\rangle_{L^2} = \lim_{j\to\infty} \langle u, (A^* - B^*)v_j\rangle_{L^2}. \tag{6.64}$$

We wish to show that this equals $\langle u, (A^* - B^*)v\rangle_{L^2}$. We have $(A^* - B^*)v_j \to (A^* - B^*)v$ in $\mathscr{D}'(\mathbb{R}^n)$; but since $u \in \mathscr{E}'(\mathbb{R}^n)$, this is not enough to conclude. Pick thus $Q \in \Psi^0(\mathbb{R}^n)$, with compactly supported Schwartz kernel, satisfying $\mathrm{WF}'(Q) \cap \mathrm{WF}(u) = \emptyset$ and $\mathrm{WF}'(I - Q) \cap (\mathrm{WF}'(A) \cup \mathrm{WF}'(B)) = \emptyset$. Then we can further write

$$\langle u, (A^* - B^*)v_j\rangle_{L^2} = \langle u, (I - Q)(A^* - B^*)v_j\rangle_{L^2} + \langle u, Q(A^* - B^*)v_j\rangle_{L^2}. \tag{6.65}$$

Since $(I - Q)(A^* - B^*) \in \Psi^{-\infty}(\mathbb{R}^n)$, we have $(I - Q)(A^* - B^*)v_j \to (I - Q)(A^* - B^*)v$ in $\mathcal{C}^\infty(\mathbb{R}^n)$, and hence, the first pairing converges to $\langle u, (I - Q)(A^* - B^*)v\rangle_{L^2}$. The second pairing equals

$$\langle Q^*u, (A^* - B^*)v_j\rangle_{L^2} \to \langle Q^*u, (A^* - B^*)v\rangle_{L^2}, \quad j \to \infty, \tag{6.66}$$

since $Q^*u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$. Since $(A^* - B^*)v \in \mathcal{C}^\infty(\mathbb{R}^n)$, we can move Q^* back to the second factor.

Altogether, we have proved that the limit in (6.64) is indeed equal to $\langle u, (A^* - B^*)v\rangle_{L^2}$, and hence (6.63) vanishes. □

A more precise form of Proposition 6.41 will be useful in positive commutator arguments in Chapters 8 and 10. First, recall from Lemma 2.16 that the L^2-pairing (6.59) extends to a sesquilinear pairing $H^s(\mathbb{R}^n) \times H^{-s}(\mathbb{R}^n) \to \mathbb{C}$. Then:

Lemma 6.42 (L^2-Pairings with H^s Wave Front Set Conditions). *Let $s \in \mathbb{R}$. Let $u \in H^s(\mathbb{R}^n)$, $v \in \mathscr{E}'(\mathbb{R}^n)$, and suppose that $\mathrm{WF}(u) \cap \mathrm{WF}^{-s}(v) = \emptyset$ (i.e., v is microlocally in H^{-s} wherever u is not smooth). Let $A \in \Psi^0(\mathbb{R}^n)$ be such that*

$$\mathrm{WF}(u) \cap \mathrm{WF}'(A) = \emptyset, \quad \mathrm{WF}^{-s}(v) \cap \mathrm{WF}'(I - A) = \emptyset. \tag{6.67}$$

Then $\langle u, v\rangle_{L^2} := \langle Au, v\rangle_{L^2} + \langle u, (I - A^*)v\rangle_{L^2}$ *is independent of* A.

The proof is left to the reader.
For a distribution $v \in \mathscr{D}'(\mathbb{R}^n)$, we have

$$\mathrm{WF}(\bar{v}) = -\mathrm{WF}(v) := \{(x, -\xi) \colon (x, \xi) \in \mathrm{WF}(v)\}. \tag{6.68}$$

We thus deduce from Proposition 6.41 that we can define a bilinear pairing

$$\langle u, v\rangle = \langle u, \bar{v}\rangle_{L^2}, \quad u, v \in \mathscr{E}'(\mathbb{R}^n),\ \mathrm{WF}(u) \cap (-\mathrm{WF}(v)) = \emptyset. \tag{6.69}$$

Corollary 6.43 (Product of Distributions). *Let* $u, v \in \mathscr{E}'(\mathbb{R}^n)$, *and suppose that*

$$\mathrm{WF}(u) \cap (-\mathrm{WF}(v)) = \emptyset. \tag{6.70}$$

Then

$$\mathcal{C}^\infty(\mathbb{R}^n) \ni \phi \mapsto \langle u, \phi v\rangle \in \mathbb{C} \tag{6.71}$$

given by (6.69) *defines a distribution* $uv \in \mathscr{E}'(\mathbb{R}^n)$.

More generally, the condition (6.70) permits one to multiply $u, v \in \mathscr{D}'(\mathbb{R}^n)$ and obtain $uv \in \mathscr{D}'(\mathbb{R}^n)$. The condition (6.70) is more precise than the condition $\operatorname{sing\,supp} u \cap \operatorname{sing\,supp} v = \emptyset$ under which the product uv can be defined easily using a partition of unity on $\mathbb{R}^n$. For the sense in which Corollary 6.43 is the *unique* extension of the pointwise product uv for smooth u, v, see Exercise 6.15.

Proof of Corollary 6.43. Pick a properly supported operator $A \in \Psi^0(\mathbb{R}^n)$ with $\mathrm{WF}(u) \cap \mathrm{WF}'(A) = \emptyset$ and $(-\mathrm{WF}(v)) \cap \mathrm{WF}'(I - A) = \emptyset$. Then

$$|\langle u, \phi v\rangle| \le |\langle Au, \phi v\rangle| + |\langle u, (I - A^T)(\phi v)\rangle|. \tag{6.72}$$

Since $Au \in \mathcal{C}^\infty(\mathbb{R}^n)$, the first summand is continuous in ϕ. Bounding the second summand requires more effort. Choose a properly supported operator $B \in \Psi^0(\mathbb{R}^n)$ such that $\mathrm{WF}'(B) \cap \mathrm{WF}'(I - A^T) = \emptyset$ and $\mathrm{WF}'(I - B) \cap \mathrm{WF}(v) = \emptyset$. Then

$$v = Bv + w, \quad w = (I - B)v \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n). \tag{6.73}$$

We can bound the first term in

$$|\langle u, (I - A^T)(\phi v)\rangle| \le |\langle u, (I - A^T)(\phi w)\rangle| + |\langle u, (I - A^T)\phi B v\rangle| \tag{6.74}$$

by $|\langle u, (I - A^T)(\phi w)\rangle| \le C\|\chi(I - A^T)(\phi w)\|_{\mathcal{C}_b^k(\mathbb{R}^n)}$ for some $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ in view of the continuity of $u \in \mathscr{E}'(\mathbb{R}^n)$. By Sobolev embedding, this is bounded by $C'\|\chi(I - A^T)(w\phi)\|_{H^s(\mathbb{R}^n)}$ where we fix $s > \frac{n}{2} + k$. Since the Schwartz kernel of $\chi(I - A^T) \circ (w\cdot) \in \Psi^0(\mathbb{R}^n)$ is compactly supported, this is further bounded by $C''\|\tilde{\chi}\phi\|_{H^s(\mathbb{R}^n)}$ for some $\tilde{\chi} \in \mathcal{C}_c^\infty(\mathbb{R}^n)$. Fixing $\mathbb{N} \ni k' \ge s$, this is finally bounded by $C'''\|\phi\|_{\mathcal{C}_b^{k'}(\mathbb{R}^n)}$.

Consider the second term in (6.74). Since $I - A^T$ and B are properly supported, the ps.d.o. $(I - A^T) \circ (\phi\cdot) \circ B \in \Psi^{-\infty}(\mathbb{R}^n)$ depends continuously on $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$. Therefore,

$$(I - A^T)\phi B v \in \mathcal{C}^\infty(\mathbb{R}^n) \tag{6.75}$$

depends continuously on $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$ as well. □

For example, $(x+i0)^{s_1} \cdot (x+i0)^{s_2} = (x+i0)^{s_1+s_2}$. On the other hand, $\delta \cdot \delta$ is not well-defined.

Lastly, we consider restrictions to submanifolds. We start with the local model

$$Y = \{(x, 0) \colon x \in \mathbb{R}^k\} \subset \mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^{n-k}_y. \tag{6.76}$$

Denote by $\iota \colon Y \hookrightarrow \mathbb{R}^n$ the inclusion map. The restriction of $u \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ to Y is given by $\iota^* u \in \mathcal{C}_c^\infty(Y)$. Regarded as a compactly supported distribution, this acts on test functions via

$$\mathcal{C}^\infty(\mathbb{R}^k) \ni \phi \mapsto (\iota^* u)(\phi) = \langle u \cdot \delta(y), \tilde{\phi}\rangle, \quad \tilde{\phi} \in \mathcal{C}^\infty(\mathbb{R}^n),\ \iota^* \tilde{\phi} = \phi. \tag{6.77}$$

If $u \in \mathscr{E}'(\mathbb{R}^n)$ is such that

$$\mathrm{WF}(u) \cap \{(x, 0, 0, \eta) \colon x \in \mathbb{R}^k,\ \eta \in \mathbb{R}^{n-k} \setminus \{0\}\} = \emptyset, \tag{6.78}$$

then the product $u(x, y)\delta(y) \in \mathscr{E}'(\mathbb{R}^n)$ is well-defined by Corollary 6.43 and Example 6.25, and hence, we get the first part of the following result:

Proposition 6.44 (Restriction of Distributions to Linear Subspaces). *Suppose $u \in \mathscr{E}'(\mathbb{R}^n)$ satisfies (6.78). Then (6.77) defines an element $\iota^* u \in \mathscr{E}'(\mathbb{R}^n)$, and*

$$\mathrm{WF}(\iota^* u) \subset \{(x, \xi) \in \mathbb{R}^k \times (\mathbb{R}^k \setminus \{0\}) \colon \exists\, \eta \in \mathbb{R}^{n-k} \text{ with } (x, 0, \xi, \eta) \in \mathrm{WF}(u)\}. \tag{6.79}$$

Proof. We have $\iota^*(\chi u) = \iota^* u$ for any $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^{n-k})$ which is identically 1 near 0. The assumption (6.78) implies that if χ has sufficiently small support, then upon replacing u by χu,

$$\hat{u}(\xi, \eta) \text{ is rapidly decaying in a cone around } \{0\} \times (\mathbb{R}^{n-k} \setminus \{0\}). \tag{6.80}$$

(This follows from the compactness of $\operatorname{supp} u$ and a partition of unity in x.) When $u = u(x,y) \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, the Fourier inversion formula gives

$$\widehat{\iota^* u}(\xi) = (\mathcal{F}_1 u)(\xi, 0) = (2\pi)^{-(n-k)} \int_{\mathbb{R}^{n-k}} \hat{u}(\xi, \eta)\, \mathrm{d}\eta, \tag{6.81}$$

where $\mathcal{F}_1$ denotes the Fourier transform in the first argument of u. We claim that the same formula holds for $u \in \mathscr{E}'(\mathbb{R}^n)$ subject to (6.80). To prove this, we use the definition of $u \cdot \delta(y)$ in Corollary 6.43 via (6.61) with $A = \mathcal{F}^{-1}\psi\mathcal{F}$ where $\psi \in S^0(\mathbb{R}^n_{\xi,\eta})$ is equal to 1 in a conic neighborhood of $\{0\} \times (\mathbb{R}^{n-k}_\eta \setminus \{0\})$ and equal to 0 outside of a slightly larger conic neighborhood such that $\psi\hat{u}$ is rapidly decaying in *all* directions. The value of the smooth function $\mathcal{F}(\iota^* u)$ at $\xi_0 \in \mathbb{R}^k$ is given by the distributional pairing of $u \cdot \delta(y)$ with $e^{-i\xi_0 \cdot}$, so

$$\mathcal{F}(\iota^* u)(\xi_0) = \langle Au, e^{-ix\cdot\xi_0}\delta(y)\rangle + \langle u, (I - A^T)(e^{-ix\cdot\xi_0}\delta(y))\rangle. \tag{6.82}$$

(We insert the variables for better readability.) The first pairing equals

$$\begin{aligned} \langle \mathcal{F}^{-1}\psi\mathcal{F}u, e^{-ix\cdot\xi_0}\delta(y)\rangle &= \langle \psi(\xi,\eta)\hat{u}(\xi,\eta), \mathcal{F}^{-1}(e^{-ix\cdot\xi_0}\delta(y))\rangle \\ &= \langle \psi(\xi,\eta)\hat{u}(\xi,\eta), (2\pi)^{-(n-k)}\delta(\xi - \xi_0)\rangle \\ &= (2\pi)^{-(n-k)} \int_{\mathbb{R}^{n-k}} \psi(\xi_0,\eta)\hat{u}(\xi_0,\eta)\, \mathrm{d}\eta. \end{aligned} \tag{6.83}$$

The second pairing equals

$$\begin{aligned} &\langle u, \mathcal{F}(1-\psi)\mathcal{F}^{-1}(e^{-ix\cdot\xi_0}\delta(y))\rangle \\ &\quad = \langle u, \mathcal{F}(1-\psi)(2\pi)^{-(n-k)}\delta(\xi - \xi_0)\rangle \\ &\quad = \left\langle u, (2\pi)^{-(n-k)} e^{-ix\cdot\xi_0} \int_{\mathbb{R}^{n-k}} e^{-iy\cdot\eta}(1 - \psi(\xi_0,\eta))\, \mathrm{d}\eta \right\rangle \\ &\quad = (2\pi)^{-(n-k)} \int_{\mathbb{R}^{n-k}} \langle u, e^{-ix\cdot\xi_0} e^{-iy\cdot\eta}\rangle (1 - \psi(\xi_0,\eta))\, \mathrm{d}\eta \\ &\quad = (2\pi)^{-(n-k)} \int_{\mathbb{R}^{n-k}} (1 - \psi(\xi_0,\eta))\hat{u}(\xi_0,\eta)\, \mathrm{d}\eta. \end{aligned} \tag{6.84}$$

Adding (6.83) and (6.84) gives (6.81).

To prove (6.79), we now apply (6.81) to a localized version of u. Suppose $(x_0, \xi_0) \in \mathbb{R}^k \times (\mathbb{R}^k \setminus \{0\})$ is such that for all $\eta \in \mathbb{R}^{n-k}$ we have $(x_0, 0, \xi_0, \eta) \notin \mathrm{WF}(u)$. Then for all $\eta \in \mathbb{R}^{n-k}$, there exists $\epsilon_\eta > 0$ such that for all $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(B(x_0, \epsilon_\eta)) \subset \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(B(0, \epsilon_\eta)) \subset \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{n-k})$, the Fourier transform of $\psi(x)\chi(y)u(x,y)$ is rapidly decaying for (ξ, η) in a conic neighborhood $C_\eta \subset \mathbb{R}^n \setminus \{0\}$ of (ξ_0, η). By (6.80), the Fourier transform

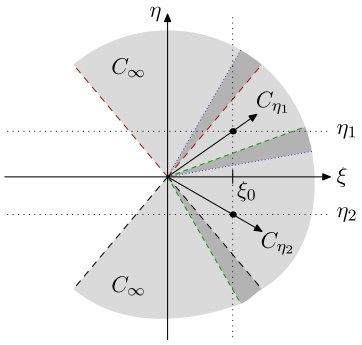


Figure 6.5 Some cones in which $\mathcal{F}(\psi(x)\chi(y)u(x,y))$ is rapidly decaying; drawn here is the setting $n = 2$, $k = 1$.

of $\psi(x)\chi(y)u(x,y)$ is also rapidly decaying in a conic neighborhood C_∞ of $\{0\} \times \mathbb{R}^{n-k}$ for all $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{n-k})$.

The bounded set of $\eta \in \mathbb{R}^{n-k}$ such that $(\xi_0, \eta) \notin C_\infty$ can be covered by finitely many sets $\{\eta \in \mathbb{R}^{n-k} \colon (\xi_0, \eta) \in C_{\eta_i}\}$ for suitable $\eta_i \in \mathbb{R}^{n-k}$, $i = 1, \dots, N$. See Figure 6.5. Set $\epsilon := \min\{\epsilon_1, \dots, \epsilon_N\}$. Then for $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(B(x_0, \epsilon))$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(B(0, \epsilon))$, we conclude that there exists $\delta > 0$ such that

$$|\widehat{\psi\chi u}(\xi, \eta)| \le C_N \langle \xi \rangle^{-N} \langle \eta \rangle^{-N}, \quad \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \delta, \tag{6.85}$$

for all N. Using (6.81) for χu (which satisfies $\iota^*(\chi u) = \iota^* u$), we conclude that

$$|\widehat{\psi \iota^* u}(\xi)| \le C'_N \langle \xi \rangle^{-N}, \quad \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \delta, \tag{6.86}$$

for all N. Therefore, $(x_0, \xi_0) \notin \mathrm{WF}(\iota^* u)$. □

Example 6.45 (Restriction of a δ-Distribution). On $\mathbb{R}^2_{x,y}$, consider $u_a = \delta(y - ax)$, $a \in \mathbb{R}$. Then the restriction of u_a to $Y = \{y = 0\}$ is well-defined for $a \neq 0$. We have $\iota^* u_a = |a|^{-1}\delta$. Note that this blows up when $|a| \to 0$; in this limit, $\mathrm{WF}(u_\alpha)$ converges to $N^*\{y = 0\} \setminus o$.

Example 6.46 (Wave Front Set of the Restriction). In general, the inclusion (6.79) is not an equality. For example, on $\mathbb{R}^2_{x,y}$, the restriction of $u(x,y) = y\delta(x)$ to $y = 0$ is 0 even though $\mathrm{WF}(u) = \{(x, y, \xi, \eta) \in \mathbb{R}^2 \times (\mathbb{R}^2 \setminus \{0\}) \colon x = 0,\ \eta = 0\}$.

Corollary 6.47 (Restriction of Distributions to Submanifolds). *Let M be a smooth manifold and let $Y \subset M$ be a smooth submanifold. Then there exists a linear restriction map*

$$\iota^* \colon \{u \in \mathscr{D}'(M) \colon \mathrm{WF}(u) \cap N^*Y = \emptyset\} \to \mathscr{D}'(Y), \tag{6.87}$$

and $\mathrm{WF}(\iota^* u) \subset T^*Y \setminus o$ is contained in the image of $\mathrm{WF}(u) \cap T^*_Y M$ in $T^*_Y M / N^*Y \cong T^*Y$.

6.5 Intermezzo: Radial Compactification of Phase Space

We have already seen that it is convenient to work with the cosphere bundle (6.8). Another perspective on this is the following. Suppose $a \in S^0_{\mathrm{cl}}(T^*M)$ is a classical symbol; then for every $0 \neq \xi \in T^*_pM$, $p \in M$, the limit

$$a_0(p,[\xi]) := \lim_{\lambda\to\infty} a(p,\lambda\xi) \tag{6.88}$$

is well-defined, and indeed $T^*M \setminus o \ni \xi \mapsto a_0(p,[\xi])$ is equal to the homogeneous principal symbol of a. While one can think of $[\xi]$ as the equivalence class of ξ in the quotient $(T^*_pM \setminus o)/\mathbb{R}_+$, it is more natural to regard it as a "point at infinity" in the direction ξ. Thus, we wish to attach a "sphere at infinity" of directions to T^*_pM. The basic construction takes place on $\mathbb{R}^n$:

Definition 6.48 (Radial Compactification). The *radial* (or *projective*) *compactification* of $\mathbb{R}^n$ is the set $\overline{\mathbb{R}^n} = \mathbb{R}^n \sqcup \mathbb{S}^{n-1}$ equipped with the structure of a smooth manifold with boundary as follows: writing $0 \neq x \in \mathbb{R}^n$ in polar coordinates as $x = r\omega$, $r > 0$, $\omega \in \mathbb{S}^{n-1}$, then

$$\overline{\mathbb{R}^n} = \Big(\mathbb{R}^n \sqcup \big([0,\infty)_\rho \times \mathbb{S}^{n-1}_\omega\big)\Big)/\sim \tag{6.89}$$

where $\mathbb{R}^n \setminus \{0\} \ni r\omega \sim (r^{-1},\omega)$. Thus, $\rho^{-1}(0) \cong \mathbb{S}^{n-1}$ is the "sphere at infinity," and $\mathbb{R}^n \subset \overline{\mathbb{R}^n}$ is the interior.

The manifold $\overline{\mathbb{R}^n}$ is diffeomorphic to the closed unit ball in $\mathbb{R}^n$; an explicit diffeomorphism is given by the continuous extension of $\mathbb{R}^n \ni x \mapsto \frac{2\arctan|x|}{\pi}\frac{x}{|x|} \in B(0,1)$. In order to understand the relationship between open subsets of $\overline{\mathbb{R}^n}$ and those of $\mathbb{R}^n$, consider a point $\alpha \in \partial\overline{\mathbb{R}^n}$ given by $(\rho,\omega) = (0,\omega_0)$. Then the sets

$$U_\delta := \{\rho < \delta,\ |\omega - \omega_0| < \delta\} \tag{6.90}$$

for $\delta > 0$ are a neighborhood basis of α. The intersection of U_δ with $\mathbb{R}^n$ is given by

$$U_\delta \cap \mathbb{R}^n = \{x \in \mathbb{R}^n \colon |x| > \delta^{-1}\} \cap \left\{0 \neq x \in \mathbb{R}^n \colon \left|\frac{x}{|x|} - \omega_0\right| < \delta\right\}, \tag{6.91}$$

which is thus the far end of a cone with aperture δ around ω_0. See Figure 6.6.

We have $\mathcal{C}^\infty(\overline{\mathbb{R}^n}) = S^0_{\mathrm{cl}}(\mathbb{R}^n)$, in the sense that the space of restrictions of elements of $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$ to (the dense subset) $\mathbb{R}^n$ is equal to $S^0_{\mathrm{cl}}(\mathbb{R}^n)$. Indeed,

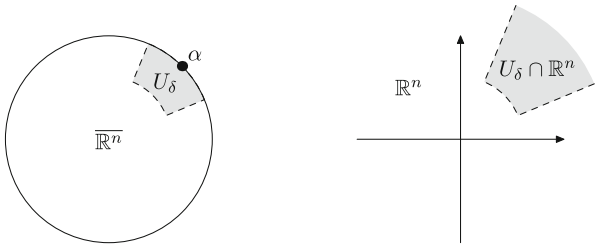


Figure 6.6 *On the left:* the open neighborhood $U_\delta \subset \overline{\mathbb{R}^n}$ (see (6.90)) of a point $\alpha \in \partial\overline{\mathbb{R}^n}$. *On the right:* the intersection of U_δ with $\mathbb{R}^n$.

being smooth on $\overline{\mathbb{R}^n_\xi}$ precisely means having a Taylor expansion in $\rho = |\xi|^{-1}$ at $\rho = 0$, in addition to being smooth for bounded ξ. From this perspective, Exercise 3.4(2) is essentially trivial: when $\chi \in S^0_{\mathrm{cl}}(\mathbb{R}^n)$, $a \in S^0_{\mathrm{cl}}(\mathbb{R}^n)$ (the case $a \in S^m_{\mathrm{cl}}(\mathbb{R}^n)$ requiring only minor modifications), and $a \neq 0$ on $\operatorname{supp}\chi$, then χ/a is a smooth function on $\overline{\mathbb{R}^n}$ and thus a classical symbol on $\mathbb{R}^n$.

Remark 6.49 (Asymptotic Summation and Borel's Lemma). The asymptotic summation of classical symbols is essentially Borel's lemma (on the existence of smooth functions with prescribed Taylor coefficients) at infinity. As a concrete example, let $\chi \in \mathcal{C}^\infty(\mathbb{R})$ be equal to 0 on $(-\infty, 1]$ and equal to 1 on $[2, \infty)$. Let $c_j \in \mathbb{C}$ be arbitrary numbers and set $a_j(\xi) := c_j \xi^{-j} \chi(\xi)$, so $a_j \in S^{-j}_{\mathrm{cl}}(\mathbb{R})$. Then an asymptotic sum $S^0_{\mathrm{cl}}(\mathbb{R}) \ni a \sim \sum_{j\geq 0} a_j$ is a smooth function on $\overline{\mathbb{R}}$ whose j-th Taylor coefficient at $\xi = +\infty$, i.e., the coefficient of $(\xi^{-1})^j$, is precisely c_j.

Local coordinates near $\partial\overline{\mathbb{R}^n}$ which are particularly convenient for computations are projective coordinates: write $x = (x_1, \ldots, x_n)$, and consider the subset $\{x_1 > 0\} \subset \mathbb{R}^n$. We then let

$$\rho_1 := \frac{1}{x_1}, \quad \hat{x}_j := \frac{x_j}{x_1},\ j = 2, \ldots, n. \tag{6.92}$$

Then $(\rho_1, \hat{x}_2, \ldots, \hat{x}_n)$ is system of local coordinates on $\mathbb{R}^n$ which by continuity extends to a local coordinate system

$$[0, \infty)_{\rho_1} \times \mathbb{R}^{n-1}_{\hat{x}_2, \ldots, \hat{x}_n} \tag{6.93}$$

on $\overline{\mathbb{R}^n}$; this follows from the fact that $(\hat{x}_2, \ldots, \hat{x}_n)$ is a smooth local coordinate system on the subset of the unit sphere $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ where $x_1 > 0$. Together with the standard coordinate system on $\mathbb{R}^n$, such coordinate systems (upon switching the sign of x_1 and permuting indices) cover $\overline{\mathbb{R}^n}$. See Figure 6.7.

The manifold $\overline{\mathbb{R}^n}$ has the following invariance property under linear changes of coordinates:

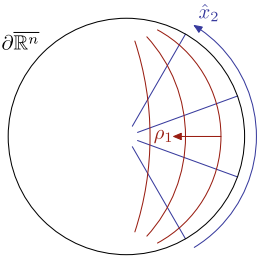


Figure 6.7 The radial compactification $\overline{\mathbb{R}^n}$ of $\mathbb{R}^n$, here shown for $n = 2$, and some level sets of the projective coordinates ρ_1 and $\hat{x}_2$ from (6.92).

Lemma 6.50 (Invertible Linear Maps and Radial Compactifications). *Let $A \in GL(n, \mathbb{R})$. Then matrix-vector multiplication $\mathbb{R}^n \ni x \mapsto Ax \in \mathbb{R}^n$ extends, by continuity, to a diffeomorphism $A\colon \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$.*

Proof. Write $A = (a_{ij})_{i,j=1,\dots,n}$. Let $e_1 = (1, 0, \dots, 0) \in \mathbb{R}^n$. By relabeling the standard coordinates in the target $\mathbb{R}^n$ and possibly switching the sign of one of them, we may assume that for $Ae_1 = (a_{11}, a_{21}, \dots, a_{n1})$ we have $a_{11} > 0$. We work in the coordinates (6.92) in a conic neighborhood of e_1 in the domain $\mathbb{R}^n$. Denoting points in the target $\mathbb{R}^n$ by $y = (y_1, \dots, y_n)$, we can then use the coordinates

$$\tau_1 := \frac{1}{y_1}, \quad \hat{y}_j := \frac{y_j}{y_1},\ j = 2, \dots, n, \tag{6.94}$$

in a conic neighborhood of Ae_1. Now $x = \rho_1^{-1}(1, \hat{x}_2, \dots, \hat{x}_n)$, so

$$Ax = \rho_1^{-1}\left(a_{11} + \sum_{\ell=2}^{n} a_{1\ell}\hat{x}_\ell, \dots,\ a_{n1} + \sum_{\ell=2}^{n} a_{n\ell}\hat{x}_\ell\right). \tag{6.95}$$

For sufficiently small $\hat{x}_2, \dots, \hat{x}_n$, the first coefficient is positive. Therefore, in the coordinates (6.94), A maps $(\rho_1, \hat{x}_2, \dots, \hat{x}_n) \mapsto (\tau_1, \hat{y}_2, \dots, \hat{y}_n)$ where

$$\tau_1 = \frac{\rho_1}{a_{11} + \sum_{\ell=2}^{n} a_{1\ell}\hat{x}_\ell}, \quad \hat{y}_j = \frac{a_{j1} + \sum_{\ell=2}^{n} a_{j\ell}\hat{x}_j}{a_{11} + \sum_{\ell=2}^{n} a_{1\ell}\hat{x}_\ell},\ j = 2, \dots, n, \tag{6.96}$$

are smooth functions of $(\rho_1, \hat{x}_2, \dots, \hat{x}_n)$. Thus, A extends to a smooth map $\overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$. Applying the same considerations to A^{-1} shows that also $A^{-1}\colon \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ is smooth. □

This lemma allows us to define radial compactifications of finite-dimensional real vector spaces and thus of real vector bundles.

Definition 6.51 (Radial Compactification of Vector Bundles). Let $E \to M$ be a real rank k vector bundle. Then the radial compactification $\overline{E} \to M$ is the smooth fiber bundle (with typical fiber $\overline{\mathbb{R}^k}$) obtained by radially

compactifying each fiber of E. (In local trivializations of E, the transition maps of $\overline{E}$ are the continuous extensions of those of E using Lemma 6.50.)

As microlocal analysts, we are interested in the radially compactified cotangent bundle

$$\overline{T^*}M \to M. \tag{6.97}$$

Note that for $p \in M$, we can identify S_p^*M with the sphere at infinity of $\overline{T_p^*}M$ by mapping $S_p^*M \ni [\xi] \mapsto \lim_{\lambda\to\infty} \lambda\xi \in \partial\overline{T_p^*}M$. This embeds

$$S^*M \subset \overline{T^*}M \tag{6.98}$$

as a submanifold (and indeed fiber subbundle), called *fiber infinity*.

Since

$$\mathcal{C}^\infty(\overline{T^*}M) = S^0_{\mathrm{cl}}(T^*M), \tag{6.99}$$

we can now identify the principal symbol of $a \in S^0_{\mathrm{cl}}(T^*M)$ with the *restriction* $a|_{S^*M} = a|_{\partial\overline{T^*}M}$ of a to fiber infinity. The elliptic set of a is the subset $\{\alpha \in S^*M \colon a(\alpha) \neq 0\}$ (Exercise 6.18). Even for non-classical symbols $a \in S^0(T^*M)$, it is reasonable to regard $\mathrm{Ell}(a)$ as a subset of fiber infinity, due to the following observation: for $\alpha \in S^*M$, we have

$$\alpha \in \mathrm{Ell}(a) \iff \exists\, \chi \in \mathcal{C}^\infty(\overline{T^*}M),\ \chi(\alpha) \neq 0,\ \text{such that } \frac{\chi}{a} \in S^0(T^*M). \tag{6.100}$$

We can similarly characterize the essential support by

$$\alpha \notin \operatorname{ess\,supp} a \iff \exists\, \chi \in \mathcal{C}^\infty(\overline{T^*}M),\ \chi(\alpha) \neq 0,\ \text{such that } \chi a \in S^{-\infty}(T^*M). \tag{6.101}$$

We mention one more useful way to think about the relevance of fiber infinity. For the sake of brevity, we discuss this only for symbols depending on ξ. To wit, ellipticity of a symbol $a \in S^0(\mathbb{R}^n_\xi)$ is a notion that only depends on the equivalence class

$$[a] \in S^0(\mathbb{R}^n)/\langle\xi\rangle^{-1}S^0(\mathbb{R}^n); \tag{6.102}$$

note that $\langle\xi\rangle^{-1}S^0 = S^{-1}$. But $\langle\xi\rangle^{-1} \in \mathcal{C}^\infty(\overline{\mathbb{R}^n})$ vanishes precisely at $\partial\overline{\mathbb{R}^n}$, and it vanishes simply there. In the classical case, this is intimately related to the following algebraic way of encoding restrictions of smooth functions on a manifold $(\overline{\mathbb{R}^n})$ to a submanifold $(\partial\overline{\mathbb{R}^n})$: consider the quotient $\mathcal{C}^\infty(\overline{\mathbb{R}^n})/\mathcal{I}$ where $\mathcal{I} \subset \mathcal{C}^\infty(\overline{\mathbb{R}^n})$ is the ideal of smooth functions vanishing at $\partial\overline{\mathbb{R}^n}$. This quotient is exactly $S^0_{\mathrm{cl}}(\mathbb{R}^n)/S^{-1}_{\mathrm{cl}}(\mathbb{R}^n) \cong \mathcal{C}^\infty(\partial\overline{\mathbb{R}^n})$, with the isomorphism given by restriction to $\partial\overline{\mathbb{R}^n}$.

6.6 Semiclassical Microlocal Analysis

Semiclassical ellipticity of a symbol $a \in S^0_\hbar(\mathbb{R}^N)$ (Definition 3.21) concerns the invertibility of a modulo the space $hS^{-1}_\hbar = h\langle\xi\rangle^{-1}S^0_\hbar$ of symbols which have an additional factor of both $\langle\xi\rangle^{-1}$ and h. Mirroring the discussion following (6.102), we are led to devise a space with the property that $h\langle\xi\rangle^{-1}$ vanishes on its boundary hypersurfaces. A natural choice is the manifold with corners

$$[0,1)_h \times \overline{\mathbb{R}^N_\xi}. \tag{6.103}$$

A classical symbol $a = (a_h)_{h\in(0,1)} \in S^0_{\hbar,\mathrm{cl}}(\mathbb{R}^N)$ extends from $(0,1)\times\mathbb{R}^N$ to a smooth function on $[0,1)\times\overline{\mathbb{R}^N}$, and conversely the restriction of a smooth function on $[0,1)\times\overline{\mathbb{R}^N}$ to $(0,h_0)\times\mathbb{R}^N$ defines a classical symbol for every $h_0 \in (0,1)$ (needed for uniformity as $h \nearrow h_0$). That is, requiring, for the sake of clarity, semiclassical symbols to be defined for all $h \in (0,1)$ but without uniform bounds as $h \nearrow 1$, we have

$$S^0_{\hbar,\mathrm{cl}}(\mathbb{R}^N) = \mathcal{C}^\infty([0,1)\times\overline{\mathbb{R}^N}). \tag{6.104}$$

We can thus identify the quotient space $S^0_{\hbar,\mathrm{cl}}/h\langle\xi\rangle^{-1}S^0_{\hbar,\mathrm{cl}}$ with the space of restrictions of smooth functions on $[0,1)\times\overline{\mathbb{R}^N}$ to the boundary hypersurfaces of $[0,1)\times\overline{\mathbb{R}^N}$,

$$\begin{aligned} &S^0_{\hbar,\mathrm{cl}}(\mathbb{R}^N)/hS^{-1}_{\hbar,\mathrm{cl}}(\mathbb{R}^N) \\ &\quad\cong \big\{(a_\hbar, a_\partial)\colon a_\hbar \in \mathcal{C}^\infty(\{0\}\times\overline{\mathbb{R}^N}),\ a_\partial \in \mathcal{C}^\infty([0,1)\times\partial\overline{\mathbb{R}^N}), \\ &\qquad\qquad a_\hbar|_{\{0\}\times\partial\overline{\mathbb{R}^N}} = a_\partial|_{\{0\}\times\partial\overline{\mathbb{R}^N}}\big\}. \end{aligned} \tag{6.105}$$

See Figure 6.8.

This suggests defining elliptic sets etc. in the semiclassical setting as subsets of

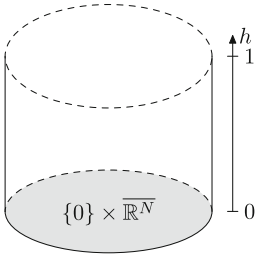


Figure 6.8 The space $[0,1)\times\overline{\mathbb{R}^N}$. We will regard the bottom boundary $\{0\}\times\overline{\mathbb{R}^N}$ as the locus of semiclassical microlocal objects (essential support, elliptic set, etc.).

$$\partial([0,1)\times\overline{\mathbb{R}^N}) = \big(\{0\}\times\overline{\mathbb{R}^N}\big)\cup\big([0,1)\times\partial\overline{\mathbb{R}^N}\big). \tag{6.106}$$

This is, however, not quite what we shall do. After all, semiclassical ellipticity concerns only the *small h behavior* of symbols; one is *not* interested in the behavior at fiber infinity for fixed positive h. Now, if a symbol $a = (a_h)_{h\in(0,1)} \in S^0_{\hbar,\mathrm{cl}}(\mathbb{R}^N)$ has an invertible restriction $a|_{h=0}$ to $\{0\}\times\overline{\mathbb{R}^N}$, then by compactness of $\overline{\mathbb{R}^N}$ there exists $h_0 > 0$ such that a is invertible for $h < h_0$ with $(a_h^{-1})_{h\in(0,h_0)} \in S^0_{\hbar,\mathrm{cl}}(\mathbb{R}^N)$. The locus of semiclassical microlocal analysis will thus be the first space on the right in (6.106) only. We identify this space with $\overline{\mathbb{R}^N}$. The final definitions for symbols and operators, immediately phrased in the manifold setting, read as follows.

Definition 6.52 (Semiclassical Essential Support, Elliptic Set, and Operator Wave Front Set). Let M be a smooth manifold. Let $a \in S^m_\hbar(T^*M)$ and $A \in \Psi^m_\hbar(M)$.

(1) (Essential support.) We say that

$$\alpha \notin \operatorname{ess\,supp}_\hbar a \subset \overline{T^*}M \iff \exists\,\chi\in S^0_{\mathrm{cl}}(T^*M),\ \chi(\alpha)\neq 0,\ \text{such that } \chi a \in h^\infty S^{-\infty}_\hbar(T^*M). \tag{6.107}$$

Here χ is regarded as an h-independent function on $[0,1)\times\overline{T^*}M$, and the membership $\chi a \in h^\infty S^{-\infty}_\hbar$ is meant in the sense that there exists an $h_0 > 0$ (depending on α, χ) such that $(\chi a_h)_{h\in(0,h_0)} \in h^\infty S^{-\infty}_\hbar$.

(2) (Elliptic set.) We say that

$$\alpha \in \mathrm{Ell}_\hbar(a) \subset \overline{T^*}M \tag{6.108}$$

(or $\mathrm{Ell}^m_\hbar(a)$ if one wants to make the order explicit) if and only if there exists $\chi \in S^0_{\mathrm{cl}}(T^*M)$ such that $\chi(\alpha)\neq 0$ and $\frac{\chi}{a} \in S^{-m}_\hbar(T^*M)$. We define $\mathrm{Char}_\hbar(a) := \overline{T^*}M \setminus \mathrm{Ell}_\hbar(a)$.

(3) (Semiclassical ps.d.o. notions.) We define the semiclassical operator wave front set $\mathrm{WF}'_\hbar(A) \subset \overline{T^*}M$ near $\overline{T^*_p}M$, $p \in M$, in local coordinates by $\operatorname{ess\,supp}_\hbar(a_0)$ where $\mathrm{Op}_\hbar(a_0)$ is the expression for A in a local coordinate chart near p. Moreover, we define

$$\mathrm{Ell}_\hbar(A) := \mathrm{Ell}_\hbar(\sigma^m_\hbar(A)),\quad \mathrm{Char}_\hbar(A) := \mathrm{Char}_\hbar(\sigma^m_\hbar(A)) \subset \overline{T^*}M. \tag{6.109}$$

These sets are in general *not* conic subsets of phase space. See Exercises 6.20–6.21 for equivalent characterizations of essential supports and elliptic sets. From the compactness of the fibers of $\overline{T^*}M$, it follows for $a \in S^m_\hbar(T^*M)$ that $\operatorname{ess\,supp}_\hbar a = \emptyset$ if and only if for all $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ one has $\chi a \in h^\infty S^{-\infty}_\hbar(T^*M)$ for $h < h_0(\chi)$. (For compact M, this means

$a \in h^\infty S^{-\infty}_\hbar(T^*M)$.) Similarly, given $A \in \Psi^m_\hbar(M)$, one has $\mathrm{WF}'_\hbar(A) = \emptyset$ if and only if $\chi A \in h^\infty \Psi^{-\infty}_\hbar(M)$ for all $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$.

Example 6.53 (Semiclassical Characteristic Sets).

(1) For a semiclassical microlocalizer $A = \psi(hD)\phi(x)$ where $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, we have $\mathrm{WF}'_\hbar(A) = \operatorname{supp}\phi \times \operatorname{supp}\psi$.
(2) Denote by Δ the Laplacian on $M = \mathbb{R}^n$ or the Laplace–Beltrami operator on a Riemannian manifold (M, g). Then $A_h = h^2\Delta - z \in \mathrm{Diff}^2_\hbar(M)$ has empty characteristic set when $z \notin [0, \infty)$, whereas for $z \in [0, \infty)$ the characteristic set is $\mathrm{Char}_\hbar(A) = \{(x, \xi) \in T^*M \colon |\xi|^2 = z\}$. Note that this is a compact subset of $T^*_x M$ for every $x \in M$.
(3) The characteristic set of $A_h = hD_x - z \in \mathrm{Diff}^1_\hbar(\mathbb{R})$ is $\{(x, \xi) \in \mathbb{R} \times \mathbb{R} \colon \xi = z\} \subset T^*\mathbb{R}$ (and thus empty unless $z \in \mathbb{R}$).
(4) Consider $A_h := h^m B$ where $B \in \mathrm{Diff}^m(M)$, $m \geq 1$, does not depend on h: in this case $\mathrm{Char}^m_\hbar(A) = \{(x, \lambda\xi) \colon (x, \xi) \in \mathrm{Char}^m(B) \subset T^*M \backslash o,\ \lambda \geq 0\}$.

The following result is proved in the same way as Proposition 6.17:

Proposition 6.54 (Semiclassical Microlocal Elliptic Parametrix). *Let $A \in \Psi^m_\hbar(M)$. Let $K \subset \mathrm{Ell}_\hbar(A)$ be a closed subset. Then there exists $B \in \Psi^{-m}_\hbar(M)$ such that*

$$K \cap \mathrm{WF}'_\hbar(AB - I) = \emptyset, \quad K \cap \mathrm{WF}'_\hbar(BA - I) = \emptyset. \tag{6.110}$$

Turning to the *semiclassical wave front set* of families $u = (u_h)_{h \in (0, h_0)}$ of distributions, we emphasize again that what one wants to describe is uniform behavior as $h \to 0$. The definition should satisfy two criteria. First, u should have empty semiclassical wave front set if, over every compact subset of M and for all $k \in \mathbb{R}$, the family $\{h^{-k} u_h \colon h \in (0, h_1)\}$ is uniformly bounded in every semiclassical Sobolev space (where $h_1 \in (0, h_0]$ may depend on the compact subset). Second, if u is *any* "sensible" family of distributions and $R \in h^\infty \Psi^{-\infty}_\hbar(M)$ is a residual operator, then Ru should have empty semiclassical wave front set. These criteria force us to restrict the types of families of distributions for which the semiclassical wave front set is defined to those that are *h-tempered*:

Definition 6.55 (Semiclassical Wave Front Set of h-Tempered Families). Let M be a smooth manifold. Let $u = (u_h)_{h \in (0, h_0)}$ be a family of distributions $u_h \in \mathscr{D}'(M)$.

(1) We say that u is *h-tempered* if for every $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$, the family $(\chi u_h)_{h \in (0, h_0)}$ is uniformly bounded in $h^{-N} H^{-N}_{h, \mathrm{loc}}(M)$ (or equivalently in $h^{-N} \dot{H}^{-N}_h(K)$ where $K = \operatorname{supp}\chi$) for some $N \in \mathbb{R}$ (depending on χ).
(2) For $s \in \mathbb{R}$ and $k \in \mathbb{R} \cup \{\infty\}$, we define the $h^k H^s_h$ *wave front set* of an h-tempered family u as follows:

$$\alpha \notin \mathrm{WF}_\hbar^{s,k}(u) \subset \overline{T^*}M \iff \exists\, A \in \Psi_\hbar^0(M),\ \alpha \in \mathrm{Ell}_\hbar(A),\ \text{such that } Au \text{ is uniformly bounded in } h^k H^s_{\hbar,\mathrm{loc}}(M). \tag{6.111}$$

(Here A is required to be properly supported.) For $k = \infty$, we denote this set by $\mathrm{WF}_\hbar^s(u) := \mathrm{WF}_\hbar^{s,\infty}(u)$.

Using Proposition 6.54 and the a priori assumption that u is h-tempered, one can show that (6.111) is equivalent to the condition that Au be uniformly bounded in $h^k H^s_{\hbar,\mathrm{loc}}$ for *all* $A \in \Psi_\hbar^0(M)$ with $\mathrm{WF}'_\hbar(A)$ contained in some fixed open neighborhood $U \subset \overline{T^*}M$ of α. See Exercise 6.22 for a hands-on characterization of semiclassical wave front sets.

Remark 6.56 (Finite Frequencies). At finite points $\alpha \in T^*M \subset \overline{T^*}M$, the differentiability order s is irrelevant; that is, $\mathrm{WF}_\hbar^{s,k}(u) \cap T^*M$ is independent of $s \in \mathbb{R}$. For $k = \infty$, one denotes this set by

$$\mathrm{WF}_\hbar(u) := \mathrm{WF}_\hbar^0(u) \cap T^*M \subset T^*M. \tag{6.112}$$

This is sometimes called the semiclassical wave front set of u *at finite frequencies*.

Roughly speaking, near a point $\xi_0 \in T^*_{x_0}M$, the semiclassical wave front set $\mathrm{WF}_\hbar^{0,k}(u)$ records whether the amplitude of frequencies $\approx h^{-1}\xi_0$ contributing to u_h near x_0 fail to be of size $\mathcal{O}(h^k)$. Thus, $\mathrm{WF}_\hbar$ detects oscillations at scales $\mathcal{O}(h)$.

Remark 6.57 (Empty Semiclassical Wave Front Set). Using the compactness of the fibers of $\overline{T^*}M$ and exploiting the mapping properties of semiclassical ps.d.o.s, one sees that

$$\mathrm{WF}_\hbar^{s,k}(u) = \emptyset \iff \forall\, \chi \in \mathcal{C}_{\mathrm{c}}^\infty(M)\colon\ \chi u \text{ is uniformly bounded in } h^k \dot{H}^s_\hbar(K),\ K := \operatorname{supp}\chi,\ h < h_0(\chi). \tag{6.113}$$

Example 6.58 (Coherent States). Fix $x_0, \xi_0 \in \mathbb{R}^n$. We consider

$$u_h = (\pi h)^{-\frac{n}{4}} e^{-|x-x_0|^2/2h} e^{i(x-x_0)\cdot\xi_0/h} \in \mathscr{S}(\mathbb{R}^n), \quad h \in (0,1). \tag{6.114}$$

This satisfies $\|u_h\|_{L^2(\mathbb{R}^n)} = 1$ for all h. We claim that for $u = (u_h)_{h\in(0,1)}$, we have

$$\mathrm{WF}_\hbar^{s,k}(u) = \begin{cases} \{(x_0,\xi_0)\}, & k > 0, \\ \emptyset, & k \le 0. \end{cases} \tag{6.115}$$

(In particular, $\mathrm{WF}_\hbar(u) = \mathrm{WF}^s_\hbar(u) = \{(x_0,\xi_0)\}$.) By shifting x, we can reduce to the case $x_0 = 0$. Using Exercise 4.13, one can moreover reduce to the case $\xi_0 = 0$. To prove (6.115), we can then test u_h with $\mathrm{Op}_h(a)$ where $a(x,\xi) = \chi(x)\psi(\xi)$ with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and $\psi \in S^0(\mathbb{R}^n)$. It suffices to show that if $(0,0) \notin \operatorname{supp} a$, then

$$\begin{aligned}\mathrm{Op}_h(a)u_h(x) &= (\pi h)^{-\frac{n}{4}}(2\pi h)^{-n}\int_{\mathbb{R}^n}\int_{\mathbb{R}^n} e^{i(x-y)\cdot\xi/h}a(x,\xi)e^{-|y|^2/2h}\,\mathrm{d}y\,\mathrm{d}\xi\\ &= (\pi h)^{-\frac{n}{4}}(2\pi h)^{-\frac{n}{2}}\chi(x)\int_{\mathbb{R}^n} e^{ix\cdot\xi/h}e^{-|\xi|^2/2h}\psi(\xi)\,\mathrm{d}\xi\end{aligned} \tag{6.116}$$

is uniformly bounded in $h^\infty H^s_h(\mathbb{R}^n)$ for all s. We only need to check this for $s = 0$ since taking derivatives of $\mathrm{Op}_h(a)u_h$ along hD_x leads to similar expressions except that ψ is then a symbol of higher order. If $0 \notin \operatorname{supp}\psi$, the integrand is bounded by $C_N h^N\langle\xi\rangle^{-n-1}$ for all N. If $0 \notin \operatorname{supp}\chi$, then we can write $e^{ix\cdot\xi/h} = (h\frac{x\cdot D_\xi}{|x|^2})^N e^{ix\cdot\xi/h}$ and integrate by parts N times to gain a factor of h^N.

We leave it to the reader to analyze the following variants of Example 6.58: if we make the Gaussian in (6.114) less localized by using $e^{-|x-x_0|^2}$, the semiclassical wave front set is all of $\{(x,\xi_0)\colon x \in \mathbb{R}^n\}$; if, on the other hand, we make the Gaussian more localized by using $e^{-|x-x_0|^2/h^2}$, then the semiclassical wave front set is equal to $\{(x_0,\xi)\colon \xi \in \mathbb{R}^n\}$.

Analogously to Proposition 6.38, we have:

Proposition 6.59 (Semiclassical Microlocality and Microlocal Elliptic Regularity). *Let $A \in \Psi^m_\hbar(M)$ and let u be an h-tempered family. Let $s \in \mathbb{R}$ and $k \in \mathbb{R}\cup\{\infty\}$. Then*

$$\begin{aligned}\mathrm{WF}^{s-m,k}_\hbar(Au) &\subset \mathrm{WF}'_\hbar(A)\cap\mathrm{WF}^{s,k}_\hbar(u),\\ \mathrm{WF}^{s,k}_\hbar(u) &\subset \mathrm{WF}^{s-m,k}_\hbar(Au)\cup\mathrm{Char}^m_\hbar(A).\end{aligned} \tag{6.117}$$

Example 6.60. We revisit Example 6.58 and note that

$$A_h := (hD_x-\xi_0)^2 + |x-x_0|^2 - hn \implies A_h u_h = 0. \tag{6.118}$$

But $A = (A_h)_{h\in(0,1)} \in \Psi^2_\hbar(M)$, $M = \mathbb{R}^n$, has semiclassical principal symbol $|\xi-\xi_0|^2 + |x-x_0|^2$, and thus $\mathrm{WF}^{s,k}_\hbar(u) \subset \mathrm{Char}(A) = \{(x_0,\xi_0)\}$. (We write $\Psi^2_\hbar(M)$ to emphasize that A fits Definition 5.84 but not Definition 2.23.) Recalling $\|u_h\|_{L^2} = 1$, one obtains (6.115).

The analogue of Proposition 6.39 reads as follows.

Proposition 6.61 (Quantitative Semiclassical Microlocal Elliptic Estimate). *Let $A \in \Psi^m_\hbar(M)$. Suppose $B, G \in \Psi^0_\hbar(M)$ are such that*

$$\mathrm{WF}'_\hbar(B) \subset \mathrm{Ell}_\hbar(G), \quad \mathrm{WF}'_\hbar(B) \subset \mathrm{Ell}_\hbar(A). \tag{6.119}$$

Assume that the Schwartz kernels of B_h and G_h are supported in $K \times K$ where $K \subset M$ is compact, and let $\tilde{K} \subset M$ be a compact set containing K such that $\operatorname{supp} K_{A_h} \cap (K \times M) \subset K \times \tilde{K}$ for all h where K_{A_h} denotes the Schwartz kernel of A_h. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ be identically 1 near $\tilde{K}$. Then there exists $h_0 > 0$ such that for all $s, N \in \mathbb{R}$ there exists $C > 0$ such that

$$\|Bu\|_{H^s_h} \leq C\Big(\|GAu\|_{H^{s-m}_h} + h^N \|\chi u\|_{H^{-N}_h}\Big), \quad h < h_0. \tag{6.120}$$

Here $\|\cdot\|_{H^s_h}$ denotes a norm on $\dot{H}^s_h(K')$ where $K' = \operatorname{supp}\chi$. The estimate (6.120) holds in the strong sense that if the right-hand side is finite, then so is the left-hand side, and the estimate holds.

Proof. The proof is essentially the same as that of Proposition 6.39: for a suitable semiclassical microlocal elliptic parametrix Q of GA on $\mathrm{WF}'_\hbar(B)$, defined for $h < h_0$ with h_0 sufficiently small, we have the identity (6.57); now BR is a residual semiclassical operator (which we can arrange to have compactly supported Schwartz kernel), and hence it is uniformly bounded as a map $h^{-N}H_h^{-N} \to H^s_h$. This gives (6.120). □

Notes

The wave front set was introduced in the present form in Hörmander [Hör71a]. Sato [Sat70] had previously introduced a related notion for his hyperfunctions. The analytic wave front set is discussed in Sjöstrand [Sjö82] and Martinez [Mar01]. For distributional sections of vector bundles, the *polarization set* introduced in Dencker [Den82] is a further refinement of the wave front set; it records the component carrying the strongest singularity at a point in phase space.

Besides the operations on distributions studied in Section 6.4, one can analyze the wave front sets of pullbacks of distributions along smooth maps and of pushforwards along smooth proper maps. More generally, one can prove rather general results on the relationship between $\mathrm{WF}(Au)$ and $\mathrm{WF}(u)$ in terms of the wave front set of the Schwartz kernel of $A\colon \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n) \to \mathscr{D}'(\mathbb{R}^m)$; see, e.g., Hörmander [Hör71a, §2.5]. Sobolev refinements of Corollary 6.43 can be found in Beals–Reed [BR84].

Our definition of the semiclassical wave front set as a subset of $\overline{T}^*M$ is essentially taken from Vasy [Vas13, §2.1]. The conceptual advantages of the radial compactification become even clearer when we discuss radial point estimates in Chapter 10. We explicitly restrict to the subset $\{h = 0\} \cong \overline{T}^*M$ of $\partial([0,1) \times \overline{T}^*M)$ since this better matches the usual usage of semiclassical

tools. This is done also in Dyatlov–Zworski [DZ16]. (An equivalent definition of $\mathrm{WF}_\hbar$ as a subset of $T^*M \sqcup S^*M$, i.e., treating finite and infinite frequencies separately, is given in Dyatlov [Dya12].)

Exercises

Exercise 6.1 (Equivalent Definition of Essential Support). Let $a \in S^m(\mathbb{R}^n;\mathbb{R}^N)$. Show that $(x_0,\xi_0) \in \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$ does *not* lie in $\operatorname{ess\,supp} a$ if and only if there exists $\epsilon > 0$ such that for all $k \in \mathbb{R}$, we have

$$|a(x,\xi)| \leq C_k \langle \xi \rangle^{-k} \quad \forall\, (x,\xi) \in \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\}),\ |x - x_0| + \left| \frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|} \right| < \epsilon. \tag{6.121}$$

Exercise 6.2 (Why Conic Sets?). The purpose of this exercise is to demonstrate two ways in which symbols of class S^0 can *only* localize in conic sets in ξ, not in any "narrower" type of set.

(1) Let $a \in S^0(\mathbb{R}^n;\mathbb{R}^N)$. Let $(x_0,\xi_0) \in \mathbb{R}^n \times (\mathbb{R}^N \setminus \{0\})$, and suppose there exist $c, C > 0$ such that $|a(x_0,\lambda\xi_0)| \geq c$ for all $\lambda \geq C$. Show that $(x_0,\xi_0) \in \mathrm{Ell}^0(a)$.

(2) Write points in $\mathbb{R}^N$ as $\xi = (\xi_1,\xi')$, $\xi_1 \in \mathbb{R}$, $\xi' \in \mathbb{R}^{N-1}$. For $\theta \in (0,1)$ and $r > 0$, set

$$C_r^\theta := \{(\xi_1,\xi') \colon \xi_1 > 0,\ |\xi'| \leq r\xi_1^\theta\}. \tag{6.122}$$

Let $0 < r < r'$. Show that there does *not* exist a symbol $a \in S^0(\mathbb{R}^N_\xi)$ such that $a = 1$ on $C_r^\theta \cap \{|\xi| > 2\}$ and $a = 0$ outside of $C_{r'}^\theta \cap \{|\xi| > 1\}$.

Exercise 6.3 (Wave Front Set of Characteristic Functions). Let $\Omega \subset \mathbb{R}^n$ be a smoothly bounded domain and denote by

$$1_\Omega(x) = \begin{cases} 1, & x \in \Omega, \\ 0, & x \notin \Omega \end{cases} \tag{6.123}$$

its characteristic function. Compute (with proof) $\mathrm{WF}(1_\Omega)$ and $\mathrm{WF}^s(1_\Omega)$ for $s \in \mathbb{R}$.

Exercise 6.4 ($\pm i0$ Distributions). Let $s \in \mathbb{C} \setminus \mathbb{N}_0$. Recall the distribution $(x \pm i0)^s \in \mathscr{S}'(\mathbb{R})$ from Exercise 2.6. Prove that

$$\mathrm{WF}((x \pm i0)^s) = \{(0,\xi) \colon \pm\xi > 0\}. \tag{6.124}$$

Compute also $\mathrm{WF}^r((x \pm i0)^s)$ for $r \in \mathbb{R}$.

Exercise 6.5 (Distribution with Special Wave Front Set). Let $n \in \mathbb{N}$. Construct a distribution $u \in \mathscr{D}'(\mathbb{R}^n)$ whose wave front set $\mathrm{WF}(u) \subset S^*\mathbb{R}^n$ consists of a single point.

Exercise 6.6 (Tensor Products with δ-Distributions). Let $s \in \mathbb{R}$ and $u \in H^s(\mathbb{R}^n_x)$. For $j \in \mathbb{N}_0$, set $v(t,x) := \delta^{(j)}(t)u(x) \in \mathscr{D}'(\mathbb{R}^{1+n}_{t,x})$.

(1) Show that $v \in H^{s'}(\mathbb{R}^{1+n})$ whenever $s' \leq s - j - \frac{1}{2}$ and $s' < -j - \frac{1}{2}$.
(2) Let $s \geq 0$. Show that $\mathrm{WF}^{s-j-\frac{1}{2}}(v) \subset N^*\{t = 0\} \setminus o = \{(t,x,\tau,\xi)\colon t = 0,\ \xi = 0,\ \tau \neq 0\}$.

Exercise 6.7 (Compact Operators). The goal of this exercise is to prove the converse of the result of Exercise 5.31.

(1) Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ be equal to 1 near 0, and let $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ be equal to 1 on a non-empty open set $S \subset \mathbb{S}^{n-1}$. Set $L := \mathrm{Op}_R((1-\chi(\xi))\psi(\frac{\xi}{|\xi|})\chi(y)) \in \Psi^0(\mathbb{R}^n)$. Show that there exist $u_j \in L^2(\mathbb{R}^n)$, $j \in \mathbb{N}$, with the following properties:

$$\begin{aligned} &\|u_j\|_{L^2} = 1\ \forall\, j; \qquad \|u_j - u_k\|_{L^2} > \frac{1}{2}\ \forall\, j \neq k; \\ &u_j - Lu_j \to 0 \text{ in } L^2; \quad u_j - Lu_j \text{ is bounded in } H^s, s > 0. \end{aligned} \tag{6.125}$$

(2) Let M be a closed manifold, $A \in \Psi^0_{\mathrm{cl}}(M)$, $a = \sigma^0(A)$. Suppose $A\colon L^2(M) \to L^2(M)$ is compact. Show that $\lim_{|\xi|\to\infty} |a(x,\xi)| = 0$ where $|\cdot|$ denotes a norm on the fibers of T^*M.
(3) Prove part (2) for $A \in \Psi^0(M)$ using a generalization of part (1) in which now $L = \mathrm{Op}_R(b(\xi)\chi(y))$ where $b \in S^0(\mathbb{R}^n_\xi)$ is equal to 1 on balls of radius $c\lambda_i$ around points $\lambda_i\xi_0$; here $\xi_0 \in \mathbb{R}^n \setminus \{0\}$ and $c > 0$ are fixed, and $\lambda_i \to \infty$.

Exercise 6.8 (Conormal Distributions). Let $a \in S^m(\mathbb{R}^k_x; \mathbb{R}^{n-k}_\zeta)$.

(1) Make sense of the oscillatory integral

$$u(x,z) := (2\pi)^{-(n-k)} \int_{\mathbb{R}^{n-k}} e^{iz\cdot\zeta} a(x,\zeta)\,\mathrm{d}\zeta \tag{6.126}$$

as a distribution on $\mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^{n-k}_z$ in such a way that for $m < -(n-k)$, your definition agrees with the convergent integral, and such that $a \mapsto u$ is continuous as a map $S^m(\mathbb{R}^k; \mathbb{R}^{n-k}) \to \mathscr{S}'(\mathbb{R}^n)$ for all $m \in \mathbb{R}$.
(2) Show that $\mathrm{WF}(u) \subset \{(x,z,\xi,\zeta)\colon z = 0,\ \xi = 0\}$.
(3) Prove that $\mathrm{WF}(u) = \{(x,0,0,\zeta)\colon (x,\zeta) \in \operatorname{ess\,supp} a\}$.

Exercise 6.9 (Wave Front Sets of Schwartz Kernels of Ps.d.o.s). Let $A \in \Psi^m(M)$ be a pseudodifferential operator, and denote its Schwartz kernel by K. Prove that

$$\mathrm{WF}(K) = \{(x, x, \xi, -\xi) \in T^*(M \times M) \colon (x, \xi) \in \mathrm{WF}'(A)\}. \tag{6.127}$$

Exercise 6.10 (Holomorphic Functions in a Half Space). The following is a partial generalization of Exercise 6.4. Denote by $\Omega = \{z \in \mathbb{C} \colon \operatorname{Im} z > 0\}$ the upper half plane, and let $F \colon \Omega \to \mathbb{C}$ be holomorphic. Suppose that for each $C > 0$ there exist $C', N \in \mathbb{R}$ such that $|F(z)| < C'|\operatorname{Im} z|^{-N}$ for $z \in \Omega$ with $|\operatorname{Re} z| < C,\ \operatorname{Im} z \in (0, 1]$.

(1) Show that the functions $F_\epsilon = F(\cdot + i\epsilon) \in \mathcal{C}^\infty(\mathbb{R})$ converge in $\mathscr{D}'(\mathbb{R})$ as $\epsilon \searrow 0$. The limit is denoted $f := F(\cdot + i0) \in \mathscr{D}'(\mathbb{R})$.
(2) Show that $\mathrm{WF}(F(\cdot + i0)) \subset \{(x, \xi) \colon \xi > 0\}$.

Exercise 6.11 (H^s Wave Front Set). Let $K \subset S^*M$ be closed. Show that the following conditions are equivalent for $u \in \mathscr{D}'(M)$:

(1) $\mathrm{WF}^s(u) \cap K = \emptyset$;
(2) there exists $G \in \Psi^0(M)$, $K \subset \mathrm{Ell}(G)$, such that $Gu \in H^s_{\mathrm{loc}}(M)$.

Exercise 6.12 (Another Microlocal Elliptic Regularity Estimate). Let $B \in \Psi^0(M)$. Suppose $B_1, \dots, B_J \in \Psi^0(M)$ are such that $\mathrm{WF}'(B) \subset \bigcup_{j=1}^J \mathrm{Ell}(B_j)$. Assume that the Schwartz kernels of B, B_j are supported in $K \times K$ where $K \subset M$ is compact. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ be identically 1 near K. Show that for all $N \in \mathbb{R}$ there exists $C > 0$ such that the estimate

$$\|Bu\|_{H^s} \leq C\left(\sum_{j=1}^J \|B_j u\|_{H^s} + \|\chi u\|_{H^{-N}}\right). \tag{6.128}$$

holds (in the strong sense). Here $\|\cdot\|_{H^s}$ is a norm on $\dot{H}^s(K')$, $K' := \operatorname{supp} \chi$.

Exercise 6.13 (Qualitative and Quantitative Microlocal Elliptic Regularity). We operate under the assumptions of Proposition 6.39 and use the notation introduced there. Show using Proposition 6.38 that the map

$$\{u \in \dot{H}^{-N}(K') \colon \operatorname{supp} u \subset \chi^{-1}(1),\ GAu \in \dot{H}^{s-m}(K')\} \ni u \mapsto Bu \in \dot{H}^s(K') \tag{6.129}$$

is well-defined. Use the Closed Graph theorem to deduce the estimate (6.56).

Exercise 6.14 (Stability of Microlocal Elliptic Estimates). Suppose $A \in \mathrm{Diff}^m(M)$, $B, G \in \Psi^0(M)$, $K = \tilde{K} \subset M$, and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ satisfy the assumptions of Proposition 6.39. Fix $s, N \in \mathbb{R}$. Show that there exist a constant C' and a neighborhood $\mathcal{U} \subset \mathrm{Diff}^m(M)$ of A such that the estimate (6.56) holds for C' and $A' \in \mathcal{U}$ in place of C and A, respectively. (The same holds for pseudodifferential A and general $K, \tilde{K}$ if one restricts to those A' whose Schwartz kernels satisfy an appropriate support condition as in Proposition 6.39.)

Exercise 6.15 (Uniqueness of Multiplication and Restriction Maps). Given a closed conic subset $\Lambda \subset T^*\mathbb{R}^n \setminus o$, we define a locally convex topology on the space

$$\mathscr{E}'_\Lambda(\mathbb{R}^n) := \{u \in \mathscr{E}'(\mathbb{R}^n) \colon \mathrm{WF}(u) \subset \Lambda\} \tag{6.130}$$

using the seminorms $u \mapsto |\langle u, \phi\rangle|$ ($\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$) and $u \mapsto \|Au\|_{\mathcal{C}^k_b(\mathbb{R}^n)}$ ($k \in \mathbb{N}_0$, $A \in \Psi^0(\mathbb{R}^n)$, $\mathrm{WF}'(A) \cap \Lambda = \emptyset$).

(1) Show that $\mathscr{E}'_\Lambda(\mathbb{R}^n)$ is sequentially complete, and $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is a sequentially dense subspace.

(2) Let $\Lambda, \Lambda' \subset T^*\mathbb{R}^n \setminus o$ be closed and conic; suppose that $\Lambda \cap (-\Lambda') = \emptyset$. Show that $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n) \times \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n) \ni (u, v) \mapsto uv \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ has a unique extension to a sequentially continuous bilinear map

$$\mathscr{E}'_\Lambda(\mathbb{R}^n) \times \mathscr{E}'_{\Lambda'}(\mathbb{R}^n) \to \mathscr{E}'(\mathbb{R}^n). \tag{6.131}$$

(3) Let $Y := \mathbb{R}^k \times \{0\} \subset \mathbb{R}^k \times \mathbb{R}^{n-k}$, and let $\Lambda \subset T^*\mathbb{R}^n \setminus o$ be a closed conic subset with $\Lambda \cap (N^*Y \setminus o) = \emptyset$. Show that the restriction map $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n) \ni u \mapsto u|_Y \in \mathcal{C}^\infty_{\mathrm{c}}(Y)$ has a unique extension to a sequentially continuous map

$$\mathscr{E}'_\Lambda(\mathbb{R}^n) \to \mathscr{E}'(Y). \tag{6.132}$$

Exercise 6.16 (Stereographic Projection and Radial Compactification). Consider the map

$$\begin{aligned} \phi \colon \mathbb{S}^n_> &:= \{x = (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \colon |x| = 1,\ x_{n+1} > 0\} \\ &\ni x \mapsto \Big(\frac{x_1}{x_n}, \dots, \frac{x_n}{x_{n+1}}\Big) \in \mathbb{R}^n. \end{aligned} \tag{6.133}$$

Show that ϕ extends by continuity to a diffeomorphism

$$\bar\phi \colon \mathbb{S}^n_\geq := \{x \in \mathbb{R}^{n+1} \colon |x| = 1,\ x_{n+1} \geq 0\} \to \overline{\mathbb{R}^n} \tag{6.134}$$

which maps the equator of $\mathbb{S}^n$ diffeomorphically to $\partial\overline{\mathbb{R}^n} = \mathbb{S}^{n-1}$. See Figure 6.9.

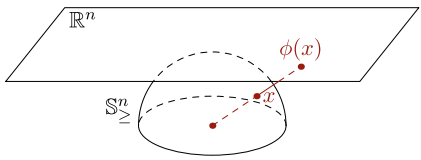


Figure 6.9 Illustration of the stereographic projection (6.133).

Exercise 6.17 (Residual Symbols and Radial Compactification). Show that $S^{-\infty}(T^*M)$ is equal to the space $\dot{\mathcal{C}}^\infty(\overline{T^*}M)$ of smooth functions which vanish to infinite order at $\partial\overline{T^*}M = S^*M$, in the sense that the restriction map $\dot{\mathcal{C}}^\infty(\overline{T^*}M) \ni u \mapsto u|_{T^*M} \in S^{-\infty}(T^*M)$ is an isomorphism.

Exercise 6.18 (More on Classical Symbols). Let $a \in S^0_{\mathrm{cl}}(T^*M) = \mathcal{C}^\infty(\overline{T^*}M)$.

(1) Show that $\mathrm{Ell}(a) = \{\alpha \in S^*M \colon a(\alpha) \neq 0\}$.
(2) Show that $S^*M \setminus \operatorname{ess\,supp} a$ is equal to the set of all $\alpha \in S^*M$ which have an open neighborhood $U \subset S^*M$ such that a vanishes to infinite order at U.

Exercise 6.19 (Regularity of Symbols on the Radial Compactification). On a smooth manifold M with boundary, write $\mathcal{V}_{\mathrm{b}}(M) \subset \mathcal{V}(M) = \mathcal{C}^\infty(M; TM)$ for the space of all vector fields V which are tangent to the boundary of M, i.e., $V_p \in T_p\partial M$ for all $p \in \partial M$.

(1) Show that ∂_{x_i} and $x_i\partial_{x_j}$, $i, j = 1, \dots, n$, define elements of $\mathcal{V}_{\mathrm{b}}(\overline{\mathbb{R}^n})$.
(2) Show, conversely, that every $V \in \mathcal{V}_{\mathrm{b}}(\overline{\mathbb{R}^n})$ can be written as $V = \sum_{i=1}^n a_i\partial_{x_i} + \sum_{i,j=1}^n a_{ij}x_i\partial_{x_j}$ with $a_i, a_{ij} \in \mathcal{C}^\infty(\overline{\mathbb{R}^n})$.
(3) Let $a \in \mathcal{C}^\infty(\mathbb{R}^n_\xi)$. Show that $a \in S^0(\mathbb{R}^n)$ if and only if $V_1 \cdots V_k u$ is bounded on $\mathbb{R}^n$ for all $V_1, \dots, V_k \in \mathcal{V}_{\mathrm{b}}(\overline{\mathbb{R}^n})$, $k \in \mathbb{N}$.

Remark. The difference between $S^0_{\mathrm{cl}}(\mathbb{R}^n)$ and $S^0(\mathbb{R}^n)$ is thus *smoothness* (remaining bounded upon application of elements of $\mathcal{V}(\overline{\mathbb{R}^n})$) versus *conormality* (remaining bounded upon application of elements of $\mathcal{V}_{\mathrm{b}}(\overline{\mathbb{R}^n})$).

Exercise 6.20 (Semiclassical Essential Support). Let $a = (a_h)_{h\in(0,h_0)} \in S^m_\hbar(\mathbb{R}^n; \mathbb{R}^N)$.

(1) Let $\alpha = (x_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^N$. Then $\alpha \notin \operatorname{ess\,supp}_\hbar a$ if and only if there exists $\epsilon > 0$ such that for all k we have

$$|a_h(x,\xi)| \leq C_k h^k \quad \forall\, h < \epsilon,\ |x - x_0| < \epsilon,\ |\xi - \xi_0| < \epsilon. \tag{6.135}$$

(2) Let $\alpha = (x_0, [\xi_0]) \in \mathbb{R}^n \times \partial\overline{\mathbb{R}^N}$. Then $\alpha \notin \operatorname{ess\,supp}_\hbar a$ if and only if there exists $\epsilon > 0$ such that for all k we have

$$|a_h(x,\xi)| \leq C_k h^k \langle\xi\rangle^{-k} \quad \forall\, h < \epsilon,\ |x - x_0| < \epsilon,\ \left|\frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|}\right| < \epsilon. \tag{6.136}$$

Exercise 6.21 (Semiclassical Elliptic Set). Let $a = (a_h)_{h\in(0,h_0)} \in S^m_\hbar(\mathbb{R}^n; \mathbb{R}^N)$.

(1) Let $\alpha = (x_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^N$. Then $\alpha \notin \mathrm{Ell}_\hbar(a)$ if and only if there exist $\epsilon, c > 0$ such that

$$|a_h(x,\xi)| \geq c \quad \forall\, h < \epsilon,\ |x - x_0| < \epsilon,\ |\xi - \xi_0| < \epsilon. \tag{6.137}$$

(2) Let $\alpha = (x_0, [\xi_0]) \in \mathbb{R}^n \times \partial\overline{\mathbb{R}^N}$. Then $\alpha \notin \mathrm{Ell}_\hbar(a)$ if and only if there exist $\epsilon, c > 0$ such that

$$|a_h(x,\xi)| \geq c\langle\xi\rangle^m \quad \forall\, h < \epsilon,\ |x - x_0| < \epsilon,\ \left|\frac{\xi}{|\xi|} - \frac{\xi_0}{|\xi_0|}\right| < \epsilon. \tag{6.138}$$

Exercise 6.22 (Characterization of Semiclassical Wave Front Sets). Let $u = (u_h)_{h\in(0,h_0)}$ be an h-tempered family of distributions on $\mathbb{R}^n$. Let $s, k \in \mathbb{R}$.

(1) Let $\alpha = (x_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^n$. Show that $\alpha \notin \mathrm{WF}_\hbar^{s,k}(u)$ if and only if there exist $\phi, \psi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$ with $\phi(x_0) \neq 0$, $\psi(\xi_0) \neq 0$, and $h_1 \in (0, h_0]$ such that

$$\{h^{-k}\psi(h\xi)\widehat{\phi u_h}(\xi)\colon h < h_1\} \text{ is uniformly bounded in } L^2(\mathbb{R}^n_\xi). \tag{6.139}$$

(2) Let $\alpha = (x_0, [\xi_0]) \in \mathbb{R}^n \times \partial\overline{\mathbb{R}^n}$. Show that $\alpha \notin \mathrm{WF}_\hbar^{s,k}(u)$ if and only if there exist $\phi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$ with $\phi(x_0) \neq 0$, $\psi \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ with $\psi(\frac{\xi_0}{|\xi_0|}) \neq 0$, $\chi \in \mathcal{C}_{\mathrm{c}}^\infty([0,1))$ with $\chi(0) \neq 0$, and $h_1 \in (0, h_0]$ such that

$$\begin{gathered}\Big\{h^{-k}\langle h\xi\rangle^s\chi(\langle h\xi\rangle^{-1})\psi\Big(\frac{\xi}{|\xi|}\Big)\widehat{\phi u_h}(\xi)\colon h < h_1\Big\} \\ \text{is uniformly bounded in } L^2(\mathbb{R}^n_\xi).\end{gathered} \tag{6.140}$$

Exercise 6.23 (Semiclassical Wave Front Sets of h-Independent Distributions). Let M be a smooth manifold. Let $u \in \mathscr{D}'(M)$; regard u as an h-independent family of distributions. Show:

(1) $\mathrm{WF}_\hbar^{0,s}(u) \cap (T^*M \setminus o) \subset \mathrm{WF}^s(u) \subset \mathrm{WF}_\hbar^{0,s+\epsilon}(u) \cap (T^*M \setminus o)$ for all $s \geq 0$ and $\epsilon > 0$. (Here we regard $\mathrm{WF}^s(u) \subset T^*M \setminus o$.)
(2) $\mathrm{WF}_\hbar^{s,s}(u) \cap S^*M = \mathrm{WF}^s(u)$ for all $s \geq 0$. (Here we regard $\mathrm{WF}^s(u) \subset S^*M$.)
(3) Let $\epsilon > 0$. Then $o_{\operatorname{supp} u} \subset \mathrm{WF}_\hbar^{0,\epsilon}(u)$ where $o_{\operatorname{supp} u} \subset T^*M$ is the zero section over $\operatorname{supp} u$.
(4) For all $s \in \mathbb{R}$, we have $\mathrm{WF}_\hbar^{0,s}(u) \cap o \subset o_{\operatorname{supp} u}$.
(5) $\mathrm{WF}_\hbar(u) \cap T^*M = \mathrm{WF}(u) \cup o_{\operatorname{supp} u}$. (Here we regard $\mathrm{WF}(u) \subset T^*M \setminus o$.)

Exercise 6.24 (Oscillatory Distributions, I). Let $a, \phi \in \mathcal{C}^\infty(\mathbb{R}^n)$. Show that

$$\mathrm{WF}_\hbar(a(x)e^{i\phi(x)/h}) = \{(x, \mathrm{d}\phi(x))\colon x \in \operatorname{supp} a\}. \tag{6.141}$$

(*Hint.* Exercise 4.13.)

Exercise 6.25 (Oscillatory Distributions, II). Let $\beta > 0$ and $0 \neq \xi_0 \in \mathbb{R}^n$. Compute $\mathrm{WF}^s_\hbar(e^{ix\cdot\xi_0/h^\beta})$.

Exercise 6.26 (Rough Estimates for Eigenfunctions). Let (M, g) be a compact Riemannian manifold, $n = \dim M$. Improve the result of Exercise 5.30 by showing that there exists a constant C such that for all $\phi \in \mathcal{C}^\infty(M)$, $\|\phi\|_{L^2} = 1$, $\Delta_g \phi = \lambda\phi$, we have

$$\|\phi\|_{L^\infty} \leq C(1+\lambda)^{\frac{n}{4}}. \tag{6.142}$$

To do this, fix $\psi_0 \in \mathcal{C}^\infty_{\mathrm{c}}([0,\infty))$ with $\psi_0 = 1$ on $[0,2]$ and set $\psi(x,\xi) = \psi_0(|\xi|^2_{g^{-1}})$. Combine the following two estimates for $h = \lambda^{-\frac{1}{2}}$ for a sequence $u_h \in \mathcal{C}^\infty(M)$ of $L^2(M)$-normalized eigenfunctions (that is, $(h^2\Delta_g - 1)u_h = 0$ and $\|u_h\|_{L^2(M)} = 1$).

(1) Show that $\|(1 - \mathrm{Op}_h(\psi))u_h\|_{L^\infty} \leq C_N h^N$ for all N.
(2) Show that $\|\mathrm{Op}_h(\psi)u_h\|_{L^\infty} \leq Ch^{-\frac{n}{2}}$. (*Hint.* Do this first on $\mathbb{R}^n$.)

Remark. One can improve the exponent in (6.142) to $\frac{n-1}{4}$. (This is sharp for suitable sequences of eigenfunctions on the unit n-sphere.) See Zworski [Zwo12, Chapter 7.4].

The following exercises continue the discussion of the scattering calculus on $\mathbb{R}^n$ introduced in Exercises 4.14–4.22. We denote by

$$\overline{{}^{\mathrm{sc}}T^*}\,\overline{\mathbb{R}^n} := \overline{\mathbb{R}^n} \times \overline{\mathbb{R}^n} \tag{6.143}$$

the *radially compactified scattering cotangent bundle*. Write $S^{m,r}({}^{\mathrm{sc}}T^*\mathbb{R}^n) := S^{m,r}(\mathbb{R}^n;\mathbb{R}^n)$.

Exercise 6.27 (Essential Support and Elliptic Set: Scattering Symbols and Operators). Give a sensible definition (i.e., one related to Definitions 6.1, 6.14, 6.52) of the essential support $\operatorname{ess\,supp}_{\mathrm{sc}} a$ and elliptic set $\mathrm{Ell}^{m,r}_{\mathrm{sc}}(a) \subset \partial(\overline{{}^{\mathrm{sc}}T^*}\,\overline{\mathbb{R}^n})$ of a scattering symbol $a \in S^{m,r}({}^{\mathrm{sc}}T^*\mathbb{R}^n)$. The definition should be such that $\operatorname{ess\,supp}_{\mathrm{sc}} a = \emptyset$ if and only if a is a residual symbol; and $\mathrm{Ell}^{m,r}_{\mathrm{sc}}(a) = \emptyset$ if and only if $a \in S^{m-1,r-1}({}^{\mathrm{sc}}T^*\mathbb{R}^n)$.

Define now $\mathrm{WF}'_{\mathrm{sc}}(A)$ and $\mathrm{Ell}_{\mathrm{sc}}(A)$ (or more precisely $\mathrm{Ell}^{m,r}_{\mathrm{sc}}(A)$) for a scattering ps.d.o. $A \in \Psi^{m,r}_{\mathrm{sc}}(\mathbb{R}^n)$ in terms of the full and principal symbol of A, respectively.

Exercise 6.28 (Scattering Wave Front Set, I). Given $u \in \mathscr{S}'(\mathbb{R}^n)$, give a sensible definition (i.e., one related to Definitions 6.18 and 6.55) of its *scattering wave front set* $\mathrm{WF}^{s,r}_{\mathrm{sc}}(u) \subset \partial(\overline{{}^{\mathrm{sc}}T^*}\,\overline{\mathbb{R}^n})$. The definition should be such that $\mathrm{WF}^{s,r}_{\mathrm{sc}}(u) = \emptyset$ if and only if $u \in \langle x\rangle^{-r} H^s(\mathbb{R}^n)$. Give a sensible definition also of $\mathrm{WF}_{\mathrm{sc}}(u) \subset \partial(\overline{{}^{\mathrm{sc}}T^*}\,\overline{\mathbb{R}^n})$ such that $\mathrm{WF}_{\mathrm{sc}}(u) = \emptyset$ if and only if $u \in \mathscr{S}(\mathbb{R}^n)$. Prove the usual properties regarding the interaction of operator wave front sets, characteristic sets, and scattering wave front sets. *Remark.*

The a priori assumption that u be a *tempered* distribution is analogous to the h-tempered condition in Definition 6.55.

Thus, the scattering wave front set tests for regularity *and decay*.

Exercise 6.29 (Scattering Wave Front Set, II). Let $u \in \mathscr{S}'(\mathbb{R}^n)$.

(1) Show that $\mathrm{WF}_{\mathrm{sc}}^{s,r}(u) \cap (\mathbb{R}^n \times \partial\overline{\mathbb{R}^n}) = \mathrm{WF}^s(u)$.
(2) For a scattering symbol $a = a(x,\xi)$, prove that $\mathcal{F}(\mathrm{Op}_L(a)u) = \mathrm{Op}_R(\mathcal{R}^* a)\mathcal{F}u$ where $\mathcal{R}(y,\xi) = (-\xi, y)$. Deduce that $\mathrm{WF}_{\mathrm{sc}}^{s,r}(\hat{u}) = \mathcal{R}^{-1}(\mathrm{WF}_{\mathrm{sc}}^{r,s}(u))$.
(3) Let $\xi_0 \in \mathbb{R}^n$ and let $u(x) := \langle x\rangle^{-\frac{n-1}{2}} e^{ix\cdot\xi_0}$. Compute $\mathrm{WF}_{\mathrm{sc}}^{0,r}(u)$ for all values of $r \in \mathbb{R}$.

Chapter 7
Hyperbolic Evolution Equations and Egorov's Theorem



As a neat application of the ps.d.o. machinery developed thus far, we prove the existence and uniqueness of solutions to hyperbolic evolution equations (Section 7.1); examples include transport and wave equations. We also prove that singularities in the initial data, in the sense of wave front sets, propagate along the Hamiltonian flow for the principal symbol of the operator governing the evolution equation (Section 7.2). We shall state results only on $\mathbb{R}^n$ and leave the purely notational modifications needed for working on compact manifolds to the reader.

7.1 Existence and Uniqueness

Let $a(t,x,\xi) \in \mathcal{C}^\infty(\mathbb{R}_t; S^1(\mathbb{R}^n;\mathbb{R}^n)^{K\times K})$ be a $K \times K$ matrix of first order symbols with smooth dependence on t; we simply write $a \in S^1$. Let $A(t) := \mathrm{Op}_L(a(t))$. Consider the initial value problem

$$\begin{cases} (D_t + A(t))u = g(t,x) \in \mathcal{C}^0\big(\mathbb{R}; H^s(\mathbb{R}^n;\mathbb{C}^K)\big), \\ u(0,x) = u_0(x) \ \in H^s(\mathbb{R}^n;\mathbb{C}^K). \end{cases} \tag{7.1}$$

We assume that (7.1) is *symmetric hyperbolic*, meaning that

$$a(t,x,\xi) - a(t,x,\xi)^* \in S^0, \tag{7.2}$$

or equivalently $A(t) - A(t)^* \in \Psi^0$.

Theorem 7.1 (Existence and Uniqueness for First-Order Symmetric Hyperbolic Systems). *The initial value problem* (7.1) *has a unique solution*

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$$u \in \mathcal{C}^0\big(\mathbb{R}; H^s(\mathbb{R}^n;\mathbb{C}^K)\big) \cap \mathcal{C}^1\big(\mathbb{R}; H^{s-1}(\mathbb{R}^n;\mathbb{C}^K)\big). \tag{7.3}$$

Proof. We drop the "bundle" $\mathbb{C}^K$ from the notation. We shall obtain u as the limit of solutions u_ϵ to the regularized equations

$$\begin{cases} D_t u_\epsilon = -J_\epsilon A J_\epsilon u_\epsilon + g, \\ u_\epsilon(0) = u_0, \end{cases} \tag{7.4}$$

where we use a Friedrichs mollifier $\phi(\epsilon D) = J_\epsilon = J_\epsilon^*$ as in (4.124), so $J_\epsilon \in \Psi^{-\infty}(\mathbb{R}^n)$ for $\epsilon > 0$, $J_\epsilon \in \Psi^0(\mathbb{R}^n)$ is uniformly bounded for $\epsilon \in (0,1]$, and $J_\epsilon \to I$ in $\Psi^{m'}(\mathbb{R}^n)$ for all $m' > 0$. For $\epsilon > 0$, $J_\epsilon A J_\epsilon$ is a continuous (in t) family of bounded operators on the Banach space $H^s(\mathbb{R}^n)$, and hence, the solvability of (7.4) with $u_\epsilon \in \mathcal{C}^1(\mathbb{R}; H^s(\mathbb{R}^n))$ follows from standard ODE theory.

We need to establish estimates on u_ϵ which are uniform as $\epsilon \to 0$. Let $\Lambda^s = \langle D_x\rangle^s$. Then

$$\begin{aligned} \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u_\epsilon(t)\|_{H^s}^2 &= -\mathrm{Re}\langle \Lambda^s J_\epsilon i A J_\epsilon u_\epsilon, \Lambda^s u_\epsilon\rangle + \mathrm{Re}\langle \Lambda^s i g, \Lambda^s u_\epsilon\rangle \\ &= -\mathrm{Re}\langle iA\Lambda^s J_\epsilon u_\epsilon, \Lambda^s J_\epsilon u_\epsilon\rangle - \mathrm{Re}\langle [\Lambda^s, iA] J_\epsilon u_\epsilon, \Lambda^s J_\epsilon u_\epsilon\rangle \\ &\quad + \mathrm{Re}\langle \Lambda^s i g, \Lambda^s u_\epsilon\rangle. \end{aligned} \tag{7.5}$$

Since $B(t) = A(t) - A(t)^* \in \Psi^0$, the first term is bounded by

$$|\langle B(t)\Lambda^s J_\epsilon u_\epsilon, \Lambda^s J_\epsilon u_\epsilon\rangle| \le C(t)\|J_\epsilon u_\epsilon\|_{H^s}^2 \le \tilde{C}(t)\|u_\epsilon\|_{H^s}^2. \tag{7.6}$$

Since $[\Lambda^s, A] \in \Psi^s$, the second term in (7.5) is bounded by $C(t)\|u_\epsilon\|_{H^s}^2$ as well. Applying Cauchy–Schwarz to the third term in (7.5), we obtain (with yet another constant $C(t)$)

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u_\epsilon(t)\|_{H^s}^2 \le C(t)\|u_\epsilon(t)\|_{H^s}^2 + \|g(t)\|_{H^s}^2. \tag{7.7}$$

This implies the ϵ-*independent* estimate

$$\|u_\epsilon(t)\|_{H^s}^2 \le \tilde{C}(t)\big(\|u_0\|_{H^s}^2 + \|g\|^2_{\mathcal{C}^0([0,t];H^s(\mathbb{R}^n))}\big). \tag{7.8}$$

Fix $T > 0$ and set $I := [-T, T]$. Then (7.8) implies the uniform boundedness of u_ϵ in $\mathcal{C}^0(I; H^s(\mathbb{R}^n))$. Equation (7.4) and the uniform boundedness of $J_\epsilon A J_\epsilon \in \mathcal{L}(H^s(\mathbb{R}^n), H^{s-1}(\mathbb{R}^n))$ imply that

$$u_\epsilon \in \mathcal{C}^0(I; H^s(\mathbb{R}^n)) \cap \mathcal{C}^1(I; H^{s-1}(\mathbb{R}^n)) \text{ is uniformly bounded.} \tag{7.9}$$

We extract a weak subsequential limit $u \in \mathcal{C}^0(I; H^{s-1}(\mathbb{R}^n))$ of u_ϵ using Lemma 7.2 below. For this limit, we have $u_\epsilon(0) = u_0 \rightharpoonup u(0)$ in $H^{s-1}(\mathbb{R}^n)$, so

$u(0) = u_0$. Moreover, u satisfies the PDE (7.1) in the following weak sense: for all $\phi \in \mathcal{C}_c^\infty(I^\circ; H^{-s+2}(\mathbb{R}^n))$,

$$\begin{aligned}\langle D_t u, \phi\rangle &= \int_I \langle D_t u(t), \phi(t)\rangle_{L^2(\mathbb{R}^n)}\,\mathrm{d}t \\ &= \langle u, -D_t\phi\rangle = \lim_{\epsilon\to 0}\langle u_\epsilon, -D_t\phi\rangle \\ &= \lim_{\epsilon\to 0}\langle D_t u_\epsilon, \phi\rangle = \lim_{\epsilon\to 0}\langle J_\epsilon A J_\epsilon u_\epsilon, \phi\rangle = \lim_{\epsilon\to 0}\langle u_\epsilon, J_\epsilon A^* J_\epsilon\phi\rangle \\ &= \langle u, A^*\phi\rangle = \langle Au, \phi\rangle,\end{aligned} \tag{7.10}$$

where in the passage to the last line we used that $J_\epsilon A^* J_\epsilon\phi \to A^*\phi$ in $\mathcal{C}_c^\infty(I^\circ; H^{-s+1}(\mathbb{R}^n))$ (Exercise 4.8). Thus, u is a weak solution of (7.1) and therefore $u \in \mathcal{C}^1(I; H^{s-2}(\mathbb{R}^n))$. The uniqueness of u in the space $\mathcal{C}^0(\mathbb{R}; H^\sigma(\mathbb{R}^n))$ (for any $\sigma \in \mathbb{R}$) follows by noting that for $u_0, g = 0$, the computations leading to (7.8) yield $\frac{\mathrm{d}}{\mathrm{d}t}\|u(t)\|_{H^{\sigma-1}}^2 \le C(t)\|u(t)\|_{H^{\sigma-1}}^2$; the initial condition $\|u(0)\|_{H^{\sigma-1}} = \|u_0\|_{H^{\sigma-1}} = 0$ then implies $u(t) = 0$ for all t.

To prove the correct regularity of u, we approximate $u_0 \in H^s(\mathbb{R}^n)$, $g \in \mathcal{C}^0(\mathbb{R}; H^s(\mathbb{R}^n))$ in these topologies by $u_{0,j} \in H^{s+1}(\mathbb{R}^n)$, $g_j \in \mathcal{C}^0(\mathbb{R}; H^{s+1}(\mathbb{R}^n))$, $j \in \mathbb{N}$. Then the earlier arguments produce solutions $u_j \in \mathcal{C}^0(I; H^s(\mathbb{R}^n)) \cap \mathcal{C}^1(I; H^{s-1}(\mathbb{R}^n))$ of (7.1) with $u_{0,j}, g_j$ in place of u_0, g. Moreover,

$$v_{jk} := u_j - u_k \tag{7.11}$$

solves (7.1) with initial data $u_{0,j} - u_{0,k}$ ($\to 0$ in $H^s(\mathbb{R}^n)$) and forcing $g_j - g_k$ ($\to 0$ in $\mathcal{C}^0(\mathbb{R}; H^s(\mathbb{R}^n))$). An estimate similar to (7.8) implies that $v_{jk} \to 0$ in $\mathcal{C}^0(I; H^s(\mathbb{R}^n))$ as $j, k \to \infty$. Therefore, u_j is a Cauchy sequence in $\mathcal{C}^0(I; H^s(\mathbb{R}^n))$, and hence its limit u satisfies (7.3). □

To complete the proof, we need:

Lemma 7.2 (Weak Subsequential Limits). *Let $(H, \langle\cdot,\cdot\rangle_H)$ denote a Hilbert space with norm $\|\cdot\|_H$. Let $a, b \in \mathbb{R}$, $a < b$, and write $\mathcal{C}^{0,1}([a,b]; H)$ for the space of Lipschitz continuous functions $u\colon [a,b] \to H$, equipped with the norm*

$$\|u\|_{\mathcal{C}^{0,1}} := \sup_{t\in[a,b]}\|u(t)\|_H + \sup_{s\neq t}\frac{\|u(t)-u(s)\|_H}{|t-s|}. \tag{7.12}$$

Let $(u_j)_{j\in\mathbb{N}}$ be a bounded sequence in $\mathcal{C}^{0,1}([a,b]; H)$. Then there exists a subsequence $(u_{j_k})_{k\in\mathbb{N}}$ converging to some $u \in \mathcal{C}^{0,1}([a,b]; H)$ in the following sense:

(1) *For all $t \in [a,b]$, we have $u_{j_k}(t) \rightharpoonup u(t)$.*
(2) *For all $\phi \in \mathcal{C}^{0,1}([a,b]; H)$, we have $\langle u_{j_k}, \phi\rangle := \int_a^b \langle u_{j_k}(t), \phi(t)\rangle_H\,\mathrm{d}t \to \langle u, \phi\rangle$.*

Proof. We rescale and shift t so that $[a,b] = [0,1]$. Fix a countable dense subset $\{t_i\} \subset [0,1]$. Since $u_j(t_1)$ is a bounded sequence in H, it has a weakly convergent subsequence $u_{j_{1,k}}(t_1)$ with weak limit denoted $\alpha_1 \in H$. The sequence $u_{j_{1,k}}(t_2)$ has a weakly convergent subsequence $u_{j_{2,k}}(t_2)$ with weak limit denoted $\alpha_2 \in H$; and so on. For the diagonal subsequence $u_{j_k} := u_{j_{k,k}}$, we then have $u_{j_k}(t_i) \rightharpoonup \alpha_i$ for all i.

Let now $C := \sup_{j\in\mathbb{N}} \|u_j\|_{\mathcal{C}^{0,1}}$. In view of

$$\|\alpha_i - \alpha_\ell\|_H \le \liminf_{k\to\infty} \|u_{j_k}(t_i) - u_{j_k}(t_\ell)\|_H \le C|t_i - t_\ell|, \tag{7.13}$$

the function $u\colon t_i \mapsto \alpha_i$ has a continuous extension with Lipschitz constant $\le C$. If $t \in [0,1]$ is arbitrary, $\phi \in H$, and $\epsilon > 0$, then

$$|\langle u(t),\phi\rangle_H - \langle u_{j_k}(t),\phi\rangle_H| \le |\langle u(t_i) - u_{j_k}(t_i),\phi\rangle_H| + 2C|t-t_i|\|\phi\|_H < \epsilon \tag{7.14}$$

provided we choose t_i such that $2C|t-t_i|\|\phi\|_H < \frac{\epsilon}{2}$ and then k large enough so that also $|\langle u(t_i) - u_{j_k}(t_i),\phi\rangle_H| < \frac{\epsilon}{2}$.

Finally, given $\phi \in \mathcal{C}^{0,1}([0,1];H)$, let L denote the Lipschitz constant of $[a,b] \ni t \mapsto \langle u(t),\phi(t)\rangle_H$. Given $\epsilon > 0$, we have

$$\left|\langle u,\phi\rangle - \frac{1}{N}\sum_{m=0}^{N-1} \langle u(m/N),\phi(m/N)\rangle\right| < \epsilon \tag{7.15}$$

provided $\frac{L}{N} < \epsilon$, i.e., $N > L\epsilon^{-1}$; similarly for u_{j_k} in place of u. Fixing such an N, we have $|\langle u(m/N),\phi(m/N)\rangle_H - \langle u_{j_k}(m/N),\phi(m/N)\rangle_H| < \epsilon$ for all $m = 0,\dots,N-1$ when k is large enough. Therefore, $|\langle u,\phi\rangle - \langle u_{j_k},\phi\rangle| < 3\epsilon$. This proves $\langle u_{j_k},\phi\rangle \to \langle u,\phi\rangle$. □

We give three examples.

Example 7.3 (Transport). Let $v \in \mathcal{C}_b^\infty(\mathbb{R}^n;\mathbb{R}^n)$. For $K = 1$ and $a(t,x,\xi) = v(x)\cdot\xi$, equation (7.1) is the transport equation $(\partial_t + v(x)\cdot\partial_x)u(t,x) = ig(t,x)$. It can be solved using the *method of characteristics*: if we define $\gamma_x(t)$ by $\gamma_x(0) = x$, $\gamma_x'(t) = v(\gamma_x(t))$, then $u(t,\gamma_x(t)) = u_0(x) + \int_0^t ig(s,\gamma_x(s))\,\mathrm{d}s$. Time-dependent v can be handled similarly.

Example 7.4 (Klein–Gordon Equation on $\mathbb{R}^n$). Consider the initial value problem

$$\begin{cases} (-D_t^2 + \Delta + 1)u(t,x) = h(t,x) \in \mathcal{C}^0(\mathbb{R};H^{s-1}(\mathbb{R}^n)), \\ u(0,x) = u_0(x) \ \in H^s(\mathbb{R}^n), \\ D_t u(0,x) = u_1(x) \ \in H^{s-1}(\mathbb{R}^n) \end{cases} \tag{7.16}$$

for the Klein–Gordon equation. Let $\Lambda := \langle D \rangle$. The problem (7.16) is then equivalent to

$$\begin{cases} (D_t + A(t))U = G, \\ U(0) = U_0, \end{cases} \tag{7.17}$$

where

$$U(t) := \begin{pmatrix} \Lambda u \\ D_t u \end{pmatrix}, \quad A := \begin{pmatrix} 0 & -\Lambda \\ -\Lambda & 0 \end{pmatrix}, \quad U_0 := \begin{pmatrix} \Lambda u_0 \\ u_1 \end{pmatrix}, \quad G := \begin{pmatrix} 0 \\ -h \end{pmatrix}. \tag{7.18}$$

Note that $U_0 \in H^{s-1}(\mathbb{R}^n; \mathbb{C}^2)$, $G \in \mathcal{C}^0(\mathbb{R}; H^{s-1}(\mathbb{R}^n; \mathbb{C}^2))$. Theorem 7.1 produces a solution $U = (U_1, U_2) \in \mathcal{C}^0 H^{s-1} \cap \mathcal{C}^1 H^{s-2}$ of (7.17). Set $u := \Lambda^{-1} U_1 \in \mathcal{C}^0 H^s \cap \mathcal{C}^1 H^{s-1}$. The first component of the equation for U implies that $U_2 = D_t u$. Therefore, $u \in \mathcal{C}^0 H^s \cap \mathcal{C}^1 H^{s-1} \cap \mathcal{C}^2 H^{s-2}$ solves (7.16).

Example 7.5 (Wave Equation on a Time-Dependent Riemannian Manifold). Let X be a closed manifold and let $h \in \mathcal{C}^\infty(\mathbb{R}_t; \mathcal{C}^\infty(X; S^2 T^* X))$ be a smooth family of Riemannian metrics on X. The initial value problem

$$\begin{cases} (-D_t^2 + \Delta_{g(t)})u(t,x) = h(t,x) & \in \mathcal{C}^0(\mathbb{R}; H^{s-1}(X)), \\ u(0,x) = u_0(x) & \in H^s(X), \\ D_t u(0,x) = u_1(x) & \in H^{s-1}(X) \end{cases} \tag{7.19}$$

for the wave equation can also be brought into the form (7.1) (with X in place of $\mathbb{R}^n$). First, pick $\Lambda_{\frac{1}{2}}(t) \in \Psi^{\frac{1}{2}}_{\mathrm{cl}}(X)$ with principal symbol $|\xi|^{\frac{1}{2}}_{g(t)_x^{-1}}$ and smooth dependence on $t \in \mathbb{R}$. Set then

$$\Lambda(t) := I + \Lambda_{\frac{1}{2}}(t)^* \Lambda_{\frac{1}{2}}(t) \in \mathcal{C}^\infty(\mathbb{R}_t; \Psi^1_{\mathrm{cl}}(X)). \tag{7.20}$$

This operator has principal symbol $|\xi|_{g(t)_x^{-1}}$. It is moreover invertible with inverse $\Lambda(t)^{-1} \in \mathcal{C}^\infty(\mathbb{R}_t; \Psi^{-1}_{\mathrm{cl}}(X))$. (This is an instance of the construction (5.152).) The problem (7.19) is then equivalent to

$$\begin{cases} (D_t + A(t))U = G, \\ U(0) = U_0, \end{cases} \tag{7.21}$$

where

$$U(t) := \begin{pmatrix} \Lambda(t) u \\ D_t u \end{pmatrix}, \qquad A(t) := \begin{pmatrix} -(D_t \Lambda(t)) \circ \Lambda(t)^{-1} & -\Lambda(t) \\ -\Delta_{g(t)} \circ \Lambda(t)^{-1} & 0 \end{pmatrix},$$

$$U_0 := \begin{pmatrix} \Lambda(0) u_0 \\ u_1 \end{pmatrix} \in H^{s-1}(X; \mathbb{C}^2), \qquad G := \begin{pmatrix} 0 \\ -h \end{pmatrix} \in \mathcal{C}^0\big(\mathbb{R}; H^{s-1}(X; \mathbb{C}^2)\big). \tag{7.22}$$

Theorem 7.1 produces a solution $U = (U_1, U_2) \in \mathcal{C}^0 H^{s-1} \cap \mathcal{C}^1 H^{s-2}$ of (7.21). The function $u := \Lambda(t)^{-1} U_1 \in \mathcal{C}^0 H^s \cap \mathcal{C}^1 H^{s-1} \cap \mathcal{C}^2 H^{s-2}$ then solves (7.19).— Initial value problems for the wave equation $(-D_t^2 + \Delta)u(t,x) = h$ on $\mathbb{R}_t \times \mathbb{R}_x^n$ can be treated in the same manner, now using $\Lambda(t) = \langle D \rangle$. We discuss the solvability of general wave equations and the regularity of their solutions in Chapter 9.

7.2 Egorov's Theorem; Propagation of Singularities

We begin the study of the microlocal behavior of solutions of hyperbolic evolution equations in the scalar setting (i.e., $K = 1$ in (7.1)). Let us thus consider

$$(D_t + A(t))u(t,x) = 0, \tag{7.23}$$

where we assume that $A \in \mathcal{C}^\infty(\mathbb{R}_t; \Psi^1_{\mathrm{cl}}(\mathbb{R}^n))$ is classical. Denote by $a \in \mathcal{C}^\infty(\mathbb{R}_t; S^1_{\mathrm{hom}}(\mathbb{R}^n; \mathbb{R}^n \setminus \{0\}))$ the homogeneous principal symbol of A. (Beware of this minor change in the meaning of a compared to the previous section.) The requirement of symmetric hyperbolicity means that

$$a(t,x,\xi) \text{ is real-valued.} \tag{7.24}$$

We denote the solution operator for (7.23) by

$$S(t,s)\colon u(s) \mapsto u(t). \tag{7.25}$$

By Theorem 7.1, $S(t,s) \in \mathcal{L}(H^\sigma(\mathbb{R}^n))$ for all $\sigma \in \mathbb{R}$. Moreover, $S(t,s)$ is invertible with $S(t,s)^{-1} = S(s,t)$.

Theorem 7.6 (Egorov's Theorem). *Let $P_0 = \mathrm{Op}(p_0) \in \Psi^m(\mathbb{R}^n)$ and define*

$$P(t) := S(t,0) \circ P_0 \circ S(0,t), \quad t \in \mathbb{R}. \tag{7.26}$$

Then $P(t)$ is an m-th order ps.d.o. plus a smoothing operator: there exists an operator $R \in \bigcap_{N \in \mathbb{R}} \mathcal{C}^\infty(\mathbb{R}; \mathcal{L}(H^{-N}(\mathbb{R}^n), H^N(\mathbb{R}^n)))$ such that $\tilde{P}(t) := P(t) - R(t) \in \Psi^m(\mathbb{R}^n)$, with smooth dependence on $t \in \mathbb{R}$. The principal symbol of $\tilde{P}(t)$ is given by

$$\sigma^m(\tilde{P}(t))\big(\mathcal{C}(t,0)(x_0,\xi_0)\big) = \sigma^m(P_0)(x_0,\xi_0), \tag{7.27}$$

where $\mathcal{C}(t_1,t_0)$ is the flow of the time-dependent Hamiltonian vector field $H_{a(t,x,\xi)}$ from t_0 to t_1; that is, $\mathcal{C}(t_1,t_0)(x_0,\xi_0) = \gamma(t_1)$ where $\gamma(t_0) = (x_0,\xi_0)$, $\gamma'(s) = H_{a(s)}|_{\gamma(s)}$.

On a closed manifold X, smoothing operators are residual ps.d.o.s. The conclusion of the analogue of Theorem 7.6 for scalar hyperbolic equations on X is thus that $P(t) \in \Psi^m(X)$.

We first prove a technical result related to (7.27):

Lemma 7.7 (Transport of Symbols). *Let $a \in \mathcal{C}^\infty(\mathbb{R}_t; S^1(\mathbb{R}^n_x; \mathbb{R}^n_\xi))$. Let $m \in \mathbb{R}$, $q_0 \in S^m(\mathbb{R}^n; \mathbb{R}^n)$, and $e \in \mathcal{C}^\infty(\mathbb{R}; S^m(\mathbb{R}^n; \mathbb{R}^n))$. Let $q = q(t, x, \xi)$ be the solution of*

$$\begin{cases} (\partial_t + H_{a(t)})q(t, x, \xi) = e(t, x, \xi), \\ q(0, x, \xi) = q_0(x, \xi). \end{cases} \tag{7.28}$$

Then $q \in \mathcal{C}^\infty(\mathbb{R}; S^m(\mathbb{R}^n; \mathbb{R}^n))$.

Proof. Define $\gamma_{t,x,\xi}(s) = (X_{t,x,\xi}(s), \Xi_{t,x,\xi}(s))$ by

$$\gamma_{t,x,\xi}(0) = (x, \xi), \quad \gamma'_{t,x,\xi}(s) = H_{a(t+s)}|_{\gamma_{t,x,\xi}(s)}. \tag{7.29}$$

Consider fixed values of t, x, ξ and write $X = X_{t,x,\xi}$, $\Xi = \Xi_{t,x,\xi}$ for brevity. Then equation (7.29) reads

$$\begin{aligned} X(0) &= x, \quad X'(s) = (\partial_\xi a)(t + s, X(s), \Xi(s)), \\ \Xi(0) &= \xi, \quad \Xi'(s) = -(\partial_x a)(t + s, X(s), \Xi(s)). \end{aligned} \tag{7.30}$$

When t, s remain in a bounded subset of $\mathbb{R}$, there exists a constant C such that

$$|X'(s)| \le C, \quad \frac{|\Xi'(s)|}{\langle \Xi(s) \rangle} \le C. \tag{7.31}$$

(Here $\langle \Xi(s) \rangle = (1 + |\Xi(s)|^2)^{\frac{1}{2}}$ as usual.) This implies the bounds

$$|X(s) - X(0)|, \ \frac{\langle \Xi(s) \rangle}{\langle \Xi(0) \rangle}, \ \frac{\langle \Xi(0) \rangle}{\langle \Xi(s) \rangle} \le C \tag{7.32}$$

for a new constant C. In particular, $\gamma_{t,x,\xi}(s)$ is defined for all $t, s \in \mathbb{R}$ and $x, \xi \in \mathbb{R}^n$.

Returning to the ODE (7.28), we can write its unique solution as

$$\begin{aligned} q(t, x, \xi) = {} & q_0(X_{t,x,\xi}(-t), \Xi_{t,x,\xi}(-t)) \\ & + \int_0^t e(s, X_{t,x,\xi}(-t + s), \Xi_{t,x,\xi}(-t + s))\, \mathrm{d}s. \end{aligned} \tag{7.33}$$

This defines a smooth function of $(t,x,\xi) \in \mathbb{R}\times\mathbb{R}^n\times\mathbb{R}^n$. For any fixed $T>0$, we shall show that $q \in \mathcal{C}^\infty((-T,T);S^m(\mathbb{R}^n;\mathbb{R}^n))$. The bounds (7.32) together with the symbol bounds on q_0, e imply

$$|q(t,x,\xi)| \le C\langle\xi\rangle^m, \quad t\in(-T,T). \tag{7.34}$$

In order to control derivatives of q, we first need to control the derivatives of $\gamma_{t,x,\xi}(s)$ in x,ξ. Differentiating (7.30) in $z = x_i, \xi_i$ gives the system

$$\begin{aligned}(\partial_z X)' &= \sum_{j=1}^n \big((\partial_\xi\partial_{x_j}a)\partial_z X_j + (\partial_\xi\partial_{\xi_j}a)\partial_z\Xi_j\big),\\ (\partial_z\Xi)' &= -\sum_{j=1}^n \big((\partial_x\partial_{x_j}a)\partial_z X_j + (\partial_x\partial_{\xi_j}a)\partial_z\Xi_j\big).\end{aligned} \tag{7.35}$$

The symbol bounds on a and the bounds (7.32) imply

$$|(\partial_z X)'|,\ \frac{|(\partial_z\Xi)'|}{\langle\Xi\rangle} \le C\Big(|\partial_z X| + \frac{|\partial_z\Xi|}{\langle\Xi\rangle}\Big) \tag{7.36}$$

when t,s remain in a bounded subset of $\mathbb{R}$. Upon integration, this implies

$$|\partial_z X|,\ \frac{|\partial_z\Xi|}{\langle\Xi\rangle} \le C\Big(|\partial_z X(0)| + \frac{|\partial_z\Xi(0)|}{\langle\Xi(0)\rangle}\Big). \tag{7.37}$$

Therefore,

$$|\partial_{x_i}X| \le C,\ |\partial_{x_i}\Xi| \le C\langle\Xi(0)\rangle, \quad |\partial_{\xi_i}X| \le C\langle\Xi(0)\rangle^{-1},\ |\partial_{\xi_i}\Xi| \le C. \tag{7.38}$$

Differentiating (7.33) in x_i,ξ_i and using (7.32) and (7.38), we obtain

$$|\partial_{x_i}q(t,x,\xi)| \le C\langle\xi\rangle^m, \quad |\partial_{\xi_i}q(t,x,\xi)| \le C\langle\xi\rangle^{m-1}, \qquad t\in(-T,T). \tag{7.39}$$

Since $\partial_t q = e - H_{a(t)}q$ by (7.28), this then also implies $|\partial_t q(t,x,\xi)| \le C\langle\xi\rangle^m$. Higher derivatives are treated in the same fashion; we leave the detailed verification to the reader. □

Proof of Theorem 7.6. Differentiating (7.26) in t gives the equation

$$P'(t) + i[A(t),P(t)] = 0, \quad P(0) = P_0. \tag{7.40}$$

We will first construct an approximate solution $Q(t) = \mathrm{Op}(q(t))$, $q(t,x,\xi)\in S^m$, of this ps.d.o.-valued ODE, i.e.,

$$Q'(t) + i[A(t),Q(t)] = R_1(t), \quad Q(0) = P_0, \qquad R_1 \in \mathcal{C}^\infty(\mathbb{R}_t;\Psi^{-\infty}(\mathbb{R}^n)); \tag{7.41}$$

we will then show that Q and P differ by a smoothing operator. We make the ansatz

$$q(t) \sim \sum_{k=0}^{\infty} q_k(t), \quad q_k(t) \in S^{m-k}. \tag{7.42}$$

Taking the principal symbol of (7.41) gives

$$(\partial_t + H_{a(t)})q_0(t,x,\xi) = 0, \quad q_0(0,x,\xi) = p_0(x,\xi). \tag{7.43}$$

Thus, $q_0(t, \mathcal{C}(t)(x,\xi)) = p_0(x,\xi)$. Lemma 7.7 gives $q_0 \in S^m$. Proceeding iteratively, we take $q_j \in S^{m-j}$, $j \geq 1$, to be the solution of a transport equation (with initial data determined by P_0)

$$(\partial_t + H_{a(t)})q_j(t,x,\xi) = e_j(t,x,\xi), \tag{7.44}$$

where $e_j \in S^{m-j}$ is an expression involving only the full symbol of A and $q_0, \ldots, q_{j-1}$.

Having thus arranged (7.41), we now prove that for all $N \in \mathbb{R}$, the difference $R(t) = P(t) - Q(t)$ maps any $\tilde{f} \in H^{-N}(\mathbb{R}^n)$ into $H^N(\mathbb{R}^n)$. Taking $\tilde{f} = S(t,0)f$, this is equivalent to

$$v(t) := S(t,0)P_0 f, \ w(t) := Q(t)S(t,0)f \implies v(t) - w(t) \in H^N(\mathbb{R}^n). \tag{7.45}$$

Note that $v(t)$ and $w(t)$ solve the equations

$$\begin{aligned} (D_t + A(t))v &= 0, & v(0) &= P_0 f, \\ (D_t + A(t))w &= -iR_1 S(t,0)f, & w(0) &= P_0 f. \end{aligned} \tag{7.46}$$

Therefore, putting $g(t) := iR_1 S(t,0)f \in \bigcap_{N\in\mathbb{R}} \mathcal{C}^\infty(\mathbb{R}_t; H^N(\mathbb{R}^n))$, we have

$$(D_t + A(t))(v - w) = g, \quad (v-w)(0) = 0. \tag{7.47}$$

Theorem 7.1 gives $v - w \in \mathcal{C}^0(\mathbb{R}; H^N(\mathbb{R}^n))$ for all N. Since $D_t(v-w) = g - A(t)(v-w)$, this can now be bootstrapped to $v - w \in \mathcal{C}^k(\mathbb{R}; H^N(\mathbb{R}^n))$ for all $N \in \mathbb{R}$, $k \in \mathbb{N}_0$. □

We next show how to use Theorem 7.6 to track the wave front set of solutions of scalar evolution equations.

Theorem 7.8 (Propagation of Wave Front Sets: Scalar Case). *Let A be as in (7.23)–(7.24). Let $s \in \mathbb{R}$. Let $u_0 \in H^{-N}(\mathbb{R}^n)$, $g \in \mathcal{C}^0(\mathbb{R}_t; H^s(\mathbb{R}^n))$, and let u denote the solution of*

$$\begin{cases} (D_t + A(t))u = g, \\ u(0) = u_0. \end{cases} \tag{7.48}$$

Let $\mathcal{C}(t_1, t_0)$ be as in the statement of Theorem 7.6. Then

$$\mathrm{WF}^s(u(t)) = \mathcal{C}(t, 0)\mathrm{WF}^s(u_0), \tag{7.49}$$

and more generally $\mathrm{WF}^s(u(t_1)) = \mathcal{C}(t_1, t_0)\mathrm{WF}^s(u(t_0))$ for all $t_0, t_1 \in \mathbb{R}$.

Proof. By subtracting from u the solution $v \in \mathcal{C}^0(\mathbb{R}; H^s(\mathbb{R}^n))$ of $(D_t + A(t))v = g$, $v(0) = 0$, we may reduce to the case $g = 0$.

It now suffices to prove the inclusion "$\subseteq$" in (7.49) since reversing the direction of time then also proves "$\supseteq$." Suppose thus that $\alpha \notin \mathrm{WF}^s(u_0)$. Pick an operator $P_0 \in \Psi^0(\mathbb{R}^n)$, elliptic at α, such that $P_0 u_0 \in H^s(\mathbb{R}^n)$. Then, in the notation of Theorem 7.6,

$$P_0 u_0 = S(0, t)P(t)S(t, 0)u_0 \in H^s(\mathbb{R}^n), \tag{7.50}$$

so $P(t)u(t) \in H^s(\mathbb{R}^n)$ and thus $\tilde{P}(t)u(t) \in H^s(\mathbb{R}^n)$. But $\tilde{P}(t)$ is elliptic at $\mathcal{C}(t, 0)\alpha$, so $\mathcal{C}(t, 0)\alpha \notin \mathrm{WF}^s(u(t))$. □

Remark 7.9 (Homogeneous Principal Symbol). The above arguments apply without change to general operators $A \in \Psi^1$ with real-valued homogeneous principal symbols $a \in S^1_{\mathrm{hom}}$. (That is, if $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n_\xi)$ equals 1 near 0, then $(1 - \chi(\xi))a(t, x, \xi)$ is a representative of $\sigma^1(A(t))$.)

Remark 7.10 (Generalization to Weighted Sobolev Spaces). Theorem 7.1, and thus also Theorem 7.8, can be generalized (with purely notational modifications) to the case of initial data and forcing terms in *weighted* Sobolev spaces. By Theorem 2.18, one can thus allow u_0 in Theorem 7.8 to be any tempered distribution $u_0 \in \mathscr{S}'(\mathbb{R}^n)$.

Example 7.11 (Simple Transport). For $A = D_x \in \Psi^1(\mathbb{R})$, the solution operator is $(S(t, 0)u)(x) = u(x-t)$, and $\mathcal{C}(t, 0)(x_0, \xi_0) = (x_0+t, \xi_0)$. Moreover, $P(t) = \mathrm{Op}(p(t))$, $p(t, x, \xi) = p_0(x - t, \xi)$.

Example 7.12 (Half-Klein–Gordon Equation). Consider the half-Klein–Gordon equation

$$(D_t + \langle D_x \rangle)u = 0 \tag{7.51}$$

on $\mathbb{R}_t \times \mathbb{R}^n_x$. (This arises from the factorization $-D_t^2 + (\Delta + 1) = -(D_t - \langle D_x \rangle)(D_t + \langle D_x \rangle)$ of the Klein–Gordon operator.) In this case, $a(x, \xi) = |\xi|$, whose Hamiltonian vector field is $H_a = \frac{\xi}{|\xi|} \cdot \partial_x$. Thus, the wave front set of u gets transported along straight lines with direction determined by the momentum variable ξ. (Equation (7.51) has the explicit solution $u(t) = e^{-it\langle D \rangle}u(0)$, and the wave front set statement can be checked explicitly from this. The true power of Theorems 7.6 and 7.8 of course lies in the fact that they apply to equations with non-constant coefficients as well.)

Since examples as simple as the wave or Klein–Gordon equations on $\mathbb{R}\times\mathbb{R}^n$ are not covered by Theorem 7.8, some readers may desire to go further. We proceed to prove a general result which, indeed, allows one to control wave front sets of solutions to a large class of symmetric hyperbolic *systems*. This proceeds via a reduction to the scalar case using a *microlocal decoupling* (or diagonalization) argument. (We shall revisit the propagation of wave front sets of solutions to wave equations *jointly* in t and x and the corresponding canonical momentum variables in Chapter 8: that will be a simpler, *scalar*, theory compared to the *non-scalar* arguments which follow now.)

For the remainder of this section, we let

$$A \in \mathcal{C}^\infty\big(\mathbb{R}_t; \Psi^1_{\mathrm{cl}}(\mathbb{R}^n)^{K\times K}\big), \quad a := \sigma^1(A) \in \mathcal{C}^\infty\big(\mathbb{R}; S^1_{\hom}(\mathbb{R}^n; \mathbb{R}^n\setminus\{0\})^{K\times K}\big). \tag{7.52}$$

Definition 7.13 (Strict Hyperbolicity). We say that the system $(D_t + A(t))u = g$ (or the operator A) is *strictly hyperbolic* if $a(t,x,\xi)$ has K distinct real eigenvalues for all $t\in\mathbb{R}$, $x\in\mathbb{R}^n$, $\xi\in\mathbb{R}^n\setminus\{0\}$.

Our goal is to decouple a strictly hyperbolic system into K scalar equations up to smoothing errors. We motivate the construction with the following formal computation: if $S(t)$ is an invertible operator of order 0, then $(D_t + A(t))u = 0$ is equivalent to an equation for $S(t)u$ which we find by writing

$$\begin{aligned} S(D_t + A)u &= S(D_t + A)S^{-1}Su \\ &= \big(D_t + SAS^{-1} + S(D_tS^{-1})\big)Su. \end{aligned} \tag{7.53}$$

For the operator $SAS^{-1} + S(D_tS^{-1})$ to be diagonal, we first of all need $SAS^{-1}\in\Psi^1_{\mathrm{cl}}$ to be diagonal; this is a linear algebra problem for the principal symbol of S. The term $S(D_tS^{-1})\in\Psi^0_{\mathrm{cl}}$ is of lower order. The full decoupling requires the construction of suitable lower order corrections for S.

Proposition 7.14 (Microlocal Decoupling of Strictly Hyperbolic Systems). *Suppose that A is strictly hyperbolic. Denote the eigenvalues of $a(t,x,\xi)$ by*

$$\lambda_1(t,x,\xi) < \cdots < \lambda_K(t,x,\xi), \quad t\in\mathbb{R},\ x\in\mathbb{R}^n,\ \xi\in\mathbb{R}^n\setminus\{0\}. \tag{7.54}$$

Let $t_0\in\mathbb{R}$, $x_0\in\mathbb{R}^n$, $\xi_0\in\mathbb{R}^n\setminus\{0\}$.

(1) (Existence.) *There exist a conic (in ξ) neighborhood $\mathcal{N}\subset\mathbb{R}_t\times\mathbb{R}^n_x\times(\mathbb{R}^n_\xi\setminus\{0\})$ of (t_0,x_0,ξ_0) and an operator $S\in\mathcal{C}^\infty(\mathbb{R};\Psi^0_{\mathrm{cl}}(\mathbb{R}^n)^{K\times K})$ which is elliptic on $\overline{\mathcal{N}}$, with microlocal parametrix denoted $S^-\in\mathcal{C}^\infty(\mathbb{R};\Psi^0_{\mathrm{cl}}(\mathbb{R}^n)^{K\times K})$, such that for the operator*

$$\mathcal{D} = (\mathcal{D}_{kl})_{k,l=1,\dots,K} := SAS^- + S(D_tS^-) \in \mathcal{C}^\infty\big(\mathbb{R};\Psi^1_{\mathrm{cl}}(\mathbb{R}^n)^{K\times K}\big) \tag{7.55}$$

we have

$$\sigma^1(\mathcal{D}_{kk}) = \lambda_k \text{ near } \overline{\mathcal{N}},\ k = 1, \dots, K; \quad \mathrm{WF}'(\mathcal{D}_{kl}) \cap \overline{\mathcal{N}} = \emptyset\ \forall\, k \neq l. \tag{7.56}$$

(2) (Uniqueness.) *If S, S' are two operators for which the conclusions of part (1) hold for the same $\mathcal{N}$, then there exist operators $B_1, \dots, B_K \in \mathcal{C}^\infty(\mathbb{R}; \Psi^0_{\mathrm{cl}}(\mathbb{R}^n))$, elliptic on $\overline{\mathcal{N}}$, such that*

$$\mathrm{WF}'(S'_{kl} - B_k S_{kl}) \cap \overline{\mathcal{N}} = \emptyset, \quad k, l = 1, \dots, K. \tag{7.57}$$

Example 7.15 (Klein–Gordon Equation). Consider the reduction (7.17)–(7.18) of the Klein–Gordon equation to a first order 2×2 system. Then

$$S := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \implies \mathcal{D} := SAS^{-1} + S(D_t S^{-1}) = \begin{pmatrix} -\Lambda & 0 \\ 0 & \Lambda \end{pmatrix}. \tag{7.58}$$

Moreover, $\lambda_1(t, x, \xi) = -|\xi|$ and $\lambda_2(t, x, \xi) = |\xi|$.

Proof of Proposition 7.14.

• Existence. *Step 1: diagonalization.* Since the eigenvalues of $a(t, x, \xi)$ have algebraic multiplicity 1, they depend smoothly on t, x, ξ. The homogeneity of a in ξ implies that $\lambda_k(t, x, \xi) = |\xi| \lambda_k(t, x, \frac{\xi}{|\xi|})$. Therefore, λ_k is a homogeneous symbol of order 1 in a conic neighborhood $\mathcal{N}_0$ of (t_0, x_0, ξ_0). Similarly, the projection $\pi_k(t, x, \xi)$ to the corresponding eigenspace is a homogeneous symbol of order 0 with values in $\mathbb{C}^{K \times K}$. Picking $v_k \in \mathbb{C}^K$ such that $\pi_k(t_0, x_0, \xi_0) v_k \neq 0$, define then the $\mathbb{C}^{K \times K}$-valued symbol

$$\tilde{s}_0(t, x, \xi) = (\tilde{s}_{0,kl}(t, x, \xi))_{k,l=1,\dots,K}, \quad \tilde{s}_{0,kl}(t, x, \xi) := (\pi_l(t, x, \xi) v_l)_k, \tag{7.59}$$

in $\mathcal{N}_0$; it is homogeneous of degree 0. The columns of $\tilde{s}_0$ form a basis of $\mathbb{C}^K$ consisting of eigenvectors of $a(t, x, \xi)$, and hence

$$\tilde{s}_0^{-1} a \tilde{s}_0 = d_1 := \mathrm{diag}(\lambda_1, \dots, \lambda_K) \tag{7.60}$$

where diag denotes the diagonal matrix with the given diagonal entries.

Let $\mathcal{N} \subset \mathcal{N}' \subset \mathcal{N}_0$ be conic neighborhoods of (t_0, x_0, ξ_0) with $\overline{\mathcal{N}} \subset \mathcal{N}'$, $\overline{\mathcal{N}'} \subset \mathcal{N}_0$. Fix $\chi \in S^0_{\mathrm{hom}}$ to be equal to 1 near $\overline{\mathcal{N}'}$ and equal to 0 near the complement of $\mathcal{N}_0$. Set then

$$s_0(t, x, \xi) := \chi(t, x, \xi) \tilde{s}_0(t, x, \xi)^{-1}. \tag{7.61}$$

Pick $S_0 \in \Psi^0_{\mathrm{cl}}$ with $\sigma^0(S_0) = s_0$. *From this point onward, we write '$A \equiv B$' (and '$A \equiv B \bmod X$') when $A - B$ is equal to a symbol, resp. operator with*

essential support, resp. operator wave front set disjoint from $\overline{\mathcal{N}'}$ *(plus an element of* X*).* Let

$$\mathcal{D}_1 = \operatorname{diag}(\Lambda_1, \dots, \Lambda_K), \quad \Lambda_k \in \Psi^1_{\mathrm{cl}},\ \sigma^1(\Lambda_k) = \chi\lambda_k,\ k = 1, \dots, K. \tag{7.62}$$

By construction of s_0, we then have

$$s_0 a s_0^{-1} = d_1 \text{ on } \overline{\mathcal{N}'}, \quad R_0 := S_0 A S_0^- + S_0(D_t S_0^-) - \mathcal{D}_1 \equiv 0 \bmod \Psi^0_{\mathrm{cl}}. \tag{7.63}$$

Here and later on, S^- *denotes a microlocal elliptic parametrix of an operator* $S \in \Psi^1$ *on* $\overline{\mathcal{N}'}$*.*

Step 2: the first lower order correction. We aim to show that if we choose $E_{-1} \in \Psi^{-1}_{\mathrm{cl}}$ suitably, then

$$S_1 A S_1^- + S_1(D_t S_1^-) - (\mathcal{D}_1 + \mathcal{D}_0) \equiv 0 \bmod \Psi^{-1}_{\mathrm{cl}}, \quad S_1 := (I + E_{-1}) S_0, \tag{7.64}$$

for some diagonal $\mathcal{D}_0 \in \Psi^0_{\mathrm{cl}}$. In order to determine the equation that $e_{-1} := \sigma^{-1}(E_{-1}) \in S^{-1}_{\mathrm{hom}}$ needs to satisfy, note first that $S_1^- \equiv S_0^-(I - E_{-1}) \bmod \Psi^{-2}_{\mathrm{cl}}$. Taking principal symbols of order 0 in (7.64) thus shows that we need to find e_{-1} and d_0 such that

$$r_0 + [e_{-1}, d_1] - d_0 = 0; \tag{7.65}$$

here $r_0 \in S^0_{\mathrm{hom}}$ is the principal symbol of the error from (7.63) near $\overline{\mathcal{N}'}$. (More precisely, let $\chi \in S^0_{\mathrm{cl}}$ be equal to 1 near $\overline{\mathcal{N}'}$ and such that $\operatorname{Op}(\chi) R_0 \in \Psi^0_{\mathrm{cl}}$; then $r_0 := \sigma^0(\operatorname{Op}(\chi) R_0)$.) For fixed t, x, ξ, the linear map

$$\mathbb{C}^{K\times K} \ni e \mapsto [e, d_1(t,x,\xi)] = \big((\lambda_l(t,x,\xi) - \lambda_k(t,x,\xi)) e_{kl}\big)_{k,l=1,\dots,K} \tag{7.66}$$

has kernel equal to the space of diagonal matrices and range equal to the space of matrices with vanishing diagonal entries. Therefore, we can solve (7.65) near $\overline{\mathcal{N}'}$ as follows: first let $d_0 = \operatorname{diag}((r_0)_{11}, \dots, (r_0)_{KK}) \in S^0_{\mathrm{cl}}$; then let $e_{-1} \in S^{-1}_{\mathrm{cl}}$ be the unique solution of (7.65) for which $(e_{-1})_{kk} = 0$ for all $k = 1, \dots, K$. Fix a diagonal operator $\mathcal{D}_0$ with principal symbol χd_0 and an operator E_{-1} with principal symbol χe_{-1}; here $\chi \in S^0_{\mathrm{cl}}$ with $\chi = 1$ near $\overline{\mathcal{N}'}$ is supported in a small neighborhood of $\overline{\mathcal{N}'}$. We have now arranged (7.64).

Step 3: remaining corrections; conclusion. Proceeding in this fashion produces diagonal operators $\mathcal{D}_{-J+1} \in \Psi^{-J+1}_{\mathrm{cl}}$ and operators $E_{-J} \in \Psi^{-J}_{\mathrm{cl}}$ for $J = 2, 3, 4, \dots$ such that for $S_J := (I + E_{-J}) \circ \cdots \circ (I + E_{-1}) \circ S_0$,

$$S_J A S_J^- + S_J(D_t S_J^-) - (\mathcal{D}_1 + \cdots + \mathcal{D}_{-J+1}) \equiv 0 \bmod \Psi^{-J}_{\mathrm{cl}}. \tag{7.67}$$

Pick then $E \in \Psi^{-1}_{\mathrm{cl}}$ such that

$$I + E - (I + E_{-J}) \circ \cdots \circ (I + E_{-1}) \in \Psi_{\mathrm{cl}}^{-J-1} \tag{7.68}$$

for all $J \in \mathbb{N}$, i.e., $I + E$ is an *asymptotic product* $\cdots \circ (I + E_{-2}) \circ (I + E_{-1})$. Set $S := (I + E)S_0$ and pick $\mathcal{D} \in \Psi_{\mathrm{cl}}^1$ to be a diagonal operator with $\mathcal{D} \sim \mathcal{D}_1 + \mathcal{D}_0 + \cdots$. Then (7.67) implies $\mathrm{WF}'(SAS^- + S(D_tS^-) - \tilde{\mathcal{D}}) \cap \overline{\mathcal{N}} = \emptyset$. We have thus arranged (7.55)–(7.56).

• Uniqueness. The transformation

$$\Phi\colon (A, S) \mapsto SAS^- + S(D_tS^-) \tag{7.69}$$

has the property that $\Phi(\Phi(A, S), T) \equiv \Phi(A, T \circ S)$. Set $\mathcal{D} := \Phi(A, S)$, $\mathcal{D}' := \Phi(A, S')$, then $\Phi(\mathcal{D}, S'S^-) \equiv \mathcal{D}'$. If both $\mathcal{D}$ and $\mathcal{D}'$ satisfy (7.56), we need to prove that $S'S^-$ is, microlocally near $\overline{\mathcal{N}}$, equal to $\mathrm{diag}(B_1, \ldots, B_K)$ with $B_k \in \Psi_{\mathrm{cl}}^0$.

In new notation, we must therefore show: given a strictly hyperbolic operator A which satisfies (7.56) with A_{kl} in place of $\mathcal{D}_{kl}$ and on $\overline{\mathcal{N}'}$ where $\mathcal{N}' \supset \overline{\mathcal{N}}$, and given $S \in \Psi_{\mathrm{cl}}^0$ such that $\mathcal{D} = \Phi(A, S)$ satisfies (7.56) as well, the operator S is microlocally diagonal near $\overline{\mathcal{N}}$.

First, note that A and $\mathcal{D}$ have the same principal symbol $\mathrm{diag}(\lambda_1, \ldots, \lambda_K)$ near $\overline{\mathcal{N}}$. Since the eigenvalues λ_k of A are pairwise distinct, this forces $\sigma^0(S)$ to be a diagonal matrix. Let $S_0 \in \Psi_{\mathrm{cl}}^0$ be the diagonal operator with principal symbol $\sigma^0(S)$. Write $S_0^- S =: I + \tilde{S}_{-1}$; then $\tilde{S}_{-1} \equiv 0 \bmod \Psi_{\mathrm{cl}}^{-1}$. Therefore, also

$$\begin{aligned} \Phi(\mathcal{D}, S_0^-) &\equiv \Phi(A, S_0^- S) \\ &\equiv \Phi(A, I + \tilde{S}_{-1}) \\ &\equiv (I + \tilde{S}_{-1})A(I + \tilde{S}_{-1})^- + (I + \tilde{S}_{-1})D_t((I + \tilde{S}_{-1})^-) \\ &\equiv A - [A, \tilde{S}_{-1}] \bmod \Psi_{\mathrm{cl}}^{-1} \end{aligned} \tag{7.70}$$

is microlocally diagonal, and thus so is $[A, \tilde{S}_{-1}]$. Recalling the properties of the map (7.66), this in turn forces $\sigma^{-1}(\tilde{S}_{-1})$ to be diagonal. Let $S_{-1} \in \Psi_{\mathrm{cl}}^{-1}$ be the diagonal operator with principal symbol $\sigma^{-1}(\tilde{S}_{-1})$.

Since $(I - S_{-1}) \circ (I + \tilde{S}_{-1}) =: I + \tilde{S}_{-2}$ where $\tilde{S}_{-2} \equiv 0 \bmod \Psi_{\mathrm{cl}}^{-2}$, we now conclude that

$$\Phi(\mathcal{D}, (I - S_{-1}) \circ S_0^-) \equiv \Phi(A, I + \tilde{S}_{-2}) \equiv A - [A, \tilde{S}_{-2}] \bmod \Psi_{\mathrm{cl}}^{-2} \tag{7.71}$$

is microlocally diagonal, and thus so is $\sigma^{-2}(\tilde{S}_{-2})$; and so on. In this fashion, we determine diagonal operators $S_{-j} \in \Psi_{\mathrm{cl}}^{-j}$, $j = 2, 3, \ldots$, such that

$$(I - S_{-J+1}) \circ \cdots \circ (I - S_{-1}) \circ S_0^- \circ S =: I + \tilde{S}_{-J}, \quad \tilde{S}_{-J} \equiv 0 \bmod \Psi_{\mathrm{cl}}^{-J}. \tag{7.72}$$

Picking $E \in \Psi_{\mathrm{cl}}^{-1}$ with $I + E - (I - S_{-J+1}) \circ \cdots \circ (I - S_{-1}) \in \Psi_{\mathrm{cl}}^{-J}$ for all $J \in \mathbb{N}$, we thus conclude that $S \equiv S_0 \circ (I + E)^-$ is microlocally diagonal near $\overline{\mathcal{N}}$. □

Returning to (7.53), the expectation, given (7.55)–(7.56), is now that each component $(S(t)u(t))_k$, $k = 1, \ldots, K$, of $S(t)u(t)$, microlocally near $\overline{\mathcal{N}}$, satisfies a decoupled *scalar* hyperbolic equation governed by the operator $\mathcal{D}_{kk}$. In the notation of (7.57), we have $(S'(t)u)_k \equiv B_k(S(t)u)_k$ on $\overline{\mathcal{N}}$ for all $u \in H^{-N}(\mathbb{R}^n; \mathbb{C}^K)$ in the sense that the wave front set of the difference of both sides is disjoint from $\overline{\mathcal{N}}$. Since the operators B_k are elliptic on $\overline{\mathcal{N}}$, the wave front set of $(S(t)u(t))_k$ on $\overline{\mathcal{N}}$ is therefore independent of the choice of the operator $S(t)$ in Proposition 7.14(1). This allows us to unambiguously write

$$\mathrm{WF}(u(t); k) := \mathrm{WF}\big((S(t)u(t))_k\big) \text{ on } \overline{\mathcal{N}}. \tag{7.73}$$

Theorem 7.16 (Propagation of Wave Front Sets: Systems Case). *Let $u_0 \in H^{-N}(\mathbb{R}^n; \mathbb{C}^K)$, $g \in \mathcal{C}^0(\mathbb{R}; H^\infty(\mathbb{R}^n; \mathbb{C}^K))$, and let u be the solution of the symmetric and strictly hyperbolic system*

$$\begin{cases} (D_t + A(t))u = g, \\ u(0) = u_0. \end{cases} \tag{7.74}$$

For $k = 1, \ldots, K$, let $\mathcal{C}_k(t_1, t_0)$ be the flow of the time-dependent Hamiltonian vector field $H_{\lambda_k(t)}$ from t_0 to t_1. Then

$$\mathrm{WF}(u(t_1); k) = \mathcal{C}_k(t_1, t_0)\mathrm{WF}(u(t_0); k), \quad k = 1, \ldots, K. \tag{7.75}$$

In particular, we have

$$\mathrm{WF}(u(t)) \subset \bigcup_{k=1}^{K} \mathcal{C}_k(t, 0)\mathrm{WF}(u_0). \tag{7.76}$$

The assumption of symmetric hyperbolicity can be dropped (Exercise 7.4). We leave the statement and proof of a Sobolev refinement of (7.75)–(7.76) to the interested reader.

Proof of Theorem 7.16. As in the proof of Theorem 7.8, it suffices to consider the case $g = 0$. By Theorem 7.1, equation (7.74) has a solution $u \in \mathcal{C}^0(\mathbb{R}; H^{-N}(\mathbb{R}^n)) \cap \mathcal{C}^1(\mathbb{R}; H^{-N-1}(\mathbb{R}^n))$.

• Reduction to a scalar equation. At a point $(t_0, x_0, \xi_0) \in \mathbb{R} \times \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$, let $S, S^-, \mathcal{N}$ be as in Proposition 7.14. Our first goal is to reduce the equation for u to a scalar equation for one component of $S(t)u(t)$. We have $S^-S = I - R$ where $R \in \Psi_{\mathrm{cl}}^0$, $\mathrm{WF}'(R) \cap \overline{\mathcal{N}} = \emptyset$. A rigorous version of the formal

computation (7.53) then reads

$$\begin{aligned} 0 = S(D_t + A)u &= S(D_t + A)(S^- S + R)u \\ &= (D_t + SAS^- + S(D_t S^-))Su + S(D_t + A)Ru. \end{aligned} \tag{7.77}$$

With $\mathcal{D} := SAS^- + S(D_t S^-) \in \Psi^1_{\mathrm{cl}}$ and $\tilde{h} := -S(D_t + A)Ru$, this becomes

$$(D_t + \mathcal{D})(S(t)u(t)) = \tilde{h}. \tag{7.78}$$

Let now $Q \in \Psi^0_{\mathrm{cl}}$ be a scalar operator (i.e., it acts component-wise on $\mathbb{C}^K$-valued distributions) with real principal symbol and with $\mathrm{WF}'(I-Q) \cap \overline{\mathcal{N}} = \emptyset$ such that $\mathrm{WF}'(Q)$ is contained in the set where $\sigma^1(\mathcal{D}_{kk}) = \lambda_k$ (see (7.56)) and $\mathrm{WF}'(Q) \cap \mathrm{WF}'(\mathcal{D}_{kl}) = \emptyset$ for all $k \neq l$. Then

$$(D_t + Q\mathcal{D})(Su) = h := \tilde{h} - (I - Q)\mathcal{D}Su = -S(D_t + A)Ru - (I - Q)\mathcal{D}Su. \tag{7.79}$$

Finally, we consider the k-th component of this equation. Since the off-diagonal terms of $Q\mathcal{D}$ are residual operators, we obtain the *scalar* symmetric hyperbolic equation

$$(D_t + Q\mathcal{D}_{kk})v(t) = h_{(k)}, \quad v(t) := (S(t)u(t))_k,\ h_{(k)} - h_k \in \mathcal{C}^0(\mathbb{R}; H^\infty(\mathbb{R}^n)). \tag{7.80}$$

Since the wave front set of h, and thus of $h_{(k)}$, is disjoint from $\overline{\mathcal{N}}$ (since $\mathrm{WF}'(R)$ and $\mathrm{WF}'(I - Q)$ are disjoint from $\overline{\mathcal{N}}$), we expect $\mathrm{WF}(v(t))$ to propagate along the flow of $\partial_t + H_{\lambda_k(t)}$ at least in a conic neighborhood of (t_0, x_0, ξ_0).

• Propagation of wave front sets. Denote the solution operator for the homogeneous problem $(D_t + Q\mathcal{D}_{kk})w(t) = 0$ by $S_k(t_1, t_0)$. Let $\epsilon > 0$ be such that the following holds: for all $t \in [t_0 - \epsilon, t_0 + \epsilon]$ and

$$(x, \xi) \in \mathcal{N}_\epsilon(x_0, \xi_0) := \left\{ (z, \zeta) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\}) \colon |z - x_0| < \epsilon,\ \left| \frac{\zeta}{|\zeta|} - \frac{\xi_0}{|\xi_0|} \right| < \epsilon \right\} \tag{7.81}$$

we have $\mathcal{C}_k(t, s)(x, \xi) \in \mathcal{N}$ for all $s \in [t_0 - \epsilon, t_0 + \epsilon]$. Suppose now $(x_1, \xi_1) \in \mathcal{N}_\epsilon(x_0, \xi_0)$, $(x_1, \xi_1) \notin \mathrm{WF}(v(t_0))$. Let $P_0 \in \Psi^0(\mathbb{R}^n)$ be elliptic at (x_1, ξ_1) such that $P_0 v(t_0) \in H^\infty(\mathbb{R}^n)$ and $\mathrm{WF}'(P_0) \subset \mathcal{N}_\epsilon(x_0, \xi_0)$. Define

$$P(t) := S_k(t, t_0) \circ P_0 \circ S_k(t_0, t), \quad t \in [t_0 - \epsilon, t_0 + \epsilon], \tag{7.82}$$

using Theorem 7.6. By the Duhamel formula, we can solve (7.80) by

$$v(t) = S_k(t, t_0)v(t_0) + \int_{t_0}^t i S_k(t, s) h_{(k)}(s)\, \mathrm{d}s. \tag{7.83}$$

Therefore,

$$P(t)v(t) = S_k(t,t_0)\big(P_0 v(t_0)\big) + \int_{t_0}^{t} iS_k(t,s)\big(P(s)h_{(k)}(s)\big)\,\mathrm{d}s. \tag{7.84}$$

The construction of $P(s)$ in the proof of Theorem 7.6 shows that $P(s)$, $s \in [t_0-\epsilon, t_0+\epsilon]$, is the sum of a smoothing operator and a ps.d.o. whose operator wave front set $\mathrm{WF}'(P(s)) = \mathcal{C}_k(s,t_0)(\mathrm{WF}'(P_0))$ is a subset of $\mathcal{N}$. (We use here that the principal symbols of $\mathcal{D}_{kk}$ and $Q\mathcal{D}_{kk}$ are equal over $\mathcal{N}$.) Therefore, the right-hand side of (7.84) lies in $H^\infty(\mathbb{R}^n)$, and thus $\mathcal{C}_k(t,t_0)(x_1,\xi_1) \notin \mathrm{WF}(v(t))$, $t \in [t_0-\epsilon, t_0+\epsilon]$.

We have now proved the validity of (7.75) in a conic neighborhood of every point in $\mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$ when $|t_1 - t_0|$ is sufficiently small. Concatenating these local results gives (7.75) for all t_0, t_1. The inclusion (7.76) finally follows from

$$\mathrm{WF}(u(t)) = \bigcup_{k=1}^{K} \mathrm{WF}(u(t);k), \quad \mathrm{WF}(u(0);k) \subset \mathrm{WF}(u_0)\ (k=1,\dots,K). \tag{7.85}$$

The proof is complete. □

Example 7.17 (Klein–Gordon Equation). Recall the system (7.17)–(7.18), which is diagonalized using (7.58). Theorem 7.16 shows that for smooth source terms h,

$$\begin{aligned}\mathrm{WF}(U(t);1) = \mathrm{WF}\big((SU(t))_1\big) &= \mathrm{WF}(\Lambda u(t) + D_t u(t)) \\ &= \mathcal{C}_-(t,0)\mathrm{WF}(\Lambda u_0 + u_1),\end{aligned} \tag{7.86}$$

and similarly

$$\mathrm{WF}(U(t);2) = \mathrm{WF}(\Lambda u(t) - D_t u(t)) = \mathcal{C}_+(t,0)\mathrm{WF}(\Lambda u_0 - u_1); \tag{7.87}$$

here $\mathcal{C}_\pm(t,0)$ is the time t flow of $H_{\pm|\xi|} = \pm\frac{\xi}{|\xi|}\cdot\partial_x$. As a consistency check for (7.87), note that if $(-D_t^2+\Delta+1)u = 0$, then $v := \Lambda u - D_t u = -(D_t - \langle D\rangle)u$ solves the equation $(D_t+\langle D\rangle)v = 0$; the propagation of singularities of v along $\frac{\xi}{|\xi|}\cdot\partial_x$ was already discussed in Example 7.12.

One can similarly study the wave front sets of solutions of the wave equation (7.19): they propagate along the integral curves of $\partial_t + H_{\lambda_\pm(t)}$, $\lambda_\pm(t,x,\xi) := \pm|\xi|_{g(t)_x^{-1}}$.

Notes

Our presentation of Theorems 7.1–7.8 is inspired by [Tay11b, §§7.7–7.8]. Another method for proving Theorem 7.1 is via a priori energy estimates for the adjoint problem $(D_t + a(t,x,D_x)^*)v = h(t,x)$ combined with a duality argument. This is the approach taken in Hörmander [Hör07, Chapter XXIII], and also in Dyatlov–Zworski [DZ19, Appendix E.5] in the semiclassical setting. The results can be extended to semiglobal existence and uniqueness statements for m-th order strictly hyperbolic differential operators; the passage from spaces such as $\mathcal{C}^0_t H^s_x$ to Sobolev spaces on "spacetime" requires the use of intermediate spaces which have independent "spacetime" and "spatial" regularity indices. We discuss a more direct approach to "spacetime" estimates for wave equations in Chapter 9.

That the conjugation of ps.d.o. by the flow of a scalar symmetric hyperbolic equation is a ps.d.o. (Theorem 7.6) is an important observation due to Egorov [Ego69].

A variant of the decoupling argument in Proposition 7.14 was used in Taylor [Tay75] to prove the reflection of singularities for hyperbolic equations on domains with boundary.

Exercises

Exercise 7.1 (Necessity of a Symmetry Assumption). On $\mathbb{R}_t \times \mathbb{R}_x$, consider the equation $(D_t + iD_x)u = 0$. (This is the Cauchy–Riemann equation for $u = u(t,x)$, regarded as a function of the complex variable $t+ix$.) Suppose that u is a solution of this equation with $u \in \mathcal{C}^0(\mathbb{R}; H^s(\mathbb{R}^n))$. What can you say about $u(0,x)$? Conclude that Theorem 7.1 fails for $a(t,x,D_x) = iD_x$.

Exercise 7.2 (Symmetrizable Hyperbolic Systems). Let $A \in \mathcal{C}^\infty(\mathbb{R}_t; \Psi^1(\mathbb{R}^n)^{K\times K})$. Suppose the $K \times K$ system

$$\begin{cases} (D_t + A(t))u = g \ \in \mathcal{C}^0\big(\mathbb{R}; H^s(\mathbb{R}^n;\mathbb{C}^K)\big), \\ u(0) = u_0 \in H^s(\mathbb{R}^n;\mathbb{C}^K) \end{cases} \tag{7.88}$$

is a *symmetrizable hyperbolic system*: this means that there exists a $K \times K$-matrix-valued symbol $s(t,x,\xi) \in S^0$ which is positive definite and such that $s(t,x,\xi)a(t,x,\xi)$ is symmetric modulo S^0. Prove that (7.88) has a unique solution

$$u \in \mathcal{C}^0\big(\mathbb{R}; H^s(\mathbb{R}^n;\mathbb{C}^K)\big) \cap \mathcal{C}^1\big(\mathbb{R}; H^{s-1}(\mathbb{R}^n;\mathbb{C}^K)\big). \tag{7.89}$$

Exercise 7.3 (Strictly Hyperbolic Systems, I). We call a $K \times K$ system of the form (7.88) *strictly hyperbolic* if the principal symbol of A has a homogeneous representative $a \in S^1_{\hom}$ which, moreover, for all $t \in \mathbb{R}$, $x \in \mathbb{R}^n$, and $\xi \in \mathbb{R}^n \setminus \{0\}$, has K distinct real eigenvalues. (This is consistent with Definition 7.13.) Show that a strictly hyperbolic system is symmetrizable.

Exercise 7.4 (Strictly Hyperbolic Systems, Ii). Show using Exercise 7.3 that Theorem 7.16 holds for all strictly hyperbolic systems. (Thus, the symmetric hyperbolicity assumption is not needed.)

Exercise 7.5 (Higher-Order Strictly Hyperbolic Equations). On $\mathbb{R}_t \times \mathbb{R}^n_x$, consider an m-th order operator

$$L = D_t^m + \sum_{j=0}^{m-1} a_j(t, x, D_x) D_t^j \tag{7.90}$$

where $a_j(t, x, D_x) \in \mathrm{Diff}^{m-j}(\mathbb{R}^n_x)$ depends smoothly on t. We study the system

$$\begin{cases} Lu = g \quad \in \mathcal{C}^0(\mathbb{R}_t; H^{s-m+1}(\mathbb{R}^n)), \\ D_t^j u(0, \cdot) = u_j \in H^{s-j}(\mathbb{R}^n), \quad j = 0, \dots, m-1. \end{cases} \tag{7.91}$$

(1) Reduce (7.91) to an $m \times m$ system of first-order evolution equations for the $\mathbb{C}^m$-valued function $(\Lambda^{m-1} u, \Lambda^{m-2} D_t u, \dots, D_t^{m-1} u)$ where $\Lambda = \langle D_x \rangle$.
(2) Give a condition on the principal symbol of L which is equivalent to the strict hyperbolicity of this $m \times m$ system.
(3) Under the strict hyperbolicity assumption of part (2), prove the existence and uniqueness of a solution $u \in \bigcap_{j=0}^m \mathcal{C}^j(\mathbb{R}_t; H^{s-j}(\mathbb{R}^n))$ of (7.91).

Chapter 8
Real Principal Type Propagation of Singularities



We now free ourselves from the restrictive setting of equations which, as in the previous chapter, are explicitly given in evolution form. In Sections 8.1–8.6, we prove the propagation of singularities (wave front sets), or equivalently, the propagation of regularity, for solutions of general non-elliptic (pseudo)differential equations

$$Pu = f, \quad P \in \Psi^m(M), \tag{8.1}$$

where $M = \mathbb{R}^n$ or some other manifold, and $m \in \mathbb{R}$. We assume that the principal symbol

$$p(x,\xi) := \sigma^m(P)(x,\xi) \in S^m_{\hom}(T^*M \setminus o) \tag{8.2}$$

is *homogeneous* of degree m (this being automatic when P is a *differential* operator) and *real*. (By the membership (8.2), we mean that $(1-\chi)p$ is a representative of $\sigma^m(P) \in S^m(T^*M)/S^{m-1}(T^*M)$ where $\chi \in \mathcal{C}^\infty(T^*M)$ equals 1 near the zero section and 0 near fiber infinity.) The semiclassical setting is treated in Section 8.7.

8.1 Statement of the Theorem; a First Proof

We first give some examples of operators for the reader to keep in mind in this chapter. The first two examples *are* in evolution form. Even for these, the propagation of singularities result we prove here will be qualitatively different from Theorem 7.8, as we will now track the wave front set of u on M, rather than wave front sets of restrictions of u to level sets of a time function.

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Example 8.1 (Operators with Real Principal Symbols).

(1) (Transport operators.) On $M = \mathbb{R}_t \times \mathbb{R}^{n-1}_x$ and for real-valued $a_j \in \mathcal{C}^\infty(M)$, $j = 1, \dots, n-1$, the operators

$$P_1 := D_t, \quad P_2 := D_t + \sum_{j=1}^{n-1} a_j(t,x) D_{x_j} \in \mathrm{Diff}^1(M) \tag{8.3}$$

have real principal symbols. Writing covectors as $\tau\,\mathrm{d}t + \sum_{j=1}^{n-1} \xi_j\,\mathrm{d}x_j$, they are

$$p_1(t,x,\tau,\xi) = \tau, \quad p_2(t,x,\tau,\xi) = \tau + \sum_{j=1}^{n-1} a_j(t,x)\xi_j. \tag{8.4}$$

(2) (Wave operator.) Another differential operator with real principal symbol is the wave operator

$$P_3 := -D_t^2 + \sum_{j=1}^{n-1} D_{x_j}^2 \in \mathrm{Diff}^2(M), \quad p_3(t,x,\tau,\xi) = -\tau^2 + |\xi|^2. \tag{8.5}$$

(3) (Multiplication operators.) On any smooth manifold M, let $p_4 \in \mathcal{C}^\infty(M;\mathbb{R})$ and consider the operator $P_4 \colon u \mapsto p_4 u$. Then $P_4 \in \Psi^0(M)$ has real principal symbol p_4.

Definition 8.2 (Null-Bicharacteristics). A (forward) *null-bicharacteristic* of P is an integral curve of the Hamiltonian vector field H_p lying in $\mathrm{Char}(P) \subset T^*M \setminus o$. We use the same terminology also for smooth curves in $\mathrm{Char}(P) \subset S^*M$ which are images of null-bicharacteristics in $S^*M = (T^*M \setminus o)/\mathbb{R}_+$, or smooth monotone reparameterizations thereof. A backward null-bicharacteristic of P is a forward null-bicharacteristic of $-P$.

Note that $H_p p = 0$; hence, an integral curve of H_p with initial condition $\alpha \in \mathrm{Char}(P)$ is automatically a null-bicharacteristic, and all null-bicharacteristics arise in this fashion.

Example 8.3 (Null-Bicharacteristics). We consider the operators from Example 8.1.

(1) (Transport operators.) Since $\mathrm{Char}(P_1) = \{(t,x,\tau,\xi) \colon \tau = 0,\ \xi \neq 0\}$ and $H_{p_1} = \partial_t$, null-bicharacteristics of P_1 are curves $s \mapsto (t_0 + s, x_0; 0, \xi_0)$. For the more general operator P_2 in (8.3), we have

$$H_{p_2} = \Big(\partial_t + \sum_{j=1}^{n-1} a_j(t,x)\partial_{x_j}\Big) - \sum_{j=1}^{n-1} \xi_j \Big(\partial_t a_j(t,x)\partial_\tau + \sum_{k=1}^{n-1} \partial_{x_k} a_j(t,x) \partial_{\xi_k}\Big). \tag{8.6}$$

Thus, null-bicharacteristics (and indeed all integral curves of H_{p_2}) project to M as integral curves of the transport vector field $\partial_t + \sum_{j=1}^{n-1} a_j(t,x)\partial_{x_j}$.

(2) (Wave operator.) We have $\mathrm{Char}(P_3) = \{(t, x, \tau, \xi) \colon |\xi| = |\tau| \neq 0\}$. Thus, $\mathrm{Char}(P_3) \cap (T^*_{(t_0,x_0)} M \setminus o)$ is a double cone (minus the vertex), called the *(dual) light cone*. We have

$$H_{p_3} = -2\tau\partial_t + 2\xi \cdot \partial_x, \tag{8.7}$$

where $\xi \cdot \partial_x := \sum_{j=1}^{n-1} \xi_j \partial_{x_j}$. Note that $H_p\tau = 0$, so τ is constant along null-bicharacteristics. Null-bicharacteristics along which $\tau = -1$ are given by

$$s \mapsto (t_0 + 2s, x_0 + 2\xi_0 s; -1, \xi_0), \quad |\xi_0| = 1. \tag{8.8}$$

Their projections to M are *null-geodesics*. See the left panel of Figure 8.1.

(3) (Multiplication operator.) In local coordinates $x \in \mathbb{R}^n$ on M, we have $\mathrm{Char}(P_4) = \{(x, \xi) \colon p_4(x) = 0,\ \xi \neq 0\}$ and

$$H_{p_4} = -(\partial_x p_4) \cdot \partial_\xi. \tag{8.9}$$

At points $x_0 \in \mathbb{R}^n$ where $p_4(x_0) = 0$, null-bicharacteristics thus stay in the fiber $T^*_{x_0}\mathbb{R}^n \setminus o$ but move in the momentum variables in the direction $-\partial_x p_4(x_0)$. See the right panel of Figure 8.1.

A first version of the main result of this chapter reads as follows.

Theorem 8.4 (Propagation of Singularities: Smooth Case). *Suppose $P \in \Psi^m(M)$ (properly supported) has a real-valued homogeneous principal symbol. Let $u \in \mathscr{D}'(M)$ and suppose that $Pu \in \mathcal{C}^\infty(M)$. Then $\mathrm{WF}(u) \subset \mathrm{Char}(P)$ is a union of maximally extended null-bicharacteristics of P.*

The statement $\mathrm{WF}(u) \subset \mathrm{Char}(P)$ follows from microlocal elliptic regularity (Proposition 6.33). The theorem asserts that within $\mathrm{Char}(P)$, the wave

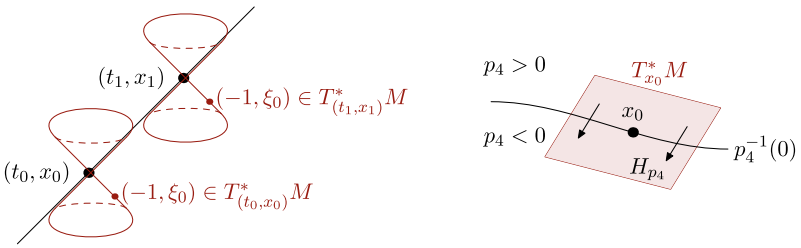


Figure 8.1 *On the left:* We illustrate the characteristic set of the wave operator P_3 over two points (t_0, x_0), (t_1, x_1) (these are the dual light cones) and mark two points on these lying along a single null-bicharacteristic of P_3. The solid line is the base projection of the null-bicharacteristic. *On the right:* an analogous illustration of a null-bicharacteristic of P_4.

front set of a "microlocal solution" u (i.e., $\mathrm{WF}(Pu) = \emptyset$) is invariant under the H_p-flow. Rather than calling Theorem 8.4 a result on the propagation of *singularities*, one often (and more usefully) regards it as a result on the propagation of *regularity*, since $\alpha \notin \mathrm{WF}(u) \cap \mathrm{Char}(P)$ implies that the maximally extended null-bicharacteristic of P through α is disjoint from $\mathrm{WF}(u)$.

Proof of Theorem 8.4. We reduce this to a local result. If $\chi, \tilde{\chi} \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ are two cutoffs and $\tilde{\chi} \equiv 1$ near $\operatorname{supp}\chi$, then

$$\chi P \tilde{\chi} u = \chi P u + \chi[P, \tilde{\chi}]u \in \mathcal{C}^\infty_{\mathrm{c}}(M). \tag{8.10}$$

We use here that $\chi[P, \tilde{\chi}] \in \Psi^{-\infty}(M)$. Thus, if $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ is identically 1 near $\operatorname{supp}\tilde{\chi}$, we can replace P by $\chi P \tilde{\chi}$ and u by ϕu, and we then still have $Pu \in \mathcal{C}^\infty(M)$; moreover, these replacements do not alter null-bicharacteristics of P and the wave front set of u in phase space over $\chi^{-1}(1)$. Localizing in this manner to a coordinate patch, we can thus assume

$$P \in \Psi^m(\mathbb{R}^n),\ u \in \mathscr{E}'(\mathbb{R}^n),\ Pu = f \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n). \tag{8.11}$$

We normalize this using $\Lambda := \langle D \rangle^{m-1}$ by replacing (P, u) by $(P\Lambda^{-1}, \Lambda u)$ and applying the aforementioned localization argument once more; thus, we can assume that $m = 1$. After these replacements, the principal symbol of P is still homogeneous and real.

We now introduce an artificial time variable t and set

$$\tilde{u}(t, x) := u(x), \quad \tilde{f}(t, x) := f(x). \tag{8.12}$$

Then $\tilde{u}$ solves the equation

$$(D_t + P)\tilde{u} = \tilde{f}. \tag{8.13}$$

Theorem 7.8 and Remark 7.9 imply

$$\mathrm{WF}(u) = \mathrm{WF}(\tilde{u}(t)) = \mathcal{C}(t, 0)\mathrm{WF}(\tilde{u}(0)) = \mathcal{C}(t, 0)\mathrm{WF}(u) \tag{8.14}$$

where $\mathcal{C}(t, 0)$ is the time t flow of H_p. Thus, $\mathrm{WF}(u)$ is invariant under the H_p-flow. □

We illustrate Theorem 8.4 using the operators of Example 8.1.

Example 8.5 (Propagation of Singularities: Transport Operators). Theorem 8.4 can be verified by hand for the operator P_1 (Exercise 8.1). Solutions u of $P_2 u = 0$ are of the form $u(t, x) = u_0(x_0)$ where x_0 is such that the time t flow of $V := \partial_t + \sum_{j=1}^{n-1} a_j(t, x)\partial_{x_j}$ starting at $(0, x_0)$ ends at (t, x). Singularities of u thus occur in directions conormal to the level sets

of the map $(t,x)\mapsto x_0$, and hence in $\mathrm{Char}(P_2)$ indeed, and they propagate along V in the base (and along H_{p_2} in phase space by Theorem 8.4, as the reader may wish to check explicitly). By contrast, Theorem 7.8 concerns the singularities of $u(t,\cdot)$ for each t: it states that $(x,\xi)\in \mathrm{WF}(u(t,\cdot))$ if and only if $(x_0,\xi_0)\in\mathrm{WF}(u_0)$ where $(x,\xi)=\mathcal{C}(t,0)(x_0,\xi_0)$ in the notation of Theorem 7.6, which in turn happens if and only if $(0,x_0,\tau_0,\xi_0)\in\mathrm{WF}(u)$ for $\tau_0+\sum_{j=1}^{n-1}a_j(t,x)(\xi_0)_j=0$. We shall not explore the relationship between Theorems 7.6 and 8.4 any further, and only note that the wave front set relationship between u and $u(t,\cdot)$ is consistent with Proposition 6.44.

Example 8.6 (Propagation of Singularities: Wave Operator). The "retarded fundamental solution" $u(t,x)=\frac{1}{2}H(t)1_{|x|<t}$ of P_3 in $2=1+1$ dimensions satisfies

$$\begin{aligned}\mathrm{WF}(u)=\{(t,x,\tau,\xi)\colon |x|=t>0,\ 0\neq\tau=-(\operatorname{sgn}x)\xi\}\\ \cup\{(0,0,\tau,\xi)\colon(\tau,\xi)\neq(0,0)\}.\end{aligned}\tag{8.15}$$

(See Examples 6.28 and 6.34(3).) Theorem 8.4 implies that in all dimensions, the wave front set of the retarded fundamental solution u of P_3, i.e., $P_3u=\delta(t,x)$ and $u|_{t<0}=0$, must satisfy

$$\begin{aligned}\mathrm{WF}(u)\subseteq\left\{(t,x,\tau,\xi)\colon |x|=t>0,\ 0\neq\tau=-\frac{x}{|x|}\cdot\xi\right\}\\ \cup\{(0,0,\tau,\xi)\colon(\tau,\xi)\neq(0,0)\}.\end{aligned}\tag{8.16}$$

(We shall not prove here that it is, in fact, equal to this set.) Indeed, if $(t_0,x_0,\tau_0,\xi_0)\in\mathrm{Char}(P_3)$ is *not* in this set, there exists a null-bicharacteristic segment through (t_0,x_0,τ_0,ξ_0) which contains a point (t,x,τ,ξ) with $t<0$ and does not pass through $\mathrm{WF}(\delta)=N^*\{(0,0)\}\setminus o$. Since $\mathrm{WF}(u)\cap\{t<0\}=\emptyset$, (8.16) follows.—We can use this example to demonstrate an advantage of working microlocally. To wit, consider the initial value problem $P_3v=f$, $v(0,x)=\partial_tv(0,x)=0$, where $f:=P_3(\chi u)$ with $\chi\in\mathcal{C}^\infty(\mathbb{R})$, $\chi(t)=0$ for $t<1$ and $\chi(t)=1$ for $t\geq 2$. Looking only at the singular support of the retarded fundamental solution, one might naively expect that v will be singular at all (t,x) for which there exists (t_0,x_0) with $|x_0|=t_0>2$ (where χu is certainly singular) and $|t-t_0|=|x-x_0|$. This is not the case: the unique solution is $v=\chi u$ and thus singular only on the light cone $|x|=t$. It is *microlocal* information, namely $\mathrm{WF}(f)$ (which is contained in (8.16)), which allows one to determine the correct singular support (and indeed wave front set) of v via Theorem 8.4.

Example 8.7 (Propagation of Singularities: Multiplication Operator). Suppose $P_4u=p_4u\in\mathcal{C}^\infty$. Then $\operatorname{sing\,supp}u\subset p_4^{-1}(0)$. Let $x_0\in p_4^{-1}(0)$, $\xi_0\neq 0$. Set $\tilde{\xi}:=-\partial_xp_4(x_0)$. Theorem 8.4 tells us that if $\xi_0\notin\mathbb{R}\tilde{\xi}$, we have $(x_0,\xi_0)\notin\mathrm{WF}(u)$ if and only if $(x_0,\xi_0+s\tilde{\xi})\notin\mathrm{WF}(u)$ for some (and thus all)

$s \in \mathbb{R}$. When $\tilde{\xi} = 0$, this is a trivial conclusion. Also when $\tilde{\xi} \neq 0$, and $\xi_0 = \lambda\tilde{\xi}$ for some $\lambda > 0$ (resp. $\lambda < 0$), this is a trivial conclusion since ξ_0 and $\xi_0 + s\tilde{\xi}$ are positive multiples of each other and thus define the same element in S^*M for $s \in (-\lambda, \infty)$ (resp. $s \in (-\infty, -\lambda)$).

Example 8.7 suggests the following notion: $\alpha \in T^*M \setminus o$ is a *radial point* for P if $H_p|_\alpha$ is a multiple of the generator of dilations in the fibers of T^*M, i.e., there exists $c \in \mathbb{R}$ such that

$$H_p|_\alpha = c\xi \cdot \partial_\xi. \tag{8.17}$$

At radial points, Theorem 8.4 is vacuous. We shall see in Chapter 10 that at certain sets of radial points, one can still draw strong conclusions about the wave front sets of solutions of $Pu = f$.

An extreme case of a radial point is a point $\alpha \in T^*M \setminus o$ where $H_p|_\alpha = 0$, or equivalently $\mathrm{d}p = 0$ at α: the null-bicharacteristic starting at α is the constant curve remaining at α. A natural condition for operators P with real homogeneous principal symbol p is to require $\mathrm{d}p \neq 0$ on $\mathrm{Char}(P)$. Not only does this exclude the possibility of constant null-bicharacteristics in $T^*M \setminus o$; it also implies by the implicit function theorem that $\mathrm{Char}(P) \subset T^*M \setminus o$ is a smooth conic codimension 1 submanifold. One often uses the terminology

real principal type propagation

to refer to applications of Theorem 8.4 on subsets of $\mathrm{Char}(P) \subset T^*M \setminus o$ on which H_p is never radial.

One of the main drawbacks of the above proof of Theorem 8.4 (apart from it being rather ad hoc) is that it ultimately rests on the solvability theory for the (auxiliary) equation (8.13). But solving PDEs is difficult, and hence, one should not attempt to solve one unless absolutely necessary. We shall thus present another proof of Theorem 8.4 which is longer and looks more complicated, but is at its core very simple: it is the prototypical example of a *positive commutator argument*. We will prove the following strengthening of Theorem 8.4:

Theorem 8.8 (Propagation of Singularities: Sobolev Case). *Suppose $P \in \Psi^m(M)$ (properly supported) has a real-valued homogeneous principal symbol. Let $u \in \mathscr{D}'(M)$ and $f := Pu \in \mathscr{D}'(M)$. Let $s \in \mathbb{R}$. Then*

$$\mathrm{WF}^s(u) \subset \mathrm{Char}(P) \cup \mathrm{WF}^{s-m}(f). \tag{8.18}$$

Moreover, within $\mathrm{Char}(P) \setminus \mathrm{WF}^{s-m+1}(f)$,

$$\mathrm{WF}^s(u) \setminus \mathrm{WF}^{s-m+1}(f) \tag{8.19}$$

is a union of maximally extended null-bicharacteristics of P.

Remark 8.9 ("Loss" of One Derivative). Note that in order to obtain the propagation of microlocal H^s-regularity of u in $\mathrm{Char}(P)$, one needs to assume that f lies in H^{s-m+1} microlocally. This is one degree more smoothness than what microlocal elliptic regularity results would require. Thus, on $\mathrm{Char}(P)$, u "loses" one derivative relative to elliptic regularity.

We will really prove *quantitative estimates* (cf. Propositions 6.38 and 6.39):

Theorem 8.10 (Propagation of Singularities: Estimate Version). *Suppose $P \in \Psi^m(M)$ (properly supported) has a real-valued homogeneous principal symbol. Let $u \in \mathscr{D}'(M)$. Let $\gamma\colon [0, s_0] \to \mathrm{Char}(P)$ be a null-bicharacteristic of P. Let $K \subset \tilde{K} \subset M$ be compact subsets such that K° contains the projection of $\gamma([0, s_0])$ to M, and the support of the Schwartz kernel of P intersects $K \times M$ in a subset of $K \times \tilde{K}$. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ be equal to 1 near $\tilde{K}$. Let $\mathcal{U}_0 \subset S^*M$ and $\mathcal{U} \subset S^*M$ be arbitrary neighborhoods of $\gamma(0)$ and $\gamma([0, s_0])$, respectively. Then there exist*

$$B, G, E \in \Psi^0(M), \tag{8.20}$$

with Schwartz kernels supported in $K \times K$,

(1) $\mathrm{Ell}(B) \supset \gamma([0, s_0])$,
(2) $\mathrm{WF}'(B) \subset \mathrm{Ell}(G)$, $\mathrm{WF}'(G) \subset \mathcal{U}$,
(3) $\mathrm{WF}'(E) \subset \mathcal{U}_0$, *and E is elliptic at $\gamma(0)$,*

such that the following estimate holds for arbitrary $N \in \mathbb{R}$ and a constant $C = C(N) > 0$:

$$\|Bu\|_{H^s} \leq C\big(\|Eu\|_{H^s} + \|GPu\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}\big). \tag{8.21}$$

Moreover, this holds in the strong sense that if $u \in \mathscr{D}'(M)$ is such that the right-hand side is finite, then so is the left-hand side, and the estimate holds.

One reads the estimate (8.21) as follows: *assuming* a priori microlocal H^s-control of u on $\mathrm{Ell}(E)$, we *conclude* microlocal H^s-control of u on $\mathrm{Ell}(B)$ by propagation along null-bicharacteristics, provided Pu remains microlocally in H^{s-m+1} along the way. See Figure 8.2.

Simple applications of Theorem 8.10 are discussed in Exercises 8.2–8.4. A more substantial application is described in Chapter 9, where the regularity of solutions of wave equations is studied using Theorem 8.10.

Remark 8.11 (Generalization to Operators on Vector Bundles). Theorems 8.8 and 8.10 also hold for operators $P \in \Psi^m(M; E)$ acting on sections of a vector bundle E, provided P has a scalar, real-valued, homogeneous principal symbol (Definition 5.47). (To extend the following proof to this case, one fixes an arbitrary smooth fiber inner product on E. The main change is then that $p_1 = \sigma^{m-1}(\frac{P-P^*}{2i})$ is now endomorphism-

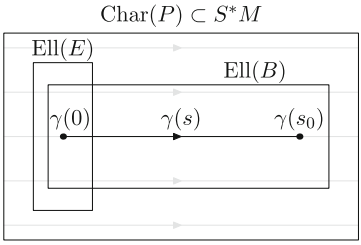


Figure 8.2 Illustration of the real principal type propagation estimate (8.21): microlocal H^s-control of u on $\mathrm{Ell}(E)$ implies microlocal H^s-bounds on u on the set $\mathrm{Ell}(B)$, provided Pu is microlocally in H^{s-m+1} in a neighborhood thereof.

valued. The square roots in (8.47), (8.51), and (8.65) are still well-defined using the power series expansion for $\sqrt{1-S}$ for $S \in \mathrm{End}(E_x)$ with $\|S\| < 1$.)

Remark 8.12 (Supports and Cutoffs). For *differential* operators P, one can of course take $\tilde{K} = K$. When M is compact, one can take $K = \tilde{K} = M$ and $\chi = 1$. The only reason for keeping track of the supports of Schwartz kernels is to ensure that all distributions appearing in (8.21) are supported in a fixed compact set (namely, $\operatorname{supp}\chi$), and thus, the Sobolev norms are well-defined (Proposition 5.58).

Theorem 8.8 is an immediate consequence of Theorem 8.10. Indeed, if $\alpha \in \mathrm{Char}(P) \setminus \mathrm{WF}^s(u)$, then Theorem 8.10 implies that the *forward* null-bicharacteristic of P with initial condition α remains disjoint from $\mathrm{WF}^s(u)$ as long as it does not intersect $\mathrm{WF}^{s-m+1}(f)$. Applying Theorem 8.10 to $-P$ gives the *backward* propagation of regularity.

We begin the proof of Theorem 8.10 with a preliminary simplification. Namely, we claim that *it suffices to prove* (8.21) *for* u *supported in* $\chi^{-1}(1)$. To show this, let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ be equal to 1 near K and supported in $\chi^{-1}(1)$. Suppose that (8.21) holds for ϕu in place of u. Note then that

$$\|Bu\|_{H^s} \leq \|B(\phi u)\|_{H^s} + \|B(1-\phi)u\|_{H^s} \leq \|B(\phi u)\|_{H^s} + C\|\chi u\|_{H^{-N}} \tag{8.22}$$

since $B(1-\phi) \in \Psi^{-\infty}$. Similarly, since $E(1-\phi) \in \Psi^{-\infty}$, we have

$$\|E(\phi u)\|_{H^s} \leq \|Eu\|_{H^s} + \|E(1-\phi)u\|_{H^s} \leq \|Eu\|_{H^s} + C\|\chi u\|_{H^{-N}}. \tag{8.23}$$

Similarly, $GP(\phi u) = GPu - GP(1-\phi)u$; by the assumptions on the Schwartz kernels of G and P, we have $GP(1-\phi)u = GP(1-\phi)\chi u$. Since $GP(1-\phi) \in \Psi^{-\infty}$, this proves (8.21) for u.

8.2 Positive Commutators I: Sketch

Let us consider a basic example of the setup of Theorem 8.10: $P = D_{x_1}$ on $\mathbb{R}^n_x$, so $p = \xi_1$ and $H_p = \partial_{x_1}$; we write points in $\mathbb{R}^n$ as $x = (x_1, x')$ here. We take B, E, G to be the characteristic functions of $Q_B := (-2,2)_{x_1} \times (-1,1)^{n-1}_{x'}$, $Q_E := (-2,-1)\times(-2,2)^{n-1}$, and $Q_G := (-3,3)\times(-3,3)^{n-1}$. (Smoothed-out versions of these are 0-th order ps.d.o.s.) Take $s = 0$; let

$$D_{x_1} u = f \in \mathscr{D}'(\mathbb{R}^n). \tag{8.24}$$

Then (8.21) asserts that $u|_{Q_B} \in L^2$, provided $u|_{Q_E} \in L^2$ and $f|_{Q_G} \in L^2$. But this follows easily from the formula $u(x_1, x') = u(x_1 - j, x') + \int_{x_1-j}^{x_1} if(s, x')\,\mathrm{d}s$, $j = 0, 1, 2, 3$ for $x_1 \in [-2 + j, -1 + j]$, for the solution of (8.24).

A better proof does not require the explicit solution of (8.24). (This "better proof" does use the fundamental theorem of calculus, but in the form of integration by parts, and in a way that generalizes readily to general operators.) To explain it, let us take $n = 1$, $x = x_1$, for simplicity, and let us, for smooth u, estimate $|u(1)|^2$ in terms of $|u(-1)|^2$ and Pu, $P := D_x$. Let $\chi = 1_{[-1,1]}$ denote the characteristic function of $[-1, 1]$. Then

$$\begin{aligned} -|u(1)|^2 + |u(-1)|^2 &= \int i[D_x, \chi]u \cdot \bar{u}\,\mathrm{d}x \\ &= i^{-1}\left(\int D_x u \cdot \chi\bar{u}\,\mathrm{d}x - \int \chi u \cdot \overline{D_x u}\,\mathrm{d}x\right) \\ &= 2\,\mathrm{Im}\langle D_x u, \chi u\rangle. \end{aligned} \tag{8.25}$$

(Here and for the remainder of the chapter, we write $\langle\cdot,\cdot\rangle$ for the L^2-pairing.) If $D_x u = 0$ (in which case, of course, $|u(1)|^2 = |u(-1)|^2$), we see from the first line that what *really* provides control of $|u(1)|^2$ is the fact that the cutoff χ has a negative (the "good" sign) derivative along $H_p = \partial_x$ ($p := \xi$) at 1. (Note that the principal symbol of $i[D_x, \chi]$ is $H_p\chi$.) Since we are proving a localized estimate, the function χ must be compactly supported, and hence, it must have a positive (the "bad" sign) derivative somewhere else, here at -1, which necessitates a priori control of u there.

If $D_x u \neq 0$ and if one wants an estimate of $\|u\|_{L^2([-1,1])}$, one needs more "negativity" of the commutator $i[D_x, \chi]$; one can, e.g., take $e^{-x}\chi$ instead of χ: starting with

$$2\,\mathrm{Im}\langle D_x u, e^{-x}\chi u\rangle = -\int_{-1}^{1} e^{-x}|u|^2\,\mathrm{d}x - e^{-1}|u(1)|^2 + e|u(-1)|^2, \tag{8.26a}$$

we estimate the left-hand side using Cauchy–Schwarz by

$$2\operatorname{Im}\langle D_x u, e^{-x}\chi u\rangle \geq -2\|\chi e^{-x/2}D_x u\|^2_{L^2([-1,1])} - \frac{1}{2}\|e^{-x/2}u\|^2_{L^2([-1,1])}. \tag{8.26b}$$

Combining (8.26a)–(8.26b) gives

$$|u(1)|^2 + \|u\|^2_{L^2([-1,1])} \leq C\big(|u(-1)|^2 + \|D_x u\|^2_{L^2([-1,1])}\big). \tag{8.27}$$

This is a typical positive commutator argument; the function $e^{-x}\chi$ is called the *commutant*.

Remark 8.13 (Negative and Positive). Strictly speaking, one should call this a *negative* commutator argument. It can be turned into a positive commutator argument by switching the sign of the commutant; this is, however, not how such arguments are typically phrased.

The proof of Theorem 8.10 will be based on similar considerations. We fix a volume density on M to define adjoints. The rough, formal sketch goes as follows. (We encourage the reader to assume at first reading that $P = P^*$.) We formally compute for a commutant $A = \operatorname{Op}(a) \in \Psi^{2s-m+1}$ (to be chosen), $A = A^*$:

$$2\operatorname{Im}\langle Pu, Au\rangle = i\big(\langle Au, Pu\rangle - \langle Pu, Au\rangle\big) = \Big\langle \Big(i[P,A] + 2\frac{P-P^*}{2i}A\Big)u, u\Big\rangle. \tag{8.28}$$

Let

$$p := \sigma^m(P), \quad p_1 := \sigma^{m-1}\Big(\frac{P-P^*}{2i}\Big). \tag{8.29}$$

Then

$$\sigma^{2s}\Big(i[P,A] + 2\frac{P-P^*}{2i}A\Big) = H_p a + 2p_1 a. \tag{8.30}$$

Suppose we can arrange

$$H_p a + 2p_1 a = -b^2 + e' \tag{8.31}$$

where $b \in S^s$ is elliptic on $\gamma([0,s_0])$ (the "conclusion region") and $e' \in S^{2s}$ has essential support contained in a neighborhood of $\gamma(0)$ (the "a priori control region"). Setting $B := \operatorname{Op}(b)$, $E' := \operatorname{Op}(e')$, we then have

$$i[P,A] + 2\frac{P-P^*}{2i}A = -B^*B + E' + R, \quad R \in \Psi^{2s-1}, \tag{8.32}$$

and hence (8.28) implies

$$\|Bu\|^2 = -2\operatorname{Im}\langle Pu, Au\rangle + \langle E'u, u\rangle + \langle Ru, u\rangle. \tag{8.33}$$

If $Pu = 0$, this gives a microlocal H^s-bound on u near $\gamma([0, s_0])$ in terms of a microlocal H^s-norm of u near $\gamma(0)$. Note that, using Lemma 6.42, $\langle E'u, u\rangle$ is finite when $\mathrm{WF}'(E') \cap \mathrm{WF}^s(u) = \emptyset$. (We caution the reader that the fact that $\langle E'u, u\rangle$ can be given a meaning does not imply that (8.33), which we only derived formally, is actually correct. If $u \in \mathcal{C}^\infty$, this is easily justified (Section 8.4). Otherwise, a regularization argument is needed.) The term $\langle Ru, u\rangle$ is of lower order; it is finite provided we already have obtained $H^{s-\frac{1}{2}}$-control of u microlocally near $\gamma([0, s_0])$. Thus, starting with $s = -N + \frac{1}{2}$, we can iteratively improve the control of u near $\gamma([0, s_0])$ by half a derivative until, after finitely many iterations, we reach the desired level of regularity.

If $Pu \neq 0$, we arrange for more negativity in (8.31) and estimate the first term in (8.33) using Cauchy–Schwarz, similarly to (8.26a)–(8.26b) above.

To make this into a rigorous positive commutator argument, we need to:

(1) Construct the commutant (this is the "interesting" part of the argument), i.e., construct a satisfying (8.31) (Section 8.3).
(2) Regularize the argument (this is the "technical" but straightforward part of the argument): we need to ensure that the integrations by parts in (8.28) and (8.33) as well as various Sobolev norms are well-defined (Section 8.5).

8.3 Positive Commutators II: Construction of the Commutant

We work near a null-bicharacteristic segment

$$\gamma\colon [0, s_0] \ni s \mapsto \gamma(s) \in \mathrm{Char}(P), \quad s_0 > 0. \tag{8.34}$$

(For now, the reader may think of γ as being a curve in $T^*M \setminus o$. We will soon pass to S^*M, however.) We will show that $\gamma(0) \notin \mathrm{WF}^s(u)$ and $\gamma([0, s_0]) \cap \mathrm{WF}^{s-m+1}(Pu) = \emptyset$ implies $\gamma([0, s_0]) \cap \mathrm{WF}^s(u) = \emptyset$, with estimates.

Lemma 8.14 (Homogeneity of the Hamiltonian Vector Field). *Let $p \in S^m_{\mathrm{hom}}(T^*M \setminus o)$. Then H_p is homogeneous of degree $m - 1$. That is, denoting fiber-dilations in T^*M by $M_\lambda\colon (x, \xi) \mapsto (x, \lambda\xi)$, $\lambda > 0$, we have*

$$M_\lambda^* H_p = \lambda^{m-1} H_p. \tag{8.35}$$

Proof. We work in local coordinates. We have $\partial_{x_j} p \in S^m_{\mathrm{hom}}$ and $\partial_{\xi_i} p \in S^{m-1}_{\mathrm{hom}}$ (Exercise 3.5). Let $f \in \mathcal{C}^\infty(T^*M)$ and $(x_0, \xi_0) \in T^*M \setminus o$. Then

$$\begin{aligned}(M_\lambda^* H_p)|_{(x_0,\xi_0)}(f) &= H_p|_{(x_0,\lambda\xi_0)}(f \circ M_\lambda^{-1}) \\ &= (\partial_\xi p)(x_0,\lambda\xi_0)\cdot\big(\partial_x(f\circ M_\lambda^{-1})\big)(x_0,\lambda\xi_0) \\ &\qquad - (\partial_x p)(x_0,\lambda\xi_0)\cdot\big(\partial_\xi(f\circ M_\lambda^{-1})\big)(x_0,\lambda\xi_0) \\ &= \lambda^{m-1}(\partial_\xi p)(x_0,\xi_0)\cdot(\partial_x f)(x_0,\xi_0) \\ &\qquad - \lambda^m(\partial_x p)(x_0,\xi_0)\cdot\lambda^{-1}(\partial_\xi f)(x_0,\xi_0) \\ &= \lambda^{m-1}H_p|_{(x_0,\xi_0)}f.\end{aligned}$$

This completes the proof. □

Fix an elliptic symbol

$$|\xi| \in S^1_{\mathrm{hom}}(T^*M \setminus o) \tag{8.36}$$

and define

$$\tilde{H}_p := |\xi|^{-m+1} H_p \in \mathcal{V}(T^*M \setminus o). \tag{8.37}$$

This is homogeneous of degree 0 and hence descends to a smooth vector field $\tilde{H}'_p \in \mathcal{V}(S^*M)$ on the cosphere bundle. (Indeed, for $f \in \mathcal{C}^\infty(S^*M)$, one defines $\tilde{H}'_p f \in \mathcal{C}^\infty(S^*M)$ by $q^*(\tilde{H}'_p f) := \tilde{H}_p(q^* f)$ where $q\colon T^*M \setminus o \to S^*M$ is the quotient map.) We immediately simplify notation and denote $\tilde{H}'_p$ by $\tilde{H}_p$ simply. An integral curve of $\tilde{H}_p$ is the image of an integral curve of H_p in S^*M, up to reparameterization.

Remark 8.15 (Passage to S^*M: Information Loss). The passage from $\tilde{H}_p \in \mathcal{V}(T^*M \setminus o)$ to $\tilde{H}_p \in \mathcal{V}(S^*M)$ does lose information, namely the fiber-radial component of $\tilde{H}_p$. For example, the vector field $\xi\cdot\partial_\xi \in \mathcal{V}(T^*\mathbb{R}^n)$, which is homogeneous of degree 0, descends to the trivial vector field $0 \in \mathcal{V}(S^*\mathbb{R}^n)$ on $S^*\mathbb{R}^n$. Keeping track of the radial component will be crucial in Chapter 10.

Working on S^*M, we take γ in (8.34) to be an integral curve of $\tilde{H}_p$; we may assume that $s_0 = 1$. Note that there is nothing to prove if γ is stationary. Otherwise, $\tilde{H}_p$ is nonzero along γ. By basic ODE theory, we can then straighten out $\tilde{H}_p$ locally: there exist local coordinates on S^*M near $\gamma([0,1])$,

$$(z_1, z'), \quad z_1 \in (-2,2),\ z' \in \mathbb{R}^{2n-2},\ |z'| < 1, \tag{8.38}$$

such that $\gamma(s) = (s,0)$ for $s \in [0,1]$ and $\tilde{H}_p = \partial_{z_1}$.

We now construct a commutant a; we need to ensure that a is supported in any given neighborhood of $\gamma([0,1])$ and that e' in (8.31) is supported in any given neighborhood of $\gamma(0)$. Suppose $\epsilon > 0$ is such that

$$\begin{aligned}\big([-2\epsilon, 1+2\epsilon]\times\{|z'|<2\epsilon\}\big)\cap \mathrm{WF}^{s-m+1}(Pu) &= \emptyset,\\ \big([-2\epsilon, 2\epsilon]\times\{|z'|<2\epsilon\}\big)\cap \mathrm{WF}^{s}(u) &= \emptyset.\end{aligned} \tag{8.39}$$

Fix a cutoff (in the transverse coordinates z')

$$\psi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n-2}),\quad \operatorname{supp}\psi\subset\{|z'|<2\epsilon\},\quad \psi(z')=1 \text{ for } |z'|<\epsilon, \tag{8.40}$$

and a "turn-on" function (in the z_1 coordinate)

$$\chi_1\in\mathcal{C}^\infty(\mathbb{R}),\quad \chi_1=0 \text{ on } (-\infty,-\epsilon],\quad \chi_1=1 \text{ on } [0,\infty). \tag{8.41}$$

The main term of the commutant arises from

$$\chi_0(x):=\begin{cases} e^{-F/x}, & x>0,\\ 0, & x\le 0,\end{cases} \tag{8.42}$$

where the constant $F>1$ will be chosen later. Note that

$$\chi_0'(x)=Fx^{-2}\chi_0(x), \tag{8.43}$$

so for x in any fixed compact subset $I\subset[0,\infty)$, and for any given $C>0$, we can choose F such that

$$\chi_0'(x)\ge C\chi_0(x),\quad x\in I. \tag{8.44}$$

We then set

$$\chi(z_1):=\chi_0(1+\epsilon-z_1),\quad \tilde a:=\chi(z_1)\chi_1(z_1)^2\psi(z')^2. \tag{8.45}$$

Thus, $\tilde a$ is supported in a 2ϵ-neighborhood of $\gamma([0,1])$. See Figure 8.3.

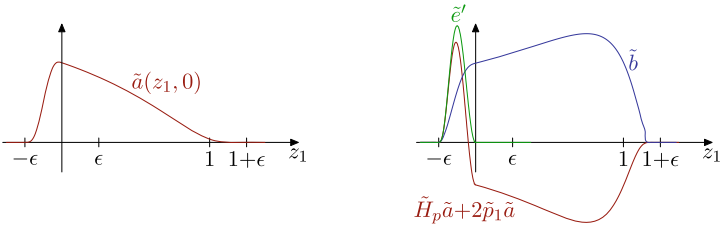


Figure 8.3 *On the left:* graph of the commutant $\tilde a(z_1,z')$ at $z'=0$. *On the right:* graphs of $\tilde H_p\tilde a+2\tilde p_1\tilde a$ and $\tilde b,\tilde e'$ in (8.46).

Recalling (8.29), set $\tilde{p}_1 := |\xi|^{-m+1}p_1$, which is homogeneous of degree 0; we regard it as an element of $\mathcal{C}^\infty(S^*M)$. We then compute

$$\begin{aligned}\tilde{H}_p\tilde{a} + 2\tilde{p}_1\tilde{a} &= 2\chi(z_1)\chi_1(z_1)\chi_1'(z_1)\psi(z')^2 \\ &\quad + \chi'(z_1)\chi_1(z_1)^2\psi(z')^2 + 2\tilde{p}_1\chi(z_1)\chi_1(z_1)^2\psi(z')^2 \\ &= -\tilde{b}^2 + \tilde{e}',\end{aligned} \tag{8.46}$$

where

$$\begin{aligned}\tilde{e}' &= 2\chi(z_1)\chi_1(z_1)\chi_1'(z_1)\psi(z')^2, \\ \tilde{b} &= \chi_1(z_1)\psi(z')\sqrt{-\chi'(z_1) - 2\tilde{p}_1\chi(z_1)} \\ &= \chi_1(z_1)\psi(z')\sqrt{F(1+\epsilon-z_1)^{-2}\chi(z_1) - 2\tilde{p}_1\chi(z_1)}.\end{aligned} \tag{8.47}$$

For sufficiently large F, we have $\tilde{b} \in \mathcal{C}^\infty(S^*M)$. Moreover, $\operatorname{supp}\tilde{e}' \subset \{|z_1|, |z'| < 2\epsilon\}$, and $\tilde{b}$ is positive on $[0,1]\times\{0\}$.

This almost arranges (8.31); we need to put the differential order back. Thus, we set

$$a := |\xi|^{2s-m+1}\tilde{a} \in S^{2s-m+1}_{\mathrm{hom}}(T^*M \setminus o) \tag{8.48}$$

and compute

$$H_p a + 2p_1 a = |\xi|^{2s}\big(\tilde{H}_p\tilde{a} + \tilde{p}_2\tilde{a}\big), \quad \tilde{p}_2 := 2\tilde{p}_1 + \big(|\xi|^{-2s+m-1}\tilde{H}_p|\xi|^{2s-m+1}\big). \tag{8.49}$$

Giving ourselves some extra room to deal with the possibility of $Pu \neq 0$, we have

$$H_p a + 2p_1 a = -|\xi|^{2m-2s-2}a^2 - b^2 + e' \tag{8.50}$$

for

$$\begin{aligned}e' &= |\xi|^{2s}\tilde{e}', \\ b &= |\xi|^s\chi_1(z_1)\psi(z')\sqrt{-\chi'(z_1) - \tilde{p}_2\chi(z_1) - \chi(z_1)^2\chi_1(z_1)^2\psi(z')^2} \\ &= |\xi|^s\chi_1(z_1)\psi(z') \\ &\qquad \times \sqrt{F(1+\epsilon-z_1)^{-2}\chi(z_1) - \tilde{p}_2\chi(z_1) - \chi(z_1)^2\chi_1(z_1)^2\psi(z')^2};\end{aligned} \tag{8.51}$$

we have $e' \in S^{2s}$ and $b \in S^s$ for sufficiently large $F > 1$.

8.4 Positive Commutators III: A Priori Estimate

Let us quantize the symbols (8.48), (8.51) (multiplied by a cutoff which equals 0 near $o \subset T^*M$ and 1 near fiber infinity) similarly to (8.32), giving $A \in \Psi^{2s-m+1}$, $B \in \Psi^s$, $E' \in \Psi^{2s}$; we can also arrange $\mathrm{WF}'(A) = \operatorname{ess\,supp} a$ etc. (Example 6.11) and that A, B, E have compactly supported Schwartz kernels. Assuming $u \in \mathcal{C}^\infty_{\mathrm{c}}(M)$, integrations by parts are never a concern, and we then have the following slight improvement over (8.33):

$$\|Bu\|^2 + \|\Lambda Au\|^2 = -2\,\mathrm{Im}\langle Pu, Au\rangle + \langle E'u, u\rangle + \langle Ru, u\rangle, \quad R \in \Psi^{2s-1}, \tag{8.52}$$

where $\Lambda \in \Psi^{m-s-1}$ is elliptic with principal symbol $|\xi|^{m-s-1}$; and $\mathrm{WF}'(R) \subset \mathrm{WF}'(A)$. Let $\Lambda^- \in \Psi^{-m+s+1}$ denote an elliptic parametrix of Λ, with $I = \Lambda^-\Lambda + \tilde{R}$, $\tilde{R} \in \Psi^{-\infty}$. Fix an operator $G \in \Psi^0$ with $\mathrm{WF}'(I-G) \cap \mathrm{WF}'(A) = \emptyset$; in particular, G is elliptic on $\mathrm{WF}'(A)$. We then have $\tilde{R}A$, $(I - G^*)A \in \Psi^{-\infty}$ and thus

$$\begin{aligned} 2|\mathrm{Im}\langle Pu, Au\rangle| &\le 2|\mathrm{Im}\langle GPu, (\Lambda^-\Lambda + \tilde{R})Au\rangle| + 2|\mathrm{Im}\langle Pu, (I-G^*)Au\rangle| \\ &\le \|(\Lambda^-)^* GPu\|^2 + \|\Lambda Au\|^2 + C\|u\|^2_{H^{-N}}. \end{aligned} \tag{8.53}$$

Let $E \in \Psi^0$ be elliptic on $\mathrm{WF}'(E')$; then

$$|\langle E'u, u\rangle| \le C\big(\|Eu\|^2_{H^s} + \|u\|^2_{H^{-N}}\big). \tag{8.54}$$

Moreover,

$$|\langle Ru, u\rangle| \le C\Big(\|Gu\|^2_{H^{s-\frac{1}{2}}} + \|u\|^2_{H^{-N}}\Big). \tag{8.55}$$

Plugging (8.53)–(8.55) into (8.52), we get the estimate

$$\|Bu\|_{L^2} \le C\Big(\|GPu\|_{H^{s-m+1}} + \|Eu\|_{H^s} + \|Gu\|_{H^{s-\frac{1}{2}}} + \|u\|_{H^{-N}}\Big). \tag{8.56}$$

(This positive commutator estimate is the analogue of (8.27): $\|u\|^2_{L^2([-1,1])}$, $|u(-1)|^2$, and $\|D_x u\|^2_{L^2([-1,1])}$ there correspond to $\|Bu\|_{L^2}$, $\|Eu\|_{H^s}$, and $\|GPu\|_{H^{s-m+1}}$ here, respectively.) If $s - \frac{1}{2} \le -N$, we simply estimate $\|Gu\|_{H^{s-\frac{1}{2}}} \le C\|u\|_{H^{-N}}$, obtaining the desired estimate (8.21). For $s > -N + \frac{1}{2}$, we can control $\|Gu\|_{H^{s-\frac{1}{2}}}$ inductively. Indeed, if $\mathrm{WF}'(G)$ lies in an $\epsilon 2^{-2|s|-2}$ neighborhood (defined using some fixed Riemannian metric on S^*M) of $\mathrm{WF}'(A)$, one can control $\|Gu\|_{H^{s-\frac{1}{2}}}$ by the right hand side of (8.56) with $s - \frac{1}{2}$ in place of s and E, G replaced by operators $\tilde{E}, \tilde{G}$ which are elliptic on $\mathrm{WF}'(E), \mathrm{WF}'(G)$ and have operator wave front sets contained in $\epsilon 2^{-2|s|-1}$-neighborhoods of $\mathrm{WF}'(E), \mathrm{WF}'(G)$. After finitely many iterations,

we thus obtain the desired estimate

$$\|Bu\|_{L^2} \leq C\big(\|\tilde{G}Pu\|_{H^{s-m+1}} + \|\tilde{E}u\|_{H^s} + \|u\|_{H^{-N}}\big), \tag{8.57}$$

where $\tilde{E}, \tilde{G} \in \Psi^0$, with $\mathrm{WF}'(\tilde{E})$ contained in a 5ϵ-neighborhood of $\gamma(0)$ and $\mathrm{WF}'(\tilde{G})$ contained in a 5ϵ-neighborhood of $\gamma([0,1])$. Starting out with ϵ replaced by $\frac{2}{5}\epsilon$, we have thus proved the desired a priori estimate (8.21) for $u \in \mathcal{C}^\infty_{\mathrm{c}}(M)$.

8.5 Positive Commutators IV: Regularization, Conclusion of the Proof

We now regularize the argument so that it applies to $u \in H^{-N}$. By an inductive argument as in Section 8.4, we may assume that $\mathrm{WF}^{s-\frac{1}{2}}(u)$ is disjoint from a 2ϵ-neighborhood of $\gamma([0,1])$. We still assume that $\mathrm{WF}^s(u)$ is disjoint from a 2ϵ-neighborhood of $\gamma(0)$.

The regularization argument replaces a, b, e' by symbols a_r, b_r, e'_r, $r \in (0,1]$, of (much) lower symbolic order which converge to a, b, e' as $r \to 0$ (or rather, to a, b, e' multiplied by a cutoff which cuts away the singularity at $\xi = 0$) in slightly weaker symbol classes. The regularized symbol a_r will be uniformly bounded in S^{2s-m+1}; we shall write this as

$$(a_r)_{r\in(0,1]} \in L^\infty((0,1]_r; S^{2s-m+1}), \quad \text{or for short } a_r \in L^\infty((0,1]_r; S^{2s-m+1}), \tag{8.58}$$

where for a locally convex vector space V we write

$$L^\infty((0,1]_r; V) := \big\{(v_r)_{r\in(0,1]} \colon \{v_r\} \subset V \text{ is a bounded subset}\big\}. \tag{8.59}$$

(Measurability is irrelevant, the notation notwithstanding.)

For $\Theta > 1$, define

$$\phi_r(\tau) = (1 + r\tau)^{-\Theta}, \quad r \in (0,1]. \tag{8.60}$$

See Figure 8.4. The symbol $\phi_r(|\xi|^2) \in S^{-2\Theta}(T^*M)$ is uniformly bounded in $S^0(T^*M)$ and converges to $\phi_0 \equiv 1$ in the topology of $S^\delta(T^*M)$ for all $\delta > 0$. Note moreover that

$$\tau\phi'_r(\tau) = f_r(\tau)\phi_r(\tau), \quad f_r(\tau) = -\Theta\frac{r\tau}{1+r\tau}, \tag{8.61}$$

so in particular $|f_r(\tau)| \leq \Theta$. Fix a function $\eta \in \mathcal{C}^\infty([0,\infty))$ that vanishes near 0 and equals 1 outside of $[0,1)$, with $\sqrt{\eta} \in \mathcal{C}^\infty$. We then define the *regularized commutant*

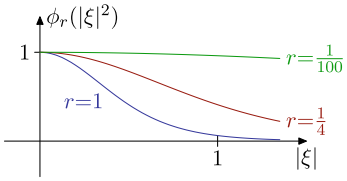


Figure 8.4 Graphs of $|\xi| \mapsto \phi_r(|\xi|^2)$ (here with $\Theta = 4$) for various values of r.

$$\begin{aligned} a_r &:= \phi_r(|\xi|^2)\eta(|\xi|) \cdot a \\ &= \phi_r(|\xi|^2)\eta(|\xi|) \cdot |\xi|^{2s-m+1}\chi(z_1)\chi_1(z_1)^2\psi(z')^2. \end{aligned} \tag{8.62}$$

Thus, $a_r \in L^\infty((0,1]_r; S^{2s-m+1}(T^*M))$, and $a_r \in S^{2s-m+1-2\Theta}(T^*M)$ for $r > 0$. In addition to analogues of the terms in (8.50), the computation of $H_p a_r$ produces two extra terms: for H_p falling on ϕ_r, we get a term involving

$$\tilde{H}_p\big(\phi_r(|\xi|^2)\big) = \tilde{f}_r\phi_r(|\xi|^2), \quad \tilde{f}_r := (|\xi|^{-2}\tilde{H}_p|\xi|^2)f_r(|\xi|^2). \tag{8.63}$$

We record $\eta(|\xi|)\tilde{f}_r \in L^\infty((0,1]_r; S^0(T^*M))$. When H_p falls on $\eta(|\xi|)$, we get a symbol with compact support in ξ, which is thus of order $-\infty$.

Using the notation of (8.46) and (8.49), we then compute

$$\begin{aligned} H_p a_r + 2p_1 a_r &= |\xi|^{2s}\phi_r(|\xi|^2)\eta(|\xi|)(\tilde{H}_p\tilde{a} + \tilde{p}_2\tilde{a}) \\ &\qquad + \tilde{f}_r\phi_r(|\xi|^2)\eta(|\xi|)|\xi|^{m-1}a + \phi_r(|\xi|^2)(H_p\eta(|\xi|))a \\ &= -|\xi|^{2m-2s-2}a_r^2 - b_r^2 + e_r' \end{aligned} \tag{8.64}$$

where

$$\begin{aligned} e_r' &= \phi_r(|\xi|^2)\big(\eta(|\xi|) \cdot 2|\xi|^{2s}\chi(z_1)\chi_1(z_1)\chi_1'(z_1)\psi(z')^2 + (H_p\eta(|\xi|))a\big), \\ b_r &= |\xi|^s\sqrt{\phi_r(|\xi|^2)\eta(|\xi|)}\chi_1(z_1)\psi(z') \\ &\quad \times \Big(F(1+\epsilon-z_1)^{-2}\chi(z_1) - \tilde{p}_2\chi(z_1) \\ &\qquad - \tilde{f}_r\chi(z_1) - \phi_r(|\xi|^2)\eta(|\xi|)\chi(z_1)^2\chi_1(z_1)^2\psi(z')^2\Big)^{\frac{1}{2}}. \end{aligned} \tag{8.65}$$

Since $\tilde{f}_r$ is uniformly bounded, the extra term $\tilde{f}_r\chi(z_1)$ here is harmless: choosing $F > 1$ sufficiently large makes the square root well-defined, and then

$$b_r \in L^\infty((0,1]_r; S^s(T^*M)), \quad e_r' \in L^\infty((0,1]_r; S^{2s}(T^*M)). \tag{8.66}$$

Moreover, by construction, $\operatorname{supp} a_r, \operatorname{supp} b_r$ and $\operatorname{supp} e_r'$ are contained in 2ϵ-neighborhoods of $\gamma([0,1])$ and $\gamma(0)$, respectively.

The quantization of (8.65) requires a bit of care since we need more precision than that afforded by a quantization which only respects principal symbols and essential supports. We thus fix a linear continuous quantization map

$$\mathrm{Op}\colon S^m(T^*M) \to \Psi^m(M) \tag{8.67}$$

using the construction in Section 5.6. Thus, Op is a quantization map in the sense that $\sigma^m(\mathrm{Op}(a)) = [a]$ for $a \in S^m(T^*M)$, and Op is surjective modulo $\Psi^{-\infty}(M)$. This definition also ensures that Op is continuous, and $\mathrm{WF}' \circ \mathrm{Op} = \operatorname{ess\,supp}$ (Example 6.11).

Let then

$$\begin{aligned} A_r &:= \frac{1}{2}(\mathrm{Op}(a_r) + \mathrm{Op}(a_r)^*) \in L^\infty((0,1]_r; \Psi^{2s-m+1}), \\ B_r &:= \mathrm{Op}(b_r) \in L^\infty((0,1]_r; \Psi^s), \\ E_r' &:= \mathrm{Op}(e_r') \in L^\infty((0,1]_r; \Psi^{2s}), \end{aligned} \tag{8.68}$$

where we again use the notation (8.59). (The definition of A_r ensures that A_r is formally self-adjoint.) The Schwartz kernels of A_r, B_r, E_r' are contained in $K' \times K'$ for some compact $K' \subset M$. Let $\Lambda := \mathrm{Op}(\langle\xi\rangle^{m-s-1})$. Then:

Lemma 8.16 (Regularized Commutator Calculation). *We have*

$$\begin{aligned} i[P, A_r] + 2\frac{P - P^*}{2i} A_r = -(\Lambda A_r)^*(\Lambda A_r) - B_r^* B_r + E_r' + R_r, \\ R_r \in L^\infty((0,1]_r; \Psi^{2s-1}). \end{aligned} \tag{8.69}$$

For $r > 0$, the orders of A_r, E_r', R_r and B_r are lower by 2Θ and Θ, respectively.

Proof. Let us call families $F_r \in L^\infty((0,1]_r; \Psi^{2s-1})$ with $F_r \in \Psi^{2s-1-2\Theta}$ for $r > 0$ *remainder terms*. We can replace A_r by $\mathrm{Op}(a_r)$ since $A_r - \mathrm{Op}(a_r) \in L^\infty((0,1]_r; \Psi^{2s-m})$ lies in $\Psi^{2s-m-2\Theta}$ for $r > 0$, so $i[P, A_r - \mathrm{Op}(a_r)]$ and $\frac{P-P^*}{2i}(A_r - \mathrm{Op}(a_r))$ are remainder terms.

Suppose that $M = \mathbb{R}^n$, and define $\mathrm{Op} := \mathrm{Op}_L$. Let $p_{\mathrm{full}} \in S^m$ be such that $P = \mathrm{Op}(p_{\mathrm{full}})$. Then Theorem 4.22 gives

$$P \circ \mathrm{Op}(a_r) = \mathrm{Op}(p_{\mathrm{full}} a_r + \partial_\xi p_{\mathrm{full}} \cdot D_x a_r) + R_r' \tag{8.70}$$

where R_r' is a remainder term. An analogous expression for $\mathrm{Op}(a_r) \circ P$ implies that $i[P, \mathrm{Op}(a_r)]$ equals $\mathrm{Op}(\{p_{\mathrm{full}}, a_r\})$ plus a remainder term, and thus $\mathrm{Op}(H_p a_r)$ plus a remainder term. Arguing similarly for $\frac{P-P^*}{2i}\mathrm{Op}(a_r)$ implies that $i[P, A_r] + 2\frac{P-P^*}{2i}A_r$ equals $\mathrm{Op}(H_p a_r + 2p_1 a_r)$ plus a remainder term. By (8.64), this implies (8.69) upon using analogous arguments for the right-hand side.

The validity of (8.69) in the manifold setting can be deduced from these local considerations; we leave the detailed arguments to the reader. □

Thus, if we take Θ large enough depending on s, N, we can safely compute

$$\begin{aligned} 2\,\mathrm{Im}\langle Pu, A_r u\rangle &= \Big\langle \Big(i[P, A_r] + 2\frac{P - P^*}{2i} A_r \Big) u, u \Big\rangle \\ &= -\|\Lambda A_r u\|^2 - \|B_r u\|^2 + \langle E_r' u, u\rangle + \langle R_r u, u\rangle. \end{aligned} \tag{8.71}$$

We need to show that the final two terms are uniformly bounded for $r \in (0, 1]$. To this end, we introduce:

Definition 8.17 (Uniform Operator Wave Front Set). Suppose $\mathcal{A} = \{A_r\} \in L^\infty((0,1]_r; \Psi^N(M))$ is a bounded family (for some $N \in \mathbb{R}$) of ps.d.o.s on M. Then $\alpha \in S^*M$ does *not* lie in the *uniform operator wave front set* $\mathrm{WF}'_{L^\infty}(\mathcal{A}) \subset S^*M$ if and only if there exists an operator $B \in \Psi^0(M)$, elliptic at α, such that $\{BA_r\}$ is bounded in $\Psi^{-\infty}(M)$.

(This generalizes WF′: if $A_r = A$ is r-independent, then $\mathrm{WF}'_{L^\infty}(\mathcal{A}) = \mathrm{WF}'(A)$.) We then have the following extension of microlocal elliptic regularity:

Lemma 8.18 (Uniform Microlocal Elliptic Regularity). *Let $\mathcal{A} = \{A_r\} \in L^\infty((0,1]_r; \Psi^m(M))$. Suppose $B \in \Psi^0(M)$ is such that $\mathrm{WF}'_{L^\infty}(\mathcal{A}) \subset \mathrm{Ell}(B)$. Assume that the Schwartz kernels of A_r, B are supported in $K \times K$ for some compact subset $K \subset M$. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ be equal to 1 near K. Let $s, N \in \mathbb{R}$. Then there exists a constant C (independent of r) such that*

$$\|A_r u\|_{H^{s-m}} \le C\big(\|Bu\|_{H^s} + \|\chi u\|_{H^{-N}}\big). \tag{8.72}$$

Proof. Writing $I = QB + R$ with $Q, R \in \Psi^0(M)$, $\mathrm{WF}'(R) \cap \mathrm{WF}'_{L^\infty}(\mathcal{A}) = \emptyset$, we have

$$A_r u = A_r Q B u + A_r R u. \tag{8.73}$$

Since $A_r Q \in L^\infty((0,1]_r; \Psi^m(M))$ and $A_r R \in L^\infty((0,1]_r; \Psi^{-\infty}(M))$ are uniformly bounded, the estimate (8.72) follows from Corollary 4.38. □

By construction, we have $\mathrm{WF}'_{L^\infty}(\{A_r\}) = \operatorname{ess\,supp} a$ etc. Pick $G \in \Psi^0$, elliptic near $\mathrm{WF}'_{L^\infty}(\{A_r\})$ and with $\mathrm{WF}'(G)$ contained in a 2ϵ-neighborhood of $\gamma([0,1])$, and $E \in \Psi^0(M)$ elliptic near $\mathrm{WF}'_{L^\infty}(\{E_r'\})$ and with $\mathrm{WF}'(E)$ contained in a 2ϵ-neighborhood of $\gamma(0)$; we pick G, E such that their Schwartz kernels are supported in $K \times K$. We then conclude that

$$\begin{aligned} |\langle E_r' u, u\rangle| &\le C\big(\|Eu\|^2_{H^s} + \|u\|^2_{H^{-N}}\big), \\ |\langle R_r u, u\rangle| &\le C\Big(\|Gu\|^2_{H^{s-\frac{1}{2}}} + \|u\|^2_{H^{-N}}\Big). \end{aligned} \tag{8.74}$$

Plugging this into (8.71) and arguing as in (8.53)–(8.56), we thus obtain a *uniform* estimate

$$\|B_r u\|_{L^2} \leq C\Big(\|Eu\|_{H^s} + \|GPu\|_{H^{s-m+1}} + \|Gu\|_{H^{s-\frac{1}{2}}} + \|u\|_{H^{-N}}\Big), \quad r \in (0,1]. \tag{8.75}$$

Since the unit ball in the Hilbert space $L^2(K)$ is weakly compact, $B_r u$ has a weakly convergent subsequence with limit $v \in L^2$. On the other hand, $B_r u \to B_0 u$ in $\mathscr{D}'(M)$ by the generalization of Lemma 4.39 to manifolds; hence $B_0 u = v \in L^2$. Therefore, $\mathrm{WF}^s(u) \cap \mathrm{Ell}(B_0) = \emptyset$. This proves the microlocal H^s-regularity of u near $\gamma([0,1])$ and at the same time gives an estimate for $\|B_0 u\|_{L^2} \leq \liminf_{r\to 0} \|B_r u\|_{L^2}$ by the right hand side of (8.75).

The proof of Theorem 8.10 is complete.

8.6 Semiglobal Estimate

Theorem 8.10, as stated, concerns the propagation of microlocal regularity only along a single null-bicharacteristic segment. It is convenient to have a version of this result that applies more generally under a geometric control condition:

Corollary 8.19 (Semiglobal Propagation Estimate). *Suppose $P \in \Psi^m(M)$ (properly supported) has a real-valued homogeneous principal symbol p. Suppose $B, G, E \in \Psi^0(M)$ satisfy the following conditions:*

(1) *There exists a compact set $K \subset M$ such that the Schwartz kernels of B, G, E are supported in $K \times K$.*
(2) $\mathrm{WF}'(B) \subset \mathrm{Ell}(G)$.
(3) *All backward null-bicharacteristics of P starting from a point in $\mathrm{WF}'(B) \cap \mathrm{Char}(P)$ enter $\mathrm{Ell}(E)$ in finite time while remaining in $\mathrm{Ell}(G)$.*

Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ with $\chi = 1$ near a compact set $\tilde{K} \subset M$ with the property that $\operatorname{supp} K_P \cap (K \times M) \subset K \times \tilde{K}$ where K_P is the Schwartz kernel of P. Let $s, N \in \mathbb{R}$. Then there exists a constant C such that for all $u \in \mathscr{D}'(M)$, the estimate

$$\|Bu\|_{H^s} \leq C(\|Eu\|_{H^s} + \|GPu\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}) \tag{8.76}$$

holds in the strong sense that if $u \in \mathscr{D}'(M)$ is such that the right-hand side is finite, then so is the left-hand side, and the estimate holds.

See Figure 8.5.

Proof of Corollary 8.19. Apply Theorem 8.10 to a finite set of null-bicharacteristics such that the elliptic sets of the operators B in Theorem 8.10 cover an open neighborhood of $\mathrm{Char}(P) \cap \mathrm{WF}'(B)$ for the present B.

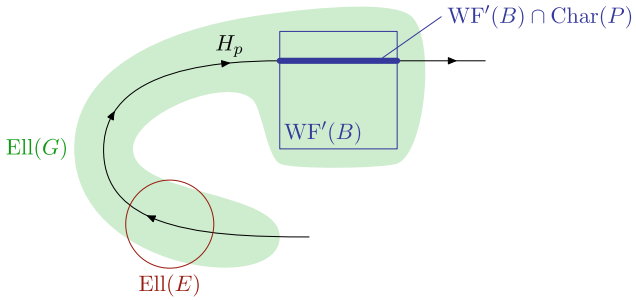


Figure 8.5 Illustration of Corollary 8.19: Bu can be bounded in H^s by Eu in H^s and GPu in H^{s-m+1}.

Microlocal elliptic estimates (Proposition 6.39) provide control of u on the complement of a neighborhood of $\mathrm{Char}(P) \cap \mathrm{WF}'(B)$. The detailed argument is as follows.

For every $\alpha \in \mathrm{WF}'(B) \cap \mathrm{Char}(P)$, let $\gamma_\alpha \colon [0, s_0(\alpha)] \to \mathrm{Char}(P)$ be a null-bicharacteristic segment of P with $\gamma_\alpha(0) \in \mathrm{Ell}(E)$, $\gamma_\alpha(s_0(\alpha)) = \alpha$, and $\gamma_\alpha([0, s_0(\alpha)]) \subset \mathrm{Ell}(G)$. Apply Theorem 8.10 to γ_α, with $\mathcal{U}_0$ and $\mathcal{U}$ there given by $\mathrm{Ell}(E)$ and $\mathrm{Ell}(G)$ in present notation; this produces $B_\alpha, G_\alpha, E_\alpha \in \Psi^0(M)$, with Schwartz kernels supported in $K \times K$, such that

$$\begin{aligned} \gamma_\alpha([0, s_0(\alpha)]) &\subset \mathrm{Ell}(B_\alpha), & \mathrm{WF}'(B_\alpha) &\subset \mathrm{Ell}(G_\alpha), \\ \mathrm{WF}'(G_\alpha) &\subset \mathrm{Ell}(G), & \mathrm{WF}'(E_\alpha) &\subset \mathrm{Ell}(E), \end{aligned} \tag{8.77}$$

and such that the estimate

$$\|B_\alpha u\|_{H^s} \le C_\alpha(\|E_\alpha u\|_{H^s} + \|G_\alpha P u\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}) \tag{8.78}$$

holds (in the strong sense). Using microlocal elliptic estimates (Proposition 6.39), we can estimate the terms involving $E_\alpha u$ and $G_\alpha(Pu)$ by Eu and $G(Pu)$, respectively, so

$$\|B_\alpha u\|_{H^s} \le C_\alpha(\|Eu\|_{H^s} + \|GPu\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}). \tag{8.79}$$

Since $\mathrm{WF}'(B) \cap \mathrm{Char}(P) \subset S^*M$ is compact and contained in the union of the open sets $\mathrm{Ell}(B_\alpha)$, $\alpha \in \mathrm{WF}'(B) \cap \mathrm{Char}(P)$, there exists a finite subset $\{\alpha_1, \dots, \alpha_J\} \subset \mathrm{WF}'(B) \cap \mathrm{Char}(P)$ such that $\bigcup_{j=1}^J \mathrm{Ell}(B_{\alpha_j}) \supset \mathrm{WF}'(B) \cap \mathrm{Char}(P)$. The operator $B_0 := \sum_{j=1}^J B_{\alpha_j}^* B_{\alpha_j} \in \Psi^0(M)$ is thus elliptic on $\mathrm{WF}'(B) \cap \mathrm{Char}(P)$, and we have

$$\|B_0 u\|_{H^s} \le C(\|Eu\|_{H^s} + \|GPu\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}). \tag{8.80}$$

It remains to estimate u on the elliptic set of P. The compact set $\mathrm{WF}'(B)\setminus \mathrm{Ell}(B_0)$ is disjoint from $\mathrm{Char}(P)$. Fix $B_1 \in \Psi^0(M)$, with Schwartz kernel supported in $K \times K$, such that

$$\mathrm{Ell}(B_1) \supset \mathrm{WF}'(B)\setminus\mathrm{Ell}(B_0), \quad \mathrm{WF}'(B_1)\cap\mathrm{Char}(P) = \emptyset, \quad \mathrm{WF}'(B_1) \subset \mathrm{Ell}(G). \tag{8.81}$$

By microlocal elliptic regularity,

$$\|B_1 u\|_{H^s} \leq C(\|GPu\|_{H^{s-m}} + \|\chi u\|_{H^{-N}}). \tag{8.82}$$

We can combine (8.80) and (8.82) to get for $B_2 := B_0^* B_0 + B_1^* B_1$ the estimate

$$\|B_2 u\|_{H^s} \leq C(\|Eu\|_{H^s} + \|GPu\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}). \tag{8.83}$$

Since $\mathrm{WF}'(B) \subset \mathrm{Ell}(B_2)$, a final application of microlocal elliptic regularity (Proposition 6.39) yields (8.76). □

8.7 Semiclassical Propagation Estimates

We now study the propagation of singularities of h-tempered solutions $u = (u_h)_{h\in(0,h_0)}$ (Definition 6.55) of semiclassical equations

$$Pu = f, \quad P = (P_h)_{h\in(0,h_0)} \in \Psi_h^m(M), \tag{8.84}$$

on smooth manifolds M. Here, we mean "singularities" in the sense of the semiclassical wave front set $\mathrm{WF}_h^{s,0}(u) \subset \overline{T^*}M$; and the propagation will happen along integral curves of a rescaling of H_p, $p := \sigma_h^m(P)$, inside of $\overline{T^*}M$.

Before stating the main results (Theorems 8.25 and 8.27), we illustrate the semiclassical setting with a few examples.

Example 8.20 (hD_x). The operator $P = hD_x \in \mathrm{Diff}_h^1(\mathbb{R})$ has semiclassical principal symbol $p(x,\xi) = \xi$, so $\mathrm{Char}_h(P) = \mathbb{R}\times\{0\} \subset \mathbb{R}_x \times \mathbb{R}_\xi$ and $H_p = \partial_x$. The equation $Pu = f$ can be solved explicitly via

$$u(x) = u(x_0) + \frac{i}{h}\int_{x_0}^{x} f(t)\,\mathrm{d}t. \tag{8.85}$$

This allows us to propagate semiclassical regularity from a neighborhood of $x = 0$ to a neighborhood of $x = 4$, say, via

$$\begin{aligned}\int_3^5 |u(x)|^2\,\mathrm{d}x &= \int_3^5 \Big|u(x-4)+\frac{i}{h}\int_{x-4}^x f(t)\,\mathrm{d}t\Big|^2\,\mathrm{d}x\\ &\le 2\Big(\int_{-1}^1 |u(x)|^2\,\mathrm{d}x+\frac{1}{h^2}\int_{-1}^5 2|f(t)|^2\,\mathrm{d}t\Big);\end{aligned}\tag{8.86}$$

therefore,

$$\|u\|_{L^2([3,5])}\le C\big(\|u\|_{L^2([-1,1])}+h^{-1}\|Pu\|_{L^2([-1,5])}\big).\tag{8.87}$$

One can write down a microlocalized version of this estimate near $\mathrm{Char}(P)$; then the semiclassical Sobolev regularity (here 0) is irrelevant since $\mathrm{Char}(P)\cap S^*M=\emptyset$. Note that the estimate (8.87) "loses" one semiclassical order compared to an elliptic estimate (Proposition 6.61).

Example 8.21 (hD_{x_1}). Let $n\ge 2$; then $P=hD_{x_1}\in\mathrm{Diff}_h^n(\mathbb{R}^n)$, $p=\xi_1$, and $H_p=\partial_{x_1}$. Integrating (8.86) in the remaining variables $x_2,\dots,x_n$ gives for any (say, compact) $K\subset\mathbb{R}^{n-1}$ the estimate

$$\|u\|_{H_h^s([3,5]\times K)}\le C\big(\|u\|_{H_h^s([-1,1]\times K)}+h^{-1}\|Pu\|_{H_h^s([-1,5]\times K)}\big)\tag{8.88}$$

for $s=0$, or by direct differentiation of $Pu=f$ along hD_{x_j}, $j=1,\dots,n$, for all integer $s\in\mathbb{N}_0$. The important difference to Example 8.20 is that the characteristic set $\overline{\{\xi_1=0\}}\subset\overline{T}^*\mathbb{R}^n$ of P now intersects fiber infinity $S^*\mathbb{R}^n$. In particular, the differential orders in (8.88) matter. In addition to the h^{-1} "loss" already noted, we thus also have the same "loss" in the regularity order as in the classical setting of Theorem 8.10.

Example 8.22 (hD_x-a). For the operator $P=hD_x-a(x)\in\mathrm{Diff}_h^1(\mathbb{R})$ where $a\in\mathcal{C}_b^\infty(\mathbb{R})$ is real-valued, the characteristic set is given by $\mathrm{Char}_h(P)=\{(x,\xi)\colon x\in\mathbb{R},\ \xi=a(x)\}$. We expect the semiclassical wave front set of solutions u of $Pu=0$ to propagate along $H_p=\partial_x+a'(x)\partial_\xi$ (where $p(x,\xi)=\xi-a(x)$) inside of $\mathrm{Char}_h(P)$; thus, we expect $\mathrm{WF}_h(u)=\mathrm{Char}_h(P)$ or $\mathrm{WF}_h(u)=\emptyset$. See Figure 8.6. This is easily checked explicitly: $Pu=0$ implies

$$u(x)=e^{iA(x)/h}u(0),\quad A(x):=\int_0^x a(t)\,\mathrm{d}t.\tag{8.89}$$

Thus, $\mathrm{WF}_h(u)=\mathrm{Char}_h(P)$ by Exercise 6.24, unless $u(0)=\mathcal{O}(h^\infty)$ in which case $\mathrm{WF}_h(u)=\emptyset$.

Example 8.23 (Laplacians). The operator $P=h^2\Delta\in\mathrm{Diff}_h^2(\mathbb{R}^n)$ has semiclassical principal symbol $p(x,\xi)=|\xi|^2$, so $\mathrm{Char}_h(P)=\mathbb{R}^n\times\{0\}$ is the zero section of $T^*\mathbb{R}^n$ and $H_p=2\xi\cdot\partial_x=0$ at $\mathrm{Char}_h(P)$. There is thus no real principal type propagation of singularities. On the other

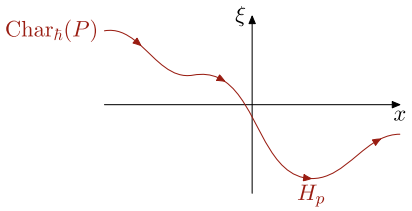


Figure 8.6 Illustration of Example 8.22. The characteristic set of $P = hD_x - a$ and the flow of the Hamiltonian vector field H_p of its principal symbol $p(x,\xi) = \xi - a(x)$.

hand, for $h^2\Delta - 1$, or more generally $P = h^2\Delta_g - 1 \in \mathrm{Diff}^2_\hbar(M)$ on a Riemannian manifold (M,g), the characteristic set is the unit cosphere bundle: $\mathrm{Char}_\hbar(P) = \{(x,\xi) \in T^*M \colon |\xi|^2_{g_x^{-1}} = 1\}$. One can show that integral curves of $\frac{1}{2}H_p$, $p(x,\xi) := \sigma^2_\hbar(P)(x,\xi) = g_x^{-1}(\xi,\xi)$, inside of $\mathrm{Char}_\hbar(P)$ are lifts of unit speed geodesics on (M,g) to phase space. (The reader can look ahead to Proposition 9.9 for the relevant computation.) On $\mathbb{R}^n$, we have $(h^2\Delta - 1)u_h = 0$ for $u_h(x) := e^{ix\cdot\xi_0/h}$, $|\xi_0| = 1$; and indeed $\mathrm{WF}_\hbar(u) = \{(x,\xi) \colon \xi = \xi_0\}$ is invariant under the flow of $H_p = 2\xi\cdot\partial_x$.

For the purpose of shifting differential orders below, fix a nowhere vanishing elliptic symbol

$$\rho \in S^{-1}_{\mathrm{cl}}(T^*M). \tag{8.90}$$

For example, one can take $\rho = (1+|\xi|^2_{g^{-1}})^{-\frac{1}{2}}$ where g is a Riemannian metric on M.

Lemma 8.24 (Rescaled Hamiltonian Vector Field). *Let $p \in S^m_{\mathrm{cl}}(T^*M)$. Then $\rho^{m-1}H_p \in \mathcal{V}(\overline{T^*}M)$, and moreover, this vector field is tangent to $S^*M = \partial\overline{T^*}M$.*

Proof. We work in local coordinates $x \in \mathbb{R}^n$ on M and in projective coordinates on the fibers of $\overline{T^*}M$ similarly to (6.92). In the region where $\xi_1 > \epsilon\max(|\xi_2|,\dots,|\xi_n|)$, $\epsilon > 0$, we may thus assume without loss of generality that $\rho = \frac{1}{\xi_1}$, and we introduce the coordinate system

$$x \in \mathbb{R}^n, \quad \rho = \frac{1}{\xi_1}, \quad \hat\xi_j := \frac{\xi_j}{\xi_1},\ j = 2,\dots,n. \tag{8.91}$$

We furthermore write $\hat\xi' = (\hat\xi_2,\dots,\hat\xi_n)$ and

$$p = \rho^{-m}p_0(x,\rho,\hat\xi'), \quad p_0 \in \mathcal{C}^\infty\big(\mathbb{R}^n_x \times [0,1)_\rho \times \mathbb{R}^{n-1}_{\hat\xi'}\big). \tag{8.92}$$

Writing $x' = (x_2,\dots,x_n)$ and $\xi' = (\xi_2,\dots,\xi_n)$, we then compute

$$\begin{aligned} H_p &= (\partial_{\xi_1} p)\partial_{x_1} + (\partial_{\xi'} p)\cdot\partial_{x'} - (\partial_{x_1} p)\partial_{\xi_1} - (\partial_{x'} p)\cdot\partial_{\xi'} \\ &= \rho\Big[-\big((\rho\partial_\rho + \hat\xi'\partial_{\hat\xi'})p\big)\partial_{x_1} + (\partial_{\hat\xi'} p)\cdot\partial_{x'} \\ &\qquad + (\partial_{x_1} p)(\rho\partial_\rho + \hat\xi'\partial_{\hat\xi'}) - (\partial_{x'} p)\cdot\partial_{\hat\xi'}\Big]. \end{aligned} \tag{8.93}$$

This implies

$$\begin{aligned} \rho^{m-1} H_p = \big(&-(\rho\partial_\rho - m + \hat\xi'\partial_{\hat\xi'})p_0\big)\partial_{x_1} + (\partial_{\hat\xi'} p_0)\cdot\partial_{x'} \\ &+ (\partial_{x_1} p_0)(\rho\partial_\rho + \hat\xi'\partial_{\hat\xi'}) - (\partial_{x'} p_0)\cdot\partial_{\hat\xi'}\big). \end{aligned} \tag{8.94}$$

From this expression, one can read off the claim. □

The qualitative version of the semiclassical real principal type propagation result is:

Theorem 8.25 (Propagation of Semiclassical Wave Front Sets). *Suppose $P = (P_h)_{h\in(0,h_0)} \in \Psi_\hbar^m(M)$ (properly supported) has a real-valued semiclassical principal symbol $p \in S^m_{\mathrm{cl}}(T^*M)$. (That is, p is a representative of $\sigma_\hbar^m(P) \in S_\hbar^m/hS_\hbar^{m-1}$.) Let $u = (u_h)_{h\in(0,h_0)}$ be an h-tempered family of distributions $u_h \in \mathscr{D}'(M)$, and set $f := Pu$, i.e., $f = (f_h)_{h\in(0,h_0)}$, $f_h := P_h u_h$. Let $s \in \mathbb{R}$. Then*

$$\mathrm{WF}_\hbar^{s,0}(u) \subset \mathrm{Char}_\hbar(P) \cup \mathrm{WF}_\hbar^{s-m,0}(f). \tag{8.95}$$

Moreover, within $\mathrm{Char}_\hbar(P)\backslash\mathrm{WF}_\hbar^{s-m+1,1}(f) = \mathrm{Char}_\hbar(P)\backslash\mathrm{WF}_\hbar^{s-m+1,0}(h^{-1}f)$,

$$\mathrm{WF}_\hbar^{s,0}(u) \setminus \mathrm{WF}_\hbar^{s-m+1,1}(f) \tag{8.96}$$

is a union of maximally extended integral curves of $\rho^{m-1}H_p$.

Remark 8.26 (h-Independence of the Semiclassical Principal Symbol). The assumption on the principal symbol of P in particular implies that the characteristic set $\mathrm{Char}(P_h)$ of $P_h \in \Psi^m(M)$, $h \in (0, h_0)$, is independent of h, and so is the null-bicharacteristic flow (suitably rescaled). This is the typical situation in applications and indeed in all applications considered in this book. It is in any case a simple matter to extend the results in this section to *classical* semiclassical operators $P \in \Psi^m_{\hbar,\mathrm{cl}}(M)$ whose principal symbols admit a real-valued representative of class $\mathcal{C}^\infty([0,h_0); S^m_{\mathrm{cl}}(T^*M))$; we leave this to the interested reader.

We phrase the quantitative version of this result immediately in a semiglobal version as in Corollary 8.19.

Theorem 8.27 (Propagation of Semiclassical Singularities: Semiglobal Estimate Version). *Suppose $P \in \Psi_\hbar^m(M)$ (properly supported) has a real-valued semiclassical principal symbol $p \in S^m_{\mathrm{cl}}(T^*M)$. Suppose $B, G, E \in \Psi_\hbar^0(M)$ satisfy the following conditions:*

(1) *There exists a compact set $K \subset M$ such that the Schwartz kernels of B_h, G_h, E_h are supported in $K \times K$.*
(2) $\mathrm{WF}'_\hbar(B) \subset \mathrm{Ell}_\hbar(G)$.
(3) *All integral curves of $-\rho^{m-1}H_p$ starting from a point in $\mathrm{WF}'_\hbar(B) \cap \mathrm{Char}_\hbar(P)$ enter $\mathrm{Ell}_\hbar(E)$ in finite time while remaining in $\mathrm{Ell}_\hbar(G)$.*

Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ with $\chi = 1$ near a compact set $\tilde{K} \subset M$ with the property that $\operatorname{supp} K_{P_h} \cap (K \times M) \subset K \times \tilde{K}$ for all $h < h_0$ where K_{P_h} is the Schwartz kernel of P_h. Let $s, N \in \mathbb{R}$. Then there exist $h_1 > 0$ and a constant C such that for all $u \in \mathscr{D}'(M)$, the following estimate holds (in the strong sense):

$$\|Bu\|_{H^s_\hbar} \leq C\Big(\|Eu\|_{H^s_\hbar} + h^{-1}\|GPu\|_{H^{s-m+1}_\hbar} + h^N\|\chi u\|_{H^{-N}_\hbar}\Big), \quad h < h_1. \tag{8.97}$$

For the proof, it suffices to show an analogue of Theorem 8.10 (and then follow the proof of Corollary 8.19): *given a null-bicharacteristic segment $\gamma\colon [0, s_0] \to \mathrm{Char}(P)$ (by which we mean an integral curve of $\rho^{m-1}H_p$) and neighborhoods $\mathcal{U}_0, \mathcal{U} \subset \overline{T^*}M$ of $\gamma(0)$ and $\gamma([0, s_0])$, respectively, there exist $B, G, E \in \Psi^0_\hbar$ with*

$$\gamma([0, s_0]) \subset \mathrm{Ell}_\hbar(B), \quad \mathrm{WF}'_\hbar(B) \subset \mathrm{Ell}_\hbar(G), \quad \mathrm{WF}'_\hbar(G) \subset \mathcal{U}, \quad \mathrm{WF}'_\hbar(E) \subset \mathcal{U}_0, \tag{8.98}$$

such that the estimate (8.97) holds. We may assume that $s_0 = 1$. Furthermore, we may again reduce to the case that u and K_{P_h} are supported in fixed compact subsets of M and $M \times M$, respectively.

To prove the propagation of regularity along the null-bicharacteristic segment γ, we proceed as in Sections 8.3–8.5. We first straighten out the rescaled Hamiltonian vector field

$$\tilde{H}_p := \rho^{m-1}H_p; \tag{8.99}$$

we distinguish two cases:

(T^*M) $\gamma(0) \in T^*M$ and thus $\gamma([0, s_0]) \subset T^*M$.
(S^*M) $\gamma(0) \in S^*M = \partial\overline{T^*}M$ and thus $\gamma([0, s_0]) \subset S^*M$.

Namely, we choose local coordinates as follows:

$$\text{case } (T^*M): \qquad z_1 \in (-2, 2),\ z' \in \mathbb{R}^{2n-1},\ |z'| < 1,\ \gamma(s) = (s, 0), \tag{8.100a}$$

$$\text{case } (S^*M): \ \rho \geq 0,\ z_1 \in (-2, 2),\ z' \in \mathbb{R}^{2n-2},\ |z'| < 1,\ \gamma(s) = (0, s, 0), \tag{8.100b}$$

with $\tilde{H}_p = \partial_{z_1}$ in both cases. The existence of the coordinates (8.100b) relies the tangency of $\tilde{H}_p$ to $\partial\overline{T^*}M$. In the first case, all coordinates other than z_1 can be treated in the same fashion; in the second case, the defining function ρ of fiber infinity needs to be treated separately, as we need to employ weights in ρ for the purpose of shifting the differential order. Fix $\epsilon > 0$ such that

- in the case (T^*M):

$$\begin{aligned} \big([-2\epsilon, 1+2\epsilon] \times \{|z'| < 2\epsilon\}\big) \cap \mathrm{WF}_\hbar^{s-m+1,1}(Pu) &= \emptyset, \\ \big([-2\epsilon, 2\epsilon] \times \{|z'| < 2\epsilon\}\big) \cap \mathrm{WF}_\hbar^{s,0}(u) &= \emptyset, \end{aligned} \tag{8.101a}$$

- in the case (S^*M):

$$\begin{aligned} \big([0, 2\epsilon] \times [-2\epsilon, 1+2\epsilon] \times \{|z'| < 2\epsilon\}\big) \cap \mathrm{WF}_\hbar^{s-m+1,1}(Pu) &= \emptyset, \\ \big([0, 2\epsilon] \times [-2\epsilon, 2\epsilon] \times \{|z'| < 2\epsilon\}\big) \cap \mathrm{WF}_\hbar^{s,0}(u) &= \emptyset. \end{aligned} \tag{8.101b}$$

We define a commutant much as in (8.45), with regularization similarly to (8.62): for $F, \Theta > 1$ to be determined, set

$$\phi_r(\rho) = (1 + r\rho^{-1})^{-\Theta}, \quad r \in (0, 1], \tag{8.102}$$

and fix

$$\begin{aligned} &\chi_0(x) := H(x)e^{-F/x}, && \chi(z_1) := \chi_0(1 + \epsilon - z_1); \\ &\psi \in \mathcal{C}_{\mathrm{c}}^\infty(\{|z'| < 2\epsilon\}), && \psi = 1 \text{ for } |z'| < \epsilon; \\ &\chi_1 \in \mathcal{C}^\infty(\mathbb{R}), && \chi_1 = 0 \text{ on } (-\infty, -\epsilon], \quad \chi_1 = 1 \text{ on } [0, \infty); \\ &\chi_2 \in \mathcal{C}_{\mathrm{c}}^\infty([0, 2\epsilon)), && \chi_2 = 1 \text{ on } [0, \epsilon]. \end{aligned} \tag{8.103}$$

We then set:

$$\text{case } (T^*M) : a := \chi(z_1)\chi_1(z_1)^2\psi(z')^2, \tag{8.104a}$$

$$\text{case } (S^*M) : a_r := \phi_r(\rho)^2\rho^{-2s+m-1}\chi_2(\rho)^2\chi(z_1)\chi_1(z_1)^2\psi(z')^2, \quad r \in (0, 1]. \tag{8.104b}$$

Note the absence of weights controlling the differential order in (8.104a); indeed $a \in S_\hbar^{-\infty}$ then. In (8.104b), note the presence of a cutoff $\chi_2(\rho)$ which localizes to a neighborhood of fiber infinity $\rho = 0$. Moreover, we immediately implement the regularization in the differential order in the case (S^*M); in the case (T^*M), we do not need any regularization. Finally, due to our choice of coordinates in (8.100b), the coordinate ρ is neutral for $\tilde{H}_p$, i.e., $\tilde{H}_p\rho = 0$. (We had not arranged anything similar in (8.48).)

We capture the skew-adjoint part of P via

$$p_1 := \sigma_\hbar^{m-1}\Big(\frac{P-P^*}{2ih}\Big), \quad \tilde{p}_1 := \rho^{m-1}p_1. \tag{8.105}$$

Analogously to (8.50) and (8.64), the main calculations for the positive commutator argument are now as follows.

(1) Case (T^*M): we may redefine ρ so that $\rho = 1$ on $\operatorname{supp} a$. Then $H_p a + 2p_1 a = -a^2 - b^2 + e'$, where

$$\begin{aligned} e' &= 2\chi(z_1)\chi_1(z_1)\chi_1'(z_1)\psi(z')^2, \\ b &= \chi_1(z_1)\psi(z') \\ &\quad \times \sqrt{F(1+\epsilon-z_1)^{-2}\chi(z_1) - 2p_1\chi(z_1) - \chi(z_1)^2\chi_1(z_1)^2\psi(z')^2}\,. \end{aligned} \tag{8.106}$$

Thus, b is smooth if we take F in (8.103) large enough.

(2) Case (S^*M): $H_p a_r + 2p_1 a_r = -\rho^{2s-2m+2}a_r^2 - b_r^2 + e_r'$, where

$$\begin{aligned} e_r' &= 2\rho^{-2s}\phi_r(\rho)^2\chi_2(\rho)^2\chi(z_1)\chi_1(z_1)\chi_1'(z_1)\psi(z')^2, \\ b_r &= \rho^{-s}\phi_r(\rho)\chi_2(\rho)\chi_1(z_1)\psi(z') \\ &\quad \times \Big(F(1+\epsilon-z_1)^{-2}\chi(z_1) - 2\tilde{p}_1\chi(z_1) \\ &\qquad - \phi_r(\rho)^2\chi_2(\rho)^2\chi(z_1)^2\chi_1(z_1)^2\psi(z')^2\Big)^{\frac{1}{2}}. \end{aligned} \tag{8.107}$$

In order to semiclassically quantize parameter-dependent symbols such as a_r, we fix (analogously to (8.67)) a quantization map $\operatorname{Op}_\hbar$ as in (5.200) via the semiclassical version of (5.92); this map has the additional property that $\mathrm{WF}_\hbar'(\operatorname{Op}_\hbar(a)) = \operatorname{ess\,supp}_\hbar a$ for all semiclassical symbols a. *All operators from now on will have Schwartz kernels supported in a fixed compact set $K \times K$.*

From this point on, we restrict our discussion to the case (S^*M). (The proof in the case (T^*M) is the same except without the regularization.) We again have the uniform bounds (8.66) and thus

$$\begin{aligned} A_r &:= \frac{1}{2}(\operatorname{Op}_\hbar(a_r) + \operatorname{Op}_\hbar(a_r)^*) \in L^\infty\big((0,1]_r; \Psi_\hbar^{2s-m+1}\big), \\ B_r &:= \operatorname{Op}_\hbar(b_r) \qquad\qquad\quad \in L^\infty\big((0,1]_r; \Psi_\hbar^{s}\big), \\ E_r' &:= \operatorname{Op}_\hbar(e_r') \qquad\qquad\quad \in L^\infty\big((0,1]_r; \Psi_\hbar^{2s}\big), \end{aligned} \tag{8.108}$$

with $A_r \in \Psi_\hbar^{2s-m+1-2\Theta}$, $B_r \in \Psi_\hbar^{s-\Theta}$, $E_r \in \Psi_\hbar^{2s-2\Theta}$ for every $r > 0$.

Put $\Lambda := \operatorname{Op}_\hbar(\rho^{-m+s+1}) \in \Psi_\hbar^{m-s-1}$. Analogously to (8.69) but using the fact that the principal symbol in the semiclassical calculus captures

operators modulo operators with one differential *and semiclassical* order lower (see (5.199) and Proposition 5.87), we then have

$$\frac{i}{h}[P, A_r] + 2\frac{P - P^*}{2ih}A_r = -(\Lambda A_r)^*(\Lambda A_r) - B_r^* B_r + E_r' + hR_r, \qquad R_r \in L^\infty\big((0,1]_r; \Psi_\hbar^{2s-1}\big), \tag{8.109}$$

with $R_r \in \Psi_\hbar^{2s-1-2\Theta}(M)$ for every $r > 0$. For Θ large enough depending on s, N, we then have (analogously to (8.71))

$$2h^{-1}\operatorname{Im}\langle Pu, A_r u\rangle = -\|\Lambda A_r u\|^2 - \|B_r u\|^2 + \langle E_r' u, u\rangle + h\langle R_r u, u\rangle. \tag{8.110}$$

In order to prove uniform bounds for the final two terms, we need the following analogues of Definition 8.17 and Lemma 8.18:

Definition 8.28 (Uniform Semiclassical Operator Wave Front Set). Let $N \in \mathbb{R}$. Suppose $\mathcal{A} = \{A_r\} \in L^\infty((0,1]_r; h^{-N}\Psi_\hbar^N(M))$ is a bounded family of semiclassical ps.d.o.s on M. Then $\alpha \in \overline{T^*}M$ does *not* lie in the *uniform semiclassical operator wave front set* $\mathrm{WF}'_{\hbar,L^\infty}(\mathcal{A}) \subset \overline{T^*}M$ if and only if there exists an operator $B \in \Psi_\hbar^0(M)$, elliptic at α, such that $\{BA_r\}$ is bounded in $h^\infty\Psi_\hbar^{-\infty}(M)$.

Lemma 8.29 (Uniform Semiclassical Microlocal Elliptic Regularity). *Let $\mathcal{A} = \{A_r\} \in L^\infty((0,1]_r; \Psi_\hbar^m(M))$. Suppose $B \in \Psi_\hbar^0(M)$ is such that $\mathrm{WF}'_{\hbar,L^\infty}(\mathcal{A}) \subset \mathrm{Ell}_\hbar(B)$. We assume that the Schwartz kernels of A_r and B are supported in $K \times K$ for some fixed compact subset $K \subset M$. Let $s, N \in \mathbb{R}$, and let $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(M)$ with $\chi = 1$ near K. Then there exist $h_0 > 0$ and a constant C (independent of r and h) such that*

$$\|A_r u\|_{H_\hbar^{s-m}} \le C\Big(\|Bu\|_{H_\hbar^s} + h^N\|\chi u\|_{H_\hbar^{-N}}\Big), \quad 0 < h < h_0. \tag{8.111}$$

The proof is left to the reader. Pick now $E, G \in \Psi_\hbar^0$ such that $\mathrm{Ell}_\hbar(E) \supset \mathrm{WF}'_{\hbar,L^\infty}(\{E_r'\}) = \operatorname{supp} e_r'$, $\mathrm{Ell}_\hbar(G) \supset \mathrm{WF}'_{\hbar,L^\infty}(\{A_r\}) = \operatorname{supp} a_r$, and with $\mathrm{WF}'_\hbar(G)$, resp. $\mathrm{WF}'_\hbar(E)$ contained in a 2ϵ-neighborhood of $\gamma([0,1])$, resp. $\gamma(0)$. Then

$$\begin{aligned} |\langle E_r' u, u\rangle| &\le C\big(\|Eu\|_{H_\hbar^s}^2 + h^N\|u\|_{H^{-N}}^2\big), \\ h|\langle R_r u, u\rangle| &\le C\Big(h\|Gu\|_{H_\hbar^{s-\frac{1}{2}}}^2 + h^N\|u\|_{H^{-N}}^2\Big). \end{aligned} \tag{8.112}$$

We use these estimates in (8.110), and write $A_r u$ on the left as $\Lambda^-\Lambda A_r u + R'A_r u$ where $\Lambda^- \in \Psi_\hbar^{-m+s+1}$ is an elliptic parametrix of Λ and $R' \in h^\infty\Psi_\hbar^{-\infty}$. Using Cauchy–Schwartz on the left-hand side of (8.110), we obtain the uniform (in r, h) estimate

$$\|B_r u\|_{L^2} \leq C\Big(\|Eu\|_{H_h^s} + \|GPu\|_{H_h^{s-m+1}} + h^{\frac{1}{2}}\|Gu\|_{H_h^{s-\frac{1}{2}}} + h^N\|u\|_{H_h^{-N}}\Big). \tag{8.113}$$

The weak compactness of the unit ball in L^2 and the distributional convergence $B_r u \to B_0 u$ for all $h > 0$ imply that $B_0 u \in L^2$, with norm bounded by the right hand side of (8.113). As in Section 8.4, one can now use an inductive argument in which one enlarges semiclassical elliptic sets and operator wave front sets by increasingly small amounts to estimate the third term on the right in (8.113). This completes the proof of the estimate (8.97) in the setting (8.98) which, as we recall, implies Theorems 8.25 and 8.27 as stated.

Notes

The first proof of the microlocal propagation of singularities was given by Hörmander in [Hör71b]. In fact, [Hör71b, §3.5] gives a positive commutator proof in the more general setting when the imaginary part of the principal symbol has a sign; this is also discussed in Exercise 8.6. Another proof was given by Duistermaat–Hörmander [DH72] using the theory of *Fourier integral operators* (FIOs) introduced by Hörmander [Hör71a]. (Precursors include works by Lax [Lax57], Ludwig [Lud60], Hörmander [Hör68], and Egorov [Ego69].) See [Hör09] for a detailed exposition. The idea for $P \in \Psi^1$ is the following. Near a point $\alpha \in T^*M \setminus o$ where H_p is not radial, one can find a symplectomorphism transforming $\sigma^1(P)(x,\xi)$ to ξ_1, and a "quantization" of this symplectomorphism to an FIO allows one to "microlocally conjugate" P to D_{x_1} up to smoothing errors. The propagation of singularities for D_{x_1}, which can be verified without any technology (Exercise 8.1), then implies the corresponding result for P. The positive commutator proof has an important technical benefit compared to this FIO-based proof: it only utilizes pseudodifferential operators.

Notice an important shift in perspective relative to Chapters 4–5: in the elliptic setting, ps.d.o.s arose, in particular, as approximate inverses of elliptic partial differential operators. In this chapter on the other hand, we use ps.d.o.s as *tools* to study a (non-elliptic) operator P (which in applications is often a differential operator) whose approximate inverse (should it even exist) would certainly fall outside the class of ps.d.o.s. For example, the integral kernel of the inverse of perhaps the simplest non-elliptic operator, $D_{x_1} \in \mathrm{Diff}^1(\mathbb{R}^n)$, $n \geq 2$, is already a rather nontrivial object known as a *paired Lagrangian distribution* [MU79].

Hörmander's definition of a *real principal type operator* requires that $P \in \Psi^m(M)$ have a real and homogeneous principal symbol and that no maximally extended null-bicharacteristic stay over a compact subset of M. See, e.g., [DH72, Definition 6.3.2]. This allows for a clean semiglobal

solvability theory, but it is not in the spirit of the microlocal black box results we are developing in this book.

In our proof of the propagation of singularities, we straighten out the (rescaled) H_p-flow along the entire null-bicharacteristic segment of interest. It is not necessary to do this if one wants to propagate for short times: only H_p at the starting point matters. Such a robust setup was introduced in Melrose–Sjöstrand [MS78, MS82]; see also [dHUV15, §2].

The semiclassical propagation of singularities in T^*M, i.e., at finite frequencies, is closely related to properties of *WKB solutions*; see Dimassi–Sjöstrand [DS99, Chapters 1 and 2] for a textbook account. (The function (8.89) is a very simple instance of a WKB solution.) A proof of the semiclassical propagation result based on conjugation to a normal form using semiclassical FIOs is given in Zworski [Zwo12, Chapter 12.3].

A detailed exposition of propagation estimates in the scattering setting is given in Vasy [Vas18].

Exercises

Exercise 8.1 (Hands-on Propagation of Singularities). Prove the following statements *without* using any of the machinery developed in this section. We work on $\mathbb{R}^n = \mathbb{R}_x \times \mathbb{R}^{n-1}_y$ and study the equation $D_x u(x,y) = f(x,y)$.

(1) Write covectors as $\xi\,\mathrm{d}x + \eta\,\mathrm{d}y$. Compute $\mathrm{Char}(D_x)$.
(2) Suppose $f \in \mathcal{C}^\infty(\mathbb{R}^n)$. Show that $\mathrm{WF}(u) \subset \mathrm{Char}(D_x)$.
(3) Suppose $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ and $(x,y,0,\eta) \in \mathrm{WF}(u)$. Show that $(x+s,y,0,\eta) \in \mathrm{WF}(u)$ for all $s \in \mathbb{R}$.
(4) Let $f \in \mathscr{D}'(\mathbb{R}^n)$ and $(x,y,0,\eta) \in \mathrm{WF}(u)$. Suppose that $s_1 < 0 < s_2$ are such that $(x+s,y,0,\eta) \notin \mathrm{WF}(f)$ for all $s \in (s_1,s_2)$. Show that $(x+s,y,0,\eta) \in \mathrm{WF}(u)$ for $s \in (s_1,s_2)$.
(5) Let $u(x,y) = 1$ for $(x,y) \in [0,1]^2$ and $u(x,y) = 0$ otherwise. Compute $\mathrm{WF}(u)$ and $\mathrm{WF}(f)$ for $f = D_x u$. Describe the wave front set of u over $y = 0$ and its relationship to $\mathrm{WF}(f)$.

Exercise 8.2 (Keldysh Equation). Suppose $u \in \mathscr{D}'(\mathbb{R}^2)$ solves the *Keldysh equation*

$$(xD_x^2 + D_y^2)u = f \in \mathcal{C}^\infty(\mathbb{R}^2). \tag{8.114}$$

Assume that

$$\mathrm{WF}(u) \cap N^*\{x = 0\} = \emptyset. \tag{8.115}$$

Show that $u \in \mathcal{C}^\infty(\mathbb{R}^2)$. Show also that there exist solutions of the equation (8.114) which are not smooth (and which thus necessarily violate (8.115)).

Exercise 8.3 (Tricomi Equation). Suppose $u \in \mathscr{D}'(\mathbb{R}^2)$ solves the *Tricomi equation* $(D_x^2 + xD_y^2)u = f \in \mathcal{C}^\infty(\mathbb{R}^2)$. Assume that $u = u(x, y)$ is smooth for $x < -1$. Show that $u \in \mathcal{C}^\infty(\mathbb{R}^2)$.

Exercise 8.4 (A Simple System of Equations). Write points in $\mathbb{R}^2$ as (x, y). Suppose $u \in \mathscr{D}'(\mathbb{R}^2)$ satisfies $xu \in \mathcal{C}^\infty(\mathbb{R}^2)$ and $yu \in \mathcal{C}^\infty(\mathbb{R}^2)$.

(1) Show that $\mathrm{WF}(u) \subset T_0^*\mathbb{R}^2 \setminus o = \{(x, y, \xi, \eta)\colon (x, y) = (0, 0), (\xi, \eta) \neq (0, 0)\}$.
(2) Suppose there exists $\alpha \in T_0^*\mathbb{R}^2 \setminus o$ with $\alpha \notin \mathrm{WF}(u)$. Show that $\mathrm{WF}(u) = \emptyset$.

Exercise 8.5 (Stability of Real Principal Type Propagation Estimates). Suppose that $P \in \mathrm{Diff}^m(M)$, $B, G, E \in \Psi^0(M)$, $K = \tilde{K} \subset M$, and $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(M)$ satisfy the assumptions of Theorem 8.10; fix $s, N \in \mathbb{R}$. Show that there exist a constant C' and a neighborhood $\mathcal{U}$ of P in $\{B \in \mathrm{Diff}^m(M)\colon B \text{ has real principal symbol}\}$ such that the estimate (8.76) holds for C' and $P' \in \mathcal{U}$ in place of C and P, respectively.

The following set of exercises concerns *complex absorbing (pseudodifferential) potentials*. These are (often artificially chosen) operators Q with $\sigma^m(Q) \geq 0$, used to modify a given real principal type operator $P \in \Psi^m$ to the operator $P - iQ$ with complex absorption (the term $-iQ$).

Exercise 8.6 (Complex Absorption). Let $P \in \Psi_{\mathrm{cl}}^m(M)$, with real principal symbol. Fix a volume density on M, let $Q' \in \Psi_{\mathrm{cl}}^{\frac{m}{2}}(M)$, and set $Q := (Q')^*Q'$. Let $\gamma\colon [0, s_0] \to \mathrm{Char}(P)$ be a null-bicharacteristic segment of P. Suppose $u \in \mathscr{D}'(M)$ satisfies $\gamma(0) \notin \mathrm{WF}^s(u)$. Assuming that

$$\mathrm{WF}^{s-m+1}((P - iQ)u) \cap \gamma([0, s_0]) = \emptyset, \tag{8.116}$$

show that $\mathrm{WF}^s(u) \cap \gamma([0, s_0]) = \emptyset$. (*Hint.* Run the positive commutator argument for P, and show that replacing P by $P - iQ$ in the relevant L^2-pairings produces error terms that have the same sign as the main commutator term and can thus be dropped.)

Remark. At places where Q is elliptic (and thus also $P - iQ$ is elliptic), microlocal control of u in terms of $(P - iQ)u$ follows from elliptic regularity. On open subsets of S^*M where $\sigma^m(Q) = 0$ on the other hand, the desired conclusion follows from Theorem 8.8. The interesting part is thus only where Q "turns on/off" along γ.

Exercise 8.7 (A Non-elliptic Fredholm Problem). Let M be compact with a fixed volume density. Let $P \in \Psi_{\mathrm{cl}}^m(M)$ and $Q = (Q')^*Q'$,

$Q' \in \Psi_{\mathrm{cl}}^{m/2}(M)$. Suppose that for all $\alpha \in \mathrm{Char}(P)$, the backward null-bicharacteristic of P starting at α reaches $\mathrm{Ell}(Q)$ in finite time. Let $s \in \mathbb{R}$.

(1) Let $-N < s$. Prove the estimate

$$\|u\|_{H^s} \leq C(\|(P - iQ)u\|_{H^{s-m+1}} + \|u\|_{H^{-N}}). \tag{8.117}$$

(2) From (8.117), conclude that the map

$$P - iQ \colon \mathcal{X}^s := \{u \in H^s(M) \colon (P - iQ)u \in H^{s-m+1}(M)\} \to H^{s-m+1}(M) \tag{8.118}$$

has finite-dimensional kernel and closed range.

(3) Prove the following "dual" estimate to (8.117), with $-N < -s + m - 1$:

$$\|v\|_{H^{-s+m-1}} \leq C(\|(P^* + iQ^*)v\|_{H^{-s}} + \|v\|_{H^{-N}}). \tag{8.119}$$

(4) Use (8.119) to show that the map (8.118) is Fredholm, with range given by the $L^2(M)$-orthogonal complement of $\ker_{H^{-s+m-1}}(P^* + iQ^*)$. (*Hint.* Given $f \in H^{s-m+1}(M)$ which is orthogonal to $\ker_{H^{-s+m-1}}(P^* + iQ^*)$, show that

$$(P^* + iQ^*)v \mapsto \langle v, f\rangle, \quad v \in H^{-s+m}(M), \tag{8.120}$$

is a well-defined linear map which is bounded in the H^{-s}-norm. Use Theorem 5.63 to conclude.)

Remark. Non-elliptic Fredholm theory will play a central role in Chapter 11.

Exercise 8.8 (Complex Absorption: The Wrong Sign). Consider the operator $P = D_t - iq(t)\langle D_y\rangle$ on $\mathbb{R} \times \mathbb{R}^n$, $n \geq 1$, where $q \in \mathcal{C}^\infty(\mathbb{R})$, $q < 0$. Let $\chi \in \mathcal{C}_{\mathrm{c}}^\infty((-1, 1))$ be equal to 1 on $[-\frac{1}{2}, \frac{1}{2}]$, and let $\tilde{\chi} \in \mathcal{C}_{\mathrm{c}}^\infty((-2, 2))$ be equal to 1 on $[-1, 1]$. Show that there do *not* exist constants $C > 0$, $N, s \in \mathbb{R}$ with $-N < s$ such that the estimate

$$\|\chi(t - 4)\chi(|y|)u\|_{H^s} \leq C\big(\|\tilde{\chi}(t)\tilde{\chi}(|y|)u\|_{H^s} + \|u\|_{H^{-N}}\big) \tag{8.121}$$

holds for all solutions $u \in \mathscr{D}'(\mathbb{R}^{1+n})$ of $Pu = 0$. *Remark.* P is not a pseudodifferential operator. We encourage the reader to work out a pseudodifferential example (also in the semiclassical setting).

Chapter 9
Solving Wave-Type Equations



As an illustration of the results on the propagation of singularities (or rather the propagation of regularity) proved in Chapter 8, we shall solve wave-type equations $Pu = f$ on suitable spacetime domains (Theorem 9.33). This substantially generalizes Example 7.5. An important feature of the approach used here, which distinguishes it from that of Chapter 7, is that neither the operator nor the function spaces are defined with respect to any splitting into time and space variables. Moreover, we deduce the existence of solutions of $Pu = f$ by duality from an a priori (energy) estimate for P^*. (Duality methods are powerful tools for solving PDEs and will feature again in Chapter 11.) Energy methods are often used in the study of wave equations; we shall only develop the basics here.

We give a brief introduction to Lorentzian metrics (Section 9.1) and prove standard energy estimates (Section 9.2). After presenting some functional analytic background on Sobolev spaces with supported and extendible character (Section 9.3), we discuss the existence, uniqueness, and higher regularity of solutions to wave-type equations (Section 9.4). Finally, we prove analogous results for semiclassical wave-type equations (Section 9.5).

We assume familiarity with basic concepts in (semi-)Riemannian geometry, specifically the Levi-Civita connection, geodesics, and Stokes' theorem. Standard references covering these topics are [HE73, O'N83, Lee18].

9.1 Lorentzian Geometry

The geometric setting for the wave-type equations studied in this chapter is:

Definition 9.1 (Lorentzian Manifold). Let M be an n-dimensional manifold, $n \geq 2$. Let $g \in \mathcal{C}^\infty(M; S^2 T^*M)$, so $g_p = g(p)$, $p \in M$, is a bilinear form on T_pM depending smoothly on p. Then g is a *Lorentzian metric* if g_p

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has signature $(n-1,1)$ (sometimes written $(-,+,\dots,+)$) for all p. We call (M,g) a *Lorentzian manifold* or *spacetime*.

By a basic result in linear algebra, there exists, for all $p \in M$, an orthonormal basis $V_1,\dots,V_n$ of T_pM. That is,

$$g(V_1,V_1) = -1; \quad g(V_j,V_j) = 1,\ j = 2,\dots,n; \quad g(V_i,V_j) = 0,\ i \neq j. \tag{9.1}$$

Since a Lorentzian metric g is a *non-degenerate* bilinear form on T_pM, it induces an isomorphism $T_pM \to T_p^*M$ via

$$T_pM \ni V \mapsto V^\flat := g_p(V,-) \in T_p^*M. \tag{9.2}$$

Thus, g induces a signature $(n-1,1)$ bilinear form $g^{-1} \in \mathcal{C}^\infty(M; S^2TM)$ on T^*M called the *dual metric*. The inverse of (9.2) is the map

$$T_p^*M \ni \omega \mapsto \omega^\sharp := g_p^{-1}(\omega,-) \in (T_p^*M)^* = T_pM. \tag{9.3}$$

Example 9.2 (Minkowski Metric). Let $M = \mathbb{R}^n = \mathbb{R}_t \times \mathbb{R}_x^{n-1}$. Then the *Minkowski metric* on M is defined by

$$g = -\mathrm{d}t^2 + \sum_{j=1}^{n-1} (\mathrm{d}x^j)^2. \tag{9.4}$$

An orthonormal basis of every tangent space is given by $\partial_t, \partial_{x^1}, \dots, \partial_{x^{n-1}}$. The dual metric is given by

$$g^{-1} = -\partial_t^2 + \sum_{j=1}^{n-1} \partial_{x^j}^2 = -\partial_t \otimes \partial_t + \sum_{j=1}^{n-1} \partial_{x^j} \otimes \partial_{x^j}. \tag{9.5}$$

For instance, $g^{-1}(\mathrm{d}t, \mathrm{d}t) = -1$.

Definition 9.3 (Causal Character of Tangent Vectors). Let (M,g) be a Lorentzian manifold.

(1) Let $p \in M$. Then we say that a tangent vector $0 \neq V \in T_pM$ is

(i) *timelike* if $g_p(V,V) < 0$,
(ii) *spacelike* if $g_p(V,V) > 0$,
(iii) *null* (or *lightlike*) if $g_p(V,V) = 0$,
(iv) *causal* if V is timelike or lightlike.

Similarly, we classify covectors $\zeta \in T_p^*M$ as timelike, spacelike, null, or causal depending on the sign of $g_p^{-1}(\zeta,\zeta)$.

(2) A linear subspace $E \subset T_pM$ is called spacelike if all nonzero vectors in E are spacelike (equivalently, $g_p|_{E\times E}$ is positive definite). It is called

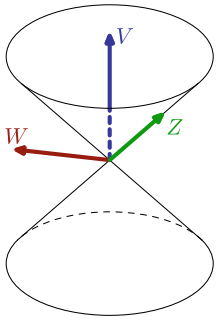


Figure 9.1 The light cone in T_pM. The vectors V, W, and Z are timelike, spacelike, and null, respectively.

Lorentzian if $g_p|_{E\times E}$ has Lorentzian signature, and null if $g_p|_{E\times E}$ is degenerate.

(3) Let $S \subset M$ be a smooth hypersurface. Then S is *spacelike*, resp. *null* if for all $p \in S$, T_pS is spacelike, resp. null.

The physical relevance of these notions is the following: massive observers travel along timelike curves in M, i.e., curves with timelike tangent vectors, whereas massless particles travel along lightlike curves. See Figure 9.1.

Example 9.4 (Causal Character). In the notation of Example 9.2, the vector $\partial_t + v_1\partial_{x^1}$ is timelike if and only if $|v_1| < 1$, null if and only if $|v_1| = 1$, and spacelike if and only if $|v_1| > 1$. The hypersurface $\{t = 0\}$ is spacelike; indeed, its tangent spaces are spanned by $\partial_{x^1}, \dots, \partial_{x^{n-1}}$. More generally, for $v \in \mathbb{R}^{n-1}$, the hypersurface $\{t = v \cdot x\}$ is spacelike if and only if $|v| < 1$.

The following characterization of spacelike and Lorentzian subspaces will be useful.

Lemma 9.5 (Spacelike Subspaces). *A linear subspace $E \subset T_pM$ is spacelike, resp. Lorentzian if and only if $E^\perp := \{v \in T_pM : g_p(v, w) = 0 \,\forall\, w \in E\}$ is Lorentzian, resp. spacelike.*

Proof. Extend a g_p-orthonormal basis $V_1, \dots, V_k$ of E to an orthonormal basis $V_1, \dots, V_n$ of T_pM. Then $E^\perp$ is spanned by $V_{k+1}, \dots, V_n$. Among the vectors $V_1, \dots, V_n$, there is only one, say V_j, with $g_p(V_j, V_j) = -1$, while $g_p(V_i, V_i) = 1$ for all $i \neq j$. When E is spacelike, then $j \geq k + 1$ and thus $E^\perp$ is Lorentzian, whereas when E is Lorentzian, then $j \leq k$ and thus $E^\perp$ is spacelike. □

Regarding Definition 9.3(3), we conclude that S is spacelike if and only if $(T_pS)^\perp$ is timelike for all $p \in S$. Since $\{V^\flat : V \in (T_pS)^\perp\} = N_p^*S$, this is in turn equivalent to all $0 \neq \zeta \in N_p^*S$ being timelike. In particular, if $S = f^{-1}(0)$ where $f \in \mathcal{C}^\infty(M)$ is such that $\mathrm{d}f$ is timelike whenever $f = 0$, then S is spacelike (since $N_p^*S = \mathrm{span}\{\mathrm{d}f|_p\}$ for $p \in S$).

Lemma 9.6 (Timelike and Causal Vectors). *Let (M, g) be a Lorentzian manifold, $p \in M$. Suppose $V \in T_pM$ is timelike and $W \in T_pM$ is causal. Then $g_p(V, W) \neq 0$.*

Proof. Choosing an orthonormal basis of T_pM as in (9.1), we may assume that $V = V_1$ and $W = \sum_{i=1}^n W^i V_i$ with $(W^1)^2 \geq \sum_{i=2}^n (W^i)^2$. Thus $g_p(V, W) = -W^1$ is nonzero since $W \neq 0$. □

The set of timelike vectors is an open double cone (without the vertex) in T_pM, and thus, it has two connected components. A continuous choice of one of them is called a *time orientation* of (M, g), and (M, g) is then called *time-oriented*. A time orientation exists if and only if there exists a smooth vector field V on M which is everywhere timelike; such a vector field V selects, at every point of M, one of the two connected components of the set of timelike vectors. (This follows from a partition of unity argument.)

Definition 9.7 (Time Orientation). Given a fixed smooth timelike vector field V on M, we say that a timelike/null/causal vector W is *future* timelike/null/causal if $g(V, W) < 0$, and *past* timelike/null/causal if $g(V, W) > 0$. We call $\omega \in T^*M$ future/past timelike/null/causal if $\omega^\sharp \in TM$ is future/past timelike/null/causal.

In particular, V itself is future timelike and $-V$ is past timelike. We caution the reader that ω being future timelike means that $g^{-1}(\omega, \omega) < 0$ and $\omega(V) < 0$. Some authors define a covector to be future timelike/null if instead $\omega(V) > 0$. (Yet other authors use a different signature convention for Lorentzian metrics.) In the Minkowski spacetime of Example 9.2, the standard time orientation is induced by $V = \partial_t$, and $\mathrm{d}t$ is then past timelike.

9.2 Wave-Type Operators, Energy Estimates, Finite Speed of Propagation

Fix an n-dimensional Lorentzian manifold (M, g). We define the *wave operator* $\Box_g \in \mathrm{Diff}^2(M)$ like the Laplace operator on a Riemannian manifold: in local coordinates $(z^1, \ldots, z^n)$ on M, write $g_{ij} := g(\partial_{z^i}, \partial_{z^j})$, $g^{ij} := g^{-1}(\mathrm{d}z^i, \mathrm{d}z^j)$, and $|g| := |\det(g_{ij})|$; then

$$\Box_g u := \sum_{i,j=1}^n |g|^{-\frac{1}{2}} D_{z^i} \big(|g|^{\frac{1}{2}} g^{ij} D_{z^j} u \big). \tag{9.6}$$

Its principal symbol is the *dual metric function*

$$G(z,\zeta) := \sigma^2(\Box_g)(z,\zeta) = \sum_{i,j=1}^{n} g(z)^{ij}\zeta_i\zeta_j = g(z)^{-1}(\zeta,\zeta), \quad \zeta \in T_z^*M. \tag{9.7}$$

We may more succinctly denote this by $G(\zeta)$. The characteristic set $\mathrm{Char}(\Box_g) = \{\zeta \in T^*M \setminus o\colon G(\zeta) = 0\}$ consists of all lightlike covectors.

In the sequel, we will use the shorthand notation $\partial_j = \partial_{z^j}$, $D_j = D_{z^j}$. Given a tensor $T \in \mathcal{C}^\infty(M; TM \otimes T^*M)$, we write its components in local coordinates as $T^i{}_j := T(\mathrm{d}z^i, \partial_j)$, similarly for other types of tensors. Index raising and lowering are effected using the metric g, so

$$T_{ij} = g_{ii'}T^{i'}{}_j, \quad T^{ij} = g^{jj'}T^i{}_{j'}. \tag{9.8}$$

We use the *summation convention* here, which means that summation over repeated indices is not spelled out; further examples are $(V^\flat)_i = g_{ii'}V^{i'}$ and $(\omega^\sharp)^i = g^{ii'}\omega_{i'}$. We define the *(metric) contraction* of two symmetric 2-tensors T, S as $T \cdot S := T_{ij}S^{ij}$. We use index notation also for covariant derivatives: if ∇ denotes the Levi-Civita connection of g, then we write

$$T^i{}_{j;k} := (\nabla T)(\partial_k, \mathrm{d}x^i, \partial_j) = (\nabla_k T)(\mathrm{d}x^i, \partial_j). \tag{9.9}$$

Higher derivatives are denoted $T^i{}_{j;kl}$, etc. Important special cases are

$$\begin{aligned}
\Box_g u &= -g^{ij}u_{;ij} && \text{(scalar wave operator)},\\
\mathrm{div}_g V &= V^i{}_{;i} && \text{(divergence of a vector field)},\\
(\Box_g T)_{i_1\dots i_k}{}^{j_1\dots j_l} &= -g^{ij}T_{i_1\dots i_k}{}^{j_1\dots j_l}{}_{;ij} && \text{(tensor wave operator)},\\
(\mathrm{div}_g T)_i &= g^{jk}T_{ij;k} && \text{(divergence of a 2-tensor)}.
\end{aligned} \tag{9.10}$$

Definition 9.8 (Wave-Type Operators). Let $E \to M$ be a vector bundle over the Lorentzian manifold (M, g). We say that $P \in \mathrm{Diff}^2(M; E)$ is a *wave-type operator* if P is principally scalar with $\sigma^2(P) = G$, where $G(\zeta) = g^{-1}(\zeta, \zeta)$ is the dual metric function.

Examples include scalar and tensor wave operators $\Box_g$, and sums of such operators (acting on sections of a bundle E) and arbitrary elements of $\mathrm{Diff}^1(M; E)$. An example of the latter type, for $E = M \times \mathbb{C}$ the trivial bundle, is $\Box_g + X + V$ where $X \in \mathcal{V}(M)$ is a vector field and $V \in \mathcal{C}^\infty(M)$ is a smooth function. We shall see that wave-type operators on time-orientable Lorentzian manifolds possess a well-behaved Cauchy problem. For a characterization of wave-type operators in terms of the notion of strict hyperbolicity, see Exercise 9.1.

The connection between Lorentzian geometry and the microlocal analysis of wave-type operators, which we will exploit in Section 9.4, is the following; we write $\pi\colon T^*M \to M$ for the projection.

Proposition 9.9 (Bicharacteristics and Geodesics). *We use the notation* (9.7).

(1) *Let $\gamma\colon I \subseteq \mathbb{R} \to T^*M$ be an integral curve of $H_{\frac{1}{2}G}$. (Thus, γ is a constant reparameterization of an integral curve of H_G.) Denote by $\alpha := \pi\circ\gamma\colon I \to M$ its projection to the base. Then α is a geodesic in (M, g), with $\alpha(s) = \pi(\gamma(s))$ and $\alpha'(s) = \gamma(s)^\sharp \in T_{\alpha(s)}M$.*

(2) *Conversely, if $\alpha\colon I \to M$ is a geodesic, then $s \mapsto \gamma(s) := \alpha'(s)^\flat \in T^*_{\alpha(s)}M$ defines an integral curve of $H_{\frac{1}{2}G}$.*

In particular, null-bicharacteristics of $\Box_g$ are, in this sense, lifts of null-geodesics (i.e., geodesics α such that $\alpha'(s)$ is null for all s) on (M, g) to T^*M.

Proof. Since $\frac{1}{2}G = \frac{1}{2}g^{ij}\zeta_i\zeta_j$, we have

$$H_{\frac{1}{2}G} = g^{ik}\zeta_i\partial_k - \frac{1}{2}(\partial_i g^{jk})\zeta_j\zeta_k\partial_{\zeta_i}. \tag{9.11}$$

We first prove part (1). Note that $\alpha'(s) = \pi_*(H_{\frac{1}{2}G}|_{\gamma(s)}) = \gamma(s)^\sharp \in T_{\gamma(s)}M$. Write $\alpha(s) = (\alpha^1(s), \dots, \alpha^n(s))$, and denote the fiber coordinates of $\gamma(s)$ by $(\zeta_1(s), \dots, \zeta_n(s))$. Then $(\alpha^k)'(s) = g^{ik}(\alpha(s))\zeta_i(s)$. Let $s_0 \in I$ and work in normal coordinates centered at $\alpha(s_0)$. Then $\partial_j g^{ik} = 0$ at $\alpha(s_0)$, and hence,

$$(\alpha^k)''(s_0) = g^{ik}(\alpha(s_0))\zeta_i'(s_0). \tag{9.12}$$

But the ∂_{ζ_i}-component of (9.11) vanishes at $\gamma(s_0)$, which gives $(\alpha^k)''(s_0) = 0$. Since the Christoffel symbols of g vanish at $\alpha(s_0)$, this is in turn equivalent to the validity of the geodesic equation $\nabla_{\alpha'}\alpha' = 0$ at $\alpha(s_0)$. Part (2) follows by reversing the argument. □

According to Theorem 8.4, singularities of solutions of wave-type equations $Pu = f \in \mathcal{C}^\infty(M)$ thus propagate along lifts of null-geodesics inside of $\mathrm{Char}(P) = \mathrm{Char}(\Box_g)$.

We now introduce *energy methods* which will allow us to prove sharply localized estimates for solutions of general wave-type equations.

Definition 9.10 (Stress–Energy–Momentum Tensor). We define the *(abstract) stress–energy–momentum tensor* $T \in \mathcal{C}^\infty(M; S^2T^*M \otimes S^2TM)$ by

$$T(X, Y) := X \otimes_s Y - \frac{1}{2}g(X, Y)g^{-1} \in S^2T_pM, \quad X, Y \in T_pM; \tag{9.13}$$

here $X \otimes_s Y := \frac{1}{2}(X \otimes Y + Y \otimes X)$ is the symmetric tensor product. Given $u \in \mathcal{C}^\infty(M; \mathbb{C})$, the stress–energy–momentum tensor $T[u] \in \mathcal{C}^\infty(M; S^2T^*M)$ of u is defined by

$$\begin{aligned} T[u](X,Y) &:= T(X,Y)(\mathrm{d}u, \mathrm{d}\bar{u}) \\ &= \frac{1}{2}\big((Xu)(Y\bar{u}) + (Yu)(X\bar{u}) - g(X,Y)g^{-1}(\mathrm{d}u, \mathrm{d}\bar{u})\big). \end{aligned} \tag{9.14}$$

For a timelike vector $V \in T_pM$ with $g(V,V) = -1$, the quantity $T[u](V,V)$ has an interpretation as the energy density of u as measured by an observer traveling in the direction V. For example, on Minkowski space (Example 9.2) and for $V = \partial_t$, we have

$$T[u](\partial_t, \partial_t) = \frac{1}{2}\big(|\partial_t u|^2 + |\nabla_x u|^2\big). \tag{9.15}$$

The positive definiteness of this expression in $\mathrm{d}u$ is a special case of the following general result.

Proposition 9.11 (Positive Definiteness of T). *If $X, Y \in T_pM$ are timelike and in the same time cone, i.e., $g_p(X,Y) < 0$, then $T(X,Y)$ is positive definite. If X, Y lie in the same causal cone, i.e., in the closure of the same time cone, then $T(X,Y)$ is positive semidefinite.*

Proof. We only need to prove the first statement, as the second one then follows by continuity. Upon replacing X by $X/\sqrt{-g_p(X,X)}$, we may assume that $g_p(X,X) = -1$. We extend X to an orthonormal basis $X, V_1, \dots, V_{n-1}$ of T_pM in such a way that $Y = \lambda X + \mu V_1$ for some $\lambda > 0$ and $\mu \in \mathbb{R}$. Upon rescaling Y, we may assume that $Y = X + \mu V_1$. We then compute

$$\begin{aligned} T(X,Y) &= T(X, X + \mu V_1) \\ &= X \otimes_s (X + \mu V_1) - \frac{1}{2} \cdot (-1)\left(-X \otimes X + \sum_{j=1}^{n-1} V_j \otimes V_j\right) \\ &= \frac{1}{2}(X \otimes X + 2\mu X \otimes_s V_1 + V_1 \otimes V_1) + \frac{1}{2}\sum_{j=2}^{n-1} V_j \otimes V_j. \end{aligned} \tag{9.16}$$

Now,

$$X \otimes X + 2\mu X \otimes_s V_1 + V_1 \otimes V_1 = (X + \mu V_1) \otimes (X + \mu V_1) + (1 - \mu^2) V_1 \otimes V_1. \tag{9.17}$$

Since $|\mu| < 1$ due to the timelike nature of Y, the desired conclusion follows. □

Definition 9.12 (J- and K-Currents). Let $X \in \mathcal{V}(M)$ and $u \in \mathcal{C}^\infty(M)$.

(1) The *J-current* associated with X and u is ${}^{(X)}J[u] := (T[u](X,\cdot))^\sharp \in \mathcal{V}(M)$.
(2) The *K-current* associated with X and u is

$$ {}^{(X)}K[u] := {}^{(X)}\pi \cdot T[u] \in \mathcal{C}^\infty(M) \tag{9.18} $$

where ${}^{(X)}\pi := \frac{1}{2}\mathcal{L}_X g$ is the *deformation tensor* of X (i.e., $\frac{1}{2}$ times the Lie derivative $\mathcal{L}_X g$ of g along X), and "$\cdot$" denotes the metric contraction. The *abstract K-current* is ${}^{(X)}K := {}^{(X)}\pi \cdot T \in \mathcal{C}^\infty(M; S^2TM)$.

Thus, ${}^{(X)}K[u] = {}^{(X)}K(\mathrm{d}u, \mathrm{d}\bar{u})$. The tensors T, J, K satisfy the following important differential identities:

Proposition 9.13 (Identities for T, J, K). *Let $X \in \mathcal{V}(M)$, $u \in \mathcal{C}^\infty(M;\mathbb{C})$, and $w \in \mathcal{C}^\infty(M)$. Then ${}^{(X)}\pi_{ij} = \frac{1}{2}(X_{i;j} + X_{j;i})$ and*

$$ \operatorname{div}_g T[u] = -\operatorname{Re}(\Box_g u \, \mathrm{d}\bar{u}), \tag{9.19} $$

$$ \operatorname{div}_g {}^{(X)}J[u] = -\operatorname{Re}(\Box_g u \, X\bar{u}) + {}^{(X)}K[u], \tag{9.20} $$

$$ {}^{(wX)}K[u] = w \, {}^{(X)}K[u] + T[u](\nabla w, X). \tag{9.21} $$

Equation (9.21) can also be written as ${}^{(wX)}K = w \, {}^{(X)}K + T(\nabla w, X)$.

Proof. Since $g_{ij;k} = 0$, we compute

$$ \begin{aligned} (\operatorname{div}_g T[u])_i = T[u]_{ij}{}^{;j} &= \operatorname{Re}\Big(u_{;i}\bar{u}_{;j} - \frac{1}{2} g_{ij} u_{;k}\bar{u}^{;k}\Big)^{;j} \\ &= \operatorname{Re}\big((u_{;i}\bar{u}_{;j}{}^{j} + u_{;i}{}^{j}\bar{u}_{;j}) - u_{;ki}\bar{u}^{;k}\big) \\ &= -\operatorname{Re}(\Box u \, \bar{u}_{;i}), \end{aligned} \tag{9.22} $$

where we used that $u_{;ki} = u_{;ik}$. This proves (9.19).

The identity (9.20) now follows from the symmetry $T[u]_{ij} = T[u]_{ji}$ and

$$ \operatorname{div}_g {}^{(X)}J[u] = (T[u]_{ij}X^i)^{;j} = T[u]_{ij}{}^{;j}X^i + T[u]_{ij} \cdot \frac{1}{2}(X^{i;j} + X^{j;i}) \tag{9.23} $$

once we show that ${}^{(X)}\pi_{ij} = \frac{1}{2}(X_{i;j} + X_{j;i})$. To do this, we work near a point $p \in M$ in normal coordinates centered at p. Then at p we have $Xg_{ij} = 0$ and $\nabla_j \partial_i = 0$ (and thus also $\nabla_X \partial_i = 0$), and therefore,

$$\begin{aligned}(\mathcal{L}_X g)_{ij} &= (\mathcal{L}_X g)(\partial_i, \partial_j)\\ &= X(g(\partial_i, \partial_j)) - g(\mathcal{L}_X \partial_i, \partial_j) - g(\partial_i, \mathcal{L}_X \partial_j)\\ &= -g([X, \partial_i], \partial_j) - g(\partial_i, [X, \partial_j])\\ &= -g(\nabla_X \partial_i - \nabla_i X, \partial_j) - g(\partial_i, \nabla_X \partial_j - \nabla_j X)\\ &= g(\nabla_i X, \partial_j) + g(\partial_i, \nabla_j X)\\ &= X_{j;i} + X_{i;j}.\end{aligned} \tag{9.24}$$

Using (9.24), we further compute

$$2\,{}^{(wX)}\pi_{ij} = (wX)_{i;j} + (wX)_{j;i} = w(X_{i;j} + X_{j;i}) + w_{;j}X_i + w_{;i}X_j, \tag{9.25}$$

so ${}^{(wX)}\pi = w\,{}^{(X)}\pi + \mathrm{d}w \otimes_s X^\flat$. This implies (9.21). □

The application of the divergence theorem to the J-current associated with any vector field X will produce an *energy identity* on suitable domains $\Omega \subset M$.

Definition 9.14 (Admissible Spacetime Domains). We call a closed subset $\Omega \subset M$ an *admissible (spacetime) domain* if there exist functions $f_1, \dots, f_N \in \mathcal{C}^\infty(M)$ such that the following conditions hold:

(1) We have

$$\Omega = \bigcap_{j=1}^{N} \{p \in M \colon f_j(p) \geq 0\}. \tag{9.26}$$

(2) At every $p \in \Omega$, the differentials of all f_j vanishing at p are linearly independent.
(3) $\Omega \cap f_j^{-1}(0)$ is connected (or empty) for all j, and $\mathrm{d}f_j(p)$ is timelike for all $p \in \Omega \cap f_j^{-1}(0)$.

If, moreover, a time orientation is fixed on M, we call a non-empty hypersurface $f_j^{-1}(0) \cap \Omega$ *initial*, resp. *final* if $\mathrm{d}f_j$ is past, resp. future timelike on it.

By removing those f_j which do not vanish on Ω, we may assume that $H_j := \Omega \cap f_j^{-1}(0)$ is non-empty for all j; each H_j is spacelike by condition (3). Condition (2) implies that all H_j intersect transversally. An application of the inverse function theorem moreover shows that Ω is a submanifold with corners, and the H_j are its (embedded) boundary hypersurfaces (Exercise 9.2). The future timelike unit normal at H_j is given by

$$\nu_j := \mp \frac{(\mathrm{d}f_j)^\sharp}{\sqrt{-g^{-1}(\mathrm{d}f_j, \mathrm{d}f_j)}}, \tag{9.27}$$

with "−," resp. "+" at initial, resp. final hypersurfaces. Thus, ν_j is inward, resp. outward pointing at initial, resp. final hypersurfaces. There exist unique volume densities

$$|\mathrm{d}\sigma_j| \in \mathcal{C}^\infty(H_j; \Omega H_j) \tag{9.28}$$

such that, for all $V_1, \ldots, V_{n-1} \in T_pM$,

$$|\mathrm{d}g|(\nu_j, V_1, \ldots, V_{n-1}) = |\mathrm{d}\sigma_j|(V_1, \ldots, V_{n-1}). \tag{9.29}$$

Example 9.15 (Admissible Domains in Minkowski Space). Consider $M = \mathbb{R}_t \times \mathbb{R}_x^{n-1}$ with the metric $g = -\mathrm{d}t^2 + \sum_{j=1}^{n-1}(\mathrm{d}x^j)^2$ and the standard time orientation.

(1) The domain $\{0 \le t \le 1\}$ is admissible: take $f_1(t,x) = t$ and $f_2(t,x) = 1-t$. The hypersurface $\{t = 0\}$ is initial, and $\{t = 1\}$ is final. The volume density of both hypersurfaces is $|\mathrm{d}x| = |\mathrm{d}x^1 \cdots \mathrm{d}x^{n-1}|$.

(2) The domain $\Omega_1 := \{t \ge 0,\ |x| < 1 - t\}$ is not quite admissible since it is neither of the required form near the tip $(t,x) = (1,0)$, nor is the hypersurface $0 < |x| = 1 - t \le 1$ spacelike. However, for any $0 < \theta' < \theta < 1$, the slight modification

$$\Omega_{\theta,\theta'} := \{0 \le t \le \theta',\ |x| < 1 - \theta^{-1}t\} \tag{9.30}$$

is admissible. It has one initial hypersurface, $\{t = 0\}$, and two final hypersurfaces, $\{t = \theta'\}$ and $\{|x| = 1 - \theta^{-1}t\}$. See Figure 9.2.

Theorem 9.16 (Energy Identity). *Let (M, g) be time-oriented. Let $\Omega \subset M$ be an admissible spacetime domain of the form (9.26). Order the boundary hypersurfaces $H_j := \Omega \cap f_j^{-1}(0)$ of Ω such that $H_1, \ldots, H_{N_\mathrm{I}}$ are the initial*

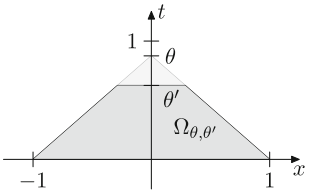


Figure 9.2 The admissible domain $\Omega_{\theta,\theta'}$ in (1+1)-dimensional Minkowski space $\mathbb{R}_t \times \mathbb{R}_x$. We prove an energy identity for solutions of wave-type equations on domains such as this one (Theorem 9.16). In particular, we show that if $u = \partial_t u = 0$ at $t = 0$, then $u = 0$ throughout $\Omega_{\theta,\theta'}$ (Example 9.18).

hypersurfaces and $H_{N_\mathrm{I}+1}, \dots, H_{N_\mathrm{I}+N_\mathrm{F}}$ *are the final hypersurfaces. Denote the future timelike unit normal at* H_j *by* ν_j *(see* (9.27)*). Let* $X \in \mathcal{V}(M)$*. Then for all* $u \in \mathcal{C}^2_\mathrm{c}(\Omega)$*, we have the identity*

$$\begin{aligned} &\sum_{j=1}^{N_\mathrm{F}} \int_{H_{N_\mathrm{I}+j}} T[u](X, \nu_{N_\mathrm{I}+j}) \, |\mathrm{d}\sigma_{N_\mathrm{I}+j}| + \int_\Omega {}^{(X)}K[u] \, |\mathrm{d}g| \\ &\qquad = \sum_{j=1}^{N_\mathrm{I}} \int_{H_j} T[u](X, \nu_j) \, |\mathrm{d}\sigma_j| + \int_\Omega \mathrm{Re}(\Box_g u \, X\bar{u}) \, |\mathrm{d}g|. \end{aligned} \tag{9.31}$$

This is particularly useful when the bulk term ${}^{(X)}K[u]$ is non-negative; see Examples 9.17–9.18 and Theorem 9.19 below.

Proof of Theorem 9.16. This is a standard application of Stokes' theorem; we present the computations in detail to verify the signs in (9.31).

Using a partition of unity, it suffices to consider the case that X is supported in a coordinate chart $z \in \mathbb{R}^n$, with the initial hypersurfaces given by $H_i = \{z^i = 0\}$, $i = 1, \dots, N'_\mathrm{I} \le N_\mathrm{I}$, the final hypersurfaces (upon relabeling) by $H_i = \{z^i = 0\}$, $i = N'_\mathrm{I} + 1, \dots, N'_\mathrm{I} + N'_\mathrm{F} =: N \le n$ where $N'_\mathrm{F} \le N_\mathrm{F}$, and with $\operatorname{supp} X$ contained in the set $\{0 \le z^i < 1, \ i = 1, \dots, N, \ |z^j| < 1, \ j = N+1, \dots, n\}$. By multiplying z_i, $i = 1, \dots, N$, by suitable positive smooth functions, we may moreover assume that $\mathrm{d}z^i$ is a unit covector at $z^i = 0$ for $i = 1, \dots, N$, and thus,

$$\nu_i = -(\mathrm{d}z^i)^\sharp \ (i = 1, \dots, N'_\mathrm{I}), \quad \nu_i = (\mathrm{d}z^i)^\sharp \ (i = N'_\mathrm{I} + 1, \dots, N), \tag{9.32}$$

and $(\mathrm{d}z^i)^\sharp$ is an outward pointing unit normal at H_i. In this chart, the metric volume form (with the standard orientation) is

$$\mathrm{d}g = \sqrt{|g|} \, \mathrm{d}z^1 \wedge \cdots \wedge \mathrm{d}z^n, \qquad |g| := |\det(g_{ij})|. \tag{9.33}$$

For any vector field J, we have $\mathrm{d}(J \lrcorner \mathrm{d}g) = (\mathrm{div}_g \, J) \, \mathrm{d}g$, as is easily checked in normal coordinates. The area forms on H_i, $i = 1, \dots, N$, with the induced orientations, are

$$\mathrm{d}\sigma_i = (\mathrm{d}z^i)^\sharp \lrcorner \mathrm{d}g. \tag{9.34}$$

We abbreviate $J = {}^{(X)}J[u]$. Define the $(n-1)$-form $\omega := J \lrcorner \mathrm{d}g$. Then $\mathrm{d}\omega = (\mathrm{div}_g \, J) \, \mathrm{d}g$, so the identity (9.20) and Stokes' theorem give

$$\int_\Omega {}^{(X)}K[u] \, \mathrm{d}g - \int_\Omega \mathrm{Re}(\Box_g u \, X\bar{u}) \, \mathrm{d}g = \int_\Omega \mathrm{d}\omega = \int_{\partial\Omega} \omega = \sum_{i=1}^{N} \int_{H_i} \iota_{H_i}^* \omega \tag{9.35}$$

where $\iota_{H_i}: H_i \hookrightarrow M$ is the inclusion. We relate $\iota_{H_i}^* \omega$ to $\mathrm{d}\sigma_i$ by computing

$$\begin{aligned}(\partial_1, \dots, \widehat{\partial_i}, \dots, \partial_n) &= (-1)^{i-1} \mathrm{d}g(\partial_1, \dots, J, \dots, \partial_n) \\ &= (-1)^{i-1} \sqrt{|g|} \mathrm{d}z^i(J), \\ \mathrm{d}\sigma_i(\partial_1, \dots, \widehat{\partial_i}, \dots, \partial_n) &= (-1)^{i-1} \mathrm{d}g(\partial_1, \dots, (\mathrm{d}z^i)^\sharp, \dots, \partial_n) \\ &= (-1)^{i-1} \sqrt{|g|} \mathrm{d}z^i((\mathrm{d}z^i)^\sharp).\end{aligned} \tag{9.36}$$

Since $\mathrm{d}z^i((\mathrm{d}z^i)^\sharp) = g^{-1}(\mathrm{d}z^i, \mathrm{d}z^i) = -1$ at H_i, we conclude that

$$\iota_{H_i}^* \omega = -\mathrm{d}z^i(J)\, \mathrm{d}\sigma_i = \pm g(J, \nu_i)\, \mathrm{d}\sigma_i, \tag{9.37}$$

with the "+" sign for $i = 1, \dots, N_{\mathrm{I}}'$ and the "−" sign for $i = N_{\mathrm{I}}' + 1, \dots, N$. Plugging this into (9.35) yields (9.31). □

The proof yields the following identity for all $J \in \mathcal{V}(M)$:

$$\int_\Omega \mathrm{div}_g\, J\, |\mathrm{d}g| = \sum_{j=1}^{N_{\mathrm{I}}} \int_{H_j} g(J, \nu_j)\, |\mathrm{d}\sigma_j| - \sum_{j=1}^{N_{\mathrm{F}}} \int_{H_{N_{\mathrm{I}}+j}} g(J, \nu_{N_{\mathrm{I}}+j})\, |\mathrm{d}\sigma_{N_{\mathrm{I}}+j}|. \tag{9.38}$$

Example 9.17 (Energy Identity on Minkowski Space). Continuing Example 9.15, consider $X = \partial_t$ and $\Omega = \{0 \le t \le 1\}$. Then $\nu_1 = \nu_2 = \partial_t$ and ${}^{(X)}K[u] = 0$. If $\Box_g u = 0$, the identity (9.31) thus gives for the energy of u on a t-level set, defined by

$$E[u](t) := \int_{\mathbb{R}^n} T[u]|_{(t,x)}(\partial_t, \partial_t)\, \mathrm{d}x = \int_{\mathbb{R}^n} \frac{1}{2} \left(|\partial_t u(t,x)|^2 + |\nabla_x u(t,x)|^2 \right) \mathrm{d}x \tag{9.39}$$

(see (9.15)), the identity $E[u](1) = E[u](0)$.

Example 9.18 (Finite Speed of Propagation). Consider on Minkowski space the domain $\Omega_{\theta,\theta'}$ in (9.30), with $0 < \theta' < \theta < 1$ fixed. Suppose that $\Box_g u = 0$ in $\Omega_{\theta,\theta'}$ and $u, \partial_t u = 0$ at $t = 0$. Then (9.31) for $X = \partial_t$ (thus, ${}^{(X)}K = 0$), applied separately on $\Omega_{\theta,\theta'} \cap \{t \le \theta_0\} = \Omega_{\theta,\theta_0}$ for all $\theta_0 \in (0, \theta']$, implies $\nabla u = 0$ in $\Omega_{\theta,\theta'}$, and thus, $u = 0$. This means that even if u or $\partial_t u$ are nonzero at $t = 0$ for $|x| > 1$, a smooth solution of the wave equation nonetheless vanishes in $\Omega_{\theta,\theta'}$. (As $\theta', \theta \nearrow 1$, this implies $u = 0$ in $\Omega_1 = \{0 \le |x| \le 1 - t\}$.) This is a typical instance of the *finite speed of propagation* for solutions of the wave equation. A slight modification of the vector field X allows one to draw the conclusion $\nabla u|_{\Omega_{\theta,\theta'}} = 0$ from a *single* application of (9.31) to $\Omega_{\theta,\theta'}$. To wit, consider $X = e^{-\alpha t}\partial_t$ where $\alpha > 0$. This is still future timelike; but its K-current is now nonzero due to the presence of the decreasing weight $e^{-\alpha t}$. In fact, by (9.21),

$$ {}^{(X)}K[u] = T[u](\nabla e^{-\alpha t}, \partial_t) = \alpha T[u](\partial_t, \partial_t), \tag{9.40} $$

which is a *positive definite* quadratic form in du. (Equivalently, ${}^{(X)}K \in \mathcal{C}^\infty(M; S^2TM)$ is positive definite.) Applying (9.31) on $\Omega_{\theta,\theta'}$ with this vector field X implies $\int_{\Omega_{\theta,\theta'}} {}^{(X)}K[u]\,|\mathrm{d}g| \leq 0$; this directly gives $\nabla u = 0$ in $\Omega_{\theta,\theta'}$.

The second half of this example illustrates the important idea of using weights which are decreasing in time (i.e., weights of the form $0 < f(t)$ where $t \in \mathcal{C}^\infty(M)$ has past timelike differential and $f' < 0$ or even $f' < -cf$) in order to gain positivity of the bulk term $K[u]$. We use the same idea in the proof of the following general energy estimate.

Theorem 9.19 (Energy Estimate). *Let Ω be a compact admissible spacetime domain in the time-oriented spacetime (M, g), with initial and final hypersurfaces $H_1, \dots, H_{N_\mathrm{I}}$ and $H_{N_\mathrm{I}+1}, \dots, H_{N_\mathrm{I}+N_\mathrm{F}}$, respectively. Suppose there exists a function $t \in \mathcal{C}^\infty(\Omega)$ such that* dt *is past timelike. Set*

$$ E_\mathrm{I}[u] := \sum_{j=1}^{N_\mathrm{I}} \big(\|u|_{H_j}\|_{H^1(H_j;E)} + \|\nu_j(u)\|_{L^2(H_j;E)} \big), \tag{9.41} $$

where $\nu_j(u)$ is the derivative of u along ν_j with respect to a smooth connection on E. Define $E_\mathrm{F}[u]$ similarly, now using $H_{N_\mathrm{I}+j}$, $j = 1, \dots, N_\mathrm{F}$. Let $P \in \mathrm{Diff}^2(\Omega; E)$ be a wave-type operator on (Ω, g). Then there exists a constant C such that for all $u \in \mathcal{C}^2(\Omega; E)$, the estimate

$$ \|u\|_{H^1(\Omega;E)} + E_\mathrm{F}[u] \leq C(\|Pu\|_{L^2(\Omega;E)} + E_\mathrm{I}[u]) \tag{9.42} $$

holds. A fortiori, if u vanishes to second order at all initial hypersurfaces of Ω, then

$$ \|u\|_{H^1(\Omega;E)} \leq C\|Pu\|_{L^2(\Omega;E)}. \tag{9.43} $$

The L^2- and H^1-norms in Theorem 9.19 can be defined using local trivializations of E and a partition of unity. Equivalently (up to equivalence of norms), the norm on $L^2(\Omega; E)$ can be defined using any smooth volume density (e.g., $|\mathrm{d}g|$) and positive definite fiber inner product on E; to define the norm on $H^1(\Omega; E)$, fix a finite set $V_1, \dots, V_N \in \mathrm{Diff}^1(\Omega; E)$ of operators which span $\mathrm{Diff}^1(\Omega; E)$ over $\mathcal{C}^\infty(\Omega)$ and set $\|u\|^2_{H^1(\Omega;E)} := \|u\|^2_{L^2(\Omega;E)} + \sum_{i=1}^N \|V_i u\|_{L^2(\Omega;E)}$. The norms on H_j can be defined analogously. (Sobolev spaces on domains with boundary or corners will be discussed in more detail in Section 9.3.)

Note that the estimates (9.42)–(9.43) lose one derivative compared to estimates for elliptic operators.

Proof of Theorem 9.19. We only consider the case that $E = M \times \mathbb{C}$ is the trivial complex rank 1 vector bundle. The general case is notationally more burdensome but conceptually the same and thus left to the reader. Write

$$P = \Box_g + V + a, \quad V \in \mathcal{V}(\Omega),\ a \in \mathcal{C}^\infty(\Omega), \tag{9.44}$$

where we allow V and a to be complex. Fix a future timelike vector field $X_0 \in \mathcal{V}(M)$. (An explicit choice is $X_0 = -\nabla t$.) We plan to use the energy identity (9.31) for the vector field

$$X := w^2 X_0, \quad w := e^{-\alpha t},\ \alpha > 0. \tag{9.45}$$

Using (9.21) and $\frac{\nabla w}{w} = -\alpha \nabla t$, its K-current is given by

$${}^{(X)}K = w^2\Big({}^{(X_0)}K + 2T\Big(\frac{\nabla w}{w}, X_0\Big)\Big) = w^2\big({}^{(X_0)}K + 2\alpha T(-\nabla t, X_0)\big). \tag{9.46}$$

By Proposition 9.11, $T(-\nabla t, X_0)$ is positive definite on Ω. Therefore, the symmetric 2-cotensor ${}^{(X)}K \in \mathcal{C}^\infty(\Omega; S^2T\Omega)$ is positive definite provided we choose α sufficiently large. In order to deal with the subprincipal terms of P, we need to arrange for more positivity; fix thus α_0 such that for all $\alpha \geq \alpha_0$,

$${}^{(X)}K + \frac{1}{2}(V \otimes X + X \otimes \bar{V}) = w^2\Big({}^{(X_0)}K + 2\alpha T(-\nabla t, X_0) + \frac{1}{2}(V \otimes X_0 + X_0 \otimes \bar{V})\Big) \tag{9.47}$$

is positive definite.

We first prove the estimate (9.43). Applying the identity (9.31) to u vanishing to second order at all initial hypersurfaces, the sum on the right-hand side vanishes. The sum on the left-hand side is a sum of integrals of positive definite quadratic forms in $\mathrm{d}u$; we drop it for now. What remains is the inequality

$$\begin{aligned}
&\int_\Omega {}^{(X)}K[u]\,|\mathrm{d}g| \\
&\quad \leq \int_\Omega \mathrm{Re}(\Box_g u\, X\bar{u})\,|\mathrm{d}g| \\
&\quad = \int_\Omega \mathrm{Re}(Pu\, X\bar{u})\,|\mathrm{d}g| - \int_\Omega \mathrm{Re}(Vu\, X\bar{u})\,|\mathrm{d}g| - \int_\Omega \mathrm{Re}(au\, X\bar{u})\,|\mathrm{d}g|.
\end{aligned} \tag{9.48}$$

Upon moving $\int_\Omega \mathrm{Re}(Vu\, X\bar{u})\,|\mathrm{d}g|$ to the left-hand side and exploiting the positive definiteness of (9.47), we conclude that there exists a constant $c > 0$ which is independent of $\alpha \geq \alpha_0$ such that

$$c\|w\,\mathrm{d}u\|^2_{L^2(\Omega;T^*\Omega)} \leq \int_\Omega w|Pu| \cdot w|X_0 u|\,|\mathrm{d}g| + \|a\|_{L^\infty} \int_\Omega w|u| \cdot w|X_0 u|\,|\mathrm{d}g|. \tag{9.49}$$

Applying the Peter–Paul inequality to both integrands and using the elementary bound $\int_\Omega w^2|X_0 u|^2\,|\mathrm{d}g| \le C\|w\,\mathrm{d}u\|^2_{L^2(\Omega;T^*\Omega)}$ (with C independent of α), the right-hand side of (9.49) is further bounded by

$$\left(\frac{1}{4\epsilon}\int_\Omega w^2|Pu|^2\,|\mathrm{d}g| + \epsilon C\|w\,\mathrm{d}u\|^2_{L^2}\right) + \left(\frac{C}{\epsilon}\int_\Omega w^2|u|^2\,|\mathrm{d}g| + \epsilon C\|w\,\mathrm{d}u\|^2_{L^2}\right) \tag{9.50}$$

where C is independent of $\alpha \ge \alpha_0$. Fixing a sufficiently small $\epsilon > 0$, the norms on $w\,\mathrm{d}u$ can be absorbed into the left-hand side of (9.49), and we obtain

$$\|w\,\mathrm{d}u\|^2_{L^2} \le C\left(\int_\Omega w^2|Pu|^2\,|\mathrm{d}g| + \int_\Omega w^2|u|^2\,|\mathrm{d}g|\right), \tag{9.51}$$

where the constant C is independent of $\alpha \ge \alpha_0$.

To complete the proof, it remains to bound $\int_\Omega w^2|u|^2\,|\mathrm{d}g|$ by a small constant ($\frac{1}{2C}$ suffices) times that of $w\,\mathrm{d}u$. We do this using the fundamental theorem of calculus, with the smallness arising from choosing α large enough. We phrase this using the divergence theorem as follows: we have

$$\begin{aligned}\mathrm{div}_g(|u|^2X) &= |u|^2\,\mathrm{div}_g X + X(|u|^2)\\ &= |u|^2\,\mathrm{div}_g(w^2X_0) + 2w^2\,\mathrm{Re}(u\,X_0\bar{u})\\ &= w^2\Big(\Big(\mathrm{div}_g X_0 + 2\frac{X_0 w}{w}\Big)|u|^2 + 2\,\mathrm{Re}(u\,X_0\bar{u})\Big).\end{aligned} \tag{9.52}$$

Since $\mathrm{div}_g X_0 \in \mathcal{C}^\infty(\Omega)$ is bounded and $\frac{X_0 w}{w} = -\alpha(X_0 t) < 0$, there exists a constant $c > 0$ (independent of α) such that, for all sufficiently large α,

$$\mathrm{div}_g(-|u|^2X) \ge \alpha c w^2|u|^2 - 2w^2\,\mathrm{Re}(u\,X_0\bar{u}). \tag{9.53}$$

We apply the divergence theorem (9.38) to the vector field $-|u|^2X$ on Ω. The boundary terms at the initial hypersurfaces of Ω vanish by our present assumption on u, while the boundary terms at the final hypersurfaces are equal to $-\int_{H_{N_1+j}} g(-|u|^2X, \nu_{N_1+j})\,|\mathrm{d}\sigma_{N_1+j}|$, and thus, integrals of negative definite quadratic forms in u; we drop the latter for now. Therefore,

$$c\alpha\int_\Omega w^2|u|^2\,|\mathrm{d}g| - 2\int_\Omega \mathrm{Re}(wu\,wX_0\bar{u})\,|\mathrm{d}g| \le \int_\Omega \mathrm{div}_g(-|u|^2X)\,|\mathrm{d}g| \le 0. \tag{9.54}$$

The Peter–Paul inequality gives

$$c\alpha\int_\Omega w^2|u|^2\,|\mathrm{d}g| \le \int_\Omega w^2|u|^2\,|\mathrm{d}g| + C\|w\,\mathrm{d}u\|^2_{L^2}. \tag{9.55}$$

For $\alpha \geq \frac{2}{c}$, we thus obtain

$$\int_\Omega w^2 |u|^2 \, |\mathrm{d}g| \leq C' \alpha^{-1} \|w \, \mathrm{d}u\|_{L^2}^2 \tag{9.56}$$

for an α-independent constant C'. Adding $(C+1)$ times this estimate to (9.51) and choosing α so large that $(C+1)C'\alpha^{-1} < \frac{1}{2}$, we obtain $\|wu\|_{L^2}^2 + \|w \, \mathrm{d}u\|_{L^2}^2 \leq C \|w \, Pu\|_{L^2}^2$. This implies the desired estimate (9.43).

In order to prove the full estimate (9.42), it is only necessary to keep the boundary terms which we dropped earlier. Concretely, writing $E_{\mathrm{I}}^1[w,u] := \sum_{j=1}^{N_{\mathrm{I}}} \|w \, \mathrm{d}u|_{H_j}\|_{L^2(H_j;T^*\Omega)}^2$ and defining $E_{\mathrm{F}}^1[w,u]$ analogously, the estimate (9.51) can be sharpened to

$$E_{\mathrm{F}}^1[w,u] + \|w \, \mathrm{d}u\|_{L^2}^2 \leq C \left(E_{\mathrm{I}}^1[w,u] + \int_\Omega w^2 |Pu|^2 \, |\mathrm{d}g| + \int_\Omega w^2 |u|^2 \, |\mathrm{d}g| \right). \tag{9.57}$$

Writing $E_{\mathrm{I}}^0[w,u] := \sum_{j=1}^{N_{\mathrm{I}}} \|wu|_{H_j}\|_{L^2(H_j)}^2$ and defining $E_{\mathrm{F}}^0[w,u]$ analogously, the estimate (9.56) can similarly be sharpened to

$$\alpha^{-1} E_{\mathrm{F}}^0[w,u] + \int_\Omega w^2 |u|^2 \, |\mathrm{d}g| \leq C' \alpha^{-1} \left(\|w \, \mathrm{d}u\|_{L^2}^2 + E_{\mathrm{I}}^0[w,u] \right) \tag{9.58}$$

for some constant $C' < \infty$ which is independent of α. Adding $(C+1)$ times this estimate to (9.57) yields (9.42) when α is large enough. □

Generalizing Example 9.18, we conclude in particular that if u vanishes to second order at the initial hypersurfaces of Ω and satisfies the *homogeneous* wave-type equation $Pu = 0$, then $u = 0$ throughout Ω. (Applying this to the difference of two solutions u, v of $Pu = f = Pv$ with the same initial data at the initial hypersurfaces, i.e., $u - v$ vanishes to second order there, we see that $u = v$: smooth solutions of initial value problems for wave-type equations are unique.) This is again a statement of *finite speed of propagation*: waves cannot propagate into Ω through final hypersurfaces. Informally:

Waves do not travel in spacelike directions, i.e. 'not faster than the speed of light.' (9.59)

Note that (9.42) is an *a priori* estimate: it applies only to $u \in \mathcal{C}^\infty(\Omega; E)$. (By a continuity and density argument, it holds also for $u \in H^2(\Omega; E)$.) In particular, it does not, by itself, imply the existence of solutions to $Pu = f$. We shall prove the existence of solutions by duality, starting with the a priori estimate (9.42) applied to P^* (Section 9.4). We address the necessary functional analytic preparations in Section 9.3. Once solvability and uniqueness have been established, the estimate (9.42) can be upgraded

from an a priori estimate to an estimate for the solution of the wave-type equation; see (9.95) for such an estimate when the initial data of u are trivial (so $E_{\mathrm{I}}[u] = 0$) and the final energies $E_{\mathrm{F}}[u] \geq 0$ are dropped.

We end this section by outlining how the sign of the K-current exploited in the heart of the proof of the energy estimate (9.42) relates to the positive commutator arguments of Chapter 8. We consider the scalar wave operator $\Box_g$. For a vector field X, consider then the "commutator" (cf. (8.28))

$$2\,\mathrm{Im}\langle \Box_g u, i^{-1} X u\rangle_{L^2(\Omega)}. \tag{9.60}$$

Ignoring boundary terms, this is equal to $\langle Qu, u\rangle$ where

$$Q := \Box_g X + X^* \Box_g = i[\Box_g, i^{-1}X] - (\mathrm{div}_g\, X)\Box_g \tag{9.61}$$

in view of $X^* = -X - \mathrm{div}_g\, X$. Writing $X = X^j \partial_j$, the principal symbol of Q equals

$$\begin{aligned} \sigma^2(Q)(\zeta) &= H_G(X^j\zeta_j) - (\mathrm{div}_g\, X) g^{-1}(\zeta,\zeta) \\ &= \zeta_j\zeta_k(2g^{k\ell}\partial_\ell X^j - X^\ell \partial_\ell g^{jk}) - (\mathrm{div}_g\, X) g^{jk}\zeta_j\zeta_k \\ &= \zeta_j\zeta_k(g^{k\ell}\partial_\ell X^j + g^{j\ell}\partial_\ell X^k - X^\ell \partial_\ell g^{jk}) - \zeta_j\zeta_k g^{jk} X_{\ell;}{}^{\ell} \\ &= (X^{j;k} + X^{k;j} - X_{\ell;}{}^{\ell} g^{jk})\zeta_j\zeta_k. \end{aligned} \tag{9.62}$$

This is related to the abstract K-current as follows: we have ${}^{(X)}K(\zeta,\zeta) = {}^{(X)}\pi^{jk} T_{jk}(\zeta,\zeta)$ with $T_{jk}(\zeta,\zeta) = \zeta_j\zeta_k - \frac{1}{2} g^{\ell m}\zeta_\ell\zeta_m g_{jk}$, and therefore, using Proposition 9.13,

$$\sigma^2(Q)(\zeta) = 2\,{}^{(X)}K(\zeta,\zeta). \tag{9.63}$$

Unlike in the real principal type setting, however, it is not only the positivity of this principal symbol on the characteristic set that matters in the proof of the energy estimate (9.42), but the positivity for *all* covectors ζ, including those in the elliptic set of $\Box_g$.

9.3 Sobolev Spaces with Extendible and Supported Character

Let M be a smooth manifold. (In this section, we do not need any further structures on M.) We only discuss complex-valued distributions here and leave the notational modifications required to treat bundle-valued distributions to the reader.

Definition 9.20 (Supported and Extendible Distributions). Let $\mathcal{F}(M)$ denote a locally convex space of distributions, with continuous inclusion $\mathcal{F}(M) \to \mathscr{D}'(M)$.

(1) (Supported distributions.) Let $K \subset M$ be closed. Then

$$\dot{\mathcal{F}}(K) := \{u \in \mathcal{F}(M) \colon \operatorname{supp} u \subset K\} \subset \mathscr{D}'(M) \tag{9.64}$$

denotes the space of *supported distributions on* K (of class $\mathcal{F}$).

(2) (Extendible distributions.) Let $U \subset M$ be open. Then

$$\bar{\mathcal{F}}(U) := \{u|_U \colon u \in \mathcal{F}(M)\} \subset \mathscr{D}'(U) \tag{9.65}$$

denotes the space of *extendible distributions on* U (of class $\mathcal{F}$).

Such spaces arise naturally in the study of evolution equations $Pu = f$ on $\mathbb{R}_t \times X$: if $f = 0$ for $t < 0$, and one seeks a solution u with $u = 0$ for $t < 0$, then it is natural to work in $\dot{\mathcal{F}}([0,\infty) \times X)$ for some space $\mathcal{F}$. If one only seeks a solution up to time $t = T > 0$, then near $t = T$ one should work with a $\bar{\mathcal{F}}((-\infty, T) \times X)$ type space. Mixed spaces that encode both support properties (at $t = 0$) and extendible properties (at $t = T$) are introduced in Definition 9.28 below.

We equip $\dot{\mathcal{F}}(K) \subset \mathcal{F}(M)$ with the subspace topology. The restriction map $\mathcal{F}(M) \to \bar{\mathcal{F}}(U)$ has kernel $\dot{\mathcal{F}}(M \setminus U)$ and thus induces an isomorphism

$$\bar{\mathcal{F}}(U) \cong \mathcal{F}(M)/\dot{\mathcal{F}}(M \setminus U); \tag{9.66}$$

we equip $\bar{\mathcal{F}}(U)$ with the quotient topology. When $\Omega = \overline{\Omega^\circ} \subset M$ is a smooth submanifold with corners, one sometimes says that elements of $\dot{\mathcal{F}}(\Omega)$ have *supported character at* $\partial\Omega$, and elements of $\bar{\mathcal{F}}(\Omega^\circ)$ have *extendible character at* $\partial\Omega$. In the sequel, we mainly consider the case

$$\mathcal{F}(M) = H^s_{\mathrm{loc}}(M). \tag{9.67}$$

When $M = \mathbb{R}^n$, one can also consider $\mathcal{F}(\mathbb{R}^n) = H^s(\mathbb{R}^n)$; some properties of $\dot{H}^s(K)$ and $\bar{H}^s(U)$ in the latter case are the subject of Exercises 9.4–9.6. Another case one can consider is $\mathcal{F}(M) = \mathcal{C}^\infty(M)$; in this case, $\bar{\mathcal{C}}^\infty(U)$ is the space of all smooth functions on U which are smoothly extendible to M, and $\dot{\mathcal{C}}^\infty(K)$ is the space of all smooth functions vanishing outside K (and thus, in particular, vanishing to infinite order at ∂K).

Example 9.21 (Extendible and Supported Distributions on the Interval). The function u defined by $u(x) := 1$ for $x \in (0,1)$ is the restriction of a function in $L^2(\mathbb{R})$ to $(0,1)$ and thus defines an element $u \in \bar{L}^2((0,1))$. Its derivative is $u' = 0 \in \bar{H}^{-1}((0,1))$. On the other hand, the characteristic

function $v := 1_{[0,1]} \in L^2(\mathbb{R})$ has support in $[0,1]$, so $v \in \dot{L}^2([0,1])$. Its derivative $v' = \delta_0 - \delta_1 \in \dot{H}^{-1}([0,1])$ is nonzero.

Standard density results hold in the supported and extendible setting:

Proposition 9.22 (Density). *Let $s \in \mathbb{R}$.*

(1) (Extendible case.) *Let $U \subset M$ be open. Then $\bar{\mathcal{C}}^\infty(U) \subset \bar{H}^s_{\mathrm{loc}}(U)$ is dense.*
(2) (Supported case.) *Let $\Omega \subset M$ be a smooth submanifold with corners. Then $\dot{\mathcal{C}}^\infty(\Omega) \subset \dot{H}^s_{\mathrm{loc}}(\Omega)$ is dense.*

Proof. For part (1), let $\{\phi_i\}$, $\phi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$, be a partition of unity subordinate to a cover $M = \bigcup_i U_i$ by coordinate charts $F_i \colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ with $\overline{U_i}$ compact. Fix $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $\int_{\mathbb{R}^n} \psi_i(x)\,\mathrm{d}x = 1$, with support in $\{|x| < \epsilon_i\}$ where $\epsilon_i > 0$ is fixed such that $z \in \operatorname{supp}\phi_i$, $|x| < \epsilon_i$ implies $F_i(z) + x \in F_i(U_i)$. Set $\psi_{i,\epsilon}(x) := \epsilon^{-n}\psi_i(x/\epsilon)$. For every $\tilde{u} \in H^s_{\mathrm{loc}}(M)$, we then have

$$\tilde{u}_\epsilon := \sum_i F_i^*(\psi_{i,\epsilon} * (F_i^{-1})^*(\phi_i \tilde{u})) \to \tilde{u} \tag{9.68}$$

in $H^s_{\mathrm{loc}}(M)$ as $\epsilon \to 0$. Given $u \in \bar{H}^s_{\mathrm{loc}}(U)$ and an extension $\tilde{u} \in H^s_{\mathrm{loc}}(M)$ of u, we therefore have $u_\epsilon := \tilde{u}_\epsilon|_U \in \bar{\mathcal{C}}^\infty(U)$ and $u_\epsilon \to u$ in $\bar{H}^s_{\mathrm{loc}}(U)$.

For part (2), one can again reduce the proof to a local result for $\Omega = [0,\infty)^k \times \mathbb{R}^{n-k} \subset \mathbb{R}^n$ using a partition of unity. Let thus $u \in H^s(\mathbb{R}^n)$ with $\operatorname{supp} u \subset \Omega$; then

$$u_\delta(x_1, \dots, x_k, y_1, \dots, y_{n-k}) := u(x_1 - \delta, \dots, x_k - \delta, y_1, \dots, y_{n-k}) \tag{9.69}$$

lies in $H^s(\mathbb{R}^n)$, is supported in $[\delta,\infty)^k \times \mathbb{R}^{n-k}$, and converges to u in $H^s(\mathbb{R}^n)$ as $\delta \to 0$. Fix $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(B(0,1))$ with $\int_{\mathbb{R}^n} \psi(x)\,\mathrm{d}x = 1$ and set $\psi_\epsilon(x) := \epsilon^{-n}\psi(x/\epsilon)$. Then $u_{\delta,\epsilon} := \psi_\epsilon * u_\delta \to u_\delta$ in $H^s(\mathbb{R}^n)$ as $\epsilon \to 0$; but $\operatorname{supp} u_{\delta,\epsilon} \subset \Omega$ when $\epsilon \le \delta$, so $u_{\delta,\epsilon} \in \dot{\mathcal{C}}^\infty(\Omega)$ then. □

For (pre)compact $U, K \subset M$, the spaces $\dot{H}^s(K)$ and $\bar{H}^s(U)$ can be given the structure of Hilbert spaces. For the former space, this was already shown in Proposition 5.58.

Proposition 9.23 (Hilbert Space Structure: Extendible Case). *Let $U \subset M$ be a precompact open subset. Then $\bar{H}^s(U) := \bar{H}^s_{\mathrm{loc}}(U)$ is Hilbertizable.*

Proof. Fix a compact subset $K \subset M$ containing the closure of U in its interior. Fix $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(K^\circ)$ with $\chi = 1$ on $\bar{U}$. The continuous map $\dot{H}^s(K) \to \bar{H}^s(U)$ given by restriction to U is surjective, for if $u \in \bar{H}^s(U)$ and $\tilde{u} \in H^s_{\mathrm{loc}}(M)$ is an extension of u, then $\chi\tilde{u} \in \dot{H}^s(K)$ is an extension as well. Since this map has kernel $\dot{H}^s(K \setminus U)$, we obtain an isomorphism

$$\dot{H}^s(K)/\dot{H}^s(K \setminus U) \cong \bar{H}^s(U) \tag{9.70}$$

of Fréchet spaces. By Proposition 5.58, the left-hand side is a Hilbertizable space, and thus so is $\bar{H}^s(U)$. □

It is useful to know that for non-negative integer orders $s \in \mathbb{N}_0$ and domains with smooth boundary or corners, the spaces $\dot{H}^s(K)$ and $\bar{H}^s(U)$ have an intrinsic characterization, i.e., one that does not refer to the ambient manifold M. This is straightforward for $\dot{H}^s(K)$: a function $u \in \dot{L}^2(K)$ lies in $\dot{H}^s(K)$ if and only if $Pu \in \dot{L}^2(K)$ for all $P \in \mathrm{Diff}^s(M)$. The result for $\bar{H}^s(U)$ reads as follows.

Proposition 9.24 (Intrinsic Characterization: Integer Orders). *Suppose $\Omega = \overline{\Omega^\circ} \subset M$ is a smooth compact submanifold with corners. Fix a volume density on Ω to define $L^2(\Omega)$. Let $s \in \mathbb{N}_0$, and let $\{P_i \colon i = 1, \dots, N\} \subset \mathrm{Diff}^s(\Omega)$ be a set of differential operators which spans $\mathrm{Diff}^s(\Omega)$ as a $\mathcal{C}^\infty(\Omega)$-module. Then $\bar{H}^s(\Omega^\circ)$ is equal to the space $H^s(\Omega^\circ)$ of distributions on Ω° with finite squared norm*

$$\|u\|^2_{H^s(\Omega^\circ)} := \sum_{i=1}^{N} \|P_i u\|^2_{L^2(\Omega^\circ)}, \tag{9.71}$$

and the topology induced by this norm is equal to the standard topology on $\bar{H}^s(\Omega^\circ)$.

The idea of the proof is to extend elements of $H^s(\Omega^\circ)$ to H^s-functions defined and supported in a neighborhood of Ω. We begin with a local extension result, stated here in a strong form due to Seeley [See64]:

Theorem 9.25 (Extension Operators: Local). *Let $c > 0$. There exists a continuous linear map $E \colon \mathcal{C}^\infty([0,\infty)) \to \mathcal{C}^\infty(\mathbb{R})$ with the following properties.*

(1) (Extension and support.) *For $u \in \mathcal{C}^\infty([0,\infty))$, we have $(Eu)|_{[0,\infty)} = u$ and $\mathrm{supp}(Eu) \subset [-c, \infty)$.*
(2) (Boundedness.) *For all $s \in \mathbb{N}_0$, the map E extends by continuity and density from a map $\mathcal{C}^\infty_{\mathrm{c}}([0,\infty)) \to \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ to a bounded linear map*

$$\begin{aligned} H^s((0,\infty)) &:= \{u \in L^2((0,\infty)) \colon u^{(j)} \in L^2((0,\infty)) \ \forall\, j = 0, \dots, s\} \\ &\to H^s(\mathbb{R}). \end{aligned} \tag{9.72}$$

For $0 \le j \le k \le n$, there exist analogous extension operators $E_{k,j}$ mapping functions on $[0,\infty)^k \times \mathbb{R}^{n-k}$ to functions on $[0,\infty)^j \times \mathbb{R}^{n-j}$ with support in $[0,\infty)^j \times [-c,\infty)^{k-j} \times \mathbb{R}^{n-k}$.

Proof. Fix a function $\chi \in \mathcal{C}^\infty([0,c))$ which equals 1 near 0. For constants $c_\ell \in \mathbb{R}$ and $\epsilon_\ell \in (0,1]$, with $\epsilon_\ell \to 0$, specified later, set

$$(Eu)(x) := \begin{cases} u(x), & x \geq 0, \\ \sum_{\ell=0}^{\infty} c_\ell (\chi u)\big(-\frac{x}{\epsilon_\ell}\big), & x < 0. \end{cases} \tag{9.73}$$

Since for each $x < 0$ the sum defining $(Eu)(x)$ is finite, Eu is smooth in $x \neq 0$. Furthermore, the j-th derivative of the ℓ-th summand obeys the bound

$$\sup_{x<0} \left| c_\ell (-\epsilon_\ell)^{-j} \frac{\mathrm{d}^j}{\mathrm{d}x^j} (\chi u)\Big(-\frac{x}{\epsilon_\ell}\Big) \right| \leq C_j |c_\ell| \epsilon_\ell^{-j}, \quad C_j := \|\chi u\|_{\mathcal{C}^j([0,\infty))} < \infty. \tag{9.74}$$

Thus, if the c_ℓ satisfy

$$\sum_{\ell=0}^{\infty} |c_\ell| \epsilon_\ell^{-j} < \infty, \quad j \in \mathbb{N}_0, \tag{9.75}$$

then $(Eu)|_{x<0} \in \mathcal{C}^\infty((-\infty, 0))$ extends smoothly to $(-\infty, 0]$. We compute its j-th derivative at 0 to be equal to

$$\frac{\mathrm{d}^j}{\mathrm{d}x^j} (Eu)(x)\Big|_{x=0-} = \sum_{\ell=0}^{\infty} c_\ell (-\epsilon_\ell)^{-j} \frac{\mathrm{d}^j u}{\mathrm{d}x^j}(0). \tag{9.76}$$

For this to equal $\frac{\mathrm{d}^j u}{\mathrm{d}x^j}(0)$, we must require

$$\sum_{\ell=0}^{\infty} c_\ell \epsilon_\ell^{-j} = (-1)^j, \quad j \in \mathbb{N}_0. \tag{9.77}$$

We arrange (9.75) and (9.77) by setting $\epsilon_\ell := 3^{-\ell}$ and defining c_ℓ such that

$$\sin\left(\frac{\pi}{2} z\right) = \sum_{\ell=0}^{\infty} c_\ell z^\ell. \tag{9.78}$$

(Explicitly, $c_{2m} = 0$ and $c_{2m+1} = (-1)^m \frac{(\pi/2)^{2m+1}}{(2m+1)!}$.) Then $\sum_{\ell=0}^{\infty} c_\ell \epsilon_\ell^{-j} = \sin(\frac{\pi}{2} 3^j) = (-1)^j$ indeed since $3^j \equiv (-1)^j \bmod 4$, so (9.77) holds; and (9.75) follows from the absolute convergence of (9.78) for all $z \in \mathbb{C}$ (in particular for $z = 3^j$).

Turning to part (2), we estimate the j-th derivative of the ℓ-th summand of $(Eu)|_{x<0}$ for $u \in \mathcal{C}_{\mathrm{c}}^\infty([0,\infty))$ by

$$\left(\int_{-\infty}^{0} \left| c_\ell (-\epsilon_\ell)^{-j} \frac{\mathrm{d}^j}{\mathrm{d}x^j} (\chi u)\Big(-\frac{x}{\epsilon_\ell}\Big) \right|^2 \mathrm{d}x \right)^{\frac{1}{2}} \leq |c_\ell| \epsilon_\ell^{-j+\frac{1}{2}} \|u\|_{H^j((0,c))}. \tag{9.79}$$

Therefore, $\|(Eu)|_{x<0}\|_{H^s((-\infty,0))} \le C_s\|u\|_{H^s((0,c))}$ and thus

$$\|Eu\|_{H^s(\mathbb{R})} \le C_s\|u\|_{H^s((0,\infty))}. \tag{9.80}$$

The density of $\mathcal{C}^\infty_{\mathrm{c}}([0,\infty)) \subset H^s((0,\infty))$ follows easily from this construction by *defining* $E\colon H^s((0,\infty)) \to H^s(\mathbb{R})$ via (9.73), using the density of $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}) \subset H^s(\mathbb{R})$ to approximate Eu in $H^s(\mathbb{R})$ by compactly supported smooth functions, and then restricting these smooth functions back to $[0,\infty)$.

Extension operators $E_k\colon \mathcal{C}^\infty([0,\infty)^k \times \mathbb{R}^{n-k}) \to \mathcal{C}^\infty([0,\infty)^{k-1} \times \mathbb{R}^{n-k+1})$ can be defined via

$$\begin{aligned}&(E_k u)(x_1,\dots,x_{k-1},x,y_1,\dots,y_{n-k}) := (Eu_{x',y})(x),\\ &u_{x',y}(x) := u(x',x,y) \in \mathcal{C}^\infty([0,\infty)),\ x' = (x_1,\dots,x_{k-1}),\ y = (y_1,\dots,y_{n-k}).\end{aligned} \tag{9.81}$$

Since the aforementioned estimates on E apply verbatim with additional smooth dependence on x', y, this indeed maps $\mathcal{C}^\infty([0,\infty)^k \times \mathbb{R}^{n-k}) \to \mathcal{C}^\infty([0,\infty)^{k-1} \times \mathbb{R}^{n-k+1})$ and is bounded on every Sobolev space of non-negative integer order. The concatenation $E_{k,j} := E_{j+1} \circ E_{j+2} \circ \cdots \circ E_{k-1} \circ E_k$ is the desired map $\mathcal{C}^\infty([0,\infty)^k \times \mathbb{R}^{n-k}) \to \mathcal{C}^\infty([0,\infty)^j \times \mathbb{R}^{n-j})$. □

Corollary 9.26 (Extension Operators: Submanifolds with Corners). *Let M be a smooth manifold, and let $\Omega \subset M$ be a smooth compact submanifold with corners. Let $K \subset M$ be a compact set containing Ω in its interior. Then there exists a continuous linear map $E\colon \mathcal{C}^\infty(\Omega) \to \dot{\mathcal{C}}^\infty(K)$ which extends by continuity and density to a bounded linear map*

$$E\colon H^s(\Omega^\circ) \to \dot{H}^s(K) \tag{9.82}$$

for all $s \in \mathbb{N}_0$.

Proof. Fix a partition of unity $\{\phi_i\}$, $i = 1,\dots,N$, $\phi_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$, with $\sum_{i=1}^N \phi_i = 1$ on Ω, which is subordinate to coordinate charts $F_i\colon U_i \to F_i(U_i) \subset \mathbb{R}^n$ on M in each of which Ω is a product $[0,\infty)^{k_i} \times \mathbb{R}^{n-k_i}$ (cf. (5.16)). By multiplying each ϕ_i with a cutoff function which equals 1 on Ω and is supported in K°, we can arrange that $\operatorname{supp}\phi_i \subset K^\circ$ for all i. Fix $\tilde{\phi}_i \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$ with $\tilde{\phi}_i = 1$ on $\operatorname{supp}\phi_i$ and $\operatorname{supp}\tilde{\phi}_i \subset K$. Write $E_{k_i} := E_{k_i,0}$ in the notation of Theorem 9.25. The map

$$Eu := \sum_{i=1}^N \tilde{\phi}_i F_i^* \Big(E_{k_i}\big((F_i^{-1})^*(\phi_i u)\big)\Big) \tag{9.83}$$

then satisfies all requirements. The density of $\mathcal{C}^\infty(\Omega) \subset H^s(\Omega^\circ)$ can be proved by smoothing out the extension $Eu \in \dot{H}^s(K)$ of a given element $u \in H^s(\Omega^\circ)$ in local coordinates using a partition of unity and restricting the smoothed-out functions back to Ω. □

The proof of Proposition 9.24 is now straightforward.

Proof of Proposition 9.24. The identity map

$$\bar{H}^s(\Omega^\circ) \to H^s(\Omega^\circ) \tag{9.84}$$

is continuous. Fix a compact set $K \subset M$ which contains Ω in its interior. Denote by E the extension map from Corollary 9.26. Then the inverse of (9.84) is given by the continuous map $H^s(\Omega^\circ) \ni u \mapsto (Eu)|_{\Omega^\circ} \in \bar{H}^s(\Omega^\circ)$. □

Next, we discuss dual spaces with respect to L^2. We have the following analogue of Theorem 5.63. (We leave it to the reader to formulate a version for non-compact Ω.)

Theorem 9.27 (Dual Spaces). *Fix a volume density μ on M. Let $\Omega \subset M$ be a smooth compact submanifold with corners. Let $s \in \mathbb{R}$. Then the sesquilinear form*

$$\dot{\mathcal{C}}^\infty(\Omega) \times \bar{\mathcal{C}}^\infty(\Omega^\circ) \ni (u, v) \mapsto \langle u, v\rangle_{L^2} := \int_{\Omega^\circ} u(x)\overline{v(x)}\,\mathrm{d}\mu(x) \in \mathbb{C} \tag{9.85}$$

extends, by continuity and density, to a continuous sesquilinear form $\dot{H}^s(\Omega) \times \bar{H}^{-s}(\Omega^\circ) \to \mathbb{C}$. The maps $\dot{H}^s(\Omega) \ni u \mapsto \langle u, \bar{\cdot}\rangle_{L^2} \in \bar{H}^{-s}(\Omega^\circ)^*$ and $\bar{H}^{-s}(\Omega^\circ) \ni v \mapsto \langle \cdot, \bar{v}\rangle_{L^2} \in \dot{H}^s(\Omega)^*$ are isomorphisms.

Thus,

$$\dot{H}^s(\Omega)^* = \bar{H}^{-s}(\Omega^\circ), \qquad \bar{H}^s(\Omega^\circ)^* = \dot{H}^{-s}(\Omega). \tag{9.86}$$

Proof of Theorem 9.27. Theorem 5.63 implies that the L^2-pairing

$$\dot{H}^s(\Omega) \times H^{-s}_{\mathrm{loc}}(M) \to \mathbb{C}, \quad (u, v) \mapsto \langle u, v\rangle_{L^2}, \tag{9.87}$$

is well-defined. If $\operatorname{supp} v \subset M \setminus \Omega^\circ$, we can approximate v in $H^{-s}_{\mathrm{loc}}(M)$ by elements $v_\delta \in H^{-s}_{\mathrm{loc}}(M)$ with $\operatorname{supp} v_\delta \subset M \setminus \Omega$ using a shifting argument as in (9.69). The pairing $\langle u, v_\delta\rangle_{L^2}$ vanishes since it is the limit of pairings $\langle u_\epsilon, v_{\delta,\epsilon}\rangle_{L^2}$ of smoothed-out versions $u_\epsilon \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ and $v_{\delta,\epsilon} \in \mathcal{C}^\infty(M)$ as $\epsilon \to 0$. But we can arrange that, for all sufficiently small $\epsilon > 0$, $\operatorname{supp} u_\epsilon \cap \operatorname{supp} v_{\delta,\epsilon} = \emptyset$, and hence, $\langle u_\epsilon, v_{\delta,\epsilon}\rangle_{L^2} = 0$. Therefore, $\langle u, v\rangle_{L^2} = 0$ for $v \in \dot{H}^{-s}_{\mathrm{loc}}(M \setminus \Omega^\circ)$. We conclude that (9.87) descends to a well-defined pairing $\dot{H}^s(\Omega) \times \bar{H}^{-s}(\Omega^\circ) \to \mathbb{C}$.

Consider now a continuous linear functional $\lambda\colon \dot{H}^s(\Omega) \to \mathbb{C}$. By the Hahn–Banach theorem, we can extend λ to a continuous linear functional $\tilde{\lambda}\colon H^s_{\mathrm{c}}(M) \to \mathbb{C}$ which is thus given by the L^2-pairing with some $\tilde{v} \in H^{-s}_{\mathrm{loc}}(M)$. (One could equally well extend λ to a continuous functional on $H^s_{\mathrm{loc}}(M) \to \mathbb{C}$,

which is thus given by the L^2-pairing with some $\tilde{v} \in H_{\mathrm{c}}^{-s}(M)$.) Therefore, $\lambda(u) = \langle u, \tilde{v}\rangle_{L^2}$ equals $\langle u, v\rangle_{L^2}$ where we set $v := \tilde{v}|_{\Omega^\circ} \in \bar{H}^{-s}(\Omega^\circ)$.

Finally, consider a continuous linear functional $\lambda\colon \bar{H}^{-s}(\Omega^\circ) \to \mathbb{C}$. The composition

$$\tilde{\lambda}\colon H_{\mathrm{loc}}^{-s}(M) \xrightarrow{u\mapsto u|_{\Omega^\circ}} \bar{H}^{-s}(\Omega^\circ) \xrightarrow{\lambda} \mathbb{C} \tag{9.88}$$

is a continuous linear functional and thus given by the L^2-pairing with some $u \in H_{\mathrm{c}}^{s}(M)$. But since $\langle u, v\rangle_{L^2} = 0$ for all $v \in \dot{H}_{\mathrm{loc}}^{-s}(M \setminus \Omega^\circ) \subset \ker\tilde{\lambda}$, we conclude that $\operatorname{supp} u \subset \Omega$, so $u \in \dot{H}^s(\Omega)$. □

One can avoid working with $H_{\mathrm{loc}}^{-s}(M)$ and $H_{\mathrm{c}}^{s}(M)$ in the proof above by replacing M with a compact manifold containing the diffeomorphic image of a neighborhood of Ω. Such a replacement does not affect the topologies on $\dot{H}^s(\Omega)$ and $\bar{H}^{-s}(\Omega^\circ)$, and one can now work with (the Hilbert spaces) $H^{\pm s}(M)$ whose L^2-duals are $H^{\mp s}(M)$ by (5.154).

As motivated after Definition 9.20, we finally introduce spaces with supported and extendible characters at different boundary hypersurfaces:

Definition 9.28 (Mixed Spaces). Let $\Omega \subset M$ be a compact subset of the form $\Omega = \bigcap_{j=1}^{N}\{f_j \geq 0\}$ where $f_1, \dots, f_N \in \mathcal{C}^\infty(M)$ are such that at every $p \in \Omega$, the differentials of all f_j vanishing at p are linearly independent, and such that $H_j := \Omega \cap f_j^{-1}(0)$ is connected for all j. Let $N_{\mathrm{I}} \in \{0, \dots, N\}$. Then the space

$$H^s(\Omega)^{\bullet,-} \tag{9.89}$$

of *distributions of class* H^s *which are of supported character at* $H_1, \dots, H_{N_{\mathrm{I}}}$ *and of extendible character at* $H_{N_{\mathrm{I}}+1}, \dots, H_N$ is defined as follows. Let $\delta > 0$ be such that the conditions on the f_j are valid also for $f_{j,\delta} = f_j + \delta$ and such that $\Omega_\delta := \bigcap_{j=1}^{N}\{f_{j,\delta} \geq 0\}$ is compact. Let $U := \bigcap_{j=N_{\mathrm{I}}+1}^{N}\{f_j > 0\}$. Then

$$H^s(\Omega)^{\bullet,-} := \{u|_U\colon u \in \dot{H}^s(\Omega_\delta)\}, \tag{9.90}$$

equipped with the quotient topology.

The topology on $H^s(\Omega)^{\bullet,-}$ is independent of the choice of δ. Moreover, since $\dot{H}^s(\Omega_\delta)$ is Hilbertizable, so is $H^s(\Omega)^{\bullet,-}$.

If $N_{\mathrm{I}} = 0$, then $H^s(\Omega)^{\bullet,-} = \bar{H}^s(\Omega^\circ)$, whereas if $N_{\mathrm{I}} = N$, then $H^s(\Omega)^{\bullet,-} = \dot{H}^s(\Omega)$. Pedantic readers may prefer to write $H^s(\Omega')^{\bullet,-}$ for the space (9.90), where $\Omega' := \bigcap_{j=1}^{N_{\mathrm{I}}}\{f_j \geq 0\} \cap \bigcap_{j=N_{\mathrm{I}}+1}^{N}\{f_j > 0\}$, but we shall not do so to simplify the notation.

Example 9.29 (Initial Value Problem for an ODE). We illustrate how to use mixed spaces to encode the direction in which one solves an evolution equation. Consider $\Omega = [0, 1]$ with initial hypersurface $\{0\}$ and

final hypersurface $\{1\}$. Consider the ODE

$$u' = f \in H^{s-1}([0,1])^{\bullet,-} \tag{9.91}$$

(or, more pedantically, $f \in H^{s-1}([0,1))^{\bullet,-}$). Upon extending f to $\tilde{f} \in H^{s-1}(\mathbb{R})$ with $\operatorname{supp}\tilde{f} \subset [0,\infty)$, we can integrate $\tilde{f}$ starting at 0 to produce $\tilde{u} \in H^s_{\mathrm{loc}}(\mathbb{R})$ with $\operatorname{supp}\tilde{u} \subset [0,\infty)$ and $\tilde{u}' = \tilde{f}$. Therefore $u = \tilde{u}|_{(-\infty,1)} \in H^s([0,1])^{\bullet,-}$ solves (9.91). Moreover, we have the following strong uniqueness statement: if $v \in H^{-\infty}([0,1])^{\bullet,-}$, i.e., $v \in H^{s'}([0,1])^{\bullet,-}$ for some $s' \in \mathbb{R}$, and $v' = f$, then $v = u$. Indeed, $v-u$ vanishes on $(-\infty,0)$ and satisfies $(v-u)' = 0$ on $(-\infty,1)$, and hence $v = u$ on $(-\infty,1)$.—If $f \in \mathcal{C}^\infty_{\mathrm{c}}((0,1))$, this argument produces the unique forward solution $u(x) = \int_0^x f(y)\,\mathrm{d}y$ of (9.91). For such f, one could alternatively solve $u'_- = f$ in $H^s([0,1])^{-,\bullet}$; this would yield a solution $u_- \in H^s([0,1])^{-,\bullet}$, unique in $H^{-\infty}([0,1])^{-,\bullet}$, which is explicitly given by the *backward* solution $u_-(x) = \int_1^x f(y)\,\mathrm{d}y$.

Example 9.30 (Energy Estimate). The energy estimate (9.43) implies

$$\|u\|_{H^1(\Omega;E)^{\bullet,-}} \leq C\|Pu\|_{L^2(\Omega;E)^{\bullet,-}}, \quad u \in \mathcal{C}^2(\Omega;E)^{\bullet,-}. \tag{9.92}$$

The a priori membership $u \in \mathcal{C}^2(\Omega;E)^{\bullet,-}$ encodes the *third* order vanishing at the initial hypersurface of Ω (which is thus stronger than necessary). The norms in (9.92) are equal to the norms used in (9.43) by a simple generalization of Proposition 9.24.

A simple modification of the proof of Theorem 9.27 yields the following result, the proof of which is left to the reader.

Theorem 9.31 (Dual Spaces). *In the notation of Definition 9.28, we have*

$$(H^s(\Omega)^{\bullet,-})^* \cong H^{-s}(\Omega)^{-,\bullet}, \tag{9.93}$$

where $H^{-s}(\Omega)^{-,\bullet}$ is the space of distributions of class H^{-s} which are of extendible character at $H_1,\dots,H_{N_{\mathrm{I}}}$ and of supported character at $H_{N_{\mathrm{I}}+1},\dots,H_N$. The isomorphism is induced by any choice of a volume density on M by continuity and density from $\mathcal{C}^\infty(\Omega)^{\bullet,-} \times \mathcal{C}^\infty(\Omega)^{-,\bullet} \ni (u,v) \mapsto \langle u,v\rangle_{L^2(\Omega)}$.

For solving PDEs using duality arguments, it is important to understand how integration by parts works in L^2-pairings $H^s(\Omega)^{\bullet,-} \times H^{-s}(\Omega)^{-,\bullet} \ni (u,v) \mapsto \langle u,v\rangle_{L^2}$.

Lemma 9.32 (Integration by Parts). *We use the notation of Definition 9.28. Fix a volume density on M. Let $P \in \mathrm{Diff}^m(\Omega)$, and let $s \in \mathbb{R}$. Then*

$$\langle Pu,v\rangle_{L^2} = \langle u,P^*v\rangle_{L^2} \quad \forall\, u \in H^s(\Omega)^{\bullet,-},\ v \in H^{-s+m}(\Omega)^{-,\bullet}. \tag{9.94}$$

Proof. By continuity and density, it suffices to prove (9.94) for $u \in \mathcal{C}^\infty(\Omega)^{\bullet,-}$ and $v \in \mathcal{C}^\infty(\Omega)^{-,\bullet}$. Integration by parts then does not produce any boundary terms since at each boundary hypersurface of Ω at least one of u and v vanishes to infinite order. □

9.4 Regularity, Uniqueness, and Solvability

The main goal of this section is to prove the following result on the well-posedness of forward problems for wave-type equations.

Theorem 9.33 (Solving Wave-Type Equations). *Let Ω be a compact admissible spacetime domain (Definition 9.14) in the time-oriented spacetime (M, g). Suppose there exists a function $t \in \mathcal{C}^\infty(\Omega)$ with timelike differential. Let $P \in \mathrm{Diff}^2(M; E)$ be a wave-type operator on (M, g) (Definition 9.8). Let $s \in \mathbb{R}$ and write $H^s(\Omega; E)^{\bullet,-}$ for the space of distributional sections of E of class H^s which are of supported, resp. extendible character at the initial, resp. final hypersurfaces of Ω. Let $f \in H^{s-1}(\Omega; E)^{\bullet,-}$. Then:*

(1) (Existence.) *There exists $u \in H^s(\Omega; E)^{\bullet,-}$ with $Pu = f$. Moreover, there exists a constant C (independent of f) such that*

$$\|u\|_{H^s(\Omega;E)^{\bullet,-}} \leq C\|f\|_{H^{s-1}(\Omega;E)^{\bullet,-}}. \tag{9.95}$$

(2) (Uniqueness.) *If $v \in H^{-\infty}(\Omega; E)^{\bullet,-}$ (i.e., $v \in H^{s'}(\Omega; E)^{\bullet,-}$ for some $s' \in \mathbb{R}$) and $Pv = f$, then $v = u$.*

The solution u is called the *forward solution*. It is a simple consequence of this result and Sobolev embedding that if the source term $f \in \mathcal{C}^\infty(\Omega; E)^{\bullet,-}$ is smooth, then the forward solution is smooth as well, i.e., $u \in \mathcal{C}^\infty(\Omega; E)^{\bullet,-}$. A variant of Theorem 9.33 on the continuation of solutions and persistence of regularity will be discussed in Corollary 9.38.

For notational brevity, we will only discuss the complex-valued case, i.e., $E = M \times \mathbb{C}$ is the trivial bundle. To prepare the proof, we introduce the following notation.

(1) We write $\Omega = \bigcap_{j=1}^N \{f_j \geq 0\}$ where the $f_j \in \mathcal{C}^\infty(M)$ satisfy the conditions of Definition 9.14, with each $H_j := \Omega \cap f_j^{-1}(0)$ non-empty. We order the f_j so that $H_1, \ldots, H_{N_\mathrm{I}}$ are the initial hypersurfaces and $H_{N_\mathrm{I}+1}, \ldots, H_N$ the final hypersurfaces of Ω.

(2) Fix $\delta > 0$ such that the conditions of Definition 9.14 are satisfied also for the functions $f_{j,\delta} := f_j + \delta_j$ for any choice of $\delta_j \in [-\delta, \delta]$. For $\delta_\mathrm{I}, \delta_\mathrm{F} \in [-\delta, \delta]$, set

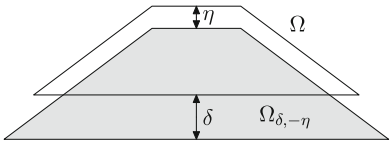


Figure 9.3 An admissible spacetime domain Ω and a modification $\Omega_{\delta,-\eta}$, $\delta, \eta > 0$.

$$\Omega_{\delta_{\mathrm{I}},\delta_{\mathrm{F}}} := \bigcap_{j=1}^{N_{\mathrm{I}}} \{f_{j,\delta_{\mathrm{I}}} \geq 0\} \cap \bigcap_{j=N_{\mathrm{I}}+1}^{N} \{f_{j,\delta_{\mathrm{F}}} \geq 0\}. \tag{9.96}$$

Thus, for $\delta_{\mathrm{I/F}} > 0$ one enlarges the domain near the initial/final hypersurfaces, whereas for $\delta_{\mathrm{I/F}} < 0$ one shrinks it. See Figure 9.3.

(3) $H^s(\Omega)^{-,\bullet}$ denotes the space of distributions of class H^s with extendible, resp. supported character at the initial, resp. final hypersurfaces of Ω; similarly for $H^s(\Omega_{\delta_{\mathrm{I}},\delta_{\mathrm{F}}})^{-,\bullet}$.

The proof strategy is as follows:

(1) We first prove existence for $s = 0$ using a duality argument (Lemma 9.34).
(2) We prove higher regularity using Theorem 8.10 (propagation of singularities). Uniqueness follows from the smoothness of elements in the nullspace of P on $H^0(\Omega)^{\bullet,-}$ and the energy estimate (9.43) (Lemma 9.35).
(3) Since Theorem 8.10 applies only on open manifolds, it does not give uniform estimates near the final hypersurfaces of Ω. To remedy this, we introduce an extension/restriction argument (Lemma 9.36 and Corollary 9.37). This means that we use the already established solvability for $s = 0$ to extend a solution of the wave equation with regular forcing to the solution of a wave equation on a larger domain, with an extended and equally regular forcing; propagation of singularities and microlocal elliptic regularity results then give H^s-control on an open neighborhood of the original domain Ω. Restriction back to Ω yields the desired higher regularity estimate.
(4) We deal with $s < 0$ by applying the uniqueness and regularity statements proved thus far to P^* and using a duality argument.

We begin with the basic existence result for $s = 0$.

Lemma 9.34 (Existence: $s = 0$). *There exists a constant $C < \infty$ such that for all $f \in H^{-1}(\Omega)^{\bullet,-}$ there exists $u \in H^0(\Omega)^{\bullet,-}$ with $Pu = f$ and*

$$\|u\|_{H^0(\Omega)^{\bullet,-}} \leq C\|f\|_{H^{-1}(\Omega)^{\bullet,-}}. \tag{9.97}$$

Proof. We apply Theorem 9.19 to the wave-type operator P^* and the opposite time orientation of (M, g), i.e., the one for which $\mathrm{d}t$ is *future* timelike; thus, initial and final hypersurfaces of Ω exchange roles. In present notation, we thus have the estimate

$$\|v\|_{H^1(\Omega)^{-,\bullet}} \leq C\|P^*v\|_{L^2(\Omega)^{-,\bullet}}, \quad v \in \mathcal{C}^2(\Omega)^{-,\bullet}. \tag{9.98}$$

(See also Example 9.30.) Consider now the linear functional

$$\lambda\colon P^*v \mapsto \langle v, f\rangle_{L^2}, \quad v \in \mathcal{C}^2(\Omega)^{-,\bullet}. \tag{9.99}$$

Since $\mathcal{C}^2(\Omega)^{-,\bullet} \subset H^2(\Omega)^{-,\bullet} \subset H^1(\Omega)^{-,\bullet}$, this is well-defined by Theorem 9.31. Moreover, using (9.98), we have

$$|\lambda(P^*v)| \leq C'\|v\|_{H^1(\Omega)^{-,\bullet}}\|f\|_{H^{-1}(\Omega)^{\bullet,-}} \leq CC'\|P^*v\|_{L^2(\Omega)^{-,\bullet}}\|f\|_{H^{-1}(\Omega)^{\bullet,-}}. \tag{9.100}$$

By the Hahn–Banach theorem, λ therefore extends to a continuous linear functional on $L^2(\Omega)^{-,\bullet}$ with norm bounded by $CC'\|f\|_{H^{-1}(\Omega)^{\bullet,-}}$. Since the dual space of $L^2(\Omega)^{-,\bullet}$ is $L^2(\Omega)^{\bullet,-}$, there thus exists $u \in L^2(\Omega)^{\bullet,-} = H^0(\Omega)^{\bullet,-}$ such that $\lambda(w) = \langle w, u\rangle_{L^2}$ for all $w \in L^2(\Omega)^{-,\bullet}$. For $w = P^*v$, $v \in \mathcal{C}^2(\Omega)^{-,\bullet}$, this means

$$\langle P^*v, u\rangle_{L^2} = \langle v, f\rangle_{L^2}. \tag{9.101}$$

By Lemma 9.32, the left-hand side is equal to $\langle v, Pu\rangle_{L^2}$. Therefore,

$$\langle v, Pu - f\rangle_{L^2} = 0 \quad \forall\, v \in \mathcal{C}^2(\Omega)^{-,\bullet}. \tag{9.102}$$

Since $Pu - f \in H^{-2}(\Omega)^{\bullet,-}$, this continues to hold (by continuity and density) for all $v \in H^2(\Omega)^{-,\bullet}$. Hence, $Pu - f = 0 \in (H^2(\Omega)^{-,\bullet})^* = H^{-2}(\Omega)^{\bullet,-}$. □

Lemma 9.35 (Uniqueness). *Suppose $u, v \in H^{-\infty}(\Omega)^{\bullet,-}$ satisfy $Pu = Pv$. Then $u = v$.*

Proof. It suffices to show that $Pu = 0$ implies $u = 0$ when $u \in H^{s'}(\Omega)^{\bullet,-}$ for some $s' \in \mathbb{R}$. We extend u by 0 to $\Omega_{\delta,0}$; thus $u \in H^{s'}(\Omega_{\delta,0})^{\bullet,-}$ solves $Pu = 0$. Our aim is to prove that for all $\eta > 0$, we have

$$u \in H^{\infty}(\Omega_{0,-\eta})^{\bullet,-}. \tag{9.103}$$

Once this is shown, then we obtain $u \in \mathcal{C}^2(\Omega_{0,-\eta})^{\bullet,-}$ by Sobolev embedding. Theorem 9.19, in the form of the estimate (9.92), implies $u = 0$ on $\Omega_{0,-\eta}$. Since $\eta > 0$ is arbitrary, this gives $u = 0$ on Ω.

Towards the proof of (9.103), consider $z \in \Omega_{0,-\eta}$ and a point $\zeta \in \mathrm{Char}(P) \cap T^*_zM$. Let

$$\gamma\colon I \to T^*\Omega_{\delta,-\eta}, \quad I \subseteq [0,\infty), \tag{9.104}$$

be the maximal past-directed null-bicharacteristic starting at $\gamma(0) = (z,\zeta)$; that is, $\gamma'(s) = \pm H_G|_{\gamma(s)}$ where we use the "$-$," resp. "$+$" sign when ζ is future, resp. past lightlike. We claim that there exists $T \in I$ with

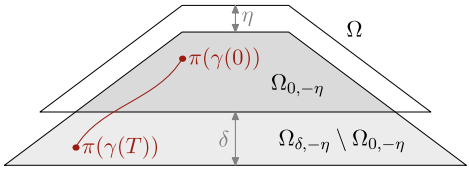


Figure 9.4 A null-geodesic $\pi \circ \gamma$ (the projection to the base of a null-bicharacteristic γ) starting at a point $\pi(\gamma(0)) \in \Omega_{\delta,-\eta}$ and ending at a point $\pi(\gamma(T)) \in \Omega_{\delta,-\eta} \setminus \Omega_{0,-\eta}$.

$$\pi(\gamma(T)) \in \Omega_{\delta,-\eta} \setminus \Omega_{0,-\eta} \tag{9.105}$$

where $\pi\colon T^*M \to M$ is the projection. (In particular, γ reaches the region where u vanishes and thus is smooth.) See Figure 9.4. To prove this, recall Proposition 9.9 and consider the past-directed null-geodesic $\alpha := \pi \circ \gamma$. Note that $t \circ \alpha$ is monotonically decreasing, for its derivative equals $\mathrm{d}t(\alpha'(s)) < 0$ since $\mathrm{d}t$ is past timelike and $\alpha'(s)$ is past null. Since $\Omega_{\delta,-\eta}$ is compact, we in fact have $c := \inf_{s\in I} |\mathrm{d}t(\alpha'(s))| > 0$. Letting $t_- := \min_{\Omega_{\delta,-\eta}} t > -\infty$, the interval I is thus a compact subinterval of $[0, (t(\gamma(0)) - t_-)/c]$; let $s_+ = \max I$. We claim that $\alpha(s_+)$ lies on an initial hypersurface of $\Omega_{\delta,-\eta}$, which would imply (9.105). If the claim were false, then $\alpha(s_+) \in \Omega_{\delta',-\eta}$ for some $\delta' \in (0, \delta)$. Note then that $\alpha(s_+)$ cannot lie on a final hypersurface of $\Omega_{\delta',-\eta}$, for otherwise $\alpha(s_+ - \epsilon)$ would lie in $\Omega_{\delta,0} \setminus \Omega_{\delta,-\eta}$ for sufficiently small $\epsilon > 0$. Therefore, $\alpha(s_+)$ lies in the interior of $\Omega_{\delta,-\eta}$, and hence, α is not inextendible, in contradiction to the maximality of its interval of definition I.

For every $z \in \Omega_{0,-\eta}$ and $\zeta \in T^*_z M \cap \mathrm{Char}(P)$, we apply the propagation of regularity (Theorem 8.8) to the past-directed null-bicharacteristic starting at (z, ζ) to conclude that $(z, \zeta) \notin \mathrm{WF}(u)$. Therefore, $\mathrm{WF}(u)$ is disjoint from $(T^*_{\Omega_{0,-\eta}} M \setminus o) \cap \mathrm{Char}(P)$. On the other hand, $\mathrm{WF}(u)$ is disjoint from $\mathrm{Ell}(P)$ by microlocal elliptic regularity (Proposition 6.33). Therefore, $\mathrm{WF}(u) \cap (T^*_{\Omega_{0,-\eta}} M \setminus o) = \emptyset$ for all $\eta > 0$, which gives (9.103).

For later purposes (concretely, for the proof of Lemma 9.36), it is useful to be more quantitative. We can apply Corollary 8.19 to suitable ps.d.o.s $B, G, E \in \Psi^0(M)$ with the following properties:

(1) The Schwartz kernels of B and G are supported in an open subset of $M \times M$ that is contained in $\Omega_{\delta,-\frac{\eta}{2}} \times \Omega_{\delta,-\frac{\eta}{2}}$; the Schwartz kernel of E is supported in $U \times U$ for some open subset $U \subset M$ that is contained in $\Omega_{\delta,-\frac{\eta}{2}} \times \Omega_{\delta,-\frac{\eta}{2}}$.
(2) B is elliptic on $S^*_{\Omega_{0,-\eta}} M$.
(3) The estimate (8.21) holds, so

$$\|Bu\|_{H^s} \le C(\|Eu\|_{H^s} + \|GPu\|_{H^{s-1}} + \|\chi u\|_{H^{s'}}). \tag{9.106}$$

Here we fix a cutoff function $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega^\circ_{\delta,0})$ which equals 1 near the base projections (to either factor) of the supports of the Schwartz kernels of B, G, E.

Using that $Eu = 0$ due to the vanishing of u on U, and using moreover that $Pu = 0$, the estimate (9.106) implies

$$\|Bu\|_{H^s} \leq C\|u\|_{H^{s'}(\Omega)^{\bullet,-}}. \tag{9.107}$$

Let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega^\circ_{\delta,0})$ be such that $\phi = 1$ on $\Omega_{0,-\eta}$ and $\mathrm{Ell}(B) \supset S^*_{\operatorname{supp}\phi} M$. Then $\phi u \in H^s$, with norm bounded by $C(\|Bu\|_{H^s} + \|\chi u\|_{H^{s'}})$, and thus,

$$\|u\|_{H^s(\Omega_{0,-\eta})^{\bullet,-}} \leq C\|\phi u\|_{H^s} \leq C'\|u\|_{H^{s'}(\Omega)^{\bullet,-}}. \tag{9.108}$$

Since s is arbitrary, we get (9.103). □

It is customary to only cite qualitative results on the absence or propagation of H^s wave front sets (as in the first half of the above proof) and omit the explicit translation into quantitative estimates (which we spell out in the second half). The reason is that all qualitative results (Proposition 6.33, Theorem 8.8) come with (and are, in fact, proved as a consequence of) quantitative estimates (Proposition 6.39, Theorem 8.10), and a combination of these quantitative estimates gives the desired quantitative H^s-control.

Lemma 9.36 (Higher Regularity: $s \geq 0$). *Let $s_0, s \geq 0$. Suppose $u \in H^{s_0}(\Omega)^{\bullet,-}$ satisfies $Pu = f \in H^{s-1}(\Omega)^{\bullet,-}$. Then $u \in H^s(\Omega)^{\bullet,-}$, and we have an estimate*

$$\|u\|_{H^s(\Omega)^{\bullet,-}} \leq C\big(\|u\|_{H^{s_0}(\Omega)^{\bullet,-}} + \|f\|_{H^{s-1}(\Omega)^{\bullet,-}}\big). \tag{9.109}$$

Proof. On $H^{s-1}(\Omega)^{\bullet,-} \cong \dot{H}^{s-1}(\Omega_{0,\delta})/\dot{H}^{s-1}(\Omega_{0,\delta}\setminus\Omega)$, let us use the quotient norm. (The pedantic reader may wish to write $\overline{\Omega_{0,\delta} \setminus \Omega}$ here.) Take then an extension $\tilde{f} \in \dot{H}^{s-1}(\Omega_{0,\delta})$ with $\|\tilde{f}\|_{\dot{H}^{s-1}(\Omega_{0,\delta})} = \|f\|_{H^{s-1}(\Omega)^{\bullet,-}}$. Denote the extension of $\tilde{f}$ by 0 to $\Omega_{\delta,\delta}$ by $\tilde{f} \in \dot{H}^{s-1}(\Omega_{\delta,\delta})$ still. We can now use Lemma 9.34 to solve

$$P\tilde{u} = \tilde{f}, \quad \tilde{u} \in H^0(\Omega_{\delta,\delta})^{\bullet,-}. \tag{9.110}$$

The extension by 0 of u to $\Omega_{\delta,0}$ solves the equation $Pu = f = \tilde{f}|_{\Omega_{\delta,0}}$; by uniqueness (Lemma 9.35), we thus have $\tilde{u} = u$ on $\Omega_{\delta,0}$. In particular, $\tilde{u}$ vanishes on $\Omega_{\delta,0} \setminus \Omega$.

We now apply the propagation of singularities (Theorem 8.8) to the equation (9.110). All past-directed null-bicharacteristics starting at a point over Ω reach a point over $\Omega_{\delta,0} \setminus \Omega$ where $\tilde{u}$ vanishes (and in particular is smooth). Combined with microlocal elliptic regularity for P on $S^*_\Omega M \setminus$

$\mathrm{Char}(P)$, we conclude that $\mathrm{WF}^s(\tilde{u}) \cap S^*_\Omega M = \emptyset$, and hence, the restriction $u = \tilde{u}|_{\Omega_{\delta,0}}$ lies in $H^s(\Omega)^{\bullet,-}$. The quantitative estimate (9.109) follows by using quantitative propagation and elliptic estimates. □

Corollary 9.37 (Existence and Uniqueness: $s \geq 0$). *Let $s \geq 0$. Then there exists a constant $C < \infty$ such that for all $f \in H^{s-1}(\Omega)^{\bullet,-}$ there exists $u \in H^s(\Omega)^{\bullet,-}$ with $Pu = f$ and*

$$\|u\|_{H^s(\Omega)^{\bullet,-}} \leq C\|f\|_{H^{s-1}(\Omega)^{\bullet,-}}. \tag{9.111}$$

Moreover, u is the unique solution of $Pu = f$ in the space $H^{-\infty}(\Omega)^{\bullet,-}$.

Proof. Lemma 9.34 produces a solution $u \in H^0(\Omega)^{\bullet,-}$ of $Pu = f$, and Lemma 9.36 shows that, in fact, $u \in H^s(\Omega)^{\bullet,-}$. Combining (9.109) for $s_0 = 0$ with (9.97) proves the estimate (9.111). Uniqueness follows from Lemma 9.35. □

We now complete the proof of Theorem 9.33.

Proof of Theorem 9.33. It remains to treat the case $s < 0$. We proceed similarly to the proof of Lemma 9.34. We apply Corollary 9.37 to the operator P^* and the opposite time orientation of (M, g); this gives the a priori estimate

$$\|v\|_{H^{-s+1}(\Omega)^{-,\bullet}} \leq C\|P^*v\|_{H^{-s}(\Omega)^{-,\bullet}}, \quad v \in H^{-s+2}(\Omega)^{-,\bullet}. \tag{9.112}$$

(Note that $-s+1 > 0$, so Corollary 9.37 indeed applies.) The linear functional

$$\lambda\colon P^*v \mapsto \langle v, f\rangle_{L^2}, \quad v \in H^{-s+2}(\Omega)^{-,\bullet}, \tag{9.113}$$

is thus well-defined and satisfies the estimate

$$\begin{aligned} |\lambda(P^*v)| &\leq C'\|v\|_{H^{-s+1}(\Omega)^{-,\bullet}}\|f\|_{H^{s-1}(\Omega)^{\bullet,-}} \\ &\leq CC'\|P^*v\|_{H^{-s}(\Omega)^{-,\bullet}}\|f\|_{H^{s-1}(\Omega)^{\bullet,-}}. \end{aligned} \tag{9.114}$$

Thus, λ can be extended to a continuous linear functional on $H^{-s}(\Omega)^{-,\bullet}$. By Theorem 9.31, it is therefore given by the L^2-pairing with some $u \in H^s(\Omega)^{\bullet,-}$. The resulting identity

$$0 = \langle P^*v, u\rangle_{L^2} - \langle v, f\rangle_{L^2} = \langle v, Pu - f\rangle_{L^2}, \quad v \in H^{-s+2}(\Omega)^{-,\bullet}, \tag{9.115}$$

implies, again by Theorem 9.31, that $Pu - f \in H^{s-2}(\Omega)^{\bullet,-}$ vanishes. This establishes the existence of a solution of $Pu = f$. The uniqueness of u was already shown in Lemma 9.35. □

As a useful corollary, we can prove H^s-estimates for solutions of wave-type equations if they are known to be of class H^s initially; see Figure 9.5. Using

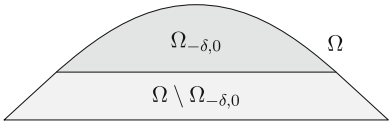


Figure 9.5 Illustration of the estimate (9.116): H^s-bounds on u in $\Omega_{-\delta,0} \setminus \Omega$ (i.e., near the initial hypersurfaces of Ω) imply H^s-bounds on u in all of Ω.

the assumptions of Theorem 9.33 and the notation introduced thereafter, the precise statement is as follows:

Corollary 9.38 (Continuation Estimates). *Let $s \in \mathbb{R}$. Let $u \in \bar{H}^{-\infty}(\Omega; E)$ be such that $Pu = f \in \bar{H}^{s-1}(\Omega; E)$ and $u|_{\Omega\setminus\Omega_{-\delta,0}} \in \bar{H}^s(\Omega \setminus \Omega_{-\delta,0}; E)$ for some $\delta > 0$, i.e., u lies in H^s near the initial hypersurfaces of Ω. Then $u \in \bar{H}^s(\Omega; E)$, and we have the estimate*

$$\|u\|_{\bar{H}^s(\Omega;E)} \leq C\big(\|f\|_{\bar{H}^{s-1}(\Omega;E)} + \|u|_{\Omega\setminus\Omega_{-\delta,0}}\|_{\bar{H}^s(\Omega\setminus\Omega_{-\delta,0};E)}\big). \tag{9.116}$$

The pedantic reader may wish to write $(\Omega \setminus \Omega_{-\delta,0})^\circ$ here.

Proof of Corollary 9.38. We drop the bundle E from the notation. Suppose $u \in \bar{H}^{s'}(\Omega)$. Let $\chi \in \mathcal{C}^\infty(\Omega)$ be equal to 1 near the initial hypersurfaces of Ω and equal to 0 near $\Omega_{-\delta,0}$. Then $u' := (1-\chi)u \in H^{s'}(\Omega)^{\bullet,-}$ is the (unique) forward solution of

$$Pu' = (1-\chi)Pu - [P,\chi]u = (1-\chi)f - [P,\chi]u =: f'. \tag{9.117}$$

For $\tilde{\chi} \in \mathcal{C}^\infty(\Omega)$ which is equal to 1 near $\operatorname{supp} \mathrm{d}\chi$ and equal to 0 near $\Omega_{-\delta,0}$, we have $[P,\chi] = [P,\chi]\tilde{\chi} \in \mathrm{Diff}^1(M)$, and therefore $[P,\chi]u = [P,\chi](\tilde{\chi}u) \in H^{s-1}(\Omega)^{\bullet,-}$. Thus, $f' \in H^{s-1}(\Omega)^{\bullet,-}$, and hence Theorem 9.33 gives $u' \in H^s(\Omega)^{\bullet,-}$. Applying (9.95) to u', f' gives

$$\begin{aligned}
\|u\|_{\bar{H}^s(\Omega)} &\leq \|\chi u\|_{\bar{H}^s(\Omega)} + \|(1-\chi)u\|_{\bar{H}^s(\Omega)} \\
&\leq C'\big(\|u|_{\Omega\setminus\Omega_{-\delta,0}}\|_{\bar{H}^s(\Omega\setminus\Omega_{-\delta,0})} + \|u'\|_{H^s(\Omega)^{\bullet,-}}\big) \\
&\leq C''\big(\|u|_{\Omega\setminus\Omega_{-\delta,0}}\|_{\bar{H}^s(\Omega\setminus\Omega_{-\delta,0})} + \|f'\|_{H^{s-1}(\Omega)^{\bullet,-}}\big) \\
&\leq C\big(\|u\|_{\Omega\setminus\Omega_{-\delta,0}}\|_{\bar{H}^s(\Omega\setminus\Omega_{-\delta,0})} + \|f\|_{\bar{H}^{s-1}(\Omega)}\big).
\end{aligned}$$

This completes the proof. □

9.5 Semiclassical Estimates

In order to state the correct semiclassical generalization of Definition 9.8, we must restrict to Lorentzian manifolds (M, g) on which there exists a smooth function $t \in \mathcal{C}^\infty(M)$ such that $\mathrm{d}t$ is timelike everywhere. We write $G(z, \zeta) := g_z^{-1}(\zeta, \zeta)$ for the dual metric function.

Definition 9.39 (Semiclassical Wave-Type Operator). Let $E \to M$ be a vector bundle. We say that a classical semiclassical operator $P = (P_h)_{(0,h_0)} \in \mathrm{Diff}_h^2(M; E)$ is a *semiclassical wave-type operator on* (M, g, t) if the principal symbol of P is principally scalar and equal to a fiber-wise second-order polynomial $p \in S^2(T^*M)$ with the following properties:

(1) $p \equiv G \bmod S^1(T^*M)$;
(2) for all $z \in M$ and $\zeta \in T_z^*M$, the equation $p(z, \zeta + \lambda\,\mathrm{d}t) = 0$ has two distinct real roots $\lambda = \lambda_\pm$.

In the scalar case, i.e., $E = M \times \mathbb{C}$, the first condition is equivalent to the requirement $P - h^2\Box_g \in \mathrm{Diff}_h^1$.

Example 9.40 (A Non-example). Write points and covectors on $\mathbb{R}^{1+n}$ as (t, x) and $\tau\,\mathrm{d}t + \xi\,\mathrm{d}x$, respectively. The operator $P := h^2(-D_t^2 + D_x^2) \in \mathrm{Diff}_h^2(\mathbb{R}^{1+n}_{t,x})$ has semiclassical principal symbol $p(t, x, \tau, \xi) = -\tau^2 + |\xi|^2$. This vanishes quadratically at the zero section $\{(\tau, \xi) = (0, 0)\} \subset T^*\mathbb{R}^{1+n}$. It is therefore *not* a semiclassical wave operator in the sense of Definition 9.39. The equation $Pu = f$ is equivalent to the non-semiclassical wave equation $(-D_t^2 + D_x^2)u = h^{-2}f$ with h-dependent source.

Example 9.41 (Wave Operators with Potentials). In the notation of the previous example, let $V \in \mathcal{C}^\infty(\mathbb{R}^{1+n})$ and consider $P := h^2(-D_t^2 + D_x^2) + V(t, x) \in \mathrm{Diff}_h^2(\mathbb{R}^{1+n})$. The semiclassical principal symbol of P is $p(t, x, \tau, \xi) = -\tau^2 + |\xi|^2 + V(t, x)$. This is a semiclassical wave operator if and only if $V > 0$ everywhere. (See Exercise 9.12 regarding the failure of semiclassical estimates when this sign condition is violated.)

The following construction, which arises naturally in Chapter 11, produces a large class of semiclassical wave-type operators. In fact, all semiclassical wave-type operators arise via this construction modulo $h^2\mathrm{Diff}_h^1$ (Proposition 9.43).

Proposition 9.42 (Spectral Families of Wave Operators At High Energies). *Let $M' = \mathbb{R}_s \times M$, and let g' be a Lorentzian metric on M' with the following properties:*

(1) $\mathcal{L}_{\partial_s} g' = 0$, *i.e., translations $(s, p) \mapsto (s + c, p)$ are isometries of (M', g') for all $c \in \mathbb{R}$.*
(2) ∂_s *is spacelike for g'; equivalently,*

$$T_z^*M := \operatorname{ann} \partial_s = \{\omega \in T^*_{(0,z)}M' \colon \omega(\partial_s) = 0\}, \quad z \in M, \tag{9.118}$$

is a Lorentzian subspace of $T^*_{(0,z)}M'$ *for* g'.

(3) *Defining the Lorentzian metric* g *on* M *by* $g_z^{-1} := (g'_{(0,z)})^{-1}|_{T_z^*M}$, *there exists a function* $t \in \mathcal{C}^\infty(M)$ *with everywhere timelike differential.*

Let $P' \in \operatorname{Diff}^2(M')$ *be a wave-type operator on* (M', g') *which is stationary, meaning that* $[\partial_s, P'] = 0$. *Denote the* spectral family *of* P' *by*

$$\widehat{P'}(\sigma) \in \operatorname{Diff}^2(M), \ \sigma \in \mathbb{C}, \quad \widehat{P'}(\sigma)u := \big(e^{i\sigma s}P'(e^{-i\sigma s}u)\big)\big|_{s=0}. \tag{9.119}$$

Finally, fix $\alpha \in \mathbb{R}$, *and define (for either choice of sign)*

$$P = (P_h)_{h\in(0,1)}, \quad P_h := h^2\widehat{P'}\big(\pm h^{-1}(1 - ih\alpha)\big). \tag{9.120}$$

Then $P \in \operatorname{Diff}^2_h(M)$ *is a classical semiclassical wave-type operator on* (M, g, t) *and*

$$\sigma^2_h(P)(z,\zeta) = \sigma^2(P')((0,z), \mp \mathrm{d}s + \zeta), \quad z \in M, \ \zeta \in T_z^*M. \tag{9.121}$$

Regarding (9.118), we use that the map $q\colon T^*_{\{0\}\times M}M' \to T^*M$ given by $q(\omega)(V) := \omega(V)$, $V \in TM$, restricts to an isomorphism $q\colon \operatorname{ann}\partial_s \xrightarrow{\cong} T^*M$. (We exploit here the product structure $M' = \mathbb{R}_s \times M$.)

Geometrically, (9.121) implies that the semiclassical characteristic set of P is equal to the intersection of the affine hyperplane $\mp \mathrm{d}s + T^*M$ (where $T^*M = \operatorname{ann}\partial_s$) with the characteristic set of P'. See Figure 9.6.

The definition (9.119) of the spectral family can be generalized to the case that P' acts on sections of a vector bundle $\pi^*E \to M'$ where $E \to M$ is a vector bundle and $\pi\colon M' \ni (s,p) \mapsto p \in M$ is the projection.

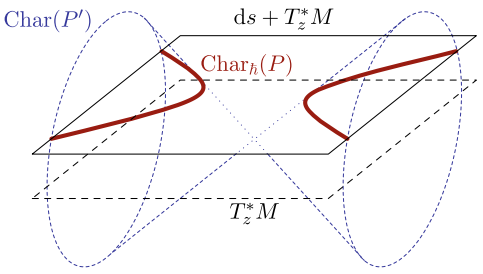


Figure 9.6 A fiber $T^*_{(0,z)}M'$ of T^*M', its linear subspace T_z^*M, and its affine subspace $\mathrm{d}s + T_z^*M$. The formula (9.121) (with the bottom sign) implies that the semiclassical characteristic set $\operatorname{Char}_\hbar(P)$ of P is the intersection of the characteristic set $\operatorname{Char}(P')$ of P' with $\mathrm{d}s + T_z^*M$.

Before giving the proof of Proposition 9.42, we give a simple example. Let $M = \mathbb{R}_t \times \mathbb{R}^{n-1}_x$ and equip $M' = \mathbb{R}_s \times M$ with the metric $g' = \mathrm{d}s^2 + g$ where $g = -\mathrm{d}t^2 + \mathrm{d}x^2$. Let $P' := \Box_{g'} = D_s^2 - D_t^2 + D_x^2$. Then $\widehat{P'}(\sigma) = -D_t^2 + D_x^2 + \sigma^2$ and therefore

$$P_h = -(hD_t)^2 + (hD_x)^2 + (1 - ih\alpha)^2. \tag{9.122}$$

The semiclassical principal symbol of $P = (P_h)_{h\in(0,1)}$ is $p = -\tau^2 + |\xi|^2 + 1$. (This is Example 9.41 with $V = 1$.) Thus, P is a semiclassical wave-type operator on (M, g, t).

Proof of Proposition 9.42. In local coordinates $z \in \mathbb{R}^n$ on M, we can write

$$P' = \sum_{j+|\alpha|=2} (g')^{j,\alpha}(z) D_s^j D_z^\alpha + \sum_{j+|\alpha|\le 1} a_{j,\alpha}(z) D_s^j D_z^\alpha, \tag{9.123}$$

where $(g')^{2,0} = g'^{-1}(\mathrm{d}s, \mathrm{d}s)$, $(g')^{1,\alpha} = g'^{-1}(\mathrm{d}s, \mathrm{d}z_j)$ for $\alpha = e_j := (0, \dots, 1, \dots, 0)$ (with 1 in the j-th slot), and $(g')^{0,\alpha} = g'^{-1}(\mathrm{d}z_j, \mathrm{d}z_k) = g^{-1}(\mathrm{d}z_j, \mathrm{d}z_k)$ for $\alpha = e_j + e_k$; moreover, $a_{j,\alpha} \in \mathcal{C}^\infty$. The spectral family of P' is obtained from (9.123) by formally replacing D_s by $-\sigma$. Therefore,

$$\begin{aligned} P_h = &\sum_{j+|\alpha|=2} (g')^{j,\alpha}(z)\big(\mp(1 - ih\alpha)\big)^j (hD_z)^\alpha \\ &+ h \sum_{j+|\alpha|\le 1} h^{1-(j+|\alpha|)} a_{j,\alpha}(z)\big(\mp(1 - ih\alpha)\big)^j (hD_z)^\alpha. \end{aligned} \tag{9.124}$$

Only the first sum contributes to the semiclassical symbol p of P which is thus given by $p\colon (z,\zeta) \mapsto \sum_{j+|\alpha|=2} (g')^{j,\alpha}(z)(\mp 1)^j \zeta^\alpha = g'^{-1}(\mp \mathrm{d}s + \zeta, \mp \mathrm{d}s + \zeta)$, as claimed. Since therefore $p(z,\zeta) \equiv (g'_z)^{-1}(\zeta,\zeta) = g_z^{-1}(\zeta,\zeta)$ modulo S^1, this verifies part (1) of Definition 9.39.

It remains to check part (2) of Definition 9.39; this is where the spacelike nature of ∂_s and thus the Lorentzian nature of $T^*_z M = \operatorname{ann} \partial_s \subset T^*_{(0,z)} M'$ will be used. Extending t to an s-independent smooth function on M', we have $\mathrm{d}t(\partial_s) = \partial_s t = 0$ and thus $\mathrm{d}t \in T^*_z M$ for all $z \in M$; this is timelike for g'. Let us introduce an orthonormal basis $\epsilon^1, \dots, \epsilon^n$ of $T^*_z M$ with $g'^{-1}(\epsilon^1, \epsilon^1) = -1$, $g'^{-1}(\epsilon^j, \epsilon^j) = 1$ $(j = 2, \dots, n)$, and $\mathrm{d}t = a\epsilon^1$, $a \neq 0$, and complete this with an appropriate $\epsilon^0 \in T^*_{(0,z)} M'$, $g'^{-1}(\epsilon^0, \epsilon^0) = 1$, to an orthonormal basis of $T^*_{(0,z)} M'$. Since $\mathrm{d}s \notin T^*_z M$, we have $\mathrm{d}s = \sum_{j=0}^n s_j \epsilon^j$ with $s_0 \neq 0$. Writing a given covector $\zeta \in T^*_z M$ as $\zeta = \sum_{j=1}^n \zeta_j \epsilon^j$, we now compute the inner product of

$$\mp \mathrm{d}s + \zeta + \lambda\,\mathrm{d}t = \mp s_0\epsilon^0 + \lambda a\epsilon^1 + \sum_{j=1}^{n}(\zeta_j \mp s_j)\epsilon^j \tag{9.125}$$

with itself with respect to $g'^{-1}_{(0,z)}$ to be

$$p(z, \zeta + \lambda\,\mathrm{d}t) = -(\lambda a + \zeta_1 \mp s_1)^2 + s_0^2 + \sum_{j=2}^{n}(\zeta_j \mp s_j)^2. \tag{9.126}$$

For $\lambda = -a^{-1}(\zeta_1 \mp s_1)$, this expression is positive since $s_0 \neq 0$. On the other hand, for $\lambda \in \mathbb{R}$ with sufficiently large absolute value, it is negative. Therefore, $\lambda \mapsto p(z, \zeta + \lambda\,\mathrm{d}t)$ has two distinct real roots. □

Conversely:

Proposition 9.43 (Characterization of Semiclassical Wave-Type Operators). *Let $P = (P_h)_{h\in(0,h_0)} \in \mathrm{Diff}^2_{\hbar}(M)$ be a classical semiclassical wave-type operator on (M, g, t). Then there exist a Lorentzian metric g' on $M' := \mathbb{R}_s \times M$ and a wave-type operator $P' \in \mathrm{Diff}^2(M')$ on (M', g') satisfying the conditions of Proposition 9.42 such that, in the notation (9.119),*

$$\begin{aligned} &P_h - h^2\widehat{P'}(h^{-1}) = h^2 V_h + h^3 r_h, \\ &V = (V_h)_{h\in(0,h_0)} \in \mathcal{V}_{\hbar}(M), \quad r = (r_h)_{h\in(0,h_0)} \in \mathrm{Diff}^0_{\hbar}(M). \end{aligned} \tag{9.127}$$

Here we allow V_h and r_h to be complex. A fortiori,

$$P_h - h^2\widehat{P'}(h^{-1}) \in h^2\mathrm{Diff}^1_{\hbar}(M). \tag{9.128}$$

Proof. Denote the semiclassical principal symbol of P by $p \in S^2(T^*M)$. Write $\langle\cdot,\cdot\rangle := g^{-1}(\cdot,\cdot)$. There exist unique $a \in \mathcal{C}^\infty(M; T^*M)$ and $b \in \mathcal{C}^\infty(M)$ such that

$$p(z,\zeta) = \langle\zeta,\zeta\rangle + 2\langle a,\zeta\rangle + b = \langle\zeta + a, \zeta + a\rangle + (b - \langle a, a\rangle). \tag{9.129}$$

For $\zeta = -a$, the requirement that $p(z, \zeta + \lambda\,\mathrm{d}t) = \lambda^2\langle\mathrm{d}t, \mathrm{d}t\rangle + (b - \langle a, a\rangle)$ have two distinct real roots forces

$$b - \langle a, a\rangle > 0 \tag{9.130}$$

since $\langle\mathrm{d}t, \mathrm{d}t\rangle < 0$.

Define $g'_{(s,z)} \in S^2T^*_{(s,z)}M'$ via its inverse by

$$g'^{-1}_{(s,z)}(-c\,\mathrm{d}s+\zeta, -c\,\mathrm{d}s+\zeta) := \langle\zeta+ca, \zeta+ca\rangle + c^2(b-\langle a,a\rangle), \quad c \in \mathbb{R},\ \zeta \in T^*_zM. \tag{9.131}$$

This is the unique quadratic form in (c,ζ) which for $c=1$ equals (9.129). Clearly $\mathcal{L}_{\partial_s} g' = 0$, and ∂_s is spacelike since $\operatorname{ann}\partial_s = T_z^*M$ is Lorentzian. Consider now the difference

$$E_h := P_h - h^2\widehat{\Box_{g'}}(h^{-1}). \tag{9.132}$$

By Proposition 9.42, $\sigma^2_\hbar(E) = 0$ for $E = (E_h)_{h\in(0,h_0)}$, and hence $E \in h\mathrm{Diff}^1_\hbar(M)$. Since E is classical, we can therefore write

$$E_h = h\big(hX + q_0 + h(V_h + q_1) + h^2 r_h\big) \tag{9.133}$$

where $X \in \mathcal{V}(M)$ (so $(hX)_{h\in(0,h_0)} \in \mathcal{V}_\hbar(M)$) and $q_0, q_1 \in \mathcal{C}^\infty(M)$, further $(V_h)_{h\in(0,h_0)} \in \mathcal{V}_\hbar(M)$, and $r_h \in \mathcal{C}^\infty(M)$ is uniformly bounded. Setting

$$P' := \Box_{g'} + X - q_0 D_s + q_1 \in \mathrm{Diff}^2(M'), \tag{9.134}$$

we therefore have $P_h - h^2\widehat{P'}(h^{-1}) = h^2 V_h + h^3 r_h$. □

From (9.129)–(9.130), we see that the semiclassical characteristic set $\mathrm{Char}_\hbar(P) \subset \overline{T}^*M$ has two components (which are connected when $n = \dim M \geq 3$) distinguished by the sign of $\langle \zeta + a, \mathrm{d}t\rangle$: declaring $\mathrm{d}t$ to be past timelike, the closure of the set of $(z,\zeta) \in \mathrm{Char}_\hbar(P) \cap T^*M$ with $g_z^{-1}(\zeta + a, \mathrm{d}t) > 0$, resp. < 0 is the future, resp. past component.

We now turn to energy estimates. We begin by proving an analogue of Theorem 9.19. We use the notation of Definition 9.14.

Theorem 9.44 (Semiclassical Energy Estimate). *Let $\Omega \subset M$ be a compact admissible spacetime domain. Fix a finite spanning set $V_1, \ldots, V_N \in \mathrm{Diff}^1(\Omega; E)$ of $\mathrm{Diff}^1(\Omega; E)$ over $\mathcal{C}^\infty(\Omega)$. Denote by $\nu_j(u)$ the derivative of u along ν_j with respect to a smooth connection on E. Set*

$$\begin{aligned}
\|u\|^2_{H^1_\hbar(\Omega;E)} &:= \|u\|^2_{L^2(\Omega;E)} + \sum_{i=1}^N \|hV_i u\|^2_{L^2(\Omega;E)},\\
E_{h,\mathrm{I}}[u] &:= \sum_{i=1}^{N_\mathrm{I}} \big(\|u|_{H_j}\|_{H^1_\hbar(H_j;E)} + \|h\nu_j(u)\|_{L^2(H_j;E)}\big),\\
E_{h,\mathrm{F}}[u] &:= \sum_{i=1}^{N_\mathrm{F}} \big(\|u|_{H_{N_\mathrm{I}+j}}\|_{H^1_\hbar(H_{N_\mathrm{I}+j};E)} + \|h\nu_{N_\mathrm{I}+j}(u)\|_{L^2(H_{N_\mathrm{I}+j};E)}\big).
\end{aligned} \tag{9.135}$$

Let $P = (P_h)_{h\in(0,h_0)} \in \mathrm{Diff}^2_\hbar(\Omega; E)$ be a semiclassical wave-type operator on (Ω, g, t). Then there exists a constant C such that for all $u \in \mathcal{C}^2(\Omega; E)$, we have the estimate

$$\|u\|_{H_h^1(\Omega;E)} + E_{h,\mathrm{F}}[u] \le C\big(h^{-1}\|P_h u\|_{L^2(\Omega;E)} + E_{h,\mathrm{I}}[u]\big). \tag{9.136}$$

A fortiori, if u vanishes to second order at all initial hypersurfaces of Ω, then

$$\|u\|_{H_h^1(\Omega;E)} \le Ch^{-1}\|P_h u\|_{L^2(\Omega;E)}. \tag{9.137}$$

Proof. We only discuss the case $E = M \times \mathbb{C}$. Let g', $P' \in \mathrm{Diff}^2(\mathbb{R}\times\Omega)$ be as in Proposition 9.43, so $P_h = h^2\widehat{P'}(h^{-1}) + R_h$, $R = (R_h)_{h\in(0,h_0)} \in h^2\mathrm{Diff}_h^1(\Omega)$. Assuming we have the estimate (9.136) for the operator $h^2\widehat{P'}(h^{-1})$, we can estimate

$$\begin{aligned} Ch^{-1}\big\|h^2\widehat{P'}(h^{-1})u\big\|_{L^2} &\le Ch^{-1}\big(\|P_h u\|_{L^2} + h^2\|u\|_{H_h^1}\big) \\ &= Ch^{-1}\|P_h u\|_{L^2} + Ch\|u\|_{H_h^1}. \end{aligned} \tag{9.138}$$

For $h < h_1 := \min(h_0, 1/2C)$, we can absorb $Ch\|u\|_{H_h^1}$, and thus deduce (9.136) for P_h. For $h \in [h_1, h_0)$, on the other hand, the norms on H_h^1 and H^1 are uniformly equivalent, and hence (9.136) for these h follows from (9.42).

For the remainder of the proof, we may thus work with $P_h := h^2\widehat{P'}(h^{-1})$. Write $\sigma := h^{-1}$. We multiply the desired estimate (9.136) by h^{-1} and thus need to show

$$\|\sigma u\|_{L^2(\Omega)} + \|\mathrm{d}u\|_{L^2(\Omega;T^*\Omega)} + E_{(\sigma),\mathrm{F}}[u] \le C\big(\big\|\widehat{P'}(\sigma)u\big\|_{L^2(\Omega)} + E_{(\sigma),\mathrm{I}}[u]\big),\ \sigma > \sigma_1, \tag{9.139}$$

for any large but fixed $\sigma_1 \ge h_1^{-1}$ and for all $u \in \mathcal{C}^2(\Omega)$, where $E_{(\sigma),\bullet}[u]$ is the sum of the L^2-norms of σu and $\mathrm{d}u$ on H_j, $j = 1, \dots, N_{\mathrm{I}}$ for $\bullet = \mathrm{I}$, resp. $j = N_{\mathrm{I}}+1, \dots, N_{\mathrm{I}}+N_{\mathrm{F}}$ for $\bullet = \mathrm{F}$. We shall deduce this from an energy estimate for

$$u'(s,z) := e^{-i\sigma s}u(z) \text{ on } \Omega' := [-1,1]_s \times \Omega. \tag{9.140}$$

Write the exterior derivative on $\mathbb{R}\times\Omega$ as

$$\mathrm{d}'f(s,z) := \partial_s f(s,z)\,\mathrm{d}s + \mathrm{d}(f(s,\cdot)) \tag{9.141}$$

where $\mathrm{d}(f(s,\cdot)) \in \mathcal{C}^\infty(M; T^*M)$ is the exterior derivative of $f(s,\cdot) \in \mathcal{C}^\infty(M)$. Then

$$\mathrm{d}'u' = e^{-i\sigma s}(-i\sigma\,\mathrm{d}s + \mathrm{d}u), \quad P'u'(s,z) = e^{-i\sigma s}(\widehat{P'}(\sigma)u)(z). \tag{9.142}$$

Since $|e^{-i\sigma s}| = 1$, the estimate (9.139) is thus equivalent to

$$\begin{aligned}\|\mathrm{d}'u'\|_{L^2(\Omega';T^*\Omega')} + E'_{\mathrm{F}}[u'] \le C\big(\|P'u'\|_{L^2(\Omega')} + E'_{\mathrm{I}}[u']\big),\\ u' \in e^{-i\sigma s}\mathcal{C}^2(\Omega) \subset \mathcal{C}^2(\Omega'),\end{aligned} \tag{9.143}$$

where, with $H'_j := [-1,1] \times H_j$,

$$E'_{\mathrm{I}}[u'] := \sum_{j=1}^{N_{\mathrm{I}}} \|(\mathrm{d}'u')|_{H'_j}\|_{L^2(H'_j;T^*\Omega')}, \tag{9.144}$$

with an analogous definition for $E'_{\mathrm{F}}[u']$. This is not quite a consequence of Theorem 9.19 since the hypersurfaces $\{\pm 1\} \times \Omega$ of Ω' appear not to be accounted for in (9.143).

To prove (9.143), we instead revisit the proof of Theorem 9.19, which is based on the identity (9.31). Consider first the case $P' = \Box_{g'}$. We apply the divergence theorem to ${}^{(X)}J'[u']$ where J' is the J-current for the metric g' and X is an s-independent vector field on Ω', i.e., $\mathcal{L}_{\partial_s} X = 0$. We claim that for $u' \in e^{-i\sigma s}\mathcal{C}^2(\Omega)$,

$$\begin{aligned}&\sum_{j=1}^{N_{\mathrm{F}}} \int_{H'_{N_{\mathrm{I}}+j}} T'[u'](X, \nu_{N_{\mathrm{I}}+j})\,|\mathrm{d}s\,\mathrm{d}\sigma_{N_{\mathrm{I}}+j}| + \int_{\Omega'} {}^{(X)}K'[u']\,|\mathrm{d}s\,\mathrm{d}g|\\ &\quad = \sum_{j=1}^{N_{\mathrm{I}}} \int_{H'_j} T'[u'](X,\nu_j)\,|\mathrm{d}s\,\mathrm{d}\sigma_j| + \int_{\Omega'} \mathrm{Re}(\Box_{g'}u'\,X\overline{u'})\,|\mathrm{d}s\,\mathrm{d}g|,\end{aligned} \tag{9.145}$$

where T', resp. K' is the stress–energy–momentum tensor, resp. K-current for g'. To prove this, we only need to show that for $u' \in e^{-i\sigma s}\mathcal{C}^2(\Omega)$, the additional boundary terms at $\{\pm 1\} \times \Omega$ cancel. To this end, we can use a partition of unity on Ω to localize X in an orientable chart; denote then by $\mathrm{d}g'$ the volume form. Define $\phi_s \colon \Omega \ni z \mapsto (s,z) \in \Omega'$ and set

$$B(s) := \int_{\Omega} \phi_s^*\big({}^{(X)}J'[u'] \lrcorner \mathrm{d}g'\big). \tag{9.146}$$

To show $B(-1) = B(1)$, it suffices to establish $\mathcal{L}_{\partial_s}({}^{(X)}J'[u'] \lrcorner \mathrm{d}g') = 0$. Since $\mathcal{L}_{\partial_s} g' = 0$ and $\mathcal{L}_{\partial_s} X = 0$, this follows from the fact that the Lie derivative of

$$\begin{aligned}T'[u'](\cdot,\cdot) &= \mathrm{d}'u' \otimes_s \mathrm{d}'\,\overline{u'} - \frac{1}{2}g'^{-1}(\mathrm{d}'u', \mathrm{d}'\,\overline{u'})g'\\ &= \mathrm{Re}\big((-i\sigma\,\mathrm{d}s + \mathrm{d}u) \otimes_s (i\sigma\,\mathrm{d}s + \mathrm{d}\bar{u})\big)\\ &\quad - \frac{1}{2}g'^{-1}\big(-i\sigma\,\mathrm{d}s + \mathrm{d}u, i\sigma\,\mathrm{d}s + \mathrm{d}\bar{u}\big)g'\end{aligned} \tag{9.147}$$

along ∂_s vanishes.

We now use (9.145) with $X = e^{-2\alpha t}X_0$, $X_0 = -g'^{-1}(\mathrm{d}t, \cdot)$, as in (9.45); this indeed satisfies $\mathcal{L}_{\partial_s} X = 0$. Moreover, since $\mathrm{d}t$ is timelike for g^{-1}, X is timelike for g'. For large α, the K-current ${}^{(X)}K'[u']$ is positive definite in $\mathrm{d}'u'$ and thus in $(\sigma u, \mathrm{d}u)$. Since each integrand in (9.145) is independent of s, we obtain the estimate (9.143) for $P' = \Box_{g'}$.

For general $P' = \Box_{g'} + V' + a$ where $a \in \mathcal{C}^\infty(M)$ (regarded as an s-independent function on M'), and $V' \in \mathcal{V}(M')$ commutes with s-translations, we rewrite the final integrand in (9.145) as

$$\mathrm{Re}(P'u'\, X\overline{u'}) - \mathrm{Re}(V'u'\, X\overline{u'}) - \mathrm{Re}(a'u'\, X\overline{u'}). \tag{9.148}$$

The sum of the second term here with ${}^{(X)}K'[u']$ in (9.145) is still positive definite when α is large enough, so we obtain (9.143) and thus (9.139) when $a' = 0$. For general a', we can start with (9.139) for $\Box_{g'} + V'$ in place of P' and estimate the difference by

$$C\|au\|_{L^2} \leq \frac{C\|a\|_{L^\infty}}{|\sigma|}\|\sigma u\|_{L^2}. \tag{9.149}$$

For sufficiently large σ, this can be absorbed into the left-hand side of (9.139). □

We can now prove a semiclassical version of Theorem 9.33.

Theorem 9.45 (Solving Semiclassical Wave-Type Equations). *Let Ω be a compact admissible spacetime domain in the time-oriented spacetime (M, g) on which there exists a function $t \in \mathcal{C}^\infty(M)$ with timelike differential. Let $P = (P_h)_{h\in(0,h_0)}$ be a semiclassical wave-type operator on (M, g, t). Let $s \in \mathbb{R}$. Then there exists a constant C such that the following holds: for all $f \in H^{s-1}(\Omega; E)^{\bullet,-}$, the unique forward solution $u \in H^s(\Omega; E)^{\bullet,-}$ of $P_h u = f$ satisfies*

$$\|u\|_{H_h^s(\Omega;E)^{\bullet,-}} \leq Ch^{-1}\|f\|_{H_h^{s-1}(\Omega;E)^{\bullet,-}}. \tag{9.150}$$

We clarify the norms used in this result: let U be an open set with compact closure, and let K be a compact set containing $\bar{U}$ in its interior. The norms on $\dot{H}_h^s(\bar{U})$ were already discussed before (5.205). We define the norm on $\bar{H}_h^s(U)$ to be the quotient norm of $\dot{H}_h^s(K)/\dot{H}_h^s(K \setminus U)$; the main point is that we use a fixed K for all h. (Different choices of K lead to *uniformly* equivalent norms.) The norms in (9.150) are defined as quotient norms analogously to Definition 9.28.

Proof of Theorem 9.45. We drop the bundle E from the notation. Solvability and uniqueness follow from Theorem 9.33, so only the uniform estimate (9.150) is at issue. It is nonetheless simplest to repeat some of the arguments used in the proof of Theorem 9.33 in order to establish (9.150).

Using the a priori estimate (9.137) for P^* as in the proof of Lemma 9.34, we obtain the estimate (9.150) for $s = 0$. To prove the estimate for higher regularity orders $s \geq 0$, we use an extension/restriction argument and the propagation of semiclassical regularity. We use the notation $\Omega_{\delta_{\mathrm{I}},\delta_{\mathrm{F}}}$ from (9.96), and fix a small $\delta > 0$. We use the quotient norm on $H_h^{s-1}(\Omega)^{\bullet,-}$ defined using the restriction map with domain $\dot{H}_h^{s-1}(\Omega_{0,\delta})$. Given $h \in (0, h_0)$ and $f \in H^{s-1}(\Omega)^{\bullet,-}$, we thus fix $\tilde{f} \in \dot{H}^{s-1}(\Omega_{0,\delta})$ with

$$\|\tilde{f}\|_{\dot{H}_h^{s-1}(\Omega_{0,\delta})} = \|f\|_{H_h^{s-1}(\Omega)^{\bullet,-}}. \tag{9.151}$$

Extend $\tilde{f}$ by 0 to $\Omega_{\delta,\delta}$. Solve $P_h\tilde{u} = \tilde{f}$ for $\tilde{u} \in H^s(\Omega_{\delta,\delta})^{\bullet,-}$. Then $\tilde{u}$ satisfies the uniform estimate

$$\|\tilde{u}\|_{L^2(\Omega_{\delta,\delta})^{\bullet,-}} \leq Ch^{-1}\|\tilde{f}\|_{H_h^{-1}(\Omega_{\delta,\delta})^{\bullet,-}} \leq C'h^{-1}\|\tilde{f}\|_{H_h^{s-1}(\Omega_{\delta,\delta})^{\bullet,-}}. \tag{9.152}$$

Fix $\phi \in \mathcal{C}_{\mathrm{c}}^\infty(\Omega^\circ_{\delta,\delta})$ with $\phi = 1$ on Ω. We claim that

$$\|\phi\tilde{u}\|_{\dot{H}_h^s(\Omega_{\delta,\delta})} \leq C\Big(h^{-1}\|\tilde{f}\|_{\dot{H}_h^{s-1}(\Omega_{\delta,\delta})} + h^N\|\tilde{u}\|_{L^2(\Omega_{\delta,\delta})^{\bullet,-}}\Big). \tag{9.153}$$

In light of (9.152), this then gives for $u := \tilde{u}|_\Omega$ the estimate (9.150).

Write $p = \sigma_h^2(P)$ and fix an elliptic and positive symbol $\rho \in S_{\mathrm{cl}}^{-1}(T^*M)$. Let $\tilde{H}_p := \rho H_p$. Then the estimate (9.153) follows from Theorem 8.27 once we prove that every integral curve of $\mp\tilde{H}_p$ starting at a point in $\mathrm{Char}_\hbar(P) \cap \overline{T^*_{\Omega^\circ_{\delta,\delta}}}M$ enters $\Omega_{\delta,\delta} \setminus \Omega_{0,\delta}$ in finite time; here the sign is "$-$," resp. "$+$" for points in the future, resp. past component of the characteristic set. Now, integral curves in S^*M are lifts of null-geodesics on (M, g) projected to S^*M, and they are analyzed as in the proof of Lemma 9.35. For integral curves in T^*M, we can take $\rho = 1$ and use the expression (9.129) for the principal symbol of P. This gives $H_p t = 2\langle \mathrm{d}t, \zeta + a\rangle$. But since $0 = p = \langle \zeta + a, \zeta + a\rangle + (b - \langle a, a\rangle)$ implies $\langle \zeta + a, \zeta + a\rangle < 0$ by (9.130), $\zeta + a$ is timelike; and thus $H_p t > 0$, resp. < 0 when $\zeta + a$ is future, resp. past timelike. Again arguing as in Lemma 9.35 gives the desired conclusion also in this case.

Finally, we treat the case $s < 0$ using a duality argument as in the final step of the proof of Theorem 9.33. □

Using the setup of Theorem 9.45, we have the following analogue of Corollary 9.38 (with the same proof):

Corollary 9.46 (Semiclassical Continuation Estimates). *Let $s \in \mathbb{R}$. Let $u \in \bar{H}^{-\infty}(\Omega; E)$ be such that $P_h u = f \in \bar{H}^{s-1}(\Omega; E)$ and $u|_{\Omega\setminus\Omega_{-\delta,0}} \in \bar{H}^s(\Omega \setminus \Omega_{-\delta,0}; E)$ for some $\delta > 0$. Then $u \in \bar{H}^s(\Omega; E)$, and we have the uniform estimate*

$$\|u\|_{\bar{H}_h^s(\Omega;E)} \leq C\Big(h^{-1}\|f\|_{\bar{H}_h^{s-1}(\Omega;E)} + \|u|_{\Omega\setminus\Omega_{-\delta,0}}\|_{\bar{H}_h^s(\Omega\setminus\Omega_{-\delta,0};E)}\Big). \tag{9.154}$$

Notes

The approach to the solution of wave equations presented in this chapter gives optimal spacetime regularity for forcing problems in a rather clean manner. An alternative approach to the existence, uniqueness, and regularity theory for wave equations, which is more closely related to the methods of Chapter 7, is given in Hörmander [Hör07, Chapter XXIII]. This approach allows for a cleaner treatment of initial value problems, but it is somewhat less efficient for forcing problems as one needs to introduce spaces with different spatial and spacetime regularity orders in order to deduce from $\mathcal{C}_t^0 H_x^s$-regularity the $H_{t,x}^s$-regularity of waves (with $H_{t,x}^{s-1}$ source terms).

Estimates for semiclassical wave-type operators are proved in Dyatlov–Zworski [DZ19, Appendix E.4] using a semiclassical version of the factorization method used in Hörmander [Hör07, Chapter XXIII]. Our presentation in Section 9.5 is more closely related to the approach taken in Vasy [Vas13, §3.3].

We have restricted to wave-type equations—which are the same as strictly hyperbolic *second order* differential operators (Exercise 9.1)—mainly due to the distinguished role they play in applications (e.g., in Chapter 11). Energy estimates for higher-order operators are discussed in Taylor [Tay11a, Chapter 2.8].

Exercises

Exercise 9.1 (Wave-Type Operators and Strict Hyperbolicity). Let M be a smooth manifold, and let $t \in \mathcal{C}^\infty(M)$ be a smooth function with nowhere vanishing differential. Let $P \in \mathrm{Diff}^2(M;E)$ be a principally scalar operator; denote by $p\colon T^*M \to \mathbb{C}$ its principal symbol. Show that the following conditions are equivalent.

(1) P is a wave-type operator with respect to a Lorentzian metric g for which $\mathrm{d}t$ is everywhere timelike.
(2) The principal symbol p is real-valued, and for all $z \in M$ and $\zeta \in T_z^*M \setminus \mathrm{span}\{\mathrm{d}t\}$, the equation $p(z, \zeta + \lambda\,\mathrm{d}t) = 0$ has two distinct real solutions λ.

Exercise 9.2 (Admissible Spacetime Domains). Let $\Omega \subset M$ be an admissible spacetime domain as in (9.26), and set $H_j := \Omega \cap f_j^{-1}(0)$.

(1) Show that Ω is a *submanifold with corners*. This means that for every $p \in \Omega$ there exist $k \in \{0, \dots, n\}$ and local coordinates $z = (z^1, \dots, z^n) \in \mathbb{R}^n$ defined in a neighborhood U of p such that $z = 0$ at p and $\Omega \cap U = \{z\colon z^1 \geq 0,\ \dots,\ z^k \geq 0\}$.
(2) Show that each H_j is the closure in Ω of a smooth hypersurface in M.

Exercise 9.3 (Higher-Order Energy Estimate). Let Ω be a compact admissible spacetime domain in the time-oriented spacetime (M, g). Define

$$E^m_{\mathrm{I}}[u] := \sum_{j=1}^{N_{\mathrm{I}}} \big(\|u|_{H_j}\|_{H^m(H_j;E)} + \|\nu_j(u)\|_{H^{m-1}(H_j;E)} \big). \tag{9.155}$$

Define $E^m_{\mathrm{F}}[u]$ similarly, now using $H_{N_{\mathrm{I}}+j}$, $j = 1, \dots, N_{\mathrm{F}}$. Suppose there exists a function $t \in \mathcal{C}^\infty(\Omega)$ such that $\mathrm{d}t$ is past timelike. Let $P \in \mathrm{Diff}^2(\Omega; E)$ be a wave-type operator on (Ω, g). Let $m \in \mathbb{N}$. Show that there exists a constant C such that for all $u \in \mathcal{C}^{m+1}(\Omega; E)$, we have the estimate

$$\|u\|_{H^m(\Omega;E)} + E^m_{\mathrm{F}}[u] \leq C(\|Pu\|_{H^{m-1}(\Omega;E)} + E^m_{\mathrm{I}}[u]). \tag{9.156}$$

You can proceed as follows.

(1) Fix a finite set $\{A_1, \dots, A_J\} \subset \mathrm{Diff}^{m-1}(\Omega)$ which spans $\mathrm{Diff}^{m-1}(\Omega)$ over $\mathcal{C}^\infty(\Omega)$. Show that for each $j = 1, \dots, J$, there exist operators $X_{j,k} \in \mathrm{Diff}^1(\Omega)$, $k = 1, \dots, J$, such that

$$[P, A_j] = \sum_{k=1}^{J} X_{j,k} A_k. \tag{9.157}$$

Deduce that if $f = Pu$, then $U := (A_j u)_{j=1,\dots,J}$ satisfies the wave-type equation $\mathcal{P}U = F := (A_j f)_{j=1,\dots,J}$ where $\mathcal{P} := (\delta_{jk} P - X_{j,k})_{j,k=1,\dots,J}$.
(2) Obtain an energy estimate for U using Theorem 9.19. Using the PDE $Pu = f$, show $E^1_{\mathrm{I}}[U] \leq C E^m_{\mathrm{I}}[u]$ and complete the proof.

Exercise 9.4 (Order Shifting Operators). Let $\mathbb{R}^n_+ := \mathbb{R}^{n-1}_{x'} \times (0, \infty)_{x_n}$. Let $s \in \mathbb{R}$.

(1) Show that the Fourier multiplier $(\langle D_{x'} \rangle + i D_{x_n})^s$ defines an isometric isomorphism $\dot{H}^s(\overline{\mathbb{R}^n_+}) \to \dot{L}^2(\overline{\mathbb{R}^n_+})$. (*Hint.* Use the Paley–Wiener theorem of Exercise 2.14.)
(2) Show that $(\langle D_{x'} \rangle - i D_{x_n})^s\colon H^s(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ induces an isometric isomorphism $\bar{H}^s(\mathbb{R}^n_+) \to \bar{L}^2(\mathbb{R}^n_+)$.

Exercise 9.5 (Relationship Between Extendible and Supported Spaces, I). We study the properties of the restriction map $R\colon \dot{H}^s([0, \infty)) \ni u \mapsto u|_{(0,\infty)} \in \bar{H}^s((0, \infty))$.

(1) Let $s > -\frac{1}{2}$. Show that R is injective.
(2) Let $0 \le s < \frac{1}{2}$. Show that R is surjective. To do this, proceed as follows.
   (i) Show that multiplication by $|x|^{-s}$ defines a bounded linear map $H^s(\mathbb{R}) \to L^2(\mathbb{R})$. (*Hint.* It suffices to show $\int_0^\infty x^{-2s}|u(x)|^2\,\mathrm{d}x \le C\|u\|^2_{H^s(\mathbb{R})}$ for $u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$. Given $u \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$, consider $v(x) := \frac{1}{x}\int_0^x (u(x) - u(y))\,\mathrm{d}y$, and show that $\|x^{-s}v\|_{L^2((0,\infty))} \le C\|u\|_{H^s(\mathbb{R})}$ using Exercise 2.10. Show moreover that $w(x) := u(x) - v(x) = \int_x^\infty \frac{v(y)}{y}\,\mathrm{d}y$, and establish the estimate $\|x^{-s}w\|_{L^2((0,\infty))} \le C\|x^{-s}v\|_{L^2((0,\infty))}$.)
   (ii) Show that if $u \in \bar{H}^s((0,\infty))$, then the L^2-function $x \mapsto H(x)u(x)$ (where H is the Heaviside function) lies in $H^s(\mathbb{R})$.

(3) Let $|s| < \frac{1}{2}$. Show that R is an isomorphism. More generally, let $\Omega \subset \mathbb{R}^n$ be a smooth compact submanifold with corners; show that $\dot{H}^s(\Omega) = \bar{H}^s(\Omega^\circ)$, i.e., the restriction map $\dot{H}^s(\Omega) \ni u \mapsto u|_{\Omega^\circ} \in \bar{H}^s(\Omega^\circ)$ is an isomorphism.
(4) Let $s < \frac{1}{2}$, $s \notin \mathbb{Z} + \frac{1}{2}$. Show that R is surjective.

Exercise 9.6 (Relationship Between Extendible and Supported Spaces, Ii). This exercise builds on Exercise 9.5. Let $s \notin \mathbb{Z} + \frac{1}{2}$.

(1) Show that $u \in \bar{H}^s((0,\infty))$ is the restriction to $(0,\infty)$ of an element of $\dot{H}^s([0,\infty))$ if and only if $u^{(j)}(0) = 0$ for all $0 \le j < s - \frac{1}{2}$. (This is a vacuous condition for $s < \frac{1}{2}$.) Note that the values $u^{(j)}(0)$ are well-defined by Sobolev embedding.
(2) Generalize this to $\bar{H}^s(\Omega)$ and $\dot{H}^s(\Omega^\circ)$ when Ω is a smooth compact submanifold with boundary inside of a smooth manifold M.

Exercise 9.7 (Further Applications of Supported/Extendible Spaces: Local Solvability, I: Elliptic Equations). Let $P \in \mathrm{Diff}^m(M)$ be an elliptic differential operator, and let $x_0 \in M$. The goal of this exercise is to show that there exists an open neighborhood $U \subset M$ of x_0 such that, for all $s \in \mathbb{R}$,

$$P\colon \bar{H}^s(U) \to \bar{H}^{s-m}(U) \text{ is surjective.} \tag{9.158}$$

(1) Let N be such that $-N < -s + m$. Let U be precompact. Prove

$$\|v\|_{\dot{H}^{-s+m}(\bar{U})} \le C\big(\|P^*v\|_{\dot{H}^{-s}(\bar{U})} + \|v\|_{\dot{H}^{-N}(\bar{U})}\big) \tag{9.159}$$

and conclude that $P^*\colon \dot{H}^{-s+m}(\bar{U}) \to \dot{H}^{-s}(\bar{U})$ has finite-dimensional kernel and closed range.

(2) Show that if U is a sufficiently small neighborhood of x_0, then $\ker_{\dot{\mathcal{C}}^\infty(\bar{U})} P^* = \{0\}$. *We fix such a U from now on.* Conclude that $\ker_{\dot{H}^{-s+m}(\bar{U})} P^* = \{0\}$ for all $s \in \mathbb{R}$.
(3) With U as in the previous part, prove the estimate

$$\|v\|_{\dot{H}^{-s+m}(\bar{U})} \leq C' \|P^* v\|_{\dot{H}^{-s}(\bar{U})}. \tag{9.160}$$

(4) Use a duality argument to deduce (9.158).
(5) Let U' be an open neighborhood of x_0 with closure contained in U. Show that $P\colon \bar{\mathcal{C}}^\infty(U') \to \bar{\mathcal{C}}^\infty(U')$ is surjective.
(6) Is $P\colon \dot{\mathcal{C}}^\infty(\bar{U}) \to \dot{\mathcal{C}}^\infty(\bar{U})$ surjective?

Exercise 9.8 (Further Applications of Supported/Extendible Spaces: Local Solvability, II: Real Principal Type Equations). Let $P \in \mathrm{Diff}^m(M)$ be a differential operator with real principal symbol p, and let $x_0 \in M$. Suppose that no maximal null-bicharacteristic P is contained in $T^*_{x_0} M \setminus o$. The goal of this exercise is to show that there exists a precompact open neighborhood $U \subset M$ of x_0 such that, for all $s \in \mathbb{R}$ and $f \in \bar{H}^{s-m+1}(U)$, there exists a solution $u \in \bar{H}^s(U)$ of $Pu = f$.

(1) Show that one can choose U such that every maximal null-bicharacteristic starting in $\mathrm{Char}(P) \cap (T^*_{\bar{U}} M \setminus o)$ enters $T^*(M \setminus U) \setminus o$. For such U, prove

$$\|v\|_{\dot{H}^{-s+m-1}(\bar{U})} \leq C\big(\|P^* v\|_{\dot{H}^{-s}(\bar{U})} + \|v\|_{\dot{H}^{-N}(\bar{U})}\big) \tag{9.161}$$

where N is fixed with $-N < -s + m$.
(2) Shrinking U further, show that $\|v\|_{\dot{H}^{-s+m-1}(\bar{U})} \leq C' \|P^* v\|_{\dot{H}^{-s}(\bar{U})}$.
(3) Conclude using a duality argument.

Remark. For this and related results, see Hörmander [Hör63, Chapters III, XIII], [Hör71b, §3.2] and Taylor [Tay81, Chapter VI, §3].

Exercise 9.9 (Rellich Compactness). Let $\Omega \subset M$ be a compact submanifold with corners of the smooth manifold M as in Definition 9.28. Let $s > s'$. Show that the inclusion $H^s(\Omega)^{\bullet,-} \to H^{s'}(\Omega)^{\bullet,-}$ is compact.

The purpose of the next exercise is to apply the theory of forcing problems developed in this chapter to the solution of initial value problems.

Exercise 9.10 (Initial Value Problems from Forcing Problems). Let $M = \mathbb{R} \times X$ where X is a closed manifold. Let g be a Lorentzian metric on M for which $\mathrm{d}t$ is past timelike. We assume that $g^{-1}(\mathrm{d}t, \mathrm{d}t) = -1$. (Without this normalization, the exercise can be solved as well; only the expression in (9.163) gets more complicated.) Let $P \in \mathrm{Diff}^2(M)$ be a wave-type operator on (M, g), so

$$P = \partial_t^2 + P_1 \partial_t + P_2 \tag{9.162}$$

where $P_1 \in \mathcal{C}^\infty(\mathbb{R}; \mathrm{Diff}^1(X))$ and $P_2 \in \mathcal{C}^\infty(\mathbb{R}; \mathrm{Diff}^2(X))$.

(1) Let $u \in \mathcal{C}^\infty(\mathbb{R} \times X)$ and set $f := Pu$. With $H = H(t)$ denoting the Heaviside function, prove the identity

$$P(Hu) = Hf + \delta'(t)u_0 + \delta(t)(P_1(0)u_0 + u_1), \quad u_0 := u(0,\cdot),\ u_1 := \partial_t u(0,\cdot). \tag{9.163}$$

(2) Given $u_0, u_1 \in \mathcal{C}^\infty(X)$ and $f \in \mathcal{C}^\infty([0,T] \times X)$, define $f' := Hf + \delta'(t)u_0 + \delta(t)(P_1(0)u_0 + u_1)$. Show that $f' \in H^{s'}([0,T] \times X)^{\bullet,-}$ for all $s' < -\frac{3}{2}$, and $\mathrm{WF}(f') \subset N^*\{t = 0\} \setminus o$.

(3) Continuing the previous part, define $u' \in H^{s'+1}([0,T] \times X)^{\bullet,-}$ as the solution of $Pu' = f'$. Show using microlocal elliptic and real principal type propagation results that $u' \in H^s([0,T'] \times X)^{\bullet,-}$ for all $s < \frac{1}{2}$ and $T' \in (0,T)$, and that u' is smooth in $(0,T) \times X$.

(4) Continuing the previous two parts, let $\tilde{f} \in \mathcal{C}^\infty([-1,T] \times X)$ be an extension of f. Show that there exists a unique $\tilde{u} \in \mathcal{C}^\infty([-1,T] \times X)$ which solves $P\tilde{u} = \tilde{f}$ and equals u' on $(0,T) \times X$. Show, finally, that $u := \tilde{u}|_{[0,T]\times X}$ solves the initial value problem

$$Pu = f, \quad u|_{t=0} = u_0, \quad \partial_t u|_{t=0} = u_1. \tag{9.164}$$

(5) Use Exercise 9.3 to extend the solvability result for (9.164) to the case that $u_0 \in H^m(X)$, $u_1 \in H^{m-1}(X)$, $f \in \bar{H}^{m-1}([0,T] \times X)$, $m \in \mathbb{N}$, with the solution lying in $\bar{H}^m([0,T] \times X)$.

Exercise 9.11 (Exponential Bounds for Stationary Wave Equations). Let X be a closed manifold, and let $M = \mathbb{R} \times X$. Let g be a smooth Lorentzian metric on M which is stationary, i.e., $\mathcal{L}_{\partial_t} g = 0$, and for which $\mathrm{d}t$ is past timelike. Let $P \in \mathrm{Diff}^2(M)$ be a wave-type operator on (M,g) which is stationary, i.e., $[\partial_t, P] = 0$. Define Sobolev norms on $\mathbb{R} \times X$ by $\|u\|^2_{H^m(\mathbb{R}\times X)} := \sum_{j=0}^m \int_{\mathbb{R}} \|\partial_t^j u(t,\cdot)\|^2_{H^{m-j}(X)}\,\mathrm{d}t$. Show that for all $m \in \mathbb{N}$ there exist constants C, α such that for the forward solution u of $Pu = f$, with $t \geq 0$ on $\operatorname{supp} u, \operatorname{supp} f$, we have

$$\|e^{-\alpha t}u\|_{\dot{H}^m([0,\infty)\times X)} \leq C\|e^{-\alpha t}f\|_{\dot{H}^{m-1}([0,\infty)\times X)}. \tag{9.165}$$

Exercise 9.12 (Not a Semiclassical Wave-Type Operator). Consider $P_h = -(hD_t)^2 + (hD_\theta)^2 - 1$ on $\mathbb{R}_t \times \mathbb{S}^1_\theta$ where $\mathbb{S}^1_\theta = \mathbb{R}/2\pi\mathbb{Z}$. Show that there are no constants C, N such that for all $f \in \mathcal{C}^\infty(\mathbb{S}^1)$ the solution $u_h = u_h(t,x)$ of the initial value problem

$$P_h u_h = 0, \quad u_h(0,x) = f(x), \quad \partial_t u_h(0,x) = 0, \tag{9.166}$$

satisfies the bound

$$\|u_h(1,x)\|_{L^2(\mathbb{S}^1)} \leq Ch^{-N}\|f\|_{H^N(\mathbb{S}^1)}. \tag{9.167}$$

Chapter 10
Propagation of Singularities at Radial Sets



The propagation theorem proved in Chapter 8 is a general-purpose tool for analyzing the regularity of solutions of general linear PDEs $Pu = f$ when $P \in \Psi^m(M)$ has a real homogeneous principal symbol, *assuming* one has information on u *somewhere* to begin with. In particular, one *cannot*, in general, control u globally *only in terms of* f (cf. the a priori control assumption of microlocal regularity of u encoded by the term Eu in the estimate (8.21)). What is needed for global control of u is the existence of a subset of phase space where one can get unconditional control of u. There are two main settings in which this happens.

(1) *Initial value problems* or *forward problems* for evolution equations. The required initial control of u comes from the regularity of the initial data (for initial value problems) or from the vanishing of u prior to the onset of the forcing term (for forward problems). This scenario was discussed at length in the previous chapter.
(2) *Radial sets* (see (8.17)) or other degeneracies of P. Generalized radial sets are the subject of this chapter.

10.1 Radial Point Estimates: A Simple Example

Consider the multiplication operator $P := x \in \Psi^0(M)$, $M := \mathbb{R}^n = \mathbb{R}_x \times \mathbb{R}^{n-1}_y$. Write ξ, η for the canonical momentum variables. The principal symbol of P is $p(x, y, \xi, \eta) = x$, the characteristic set is

$$\Sigma := \operatorname{Char}(P) = \{(x, y, \xi, \eta) \colon x = 0,\ (\xi, \eta) \neq (0, 0)\} \subset T^*\mathbb{R}^n \setminus o, \tag{10.1}$$

and the Hamiltonian vector field is $H_p = -\partial_\xi$. (This is a special case of Example 8.1(3).) Suppose that $u \in \mathscr{D}'(\mathbb{R}^n)$ solves the "PDE"

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$$Pu = xu = f \in \mathcal{C}^\infty(\mathbb{R}^n). \tag{10.2}$$

Elliptic regularity (Proposition 6.32) implies that $\mathrm{WF}(u) \subset \Sigma$. By the propagation of singularities (Theorem 8.4), $\mathrm{WF}(u)$ is a union of maximally extended null-bicharacteristics of P. Note that at $(x, 0, \xi, 0) \in \Sigma$,

$$H_p = -\partial_\xi \tag{10.3}$$

is radial; the null-bicharacteristic through $(x, 0, \xi, 0)$ remains in the half-line $\{(x, 0, c\xi, 0)\colon c > 0\}$, and hence, the propagation theorem is vacuous there. Let us thus define the following two sets of radial points:

$$\mathcal{R}_\pm = \{(0, y, \xi, 0)\colon \pm\xi > 0\} \subset \Sigma. \tag{10.4}$$

See Figure 10.1.

Now, the general solution of the PDE (10.2) is of the form

$$u(x, y) = u_+(y)(x + i0)^{-1} + u_-(y)(x - i0)^{-1} + \tilde{u}(x, y), \tag{10.5}$$

where $u_\pm \in \mathscr{D}'(\mathbb{R}^{n-1})$ and $\tilde{u} \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $u_+(y) + u_-(y) = f(0, y)$. To prove this, write $f(x, y) = f(0, y) + xg(x, y)$; then $(x + i0)^{-1} f(0, y) + g(x, y)$ is one solution of (10.2). The difference $v \in \mathscr{D}'(\mathbb{R}^n)$ of two solutions of (10.2) solves $xv = 0$, so $v(x, y) = \delta(x)w(y)$ for some $w \in \mathscr{D}'(\mathbb{R}^{n-1})$. It remains to recall Exercise 2.6(6).

Proposition 10.1 (Multiplication by x). *Suppose $u \in \mathscr{D}'(\mathbb{R}^n)$, $xu = f \in \mathcal{C}^\infty(\mathbb{R}^n)$, and $\mathrm{WF}^{s_0}(u) \cap \mathcal{R}_+ = \emptyset$ for some $s_0 > -\frac{1}{2}$. Then $\mathrm{WF}(u) \subset \mathcal{R}_-$. Moreover, $\mathrm{WF}^s(u) = \emptyset$ for all $s < -\frac{1}{2}$.*

Proof. We have $(x \pm i0)^{-1} \in H^s_{\mathrm{loc}}(\mathbb{R}^n)$ if and only if $s < -\frac{1}{2}$. (This follows from (2.72); see also Exercise 6.4.) Moreover, $\mathrm{WF}^{s_0}(u_-(y)(x - i0)^{-1})$ and

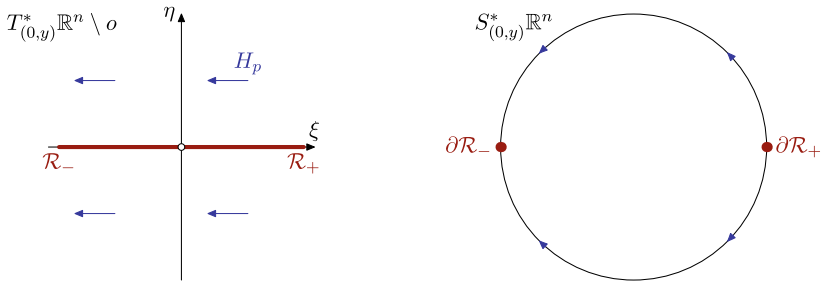


Figure 10.1 *On the left:* the radial sets $\mathcal{R}_\pm$ in one fiber of $T^*\mathbb{R}^n$ (here shown for $n = 2$), and the Hamiltonian vector field $H_p = -\partial_\xi$ of $p(x, y, \xi, \eta) = x$. *On the right:* null-bicharacteristics in a fiber of $S^*\mathbb{R}^n$. The boundaries $\partial\mathcal{R}_\pm$ of $\mathcal{R}_\pm$ at fiber infinity are drawn as thick dots.

$\mathrm{WF}^{s_0}(\tilde{u})$ are disjoint from $\mathcal{R}_+$ for all $u_- \in \mathscr{D}'(\mathbb{R}^{n-1})$ (cf. Example 6.30) and $\tilde{u} \in \mathcal{C}^\infty(\mathbb{R}^n)$, whereas $\mathrm{WF}^{s_0}(u_+(y)(x+i0)^{-1}) \cap \mathcal{R}_+ = \emptyset$ if and only if $u_+ = 0$. (These statements in fact hold for all $s_0 \geq -\frac{1}{2}$.) The assumption on u thus implies that $u_+ \equiv 0$; therefore, $u_-(y) = f(0,y)$ is smooth. The first conclusion now follows from the fact that $\mathrm{WF}(u_-(y)(x-i0)^{-1}) \subset \mathcal{R}_-$. The second conclusion follows from the fact that $(x-i0)^{-1} \in H^{-\frac{1}{2}-\epsilon}_{\mathrm{loc}}(\mathbb{R}^n)$ for all $\epsilon > 0$. □

Instead of using the explicit solution formula (10.5), a microlocal proof of Proposition 10.1 will proceed via the concatenation of three microlocal regularity results:

(1) Under the a priori regularity assumption $\mathrm{WF}^{s_0}(u) \cap \mathcal{R}_+ = \emptyset$, one can conclude that, in fact, $\mathrm{WF}(u) \cap \mathcal{R}_+ = \emptyset$, and hence, u is microlocally smooth near $\mathcal{R}_+$.
(2) By the propagation of regularity, we infer that $\mathrm{WF}(u) \subset \mathcal{R}_-$.
(3) If $\mathrm{WF}(u)$ is disjoint from a punctured neighborhood of $\mathcal{R}_-$, then u is microlocally in H^s near $\mathcal{R}_-$ for all $s < -\frac{1}{2}$.

Parts (1) and (3) are special cases of a general result on the propagation of singularities/regularity at radial points which we state and prove below (Theorem 10.5). A key feature of this result is that there is a *threshold regularity*: if the microlocal regularity of u exceeds a threshold (here $-\frac{1}{2}$) at $\mathcal{R}_+$, then u is microlocally as smooth near $\mathcal{R}_+$ as f permits. Thus, we can propagate H^s-regularity "out of $\mathcal{R}_+$" for $s > -\frac{1}{2}$. On the other hand, one can conclude microlocal regularity of u *below* a threshold (here again $-\frac{1}{2}$) when propagating *into* $\mathcal{R}_-$.

The desired radial point estimates can, in a very rough sense, be thought of as uniform versions of real principal type propagation for infinite times: after all, in the above example, the flow of H_p starting at a point $(x,0,\xi,\eta)$ with $\eta \neq 0$ tends (when projected to $S^*\mathbb{R}^n$) to $\mathcal{R}_\mp$ in the forward, resp. backward direction. The key to the proof of radial point estimates, in the context of a positive commutator argument, is thus the construction of a symbol with suitable *uniform* monotonicity properties along the H_p-flow. An obvious candidate for such a symbol in the case (10.3) and in the region $\xi > \epsilon|\eta|$, $\epsilon > 0$, is the weight ξ^{2s+1}: its derivative along H_p is $-(2s+1)\xi^{2s}$ and thus a definite multiple of ξ^{2s} when $s \neq -\frac{1}{2}$. (A non-microlocal toy model for "long time" energy estimates is given in Exercise 10.1.) Note that it is precisely the nontrivial behavior of H_p in the fiber-radial direction at $\mathcal{R}_\mp$ which is responsible for this monotonicity. Unlike in our proof of real principal type propagation estimates in Chapter 8, we must therefore *not* pass entirely to the cosphere bundle, as this loses information about the fiber-radial direction. Instead, we work on the radially compactified cotangent bundle introduced in Section 6.5:

Lemma 10.2 (Homogeneous Vector Fields and Compactifications). *Suppose $V \in \mathcal{V}(T^*M \setminus o)$ is homogeneous of degree 0. Then V extends by continuity to a smooth vector field*

$$V \in \mathcal{V}(\overline{T^*}M \setminus o) \tag{10.6}$$

which is tangent to S^*M.

Proof. In local coordinates $x_1, \dots, x_n$ on M and writing $\xi_1, \dots, \xi_n$ for the canonical momentum variables, we have

$$V = \sum_{j=1}^{n} a_j(x,\xi)\partial_{x_j} + b_{jk}(x,\xi)\xi_k\partial_{\xi_j}, \tag{10.7}$$

where $a_j(x,\lambda\xi) = a_j(x,\xi)$ and $b_{jk}(x,\lambda\xi) = b_{jk}(x,\xi)$ for all $\lambda > 0$. In the region where $\xi_1 > 0$, let us work in projective coordinates

$$\rho := \frac{1}{\xi_1}, \quad \hat{\xi}_j := \frac{\xi_j}{\xi_1},\ j = 2, \dots, n. \tag{10.8}$$

Then $a_j(x,\xi) = a_j(x,(1,\hat{\xi}_2,\dots,\hat{\xi}_n))$ is smooth down to $\rho = 0$, and so is b_{jk}. Moreover, $\partial_{x_j} \in \mathcal{V}(\overline{T^*}M)$. For $2 \le j,k \le n$, we compute

$$\begin{aligned} \xi_1\partial_{\xi_1} &= -\rho\partial_\rho - \sum_{\ell=2}^{n} \hat{\xi}_\ell\partial_{\hat{\xi}_\ell}, & \xi_1\partial_{\xi_j} &= \partial_{\hat{\xi}_j}, \\ \xi_k\partial_{\xi_1} &= -\hat{\xi}_k\rho\partial_\rho - \sum_{\ell=2}^{n} \hat{\xi}_k\hat{\xi}_\ell\partial_{\hat{\xi}_\ell}, & \xi_k\partial_{\xi_j} &= \hat{\xi}_k\partial_{\hat{\xi}_j}. \end{aligned} \tag{10.9}$$

The lemma follows from an inspection of these expressions. □

In the coordinates used in (10.1), consider again the equation $Pu := xu = f$ and the Hamiltonian vector field $H_p = -\partial_\xi$. In projective coordinates

$$\rho := \xi^{-1},\ \hat{\eta} := \frac{\eta}{\xi}, \quad \xi > \epsilon|\eta|, \tag{10.10}$$

let us rescale H_p to the vector field

$$V := \xi H_p = -\xi\partial_\xi = \rho\partial_\rho + \hat{\eta}\partial_{\hat{\eta}}, \quad \rho > 0,\ |\hat{\eta}| < \epsilon^{-1}, \tag{10.11}$$

which is homogeneous of degree 0. Restricting V to $S^*\mathbb{R}^n$, the term $\rho\partial_\rho$ disappears, and we see that H_p being radial means that $V|_{S^*\mathbb{R}^n}$ vanishes on

$$\partial\mathcal{R}_+ := \{(x,y,\rho,\hat{\eta})\colon x = 0,\ \rho = 0,\ \hat{\eta} = 0\} \subset S^*\mathbb{R}^n, \tag{10.12}$$

the boundary of $\mathcal{R}_+$ (from (10.4)) at fiber infinity. In our quest to prove microlocal estimates at $\partial\mathcal{R}_+$ via positive commutators, we therefore need to make use of the first summand in (10.11): we need to exploit that V has nontrivial behavior in the fiber-radial direction, that is, it acts nontrivially on weights $\rho^{-s} = \xi^s$. Making our previous heuristic argument more precise, we consider a commutant

$$a = \rho^{-2s-1}\psi(|\hat{\eta}|^2), \tag{10.13}$$

where $\psi \in \mathcal{C}_c^\infty([0,\infty))$ is identically 1 near 0 and satisfies $r\psi'(r) \leq 0$ for all $r \geq 0$. Then, in the projective coordinate system (10.10), we have

$$H_p a = \rho V a = \rho^{-2s}\big(-(2s+1)\psi(|\hat{\eta}|^2) + 2|\hat{\eta}|^2\psi'(|\hat{\eta}|^2)\big). \tag{10.14}$$

When $s > -\frac{1}{2}$, *both summands have the same (indefinite) sign* (namely, they are ≤ 0). Moreover, crucially, the first summand is *elliptic* at $\partial\mathcal{R}_+$. Thus, quantizing the calculation (10.14) as in Chapter 8, we can write

$$i[P, A] = -B^*B - B_1^*B_1 + R, \quad R \in \Psi^{2s-1}, \quad s > -\frac{1}{2}, \tag{10.15}$$

with $B \in \Psi^s$ elliptic at $\partial\mathcal{R}_+$ and $B_1 \in \Psi^s$ capturing the term involving ψ'. Ultimately, this gives an estimate for $\|Bu\|_{L^2}$, and hence, a microlocal H^s-estimate of u near $\partial\mathcal{R}_+$, *without any a priori control term*. Notice on the other hand that the "positivity" (meaning: the "good" sign, so currently: the negativity...) of the first term in (10.14) is delicate and limited; thus, error terms from a regularization argument (cf. the terms $f_r, \tilde{f}_r$ in (8.61) and (8.65)) can only be absorbed when the amount of regularization is limited. This will be the reason for the a priori regularity assumption at $\partial\mathcal{R}_+$.

On the other hand, when $s < -\frac{1}{2}$, the two terms in (10.14) have *opposite* signs, but the first summand is still elliptic at $\partial\mathcal{R}_+$. Thus, *assuming* H^s-control of u on the support of the second summand (which is contained in a punctured neighborhood of $\partial\mathcal{R}_+$), we can *conclude* H^s-regularity of u near $\partial\mathcal{R}_+$. (The situation at $\partial\mathcal{R}_-$ is completely analogous of course.)

10.2 Radial Point Estimates: General Setup

We now set up the general theorem on the propagation of singularities/regularity at (generalized) radial points. We need the following notion.

Definition 10.3 (Quadratic Defining Functions). Let M be a closed smooth manifold, and let $X \subset M$ be a smooth submanifold. Then $\rho \in \mathcal{C}^\infty(M)$

is a *quadratic defining function of* X if $X = \rho^{-1}(0)$ and $\mathrm{d}\rho = 0$ on X, and for all $p \in X$, $V^2\rho = V(V\rho) > 0$ on X near p for all $V \in \mathcal{V}(M)$ which are not tangent to X at p.

See Exercise 10.2 for a slightly cleaner definition. Every closed submanifold X admits a quadratic defining function (Exercise 10.3). Moreover, given any quadratic defining function $\rho \in \mathcal{C}^\infty(M)$, there exist, around every point on X, local coordinates $x_1, \dots, x_n$ on M in which X is locally given by $x_1 = \dots = x_k = 0$ (with k being the codimension of X) and $\rho = \frac{1}{2}\sum_{j=1}^k x_j^2$ (Exercise 10.4). In particular, when X is compact, $\rho \in \mathcal{C}^\infty(M)$ is a fixed quadratic defining function, and $U, V \subset M$ are open neighborhoods of X with $\bar{U}$ compact, there exists $\delta > 0$ such that $\{p \in U \colon \rho(p) < \delta\} \subset V$. Thus, one can use a quadratic defining function of X to define arbitrarily small neighborhoods of X.

Definition 10.4 (Normal Sources and Sinks). Given a vector field $V \in \mathcal{V}(M)$ tangent to the closed submanifold $X \subset M$, we say that X is a *normal source* for V if there exist an open neighborhood $U \subset M$ of X and a quadratic defining function $\rho \in \mathcal{C}^\infty(M)$ such that

$$V\rho \geq \beta\rho \text{ on } U \text{ for some } 0 < \beta \in \mathcal{C}^\infty(U). \tag{10.16}$$

A *normal sink* is defined analogously through the condition $V\rho \leq -\beta\rho$.

We now make the following assumptions.

(R.1) $P \in \Psi^m_{\mathrm{cl}}(M)$ is a classical operator with real homogeneous principal symbol p. We regard its characteristic set as a closed subset $\mathrm{Char}(P) \subset \overline{T^*M}\backslash o$ (given by the closure of the usual characteristic set in $T^*M \backslash o$).

(R.2) Fix an elliptic symbol $0 < \rho \in S^{-1}_{\mathrm{cl}}(T^*M)$ and let

$$\tilde{p} := \rho^m p \in \mathcal{C}^\infty(\overline{T^*M} \setminus o), \quad \tilde{H}_p := \rho^{m-1} H_p \in \mathcal{V}(\overline{T^*M} \setminus o). \tag{10.17}$$

We assume that

$$\mathcal{R} \subset \mathrm{Char}(P) \cap S^*M \subset S^*M \text{ is a smooth compact submanifold, and } \tilde{H}_p \text{ is tangent to } \mathcal{R}. \tag{10.18}$$

We assume moreover that $\mathrm{d}\tilde{p} \neq 0$ in a neighborhood of $\mathcal{R}$ in S^*M. (This implies that $\mathrm{Char}(P)$ is a smooth conic codimension 1 submanifold of $\overline{T^*M} \setminus o$ near $\mathcal{R}$.)

For the sake of definiteness, we assume that $\mathcal{R}$ is a *normal source* for the $\tilde{H}_p$-flow. (If one replaces P by $-P$ and thus p by $-p$, a normal source turns into a normal sink and vice versa.) This means:

(R.3) $\mathcal{R}$ is a normal source for $\tilde{H}_p$ inside of $\mathrm{Char}(P) \cap S^*M$. That is, there exist a quadratic defining function $\rho_1 \in \mathcal{C}^\infty(S^*M)$ of $\mathcal{R}$ inside of $\mathrm{Char}(P)$ (that is, $\rho_1|_{\mathrm{Char}(P)}$ is a quadratic defining function of $\mathcal{R}$ in a neighborhood of $\mathcal{R}$) and $\delta_1 > 0$ as well as a positive function $0 < \beta_1 \in \mathcal{C}^\infty(S^*M)$ such that

$$\tilde{H}_p \rho_1 \geq \beta_1 \rho_1 \text{ on } \mathrm{Char}(P) \cap \{\rho_1 < \delta_1\}. \tag{10.19}$$

(R.4) We have

$$\tilde{H}_p \rho = \beta_0 \rho, \quad \beta_0|_{\mathcal{R}} > 0. \tag{10.20}$$

(Since $\tilde{H}_p$ is tangent to S^*M, $\tilde{H}_p\rho$ vanishes there and hence is of the stated form with $\beta_0 \in \mathcal{C}^\infty(\overline{T^*}M)$ near $\mathcal{R}$; the new information is the positivity of β_0 at $\mathcal{R}$.)

Assumption (R.3) says that $\mathcal{R}$ is a source for the $\tilde{H}_p$-flow within $\mathrm{Char}(P) \cap S^*M$. Assumption (R.4) says that $\mathcal{R}$ is a source for the $\tilde{H}_p$-flow also in the fiber-radial direction. See Figure 10.2. Some authors call $\mathcal{R}$ a *generalized radial source* for P; the attribute "generalized" is used since H_p is not required to be radial on the conic subset of $T^*M \setminus o$ determined by $\mathcal{R}$ (see also Example 10.14).

The subprincipal part of P at $\mathcal{R}$ plays an important role: it causes a shift in the threshold regularity ($-\frac{1}{2}$ in (10.15)); see also Example 10.12 below. We thus introduce the following notation:

(R.5) Fix a volume density on M. Let $p_1 := \sigma^{m-1}(\frac{1}{2i}(P - P^*))$ and $\tilde{p}_1 := \rho^{m-1} p_1$. Define $\tilde{\beta} \in \mathcal{C}^\infty(S^*M)$ near $\mathcal{R}$ by

$$\tilde{p}_1 = \beta_0 \tilde{\beta}. \tag{10.21}$$

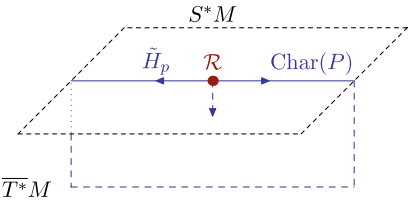


Figure 10.2 A radial set $\mathcal{R} \subset S^*M$, here a single point, which is a normal source for the rescaled Hamiltonian vector field $\tilde{H}_p$ inside of the characteristic set in S^*M and also in the fiber-radial direction.

We now state the main result of this chapter in two forms: one qualitative (analogous to Theorem 8.8) and one quantitative (analogous to Theorem 8.10).

Theorem 10.5 (Microlocal Regularity At Normal Sources: Qualitative Statement). *Suppose P and $\mathcal{R} \subset \operatorname{Char}(P) \subset S^*M$ satisfy the assumptions (R.1)–(R.5). Let $u \in \mathscr{D}'(M)$, $Pu =: f$.*

(1) (Forward propagation: out of $\mathcal{R}$.) *Let $s, s_0 \in \mathbb{R}$ with $s > s_0 > \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$. If $\mathrm{WF}^{s_0}(u) \cap \mathcal{R} = \emptyset$ and $\mathrm{WF}^{s-m+1}(f) \cap \mathcal{R} = \emptyset$, then $\mathrm{WF}^{s}(u) \cap \mathcal{R} = \emptyset$.*

(2) (Backward propagation: into $\mathcal{R}$.) *Let $s \in \mathbb{R}$ with $s < \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$. If $\mathrm{WF}^{s}(u)$ is disjoint from a punctured neighborhood of $\mathcal{R}$ and $\mathrm{WF}^{s-m+1}(f) \cap \mathcal{R} = \emptyset$, then $\mathrm{WF}^{s}(u) \cap \mathcal{R} = \emptyset$.*

Remark 10.6 (Choices; Bundles). When $\tilde{H}_p|_{\mathcal{R}} \neq 0$, the quantity $\tilde{\beta}|_{\mathcal{R}}$ depends on the choice of volume density used to define P^*. To obtain the sharpest possible result, one takes $\tilde{\beta}$ to be the infimum (for part (1)) or supremum (for part (2)) over all choices. Theorem 10.5 also holds for $P \in \Psi^m(M; E)$ with real, homogeneous, and scalar principal symbol. The threshold quantities now depend on the choice of a fiber inner product on E.

The version with estimates, which we immediately state in a semiglobal form analogous to Corollary 8.19, reads as follows.

Theorem 10.7 (Microlocal Regularity At Normal Sources: Quantitative Statement). *Suppose P and $\mathcal{R} \subset \operatorname{Char}(P) \subset S^*M$ satisfy the assumptions (R.1)–(R.5). Let $K \subset M$ be a compact set such that the base projection of $\mathcal{R}$ is contained in K°. Let $\tilde{K} \subset M$ be a compact set such that for the Schwartz kernel K_P of P, we have $\operatorname{supp} K_P \cap (K \times M) \subset K \times \tilde{K}$. Fix $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(M)$ with $\chi = 1$ near $\tilde{K}$. Let $u \in \mathscr{D}'(M)$.*

(1) (Forward propagation: out of $\mathcal{R}$.) *Let $B, G \in \Psi^0(M)$ be such that:*

(i) *The Schwartz kernels of B, G are supported in $K \times K$.*
(ii) $\mathrm{WF}'(B) \subset \mathrm{Ell}(G)$.
(iii) $\mathcal{R} \subset \mathrm{Ell}(G)$.
(iv) *All backward null-bicharacteristics of P starting from a point in $\mathrm{WF}'(B) \cap \operatorname{Char}(P)$ tend to $\mathcal{R}$ while remaining in $\mathrm{Ell}(G)$.*

Then for all $s, s_0, N \in \mathbb{R}$ with $s > s_0 > \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$, there exists $C > 0$ such that

$$\|Bu\|_{H^s} \leq C\big(\|GPu\|_{H^{s-m+1}} + \|Gu\|_{H^{s_0}} + \|\chi u\|_{H^{-N}}\big). \tag{10.22}$$

This estimate holds in the strong sense that if u is such that the right-hand side is finite, then so is the left-hand side, and the estimate holds.

(2) (Backward propagation: into $\mathcal{R}$.) *Let $B, G, E \in \Psi^0(M)$ be such that:*

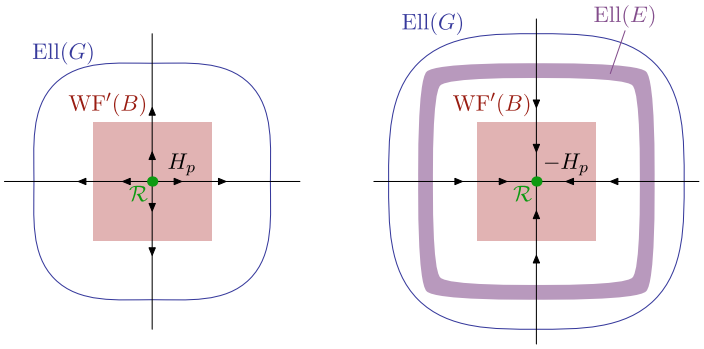


Figure 10.3 Illustration of Theorem 10.7: inside the characteristic set, we show the Hamiltonian flow (i.e., null-bicharacteristics) near the normal source $\mathcal{R}$. *On the left:* propagation out of $\mathcal{R}$. *On the right:* propagation into $\mathcal{R}$.

(i) *The Schwartz kernels of* B, G, E *are supported in* $K \times K$.
(ii) $\mathrm{WF}'(B) \subset \mathrm{Ell}(G)$.
(iii) $\mathcal{R} \subset \mathrm{Ell}(B)$.
(iv) *All forward null-bicharacteristics of* P *from* $\mathrm{WF}'(B) \cap \mathrm{Char}(P)$ *are either contained in* $\mathcal{R}$ *or enter* $\mathrm{Ell}(E)$ *in finite time, all while remaining in* $\mathrm{Ell}(G)$.

Then for all $s, N \in \mathbb{R}$ *with* $s < \frac{m-1}{2} + \tilde{\beta}$ *on* $\mathcal{R}$, *there exists* $C > 0$ *such that*

$$\|Bu\|_{H^s} \leq C\big(\|GPu\|_{H^{s-m+1}} + \|Eu\|_{H^s} + \|\chi u\|_{H^{-N}}\big). \tag{10.23}$$

This estimate holds in the usual strong sense.

See Figure 10.3. The estimates (10.22)–(10.23) are often referred to as (generalized) *radial point estimates*.

Before describing examples of the above setup, we present a convenient result which allows one to reduce the verification of the normal source condition (R.3) to the computation of the linearization of $\tilde{H}_p$.

Definition 10.8 (Linearization of a Vector Field). Let M be a smooth manifold, $X \subset M$ a closed smooth submanifold, and suppose $V \in \mathcal{V}(M)$ vanishes along X. Then the *linearization* of V is the vector bundle map

$$[V]\colon N^*X \to N^*X, \quad \mathrm{d}f \mapsto \mathrm{d}(Vf), \tag{10.24}$$

for $f \in \mathcal{C}^\infty(M)$ vanishing on X.

In local coordinates $x_1, \dots, x_n$ on M in which X is given by $x_1 = \cdots = x_k = 0$, we can write $V = \sum_{i=1}^k x_i V_i$ where $V_i \in \mathcal{V}(M)$; moreover, any $f \in$

$\mathcal{C}^\infty(M)$ vanishing on X can be written as $f = \sum_{j=1}^k x_j f_j$ where $f_j \in \mathcal{C}^\infty(M)$. At $x_1 = \cdots = x_k = 0$, we then compute

$$\mathrm{d}(Vf) = \mathrm{d}\Big(\sum_{i,j=1}^k x_i x_j V_i(f_j) + \sum_{i,j=1}^k x_i V_i(x_j) f_j \Big) = \sum_{i,j=1}^k V_i(x_j) f_j \, \mathrm{d}x_i, \tag{10.25}$$

which thus depends linearly on $\mathrm{d}f = \sum_{j=1}^k f_j \, \mathrm{d}x_j$. Concretely, if we write $V_i = \sum_{\ell=1}^n V_{i\ell}\partial_\ell$, then in the basis $\mathrm{d}x_1, \ldots, \mathrm{d}x_k$ of N^*X, the map $[V]$ is given by the matrix $(V_i(x_\ell))_{i,\ell=1}^k = (V_{i\ell})_{i,\ell=1}^k$.

Lemma 10.9 (Linearizations and Normal Sources). *For $V \in \mathcal{V}(M)$ vanishing on the closed submanifold $X \subset M$, the following conditions are equivalent.*

(1) *X is a normal source for V.*
(2) *All eigenvalues of $[V] \in \mathrm{End}(N_p^* X)$, $p \in X$, have positive real part.*

Proof. We first prove (1) $\Longrightarrow$ (2). Given a quadratic defining function ρ satisfying (10.16), pick local coordinates $x_1, \ldots, x_n$ around a point in X such that $\rho = \frac{1}{2}\sum_{j=1}^k x_j^2$ (Exercise 10.4), and X is given by $x' = 0$ where we write $x' = (x_1, \ldots, x_k)$ and $x'' = (x_{k+1}, \ldots, x_n)$. Write $V = \sum_{i=1}^k \sum_{\ell=1}^n x_i V_{i\ell}\partial_\ell$; then $V\rho = \sum_{i,\ell=1}^k V_{i\ell}(x', x'')x_i x_\ell$ differs from $\sum_{i,\ell=1}^k V_{i\ell}(0, x'')x_i x_\ell$ by a term vanishing cubically at $x' = 0$. The validity of (10.16) thus implies that for the matrix $A := [V] = (V_{i\ell}(0, x''))_{i,\ell=1}^k \in \mathbb{R}^{k\times k}$, we have

$$\sum_{i,j=1}^k A_{ij} x_i x_j = (x')^T A x' > 0, \quad 0 \neq x' = (x_1, \ldots, x_k) \in \mathbb{R}^k. \tag{10.26}$$

Note that A need not be a symmetric matrix. Suppose that $\lambda + i\mu$ is an eigenvalue of A with eigenvector $v + iw \neq 0$, where $\lambda, \mu \in \mathbb{R}$ and $v, w \in \mathbb{R}^k$. Since A is a real matrix, the eigenvalue equation $A(v+iw) = (\lambda+i\mu)(v+iw)$ yields two equations: $Av = \lambda v - \mu w$ and $Aw = \mu v + \lambda w$. Therefore, (10.26) implies

$$0 < v^T A v + w^T A w = \lambda(|v|^2 + |w|^2), \tag{10.27}$$

and hence $\lambda > 0$.

To prove (2) $\Longrightarrow$ (1), we first note that it suffices to construct a quadratic defining function ρ satisfying (10.16) locally. Indeed, if ρ_1, ρ_2 are two such functions defined on open sets U_1, U_2 and χ_1, χ_2 is a partition of unity subordinate to U_1, U_2, then for $\rho := \chi_1\rho_1 + \chi_2\rho_2$, one computes

$$V\rho = \chi_1 V\rho_1 + \chi_2 V\rho_2 + (V\chi_1)(\rho_1 - \rho_2). \tag{10.28}$$

The first and second terms are $\geq \chi_1\beta\rho_1$ and $\geq \chi_2\beta\rho_2$, respectively; the third term vanishes cubically at X. Thus, $V\rho \geq \frac{1}{2}\beta\rho$ in a neighborhood of $X \cap (U_1 \cup U_2)$.

For the local construction then, we use Lemma 10.10 below to deduce that if all eigenvalues of $A = (A_{ij}) \in \mathbb{R}^{k\times k}$ have positive real part, then there exists a positive definite matrix $B = (B_{ij}) \in \mathbb{R}^{k\times k}$ such that $v^T ABv = \frac{1}{2}v^T(AB + (AB)^T)v > 0$ for all $0 \neq v \in \mathbb{R}^k$. Then the function $\rho := \frac{1}{2}\sum_{j,\ell=1}^k B_{j\ell}x_j x_\ell$ satisfies

$$\sum_{i,j=1}^k x_i A_{ij}\partial_j\rho = \sum_{i,j,\ell=1}^k A_{ij}B_{j\ell}x_i x_\ell \geq \frac{1}{2}\beta\rho \tag{10.29}$$

where $\beta > 0$ is such that $\frac{1}{2}(AB + (AB)^T) - \beta B$ is positive definite still. □

Lemma 10.10. *Let $A \in \mathbb{R}^{k\times k}$. Then the following conditions are equivalent.*

(1) *Every eigenvalue of A has positive real part.*
(2) *There exists a positive definite matrix $B \in \mathbb{R}^{k\times k}$ such that $v^T ABv > 0$ for all $0 \neq v \in \mathbb{R}^k$. (Equivalently, $AB + (AB)^T$ is positive definite.)*

Proof. We write $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ for the Euclidean inner product and norm, respectively. We show that (1) implies (2). For $L > 0$, to be determined, set

$$\langle v, w\rangle_L := \int_0^L \langle e^{tA^T}v, e^{tA^T}w\rangle\,\mathrm{d}t, \quad v, w \in \mathbb{R}^k. \tag{10.30}$$

For $v \in \mathbb{R}^k$, we have $\frac{\mathrm{d}}{\mathrm{d}t}e^{tA^T}v = A^T e^{tA^T}v$. Hence, writing $v = e^{-LA^T}w$, we have

$$\begin{aligned}\langle v, A^T v\rangle_L &= \frac{1}{2}\int_0^L \frac{\mathrm{d}}{\mathrm{d}t}\langle e^{tA^T}v, e^{tA^T}v\rangle\,\mathrm{d}t \\ &= \frac{\|e^{LA^T}v\|^2 - \|v\|^2}{2} = \frac{\|w\|^2 - \|e^{-LA^T}w\|^2}{2}.\end{aligned} \tag{10.31}$$

Since the eigenvalues of A^T are the same as those of A, every eigenvalue of e^{-A^T} has absolute value less than 1. Therefore, by the spectral radius formula, for sufficiently large $L \in \mathbb{N}$, the operator norm of $e^{-LA^T} = (e^{-A^T})^L$ is less than 1. For such L, we thus conclude that $\langle v, A^T v\rangle_L > 0$ for all $v \neq 0$. Setting $B := \int_0^L e^{tA}e^{tA^T}\,\mathrm{d}t$, this is equivalent to $v^T ABv = v^T BA^T v = \langle v, A^T v\rangle_L > 0$.

Conversely, if condition (2) holds for A, B, then it also holds for $S^T A(S^T)^{-1}$ and $S^T BS$ for every invertible matrix S. We choose S such that $S^T BS = I$. Replacing A by $S^T A(S^T)^{-1}$, we thus have (10.26). The

positivity of $\operatorname{Re}\lambda$ for all eigenvalues λ of A was then shown in the paragraph following (10.26). □

We now give four examples of the radial point estimate setup.

Example 10.11 (Multiplication by x). Let us verify the assumptions (R.1)–(R.5) for the operator

$$P = x \in \operatorname{Diff}^0(M), \quad M = \mathbb{R} \times Y; \tag{10.32}$$

here Y is a compact manifold without boundary. In local coordinates $y \in \mathbb{R}^{n-1}$ on Y, write points in $T^*(\mathbb{R}\times Y)$ as (x, y, ξ, η) where $x, \xi \in \mathbb{R}$, $y, \eta \in \mathbb{R}^{n-1}$. Then $p(x, y, \xi, \eta) = x$ and $H_p = -\partial_\xi$. We fix the elliptic symbol $\rho \in S^{-1}_{\mathrm{cl}}(T^*\mathbb{R}^n)$ to be equal to $|\xi|^{-1}$ when $|\xi| > \max(|\eta|, 1)$. Working in $\xi > \max(|\eta|, 1)$, we use the coordinates $\rho = \xi^{-1}$, $\hat\eta = \frac{\eta}{\xi}$ from (10.10), and thus $\tilde H_p = \rho\partial_\rho + \hat\eta\partial_{\hat\eta}$ by (10.11). The characteristic set at fiber infinity in the coordinates $x, y, \rho, \hat\eta$ is given by $\{x = 0,\ \rho = 0\}$. Now, $\tilde H_p$ is tangent to $\mathcal{R} = \{x = 0,\ \rho = 0,\ \hat\eta = 0\}$. (This set is compact because Y is compact—this is the reason for working with $\mathbb{R}\times Y$ instead of the space $\mathbb{R}\times\mathbb{R}^{n-1}$ used in Proposition 10.1.) The conormal bundle of $\mathcal{R}$ in $\operatorname{Char}(P)$ is spanned by $\mathrm{d}\hat\eta$. Since

$$[\tilde H_p]\colon \mathrm{d}\hat\eta \mapsto \mathrm{d}(\tilde H_p\hat\eta) = \mathrm{d}\hat\eta \tag{10.33}$$

is the identity map, condition (R.3) is satisfied by Lemma 10.9. Condition (R.4) holds with $\beta_0 = 1$. Finally, (10.21) holds with $\tilde\beta = 0$. We can thus apply Theorem 10.5, with threshold regularity $-\frac12$, to recover Proposition 10.1 in the microlocal step-by-step procedure outlined after its proof. (In fact, one gets a Sobolev refinement of Proposition 10.1.)

Example 10.12 (A Non-symmetric Operator). In order to illustrate the relevance of the subprincipal part $\tilde\beta$ in (R.5), consider $P = xD_x - \lambda \in \operatorname{Diff}^1(M)$, $M = \mathbb{R}$. Then $Pu = 0$ for $u = \chi_+^{i\lambda}$ (Exercise 2.6), suggesting that the threshold regularity at the radial set $T^*_0\mathbb{R}\setminus o$ (which has two connected components) is $\frac12 - \operatorname{Im}\lambda$ (which is the Sobolev regularity and which $\chi_+^{i\lambda}$ barely fails to have by Exercise 6.4). Now $p(x,\xi) := \sigma^1(P)(x,\xi) = x\xi$ and $H_p = x\partial_x - \xi\partial_\xi$. Working in $\xi > 1$ for concreteness, we use $\rho = \xi^{-1}$, so $\tilde H_p = H_p = x\partial_x + \rho\partial_\rho$. At fiber infinity, we have $\operatorname{Char}(P) = \{x = \rho = 0\} = \mathcal{R}$, so condition (R.3) is vacuous, and (10.20) holds with $\beta_0 = 1$, while $p_1 = \sigma^0(\frac{1}{2i}(P - P^*)) = \frac12 - \operatorname{Im}\lambda$ implies $\tilde\beta = \frac12 - \operatorname{Im}\lambda$ in (10.21). Theorem 10.5 implies that $u \in H^{s_0}_{\mathrm{loc}}(\mathbb{R})$, $Pu = f \in H^s_{\mathrm{loc}}(\mathbb{R})$ with $s \geq s_0 > \frac12 - \operatorname{Im}\lambda$ implies $u \in H^s_{\mathrm{loc}}(\mathbb{R})$.

Example 10.13 (Keldysh Operator). Consider the operator $P := xD_x^2 + \Delta_Y \in \operatorname{Diff}^2(\mathbb{R}\times Y)$, where Y is a compact Riemannian manifold. Write covectors on $\mathbb{R}\times Y$ as $\xi\,\mathrm{d}x + \eta$, $\eta \in T^*Y$. Then the boundary at fiber infinity

of $\{\xi > 0\} \cap N^*\{x = 0\} \setminus o$ is a radial set and a normal source in the sense of (R.3)–(R.4). Notice that in $x > 0$, P is elliptic, while in $x < 0$ it is hyperbolic; the radial set over $x = 0$ lies over the interface of the two regimes. The reader is asked to supply the detailed computations for a generalization of this operator in Exercise 10.6.

Example 10.14 (Nontrivial Flow Within $\mathcal{R}$). In the examples discussed thus far, $\tilde{H}_p$ vanishes at $\mathcal{R}$. The following operator (a variant of Example 10.13) features nontrivial dynamics inside of $\mathcal{R}$: on $M = (-\frac{1}{4}, \frac{1}{4})_x \times \mathbb{S}^1_y$, let

$$P := xD_x^2 + D_xD_y + D_y^2. \tag{10.34}$$

Then $p(x, y, \xi, \eta) = x\xi^2 + \xi\eta + \eta^2$ and $H_p = (2x\xi + \eta)\partial_x + (\xi + 2\eta)\partial_y - \xi^2\partial_\xi$. In $\xi > 0$, set $\rho := \frac{1}{\xi}$, $\hat{\eta} := \frac{\eta}{\xi}$; in the coordinates $x, y, \rho, \hat{\eta}$, we then have

$$\tilde{H}_p := \rho H_p = (2x + \hat{\eta})\partial_x + (1 + 2\hat{\eta})\partial_y + \rho\partial_\rho + \hat{\eta}\partial_{\hat{\eta}}. \tag{10.35}$$

The set $\mathcal{R} := \{x = 0, \rho = 0, \hat{\eta} = 0\}$ is a normal source for $\tilde{H}_p$: inside of $\mathrm{Char}(P) \cap S^*M = \{x = -\hat{\eta} - \hat{\eta}^2,\ \rho = 0\}$, $\mathcal{R}$ is given by $\hat{\eta} = 0$, and $\tilde{H}_p\hat{\eta} = \hat{\eta}$. Finally, $\tilde{H}_p|_{\mathcal{R}} = \partial_y$ is nontrivial.

We now give the proof of Theorem 10.5.

Proof of Theorem 10.5. We use a positive commutator argument as in Sections 8.3–8.5 to prove a quantitative estimate of the form (10.22) for suitable B, G, χ. Using the arguments involving (8.22)–(8.23), we may again reduce to the case that $\chi u = u$.

- Construction of the commutant for part (1). With $\tilde{p}$ as in (10.17), the quadratic defining function ρ_1 of $\mathcal{R}$ as in (10.19), and the defining function ρ of fiber infinity as used in (10.20), we set

$$a := \rho^{-2s+m-1}\phi(\tilde{p})^2\psi(\rho_1)^2. \tag{10.36}$$

Here,

$$\begin{aligned} &\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}; [0,1]), \qquad \phi = 1 \text{ near } 0, \\ &\psi \in \mathcal{C}^\infty_{\mathrm{c}}([0,\infty); [0,1]),\ \psi = 1 \text{ near } 0, \text{ and } \sqrt{-\psi'\psi} \in \mathcal{C}^\infty([0,\infty)). \end{aligned} \tag{10.37}$$

Thus, $\phi(\tilde{p})$ localizes near $\mathrm{Char}(P)$ and $\psi(\rho_1)$ localizes further near $\mathcal{R}$. We have

$$\tilde{H}_p\tilde{p} = \tilde{q}\tilde{p}, \quad \tilde{q} := \rho^{-m}\tilde{H}_p\rho^m; \tag{10.38}$$

$\tilde{q}$ is smooth near $\mathcal{R}$. Pick $A = A^* \in \Psi^{2s-m+1}$ with $\sigma^{2s-m+1}(A) = a$. We then compute the principal symbol of $i[P, A] + 2\frac{P-P^*}{2i}A$ to be

$$\begin{aligned} H_p a + 2p_1 a &= \rho^{-m+1}(\tilde{H}_p a + 2\tilde{p}_1 a) \\ &= \rho^{-2s}\Big(\beta_0(-2s + m - 1 + 2\tilde{\beta})\phi(\tilde{p})^2\psi(\rho_1)^2 \\ &\qquad + 2(\tilde{H}_p\rho_1)\phi(\tilde{p})^2\psi'(\rho_1)\psi(\rho_1) \\ &\qquad + 2\tilde{q}\tilde{p}\phi'(\tilde{p})\phi(\tilde{p})\psi(\rho_1)^2\Big). \end{aligned} \tag{10.39}$$

When the supports of ψ and ϕ are sufficiently small, the terms in parentheses have the following properties.

(1) The first term is elliptic (and negative) at $\mathcal{R}$ under the stated assumptions on s.
(2) The second term is non-positive on the characteristic set and has essential support contained in a punctured neighborhood of $\mathcal{R}$. More precisely, on the support of the second term, ρ_1 is strictly positive, and thus so is $\tilde{H}_p\rho_1$ by (10.19) when $\tilde{p} = 0$ and $\operatorname{supp}\psi$ is sufficiently small. The positivity of $\tilde{H}_p\rho_1$ then persists on $\operatorname{supp}(\phi(\tilde{p})^2\psi'(\rho_1)\psi(\rho_1))$ when the support of ϕ is sufficiently small.
(3) The third term is supported in the elliptic set of P.

We emphasize that the signs of the terms with essential support intersecting $\operatorname{Char}(P)$ can be determined by inspecting the H_p-flow in Figure 10.2.

At this point, one can already prove a quantitative estimate of the form (10.22) for smooth u by repeating the arguments in Chapter 8, Section 8.4; we leave this to the reader.

- Regularization of the commutant for part (1). We regularize our commutant a as in Section 8.5, see equations (8.60)–(8.61), though we use slightly different notation here. Thus, for $\Theta > 0$ (fixed below) and $r \geq 0$, put

$$\phi_r(\rho) := (1 + r\rho^{-1})^{-\Theta}; \qquad \rho\phi_r'(\rho) = f_r(\rho)\phi_r(\rho), \quad f_r(\rho) = \Theta\frac{r\rho^{-1}}{1 + r\rho^{-1}}, \tag{10.40}$$

so $\phi_r \in L^\infty((0,1]_r; S^0)$ (recalling (8.59)), and $\phi_r \in S^{-\Theta}$ for $r > 0$. We then let

$$a_r := \phi_r(\rho)^2 a \in L^\infty((0,1]_r; S^{2s-m+1}) \tag{10.41}$$

and compute

$$H_p a_r + 2p_1 a_r = \rho^{-2s}\phi_r(\rho)^2\Big(\beta_0(-2s+m-1+2\tilde\beta+2f_r)\phi(\tilde p)^2\psi(\rho_1)^2 + 2(\tilde H_p\rho_1)\phi(\tilde p)^2\psi'(\rho_1)\psi(\rho_1) + 2\tilde q\tilde p\phi'(\tilde p)\phi(\tilde p)\psi(\rho_1)^2\Big). \tag{10.42}$$

Since $0 \le f_r \le \Theta$, the amount Θ of regularization is limited when propagating out of the radial set $\mathcal{R}$: in order to ensure that the first term in (10.42) is negative at $\mathcal{R}$, we need $s - \Theta > \frac{m-1}{2} + \tilde\beta$ on $\mathcal{R}$. Fix such $\Theta > 0$. For $\delta > 0$ chosen so small that still $s - \Theta - \delta\beta_0^{-1} > \frac{m-1}{2} + \tilde\beta$ on $\mathcal{R}$, we then write

$$H_p a_r + 2p_1 a_r = -2\delta\rho^{2s-2m+2}a_r^2 - b_r^2 - b_{1,r}^2 + h_r p \tag{10.43}$$

where

$$\begin{aligned} b_r &:= \rho^{-s}\phi_r(\rho)\phi(\tilde p)\psi(\rho_1) \\ &\qquad \times \sqrt{\beta_0\big(2s - \big(m-1+2\tilde\beta+2f_r+2\delta\beta_0^{-1}\phi_r(\rho)^2\phi(\tilde p)^2\psi(\rho_1)^2\big)\big)}, \\ b_{1,r} &:= \rho^{-s}\phi_r(\rho)\phi(\tilde p)\sqrt{-2(\tilde H_p\rho_1)\psi'(\rho_1)\psi(\rho_1)}, \\ h_r &:= 2\rho^{-2s+m}\phi_r(\rho)^2\tilde q\phi'(\tilde p)\phi(\tilde p)\psi(\rho_1)^2. \end{aligned} \tag{10.44}$$

Thus, $b_r, b_{1,r} \in L^\infty((0,1]_r; S^s)$ and $h_r \in L^\infty((0,1]_r; S^{2s-m})$, with orders reduced by Θ, Θ and 2Θ, respectively, for $r > 0$.

- Quantization; conclusion of the proof of part (1). Using a full quantization map as in (8.67), let $A_r := \frac{1}{2}(\mathrm{Op}(a_r) + \mathrm{Op}(a_r)^*)$, $B_r := \mathrm{Op}(b_r)$, $B_{1,r} := \mathrm{Op}(b_{1,r})$, and $H_r := \mathrm{Op}(h_r)$. Set $\Lambda := \mathrm{Op}(\rho^{s-m+1})$. Then (10.43) and (10.44) imply

$$i[P, A_r] + 2\frac{P - P^*}{2i}A_r = -2\delta(\Lambda A_r)^*(\Lambda A_r) - B_r^* B_r - B_{1,r}^* B_{1,r} + H_r P + R_r \tag{10.45}$$

where $R_r \in L^\infty((0,1]_r; \Psi^{2s-1})$, $\mathrm{WF}'_{L^\infty}(\{R_r\}) \subset \operatorname{ess\,supp} a$.

Recall now that we are assuming $\mathrm{WF}^{s_0}(u) \cap \mathcal{R} = \emptyset$; let $\tilde B \in \Psi^0$ be elliptic at $\mathcal{R}$ and such that $\tilde B u \in H^{s_0}$ and $\tilde B P u \in H^{s-m+1}$. Fix $\Theta > 0$ such that

$$\frac{m-1}{2} + \tilde\beta < s - \Theta < s_0, \tag{10.46}$$

and choose the supports of ϕ, ψ in (10.37) so small that $\operatorname{ess\,supp} a \subset \mathrm{Ell}(\tilde B)$. Since $A_r \in \Psi^{2s-m+1-2\Theta} \subset \Psi^{2s_0-m+1}$ for $r > 0$, we have $A_r u \in H^{-s_0+m-1}$. We claim that

$$\begin{aligned} 2\operatorname{Im}\langle Pu, A_r u\rangle &= i\big(\langle Pu, A_r u\rangle - \langle A_r u, Pu\rangle\big) \\ &= \Big\langle \Big(i[P, A_r] + 2\frac{P-P^*}{2i}A_r\Big)u, u\Big\rangle. \end{aligned} \tag{10.47}$$

All terms make sense individually by Lemma 6.42: this uses $\mathrm{WF}^{s_0-m+1}(Pu) \cap \mathrm{WF}'(A_r) = \emptyset$ for the first line and the fact that $i[P, A_r] + 2\frac{P-P^*}{2i}A_r \in \Psi^{2s_0}$, with operator wave front set contained in $\mathrm{Ell}(\tilde{B})$, for the second line. However, the integration by parts needs to be justified: for *general* $u \in H^{s_0}$, one only has $Pu \in H^{s_0-m}$, which is in general insufficient to justify the integration by parts. This is easily accomplished by inserting yet another regularizer $J_\epsilon \in L^\infty((0,1]_\epsilon; \Psi^0)$, with $J_\epsilon = J_\epsilon^* \in \Psi^{-\infty}$ for $\epsilon > 0$ and $J_\epsilon \to I$ in the topology of Ψ^η, $\eta > 0$. Namely,

$$\begin{aligned} &\langle Pu, A_r u\rangle - \langle A_r u, Pu\rangle \\ &= \lim_{\epsilon\to 0} \big(\langle Pu, J_\epsilon A_r u\rangle - \langle A_r u, J_\epsilon Pu\rangle\big) \\ &= \lim_{\epsilon\to 0} \langle (A_r J_\epsilon P - P^* J_\epsilon A_r)u, u\rangle \\ &= \lim_{\epsilon\to 0} \big(\langle J_\epsilon(A_r P - P^* A_r)u, u\rangle + \langle ([A_r, J_\epsilon]P - [P^*, J_\epsilon]A_r)u, u\rangle\big) \\ &= \langle (A_r P - P^* A_r)u, u\rangle. \end{aligned} \tag{10.48}$$

Note here that for fixed $r > 0$, the family $[A_r, J_\epsilon]$, $\epsilon \in (0,1]$, is uniformly bounded in Ψ^{2s_0-m} and converges to $[A_r, I] = 0$ in $\Psi^{2s_0-m+\eta}$, $\eta > 0$, as $\epsilon \to 0$; this implies $[A_r, J_\epsilon]Pu \to 0$ in H^{-s_0} by Lemma 4.39, and therefore, $\langle [A_r, J_\epsilon]Pu, u\rangle \to 0$ as $\epsilon \to 0$. Similarly, $\langle [P^*, J_\epsilon]A_r u, u\rangle \to 0$ as $\epsilon \to 0$ for fixed $r > 0$.

We proceed to rewrite the right-hand side of the pairing (10.47) by plugging in (10.45). Pick $G \in \Psi^0$ with $\mathrm{WF}'(I-G) \cap \mathrm{WF}'_{L^\infty}(\{A_r\}) = \emptyset$. Then the Peter–Paul inequality and Lemma 8.18 give the estimate

$$\begin{aligned} &\|B_r u\|^2_{L^2} + \|B_{1,r} u\|^2_{L^2} + 2\delta\|\Lambda A_r u\|^2_{L^2} \\ &\quad \le 2\delta\|\Lambda A_r u\|^2_{L^2} + C\delta^{-1}\|GPu\|^2_{H^{s-m+1}} \\ &\qquad + |\langle Pu, H_r^* u\rangle| + C\Big(\|Gu\|^2_{H^{s-\frac{1}{2}}} + \|u\|^2_{H^{-N}}\Big) \end{aligned} \tag{10.49}$$

for an r-independent constant C. Using $\mathrm{WF}'_{L^\infty}(\{H_r\}) \cap \mathrm{WF}'(I-G) = \emptyset$, the first term in the last line can be estimated by

$$\begin{aligned} |\langle Pu, H_r^* u\rangle| &\le C\big(\|GPu\|^2_{H^{s-m+1}} + \|H_r^* u\|^2_{H^{-s+m-1}} + \|u\|^2_{H^{-N}}\big) \\ &\le C\big(\|GPu\|^2_{H^{s-m+1}} + \|Gu\|^2_{H^{s-1}} + \|u\|^2_{H^{-N}}\big); \end{aligned} \tag{10.50}$$

recall here that $H_r \in L^\infty((0,1];\Psi^{2s-m})$. Combined with (10.49), we obtain the uniform estimate

$$\|B_r u\|_{L^2} \leq C\Big(\|GPu\|_{H^{s-m+1}} + \|Gu\|_{H^{s-\frac{1}{2}}} + \|u\|_{H^{-N}}\Big). \tag{10.51}$$

As in Section 8.5, we thus conclude that $B_0 u \in L^2$ (so $\mathrm{WF}^s(u) \cap \mathcal{R} = \emptyset$), and indeed $\|B_0 u\|_{L^2}$ is bounded by the right-hand side of (10.51). If $s - \frac{1}{2} \leq s_0$, we can further estimate $\|Gu\|_{H^{s-\frac{1}{2}}} \leq C(\|\tilde{B}u\|_{H^{s_0}} + \|u\|_{H^{-N}})$ and obtain (10.22) (upon relabeling the operators). Otherwise, an iterative argument allows us to improve the regularity by $\frac{1}{2}$ in each step, and we obtain (10.22) after finitely many steps.

- Modifications for part (2). The propagation of microlocal regularity *into* a normal source uses the same commutant (10.41). The amount Θ of regularization is now arbitrary. Indeed, in (10.42), the first term on the right-hand side is now *positive* at $\mathcal{R}$ regardless of the amount $\Theta > 0$ of regularization, and hence has the opposite sign of the second term. We thus now write

$$H_p a_r + 2p_1 a_r = 2\delta\rho^{2s-2m+2}a_r^2 + b_r^2 - b_{1,r}^2 + h_r p \tag{10.52}$$

where $b_{1,r}$, h_r are as in (10.44), and

$$\begin{aligned} b_r := {} & \rho^{-s}\phi_r(\rho)\phi(\tilde{p})\psi(\rho_1) \\ & \times \sqrt{\beta_0\big(-2s+m-1+2\tilde{\beta}+2f_r - 2\delta\beta_0^{-1}\phi_r(\rho)^2\phi(\tilde{p})^2\psi(\rho_1)^2\big)}. \end{aligned} \tag{10.53}$$

Upon quantization as before, we get a uniform estimate

$$\|B_r u\|_{L^2} \leq C\Big(\|GPu\|_{H^{s-m+1}} + \|B_{1,r}u\|_{L^2} + \|Gu\|_{H^{s-\frac{1}{2}}} + \|u\|_{H^{-N}}\Big). \tag{10.54}$$

Thus, we now have an a priori control term $\|B_{1,r}u\|_{L^2}$: it is uniformly bounded if $\mathrm{WF}^s(u)$ is disjoint from a punctured neighborhood of $\mathcal{R}$ containing $\mathrm{WF}'_{L^\infty}(\{B_{1,r}\})$. In the limit $r \to 0$, we obtain $B_0 u \in L^2$. Hence, $\mathrm{WF}^s(u) \cap \mathcal{R} = \emptyset$. An iterative argument (improving the regularity by $\frac{1}{2}$ at each step) implies the uniform estimate

$$\|B_0 u\|_{L^2} \leq C\big(\|GPu\|_{H^{s-m+1}} + \|Eu\|_{H^s} + \|u\|_{H^{-N}}\big) \tag{10.55}$$

for $E \in \Psi^0$ with $\mathrm{Ell}(E) \supset \mathrm{WF}'_{L^\infty}(\{B_{1,r}\})$; in particular, E can be chosen such that $\mathrm{WF}'(E) \cap \mathcal{R} = \emptyset$. □

The proof of Theorem 10.7, which is similar to that of Corollary 8.19, is left to the reader. We emphasize that the estimate (10.22) has the following microlocal building blocks.

(1) Microlocally near $\mathcal{R}$, one uses the radial point estimate, i.e., the quantitative estimate established in the above proof, to obtain a microlocal estimate for u in H^s near $\mathcal{R}$.
(2) Using real principal type propagation (Theorem 8.8, or Theorem 8.10 for the quantitative version), one can then microlocally control u in H^s near the flow-out of $\mathcal{R}$.
(3) Finally, on the elliptic set of P, one can estimate u in terms of Pu using microlocal elliptic estimates.

Remark 10.15 (No Gu Term). *Assuming* that $\mathrm{WF}^{s_0}(u)\cap\mathcal{R}=\emptyset$, one can show a stronger estimate than (10.22), namely

$$\|Bu\|_{H^s} \leq C\big(\|GPu\|_{H^{s-m+1}} + \|\chi u\|_{H^{-N}}\big). \tag{10.56}$$

This can be seen as follows. First of all, we only need to consider B which are elliptic at $\mathcal{R}$. Then if $\mathrm{WF}'(G)$ is close to being as small as possible given the requirements (1ii)–(1iv) in the statement of Theorem 10.7, then all backward null-bicharacteristics of P from $\mathrm{WF}'(G)\cap\mathrm{Char}(P)$ tend to $\mathcal{R}$, and in particular enter $\mathrm{Ell}(B)$ in finite time, while remaining in $\mathrm{Ell}(\tilde{G})$ where $\tilde{G}\in\Psi^0$ has slightly larger operator wave front set than G. Using the propagation of singularities, we can therefore estimate

$$\|Gu\|_{H^{s_0}} \leq C\big(\|\tilde{G}Pu\|_{H^{s_0-m+1}} + \|Bu\|_{H^{s_0}} + \|\chi u\|_{H^{-N}}\big). \tag{10.57}$$

Plugging this estimate in (10.22), it remains to use the interpolation inequality

$$\|Bu\|_{H^{s_0}} \leq \epsilon\|Bu\|_{H^s} + C_\epsilon\|Bu\|_{H^{-N}} \tag{10.58}$$

(Exercise 2.13). Indeed, for sufficiently small ϵ, the first term can be absorbed into the left-hand side of (10.22), yielding (10.56). We chose not to state the radial point estimate in the form (10.56) in Theorem 10.7 since (10.56) does *not* hold in the strong sense: for example, when $P=x\in\Psi^0(M)$, $M=\mathbb{R}\times Y$ (with Y compact), with points in $\mathbb{R}\times Y$ denoted (x,y), $x\in\mathbb{R}$, $y\in Y$, and $u(x,y)=\delta(x)\phi(y)$ where $0\neq\phi\in\mathcal{C}^\infty(Y)$, then $Pu=0$ and $u\in H^{-N}$ for $N>\frac{1}{2}$, but $Bu\notin H^s$ for $s>-\frac{1}{2}$ when $B\in\Psi^0$ is elliptic at the radial set $N^*\{x=0\}\setminus o$.

10.3 Semiclassical Radial Point Estimates

The general setup is very similar to that in Section 10.2. We make the following assumptions and definitions.

(R$\hbar$.1) $P = (P_h)_{h\in(0,h_0)} \in \Psi^m_{\hbar,\mathrm{cl}}(M)$ is a classical semiclassical operator (properly supported) with h-independent principal symbol $p \in S^m_{\mathrm{cl}}(T^*M)$.

(R$\hbar$.2) Fix an elliptic symbol $0 < \rho \in S^{-1}_{\mathrm{cl}}(T^*M)$ and let

$$\tilde{p} := \rho^m \rho \in \mathcal{C}^\infty(\overline{T^*}M), \quad \tilde{H}_p := \rho^{m-1} H_p \in \mathcal{V}(\overline{T^*}M). \tag{10.59}$$

(See Exercise 10.10 for the membership of $\tilde{H}_p$.) Suppose that

$$\mathcal{R} \subset \mathrm{Char}(P) \cap S^*M \subset S^*M \text{ is a smooth compact submanifold,}$$
$$\text{and } \tilde{H}_p \text{ is tangent to } \mathcal{R}. \tag{10.60}$$

Suppose moreover that $\mathrm{d}\tilde{p} \neq 0$ near $\mathcal{R}$, so $\mathrm{Char}(P)$ is a smooth codimension 1 submanifold of $\overline{T^*}M$ near $\mathcal{R}$.

(R$\hbar$.3) $\mathcal{R}$ is a normal source for $\tilde{H}_p$ inside of $\mathrm{Char}(P) \cap S^*M$. We write $\rho_1 \in \mathcal{C}^\infty(S^*M)$ for a quadratic defining function with $\tilde{H}_p\rho_1 \geq \beta_1\rho_1$ near $\mathcal{R}$ for some $0 < \beta_1 \in \mathcal{C}^\infty(S^*M)$.

(R$\hbar$.4) $\mathcal{R}$ is a source also in the fiber-radial direction, to wit,

$$\tilde{H}_p\rho = \beta_0\rho, \quad \beta_0|_{\mathcal{R}} > 0. \tag{10.61}$$

(R$\hbar$.5) We define $\tilde{\beta} \in \mathcal{C}^\infty(S^*M)$ near $\mathcal{R}$ by

$$\tilde{p}_1 := \rho^{m-1} p_1 = \beta_0\tilde{\beta}, \quad p_1 := \sigma^{m-1}_\hbar\Big(\frac{1}{2ih}(P - P^*)\Big). \tag{10.62}$$

Figure 10.2 is again a useful illustration of the setup. The main new feature of the semiclassical setting is the fact that the characteristic set is no longer conic in the fibers of T^*M, and thus, microlocalizing near $\mathcal{R}$ now also requires localizing near fiber infinity.

We state the semiclassical radial point estimates in this setting in two versions: one qualitative and one semiglobal quantitative version.

Theorem 10.16 (Semiclassical Microlocal Regularity At Normal Sources: Qualitative Statement). *Let $u = (u_h)_{h\in(0,h_0)}$ be an h-tempered family of distributions on M, and let $f = (f_h)_{h\in(0,h_0)}$, $f_h := P_h u_h$.*

(1) (Forward propagation: out of $\mathcal{R}$.) *Let $s, s_0 \in \mathbb{R}$ with $s > s_0 > \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$. If $\mathrm{WF}_\hbar^{s_0,0}(u) \cap \mathcal{R} = \emptyset$ and $\mathrm{WF}^{s-m+1,1}(f) \cap \mathcal{R} = \emptyset$, then $\mathrm{WF}_\hbar^{s,0}(u) \cap \mathcal{R} = \emptyset$.*

(2) (Backward propagation: into $\mathcal{R}$.) *Let $s \in \mathbb{R}$ with $s < \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$. If $\mathrm{WF}_\hbar^{s,0}(u)$ is disjoint from a punctured neighborhood of $\mathcal{R}$ and $\mathrm{WF}_\hbar^{s-m+1,1}(f) \cap \mathcal{R} = \emptyset$, then $\mathrm{WF}_\hbar^{s,0}(u) \cap \mathcal{R} = \emptyset$.*

Theorem 10.17 (Semiclassical Microlocal Regularity At Normal Sources: Quantitative Statement). *Let $K \subset M$ be a compact set such that the base projection of $\mathcal{R}$ is contained in K°. Let $\tilde{K} \subset M$ be a compact set such that for the Schwartz kernels $K_{P_\hbar}$ of $P_\hbar$ we have $\operatorname{supp} K_{P_\hbar} \cap (K \times M) \subset K \times \tilde{K}$. Fix $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(M)$ with $\chi = 1$ near $\tilde{K}$. Let $u \in \mathscr{D}'(M)$.*

(1) (Forward propagation: out of $\mathcal{R}$.) *Let $B, G \in \Psi_\hbar^0(M)$ be such that:*

(i) *The Schwartz kernels of B, G are supported in $K \times K$.*
(ii) $\mathrm{WF}'_\hbar(B) \subset \mathrm{Ell}_\hbar(G)$.
(iii) $\mathcal{R} \subset \mathrm{Ell}_\hbar(G)$.
(iv) *All integral curves of $-\rho^{m-1} H_p$ starting from a point in $\mathrm{WF}'_\hbar(B) \cap \mathrm{Char}_\hbar(P)$ tend to $\mathcal{R}$ while remaining in $\mathrm{Ell}_\hbar(G)$.*

Then for all $s, s_0, N \in \mathbb{R}$ with $s > s_0 > \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$, there exist $h_1, C > 0$ such that the estimate

$$\|Bu\|_{H_\hbar^s} \leq C\Big(h^{-1}\|GPu\|_{H_\hbar^{s-m+1}} + h^N\|Gu\|_{H_\hbar^{s_0}} + h^N\|\chi u\|_{H_\hbar^{-N}}\Big) \tag{10.63}$$

holds for all $h \in (0, h_1)$ in the strong sense.

(2) (Backward propagation: into $\mathcal{R}$.) *Let $B, G, E \in \Psi_\hbar^0(M)$ be such that:*

(i) *The Schwartz kernels of B, G, E are supported in $K \times K$.*
(ii) $\mathrm{WF}'_\hbar(B) \subset \mathrm{Ell}_\hbar(G)$.
(iii) $\mathcal{R} \subset \mathrm{Ell}_\hbar(B)$.
(iv) *All integral curves of $\rho^{m-1} H_p$ starting from a point in $\mathrm{WF}'_\hbar(B) \cap \mathrm{Char}_\hbar(P)$ are either contained in $\mathcal{R}$ or enter $\mathrm{Ell}_\hbar(E)$ in finite time, all while remaining in $\mathrm{Ell}_\hbar(G)$.*

Then for all $s, N \in \mathbb{R}$ with $s < \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$, there exist $h_1, C > 0$ such that the estimate

$$\|Bu\|_{H_\hbar^s} \leq C\Big(h^{-1}\|GPu\|_{H_\hbar^{s-m+1}} + \|Eu\|_{H_\hbar^s} + h^N\|\chi u\|_{H_\hbar^{-N}}\Big). \tag{10.64}$$

holds for all $h \in (0, h_1)$ in the strong sense.

We explain the proof of a quantitative estimate corresponding to Theorem 10.16 (which is microlocal near $\mathcal{R}$) following the arguments in the previous section; Theorem 10.17 then follows from arguments similar to those used in the proof of Corollary 8.19. The main changes in the argument are,

firstly, that we now need to localize to a neighborhood of fiber infinity and, secondly, that we need to put powers of h in the appropriate places. We may again reduce to the case that u is supported in $\chi^{-1}(1)$.

We begin by discussing part (1) of Theorem 10.16. Let $\rho_1 \in \mathcal{C}^\infty(\overline{T^*}M)$ be such that $\rho_1|_{\mathrm{Char}_\hbar(P)\cap S^*M}$ is a quadratic defining function of $\mathcal{R}$ inside of $\mathrm{Char}_\hbar(P)\cap S^*M$ such that the monotonicity property (10.19) holds. Let ϕ, ψ be cutoffs as in (10.37) and let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,\infty);[0,1])$ be a cutoff with $\chi = 1$ near 0 and $\sqrt{-\chi'\chi} \in \mathcal{C}^\infty$. Consider first the unregularized commutant

$$a := \rho^{-2s+m-1}\chi(\rho)^2\phi(\tilde{p})^2\psi(\rho_1)^2. \tag{10.65}$$

Define $\tilde{q}$ by (10.38). Pick $A = A^* \in \Psi_\hbar^{2s-m+1}$ with $\sigma_\hbar^{2s-m+1}(A) = a$. We then compute the semiclassical principal symbol of $\frac{i}{\hbar}[P,A] + 2\frac{P-P^*}{2i\hbar}A$ to be

$$\begin{aligned} H_p a + 2p_1 a &= \rho^{-m+1}(\tilde{H}_p a + 2\tilde{p}_1 a) \\ &= \rho^{-2s}\Big(\beta_0(-2s+m-1+2\tilde{\beta})\chi(\rho)^2\phi(\tilde{p})^2\psi(\rho_1)^2 \\ &\qquad + 2\beta_0\rho\chi'(\rho)\chi(\rho)\phi(\tilde{p})^2\psi(\rho_1)^2 \\ &\qquad + 2(\tilde{H}_p\rho_1)\chi(\rho)^2\phi(\tilde{p})^2\psi'(\rho_1)\psi(\rho_1) \\ &\qquad + 2\tilde{q}\tilde{p}\chi(\rho)^2\phi'(\tilde{p})\phi(\tilde{p})\psi(\rho_1)^2\Big). \end{aligned} \tag{10.66}$$

The four terms on the right-hand side have the following properties.

(1) The first, main, term is negative at $\mathcal{R}$, and thus, it remains negative on $\operatorname{supp} a$ when the supports of χ, ϕ, ψ are small enough.
(2) The second term is supported at finite frequencies and, in view of (10.61) and $\chi' \leq 0$, is non-positive as well when $\operatorname{supp}\chi$ is sufficiently small. It is, in fact, the negative of the square of a symbol. The signs of the first two terms are due to the source nature of $\mathcal{R}$ in the fiber-radial direction.
(3) We note that ρ_1 is positive on the support of the third term. Furthermore, within the characteristic set at fiber infinity $\{\tilde{p} = \rho = 0\}$ and on $\operatorname{supp}\psi'(\rho_1)$, the function $\tilde{H}_p\rho_1$ is positive due to assumption (R$\hbar$.3) provided $\operatorname{supp}\psi$ is sufficiently small. This persists in a neighborhood of $\{\tilde{p} = \rho = 0\}$. Thus, choosing the supports of ϕ, χ small (depending on the choice of ψ) ensures that the third term is non-positive and indeed the negative of the square of a symbol. That this term has the same sign as the second term is due to the source nature of $\mathcal{R}$ in $\mathrm{Char}_\hbar(P)$ both within fiber infinity and in the fiber-radial direction.
(4) The fourth term is supported in $\mathrm{Ell}_\hbar(P)$.

For the purpose of regularization, we use $\phi_r(\rho) = (1 + r\rho^{-1})^{-\Theta}$ as in (10.40). The amount Θ of regularization is chosen such that $s_0 > s - \Theta > \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$. Define the regularized symbol

$$a_r := \phi_r(\rho)^2 a. \tag{10.67}$$

Analogously to (10.43), we fix $\delta > 0$ such that $s - \Theta - \delta\beta_0^{-1} > \frac{m-1}{2} + \tilde{\beta}$ on $\mathcal{R}$ still and write

$$H_p a_r + 2p_1 a_r = -2\delta\rho^{2s-2m+2}a_r^2 - b_r^2 - b_{1,r}^2 - b_{2,r}^2 + h_r p \tag{10.68}$$

where, using the notation of (10.40),

$$\begin{aligned}
b_r &:= \rho^{-s}\phi_r(\rho)\chi(\rho)\phi(\tilde{p})\psi(\rho_1) \\
&\quad \times \sqrt{\beta_0\big(2s - \big(m-1+2\tilde{\beta}+2f_r + 2\delta\beta_0^{-1}\phi_r(\rho)^2\chi(\rho)^2\phi(\tilde{p})^2\psi(\rho_1)^2\big)\big)}, \\
b_{1,r} &:= \rho^{-s}\phi_r(\rho)\chi(\rho)\phi(\tilde{p})\sqrt{-2(\tilde{H}_p\rho_1)\psi'(\rho_1)\psi(\rho_1)}, \\
b_{2,r} &:= \rho^{-s}\phi_r(\rho)\phi(\tilde{p})\psi(\rho_1)\sqrt{-2\beta_0\rho\chi'(\rho)\chi(\rho)}, \\
h_r &:= 2\rho^{-2s+m}\phi_r(\rho)^2\chi(\rho)^2\tilde{q}\phi'(\tilde{p})\phi(\tilde{p})\psi(\rho_1)^2.
\end{aligned} \tag{10.69}$$

When ψ and ϕ, χ have sufficiently small supports, we have $b_r, b_{1,r}, b_{2,r} \in L^\infty((0,1]_r; S^s)$ and $h_r \in L^\infty((0,1]_r; S^{2s-m})$, with orders reduced by Θ, Θ, Θ and 2Θ, respectively, for $r > 0$.

Using a full quantization map as in (5.200), we now set $A_r := \frac{1}{2}(\mathrm{Op}_\hbar(a_r) + \mathrm{Op}_\hbar(a_r)^*)$ and $B_r := \mathrm{Op}_\hbar(b_r)$, $B_{j,r} := \mathrm{Op}_\hbar(b_{j,r})$ $(j = 1, 2)$, and $H_r := \mathrm{Op}_\hbar(h_r)$. Let $\Lambda := \mathrm{Op}_\hbar(\rho^{-m+s-1})$. Then

$$\begin{aligned}
&\frac{i}{h}[P, A_r] + 2\frac{P - P^*}{2ih}A_r \\
&\qquad = -2\delta(\Lambda A_r)^*(\Lambda A_r) - B_r^* B_r - B_{1,r}^* B_{1,r} - B_{2,r}^* B_{2,r} + H_r P + hR_r
\end{aligned} \tag{10.70}$$

where $R_r \in L^\infty((0,1]_r; \Psi_\hbar^{2s-1})$, $\mathrm{WF}'_{\hbar,L^\infty}(\{R_r\}) \subset \operatorname{ess\,supp}_\hbar a$. We can then justify the integration by parts in the L^2-pairing

$$\begin{aligned}
&2h^{-1}\operatorname{Im}\langle Pu, A_r u\rangle \\
&\qquad = \Big\langle \Big(\frac{i}{h}[P, A_r] + 2\frac{P - P^*}{2ih}A_r\Big)u, u\Big\rangle \\
&\qquad = -2\delta\|\Lambda A_r u\|^2 - \|B_r u\|^2 - \|B_{1,r}u\|^2 - \|B_{2,r}u\|^2 \\
&\qquad\qquad + \langle H_r P u, u\rangle + \langle hR_r u, u\rangle
\end{aligned} \tag{10.71}$$

by repeating the secondary regularization argument in (10.48). We now proceed as in (10.49). Fixing $G \in \Psi_\hbar^0$ with $\mathrm{WF}'_\hbar(I - G) \cap \mathrm{WF}'_{\hbar,L^\infty}(\{A_r\}) = \emptyset$, we can use the Peter–Paul inequality to estimate

$$\begin{aligned}\|B_r u\|^2 &\leq C\delta^{-1}h^{-2}\|GPu\|^2_{H_h^{s-m+1}}\\ &\quad + |\langle Pu, H_r u\rangle| + C\Big(h^{\frac{1}{2}}\|Gu\|_{H_h^{s-\frac{1}{2}}} + h^N\|u\|^2_{H_h^{-N}}\Big).\end{aligned} \tag{10.72}$$

(We have dropped the terms $\|B_{1,r}u\|^2$ and $\|B_{2,r}u\|^2$ which have the same sign as $\|B_r u\|^2$.) The second term on the right can further be estimated by

$$\begin{aligned}|\langle Pu, H_r u\rangle| &\leq C\Big(h^{-2}\|GPu\|^2_{H_h^{s-m+1}} + h^2\|H_r u\|^2_{H_h^{-s+m-1}} + h^N\|u\|^2_{H_h^{-N}}\Big)\\ &\leq C\Big(h^{-2}\|GPu\|^2_{H_h^{s-m+1}} + h^2\|Gu\|^2_{H_h^{s-1}} + h^N\|u\|^2_{H_h^{-N}}\Big).\end{aligned} \tag{10.73}$$

Altogether, we thus obtain

$$\|B_r u\| \leq C\Big(h^{-1}\|GPu\|_{H_h^{s-m+1}} + h^{\frac{1}{2}}\|Gu\|_{H_h^{s-\frac{1}{2}}} + h^N\|u\|_{H_h^{-N}}\Big). \tag{10.74}$$

We conclude that $B_0 u \in L^2$, with norm bounded by the same right-hand side. An iterative argument in which one improves the regularity and semiclassical orders by $\frac{1}{2}$ at each step yields the estimate (10.63) except with $h^{\frac{1}{2}}\|Gu\|_{H_h^{s_0}}$ on the right instead of $h^N\|Gu\|_{H_h^{s_0}}$. But using a propagation of singularities argument as in Remark 10.15, one can drop this term when h is sufficiently small. This establishes the estimate (10.63) without the Gu term. In order for (10.63) to hold in the desired strong sense, one merely needs to add this term back with any prefactor that is positive for $h > 0$; in (10.63) we choose h^N for this purpose.

The proof of part (2) of Theorem 10.16 proceeds along similar lines. The differences are as follows. First, the terms involving derivatives of χ and ψ in the second and third lines on the right in (10.66) now have the *opposite* sign compared to the main term which is in the first line on the right. Second, the amount Θ of regularization is now unlimited. Thus, we now write

$$H_p a_r + 2p_1 a_r = 2\delta\rho^{2s-2m+2}a_r^2 + b_r^2 - b_{1,r}^2 - b_{2,r}^2 + h_r p \tag{10.75}$$

where $b_{1,r}$, $b_{2,r}$, h_r are as in (10.69), while now

$$\begin{aligned}b_r &= \rho^{-s}\phi_r(\rho)\chi(r)\phi(\tilde{p})\psi(\rho_1)\\ &\quad\times\sqrt{\beta_0\big(-2s+m-1+2\tilde{\beta}+2f_r - 2\delta\beta_0^{-1}\phi_r(\rho)^2\chi(\rho)^2\phi(\tilde{p})^2\psi(\rho_1)^2\big)}.\end{aligned} \tag{10.76}$$

We then obtain

$$\begin{aligned}\|B_r u\| \leq C\Big(&h^{-1}\|GPu\|_{H_h^{s-m+1}} + \|B_{1,r}u\| + \|B_{2,r}u\|\\ &+ h^{\frac{1}{2}}\|Gu\|_{H_h^{s-\frac{1}{2}}} + h^N\|u\|_{H_h^{-N}}\Big).\end{aligned} \tag{10.77}$$

By assumption, $\mathrm{WF}_\hbar^{s,0}(u)$ is disjoint from a punctured neighborhood of $\mathcal{R}$. The terms $B_{1,r}u$ and $B_{2,r}u$ are therefore uniformly bounded when the supports of ψ, ϕ, χ are sufficiently small. For $E \in \Psi_\hbar^0$ with $\mathrm{Ell}_\hbar(E) \supset \mathrm{WF}'_{\hbar,L^\infty}(B_{1,r}) \cup \mathrm{WF}'_{\hbar,L^\infty}(B_{2,r})$, we can bound

$$\|B_{1,r}u\| + \|B_{2,r}u\| \leq C\Big(\|Eu\|_{H_\hbar^s} + h^N \|u\|_{H_\hbar^{-N}}\Big) \tag{10.78}$$

and thus obtain (10.64) except for the presence of an additional term $h^{\frac{1}{2}}\|Gu\|_{H_\hbar^{s-\frac{1}{2}}}$ on the right-hand side. This latter term can be estimated iteratively as usual, and after finitely many iterations we obtain (10.64) as stated. This completes the proof of Theorem 10.16.

Notes

The radial point estimates presented here are due to Vasy [Vas13]. They originate in the work of Melrose [Mel94] on scattering theory on asymptotically conic spaces. The first instance of radial points in the microlocal literature appears to be a paper by Guillemin–Schaeffer [GS77]; there, they form a discrete set. Discrete radial points also arise in Hassell–Melrose–Vasy [HMV08] in the analysis of scattering by non-decaying potentials. In the semiclassical setting, Dyatlov–Zworski [DZ19, Appendix E.4] prove a slightly more general estimate than (10.63)–(10.64) using their notion of "eventual positivity." For operators acting on sections of vector bundles, the notion of *subprincipal operator* introduced in Hintz [Hin17] based on Dencker's partial connection [Den82] allows for a clean determination of the optimal threshold quantities (Remark 10.6).

A symplectic (and thus, via FIOs, microlocal) normal form for *Lagrangian* submanifolds of radial points is produced in Haber [Hab14]. The microlocal normal form is $xD_x - q(y)$ on $\mathbb{R}_x \times \mathbb{R}_y^{n-1}$ where $q \in \mathcal{C}^\infty$, valid near $(x, y, \xi, \eta) = (0, 0, 1, 0)$. In this setting, one can microlocalize radial propagation estimates *within* the radial set, as shown in Haber–Vasy [HV13].

Exercises

Exercise 10.1 (Long-Time Energy Estimate). Consider the operator $P := D_x - \lambda x^{-1} \in \mathrm{Diff}^1((1,\infty)_x)$. Let $s_0 := -\frac{1}{2} + \operatorname{Im}\lambda$. For $\alpha \in \mathbb{R}$ and $I \subseteq [1,\infty)$, define the weighted L^2-norms $\|u\|^2_{x^{-\alpha}L^2(I)} := \int_I x^{2\alpha}|u(x)|^2\,\mathrm{d}x$.

(1) Let $s < -\frac{1}{2} + \operatorname{Im}\lambda$. Show that

$$\|u\|_{x^{-s}L^2([1,\infty))} \leq C\big(\|u\|_{L^2([1,2])} + \|Pu\|_{x^{-s-1}L^2([1,\infty))}\big). \tag{10.79}$$

(2) Let $s > s_0 > -\frac{1}{2} + \operatorname{Im}\lambda$. Show that

$$\|u\|_{x^{-s}L^2([1,\infty))} \leq C\big(\|u\|_{x^{-s_0}L^2([1,\infty))} + \|Pu\|_{x^{-s-1}L^2([1,\infty))}\big). \tag{10.80}$$

Remark. These estimates are instances of radial point estimates in the scattering cotangent bundle at $\mathcal{R} := \{(x,\xi) = (\infty,0)\} \subset {}^{\mathrm{sc}}T^*\overline{\mathbb{R}}$ in (10.79) we propagate weighted L^2-control into $\mathcal{R}$ (analogously to Theorem 10.5(2)), and in (10.80) we propagate weighted L^2-control out of $\mathcal{R}$ (analogously to Theorem 10.5(1)). General radial point estimates in the scattering setting are proved in Vasy [Vas13].

The following three exercises illuminate the notion of quadratic defining functions.

Exercise 10.2 (Hessian). Let M be a smooth manifold, and let $X \subset M$ be a closed smooth submanifold. Suppose $\rho \in \mathcal{C}^\infty(M)$ is such that $X = \rho^{-1}(0)$ and $\mathrm{d}\rho = 0$ on X. Show that for $V, W \in \mathcal{V}(M)$, the value $V(W(\rho))(p)$, $p \in X$, depends only on $V(p), W(p) \in T_pM$, is symmetric in V, W, and vanishes when $V(p) \in T_pX$ or $W(p) \in T_pX$. Conclude that the *Hessian* of ρ at X, defined by $\mathrm{Hess}(\rho)(V,W) := V(W(\rho))$, is well-defined as a symmetric bilinear form on $T_XM/TX = NX$ (Exercise 5.8), and thus defines an element

$$\mathrm{Hess}(\rho) \in \mathcal{C}^\infty(X; S^2N^*X). \tag{10.81}$$

Show that ρ is a quadratic defining function of X if and only if $\mathrm{Hess}(\rho)$ in (10.81) is positive definite.

Exercise 10.3 (Quadratic Defining Function: Existence). Let M be a smooth manifold.

(1) (Simple case.) Let $\rho_1, \dots, \rho_k \in \mathcal{C}^\infty(M)$ be such that

$$X = \{\rho_1 = \cdots = \rho_k = 0\} \subset M \tag{10.82}$$

is a smooth submanifold of M, and $\mathrm{d}\rho_1, \dots, \mathrm{d}\rho_k$ are linearly independent at each point of X. Show that $\rho := \frac{1}{2}\sum_{j=1}^k \rho_j^2$ is a quadratic defining function of X.

(2) (General case.) Show that every closed smooth submanifold $X \subset M$ admits a quadratic defining function.

Exercise 10.4 (Quadratic Defining Functions: Structure). Let M be a smooth manifold and let $X \subset M$ be a closed smooth submanifold of codimension k. Show that the following two conditions on $\rho \in \mathcal{C}^\infty(M)$ are equivalent.

(1) ρ is a quadratic defining function of X.

(2) For every $p \in X$, there exist local coordinates $x_1, \dots, x_n$ in an open neighborhood $U \subset M$ of p such that $X \cap U = \{x_1 = \dots = x_k = 0\}$ and $\rho = \frac{1}{2}\sum_{j=1}^{k} x_j^2$.

Exercise 10.5 (Linearizations). Let M be a smooth manifold and let $X \subset M$ be a smooth submanifold.

(1) Write $\mathcal{I} = \{u \in \mathcal{C}^\infty(M)\colon u|_X = 0\}$ for the ideal of functions vanishing at X. Show that $\mathcal{I} \ni u \mapsto (\mathrm{d}u)|_X \in \mathcal{C}^\infty(X; N^*X)$ induces a ring isomorphism $\phi\colon \mathcal{I}/\mathcal{I}^2 \to \mathcal{C}^\infty(X; N^*X)$.
(2) Show that ϕ is tensorial, i.e., $\phi(fu) = f\phi(u)$ for all $u \in \mathcal{I}$ and $f \in \mathcal{C}^\infty(M)$.
(3) Let $V \in \mathcal{V}(M)$ be a smooth vector field vanishing on X. Show that $V\colon \mathcal{I} \ni f \mapsto Vf \in \mathcal{I}$ induces a tensorial map $[V]\colon \mathcal{I}/\mathcal{I}^2 \to \mathcal{I}/\mathcal{I}^2$ which, under the identification ϕ, is equal to the linearization of V defined by (10.24).

Exercise 10.6 (Keldysh Operator). Fix a closed Riemannian manifold (Y, g). Fix $\lambda \in \mathbb{C}$ and consider the operator

$$P := D_x x D_x - \lambda D_x + \Delta_g \in \mathrm{Diff}^2(M), \quad M = \mathbb{R} \times Y. \tag{10.83}$$

Let $p(x, y, \xi, \eta) := \sigma^2(P)(x, y, \xi, \eta) = x\xi^2 + |\eta|^2_{g_y^{-1}}$.

(1) Show that the set of points in $T^*M \setminus o$ where H_p is radial is equal to $N^*\{x = 0\} \setminus o$. Denote by

$$\mathcal{R}_\pm := \partial\{(x, y, \xi, \eta)\colon x = 0,\ \eta = 0,\ \pm\xi > 0\} \subset S^*M \tag{10.84}$$

the boundaries at fiber infinity of the two components of $N^*\{x = 0\} \setminus o$.
(2) Show that $\mathrm{Char}(P) \cap S^*M$ is smooth and $\mathcal{R}_+$ is a normal source for the (rescaled) H_p-flow (condition (R.3)). Verify also condition (R.4) at $\mathcal{R}_+$ and compute the quantity $\tilde{\beta}$.
(3) Suppose $u \in H^{s_0}_{\mathrm{loc}}(M)$, $s_0 > \frac{1}{2} - \mathrm{Im}\,\lambda$, solves $Pu = f \in H^{s-1}_{\mathrm{loc}}(M)$. Show that $u \in H^s_{\mathrm{loc}}(M)$. For $s = \infty$, relate this to Exercise 8.2.

Exercise 10.7 (A Radial Vector Field). Let $P = \sum_{j=1}^{n} x_j D_{x_j} \in \mathrm{Diff}^1(M)$, $M = \mathbb{R}^n$. Suppose $u \in \mathscr{D}'(\mathbb{R}^n)$ solves $Pu = f \in \mathcal{C}^\infty(\mathbb{R}^n)$.

(1) Let $\mathcal{A} := \{x \in \mathbb{R}^n\colon |x| > 1\}$. Assume that $u|_{\mathcal{A}} \in H^s_{\mathrm{loc}}(\mathcal{A})$ for some $s \in \mathbb{R}$. Show that $u \in H^{s'}_{\mathrm{loc}}(\mathbb{R}^n)$ for all $s' \in \mathbb{R}$ with $s' \le s$ and $s' < \frac{n}{2}$.
(2) If $u \in H^s_{\mathrm{loc}}(\mathbb{R}^n)$ for some $s > \frac{n}{2}$, show that $u \in \mathcal{C}^\infty(\mathbb{R}^n)$.

Exercise 10.8 (Stability of Radial Point Estimates). Prove the analogue of Exercise 8.5 regarding the stability of the estimates in Theorem 10.7 when P is a differential operator: the estimates (10.22) and (10.23) hold for all P' in a neighborhood $\mathcal{U} \subset \mathrm{Diff}^m(M)$ of P within the space of differential

operators with real principal symbols. *Remark.* One does *not* need to require the operators in $\mathcal{U}$, other than P of course, to satisfy the normal source/sink conditions placed on P.

Exercise 10.9 (Non-classical Operators). Show that Theorem 10.5 continues to hold if we merely assume that $P \in \Psi^m(M)$ has a homogeneous principal symbol (as in the statement of Theorem 8.4!) satisfying the conditions stated in Section 10.2, with the following differences: the function $\tilde{\beta}$ in (10.21) is now a symbol $\tilde{\beta} \in S^0(T^*M)$; and in the threshold conditions in Theorem 10.5, one must replace $\tilde{\beta}$ by $\overline{\beta}(x,[\xi]) := \limsup_{\lambda\to\infty} \tilde{\beta}(x,\lambda\xi)$ and $\underline{\beta}(x,[\xi]) := \liminf_{\lambda\to\infty} \tilde{\beta}(x,\lambda\xi)$, respectively.

Exercise 10.10 (Rescaled Hamiltonian Vector Field). Fix an elliptic symbol $0 < \rho \in S^{-1}_{\mathrm{cl}}(T^*M)$. Let $p \in S^m_{\mathrm{cl}}(T^*M)$. Show that $\tilde{H}_p := \rho^{m-1} H_p \in \mathcal{V}(\overline{T^*M})$ and that $\tilde{H}_p$ is tangent to S^*M.

The next exercise gives an example of an operator which has a radial set that is not a normal source/sink for the null-bicharacteristic flow, but rather a saddle point.

Exercise 10.11 (A Saddle Point Estimate). On $\mathbb{R}^2_{x,y}$, let $P := xD_x - yD_y$, and denote by $p(x,y,\xi,\eta) = x\xi - y\eta$ its principal symbol.

(1) Show that $\mathrm{d}p \neq 0$ on $\mathrm{Char}(P) \subset T^*\mathbb{R}^2 \setminus o$. Show moreover that the set of radial points is the union of the sets

$$\mathcal{R}^{\pm}_x := \{(0,0,\xi,0)\colon \pm\xi > 0\}, \quad \mathcal{R}^{\pm}_y := \{(0,0,0,\eta)\colon \pm\eta > 0\}. \tag{10.85}$$

(2) Focusing on the set $\mathcal{R}^+_x$, introduce projective fiber coordinates $\rho := \frac{1}{\xi}$, $\hat{\eta} := \frac{\eta}{\xi}$. Compute the linearization of H_p at $\partial\mathcal{R}^+_x = \{x = y = \rho = \hat{\eta} = 0\} \subset S^*\mathbb{R}^2$ and show that $\partial\mathcal{R}^+_x$ is a saddle point for the H_p-flow in $\mathrm{Char}(P) \cap S^*\mathbb{R}^2$.

(3) Draw a picture of the null-bicharacteristic flow of P inside of $\mathrm{Char}(P) \cap S^*\mathbb{R}^2$.

(4) Let

$$a(x,y,\xi,\eta) := \rho^{-2s}\chi(x)^2\chi(y)^2\chi(|\hat{\eta}|)^2 \tag{10.86}$$

where $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,1))$ equals 1 in a small neighborhood of 0 and satisfies $\sqrt{-\chi'\chi} \in \mathcal{C}^\infty$. Express $H_p a$ as a difference of sums of squares of symbols. (Make a distinction between $s > 0$ and $s < 0$.)

(5) Let $s > 0$. Show that if

$$\begin{aligned} &\mathrm{WF}^s(Pu) \cap \mathcal{R}^+_x = \emptyset, \\ &\mathrm{WF}^s(u) \cap \{\rho = 0,\ x = \hat{\eta} = 0,\ 0 < |y| < \delta\} = \emptyset \end{aligned} \tag{10.87}$$

for some $\delta > 0$, then $\mathrm{WF}^s(u) \cap \mathcal{R}_x^+ = \emptyset$.

(6) Let $s < 0$. Show that if

$$\begin{aligned} &\mathrm{WF}^s(Pu) \cap \mathcal{R}_x^+ = \emptyset, \\ &\mathrm{WF}^s(u) \cap \{\rho = 0,\ 0 < |x| + |\hat{\eta}| < \delta,\ y = 0\} = \emptyset \end{aligned} \tag{10.88}$$

for some $\delta > 0$, then $\mathrm{WF}^s(u) \cap \mathcal{R}_x^+ = \emptyset$.

(7) State and prove a general result on the "propagation through a (generalized) set of radial saddle points."

Remark. The sets on which we assume the absence of H^s wave front set in (10.87) and (10.88) are punctured neighborhoods (i.e., with $\partial\mathcal{R}_x^+$ removed) of $\partial\mathcal{R}_x^+$ inside of the stable and unstable manifolds of $\partial\mathcal{R}_x^+$ in $\mathrm{Char}(P) \cap S^*\mathbb{R}^2$, respectively. Thus, parts (5) and (6) state that microlocal regularity above or below a certain threshold (here 0) can be propagated from the stable manifold into the radial set and then out through the unstable manifold, or vice versa.

Chapter 11
Late-Time Asymptotics of Linear Waves on de Sitter Space



In the final chapter of this book, we set off a little microlocal firework by combining almost all of the microlocal tools we have developed in order to give a precise description of the asymptotic (late-time) behavior of solutions of linear wave-type equations on a spacetime of interest in general relativity—de Sitter space, more precisely, a neighborhood of the static patch of de Sitter space. We describe this spacetime and the null-geodesic flow on it in Section 11.1. We then state the main results in Section 11.2 and sketch their proofs. The details of the proofs are given in Sections 11.3–11.5. We describe a simple nonlinear application in Section 11.6.

We do not aim for any level of generality as far as the spacetime under consideration is concerned. By working on a simple and explicit yet geometrically interesting spacetime, we can let the analytic tools take center stage.

11.1 Geometry of de Sitter Space

We shall work on the following spacetime:

Definition 11.1 ((2+1)-Dimensional de Sitter Space). Let $X := \{x \in \mathbb{R}^2 : |x| \leq 2\}$ and $M := \mathbb{R}_t \times X$. Write points in $\mathbb{R}^2$ as $x = (x_1, x_2)$. Then the *de Sitter metric* g on M is

$$g := -(1 - |x|^2)\,\mathrm{d}t^2 - 2\,\mathrm{d}t \otimes_s (x \cdot \mathrm{d}x) + \mathrm{d}x^2, \tag{11.1}$$

where $\alpha \otimes_s \beta := \frac{1}{2}(\alpha\otimes\beta + \beta\otimes\alpha)$, $x \cdot \mathrm{d}x := x_1\,\mathrm{d}x_1 + x_2\,\mathrm{d}x_2$, and $\mathrm{d}x^2 := \mathrm{d}x_1^2 + \mathrm{d}x_2^2$.

In physics, the interest in (M, g) stems from it being a solution of the Einstein vacuum equations with positive cosmological constant: $\mathrm{Ric}(g) = 2g$.

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P. Hintz, *An Introduction to Microlocal Analysis*, Graduate Texts in Mathematics 304, https://doi.org/10.1007/978-3-031-90706-7_11

Remark 11.2 (Higher Dimensions). The expression (11.1) defines the de Sitter metric also in higher dimensions, the case of $3+1$ dimensions being of particular interest in physics. Our computations and arguments below go through in all dimensions $\geq 2+1$ with only minor notational modifications. The benefit of working in $2+1$ dimensions is purely notational, as polar coordinates in 2 dimensions are particularly straightforward to work with.

The metric g is nondegenerate; the inverse metric is given by

$$g^{-1} = -\partial_t^2 - 2\partial_t \otimes_s x\partial_x + \partial_x^2 - (x \cdot \partial_x)^2. \tag{11.2}$$

In the region $\{x \neq 0\}$, it is more convenient to work with polar coordinates $r > 0$, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, so $x = (r\cos\theta, r\sin\theta)$. Then

$$\begin{aligned} g &= -(1-r^2)\,\mathrm{d}t^2 - 2\,\mathrm{d}t \otimes_s r\,\mathrm{d}r + \mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2, \\ g^{-1} &= -\partial_t^2 - 2\partial_t \otimes_s r\partial_r + (1-r^2)\partial_r^2 + r^{-2}\partial_\theta^2. \end{aligned} \tag{11.3}$$

The following features of (M, g) follow immediately from the definition.

(1) g is *stationary*, i.e., $\mathcal{L}_{\partial_t} g = 0$. That is, translations $(t, x) \mapsto (t + c, x)$ are isometries for all $c \in \mathbb{R}$.
(2) $\mathrm{d}t$ is timelike since $g^{-1}(\mathrm{d}t, \mathrm{d}t) = -1 < 0$. We define a time orientation on (M, g) by declaring $\mathrm{d}t$ to be *past* timelike. (Carefully note that the generator ∂_t of "time" translations is timelike only for $|x| < 1$. It is null for $|x| = 1$ and spacelike for $|x| > 1$.)
(3) For $r > 1$, the covector

$$\mathrm{d}r \text{ is past timelike} \tag{11.4}$$

since $g^{-1}(\mathrm{d}r, \mathrm{d}r) = 1 - r^2 < 0$ and $g^{-1}(\mathrm{d}t, \mathrm{d}r) = -r < 0$ there.
(4) The hypersurface

$$\overline{\mathcal{H}}^+ := r^{-1}(1) \subset M \tag{11.5}$$

is a null hypersurface since $\mathrm{d}r$ is lightlike at $r = 1$. It is called the *cosmological horizon*.

We can already draw simple conclusions about the causal properties of (M, g): if $\gamma\colon I \to M$ is a future null-geodesic (i.e., $\gamma'(s)$ is future null for all $s \in I$) with $0 \in I$ and $r(\gamma(0)) > 1$, then $r \circ \gamma$ is monotonically increasing. In particular, $r(\gamma(s)) > 1$ for all $s \geq 0$, and hence, γ cannot enter the region $\{r \leq 1\}$. On the other hand, one can show that for every pair $x_0, x_1 \in X$ with $|x_0| < 1$ there exists a future null-geodesic from $(0, x_0)$ to (t_1, x_1) for some $t_1 > 0$. Thus, future null-geodesics can only cross $\overline{\mathcal{H}}^+$ in the outward direction, i.e., the direction of increasing r. See Figure 11.1.

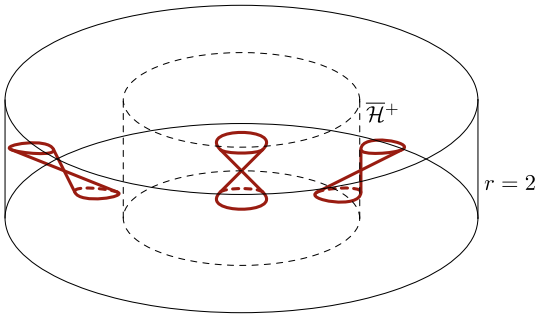


Figure 11.1 Three light cones of (M, g). The light cones at points in $\overline{\mathcal{H}}^+$ are tangent to $\overline{\mathcal{H}}^+$, and the future (upward pointing) cones at points in $r > 1$ are tilted toward even larger r.

The set of lightlike covectors

$$\Sigma^\circ := \{\zeta \in T^*M \setminus o \colon G(\zeta) = 0\}, \quad G(\zeta) := g^{-1}(\zeta, \zeta), \tag{11.6}$$

is a smooth conic codimension 1 submanifold of $T^*M \setminus o$ and has two components:

$$\Sigma^\circ = \Sigma^\circ_+ \sqcup \Sigma^\circ_-, \quad \Sigma^\circ_\pm := \{\zeta \in \Sigma^\circ \colon \pm g^{-1}(\zeta, \mathrm{d}t) < 0\}. \tag{11.7}$$

The covectors in Σ°_+, resp. Σ°_- are future, resp. past null.

We study the future null-geodesic flow on (M, g) lifted to phase space. The dynamics we describe here will be translated into microlocal real principal type propagation and radial point estimates for the spectral family associated with a stationary wave-type operator on (M, g). (The reader requiring further motivation before embarking on these computations may wish to look at Sections 11.3–11.4 first.) Recall from Proposition 9.9 that null-geodesics on (M, g) are projections to M of integral curves of $\frac{1}{2}H_G$ in $\Sigma^\circ \subset T^*M$.

We introduce fiber-linear coordinates on T^*M by writing covectors and the dual metric function G as

$$\begin{aligned}
&\zeta = -\sigma\,\mathrm{d}t + \xi \cdot \mathrm{d}x,\\
&\qquad G(t, x, \sigma, \xi) = -\sigma^2 + 2\sigma x \cdot \xi + |\xi|^2 - (x \cdot \xi)^2\\
&\qquad\qquad = -(\sigma - x \cdot \xi)^2 + |\xi|^2,\\
&\zeta = -\sigma\,\mathrm{d}t + \xi_r\,\mathrm{d}r + \xi_\theta\,\mathrm{d}\theta,\\
&\qquad G(t, r, \theta, \sigma, \xi_r, \xi_\theta) = -\sigma^2 + 2r\sigma\xi_r + (1 - r^2)\xi_r^2 + r^{-2}\xi_\theta^2\\
&\qquad\qquad = -(\sigma - r\xi_r)^2 + \xi_r^2 + r^{-2}\xi_\theta^2.
\end{aligned} \tag{11.8}$$

(The minus sign in front of σ is a matter of convention.) Correspondingly,

$$\begin{aligned}\frac{1}{2}H_G &= \frac{1}{2}\big(-(\partial_\sigma G)\partial_t + (\partial_\xi G)\cdot\partial_x + (\partial_t G)\partial_\sigma - (\partial_x G)\cdot\partial_\xi\big)\\ &= (\sigma - x\cdot\xi)\partial_t + \big((\sigma - x\cdot\xi)x + \xi\big)\cdot\partial_x - (\sigma - x\cdot\xi)\xi\cdot\partial_\xi\\ &= \frac{1}{2}\big(-(\partial_\sigma G)\partial_t + (\partial_{\xi_r} G)\partial_r + (\partial_{\xi_\theta} G)\partial_\theta + (\partial_t G)\partial_\sigma - (\partial_r G)\partial_{\xi_r} - (\partial_\theta G)\partial_{\xi_\theta}\big)\\ &= (\sigma - r\xi_r)\partial_t + \big((\sigma - r\xi_r)r + \xi_r\big)\partial_r + r^{-2}\xi_\theta\partial_\theta - \big((\sigma - r\xi_r)\xi_r - r^{-3}\xi_\theta^2\big)\partial_{\xi_r}.\end{aligned} \tag{11.9}$$

We have $0 > \pm g^{-1}(\zeta, \mathrm{d}t) = \mp\frac{1}{2}\partial_\sigma G(\zeta)$ on $\Sigma^\circ_\pm$, so

$$\pm\,(\sigma - x\cdot\xi) = \pm(\sigma - r\xi_r) > 0 \text{ on } \Sigma^\circ_\pm. \tag{11.10}$$

The stationarity of g, which is equivalent to $\partial_t G = 0$, implies that $\frac{1}{2}H_G\sigma = 0$, i.e., σ is conserved along the H_G-flow. Since the values $t(s)$ of t along an integral curve of $\frac{1}{2}H_G$ can be recovered by integrating $\sigma(s) - x(s)\cdot\xi(s)$, it suffices to study the flow of the vector field

$$\begin{aligned}\frac{1}{2}H_\sigma &:= \big((\sigma - x\cdot\xi)x + \xi\big)\cdot\partial_x - (\sigma - x\cdot\xi)\xi\cdot\partial_\xi\\ &= \big((\sigma - r\xi_r)r + \xi_r\big)\partial_r + r^{-2}\xi_\theta\partial_\theta - \big((\sigma - r\xi_r)\xi_r - r^{-3}\xi_\theta^2\big)\partial_{\xi_r}\end{aligned} \tag{11.11}$$

on the set

$$\Sigma^\circ_\sigma := \{(x,\xi)\in T^*X\colon (0,x,\sigma,\xi)\in\Sigma^\circ\} = \Sigma^\circ_{\sigma,+}\sqcup\Sigma^\circ_{\sigma,-}, \tag{11.12}$$

where $\Sigma^\circ_{\sigma,\pm} := \{(x,\xi)\in T^*X\colon (0,x,\sigma,\xi)\in\Sigma^\circ_\pm\}$. Geometrically, Σ°_σ equals the intersection of Σ° with $-\sigma\,\mathrm{d}t + T^*X \subset T^*M\cap\{t=0\}$ upon identifying $T^*X \cong -\sigma\,\mathrm{d}t + T^*X$ via $(x,\xi)\mapsto(0,x,\sigma,\xi)$. See Figure 11.2. For $\sigma = 0$, this means that Σ°_0 is equal to the set of all null covectors which are orthogonal to ∂_t (upon identifying $(x,\xi)\in T^*X$ with $(0,x,0,\xi)\in T^*M$). We point out that Σ°_0 is a conic subset of $T^*X\setminus o$, whereas for $\sigma\neq 0$, the set $\Sigma^\circ_\sigma\subset T^*X$ is not conic. In order to describe sets of radial points for $\frac{1}{2}H_\sigma$, we introduce the following notation for the closures of these sets at fiber infinity:

$$\begin{aligned}&\Sigma_0 := \overline{\Sigma^\circ_0}\cap(\overline{T^*}X\setminus o), \quad \Sigma_{0,\pm} := \overline{\Sigma^\circ_{0,\pm}}\cap(\overline{T^*}X\setminus o),\\ &\Sigma_\sigma := \overline{\Sigma^\circ_\sigma}\subset\overline{T^*}X, \qquad \Sigma_{\sigma,\pm} := \overline{\Sigma^\circ_{\sigma,\pm}}\subset\overline{T^*}X, \qquad \sigma\neq 0.\end{aligned} \tag{11.13}$$

Their intersections with fiber infinity do not depend on σ: we have

$$\Sigma_0\cap S^*X = \Sigma_\sigma\cap S^*X \quad \forall\,\sigma\in\mathbb{R}, \tag{11.14}$$

as these sets are all equal to the intersection of $\Sigma = \overline{\Sigma^\circ}\subset\overline{T^*}M\setminus o$ with the submanifold $S^*X\subset\overline{T^*}X\setminus o\subset\overline{T^*_{\{t=0\}}}M\setminus o$. (We invite the reader to check this by hand using the expressions in (11.8).)

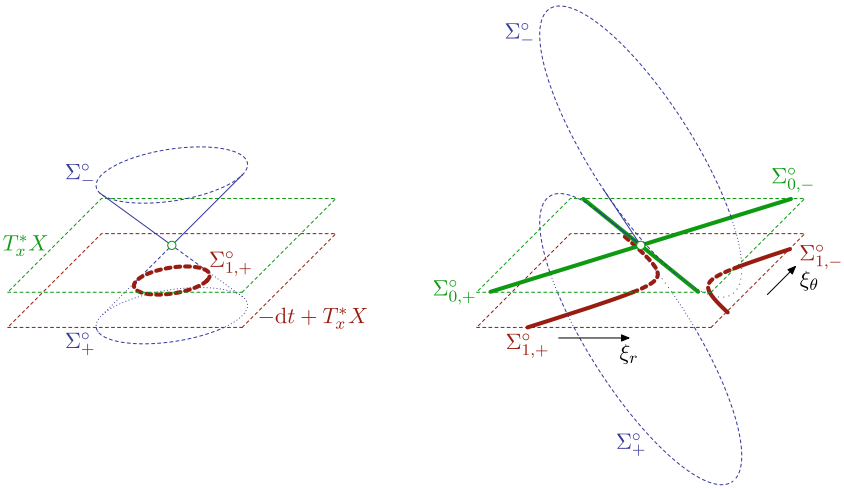


Figure 11.2 The two components $\Sigma^\circ_\pm \cap T^*_{(0,x)}M$ of dual light cones over two points $x \in X$ (left: $x = 0$, right: $|x| > 1$) and their intersections $\Sigma^\circ_{0,\pm}$ and $\Sigma^\circ_{1,\pm}$ with T^*X and $-\,\mathrm{d}t + T^*X$ (if non-empty).

For $\zeta \in T^*_z M \cap \Sigma^\circ$, write

$$\gamma_{z,\zeta} \colon I_{z,\zeta} \subseteq \mathbb{R} \to M \tag{11.15}$$

for the maximal integral curve of H_G with $\gamma_{z,\zeta}(0) = \zeta$. Note that $\gamma_{z,-\zeta}(s) = -\gamma_{z,\zeta}(-s)$ and $\gamma_{z,\lambda\zeta}(s) = \lambda\gamma_{z,\zeta}(\lambda s)$ since H_G is homogeneous of degree 1 (Lemma 8.14). It thus suffices to study the $\pm H_\sigma$-flow on $\Sigma^\circ_{\sigma,\pm}$ for $\sigma = 0$ and $\sigma = 1$, in the sense that the full $\frac{1}{2}H_G$-flow on Σ° can be recovered from it. We also record that

$$H_\sigma|_{(x,\xi)} = -H_{-\sigma}|_{(x,-\xi)}; \quad (x,\xi) \in \Sigma_{\sigma,\pm} \iff (x,-\xi) \in \Sigma_{-\sigma,\mp}. \tag{11.16}$$

Lemma 11.3 (Radial Sets for H_0). *The set of points in Σ°_0 at which the vector field H_0 is radial is equal to $N^*\{r = 1\} \setminus o$ (i.e., the conormal bundle of the intersection of the cosmological horizon $\overline{\mathcal{H}}^+$ with $\{t = 0\} \cong X$, without the zero section). There, we have $H_0 = \xi_r^2 \partial_{\xi_r}$.*

Proof. H_0 being radial at a point (x,ξ) implies, by (11.11), that $(-x \cdot \xi)x + \xi = 0$, and hence, $|\xi|^2 = (x \cdot \xi)^2 |x|^2$. Plugging this into (11.8) with $\sigma = 0$ gives

$$0 = G = (x \cdot \xi)^2(-1 + |x|^2). \tag{11.17}$$

By (11.10), this gives $|x| = 1$; and $\xi = (x \cdot \xi)x$, so $\xi \in N^*\{|x| = 1\}$. Conversely, if $r = 1$, $\xi_\theta = 0$, then (11.11) gives $H_0 = \xi_r^2 \partial_{\xi_r}$, which is radial. □

Note that $\mp\xi_r > 0$ on $\Sigma^\circ_{0,\pm} \cap N^*\{r=1\}$ by (11.10). In a conic neighborhood of $\Sigma^\circ_{0,\pm} \cap N^*\{r=1\}$, we introduce the projective coordinates

$$\rho := \frac{1}{\mp\xi_r} \geq 0, \quad \hat{\xi}_\theta := \frac{\xi_\theta}{\xi_r} \in \mathbb{R} \tag{11.18}$$

on $\overline{T^*}X$. We then define the future ('+') and past ('−') radial sets by

$$\mathcal{R}_\pm := \{r=1,\ \rho=0,\ \hat{\xi}_\theta = 0\} \subset \Sigma_{0,\pm} \cap S^*X. \tag{11.19}$$

In order to work on $\overline{T^*}X$, we rescale H_σ and introduce

$$\mathsf{H}_\sigma := (\sigma^2 + |\xi|^2)^{-\frac{1}{2}} H_\sigma, \quad \mathsf{H}_0 \in \mathcal{V}(\overline{T^*}X \setminus o), \quad \mathsf{H}_\sigma \in \mathcal{V}(\overline{T^*}X)\ (\sigma \neq 0). \tag{11.20}$$

By Lemma 10.2 (for $\sigma = 0$) and Exercise 10.10 (for $\sigma \neq 0$), the vector field H_σ is tangent to S^*X. Complementing (11.14), we have

$$\mathsf{H}_\sigma|_{S^*X} = \mathsf{H}_0|_{S^*X} \tag{11.21}$$

since, firstly, $|\xi|^{-1}/(\sigma^2+|\xi|^2)^{-\frac{1}{2}} = 1$ at $S^*X = \{|\xi|^{-1} = 0\}$ and, secondly, $|\xi|^{-1}\frac{1}{2}(H_\sigma - H_0) = \sigma|\xi|^{-1}(x\cdot\partial_x - \xi\cdot\partial_\xi)$ is $\sigma|\xi|^{-1}$ times a vector field which is smooth on $\overline{T^*}X$, and thus, its restriction to S^*X vanishes. The vector field H_0 is homogeneous of degree 0 with respect to dilations in ξ, whereas H_σ, $\sigma \neq 0$, is not. By Lemma 11.3, the vector field H_σ vanishes at $\partial\mathcal{R}_\pm$ but nowhere else in $\Sigma_\sigma \cap S^*X$.

Lemma 11.4 (Radial Sets Are Normal Sources). *$\mathcal{R}_\pm$ is a normal source for the $\pm\mathsf{H}_\sigma$-flow in $\Sigma_{\sigma,\pm} \cap S^*X$.*

Proof. We work with the smooth positive rescaling $\pm\rho\mathsf{H}_\sigma$ of $\pm\mathsf{H}_\sigma$ near $\mathcal{R}_\pm$. Explicitly, in the coordinates $r, \theta, \rho, \hat{\xi}_\theta$ from (11.18), we set $\hat{\sigma} := \frac{\sigma}{\xi_r}$ and compute

$$\pm\frac{1}{2}\rho H_\sigma = -(\hat{\sigma}+1-r^2)\partial_r - r^{-2}\hat{\xi}_\theta\partial_\theta + (r-\hat{\sigma}+r^{-3}\hat{\xi}_\theta^2)(\rho\partial_\rho + \hat{\xi}_\theta\partial_{\hat{\xi}_\theta}) \tag{11.22}$$

Restricting this to S^*X amounts to dropping the $\rho\partial_\rho$ term and setting $\hat{\sigma} = 0$. The linearization (Definition 10.8) of

$$V := \Big(\pm\frac{1}{2}\rho H_\sigma\Big)\Big|_{S^*X} = -(1-r^2)\partial_r - r^{-2}\hat{\xi}_\theta\partial_\theta + (r+r^{-3}\hat{\xi}_\theta^2)\hat{\xi}_\theta\partial_{\hat{\xi}_\theta} \tag{11.23}$$

at $\mathcal{R}_\pm = \{r-1=0,\ \hat{\xi}_\theta = 0\}$ is given by

$$[V]\colon \mathrm{d}(r-1) \mapsto 2\,\mathrm{d}(r-1), \quad \mathrm{d}\hat{\xi}_\theta \mapsto \mathrm{d}\hat{\xi}_\theta. \tag{11.24}$$

This has only positive eigenvalues. By Lemma 10.9, $\mathcal{R}_\pm$ is therefore a normal source for the V-flow in S^*X, and hence also for the V-flow in $\Sigma_{\sigma,\pm} \cap S^*X$. □

We also record

$$\pm \rho H_\sigma \rho = \beta_0 \rho, \quad \beta_0|_{\mathcal{R}_\pm} = 2 > 0. \tag{11.25}$$

We next describe the global behavior of H_0-integral curves in S^*X. (These are the ∂_t-orthogonal null-geodesics when lifted to phase space T^*M, projected to the spatial phase space $T^*X \setminus o$, and finally projected to S^*X.) By (11.14) and (11.21), these are equal to the integral curves of H_σ in S^*X.

Proposition 11.5 (H_σ-Flow at Fiber Infinity in Σ_σ). *Let $\gamma\colon I \subseteq \mathbb{R} \to \Sigma_{\sigma,\pm} \cap S^*X$ be a maximal integral curve of $\pm\mathsf{H}_\sigma$. Then either*

(1) *$\gamma(s) \in \mathcal{R}_\pm$ for all s, or*
(2) *$\gamma(s)$ tends to $\mathcal{R}_\pm$ as $s \to s_0 := \inf I$ and $r(\gamma(s)) \to 2$ as $s \to s_1 := \sup I$.*

See Figure 11.3.

Proof of Proposition 11.5. It suffices to consider $\sigma = 0$, in which case we may study the integral curves of the homogeneous vector field H_0 on $T^*X \setminus o$.

On Σ_0 we have $0 \neq |\xi|^2 = (x \cdot \xi)^2 \leq |x|^2|\xi|^2$ by (11.8), so $|x| \geq 1$. When $|x| = 1$, we must have equality, and thus, ξ is a multiple of x, so $(x, \xi) \in N^*\{|x| = 1\}$. If γ is *not* contained in $\mathcal{R}_\pm$, we must therefore have $r(\gamma(s)) > 1$ for all s. By (11.10), we have $\mp\xi_r > 0$ on $\Sigma_{0,\pm}$, so (11.11) gives $\pm\frac{1}{2}H_0 r = \pm(1 - r^2)\xi_r > 0$ on $\Sigma_{0,\pm} \cap \{r > 1\}$. This implies that $r(\gamma(s)) \to 1$ as $s \to s_0$ and $r(\gamma(s)) \to 2$ as $s \to s_1$. It remains to show that $\gamma(s)$ tends to $\mathcal{R}_\pm$ as $s \to s_0$. But in a neighborhood $\mathcal{U} \subset S^*X$ of $\Sigma_{0,\pm} \cap S^*X \cap \{r = 1\}$, we can use the coordinates $r, \theta, \hat{\xi}_\theta$ on S^*X from (11.18); then $\Sigma_{0,\pm} \cap S^*X$ is, in $\mathcal{U}$, the zero set of

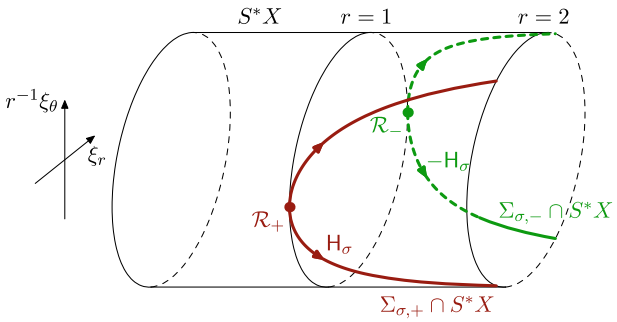


Figure 11.3 The rescaled Hamiltonian flow inside of $\Sigma_{\sigma,\pm} \cap S^*X$.

$$\xi_r^{-2} G|_{\sigma=0} = (1-r^2) + r^{-2}\hat{\xi}_\theta^2. \tag{11.26}$$

At $r = 1$, this only vanishes at $\mathcal{R}_\pm$ (i.e., for $\hat{\xi}_\theta = 0$). Since

$$\partial_r(\xi_r^{-2} G|_{\sigma=0}) = -2r - 2r^{-3}\hat{\xi}_\theta^2 = -2 \neq 0 \quad \text{at } \mathcal{R}_\pm, \tag{11.27}$$

we infer that $\Sigma_{0,\pm} \cap \mathcal{U}$ is a smooth submanifold of S^*X. The convergence $r(\gamma(s)) \to 1$ thus implies the convergence of $\gamma(s)$ to $\Sigma_{0,\pm} \cap S^*X \cap \{r = 1\} = \mathcal{R}_\pm$. □

Next, we study null-geodesics with fixed t-momentum $\sigma = 1$ and x-momentum $\xi \in \overline{T^*_x}X$. Their behavior is qualitative the same in the case $\sigma = 0$ studied in Proposition 11.5, though this is by no means automatic since H_1 is not homogeneous with respect to dilations in ξ.

Proposition 11.6 (H_1-Flow in Σ_1). *Let $\gamma\colon I \subseteq \mathbb{R} \to \Sigma_{1,\pm} \subset \overline{T^*}X$ be a maximal integral curve of $\pm\mathsf{H}_1$. Then either*

(1) *$\gamma(s) \in \mathcal{R}_\pm$ for all s, or*
(2) *$\gamma(s)$ tends to $\mathcal{R}_\pm$ as $s \to s_0 := \inf I$ and $r(\gamma(s)) \to 2$ as $s \to s_1 := \sup I$.*

See Figure 11.4.

Proof of Proposition 11.6. For clarity, we write $1 = \sigma$. Since Proposition 11.5 already describes integral curves in S^*X, we only need to consider integral curves in $\Sigma^\circ_{\sigma,\pm} \subset T^*X$ here. By (11.4), we have

$$\begin{aligned} \pm\frac{1}{2}\mathsf{H}_\sigma r &= \pm(\sigma^2 + |\xi|^2)^{-\frac{1}{2}}\partial_\xi G \\ &= \pm 2 g^{-1}\Big(\mathrm{d}r, \frac{\cdot}{(\sigma^2+|\xi|^2)^{\frac{1}{2}}}\Big) > 0 \text{ on } \Sigma_{\sigma,\pm} \cap \{r > 1\}. \end{aligned} \tag{11.28}$$

Therefore, $r(\gamma(s))$ remains in a compact subset of $[0, 2)$ as $s \to s_0$. (In other words, γ cannot escape through $r = 2$ in the backward direction.)

We next compute

$$-\frac{1}{2} H_\sigma(\sigma - x\cdot\xi) = |\xi|^2 = (\sigma - x\cdot\xi)^2 \tag{11.29}$$

on Σ°_σ. For the function $F := \pm(\sigma - x\cdot\xi)^{-1}$, which satisfies $F > 0$ on $\Sigma^\circ_{\sigma,\pm}$ by (11.10), this means $\pm\frac{1}{2}H_\sigma F = 1$. Since $\gamma(s)$ remains over a compact subset of X (and thus in a compact subset of $\overline{T^*}X$) as $s \to s_0$, we must have $F(\gamma(s)) \to 0$, and hence $\pm(\sigma - x\cdot\xi) \to \infty$. This forces $|\xi| \to \infty$. But since $|\xi|^2 = (\sigma - x\cdot\xi)^2$ on Σ°_σ, we have

$$1 = \Big(x\cdot\frac{\xi}{|\xi|} - \frac{\sigma}{|\xi|}\Big)^2 \tag{11.30}$$

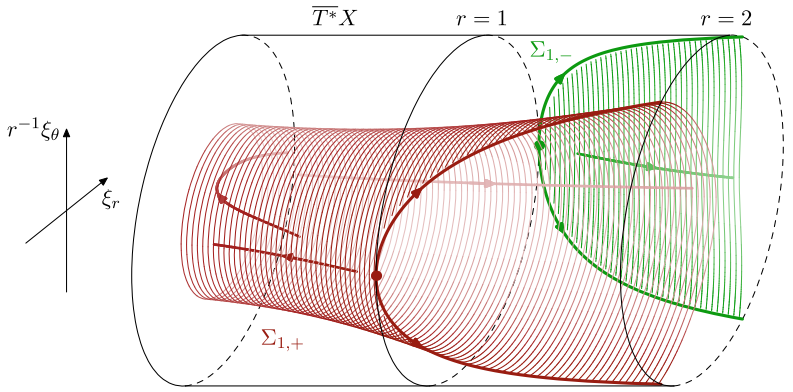


Figure 11.4 The two components of Σ_1. The future component $\Sigma_{1,+}$ is closer to the reader. The past component $\Sigma_{1,-}$ lives only over $r \geq 1$. The rescaled Hamiltonian vector field $\pm \mathsf{H}_1$ has a normal source over $r = 1$ at the radial sets $\mathcal{R}_\pm$ (fat points).

and thus $|x \cdot \frac{\xi}{|\xi|}| \to 1$. In particular, $\liminf_{s \to s_0} |x| \geq 1$. We cannot have $\liminf_{s \to s_0} |x| > 1$ in view of the monotonicity (11.28). Therefore, $|x| \to 1$, and hence, $\frac{\xi}{|\xi|} \to \mp \frac{x}{|x|}$. This implies that $\gamma(s)$ tends to $\mathcal{R}_\pm$ as $s \to s_0$.

In the other direction, we note that F has a finite upper bound on $\Sigma_{\sigma,\pm} \subset \overline{T^*}X$, and hence $\gamma(s)$ must leave every compact set of $\overline{T^*}X$ as $s \to s_1$. This means that $r(\gamma(s)) > 1$ at sufficiently late times s, and then the monotonicity of $r \circ \gamma$ implies that $r(\gamma(s)) \to 2$ as $s \to s_1$. □

11.2 Statements of the Main Results

For the remainder of this chapter, we fix a wave-type operator

$$P \in \mathrm{Diff}^2(M), \quad \mathfrak{o}^2(P) = G, \tag{11.31}$$

on de Sitter space (M, g); the dual metric function G was defined in (11.6). (All arguments presented below go through without change when P acts on sections of a vector bundle $E \to M$ that is the pullback of a vector bundle over X along the projection $M = \mathbb{R} \times X \to X$.) We assume that P commutes with t-translations, i.e., $[\partial_t, P] = 0$. This implies that we can write

$$P = P_0 + P_1 D_t + P_2 D_t^2, \quad P_j \in \mathrm{Diff}^{2-j}(X), \ j = 0, 1, 2. \tag{11.32}$$

Among the smooth solutions of $Pu = 0$, those with time dependence $e^{-i\sigma t}$, $\sigma \in \mathbb{C}$, will play a distinguished role.

Definition 11.7 (Resonant States). For $\sigma \in \mathbb{C}$, we define the space of *resonant states with frequency* σ as

$$\operatorname{Res}(P,\sigma) := \Big\{ v \in \ker P \colon \exists\, k \in \mathbb{N}_0,\ u_0, \dots, u_k \in \mathcal{C}^\infty(X), \\ v(t,x) = \sum_{j=0}^{k} u_j(x) t^j e^{-i\sigma t} \Big\}. \tag{11.33}$$

We call $\sigma \in \mathbb{C}$ a *resonance* when $\operatorname{Res}(P,\sigma) \neq \{0\}$. The set of resonances is denoted $\operatorname{Res}(P) \subset \mathbb{C}$.

Note that if $v = \sum_{j=0}^{k} u_j(x) t^j e^{-i\sigma t} \in \operatorname{Res}(P,\sigma)$ with $u_k \neq 0$, then also $0 \neq (\partial_t + i\sigma)^k v = k! u_k(x) e^{-i\sigma t} \in \operatorname{Res}(P,\sigma)$. This shows that $\operatorname{Res}(P,\sigma)$, when non-empty, contains nonzero elements with harmonic time dependence $e^{-i\sigma t}$. Furthermore, elements $e^{-i\sigma t} u_0(x) \in \operatorname{Res}(P,\sigma)$ can be characterized in terms of the spectral family $\hat{P}(\sigma)$:

$$u_0 \in \mathcal{C}^\infty(X),\ \hat{P}(\sigma) u_0 = 0, \quad \hat{P}(\sigma) := P_0 - \sigma P_1 + \sigma^2 P_2. \tag{11.34}$$

Example 11.8 (Resonances for the Scalar Wave Operator). The spectral family of the scalar wave operator $P = \Box_g$ on (M,g) is given by

$$\widehat{\Box_g}(\sigma) = \big(r^{-1} D_r r (1-r^2) D_r + r^{-2} D_\theta^2\big) + 2(r D_r - i)\sigma - \sigma^2 \tag{11.35}$$

in polar coordinates. The set of resonances can be computed to be $-i\mathbb{N}_0$, and $\operatorname{Res}(\Box_g, 0) = \operatorname{span}\{1\}$. One can, more generally, compute the resonances of $\Box_g + \mu$, $\mu \in \mathbb{C}$: they are given by $-i\sigma_\pm - in$, $n \in \mathbb{N}_0$, where $\sigma_\pm = 1 \pm \sqrt{1-\mu}$. See Exercise 11.1 for proofs of some of these statements.

Remark 11.9 (Relationship with the Keldysh Operator). If in the expression (11.35) for $\widehat{\Box_g}(\sigma)$ for $P = \Box_g$ we set $x = \frac{1-r^2}{2}$, the resulting operator is equal to twice $D_x x D_x - \sigma D_x + \frac{1}{2} D_\theta^2$ plus further terms which neither affect the qualitative behavior of null-bicharacteristics near $x = 0$ nor the quantitative threshold quantities at the radial set $N^*\{x = 0\} \setminus o$. This is essentially the Keldysh operator studied in Exercise 10.6 (see also Example 10.13). The operator $\widehat{\Box_g}(\sigma)$ transitions from being elliptic ($r < 1$) to being hyperbolic ($r > 1$) through a radial set (over $r = 1$); this is true for $\hat{P}(\sigma)$ in general.

The stated definition of $\operatorname{Res}(P,\sigma)$ makes sense also when P is the wave operator $-D_t^2 + \Delta_h$ on $\mathbb{R}_t \times X$ where (X,h) is a closed Riemannian manifold. In this case, $\operatorname{Res}(P,\sigma) = \{0\}$ unless σ^2 is an eigenvalue of Δ_h, in which case $\operatorname{Res}(P,\sigma) = \ker(\Delta_h - \sigma^2)$ is the corresponding space of eigenfunctions when $\sigma \neq 0$, and $\operatorname{Res}(P,0) = \operatorname{span}\{1, t\}$. In particular, $\operatorname{Res}(-D_t^2 + \Delta_h) \subset \mathbb{R}$ is a

discrete subset. In this setting, solutions of the wave equation $(-D_t^2+\Delta_h)v = 0$ can be written as superpositions of $e^{-i\sigma t}u_\sigma(x)$ for $0 \neq \sigma \in \operatorname{Res}(P)$ and $u_\sigma \in \operatorname{Res}(P,\sigma)$, and of constant multiples of $1, t$. The same is true for forward solutions of $(-D_t^2+\Delta_h)v = w \in \mathcal{C}_c^\infty(\mathbb{R}\times X)$ for times exceeding the maximum of t on $\operatorname{supp} w$. We shall show that forward solutions of $Pv = w \in \mathcal{C}_c^\infty(\mathbb{R}\times X)$ for the wave-type operator P on de Sitter space from (11.31) admit a similar description at late times, with resonant states replacing eigenfunctions.

Theorem 11.10 (Resonances). *For every $\alpha \in \mathbb{R}$, the set $\{\sigma \in \operatorname{Res}(P)\colon \operatorname{Im}\sigma \geq -\alpha\}$ is finite. For all $\sigma \in \operatorname{Res}(P)$, the space $\operatorname{Res}(P,\sigma)$ is finite-dimensional.*

Theorem 11.11 (Late-Time Asymptotics: $\mathcal{C}^\infty$ Version). *Let $w \in \mathcal{C}_c^\infty(\mathbb{R}\times X)$ with $t \geq 0$ on $\operatorname{supp} w$. Denote by $v \in \mathcal{C}^\infty(\mathbb{R}\times X)$ the unique forward solution of $Pv = w$. (That is, $t \geq 0$ on $\operatorname{supp} v$.) Then there exist $u_\sigma \in \operatorname{Res}(P,\sigma)$ for all $\sigma \in \operatorname{Res}(P)$ such that for every $\alpha \in \mathbb{R}$, the difference*

$$\tilde{v}_\alpha(t,x) := v(t,x) - \sum_{\substack{\sigma\in\operatorname{Res}(P)\\ \operatorname{Im}\sigma\geq-\alpha}} u_\sigma(t,x) \tag{11.36}$$

satisfies

$$|\partial_t^j\partial_x^\beta \tilde{v}_\alpha(t,x)| \leq C_{\alpha j\beta}e^{-\alpha t} \quad \forall\, j\in\mathbb{N}_0,\ \beta\in\mathbb{N}_0^2,\ t\geq 0,\ x\in X. \tag{11.37}$$

One can informally write (11.36)–(11.37) as

$$v \sim \sum_{\sigma\in\operatorname{Res}(P)} u_\sigma, \tag{11.38}$$

which expresses v as a sum of resonant states up to exponentially decaying remainders; this is a *resonance expansion* of v. The conclusion of Theorem 11.11 also holds for the solution of initial value problems (Exercise 11.2). A strengthening of Theorem 11.11 which provides quantitative bounds for v in terms of w on exponentially weighted Sobolev spaces is given in Theorem 11.24.

Remark 11.12 (All Resonant States Matter). Consider any $\sigma \in \operatorname{Res}(P)$ and $0 \neq u_\sigma \in \operatorname{Res}(P,\sigma)$. Fix $\chi \in \mathcal{C}^\infty(\mathbb{R})$ with $\chi = 0$ on $(-\infty, 0]$ and $\chi = 1$ on $[1,\infty)$. Then $w := P(\chi(t)u_\sigma) = [P,\chi]u_\sigma \in \mathcal{C}_c^\infty(\mathbb{R}\times X)$. The resonance expansion of the forward solution $v(t,x) = \chi(t)u_\sigma(t,x)$ of $Pv = w$ is $v \sim u_\sigma$. This shows that every resonant state appears in the resonance expansion of v for some source term $w \in \mathcal{C}_c^\infty(\mathbb{R}\times X)$.

The outline of the proof of Theorem 11.11 is as follows.

(1) The starting point is a weak exponential bound on v (Lemma 11.14), roughly speaking $|v| \leq C_0 e^{C_0 t}$ for some C_0. (The actual bound is in a weighted Sobolev space.)
(2) We may now take the Fourier transform in time with the convention

$$\hat{v}(\sigma, x) := \int_{-\infty}^{\infty} e^{i\sigma t} v(t, x)\, \mathrm{d}t, \quad \sigma \in \mathbb{C}. \tag{11.39}$$

Namely, $\hat{v}(\sigma) = \hat{v}(\sigma, \cdot)$ is well-defined when $\operatorname{Im}\sigma > C_0$. Furthermore, the equation $Pv = w$ implies $\hat{P}(\sigma)\hat{v}(\sigma) = \hat{w}(\sigma)$. *If* one can prove that $\hat{P}(\sigma)$ is invertible for $\operatorname{Im}\sigma > C_0$ when acting between appropriate function spaces, one thus obtains

$$\hat{v}(\sigma) = \hat{P}(\sigma)^{-1}\hat{w}(\sigma), \quad \operatorname{Im}\sigma > C_0. \tag{11.40}$$

(3) The analysis of the operator family $\hat{P}(\sigma) \in \mathrm{Diff}^2(X)$, $\sigma \in \mathbb{C}$, lies at the heart of the proof. Its principal symbol is $\sigma^2(\hat{P}(\sigma))(x, \xi) = G(0, x, 0, \xi)$, so its characteristic set is Σ_0° and its Hamiltonian vector field is H_0. The dynamical information gathered in Section 11.1 enables us to apply microlocal regularity results (elliptic estimates, real principal type propagation, radial point estimates) to prove an estimate for solutions to $\hat{P}(\sigma)u = f$ which is of the form $\|u\|_{H^s} \lesssim \|f\|_{H^{s-1}} + \|u\|_{H^{-N}}$. In fact, we shall prove that $\hat{P}(\sigma)$ is a (non-elliptic) *Fredholm* operator between suitable function spaces (Theorem 11.16). Energy estimates imply its invertibility for large $\operatorname{Im}\sigma$ (Proposition 11.18), and therefore, $\hat{P}(\sigma)^{-1}$ is a meromorphic family of Fredholm operators (see Theorem 5.77).
(4) We conclude that $\hat{v}(\sigma)$ in (11.40) has a meromorphic continuation to the complex plane. In the expression

$$v(t, \cdot) = \frac{1}{2\pi} \int_{\operatorname{Im}\sigma = C_0 + 1} e^{-i\sigma t} \hat{P}(\sigma)^{-1} \hat{w}(\sigma)\, \mathrm{d}\sigma \tag{11.41}$$

for the inverse Fourier transform of $\hat{v}(\sigma, \cdot)$, we now wish to shift the contour to $\operatorname{Im}\sigma = -\alpha$ (assuming that the final contour contains no resonances); note here that $|e^{-i\sigma t}| = e^{(\operatorname{Im}\sigma)t} = e^{-\alpha t}$ on the final contour. By the residue theorem, the poles of $\hat{P}(\sigma)^{-1}\hat{w}(\sigma)$ in $\alpha < \operatorname{Im}\sigma < C_0 + 1$ produce the resonance expansion (11.36), with the integral over the final contour producing the remainder (11.37). The details are in Section 11.5.
(5) In order to *justify* the contour shifting, we need to control the size of $\hat{P}(\sigma)^{-1}\hat{w}(\sigma)$ for bounded $\operatorname{Im}\sigma$ when $|\operatorname{Re}\sigma| \to \infty$. That is, we need to prove *high energy estimates* for $\hat{P}(\sigma)$, which we accomplish using semiclassical techniques in Section 11.4. These estimates also imply that every half space $\{\operatorname{Im}\sigma \geq -\alpha\}$ contains only finitely many resonances, and hence, the sum in (11.36) is finite for every fixed $\alpha \in \mathbb{R}$.

To begin the analysis of P in earnest, note that for all $T > 0$, the subset

$$M_T := [0,T] \times X \subset M = \mathbb{R} \times X \tag{11.42}$$

is an admissible spacetime domain (Definition 9.14) inside of $\mathbb{R} \times \tilde{X}$ where we set $\tilde{X} := \{x \in \mathbb{R}^2 \colon |x| < 3\} \supset X$ and equip $\mathbb{R} \times \tilde{X}$ with the metric g given by the expression (11.1). The initial hypersurface of M_T is $\{0\} \times X$ (since $\mathrm{d}t$ is past timelike), and the final hypersurfaces are $\{T\} \times X$ (since $\mathrm{d}(T-t) = -\mathrm{d}t$ is future timelike) and $[0,T] \times \partial X$ (since $\mathrm{d}(2-r) = -\mathrm{d}r$ is future timelike when $r = 2$). For w as in Theorem 11.11, Theorem 9.33 applies to the forward problem $Pv_T = w|_{M_T} \in \mathcal{C}^\infty(M_T)^{\bullet,-}$ on M_T and produces a unique solution $v_T \in \mathcal{C}^\infty(M_T)^{\bullet,-}$. Since for $T' < T$ we have $v_T|_{M_{T'}} = v_{T'}$ by uniqueness, the forward solution of $Pv = w$ satisfies

$$v \in \mathcal{C}^\infty(M_+)^{\bullet,-}, \quad M_+ := [0,\infty) \times X, \quad v|_{M_T} = v_T. \tag{11.43}$$

(That is, v is smooth on M and vanishes for $t \leq 0$.) For later use, we note that the same argument shows more generally that

$$w \in H^{s-1}_{\mathrm{loc}}(M_+)^{\bullet,-} \implies v \in H^s_{\mathrm{loc}}(M_+)^{\bullet,-} \tag{11.44}$$

for all $s \in \mathbb{R}$. Our goal is to describe the precise asymptotic behavior of v as $t \to \infty$. We begin with the proof of a weak exponential bound.

Definition 11.13 (Exponentially Weighted Sobolev Spaces). For $\alpha \in \mathbb{R}$ and $m \in \mathbb{N}_0$, we define $e^{\alpha t}H^m(M)$ as the closure of $\mathcal{C}^\infty_{\mathrm{c}}(M)$ with respect to the norm defined by

$$\|v\|^2_{e^{\alpha t}H^m(M)} := \sum_{j=0}^m \int_{-\infty}^\infty e^{-2\alpha t} \|\partial_t^j v(t,\cdot)\|^2_{\bar{H}^{m-j}(X)} \,\mathrm{d}t. \tag{11.45}$$

Recalling $M_+ := [0,\infty) \times X$, we moreover define

$$e^{\alpha t}H^m(M_+)^{\bullet,-} := \{v \in e^{\alpha t}H^m(M) \colon t \geq 0 \text{ on } \operatorname{supp} v\}, \tag{11.46}$$

with norm $\|v\|_{e^{\alpha t}H^m(M_+)^{\bullet,-}} := \|v\|_{e^{\alpha t}H^m(M)}$. For $a < b$, we similarly define

$$\|v\|^2_{H^m([a,b]\times X)} := \sum_{j=0}^m \int_a^b \|\partial_t^j v(t,\cdot)\|^2_{\bar{H}^{m-j}(X)} \,\mathrm{d}t. \tag{11.47}$$

The notation in (11.46) is chosen in analogy with Definition 9.28, as elements of $e^{\alpha t}H^m(M_+)^{\bullet,-}$ are of supported character at $\{t = 0\} = \{0\} \times X$ and of extendible character at $[0,\infty) \times \partial X$.

Lemma 11.14 (Weak Exponential Bound). *Let $m \in \mathbb{N}_0$. There exists a constant C_0 such that the forward solution v of $Pv = w \in \mathcal{C}^\infty_{\mathrm{c}}(M_+)^{\bullet,-}$ obeys the bound*

$$\|v\|_{e^{C_0 t} H^m(M_+)^{\bullet,-}} \leq C_0 \|w\|_{e^{C_0 t} H^{m-1}(M_+)^{\bullet,-}} . \tag{11.48}$$

Proof. By Theorem 9.33, there exists a constant C_1 such that

$$\|v\|_{H^m([0,2]\times X)^{\bullet,-}} \leq C_1 \|Pv\|_{H^{m-1}([0,2]\times X)^{\bullet,-}} . \tag{11.49}$$

By Corollary 9.38 and the time translation invariance of P, there furthermore exists a constant $C_2 > 0$ such that for all $t_0 \geq 1$,

$$\begin{aligned} E(t_0) &:= \|v\|^2_{H^m([t_0,t_0+1]\times X)} \leq C_2\big(E(t_0-1) + F(t_0-1)\big), \\ F(t_0-1) &:= \|Pv\|^2_{H^{m-1}([t_0-1,t_0+1]\times X)} . \end{aligned} \tag{11.50}$$

For $\theta := \frac{1}{2}\log C_2$, we thus have

$$e^{-2\theta t_0} E(t_0) \leq e^{-2\theta(t_0-1)}\big(E(t_0-1) + F(t_0-1)\big). \tag{11.51}$$

Fix any $\delta > 0$, multiply (11.51) by $e^{-2\delta t_0} = e^{-2\delta} e^{-2\delta(t_0-1)}$, and sum over $t_0 \in \{2, 3, 4, \ldots\}$ to obtain

$$(1-e^{-2\delta}) \sum_{t_0=2}^{\infty} e^{-2(\theta+\delta)t_0} E(t_0) \leq e^{-2\delta} e^{-2(\theta+\delta)} E(1) + e^{-2\delta} \sum_{t_0=1}^{\infty} e^{-2(\theta+\delta)t_0} F(t_0). \tag{11.52}$$

Set $C_0 := \theta + \delta$. Using that $E(1) \leq \|v\|^2_{H^m([0,2]\times X)^{\bullet,-}}$, we then obtain from (11.49)

$$\begin{aligned} &\|v\|^2_{e^{C_0 t} H^m(M_+)^{\bullet,-}} \\ &\leq C\left(\|v\|^2_{H^m([0,2]\times X)^{\bullet,-}} + \sum_{t_0=2}^{\infty} e^{-2(\theta+\delta)t_0} E(t_0)\right) \\ &\leq C'\left(\|Pv\|^2_{H^{m-1}([0,2]\times X)^{\bullet,-}} + \sum_{t_0=0}^{\infty} e^{-2C_0 t_0} \|Pv\|^2_{H^{m-1}([t_0,t_0+2]\times X)}\right) \\ &\leq C''\|Pv\|^2_{e^{C_0 t} H^{m-1}(M_+)^{\bullet,-}} . \end{aligned} \tag{11.53}$$

This proves (11.48). □

11.3 Fredholm Theory for the Spectral Family; Resonances

The key step toward defining the inverse of $\hat{P}(\sigma)$ in (11.40) is the Fredholm property of $\hat{P}(\sigma)$. The presence of radial points (Lemma 11.3) at the conormal bundle of $\{r = 1\} \subset X$ necessitates the introduction of the following quantity which concerns the subprincipal part of P at the conormal bundle of $r = 1$ in spacetime.

Definition 11.15 (A Threshold Shift). In the coordinates $(t, r, \theta, \sigma, \xi_r, \xi_\theta)$ from (11.8), we define $\tilde{\beta}_0 \in \mathcal{C}^\infty(\mathbb{S}^1)$ by

$$\tilde{\beta}_0(\theta) := \frac{1}{2}\sigma^1\Big(\frac{P - P^*}{2i}\Big)(0, 1, \theta, 0, -1, 0). \tag{11.54}$$

Here we define the formal adjoint P^* using the metric volume density $|\mathrm{d}g| = |\mathrm{d}t\,\mathrm{d}x|$. We moreover set

$$\tilde{\beta} := \sup_{\theta\in\mathbb{S}^1} \tilde{\beta}_0(\theta). \tag{11.55}$$

For example, for $P = \Box_g + \mu$, $\mu \in \mathbb{C}$, we have $\tilde{\beta}_0 = 0$. The factor of $\frac{1}{2}$ in (11.54) is equal to $\frac{1}{\beta_0}$ at $\mathcal{R}_\pm$ (see (10.20) and (11.25)).

Theorem 11.16 (Fredholm Property of $\hat{P}(\sigma)$). *Fix $\alpha \in \mathbb{R}$ and let $s > \frac{1}{2} + \alpha + \tilde{\beta}$. Then*

$$\hat{P}(\sigma)\colon \mathcal{X}^s(X) := \{u \in \bar{H}^s(X)\colon \hat{P}(0)u \in \bar{H}^{s-1}(X)\} \to \bar{H}^{s-1}(X) \tag{11.56}$$

is an analytic family of Fredholm operators for $\sigma \in \mathbb{C}$, $\operatorname{Im}\sigma > -\alpha$. Moreover:

(1) $\ker_{\bar{H}^s(X)} \hat{P}(\sigma) \subset \bar{\mathcal{C}}^\infty(X)$;
(2) $\ker_{\dot{H}^{-s+1}(X)} \hat{P}(\sigma)^* \subset \bigcap_{\epsilon>0} \dot{H}^{\frac{1}{2}-\tilde{\beta}+\operatorname{Im}\sigma-\epsilon}(X)$;
(3) *$f \in \bar{H}^{s-1}(X)$ lies in the range of (11.56) if and only if $\langle f, u^*\rangle_{L^2} = 0$ for all $u^* \in \dot{H}^{-s+1}(X)$ with $\hat{P}(\sigma)^* u^* = 0$.*

Since $\hat{P}(\sigma) - \hat{P}(0) \in \operatorname{Diff}^1(X)$, we have $\hat{P}(\sigma)u \in \bar{H}^{s-1}(X)$ for $u \in \mathcal{X}^s(X)$ indeed. We use $\hat{P}(0)$ in (11.56), and not $\hat{P}(\sigma)$, so that the natural norm $\|u\|_{\mathcal{X}^s(X)} := \|u\|_{\bar{H}^s(X)} + \|\hat{P}(0)u\|_{\bar{H}^{s-1}(X)}$ on $\mathcal{X}^s(X)$ does not depend on σ.

Toward the proof of Theorem 11.16, we first compute:

Lemma 11.17 (Threshold Shift for $\hat{P}(\sigma)$). *With $\rho = \mp\xi_r^{-1}$ near $\Sigma^\circ_{0,\pm} \cap N^*\{r = 1\}$ as in (11.18) and β_0 as in (11.25), we have*

$$\pm\rho\sigma^1\Big(\frac{1}{2i}(\hat{P}(\sigma)-\hat{P}(\sigma)^*)\Big)=\beta_0\tilde{\beta}(\sigma),\quad \tilde{\beta}(\sigma)|_{\mathcal{R}_\pm\cap S^*_{(1,\theta)}}X=\tilde{\beta}_0(\theta)-\operatorname{Im}\sigma \tag{11.57}$$

Here, we define the formal adjoint $\hat{P}(\sigma)^*$ using the volume density $|\mathrm{d}x|$.

Proof. We have

$$\hat{P}(\sigma)^*=\widehat{P^*}(\bar{\sigma}). \tag{11.58}$$

This relies on the fact that the volume density $|\mathrm{d}x|$ is the (unique) one with the property that $|\mathrm{d}t|\otimes|\mathrm{d}x|=|\mathrm{d}g|$. Therefore, $\hat{P}(0)-\hat{P}(0)^*=\widehat{P-P^*}(0)\in\mathrm{Diff}^1(X)$, the principal symbol of which is a fiber-linear function on T^*X. We thus obtain (11.57) for $\sigma=0$ from (11.54). Since $\hat{P}(\sigma)-\hat{P}(0)\in\mathrm{Diff}^1(X)$, it remains to show that

$$\operatorname{Im}\sigma^1\big(\hat{P}(\sigma)-\hat{P}(0)\big)(1,\theta,\xi_r,0)=2\xi_r\operatorname{Im}\sigma \tag{11.59}$$

in the coordinates $(r,\theta,\xi_r,\xi_\theta)$. Now, $\hat{P}(\sigma)-\hat{P}(0)\equiv\sigma\partial_\sigma\hat{P}(0)$ mod $\mathrm{Diff}^0(X)$. Identifying functions on X with t-independent functions on M, we compute

$$\partial_\sigma\hat{P}(0)u=\partial_\sigma(e^{i\sigma t}Pe^{-i\sigma t}u)|_{\sigma=0}=-i[P,t]u,\quad u\in\mathcal{C}^\infty(X). \tag{11.60}$$

Therefore,

$$\sigma^1(\partial_\sigma\hat{P}(0))=-H_Gt. \tag{11.61}$$

At $(t,r,\theta,\sigma,\xi_r,\xi_\theta)=(0,1,\theta,0,\xi_r,0)$, this equals $2\xi_r$ by (11.9). □

Proof of Theorem 11.16. The outline of the proof is as follows.

(1) We first apply Theorem 10.7. This result, as noted after its statement, combines radial point estimates (to get microlocal H^s-control near $\mathcal{R}$), real principal propagation (to deduce H^s-control of u along null-bicharacteristics starting in a punctured neighborhood of $\mathcal{R}$), and microlocal elliptic estimates (away from $\partial\Sigma_\sigma=\partial\Sigma_0$). To cap off these estimates near $r=2$, we exploit the wave equation character of $\hat{P}(\sigma)$ there by means of Corollary 9.38.

(2) For the estimate for the adjoint $\hat{P}(\sigma)^*$, the direction in which we "propagate estimates" is reversed: one starts with wave equation estimates near $r=2$, propagates the resulting microlocal H^{-s+1}-control along null-bicharacteristics using real principal type propagation, and further into the radial set $\mathcal{R}$ using radial point estimates. Microlocal elliptic estimates are again used away from the characteristic set.

Fix $\psi,\chi\in\mathcal{C}^\infty_{\mathrm{c}}(X^\circ)$ with

$$\begin{aligned} &\psi = 1,\ r \le \tfrac{3}{2}, && \operatorname{supp}\psi \subset \{r < 2 - \tfrac{1}{4}\}, \\ &\chi = 1,\ r \le 2 - \tfrac{1}{8}, && \operatorname{supp}\chi \subset \{r < 2 - \tfrac{1}{16}\}. \end{aligned} \tag{11.62}$$

- (1) Semi-Fredholm estimate for $\hat{P}(\sigma)$. Fix operators $B_\pm \in \Psi^0(X^\circ)$ with Schwartz kernels supported in $\{r \le 2 - \frac{1}{4}\} \times \{r \le 2 - \frac{1}{4}\}$ such that $B_\pm$ is elliptic on $\partial\Sigma_{\sigma,\pm} \cap S^*_{\operatorname{supp}\psi} X$ and $\mathrm{WF}'(B_\pm) \cap \partial\Sigma_{\sigma,\mp} = \emptyset$. We can then apply Theorem 10.71 with $B = B_\pm$ and $G \in \mathcal{C}^\infty_{\mathrm{c}}(\{r < 2 - \frac{1}{8}\})$, $G = 1$ for $r \le 2 - \frac{1}{4}$, to $\hat{P}(\sigma)$ (for the "+" sign) and $-\hat{P}(\sigma)$ (for the "−" sign). The dynamical assumption (1iv) was verified in Proposition 11.5, and the regularity threshold $\frac{1}{2} + \tilde{\beta}(\sigma)$ was computed in Lemma 11.17. By elliptic regularity (Exercise 6.12 with $B = \psi$, $B_1 = B_+$, $B_2 = B_-$, and $B_3 \in \Psi^0(X^\circ)$, with Schwartz kernel supported in $\{r \le 2 - \frac{1}{4}\} \times \{r \le 2 - \frac{1}{4}\}$, such that $\mathrm{WF}'(B_3) \cap \partial\Sigma_{\sigma,\pm} = \emptyset$ and $\bigcup_{j=1}^3 \mathrm{Ell}(B_j) \supset S^*_{\operatorname{supp}\psi} X$), we then obtain the estimate

$$\begin{aligned} \|u\|_{\bar{H}^s(\{r<\frac{3}{2}\})} &\le \|\psi u\|_{H^s} \\ &\le C\big(\|\chi \hat{P}(\sigma) u\|_{H^{s-1}} + \|\chi u\|_{H^{s_0}}\big) \\ &\le C\big(\|\hat{P}(\sigma) u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^{s_0}(X)}\big) \end{aligned} \tag{11.63}$$

for $s > s_0 > \frac{1}{2} + \tilde{\beta}(\sigma)$. (This holds in the strong sense that if u is an extendible distribution on X for which the right hand side is finite, then the left-hand side is finite, and the estimate holds.) We use here that $G = G\chi$ for our choice of G. Note that

$$\operatorname{Im}\sigma > -\alpha,\ s > \frac{1}{2} + \alpha + \tilde{\beta} \implies s > \frac{1}{2} + \tilde{\beta}(\sigma). \tag{11.64}$$

To estimate u in the remaining region $\{\frac{3}{2} \le r \le 2\}$ or rather its enlargement $\{\frac{5}{4} \le r \le 2\}$, we use the timelike nature of $\mathrm{d}r$: the operator $\hat{P}(\sigma)$ is a wave-type operator on the subset $\{r > 1\}$ of (X, g_X) where $g_X^{-1} := \partial_x^2 - (x \cdot \partial_x)^2 = (1 - r^2)\partial_r^2 + r^{-2}\partial_\theta^2$. (This is the metric on X for which $g_X^{-1} = g^{-1}|_{S^2 T^* X}$.) We may thus apply Corollary 9.38 with $\Omega_{-\delta,0} = \{\frac{3}{2} \le r \le 2\}$ and $\Omega = \{\frac{5}{4} \le r \le 2\}$ to obtain

$$\begin{aligned} \|u\|_{\bar{H}^s(\{\frac{5}{4}<r<2\})} &\le C\big(\|\hat{P}(\sigma) u\|_{\bar{H}^{s-1}(\{\frac{5}{4}<r<2\})} + \|u\|_{\bar{H}^s(\{\frac{5}{4}<r<\frac{3}{2}\})}\big) \\ &\le C'\big(\|\hat{P}(\sigma) u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^s(\{r<\frac{3}{2}\})}\big); \end{aligned} \tag{11.65}$$

this estimate also holds in the strong sense. Writing $1 = \chi_- + \chi_+$ for suitable $\chi_- \in \mathcal{C}^\infty_{\mathrm{c}}(\{r < \frac{3}{2}\})$ and $\chi_+ \in \mathcal{C}^\infty_{\mathrm{c}}(\{\frac{5}{4} < r \le 2\})$, the combination of (11.63) and (11.65) gives

$$\begin{aligned}\|u\|_{\bar H^s(X)} &\leq \|\chi_- u\|_{\bar H^s(X)} + \|\chi_+ u\|_{\bar H^s(X)}\\ &\leq C\big(\|u\|_{\bar H^s(\{r<\frac{3}{2}\})} + \|u\|_{\bar H^s(\{\frac{5}{4}<r<2\})}\big)\\ &\leq C'\big(\|\hat P(\sigma)u\|_{\bar H^{s-1}(X)} + \|u\|_{\bar H^{s_0}(X)}\big).\end{aligned} \tag{11.66}$$

This estimate is of the form (5.162). Since the inclusion map $\bar H^s(X) \to \bar H^{s_0}(X)$ is compact (Exercise 9.9), it implies that $\ker_{\mathcal{X}^s(X)} \hat P(\sigma) = \ker_{\bar H^s(X)} \hat P(\sigma)$ is finite-dimensional and the range of (11.56) is closed. In addition, since (11.66) holds in the strong sense for all $s > s_0$, $u \in \bar H^{s_0}(X)$ and $\hat P(\sigma)u = 0$ imply $u \in \bar H^\infty(X) = \bar{\mathcal{C}}^\infty(X)$. This proves part (1).

- (2) Semi-Fredholm estimate for $\hat P(\sigma)^*$. We apply Theorem 9.33 to the wave-type operator $\hat P(\sigma)^*$ on $\Omega = \{\frac{5}{4} \leq r \leq 2\}$, with $\{r = 2\}$ as the initial and $\{r = \frac{5}{4}\}$ as the final hypersurface. Thus,

$$\begin{aligned}\|u^*\|_{H^{-s+1}(\{\frac{5}{4}\leq r\leq 2\})^{\bullet,-}} &\leq C\|\hat P(\sigma)^* u^*\|_{H^{-s}(\{\frac{5}{4}\leq r\leq 2\})^{\bullet,-}}\\ &\leq C'\|\hat P(\sigma)^* u^*\|_{\dot H^{-s}(X)}.\end{aligned} \tag{11.67}$$

We wish to propagate H^{-s+1}-control of u^* along null-bicharacteristics to all of $\partial\Sigma_\sigma$; by Proposition 11.5, this requires propagating in the backward direction (i.e., along $-\mathsf{H}_\sigma$) in $\partial\Sigma_+$ and in the forward direction (i.e., along H_σ) in $\partial\Sigma_-$. We shall thus apply the radial point estimate Theorem 10.72 to $\pm\hat P(\sigma)^*$ with $B_\pm$ in place of B, where $B_\pm \in \Psi^0(X^\circ)$, with Schwartz kernel supported in $\{r \leq 2 - \frac{1}{4}\} \times \{r \leq 2 - \frac{1}{4}\}$, is elliptic on $\partial\Sigma_{\sigma,\pm} \cap S^*_{\operatorname{supp}\psi} X$ and satisfies $\mathrm{WF}'(B_\pm) \cap \partial\Sigma_{\sigma,\mp} = \emptyset$; moreover, we take $E \in \mathcal{C}^\infty_{\mathrm{c}}(\{\frac{5}{4} < r < 2 - \frac{1}{8}\})$ to be equal to 1 on $\{\frac{3}{2} < r < 2 - \frac{1}{4}\}$, and $G = \chi$. We thus obtain, in total,

$$\|\psi u^*\|_{H^{-s+1}} \leq C\big(\|\chi\hat P(\sigma)^* u^*\|_{H^{-s}} + \|Eu^*\|_{H^{-s+1}} + \|\chi u^*\|_{H^{s_0^*}}\big) \tag{11.68}$$

where we fix $s_0^* < -s + 1$. The required threshold condition reads

$$-s + 1 < \frac{1}{2} \pm \frac{1}{\beta_0}\rho\sigma^1\Big(\frac{1}{2i}(\hat P(\sigma)^* - \hat P(\sigma))\Big)\Big|_{\mathcal{R}_\pm}, \tag{11.69}$$

which is equivalent to $s > \frac{1}{2} + \tilde\beta(\sigma)$; this holds by (11.64). We estimate $\|Eu^*\|_{H^{-s+1}} \leq C\|u^*\|_{H^{-s+1}(\{\frac{5}{4}\leq r\leq 2\})^{\bullet,-}}$ and use (11.67) to obtain

$$\|u^*\|_{\dot H^{-s+1}(X)} \leq C\Big(\|\hat P(\sigma)^* u^*\|_{\dot H^{-s}(X)} + \|u^*\|_{\dot H^{s_0^*}(X)}\Big). \tag{11.70}$$

Since the inclusion map $\dot H^{-s+1}(X) \to \dot H^{s_0^*}(X)$ is compact, this implies in particular that $\ker_{\dot H^{-s}(X)} \hat P(\sigma)^*$ is finite-dimensional.

We also note that the estimate 11.70 holds in the strong sense as long as the threshold condition $-s+1 < \frac{1}{2} - \tilde{\beta}(\sigma)$ is satisfied. This proves part (2).

- Range of $\hat{P}(\sigma)$. Consider first $u \in \mathcal{X}^s(X)$, so $u \in \bar{H}^s(X)$ and $f := \hat{P}(\sigma)u \in \bar{H}^{s-1}(X)$. Let $u^* \in \ker_{\dot{H}^{-s+1}(X)} \hat{P}(\sigma)^*$. We then claim that

$$\langle f, u^* \rangle_{L^2} = \langle u, \hat{P}(\sigma)^* u^* \rangle_{L^2} = 0. \tag{11.71}$$

While all pairings are well-defined, the integration by parts needs justification since for general $u \in \bar{H}^s(X)$ one only has $\hat{P}(\sigma)u \in \bar{H}^{s-2}(X)$, which cannot be paired against u^*. This obstacle can be overcome using a regularization argument as in (10.48); the details are left to the reader (Exercise 11.4).

Suppose now that $f \in \bar{H}^{s-1}(X)$ satisfies $\langle f, u^* \rangle_{L^2} = 0$ for all $u^* \in \dot{H}^{-s+1}(X)$ with $\hat{P}(\sigma)^* u^* = 0$. In order to show the existence of $u \in \bar{H}^s(X)$ with $\hat{P}(\sigma)u = f$, we argue using duality as in the proof of Lemma 9.34. Define the linear functional

$$\lambda \colon \hat{P}(\sigma)^* u^* \mapsto \langle u^*, f \rangle_{L^2}, \quad u^* \in \dot{H}^{-s+2}(X). \tag{11.72}$$

This is well-defined by our assumption on f. Fix a complementary subspace $V \subset \dot{H}^{-s+1}(X)$ of $\ker_{\dot{H}^{-s+1}(X)} \hat{P}(\sigma)^*$. Then the estimate (11.70) implies the existence of a constant C' such that $\|u^*\|_{\dot{H}^{-s+1}(X)} \leq C' \|\hat{P}(\sigma)^* u^*\|_{\dot{H}^{-s}(X)}$ for all $u^* \in V$ (see the proof of Proposition 5.74). Therefore,

$$|\lambda(\hat{P}(\sigma)^* u^*)| \leq C' \|f\|_{\bar{H}^{s-1}(X)} \|\hat{P}(\sigma)^* u^*\|_{\dot{H}^{-s}(X)}. \tag{11.73}$$

Extend λ to a continuous linear functional on $\dot{H}^{-s}(X)$. By Theorem 9.27, this extension is given by the L^2-pairing with some $u \in \bar{H}^s(X)$ which satisfies

$$\langle u^*, f \rangle_{L^2} = \langle \hat{P}(\sigma)^* u^*, u \rangle_{L^2} = \langle u^*, P(\sigma)u \rangle_{L^2} \quad \forall\, u^* \in \dot{H}^{-s+2}(X). \tag{11.74}$$

Therefore, $\hat{P}(\sigma)u = f$. □

Recall now that every $0 \neq u \in \ker \hat{P}(\sigma)$ gives rise to a nontrivial solution $v(t,x) = e^{-i\sigma t} u(x)$ of the wave equation $Pv = 0$ which grows like $e^{(\operatorname{Im}\sigma)t}$. By Lemma 11.14, solutions of this equation cannot grow too quickly in time; this puts an upper bound on $\operatorname{Im}\sigma$. More precisely:

Proposition 11.18 (Invertibility for Large $\operatorname{Im}\sigma$). *There exists $C \in \mathbb{R}$ such that for all $\sigma \in \mathbb{C}$ with $\operatorname{Im}\sigma > C$, the operator $\hat{P}(\sigma)$ in (11.56) is invertible.*

Proof. Let $0 \neq u \in \ker_{\bar{\mathcal{C}}^\infty(X)} \hat{P}(\sigma)$. Set $v(t,x) := e^{-i\sigma t}u(x)$. Let $\chi \in \mathcal{C}^\infty(\mathbb{R})$ with $\chi = 0$ on $(-\infty, 0]$ and $\chi = 1$ on $[1, \infty)$. Then

$$P(\chi v) = w := [P, \chi]v \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R} \times X). \tag{11.75}$$

Lemma 11.14 gives $e^{(-C_0 + \operatorname{Im}\sigma)t}\chi(t)u(x) \in L^2(M_+)$, so $-C_0 + \operatorname{Im}\sigma < 0$, i.e., $\operatorname{Im}\sigma < C_0$. This proves the injectivity of $\hat{P}(\sigma)$ for $\operatorname{Im}\sigma \geq C_0$.

By Theorem 11.16, the surjectivity of $\hat{P}(\sigma)$ is equivalent to the injectivity of $\hat{P}(\sigma)^*$. When $\operatorname{Im}\sigma$ is sufficiently large, we have $\frac{1}{2} - \tilde{\beta} + \operatorname{Im}\sigma > 1$, so Theorem 11.16(2) shows that every $u^* \in \ker \hat{P}(\sigma)^*$ lies in $\dot{H}^1(X)$. In view of (11.58), we thus see that

$$P^* v^* = 0, \quad v^*(t,x) = e^{-i\bar{\sigma}t}u^*(x). \tag{11.76}$$

Note that $|v^*(t,x)| = e^{-(\operatorname{Im}\sigma)t}|u^*(x)|$. Given any $w \in e^{C_0 t}H^1(M_+)^{\bullet,-}$, we can solve $Pv = w$ with $v \in e^{C_0 t}H^2(M_+)^{\bullet,-}$ (here C_0 is given by Lemma 11.14 with $m = 2$). If $\operatorname{Im}\sigma > C_0$, we can thus integrate by parts to conclude

$$\langle w, v^* \rangle_{L^2(M_+)} = \langle Pv, v^* \rangle_{L^2(M_+)} = \langle v, P^* v^* \rangle_{L^2(M_+)} = 0. \tag{11.77}$$

Since w is arbitrary, this shows that $v^* = 0$ for $t > 0$ and thus for all t. Therefore, $u^* = 0$. □

Given Theorem 11.16 and Proposition 11.18, the analytic Fredholm theorem (Theorem 5.77) applies and gives:

Corollary 11.19 (Meromorphic Continuation). *Fix $\alpha \in \mathbb{R}$ and let $s > \frac{1}{2} + \alpha + \tilde{\beta}$. Then $\hat{P}(\sigma)\colon \mathcal{X}^s(X) \to \bar{H}^{s-1}(X)$ has index 0, and $\hat{P}(\sigma)^{-1}\colon \bar{H}^{s-1}(X) \to \bar{H}^s(X)$ extends from $\operatorname{Im}\sigma \gg 1$ to a finite-meromorphic family of operators on $\{\sigma \in \mathbb{C}\colon \operatorname{Im}\sigma > -\alpha\}$. The set of poles with $\operatorname{Im}\sigma > -\alpha$ is independent of the value of s as long as $s > \frac{1}{2} + \alpha + \tilde{\beta}$.*

Proof. We only need to prove the final statement. Fix $\sigma \in \mathbb{C}$. Then $\hat{P}(\sigma)\colon \mathcal{X}^s(X) \to \bar{H}^{s-1}(X)$ has index 0 for $s > \frac{1}{2} - \operatorname{Im}\sigma + \tilde{\beta}$, and it is invertible if and only if it has trivial nullspace. But the nullspace is a subspace of $\bar{\mathcal{C}}^\infty(X)$ and thus independent of s. □

Exercise 11.5 describes $\hat{P}(\sigma)^{-1}$ near a simple pole.

We can now prove a weak version of Theorem 11.10. We can moreover characterize the space of resonances in terms of the resolvent $\hat{P}(\sigma)^{-1}$:

Proposition 11.20 (Resonances as Poles; Resonant States as Residues).

(1) *The set $\operatorname{Res}(P) \subset \mathbb{C}$ is discrete. A number $\sigma_0 \in \mathbb{C}$ is a resonance if and only if $\hat{P}(\sigma)^{-1}\colon \bar{H}^{s-1}(X) \to \bar{H}^s(x)$ has a pole at $\sigma = \sigma_0$ where $s > \frac{1}{2} - \operatorname{Im}\sigma + \tilde{\beta}$.*

(2) *For all* $\sigma_0 \in \operatorname{Res}(P)$*, the space* $\operatorname{Res}(P, \sigma_0)$ *is finite-dimensional and equal to*

$$\{\operatorname{res}_{\sigma=\sigma_0}\big(e^{-i\sigma t}\hat P(\sigma)^{-1}f(\sigma)\big)\colon f \text{ is a polynomial in } \sigma \text{ with values in } \bar{\mathcal{C}}^\infty(X)\}. \tag{11.78}$$

This also holds if we allow f *to be holomorphic near* σ *with values in* $\bar{\mathcal{C}}^\infty(X)$*.*

Proof. The first part follows from Corollary 11.19 since the set $\operatorname{Res}(P) \cap \{\operatorname{Im}\sigma > -\alpha\}$ is discrete for all $\alpha \in \mathbb{R}$, and $\hat P(\sigma)^{-1}$ has a pole at $\sigma = \sigma_0$ if and only if $\hat P(\sigma_0)$ fails to be injective.

For the second part, fix $s > \frac{1}{2} - \operatorname{Im}\sigma_0 + \tilde\beta$ and write

$$\hat P(\sigma)^{-1} = \sum_{j=1}^{J} (\sigma - \sigma_0)^{-j} B_j + \operatorname{hol}(\sigma) \tag{11.79}$$

where J is the order of the pole, the operators $B_1, \dots, B_J\colon \bar H^{s-1}(X) \to \mathcal{X}^s(X)$ have finite rank, and we write $\operatorname{hol}(\sigma)$ for a family of operators which is holomorphic near $\sigma = \sigma_0$. Since $\sigma_0 \in \operatorname{Res}(P)$, we have $J \geq 1$ and $B_J \neq 0$.

We first claim that

$$\operatorname{ran} B_j \subset \bar{\mathcal{C}}^\infty(X), \quad j = 1, \dots, J. \tag{11.80}$$

To prove this, we write

$$\begin{aligned} I = \hat P(\sigma)\hat P(\sigma)^{-1} &= \sum_{j=1}^{J}\sum_{k=0}^{2} (\sigma-\sigma_0)^{-j+k} \frac{\partial_\sigma^k \hat P(\sigma_0)}{k!} B_j + \operatorname{hol}(\sigma) \\ &= \sum_{j=1}^{J} (\sigma-\sigma_0)^{-j} \sum_{k=0}^{\min(2,J-j)} \frac{\partial_\sigma^k \hat P(\sigma_0)}{k!} B_{j+k} + \operatorname{hol}(\sigma). \end{aligned} \tag{11.81}$$

The coefficient of $(\sigma-\sigma_0)^{-J}$ vanishes, so $\hat P(\sigma_0)B_J = 0$, and hence $\operatorname{ran} B_J \subset \ker_{\mathcal{X}^s(X)} \hat P(\sigma_0) \subset \bar{\mathcal{C}}^\infty(X)$. Assuming that we have established (11.80) for $j = J, J-1, \dots, J_0+1 > 1$, the coefficient of $(\sigma-\sigma_0)^{-J_0}$ in (11.81) is equal to $\hat P(\sigma_0)B_{J_0}$ plus a sum of operators with range contained in $\bar{\mathcal{C}}^\infty(X)$. We thus need to show that

$$u \in \mathcal{X}^s(X),\ \hat P(\sigma_0)u = f \in \bar{\mathcal{C}}^\infty(X) \implies u \in \bar{\mathcal{C}}^\infty(X). \tag{11.82}$$

But this follows from the fact that the estimate (11.66) holds in the strong sense. (One can also argue directly: $\operatorname{WF}(u)$ is contained in the characteristic set of $\hat P(\sigma_0)$ by microlocal elliptic regularity; $\operatorname{WF}(u)$ is disjoint from $\mathcal{R}_\pm$ by

radial point estimates; thus $\mathrm{WF}(u)$ is empty over X° by real principal type propagation, so $u \in \mathcal{C}^\infty(X^\circ)$. Corollary 9.38 gives $u \in \bar{\mathcal{C}}^\infty(X)$.)

Let now $f(\sigma)$ be a polynomial with values in $\bar{\mathcal{C}}^\infty(X)$. Then

$$v(t,\cdot) := \mathrm{res}_{\sigma=\sigma_0}\big(e^{-i\sigma t}\hat{P}(\sigma)^{-1}f(\sigma)\big) = \sum_{j=1}^{J} B_j \frac{\partial_\sigma^{j-1}(e^{-i\sigma t}f(\sigma))|_{\sigma=\sigma_0}}{(j-1)!} \tag{11.83}$$

is of the form (11.33) with $k = J-1$. Moreover, $u \in \mathrm{Res}(P,\sigma_0)$ since

$$Pv(t,\cdot) = \frac{1}{2\pi i} P \oint_{\sigma_0} e^{-i\sigma t}\hat{P}(\sigma)^{-1}f(\sigma)\,\mathrm{d}\sigma = \frac{1}{2\pi i}\oint_{\sigma_0} e^{-i\sigma t}f(\sigma)\,\mathrm{d}\sigma = 0. \tag{11.84}$$

Here $\oint_{\sigma_0}(\cdot)\,\mathrm{d}\sigma$ is the contour integral along $|\sigma-\sigma_0| = \epsilon$, $\epsilon > 0$ small, oriented counterclockwise. The same argument applies to $\bar{\mathcal{C}}^\infty(X)$-valued f which are holomorphic in σ near $\sigma = \sigma_0$.

Conversely, let $v = \sum_{j=0}^{k} e^{-i\sigma_0 t}t^j u_j(x) \in \mathrm{Res}(P,\sigma_0)$. We can then write

$$v(t,\cdot) = \mathrm{res}_{\sigma=\sigma_0}\big(e^{-i\sigma t}\tilde{u}(\sigma)\big), \quad \tilde{u}(\sigma) := \sum_{j=0}^{k}(\sigma-\sigma_0)^{-j-1}\tilde{u}_j \tag{11.85}$$

for some $\tilde{u}_j \in \bar{\mathcal{C}}^\infty(X)$ which can be computed from the u_j. Since

$$0 = Pv(t,\cdot) = \frac{1}{2\pi i}\oint_{\sigma_0} e^{-i\sigma t}\hat{P}(\sigma)\tilde{u}(\sigma)\,\mathrm{d}\sigma \tag{11.86}$$

for all t, we infer that $F(\sigma) := \hat{P}(\sigma)\tilde{u}(\sigma)$ is holomorphic. For the $(J-1)$-st order Taylor expansion $f(\sigma) := \sum_{j=0}^{J-1}\frac{\partial_\sigma^j F(\sigma_0)}{j!}(\sigma-\sigma_0)^j$ of F around $\sigma = \sigma_0$, we then have

$$\hat{P}(\sigma)^{-1}f(\sigma) = \tilde{u}(\sigma) - \hat{P}(\sigma)^{-1}\big(F(\sigma)-f(\sigma)\big) = \tilde{u}(\sigma) + \mathrm{hol}(\sigma). \tag{11.87}$$

In combination with (11.85), we thus see that $v(t,\cdot) = \mathrm{res}_{\sigma=\sigma_0}(e^{-i\sigma t}\hat{P}(\sigma)^{-1}f(\sigma))$. □

11.4 High-Energy Estimates

The proof of *high-energy estimates*, i.e., estimates for $\hat{P}(\sigma)$ when $\mathrm{Im}\,\sigma$ is bounded and $|\mathrm{Re}\,\sigma| \to \infty$, follows along the same lines as the proof of the Fredholm property of $\hat{P}(\sigma)$ (Theorem 11.16), except we now use semiclassical tools. We shall prove:

Theorem 11.21 (High-Energy Estimates). *Fix $\alpha_+ \geq \alpha_- \in \mathbb{R}$ and let $s > \frac{1}{2} + \alpha_+ + \tilde{\beta}$. Then there exist constants C, C' such that*

$$\begin{aligned} \|u\|_{\bar{H}^s_{|\operatorname{Re}\sigma|^{-1}}(X)} \leq C|\operatorname{Re}\sigma|^{-1}\|\hat{P}(\sigma)u\|_{\bar{H}^{s-1}_{|\operatorname{Re}\sigma|^{-1}}(X)}, \\ -\alpha_+ \leq \operatorname{Im}\sigma \leq -\alpha_-,\ |\operatorname{Re}\sigma| > C'. \end{aligned} \tag{11.88}$$

In particular, for all $\sigma \in \operatorname{Res}(P)$ with $-\alpha_+ \leq \operatorname{Im}\sigma \leq -\alpha_-$, we have $|\operatorname{Re}\sigma| \leq C'$.

We rephrase this in semiclassical terms: we regard

$$h := |\operatorname{Re}\sigma|^{-1} = \pm\frac{1}{\operatorname{Re}\sigma}, \quad \pm\operatorname{Re}\sigma > 0, \tag{11.89}$$

as a semiclassical parameter and prove estimates for the semiclassical rescaling

$$P_{z,h} := h^2\hat{P}(h^{-1}z), \quad z := h\sigma = \frac{\sigma}{|\operatorname{Re}\sigma|} = \pm 1 + ih\operatorname{Im}\sigma, \tag{11.90}$$

of $\hat{P}(\sigma)$ on semiclassical Sobolev spaces. For fixed $\alpha \in \mathbb{R}$, we consider $\sigma = \sigma_0 - i\alpha$, so $z = \pm 1 - i\alpha h$ for $\pm\sigma_0 > 0$. Proposition 9.42 implies that

$$\begin{aligned} P_{\pm,\alpha} := \big(P_{\pm 1 - i\alpha h, h}\big)_{h\in(0,1)} \in \operatorname{Diff}^2_{\bar{h}}(X), \\ \sigma^2_{\bar{h}}(P_{\pm,\alpha})(x,\xi) = \sigma^2(P)((0,x), \mp \mathrm{d}t + \xi) = G(0,x,\pm 1,\xi). \end{aligned} \tag{11.91}$$

The semiclassical characteristic set of $P_{\pm,\alpha}$ is the set $\Sigma_{\pm 1} \subset \overline{T^*}X$ defined in (11.13), and the Hamiltonian vector field of the semiclassical principal symbol of $P_{\pm,\alpha}$ is equal to $H_{\pm 1}$ defined in (11.11). Since $H_1|_{(x,\xi)} = -H_{-1}|_{(x,-\xi)}$ by (11.16), it suffices to consider the "+" sign, i.e., $\sigma = \sigma_0 - i\alpha$ with $\sigma_0 > 0$. The desired estimate (11.88) now reads

$$\|u\|_{\bar{H}^s_h(X)} \leq Ch^{-1}\|P_{+,\alpha}u\|_{\bar{H}^{s-1}_h(X)}, \quad h < h_0,\ \alpha \in [\alpha_-, \alpha_+], \tag{11.92}$$

where $h_0 = 1/C'$.

Proof of Theorem 11.21. Consider first a fixed value of $\alpha \in [\alpha_-, \alpha_+]$. We will follow the strategy of the proof of Theorem 11.16 to establish the estimate

$$\|u\|_{\bar{H}^s_h(X)} \leq C\Big(h^{-1}\|P_{+,\alpha}u\|_{\bar{H}^{s-1}_h(X)} + h^N\|u\|_{\bar{H}^{s_0}_h(X)}\Big). \tag{11.93}$$

(This is the semiclassical analogue of (11.66).) We begin with an application of the radial point (and real principal type and elliptic) estimate, Theorem 10.17(1), which gives, analogously to (11.63),

$$\|u\|_{\bar{H}_{\hbar}^{s}(\{r<\frac{3}{2}\})} \leq C\Big(h^{-1}\|P_{+,\alpha}u\|_{\bar{H}_{h}^{s-1}(X)} + h^{N}\|u\|_{\bar{H}_{h}^{s_0}(X)}\Big). \tag{11.94}$$

(The threshold condition for the radial point estimate is again $s > \frac{1}{2} + \alpha + \tilde{\beta}$.) This uses the dynamical information from Proposition 11.6. In order to estimate u in the remaining region $\{\frac{5}{4} \leq r \leq 2\}$, we use Corollary 9.46 to obtain, analogously to (11.65),

$$\|u\|_{\bar{H}_{\hbar}^{s}(\{\frac{5}{4}<r<2\})} \leq C\Big(h^{-1}\|P_{+,\alpha}u\|_{\bar{H}_{h}^{s-1}(X)} + \|u\|_{\bar{H}_{\hbar}^{s}(\{r<\frac{3}{2}\})}\Big). \tag{11.95}$$

Combining the two estimates as in (11.66) gives (11.93). When $h_0 > 0$ is sufficiently small, we have $Ch^{N}\|u\|_{\bar{H}_{h}^{s_0}(X)} \leq \frac{1}{2}\|u\|_{\bar{H}_{\hbar}^{s}(X)}$ for $h < h_0$, and hence, (11.93) implies the desired estimate (11.92).

The estimate (11.92) implies the injectivity of $P_{1-i\alpha h,h}$ on $\bar{H}^{s}(X)$ for $h < h_0$, and hence the injectivity of $\hat{P}(h^{-1} - i\alpha)$. Since $\hat{P}(h^{-1} - i\alpha)\colon \mathcal{X}^{s}(X) \to \bar{H}^{s-1}(X)$ has index 0 by Corollary 11.19, we conclude that $\hat{P}(h^{-1} - i\alpha)$ is invertible. Therefore, $h^{-1} - i\alpha \notin \operatorname{Res}(P)$.

We have $P_{+,\alpha} - P_{+,\alpha'} \in h\mathrm{Diff}_{\hbar}^{1}(X)$ since the semiclassical principal symbol of $P_{+,\alpha}$ is independent of α by (11.91). Furthermore,

$$P_{+,\alpha} - P_{+,\alpha'} \xrightarrow{\alpha' \to \alpha} 0 \text{ in } h\mathrm{Diff}_{\hbar}^{1}(X), \tag{11.96}$$

and hence

$$Ch^{-1}\|P_{+,\alpha'}u\|_{\bar{H}^{s-1}(X)} \leq Ch^{-1}\|P_{+,\alpha}u\|_{\bar{H}^{s-1}(X)} + \frac{1}{2}\|u\|_{\bar{H}^{s}(X)} \tag{11.97}$$

when $|\alpha' - \alpha|$ is sufficiently small. Therefore, the estimate (11.93) implies the same estimate for $2C$ and α' in place of C and α when $|\alpha' - \alpha|$ is sufficiently small. The compactness of $[\alpha_-, \alpha_+]$ then implies that there exists a constant C such that (11.93) holds for all $\alpha \in [\alpha_-, \alpha_+]$. Thus, there exist constants C, h_0 such that (11.92) holds for all $\alpha \in [\alpha_-, \alpha_+]$. (A conceptually cleaner argument uses the stability of the estimates underlying (11.93) under perturbations of $P_{+,\alpha}$ in the space of elements of $\mathrm{Diff}_{\hbar}^{2}(X)$ with real principal symbols; see Exercise 10.8 for such a stability result in the non-semiclassical setting.) □

Despite the brevity of this proof, the reader should keep in mind that it uses almost all of the semiclassical results developed in this book.

Remark 11.22 (High-Energy Estimate for the Adjoint). One can prove the analogue

$$\|u^*\|_{\dot{H}_{h}^{-s+1}(X)} \leq C\Big(h^{-1}\|P_{+,\alpha}^{*}u^*\|_{\dot{H}_{h}^{-s}(X)} + h^{N}\|u\|_{\dot{H}_{h}^{-N}(X)}\Big) \tag{11.98}$$

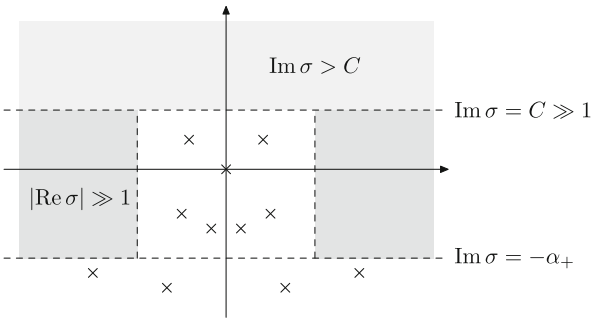


Figure 11.5 For large $\operatorname{Im}\sigma$, there are no resonances by Proposition 11.18. In any strip and for large $|\operatorname{Re}\sigma|$, there are no resonances by Theorem 11.21.

of the estimate (11.70) by first using Theorem 9.44 near $r = 2$ and then Theorem 10.17(2) over the rest of X°. For sufficiently small h, this implies the injectivity of $P^*_{1-i\alpha h,h}$, i.e., of $\hat{P}(h^{-1}-i\alpha)^*$, on $\dot{H}^{-s+1}(X)$. Together with the injectivity of $\hat{P}(h^{-1}-i\alpha)$ which follows from (11.92), we thus obtain the invertibility of $\hat{P}(h^{-1}-i\alpha)$ for small $h > 0$ without recourse to the index 0 property of $\hat{P}(\sigma)$.

We can now complete the proof of Theorem 11.10.

***Proof of Theorem* 11.10**. Let C be as in Proposition 11.18. Theorem 11.21 with $\alpha_- = -C$ and any $\alpha_+ > \alpha_-$ then implies that $\operatorname{Res}(P) \cap \{\sigma \in \mathbb{C}\colon \operatorname{Im}\sigma \geq -\alpha_+\}$ is finite. □

See Figure 11.5.

11.5 Proof of Late-Time Asymptotics

We now return to the problem $Pv = w$ on spacetime and prove Theorem 11.11. Fix $\alpha \in \mathbb{R}$ such that no resonance of P has imaginary part equal to $-\alpha$. Write $s_0 := \frac{1}{2} + \alpha + \tilde{\beta}$ for the regularity threshold in Theorems 11.16 and 11.21. Given

$$w \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R} \times X), \quad t \geq 0 \text{ on } \operatorname{supp} w, \tag{11.99}$$

Lemma 11.14, now applied with some fixed $m \in \mathbb{N}$ with $m \geq 2$ and $m > s_0$, gives the exponential bound $v \in e^{C_0 t} H^m(M_+)^{\bullet,-}$ on the forward solution of $Pv = w$. For $\operatorname{Im}\sigma \geq C_0 + \eta$, $\eta > 0$, we can then define $\hat{v}(\sigma,\cdot) = \int_{\mathbb{R}} e^{i\sigma t} v(t,\cdot)\,\mathrm{d}t$ and bound

$$\begin{aligned}\|\hat{v}(\sigma)\|_{\bar{H}^m(X)} &\leq \int_0^\infty |e^{i\sigma t}| \|v(t,\cdot)\|_{\bar{H}^m(X)}\,\mathrm{d}t \\ &\leq \int_0^\infty e^{-\eta t}\|e^{-C_0 t}v(t,\cdot)\|_{\bar{H}^m(X)}\,\mathrm{d}t \leq \frac{1}{\sqrt{2\eta}}\|v\|_{e^{C_0 t}\dot{H}^m(M_+)^{\bullet,-}}.\end{aligned} \tag{11.100}$$

Since $\mathcal{F}(\partial_t v)(\sigma) = -i\sigma\hat{v}(\sigma)$ for $\operatorname{Im}\sigma > C_0$, we have analogous bounds for $|\sigma|^j\|\hat{v}(\sigma)\|_{\bar{H}^{m-j}(X)}$ for $j = 0,\ldots,m$. Taking $\eta = 1$, we have thus proved

$$\begin{aligned}\langle\sigma\rangle^j\|\hat{v}(\sigma)\|_{\bar{H}^{m-j}(X)} &\leq C\|v\|_{e^{C_0 t}\dot{H}^m(M_+)^{\bullet,-}} \\ &\leq C'\|Pv\|_{e^{C_0 t}\dot{H}^{m-1}(M_+)^{\bullet,-}}, \quad \operatorname{Im}\sigma = C_0 + 1.\end{aligned} \tag{11.101}$$

Taking the Fourier transform of $Pv = w$ and integrating by parts in t moreover yields

$$\hat{P}(\sigma)\hat{v}(\sigma) = \hat{w}(\sigma), \quad \operatorname{Im}\sigma > C_0. \tag{11.102}$$

Since $\hat{P}(\sigma)$ is injective on $\bar{H}^m(X)$ for $\operatorname{Im}\sigma > C_0$, this implies

$$\hat{v}(\sigma) = \hat{P}(\sigma)^{-1}\hat{w}(\sigma), \quad \operatorname{Im}\sigma > C_0. \tag{11.103}$$

Since $m \geq 2$, we also conclude from (11.101) that $\|\hat{v}(\sigma)\|_{\bar{H}^{m-2}(X)} \leq C\langle\sigma\rangle^{-2}$ for $\operatorname{Im}\sigma = C_0 + 1$. The Fourier inversion formula thus gives

$$v(t,\cdot) = \frac{1}{2\pi}\int_{\operatorname{Im}\sigma = C_0+1} e^{-i\sigma t}\hat{v}(\sigma,\cdot)\,\mathrm{d}\sigma = \frac{1}{2\pi}\int_{\operatorname{Im}\sigma = C_0+1} e^{-i\sigma t}\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\,\mathrm{d}\sigma, \tag{11.104}$$

with the integral converging in $\bar{H}^{m-2}(X)$.

Set $\alpha_+ := \alpha > \alpha_- := -C_0 - 1$. We wish to shift the contour in (11.104) from $\operatorname{Im}\sigma = -\alpha_-$ to $\operatorname{Im}\sigma = -\alpha_+ = -\alpha$. Fix the constant C' from Theorem 11.21. For arbitrary $R > C'$, consider the contour

$$\begin{aligned}\gamma_R &:= \gamma_R^- \cup \gamma_{R,|}^- \cup \gamma_R^0 \cup \gamma_{R,|}^+ \cup \gamma_R^+, \\ \gamma_R^-(s) &= s - i\alpha_-, \qquad -\infty < s \leq -R, \\ \gamma_{R,|}^-(s) &= -R - is, \qquad \alpha_- \leq s \leq \alpha_+, \\ \gamma_R^0(s) &= s - i\alpha_+, \qquad -R \leq s \leq R, \\ \gamma_{R,|}^+(s) &= R + is, \qquad -\alpha_+ \leq s \leq -\alpha_-, \\ \gamma_R^+(s) &= s - i\alpha_-, \qquad R \leq s < \infty.\end{aligned} \tag{11.105}$$

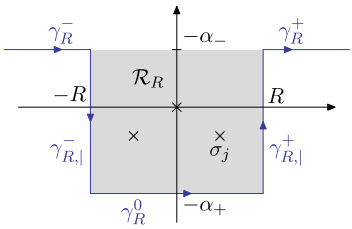


Figure 11.6 The contours from (11.105) and the resonances in the rectangle $\mathcal{R}_R$.

See Figure 11.6. The resolvent $\hat{P}(\sigma)^{-1}\colon \bar{H}^{m-1}(X) \to \bar{H}^m(X)$ exists for all $\sigma \in \gamma_R$ and it has finitely many poles in the rectangle $\mathcal{R}_R := [-R, R] \times i[-\alpha_+, -\alpha_-]$. The residue theorem thus gives

$$\begin{aligned} v(t,\cdot) = \frac{1}{2\pi}\int_{\gamma_R} e^{-i\sigma t}\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\,\mathrm{d}\sigma \\ + \sum_{\sigma_0 \in \mathrm{Res}(P)\cap\mathcal{R}_R} i\,\mathrm{res}_{\sigma=\sigma_0}\big(e^{-i\sigma t}\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\big). \end{aligned} \tag{11.106}$$

Proposition 11.20 shows that the finite sum on the right-hand side is a sum of resonant states; and it is independent of $R > C'$ since the set $\mathrm{Res}(P)\cap\mathcal{R}_R$ is.

Lemma 11.23 (Estimates for Contour Integrals). *Let $t \in \mathbb{R}$. Then the contour integrals of $e^{-i\sigma t}\hat{P}(\sigma)^{-1}\hat{w}(\sigma)$ over $\gamma_R^\pm$ and $\gamma_{R,|}^\pm$ tend to 0 in $L^2(X)$ as $R \to \infty$. Furthermore, the integral*

$$\lim_{R\to\infty}\int_{\gamma_R^0} e^{-i\sigma t}\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\,\mathrm{d}\sigma = \int_{\mathrm{Im}\,\sigma=-\alpha} e^{-i\sigma t}\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\,\mathrm{d}\sigma \tag{11.107}$$

converges in $L^2(X)$.

Proof. Write $h = |\mathrm{Re}\,\sigma|^{-1}$. Since $w \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}\times X)$, we have

$$\begin{aligned} \|\hat{w}(\sigma)\|_{\bar{H}_h^{m-1}(X)} &\le C\|\hat{w}(\sigma)\|_{\bar{H}^{m-1}(X)} \\ &\le C'(1+|\mathrm{Re}\,\sigma|)^{-2}, \quad -\alpha_+ \le \mathrm{Im}\,\sigma \le -\alpha_-. \end{aligned} \tag{11.108}$$

(See Exercise 2.23 for the first inequality.) Since $\|\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\|_{L^2(X)} \le C\|\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\|_{\bar{H}_h^m(X)}$, the high-energy estimate of Theorem 11.21 implies

$$\|\hat{P}(\sigma)^{-1}\hat{w}(\sigma)\|_{L^2(X)} \le C''|\mathrm{Re}\,\sigma|^{-3} \tag{11.109}$$

for σ on the contours $\gamma_R^\pm$, $\gamma_{R,|}^\pm$. The desired conclusion now follows from $(\alpha_+ - \alpha_-)\cdot C''R^{-3} \xrightarrow{R\to\infty} 0$ for $\gamma_{R,|}^\pm$ and $\int_R^\infty C''s^{-3}\,\mathrm{d}s \xrightarrow{R\to\infty} 0$ for $\gamma_R^\pm$.

For $\operatorname{Im}\sigma = -\alpha_+$ on the other hand, the function $\hat P(\sigma)^{-1}\hat w(\sigma)$ is bounded in $\bar H^m(X)$ and thus in $L^2(X)$ for $|\operatorname{Re}\sigma| \leq C'$, and (11.109) implies $\|\hat P(\sigma)^{-1}\hat w(\sigma)\|_{L^2(X)} \leq C'''|\operatorname{Re}\sigma|^{-3}$ for $|\operatorname{Re}\sigma| \geq C'$. This implies the final claim. □

Taking the limit $R \to \infty$ in (11.106) gives

$$\begin{aligned} v = v_1 + v_2, \quad v_1(t,\cdot) &:= \frac{1}{2\pi}\int_{\operatorname{Im}\sigma=-\alpha} e^{-i\sigma t}\hat P(\sigma)^{-1}\hat w(\sigma)\,\mathrm{d}\sigma, \\ v_2(t,\cdot) &:= \sum_{\substack{\sigma_0\in\operatorname{Res}(P)\\ \operatorname{Im}\sigma_0 > -\alpha}} i\operatorname{res}_{\sigma=\sigma_0}\big(e^{-i\sigma t}\hat P(\sigma)^{-1}\hat w(\sigma)\big). \end{aligned} \tag{11.110}$$

Writing $\sigma = \lambda - i\alpha$, $\lambda \in \mathbb{R}$, we have

$$v_1(t,\cdot) = \frac{1}{2\pi}e^{-\alpha t}\int_{\mathbb{R}} e^{-i\lambda t}\hat P(\lambda - i\alpha)^{-1}\mathcal{F}(e^{\alpha t}w)(\lambda)\,\mathrm{d}\lambda. \tag{11.111}$$

A variant of Proposition 2.27 shows that

$$\|\mathcal{F}(e^{\alpha t}w)\|_{L^2\left(\mathbb{R}_\lambda;\langle\lambda\rangle^{-(m-1)}\bar H^{m-1}_{\langle\lambda\rangle^{-1}}(X)\right)} \tag{11.112}$$

is bounded from above and below by a constant times $\|e^{\alpha t}w\|_{H^{m-1}(M)}$. The estimate (11.88) (for $\sigma = \lambda - i\alpha$ with $|\operatorname{Re}\lambda| \geq C'$) and the boundedness of $\hat P(\lambda - i\alpha)^{-1}$ (for $\lambda \in [-C', C']$) imply

$$\|\hat P(\lambda - i\alpha)^{-1}\mathcal{F}(e^{\alpha t}w)(\lambda)\|_{\langle\lambda\rangle^{-m}\bar H^m_{\langle\lambda\rangle^{-1}}(X)} \leq C\|\mathcal{F}(e^{\alpha t}w)(\lambda)\|_{\langle\lambda\rangle^{-(m-1)}\bar H^{m-1}_{\langle\lambda\rangle^{-1}}(X)}. \tag{11.113}$$

Using this information in (11.111) and using Proposition 2.27 again, we conclude that

$$\|v_1\|_{e^{-\alpha t}H^m(M)} \leq C\|w\|_{e^{-\alpha t}H^{m-1}(M)}. \tag{11.114}$$

Fix now a cutoff

$$\chi \in \mathcal{C}^\infty(\mathbb{R}), \quad \chi = 0 \text{ on } (-\infty, 1], \quad \chi = 1 \text{ on } [2,\infty). \tag{11.115}$$

We can then write

$$v = \chi v + (1-\chi)v = \chi v_2 + \big(\chi v_1 + (1-\chi)v\big). \tag{11.116}$$

The term χv_2 encodes the resonance expansion. Consider the second term: since $(1-\chi)v$ has compact support in time, we have

$$\|\chi v_1 + (1-\chi)v\|_{e^{-\alpha t}H^m(M)} \le C'\|w\|_{e^{-\alpha t}H^{m-1}(M)}. \tag{11.117}$$

We have now proved the following quantitative version of Theorem 11.11:

Theorem 11.24 (Late-Time Asymptotics: Sobolev Version). *Let $\alpha \in \mathbb{R}$ be such that $\operatorname{Im}\sigma \neq -\alpha$ for all $\sigma \in \operatorname{Res}(P)$. Define $\operatorname{Res}_\alpha(P) := \operatorname{Res}(P) \cap \{\operatorname{Im}\sigma > -\alpha\}$. Let χ be as in (11.115). Let $\mathbb{N} \ni m \ge 2$, $m > \frac{1}{2} + \alpha + \tilde\beta$. For $w \in e^{-\alpha t}H^{m-1}(M_+)^{\bullet,-}$, let v be the unique forward solution of $Pv = w$. Set*

$$\begin{aligned} u_{\sigma_0} &:= i\operatorname{res}_{\sigma=\sigma_0}\big(e^{-i\sigma t}\hat P(\sigma)^{-1}\hat w(\sigma)\big), \quad \sigma_0 \in \operatorname{Res}_\alpha(P), \\ \tilde v_\alpha(t,\cdot) &:= \chi(t)\frac{1}{2\pi}\int_{\operatorname{Im}\sigma=-\alpha} e^{-i\sigma t}\hat P(\sigma)^{-1}\hat w(\sigma)\,\mathrm{d}\sigma + (1-\chi(t))v(t,\cdot). \end{aligned} \tag{11.118}$$

Then $u_{\sigma_0} \in \operatorname{Res}(P,\sigma_0)$, $\tilde v_\alpha \in e^{-\alpha t}H^m(M_+)^{\bullet,-}$, and

$$v(t,\cdot) = \chi(t)\sum_{\sigma_0\in\operatorname{Res}_\alpha(P)} u_{\sigma_0} + \tilde v_\alpha(t,\cdot), \quad t \in \mathbb{R}. \tag{11.119}$$

Moreover, the map

$$\begin{aligned} S_\alpha\colon e^{-\alpha t}H^{m-1}(M_+)^{\bullet,-} \ni w \mapsto \big((u_{\sigma_0})_{\sigma_0\in\operatorname{Res}_\alpha(P)}, \tilde v_\alpha\big) \\ \in \bigoplus_{\sigma_0\in\operatorname{Res}_\alpha(P)} \operatorname{Res}(P,\sigma_0) \oplus e^{-\alpha t}H^m(M_+)^{\bullet,-} \end{aligned} \tag{11.120}$$

is linear and continuous.

Proof. We have proved (11.119) and the continuity of S_α when w is smooth and compactly supported. General source terms $w \in e^{-\alpha t}H^{m-1}(M_+)^{\bullet,-}$ are then handled using a density argument. □

The only reason why we have restricted m to be an integer in Theorem 11.24 is that we have not defined the space $e^{-\alpha t}H^m(M_+)^{\bullet,-}$ for non-integer m; we encourage the reader to give a treatment of the case $m \in \mathbb{R}$, $m > \frac{1}{2} + \alpha + \tilde\beta$.

Proof of Theorem 11.11. For $w \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}\times X)$ with $t \ge 0$ on $\operatorname{supp} w$, we can apply Theorem 11.24 for a sequence of $\alpha \in \mathbb{R}$ tending to ∞ and for all sufficiently large m (depending on α). The terms u_{σ_0} are independent of α, and the membership $\tilde v_\alpha \in e^{-\alpha t}H^m(M_+)^{\bullet,-}$ for all m implies, by Sobolev embedding, the pointwise bounds (11.37). □

Example 11.25 (Scalar Wave Equation). For the scalar wave operator $P = \Box_g$, all resonances σ satisfy $\operatorname{Im}\sigma \le -1$ except for $\sigma = 0$, and $\operatorname{Res}(P,0) = \operatorname{span}\{1\}$ (Example 11.8). Applying Theorem 11.24 with $\alpha \in (0,1)$ thus shows that forward solutions of $Pv = w \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}\times X)$ are of the form

$$v(t,x) = v_0 + \tilde{v}(t,x), \quad v_0 \in \mathbb{C}, \quad |\tilde{v}(t,x)| = \mathcal{O}(e^{-\alpha t}),\ t \to \infty. \tag{11.121}$$

That is, they decay exponentially fast to a constant.

11.6 A Nonlinear Toy Model

We illustrate the use of the quantitative estimates encoded in the boundedness of the map S_α in Theorem 11.24 by solving a (toy) nonlinear equation. As in Section 5.13, our main tool is the contraction mapping principle.

Theorem 11.26 (A Semilinear Wave Equation). *Let $\alpha \in (0,1)$, and let $m \geq 3$. Then there exists $\epsilon > 0$ such that the following holds. For every $w \in e^{-\alpha t} H^{m-1}(M_+)^{\bullet,-}$ with $\|w\|_{e^{-\alpha t} H^{m-1}(M_+)^{\bullet,-}} < \epsilon$, there exists a unique solution $v \in e^{-\alpha t} H^m(M_+)^{\bullet,-}$ of the semilinear wave equation*

$$\Box_g v + \langle \mathrm{d}v, \mathrm{d}v \rangle_{g^{-1}} = w. \tag{11.122}$$

For the proof, we use that for $s \geq 2$, the space $H^s(M)$ is an algebra under pointwise multiplication. (This uses $2 > \frac{\dim M}{2} = \frac{3}{2}$ and a variant of Exercise 5.29.) Therefore, $e^{-\alpha t} H^{m-1}(M_+)^{\bullet,-}$ is an algebra under pointwise multiplication for $m \geq 3$.

Proof of Theorem 11.26. For $\alpha \in (0,1)$, we have $\mathrm{Res}_\alpha(\Box_g) = \{0\}$, and $\mathrm{Res}(\Box_g, 0) = \mathrm{span}\{1\}$ (Example 11.8). We use the notation of Theorem 11.24. Keeping Example 11.25 in mind, we shall construct v from a fixed point of the map

$$T\colon (v_0, \tilde{v}) \mapsto S_\alpha\big(w - \langle \mathrm{d}(\chi v_0 + \tilde{v}), \mathrm{d}(\chi v_0 + \tilde{v}) \rangle_{g^{-1}}\big) \tag{11.123}$$

by $v = \chi v_0 + \tilde{v}$.

We claim that T maps the space

$$\mathcal{H} := \mathbb{C} \oplus e^{-\alpha t} H^m(M_+)^{\bullet,-} \tag{11.124}$$

to itself. To see this, note that each component of $\mathrm{d}(\chi v_0 + \tilde{v}) = v_0\, \mathrm{d}\chi + \mathrm{d}\tilde{v}$ in the frame $\mathrm{d}t, \mathrm{d}x_1, \mathrm{d}x_2$ lies in $e^{-\alpha t} H^{m-1}(M_+)^{\bullet,-}$. Therefore, $\langle \mathrm{d}(\chi v_0 + \tilde{v}), \mathrm{d}(\chi v_0 + \tilde{v}) \rangle_{g^{-1}} \in e^{-2\alpha t} H^{m-1}(M_+)^{\bullet,-}$ is, a fortiori, an element of the domain $e^{-\alpha t} H^{m-1}(M_+)^{\bullet,-}$ of S_α. Equipping $\mathcal{H}$ with the norm $\|(v_0, \tilde{v})\|_{\mathcal{H}} := |v_0| + \|\tilde{v}\|_{e^{-\alpha t} H^m(M_+)^{\bullet,-}}$, we moreover obtain

$$\|T(v_0, \tilde{v})\|_{\mathcal{H}} \leq C\big(\|w\|_{e^{-\alpha t} H^{m-1}(M_+)^{\bullet,-}} + \|(v_0, \tilde{v})\|_{\mathcal{H}}^2\big). \tag{11.125}$$

Let $\rho_0 > 0$ be so small that $C\rho_0^2 < \frac{1}{2}\rho_0$. Then for all $\rho_1 \in (0, \rho_0]$, and for w with norm less than $\epsilon := \frac{\rho_1}{2C}$, the map T maps the ρ_1-ball $B(0,\rho_1) \subset \mathcal{H}$ around $0 \in \mathcal{H}$ into itself.

To prove that T is a contraction on $B(0,\rho_1)$ for some $0 < \rho_1 \le \rho_0$, consider $V = (v_0, \tilde{v})$, $V' = (v_0', \tilde{v}') \in B(0,\rho_1)$. We then estimate the norm of

$$T(V) - T(V') = -S_\alpha \Big(\big\langle \mathrm{d}(\chi(v_0 - v_0') + (\tilde{v}_0 - \tilde{v}_0')), \mathrm{d}(\chi v_0 + \tilde{v}_0) \big\rangle_{g^{-1}} + \big\langle \mathrm{d}(\chi v_0' + \tilde{v}'), \mathrm{d}(\chi(v_0 - v_0') + (\tilde{v}_0 - \tilde{v}_0')) \big\rangle_{g^{-1}} \Big) \tag{11.126}$$

by $C\|V - V'\|_{\mathcal{H}} \|V + V'\|_{\mathcal{H}} \le 2C\rho_1 \|V - V'\|_{\mathcal{H}} < \frac{1}{2}\|V - V'\|_{\mathcal{H}}$ for $\rho_1 < \frac{1}{4C}$. The contraction mapping principle now applies and produces a fixed point $V = (v_0, \tilde{v})$ of T. By definition of T, the function $v := \chi v_0 + \tilde{v}$ solves (11.122).

The uniqueness of v follows by noting that the difference $u := v - v'$ of two solutions v, v' is supported in $t \ge 0$ and satisfies the homogeneous linear wave-type equation

$$Lu = 0, \quad L = \Box_g + \langle \mathrm{d}(\cdot), \mathrm{d}v \rangle_{g^{-1}} + \langle \mathrm{d}v', \mathrm{d}(\cdot) \rangle_{g^{-1}}. \tag{11.127}$$

The coefficients of the second term of L (which is of first order) are of class $\mathcal{C}^0$. This suffices to apply energy methods as in the proof of Theorem 9.19 (see the treatment of the term Vu in (9.47)–(9.49)), and hence, we conclude that the vanishing of u for $t < 0$ persists also for all $t \ge 0$. Therefore, $v = v'$. □

Notes

The proofs of Theorems 11.10 and 11.11 largely follow Vasy [Vas13], though some of the technical details differ. Vasy uses complex absorption (Exercise 8.6) beyond the horizon $r^{-1}(1)$ to "cap off" the problem beyond $r = 1$. More precisely, once the complex absorbing operator $Q_\sigma \in \mathrm{Diff}^2$ is elliptic in $r > \frac{3}{2}$, a doubling construction produces an operator $\hat{P}(\sigma) - iQ_\sigma$ on a compact manifold without boundary for which one can establish the Fredholm property (Exercise 8.7). While this approach is fully microlocal, some care is needed in the construction of Q_σ for it to be compatible with the high-energy analysis. The approach presented here, in which artificial boundaries are introduced, was first spelled out in Hintz–Vasy [HV15]. It is less microlocal near $r = 2$ in that it makes strong use of the *hyperbolic* and *differential* character of $\hat{P}(\sigma)$ in $r > 1$. It is, however, very natural from a wave equation perspective.

The arguments in Section 11.3 yielding the Fredholm property of the spectral family $\hat{P}(\sigma)$—despite $\hat{P}(\sigma)$ not being elliptic—are an instance of *non-elliptic Fredholm theory*. Another application of the same methods yields

a proof of the existence of a meromorphic continuation of so-called even asymptotically hyperbolic manifolds (M, g), via a suitable conjugation of the spectral family $\Delta_g - (\frac{n-1}{2})^2 - \sigma^2$ and an extension across the conformal boundary to a suitable smooth manifold. This extended operator shares all the features of $\hat{P}(\sigma)$ considered in this chapter. This approach to the meromorphic continuation is dubbed *Vasy's method*. Zworski gives a concise account of this method in [Zwo16]. (The meromorphic continuation was previously obtained in Mazzeo–Melrose [MM87] using a parametrix construction, with the evenness condition clarified in Guillarmou [Gui05].)

Another approach to the meromorphicity of $\hat{P}(\sigma)^{-1}$—in the setting of asymptotically anti–de Sitter spacetimes—and correspondingly to the definition of quasinormal modes is given in Warnick [War15]; it relies on energy methods. The crucial radial point estimates appear in the guise of the (enhanced) redshift estimates introduced in Dafermos–Rodnianski [DR09].

Other classes of spacetimes can be analyzed using extensions of the methods introduced in this chapter. An important example is the Kerr–de Sitter black hole spacetime, analyzed in Vasy [Vas13, §6]. The only additional feature of this spacetime is that the null-geodesic flow exhibits *normally hyperbolic trapping*. Black box microlocal estimates at such trapped sets were proved in Wunsch–Zworski [WZ11] and Dyatlov [Dya16]. Full resonance expansions were proved on Schwarzschild–de Sitter spacetimes in Bony–Häfner [BH08] using earlier work of Sá Barreto–Zworski [SBZ97], and on slowly rotating Kerr–de Sitter black hole spacetimes in Dyatlov [Dya12]. Partial resonance expansions in the full subextremal range of Kerr–de Sitter spacetimes were obtained in Petersen–Vasy [PV21]. One can also prove analogues of Theorem 11.11 on spacetimes which are not stationary but instead settle down at an exponential rate $e^{-\alpha t}$, $\alpha > 0$, to a de Sitter (or Kerr–de Sitter) spacetime. Further developments include solutions of nonlinear wave equations on such spacetimes [HV15, HV16] and the nonlinear stability of slowly rotating Kerr–de Sitter black holes [HV18]. The solution of nonlinear wave equations based on resonance expansions as in Section 11.6 was first discussed in Hintz–Vasy [HV15]. Theorem 11.26 is a special case of results proved there.

The asymptotic behavior of solutions of geometric wave equations on exact de Sitter space, such as $\Box_g v = w$ where $\Box_g = -\operatorname{tr}_g \nabla^2$ is the tensor wave operator on a tensor bundle over M, can be analyzed in a significantly simpler manner than done in this chapter, namely using energy methods and regular-singular ODE-type analysis on global de Sitter space (of which the manifold M studied in this chapter is a submanifold); see Exercise 11.1 for an indication of this approach. It originates in Hintz–Vasy [HV18, Appendix C] and Hintz–Xie [HX21, HX22] and is based on [Vas10]. Its limitation is that it does not apply to black hole spacetimes or even to general wave-type operators on (M, g) as analyzed in Theorem 11.11 (e.g., for $\Box_g + V$ where $V \in \mathcal{C}^\infty(X)$ is a generic stationary potential).

Exercises

Exercise 11.1 (Resonances of the Scalar Wave Operator). In this exercise, we prove all those claims in Example 11.8 which are needed for Example 11.25 and Theorem 11.26.

(1) Show that $0 \in \mathrm{Res}(\Box_g)$ and $1 \in \mathrm{Res}(\Box_g, 0)$.
(2) In terms of

$$\tau := e^{-t} > 0, \quad X := xe^{-t} \in \mathbb{R}^2, \tag{11.128}$$

show that

$$g = \frac{-\mathrm{d}\tau^2 + \mathrm{d}X^2}{\tau^2}, \quad \Box_g = \tau\partial_\tau(\tau\partial_\tau - 2) - (\tau\partial_{X_1})^2 - (\tau\partial_{X_2})^2. \tag{11.129}$$

(The manifold M from Definition 11.1 is thus the subset $|X| \leq 2\tau$ of $[0,\infty)_\tau \times \mathbb{R}^2_X$.)
(3) Show that the hypersurface $|X| = 2 + 2\tau$ is spacelike. Deduce that

$$\Omega := \{(\tau, X) \colon 0 < \tau \leq 1,\ |X| \leq 2 + 2\tau\} \tag{11.130}$$

is an admissible spacetime domain with initial hypersurface $\{\tau = 1\}$ and final hypersurface $\{|X| = 2 + 2\tau\}$. See Figure 11.7.
(4) We now consider g in (11.129) as a metric on $(0,1]_\tau \times \mathcal{X}$, $\mathcal{X} := (\mathbb{R}/5\mathbb{Z})_{X_1} \times (\mathbb{R}/5\mathbb{Z})_{X_2}$, and correspondingly $\Box_g$ as a wave operator on $((0,1] \times \mathcal{X}, g)$. Suppose that for all $f \in \mathcal{C}^\infty_{\mathrm{c}}((0,1) \times \mathcal{X})$, the unique forward solution u of $\Box_g u = f$ is of the form

$$u(\tau, X) = u_0(X) + \tilde{u}(\tau, X) \tag{11.131}$$

where

$$u_0 \in \mathcal{C}^\infty(\mathcal{X}), \quad |(\tau\partial_\tau)^j \partial_X^\alpha \tilde{u}(\tau, X)| \leq C_{j\alpha}\tau^\beta \quad j \in \mathbb{N}_0,\ \alpha \in \mathbb{N}_0^2, \tag{11.132}$$

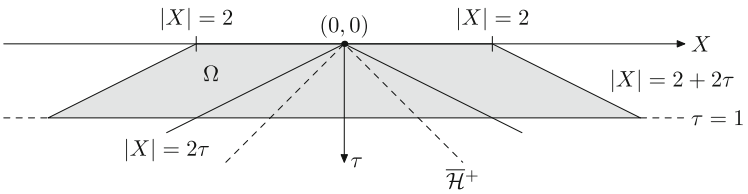


Figure 11.7 The domain Ω defined in (11.130).

for some fixed $\beta \in (1,2)$. Deduce from this that $\mathrm{Res}(\Box_g) \cap \{\mathrm{Im}\,\sigma \geq -\beta\} \subseteq \{0, -i\}$. Conclude using part (1) that

$$\mathrm{Res}(\Box_g) \cap \{\mathrm{Im}\,\sigma > -1\} = \{0\}, \quad \mathrm{Res}(\Box_g, 0) = \mathrm{span}\{1\}. \tag{11.133}$$

(5) In the remaining parts of this exercise, we establish (11.131)–(11.132) for $f \in \mathcal{C}^\infty_{\mathrm{c}}((0,1) \times \mathcal{X})$, and in fact more generally for f with support in $\tau < 1$ satisfying

$$|(\tau\partial_\tau)^j \partial_X^\alpha f(\tau, X)| \leq C_{j\alpha}\tau^\beta, \quad j \in \mathbb{N}_0,\ \alpha \in \mathbb{N}_0^2, \tag{11.134}$$

where $\beta \in (1,2)$ is fixed. Show first that the forward problem for $\Box_g u = f$ admits a unique global solution. Show then that for the energy

$$E[u](\tau) := \frac{1}{2}\int_{\mathcal{X}} |\tau\partial_\tau u(\tau, X)|^2 + |\tau\partial_X u(\tau, X)|^2 \,\mathrm{d}X, \tag{11.135}$$

we have the estimate

$$-\tau\partial_\tau E[u](\tau) \leq \int_{\mathcal{X}} |f(\tau, X)|^2 \,\mathrm{d}X. \tag{11.136}$$

Deduce that $\sup_{\tau\in(0,1]} E[u](\tau) < \infty$.

(6) Show that the operators $\partial_{X_1}, \partial_{X_2}$ and $\partial_t = -\tau\partial_\tau - X \cdot \partial_X$ commute with $\Box_g$. Deduce that

$$\sup_{\tau\in(0,1]} E[(\tau\partial_\tau)^k \partial_X^\gamma u] < \infty \quad \forall\, k \in \mathbb{N}_0,\ \gamma \in \mathbb{N}_0^2. \tag{11.137}$$

(7) Show that for every fixed $\eta < 0$ we have

$$|(\tau\partial_\tau)^j \partial_X^\alpha u(\tau, X)| \leq C_{j\alpha}\tau^\eta \quad \forall\, j \in \mathbb{N}_0,\ \alpha \in \mathbb{N}_0^2. \tag{11.138}$$

(8) We use (11.138) with $\eta = \beta - 2$. Rewrite the equation $\Box_g u = f$ as

$$\tau\partial_\tau(\tau\partial_\tau - 2)u = f + (\tau\partial_{X_1})^2 u + (\tau\partial_{X_2})^2 u. \tag{11.139}$$

Solve this ODE (with smooth parametric dependence on X) to deduce (11.131)–(11.132).

Remark. A direct approach to the computation of $\mathrm{Res}(\Box_g)$ based on separation of variables and the analysis of the resulting hypergeometric ordinary differential equation is presented in López–Ortega [LO06].

Exercise 11.2 (Initial Value Problems). Let P be a stationary wave-type operator on de Sitter space (M, g). Let $v_0, v_1 \in \mathcal{C}^\infty(X)$. Show that the

unique solution of the initial value problem $Pv = 0$, $v|_{t=0} = v_0$, $\partial_t v|_{t=0} = v_1$ admits a resonance expansion as $t \to \infty$, i.e., (11.36)–(11.37) hold.

Exercise 11.3 (Regularity of Dual Resonant States). We use the notation of Theorem 11.16. Show that if $u^* \in \dot{\mathscr{D}}'(X)$ satisfies $\hat{P}(\sigma)^* u^* = 0$, then:

(1) $u^* = 0$ for $r > 1$.
(2) $u^*|_{\{r<1\}} \in \mathcal{C}^\infty(\{r < 1\})$; more precisely, $\mathrm{WF}(u^*) \subset \mathcal{R}_+ \cup \mathcal{R}_-$.

Exercise 11.4 (Regularization). Let $N \in \mathbb{R}$.

(1) Construct a family $J_\epsilon \in \Psi^{-N}(\mathbb{R}^n)$, $\epsilon \in (0, 1]$, such that J_ϵ is bounded in $\Psi^0(\mathbb{R}^n)$, converges to I in $\Psi^\eta(\mathbb{R}^n)$ for all $\eta > 0$, and such that $\operatorname{supp} u \subset [0, \infty) \times \mathbb{R}^{n-1}$ implies $\operatorname{supp} J_\epsilon u \subset [0, \infty) \times \mathbb{R}^{n-1}$. (*Hint.* Exercise 9.4.)
(2) Let M be a closed manifold and let $\Omega \subset M$ be a smooth submanifold with boundary. Construct a family $J_\epsilon \in \Psi^{-N}(M)$, $\epsilon \in (0, 1]$, such that J_ϵ is bounded in $\Psi^0(M)$, converges to I in $\Psi^\eta(M)$ for all $\eta > 0$, and such that $\operatorname{supp} u \subset \Omega$ implies $\operatorname{supp} J_\epsilon u \subset \Omega$.
(3) We use the notation of the previous part. Fix a volume density on M. Let $P \in \mathrm{Diff}^m(M)$, and suppose $u \in \bar{H}^s(\Omega)$, $u^* \in \dot{H}^{-s+m-1}(\Omega)$ satisfy $Pu \in \bar{H}^{s-m+1}(\Omega)$ and $P^* u^* \in \dot{H}^{-s}(\Omega)$. Show that

$$\langle Pu, u^* \rangle_{L^2(\Omega)} = \langle u, P^* u^* \rangle_{L^2(\Omega)}. \tag{11.140}$$

(As a special case, this justifies the computation (11.71).)

Exercise 11.5 (Explicit Form of a Simple Pole: General Considerations). We use the notation of Theorem 11.16. Suppose $\sigma_0 \in \mathbb{C}$, $\operatorname{Im} \sigma_0 > -\alpha$, is a simple pole of $\hat{P}(\sigma)^{-1}$, i.e., in the notation used in (11.79) we have $J = 1$ and $\operatorname{rank} B_1 = 1$. Let $0 \neq u \in \ker_{\bar{H}^s(X)} \hat{P}(\sigma_0)$, $0 \neq u^* \in \ker_{\dot{H}^{-s+1}(X)} \hat{P}(\sigma_0)^*$. Show that

$$\hat{P}(\sigma)^{-1} f = (\sigma - \sigma_0)^{-1} \frac{\langle f, u^* \rangle_{L^2(X)}}{\langle \partial_\sigma \hat{P}(\sigma) u, u^* \rangle_{L^2(X)}} u + \mathrm{hol}(\sigma). \tag{11.141}$$

Exercise 11.6 (Explicit Form of a Simple Pole: Scalar Wave Operator). Continuing Exercise 11.5, consider the scalar wave operator $P := \Box_g$ and $\sigma_0 = 0 \in \mathrm{Res}(P)$. Find u^* and compute the first term in (11.141).

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